

Quantum Mechanics for People like me

Study Notes on Mathematical Foundations, Perturbation Theory, Scattering,
and Relativistic Mechanics

Department of Physics / Quantum Mechanics Notes

Personal Study Notes
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September 25, 2026

Preface

These are my personal study notes on quantum mechanics, written the way I always wanted to learn the subject: starting from foundational operator principles and working through every derivation and calculation explicitly, without skipping intermediate algebraic steps, glossing over boundary conditions, or relying on hand-waving arguments.

Throughout, the goal has been to **derive every calculation explicitly**: filling in the algebraic steps, boundary matching conditions, and physical reasoning that standard textbooks frequently leave as exercises or pass over without complete proof. This is as much for my own mastery of the material as for any other student navigating this demanding subject.

The notes span the core curriculum of advanced non-relativistic and relativistic quantum mechanics:

- **Mathematical Foundations & Pictures of QM:** Vector spaces, Dirac bra-ket notation, coordinate representations, and unitary time evolution in the Schrödinger, Heisenberg, and Dirac interaction pictures.
- **Exact Soluble Systems:** The harmonic oscillator developed through both ladder operator orderings ($a^\dagger a$ and $aa^\dagger$), the infinite and finite potential wells, and central potentials.
- **Angular Momentum & Symmetry:** Commutation relations, ladder representations, the addition of angular momenta via Clebsch–Gordan coefficients, tensor operators, and the Wigner–Eckart theorem.
- **Approximation Methods:** Time-independent perturbation theory (including non-degenerate, degenerate, fine structure, Zeeman, and Stark effects), the variational principle, and the WKB semi-classical approximation.
- **Time-Dependent Dynamics & Transitions:** Two-level dynamics, Fermi’s Golden Rule, harmonic perturbations, spontaneous emission, decaying states with non-Hermitian effective Hamiltonians, and resonance line broadening.
- **Quantum Scattering Theory:** Lippmann–Schwinger integral equations, Green’s functions, the Born approximation, partial wave analysis, phase shifts, low-energy resonances, and an exact order-by-order proof of the Optical Theorem via the Sokhotski–Plemelj formula.
- **Relativistic Wave Equations & Field Theory:** Covariant formulation of the Klein–Gordon and Dirac equations, Dirac spinor algebra, bilinear covariants, the Gordon decomposition, the Klein paradox, and the breakdown of single-particle mechanics mandating the transition to Quantum Field Theory.

Figures and diagrams are drawn using TikZ to stay faithful to the geometric and physical intuition behind sequential measurement processes, state projections, and coordinate systems.

I welcome feedback, corrections, or discussions regarding the content of these notes – please feel free to reach out by email.

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Disclaimer: These are pedagogical study notes. While I have strived for accuracy, any errors in the calculations, typos, or physical interpretations remain entirely my own. I am actively revising and correcting the notes; the latest version can always be found on the project page at lunar-photon.github.io.

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Chapter 1

Prerequisite & Mathematical Foundations

Quantum mechanics is, at its mathematical core, an eigenvalue problem formulated on a complex vector space equipped with an inner product. All physically measurable observables correspond to eigenvalues of self-adjoint (Hermitian) operators acting on this space. Before embarking on the dynamical pictures and advanced applications of quantum theory, we must establish the mathematical infrastructure that underpins the entire formalism: Dirac's bra-ket notation, the spectral theory of operators, the rigorous meaning of continuous spectra via rigged Hilbert spaces, the generation of spatial translations, the first-principles derivation of canonical commutation relations, and the coordinate representations of physical observables.

1.1 Dirac Notation and Hilbert Space

In classical mechanics, the state of a system of N particles is specified by a point in a $6N$ -dimensional phase space $(q_1, \dots, q_{3N}, p_1, \dots, p_{3N})$. In quantum mechanics, the state of an isolated physical system is represented by a vector residing in a complex vector space known as a **Hilbert space** $\mathcal{H}$, or more precisely, by a ray (an equivalence class of vectors differing only by a non-zero complex scalar).

1.1.1 Kets, Bras, and Inner Products

Following the notation introduced by P. A. M. Dirac, an element of the Hilbert space $\mathcal{H}$ is denoted as a **ket** $|\psi\rangle$. The Hilbert space is an inner product space that is complete in the metric induced by its inner product.

For every ket space $\mathcal{H}$, there exists a dual space $\mathcal{H}^*$ consisting of all continuous linear functionals that map kets to the complex numbers $\mathbb{C}$. An element of this dual space is called a **bra** and is written as $\langle\phi|$. According to the Riesz representation theorem, there is a one-to-one antilinear correspondence between $\mathcal{H}$ and $\mathcal{H}^*$:

$$|\psi\rangle \xleftrightarrow{\text{dual correspondence}} \langle\psi| = (|\psi\rangle)^\dagger. \quad (1.1.1)$$

The action of the linear functional $\langle\phi|$ on the ket $|\psi\rangle$ defines the **inner product**, written in Dirac bracket notation as:

$$\langle\phi|\psi\rangle \in \mathbb{C}. \quad (1.1.2)$$

The inner product satisfies the following fundamental mathematical axioms:

1. Skew-symmetry (Conjugate symmetry):

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*. \quad (1.1.3)$$

2. **Linearity in the ket (second argument):** For any complex numbers $c_1, c_2 \in \mathbb{C}$ and kets $|\psi_1\rangle, |\psi_2\rangle$,

$$\langle\phi|c_1\psi_1 + c_2\psi_2\rangle = c_1 \langle\phi|\psi_1\rangle + c_2 \langle\phi|\psi_2\rangle. \quad (1.1.4)$$

3. **Antilinearity in the bra (first argument):**

$$\langle c_1\phi_1 + c_2\phi_2|\psi\rangle = c_1^* \langle\phi_1|\psi\rangle + c_2^* \langle\phi_2|\psi\rangle. \quad (1.1.5)$$

4. **Positive-definiteness:** For any ket $|\psi\rangle$,

$$\langle\psi|\psi\rangle \geq 0, \quad \text{with} \quad \langle\psi|\psi\rangle = 0 \iff |\psi\rangle = 0. \quad (1.1.6)$$

The norm of a state vector is defined by $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$. Two states $|\phi\rangle$ and $|\psi\rangle$ are said to be **orthogonal** if $\langle\phi|\psi\rangle = 0$. A physical state vector is conventionally normalized to unity:

$$\langle\psi|\psi\rangle = 1. \quad (1.1.7)$$

While the bracket $\langle\phi|\psi\rangle$ is introduced here algebraically through the Riesz representation theorem, it possesses an exact operational foundation in probability theory: it is the **transition probability amplitude**. In Section 1.2.2, we prove from first principles that evaluating the expectation value of the projection operator $\hat{P}_\phi = |\phi\rangle\langle\phi|$ on state $|\psi\rangle$ yields $P(\phi | \psi) = |\langle\phi|\psi\rangle|^2$, revealing why the inner product is defined in this precise manner.

The Cauchy–Schwarz Inequality

From the positive-definiteness axiom of the inner product follows the celebrated **Cauchy–Schwarz inequality**. For any two kets $|\psi\rangle, |\phi\rangle \in \mathcal{H}$ and any arbitrary complex scalar $\lambda \in \mathbb{C}$, consider the non-negative squared norm of the linear combination $|\psi\rangle + \lambda|\phi\rangle$:

$$\| |\psi\rangle + \lambda|\phi\rangle \|^2 = \langle\psi + \lambda\phi|\psi + \lambda\phi\rangle \geq 0. \quad (1.1.8)$$

Expanding using the conjugate symmetry and linearity axioms:

$$\langle\psi + \lambda\phi|\psi + \lambda\phi\rangle = \langle\psi|\psi\rangle + \lambda \langle\psi|\phi\rangle + \lambda^* \langle\phi|\psi\rangle + |\lambda|^2 \langle\phi|\phi\rangle \geq 0. \quad (1.1.9)$$

Assuming $|\phi\rangle \neq 0$ (if $|\phi\rangle = 0$, the inequality is trivially $0 \leq 0$), we make the optimal choice for the complex scalar:

$$\lambda = -\frac{\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle}. \quad (1.1.10)$$

Substituting this choice into the expansion yields:

$$\begin{aligned} \langle\psi|\psi\rangle - \frac{\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle} \langle\psi|\phi\rangle - \frac{\langle\phi|\psi\rangle^*}{\langle\phi|\phi\rangle} \langle\phi|\psi\rangle + \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle^2} \langle\phi|\phi\rangle &\geq 0 \\ \langle\psi|\psi\rangle - \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle} - \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle} + \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle} &\geq 0 \\ \langle\psi|\psi\rangle - \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle} &\geq 0. \end{aligned} \quad (1.1.11)$$

Multiplying through by the strictly positive scalar $\langle\phi|\phi\rangle > 0$ establishes the **Cauchy–Schwarz inequality**:

$$|\langle\phi|\psi\rangle|^2 \leq \langle\phi|\phi\rangle \langle\psi|\psi\rangle \iff |\langle\phi|\psi\rangle| \leq \|\phi\| \|\psi\|. \quad (1.1.12)$$

Equality holds if and only if the two vectors are linearly dependent, $|\psi\rangle = c|\phi\rangle$ for some $c \in \mathbb{C}$.

From Inner Product Space to Normed Space and Metric Space

A profound structural insight emphasized in rigorous formulations of quantum mechanics [1] concerns the hierarchical transition from vector space to metric space:

1. **Linear Vector Space:** Endowed only with vector addition and scalar multiplication; lengths and angles do not exist.
2. **Inner Product Space:** Endowed with the scalar map $\langle \phi | \psi \rangle$ satisfying conjugate symmetry, linearity, and positive definiteness.
3. **Normed Vector Space:** The inner product naturally induces a candidate norm (length):

$$\|\psi\| \equiv \sqrt{\langle \psi | \psi \rangle}. \quad (1.1.13)$$

To qualify as a genuine norm, $\|\psi\|$ must satisfy three axioms:

- **Positivity:** $\|\psi\| \geq 0$, with $\|\psi\| = 0 \iff |\psi\rangle = 0$ (follows from positive-definiteness).
- **Absolute homogeneity:** $\|c\psi\| = |c|\|\psi\|$ for all $c \in \mathbb{C}$ (follows from linearity and conjugate symmetry).
- **The Triangle Inequality:** $\|\psi + \phi\| \leq \|\psi\| + \|\phi\|$.

Notice that **the triangle inequality is NOT an axiom of an inner product space!** Rather, it is a non-trivial theorem proven directly from the **Cauchy–Schwarz inequality**:

$$\begin{aligned} \|\psi + \phi\|^2 &= \langle \psi + \phi | \psi + \phi \rangle = \langle \psi | \psi \rangle + \langle \phi | \phi \rangle + \langle \psi | \phi \rangle + \langle \phi | \psi \rangle \\ &= \|\psi\|^2 + \|\phi\|^2 + 2\operatorname{Re} \langle \psi | \phi \rangle \\ &\leq \|\psi\|^2 + \|\phi\|^2 + 2|\langle \psi | \phi \rangle| \\ &\leq \|\psi\|^2 + \|\phi\|^2 + 2\|\psi\|\|\phi\| = (\|\psi\| + \|\phi\|)^2. \end{aligned} \quad (1.1.14)$$

Taking the non-negative square root yields the triangle inequality for the norm:

$$\boxed{\|\psi + \phi\| \leq \|\psi\| + \|\phi\|}. \quad (1.1.15)$$

4. **Metric Space:** Any normed space induces a natural distance function (metric) between any two states:

$$d(\psi, \phi) \equiv \|\psi - \phi\| = \sqrt{\langle \psi - \phi | \psi - \phi \rangle}. \quad (1.1.16)$$

A metric space (M, d) is defined by three fundamental axioms:

- $d(\psi, \phi) \geq 0$, and $d(\psi, \phi) = 0 \iff |\psi\rangle = |\phi\rangle$.
- **Symmetry:** $d(\psi, \phi) = d(\phi, \psi)$.
- **The Triangle Inequality for Distances (an essential axiom for metric spaces):**

$$d(\psi, \chi) \leq d(\psi, \phi) + d(\phi, \chi). \quad (1.1.17)$$

Evaluating the metric triangle inequality using the norm:

$$d(\psi, \chi) = \|\psi - \chi\| = \|(\psi - \phi) + (\phi - \chi)\| \leq \|\psi - \phi\| + \|\phi - \chi\| = d(\psi, \phi) + d(\phi, \chi), \quad (1.1.18)$$

we see that it holds *precisely because* the triangle inequality holds for the norm!

The Mathematical Bridge: How Schwarz Converts Normed Spaces into Metric Spaces

Notice the logical direction of the axiomatic architecture:

- The **triangle inequality is an independent axiom of a metric space**; a general distance function cannot be called a metric unless it satisfies this axiom by definition.
- Conversely, the **triangle inequality is NOT an axiom of an inner product space**; the inner product is axiomatized entirely by algebraic properties (conjugate symmetry, linearity, positive-definiteness).
- It is the **Cauchy–Schwarz inequality** that generates the triangle inequality for the norm, which in turn endows the inner product space with a rigorous metric $d(\psi, \phi) = \|\psi - \phi\|$.

When an inner product space is **complete** under the topology induced by this metric (meaning that every Cauchy sequence of state vectors $\{|\psi_n\rangle\}$ converges to a limit state vector $|\psi\rangle$ that also lies inside the space), it is designated a **Hilbert space** $\mathcal{H}$.

1.1.2 Linear Operators, Adjoints, and the Role of Self-Adjointness

In quantum mechanics, physical transformations and dynamical variables are represented by linear operators. A linear operator $\hat{A}$ is fundamentally a mapping from the space of kets to itself:

$$\hat{A} : \mathcal{H} \rightarrow \mathcal{H}, \quad |\psi\rangle \mapsto \hat{A}|\psi\rangle \equiv |\hat{A}\psi\rangle, \quad (1.1.19)$$

preserving linear superposition: $\hat{A}(c_1|\psi_1\rangle + c_2|\psi_2\rangle) = c_1\hat{A}|\psi_1\rangle + c_2\hat{A}|\psi_2\rangle$.

Operators on Bras and the Definition of the Adjoint

How does an operator act on the dual space of linear functionals (bras)? Let $\langle\phi| \in \mathcal{H}^*$ be an arbitrary bra vector. We can define the action of the operator $\hat{A}$ on the bra $\langle\phi|$ by considering the new linear functional that assigns to any ket $|\psi\rangle$ the complex number obtained by letting $\langle\phi|$ act on the transformed ket $\hat{A}|\psi\rangle$:

$$(\langle\phi|\hat{A})(|\psi\rangle) \equiv \langle\phi|\hat{A}\psi\rangle. \quad (1.1.20)$$

Because the mapping $|\psi\rangle \mapsto \langle\phi|\hat{A}\psi\rangle$ is a continuous linear functional on $\mathcal{H}$, the Riesz representation theorem guarantees that there exists a *unique* bra vector in $\mathcal{H}^*$, which we denote by $\langle\chi|$, such that:

$$(\langle\phi|\hat{A})|\psi\rangle = \langle\chi|\psi\rangle \quad \text{for all } |\psi\rangle \in \mathcal{H}. \quad (1.1.21)$$

Since $\langle\chi|$ is uniquely determined by $\langle\phi|$ (given $\hat{A}$), this defines a linear mapping on the dual space. We define the **adjoint operator** $\hat{A}^\dagger$ such that the corresponding ket to $\langle\chi|$ is $|\chi\rangle = \hat{A}^\dagger|\phi\rangle$, which means:

$$|\chi\rangle = \langle\hat{A}^\dagger\phi| \implies \langle\hat{A}^\dagger\phi|\psi\rangle = \langle\phi|\hat{A}\psi\rangle \equiv \langle\phi|\hat{A}|\psi\rangle. \quad (1.1.22)$$

This reveals the fundamental asymmetric duality of operators:

An operator $\hat{A}$ is fundamentally a map from kets to kets ($|\psi\rangle \mapsto \hat{A}|\psi\rangle$), while its adjoint $\hat{A}^\dagger$ is fundamentally a map from bras to bras ($\langle\phi| \mapsto \langle\phi|\hat{A} = \langle\hat{A}^\dagger\phi|$).

Taking the complex conjugate of the matrix element yields:

$$\langle \phi | \hat{A} | \psi \rangle^* = \langle \phi | \hat{A} \psi \rangle^* = \langle \hat{A} \psi | \phi \rangle = \langle \psi | \hat{A}^\dagger | \phi \rangle. \quad (1.1.23)$$

The algebraic properties of the adjoint operation are:

$$(c\hat{A})^\dagger = c^* \hat{A}^\dagger, \quad (1.1.24)$$

$$(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger, \quad (1.1.25)$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger, \quad (1.1.26)$$

$$(\hat{A}^\dagger)^\dagger = \hat{A}. \quad (1.1.27)$$

Self-Adjoint Operators: Symmetrical Action on Kets and Bras

An operator is said to be **Hermitian** (or symmetric) if it satisfies $\langle \phi | \hat{A} | \psi \rangle = \langle \psi | \hat{A} | \phi \rangle^*$ for all vectors in its domain. In an infinite-dimensional Hilbert space, an operator is strictly **self-adjoint** if $\hat{A}^\dagger = \hat{A}$ and their domains coincide: $\mathcal{D}(\hat{A}^\dagger) = \mathcal{D}(\hat{A})$.

Self-adjoint operators are the unique class of operators that can act both to the right on kets and to the left on bras with identical effect:

$$\boxed{\langle \phi | \hat{A} | \psi \rangle = \langle \phi | (\hat{A} | \psi \rangle) = (\langle \phi | \hat{A}) | \psi \rangle.} \quad (1.1.28)$$

In Dirac's bra-ket notation, this bidirectional freedom is what allows us to omit parentheses entirely when writing transition amplitudes $\langle \phi | \hat{A} | \psi \rangle$.

Furthermore, if $|a\rangle$ is an eigenvector of a self-adjoint operator with eigenvalue $a \in \mathbb{R}$:

$$\hat{A} |a\rangle = a |a\rangle, \quad (1.1.29)$$

taking the adjoint immediately yields the left-handed eigenvector equation:

$$\langle a | \hat{A}^\dagger = \langle a | \hat{A} = a \langle a|. \quad (1.1.30)$$

The self-adjoint property ensures that the conjugate bra $\langle a|$ to any eigenket $|a\rangle$ is simultaneously a left-eigenvector with the identical, strictly real eigenvalue a . In quantum mechanics, all physical observables (Hamiltonian, momentum, coordinate, spin) must be self-adjoint operators.

1.1.3 The Eigenvalue Problem and Spectral Decomposition

For a Hermitian operator $\hat{A}$, the eigenvalue equation takes the form:

$$\hat{A} |a_n\rangle = a_n |a_n\rangle. \quad (1.1.31)$$

Taking the Hermitian conjugate gives the corresponding bra eigenvalue equation:

$$\langle a_n | \hat{A}^\dagger = \langle a_n | \hat{A} = a_n^* \langle a_n|. \quad (1.1.32)$$

Hermitian operators possess two crucial properties that make them suitable for representing physical observables:

1. **Real Eigenvalues:** Evaluating the expectation value $\langle a_n | \hat{A} | a_n \rangle$:

$$a_n \langle a_n | a_n \rangle = \langle a_n | \hat{A} | a_n \rangle = \langle a_n | \hat{A}^\dagger | a_n \rangle = a_n^* \langle a_n | a_n \rangle. \quad (1.1.33)$$

Since $\langle a_n | a_n \rangle \neq 0$, it follows that $a_n = a_n^*$, meaning all eigenvalues of a Hermitian operator are real numbers.

2. Orthogonality of Eigenstates: Consider two eigenstates $|a_n\rangle$ and $|a_m\rangle$ with eigenvalues $a_n \neq a_m$:

$$\langle a_m | \hat{A} | a_n \rangle = a_n \langle a_m | a_n \rangle. \quad (1.1.34)$$

Acting with $\hat{A}$ to the left yields:

$$\langle a_m | \hat{A} | a_n \rangle = a_m \langle a_m | a_n \rangle. \quad (1.1.35)$$

Subtracting the two equations gives:

$$(a_n - a_m) \langle a_m | a_n \rangle = 0 \implies \langle a_m | a_n \rangle = 0 \quad (\text{for } a_n \neq a_m). \quad (1.1.36)$$

For degenerate eigenvalues ($a_n = a_m$ for $n \neq m$), the Gram-Schmidt orthogonalization procedure can always be applied to construct an orthonormal basis spanning the degenerate eigenspace. Hence, we can always choose an orthonormal set:

$$\langle a_m | a_n \rangle = \delta_{mn}. \quad (1.1.37)$$

1.1.4 Projection Operators and Completeness

The outer product of a normalized ket $|a_n\rangle$ with its dual bra $\langle a_n|$ defines a **projection operator** onto the one-dimensional subspace spanned by $|a_n\rangle$:

$$\hat{P}_n \equiv |a_n\rangle \langle a_n|. \quad (1.1.38)$$

Projection operators satisfy two characteristic algebraic identities:

$$\hat{P}_n^\dagger = (|a_n\rangle \langle a_n|)^\dagger = |a_n\rangle \langle a_n| = \hat{P}_n \quad (\text{Hermiticity}), \quad (1.1.39)$$

$$\hat{P}_n \hat{P}_m = |a_n\rangle \langle a_n | a_m \rangle \langle a_m| = \delta_{nm} \hat{P}_n \quad (\text{Idempotence / Orthogonality}). \quad (1.1.40)$$

Because the eigenstates of an observable form a complete basis for the Hilbert space $\mathcal{H}$, any arbitrary physical state $|\psi\rangle$ can be expanded as a linear superposition:

$$|\psi\rangle = \sum_n c_n |a_n\rangle. \quad (1.1.41)$$

Taking the inner product with $\langle a_m|$:

$$\langle a_m | \psi \rangle = \sum_n c_n \langle a_m | a_n \rangle = \sum_n c_n \delta_{mn} = c_m. \quad (1.1.42)$$

Thus, the expansion coefficients are precisely the inner products $c_n = \langle a_n | \psi \rangle$, representing the probability amplitude of finding the system in eigenstate $|a_n\rangle$. Substituting this back into the expansion gives:

$$|\psi\rangle = \sum_n |a_n\rangle \langle a_n | \psi \rangle = \left(\sum_n |a_n\rangle \langle a_n| \right) |\psi\rangle. \quad (1.1.43)$$

Since this identity holds for any arbitrary ket $|\psi\rangle$, we obtain the **completeness relation** (or resolution of the identity):

$$\sum_n \hat{P}_n = \sum_n |a_n\rangle \langle a_n| = \hat{I}. \quad (1.1.44)$$

Using the completeness relation, any Hermitian operator $\hat{A}$ can be expressed in its **spectral representation**:

$$\hat{A} = \hat{A} \hat{I} = \hat{A} \sum_n |a_n\rangle \langle a_n| = \sum_n a_n |a_n\rangle \langle a_n|. \quad (1.1.45)$$

This spectral representation, paired with the completeness relation $\sum_n |a_n\rangle \langle a_n| = \hat{I}$, serves as the exact algebraic mechanism connecting classical probability distributions to coordinate-free quantum expectation values, as we establish from first principles in Section 1.2.2.

1.1.5 Commuting Observables and Simultaneous Diagonalization

Let $\hat{A}$ and $\hat{B}$ be two Hermitian operators. If they commute,

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A} = 0, \quad (1.1.46)$$

then they share a simultaneous complete set of eigenstates. To see this, assume $\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle$ with non-degenerate eigenvalues. Then:

$$\hat{A}(\hat{B}|\psi_n\rangle) = \hat{B}(\hat{A}|\psi_n\rangle) = a_n(\hat{B}|\psi_n\rangle). \quad (1.1.47)$$

Thus, $\hat{B}|\psi_n\rangle$ is an eigenstate of $\hat{A}$ belonging to eigenvalue a_n . Since the eigenspace of a_n is one-dimensional, $\hat{B}|\psi_n\rangle$ must be proportional to $|\psi_n\rangle$:

$$\hat{B}|\psi_n\rangle = b_n|\psi_n\rangle. \quad (1.1.48)$$

Therefore, $|\psi_n\rangle$ is simultaneously an eigenstate of both $\hat{A}$ and $\hat{B}$.

If two operators do not commute ($[\hat{A}, \hat{B}] \neq 0$), they are termed **incompatible observables**. They cannot be simultaneously diagonalized, and there is no complete basis of mutual eigenstates. A measurement of one observable inevitably introduces an irreducible quantum uncertainty into the other, as quantified by the generalized Robertson-Schrödinger uncertainty relation:

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|. \quad (1.1.49)$$

1.2 The Postulates of Quantum Mechanics

Quantum theory is founded on a self-consistent set of mathematical and physical postulates that translate physical concepts into operators and state vectors.

- **Postulate 1 (State Space):** The physical state of an isolated quantum-mechanical system is completely specified at any instant of time by a normalized ray or state vector $|\psi\rangle$ residing in a complex Hilbert space $\mathcal{H}$ ($\langle\psi|\psi\rangle = 1$).
- **Postulate 2 (Observables):** To every physically measurable dynamical variable $\mathcal{A}$ in classical physics, there corresponds a linear Hermitian operator $\hat{A} = \hat{A}^\dagger$ acting in $\mathcal{H}$.
- **Postulate 3 (Measurement and State Collapse):**

1. The only possible outcome of a measurement of an observable $\hat{A}$ is one of the eigenvalues a_n of the operator $\hat{A}$.
2. **Born Rule:** If the system is in normalized state $|\psi\rangle$, the probability $P(a_n)$ of obtaining eigenvalue a_n in an idealized projective measurement is given by:

$$P(a_n) = \langle\psi|\hat{P}_n|\psi\rangle = |\langle a_n|\psi\rangle|^2 = |c_n|^2, \quad (1.2.1)$$

where $\hat{P}_n = |a_n\rangle\langle a_n|$ is the projection operator onto the eigenspace of a_n . For degenerate eigenvalues with orthonormal basis $\{|a_n, k\rangle\}_{k=1}^{d_n}$, the probability is:

$$P(a_n) = \sum_{k=1}^{d_n} |\langle a_n, k|\psi\rangle|^2 = \langle\psi|\hat{P}_n|\psi\rangle, \quad \hat{P}_n = \sum_{k=1}^{d_n} |a_n, k\rangle\langle a_n, k|. \quad (1.2.2)$$

3. **State Reduction (Lüders Collapse):** Measurement is an active physical intervention that changes the state vector of the system. Immediately upon obtaining the measurement result a_n , the state vector undergoes a discontinuous projection (wavefunction collapse):

$$|\psi\rangle \xrightarrow{\text{measurement yields } a_n} \frac{\hat{P}_n|\psi\rangle}{\sqrt{\langle\psi|\hat{P}_n|\psi\rangle}} = |a_n\rangle. \quad (1.2.3)$$

1.2.1 Measurement Dynamics: Eigenstate Invariance vs. Non-Commuting Collapse

Let us carefully examine the profound consequences of Postulate 3:

1. **Measurement on an Eigenstate:** Suppose the system has already been prepared in an eigenstate $|a_n\rangle$ of observable $\hat{A}$. When $\hat{A}$ is measured:

$$P(a_m) = |\langle a_m | a_n \rangle|^2 = \delta_{mn}. \quad (1.2.4)$$

The measurement yields the eigenvalue a_n with absolute certainty ($P = 1$). The post-measurement collapsed state is:

$$|a_n\rangle \xrightarrow{\text{measurement of } \hat{A}} \frac{\hat{P}_n |a_n\rangle}{\sqrt{\langle a_n | \hat{P}_n | a_n \rangle}} = |a_n\rangle. \quad (1.2.5)$$

Thus, **measuring an observable when the system is already in its eigenstate leaves the physical state completely invariant!** Repeated measurements of $\hat{A}$ will continue to yield a_n deterministically.

2. **Measurement of Non-Commuting Observables:** Now suppose that after measuring $\hat{A}$ and collapsing into $|a_n\rangle$, we immediately perform a measurement of a second observable $\hat{B}$ that does *not* commute with $\hat{A}$ ($[\hat{A}, \hat{B}] \neq 0$).

Because $[\hat{A}, \hat{B}] \neq 0$, the state $|a_n\rangle$ is generally *not* an eigenstate of $\hat{B}$. Instead, using the completeness relation for the eigenbasis $\{|b_m\rangle\}$ of $\hat{B}$, $|a_n\rangle$ must be expressed as a linear superposition:

$$|a_n\rangle = \sum_m \langle b_m | a_n \rangle |b_m\rangle. \quad (1.2.6)$$

Measuring $\hat{B}$ on the state $|a_n\rangle$ will yield eigenvalue b_m with transition probability:

$$P(b_m) = |\langle b_m | a_n \rangle|^2. \quad (1.2.7)$$

Upon obtaining result b_m , the state vector instantly collapses to the new eigenstate:

$$|a_n\rangle \xrightarrow{\text{measurement of } \hat{B}} |b_m\rangle. \quad (1.2.8)$$

Notice what has occurred! The state is now $|b_m\rangle$. If we subsequently measure $\hat{A}$ a second time, the state $|b_m\rangle$ is itself a superposition of the eigenstates of $\hat{A}$:

$$|b_m\rangle = \sum_k \langle a_k | b_m \rangle |a_k\rangle. \quad (1.2.9)$$

The second measurement of $\hat{A}$ will *no longer* yield a_n with certainty! Instead, it will yield any eigenvalue a_k with probability $|\langle a_k | b_m \rangle|^2$. The intermediate measurement of the non-commuting observable $\hat{B}$ has irreversibly disrupted the quantum coherence, scrambling the information previously established about $\hat{A}$.

Physical Realization: Sequential Stern–Gerlach Filters and Quantum Markov Chains

The successive measurement of incompatible observables described here is not a formal mathematical abstraction; it is the physical reality discovered by Otto Stern and Walther Gerlach in 1922. In Chapter 5, Section 5.3.6, we analyze the three-stage filter ($\mathbf{SG}_z \rightarrow \mathbf{SG}_x \rightarrow \mathbf{SG}_z$), demonstrate how measuring $\hat{S}_x$ fundamentally resets the state $|\psi\rangle \equiv |x+\rangle$ to regenerate a 50% probability of observing $z-$, provide a complete TikZ experimental

schematic, and formulate wavefunction collapse as a memoryless **quantum Markov process**.

1.2.2 Expectation Values, Statistical Moments, and the Foundations of the Inner Product

In introductory expositions of quantum mechanics, the expectation value of an observable $\hat{A}$ in state $|\psi\rangle$ is commonly introduced by fiat via the algebraic sandwich:

$$\langle \hat{A} \rangle_\psi \equiv \langle \psi | \hat{A} | \psi \rangle. \quad (1.2.10)$$

When presented in this manner, students frequently wonder: *Why is the expectation value defined in this particular way? Is $\langle \psi | \hat{A} | \psi \rangle$ an arbitrary definition, an independent physical postulate, or an ad hoc mathematical convention?*

Furthermore, a parallel question touches the very core of Dirac's formulation: *Why is the state space equipped with a complex inner product $\langle \phi | \psi \rangle$, and why is physical transition probability defined by the modulus squared of this bracket?*

In this section, we show that $\langle \psi | \hat{A} | \psi \rangle$ is **not an arbitrary convention or independent postulate**. Rather, it is the **exact, unique coordinate-free operator realization of the classical statistical expectation value** $\mathbb{E}[X] = \sum_i a_i P(a_i)$ (and its continuous counterpart $\int x \rho(x) dx$) under the spectral theorem and the Born rule. We then demonstrate that the Hilbert space inner product $\langle \phi | \psi \rangle$ itself arises directly from the statistical expectation value of projection operators representing yes/no experimental propositions.

Eigenvalues vs. Expectation Values: The Conceptual Divide

Before carrying out the derivation, we must sharply distinguish between two quantities that are frequently conflated:

- **Eigenvalue (a_n):** An eigenvalue is the *single, deterministic numerical outcome* registered by a measuring apparatus during an individual measurement event performed on a single quantum system. The spectrum of eigenvalues $\{a_n\}$ is an intrinsic characteristic of the operator $\hat{A}$ itself (representing the dynamical observable) and is completely independent of the prepared physical state $|\psi\rangle$.
- **Expectation Value ($\langle \hat{A} \rangle_\psi$):** An expectation value is the *statistical mean (average)* of the measurement outcomes obtained by measuring $\hat{A}$ over an identically prepared statistical ensemble of independent quantum systems, each prepared in the identical quantum state $|\psi\rangle$. Unlike eigenvalues, the expectation value depends explicitly on the state vector $|\psi\rangle$.

An individual measurement on a single system *never* yields a fractional or blurred value between eigenvalues; it yields one sharp eigenvalue a_n . The expectation value represents the center of mass of the resulting empirical histogram when the experiment is repeated across $N \rightarrow \infty$ identical systems.

First-Principles Proof: From Classical Statistics to Operator Expectation Values

Let us now prove step-by-step why the quantum expectation value must take the operator form $\langle \psi | \hat{A} | \psi \rangle$.

1. The Classical Statistical Setting. In Kolmogorov's axiomatic probability theory, let X be a discrete real random variable that can take possible values from a set $\{a_i\}$ with corresponding probabilities $P(a_i) \geq 0$, normalized such that $\sum_i P(a_i) = 1$. The mathematical expectation value (mean) of X is defined by the probability-weighted sum:

$$\mathbb{E}[X] \equiv \sum_i a_i P(a_i). \quad (1.2.11)$$

2. The Quantum Measurement Postulate. Now consider a quantum-mechanical system prepared in a normalized state vector $|\psi\rangle \in \mathcal{H}$ ($\langle\psi|\psi\rangle = 1$). Let $\hat{A}$ be a self-adjoint (Hermitian) operator with a discrete, non-degenerate spectrum:

$$\hat{A} |a_i\rangle = a_i |a_i\rangle, \quad \langle a_i | a_j \rangle = \delta_{ij}. \quad (1.2.12)$$

By the spectral theorem for self-adjoint operators (Section 1.1.4), the eigenkets $\{|a_i\rangle\}$ form a complete orthonormal basis of $\mathcal{H}$, yielding the completeness relation (resolution of the identity):

$$\hat{I} = \sum_i |a_i\rangle \langle a_i|, \quad (1.2.13)$$

and the spectral decomposition of the observable:

$$\hat{A} = \hat{A} \hat{I} = \sum_i a_i |a_i\rangle \langle a_i|. \quad (1.2.14)$$

According to Postulate 3 (the Born rule), when a measurement of $\hat{A}$ is performed on the ensemble, the outcome is a random variable $X \in \{a_i\}$, where the probability $P(a_i)$ of obtaining the specific eigenvalue a_i is given by:

$$P(a_i) = |\langle a_i | \psi \rangle|^2 = \langle a_i | \psi \rangle^* \langle a_i | \psi \rangle = \langle \psi | a_i \rangle \langle a_i | \psi \rangle, \quad (1.2.15)$$

where we have used the conjugate symmetry axiom of the inner product, $\langle a_i | \psi \rangle^* = \langle \psi | a_i \rangle$.

3. Explicit Step-by-Step Evaluation. We now substitute the physical Born probability (1.2.15) directly into the classical statistical definition of the expectation value (1.2.11):

$$\begin{aligned} \mathbb{E}[X] &= \sum_i a_i P(a_i) \\ &= \sum_i a_i \langle \psi | a_i \rangle \langle a_i | \psi \rangle. \end{aligned} \quad (1.2.16)$$

Because each eigenvalue a_i is a real scalar, it commutes with bras and kets and can be grouped directly with the eigenket $|a_i\rangle$:

$$\mathbb{E}[X] = \sum_i \langle \psi | (a_i | a_i \rangle) \langle a_i | \psi \rangle. \quad (1.2.17)$$

Now invoke the eigenvalue equation $\hat{A} |a_i\rangle = a_i |a_i\rangle$:

$$\mathbb{E}[X] = \sum_i \langle \psi | (\hat{A} | a_i \rangle) \langle a_i | \psi \rangle = \sum_i \langle \psi | \hat{A} | a_i \rangle \langle a_i | \psi \rangle. \quad (1.2.18)$$

By the linearity of the inner product and operator multiplication, we factor the state vector $\langle \psi | \hat{A}$ to the left and $|\psi\rangle$ to the right across the summation:

$$\mathbb{E}[X] = \langle \psi | \hat{A} \left(\sum_i |a_i\rangle \langle a_i| \right) | \psi \rangle. \quad (1.2.19)$$

Notice the operator inside the large parentheses! It is precisely the completeness relation (1.2.13), $\sum_i |a_i\rangle \langle a_i| = \hat{I}$:

$$\mathbb{E}[X] = \langle \psi | \hat{A} \hat{I} | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle. \quad (1.2.20)$$

Conversely, if we start from the operator sandwich $\langle \psi | \hat{A} | \psi \rangle$ and insert the spectral decomposition $\hat{A} = \sum_i a_i |a_i\rangle \langle a_i|$:

$$\langle \psi | \hat{A} | \psi \rangle = \langle \psi | \left(\sum_i a_i |a_i\rangle \langle a_i| \right) | \psi \rangle = \sum_i a_i \langle \psi | a_i \rangle \langle a_i | \psi \rangle = \sum_i a_i |\langle a_i | \psi \rangle|^2 = \sum_i a_i P(a_i) = \mathbb{E}[X]. \quad (1.2.21)$$

We have established our first primary conclusion:

The quantum expectation value $\langle \hat{A} \rangle_\psi \equiv \langle \psi | \hat{A} | \psi \rangle$ is not an arbitrary formal invention; it is the literal mathematical translation of the classical statistical mean $\sum_i a_i P(a_i)$ under the Born rule.

4. The Power of Coordinate-Free Formulation. Equation (1.2.20) illuminates the profound structural elegance of Dirac's notation:

- The classical expression $\sum_i a_i P(a_i)$ is *coordinate-dependent*: to evaluate it, one must explicitly know the eigenvalues $\{a_i\}$ and compute probabilities in the specific eigenbasis of $\hat{A}$.
- The Dirac sandwich $\langle \psi | \hat{A} | \psi \rangle$ is **coordinate-free and basis-independent**: it can be evaluated by inserting *any* convenient complete basis $\{|\chi_n\rangle\}$ (e.g., position, momentum, or energy):

$$\langle \hat{A} \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle = \sum_{n,m} \langle \psi | \chi_n \rangle \langle \chi_n | \hat{A} | \chi_m \rangle \langle \chi_m | \psi \rangle = \sum_{n,m} c_n^* A_{nm} c_m. \quad (1.2.22)$$

Regardless of the basis chosen, the matrix product yields the identical, invariant scalar expectation value.

5. Degenerate Spectra. If some eigenvalues of $\hat{A}$ are degenerate, each eigenvalue a_i corresponds to an eigenspace spanned by an orthonormal basis $\{|a_i, k\rangle\}_{k=1}^{d_i}$, where d_i is the degeneracy. The orthogonal projector onto the i -th eigenspace is:

$$\hat{P}_i = \sum_{k=1}^{d_i} |a_i, k\rangle \langle a_i, k|, \quad \sum_i \hat{P}_i = \hat{I}, \quad \hat{A} = \sum_i a_i \hat{P}_i. \quad (1.2.23)$$

According to the Born rule, the probability of obtaining eigenvalue a_i is $P(a_i) = \sum_{k=1}^{d_i} |\langle a_i, k | \psi \rangle|^2 = \langle \psi | \hat{P}_i | \psi \rangle$. Evaluating the statistical mean:

$$\mathbb{E}[X] = \sum_i a_i P(a_i) = \sum_i a_i \langle \psi | \hat{P}_i | \psi \rangle = \langle \psi | \left(\sum_i a_i \hat{P}_i \right) | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle. \quad (1.2.24)$$

The identity holds with full generality.

Continuous Spectra: First-Principles Derivation for Position and Momentum

For fundamental spatial degrees of freedom, such as the position operator $\hat{x}$ and linear momentum operator $\hat{p}$, the measurement outcomes form a continuous continuum over $\mathbb{R}$. Does the operator sandwich $\langle \psi | \hat{A} | \psi \rangle$ continue to represent the exact statistical mean?

1. The Classical Continuous Expectation Value. In classical statistics, let X be a continuous real random variable described by a probability density function $\rho(x) \geq 0$, normalized such that $\int_{-\infty}^{\infty} \rho(x) dx = 1$. The mathematical expectation value is defined by the integral:

$$\mathbb{E}[X] \equiv \int_{-\infty}^{\infty} x \rho(x) dx. \quad (1.2.25)$$

2. The Continuous Position Observable. In quantum mechanics, the position operator $\hat{x}$ acts on generalized position eigenstates according to $\hat{x} |x\rangle = x |x\rangle$, with continuous completeness:

$$\int_{-\infty}^{\infty} dx |x\rangle \langle x| = \hat{I}. \quad (1.2.26)$$

The Schrödinger wavefunction in position representation is $\psi(x) = \langle x|\psi\rangle$. According to Born's statistical interpretation, the probability of finding the particle in the spatial window $[x, x + dx]$ is $dP(x) = \rho(x) dx$, where the probability density is:

$$\rho(x) = |\psi(x)|^2 = |\langle x|\psi\rangle|^2 = \langle \psi|x\rangle \langle x|\psi\rangle. \quad (1.2.27)$$

3. Step-by-Step Continuous Integration. Substituting the Born density (1.2.27) into the classical definition (1.2.25):

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x \rho(x) dx \\ &= \int_{-\infty}^{\infty} x \langle \psi|x\rangle \langle x|\psi\rangle dx \\ &= \int_{-\infty}^{\infty} \langle \psi| (x|x\rangle) \langle x|\psi\rangle dx. \end{aligned} \quad (1.2.28)$$

Using the eigenvalue relation $x|x\rangle = \hat{x}|x\rangle$:

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} \langle \psi| (\hat{x}|x\rangle) \langle x|\psi\rangle dx \\ &= \langle \psi| \hat{x} \left(\int_{-\infty}^{\infty} |x\rangle \langle x| dx \right) |\psi\rangle. \end{aligned} \quad (1.2.29)$$

Invoking continuous completeness (1.2.26), the integral in parentheses is the identity operator $\hat{I}$:

$$\mathbb{E}[X] = \langle \psi| \hat{x} \hat{I} |\psi\rangle = \langle \psi| \hat{x} |\psi\rangle \equiv \langle \hat{x} \rangle_{\psi}. \quad (1.2.30)$$

An identical derivation applies to the linear momentum operator $\hat{p}$ using its continuous eigenbasis $\hat{p}|p\rangle = p|p\rangle$ and momentum-space wavefunction $\tilde{\psi}(p) = \langle p|\psi\rangle$:

$$\mathbb{E}[P] = \int_{-\infty}^{\infty} p |\tilde{\psi}(p)|^2 dp = \langle \psi| \hat{p} \left(\int_{-\infty}^{\infty} |p\rangle \langle p| dp \right) |\psi\rangle = \langle \psi| \hat{p} |\psi\rangle \equiv \langle \hat{p} \rangle_{\psi}. \quad (1.2.31)$$

For both discrete and continuous observables, Dirac's operator expectation value is the exact coordinate-free representation of the classical statistical expectation value.

Why the Inner Product Is Defined This Way: Projectors, Probability Amplitudes, and Quantum Interference

We now confront the deeper question: *Why does quantum mechanics represent physical states in an inner product space $\mathcal{H}$, and why is the transition probability between two states given by the squared modulus of their inner product, $P(\phi | \psi) = |\langle \phi|\psi\rangle|^2$?*

1. The Expectation Value of a Proposition Projector. In experimental physics, every measurement can be reduced to a sequence of binary yes/no questions (elementary propositions): “Did detector D register a particle or not?” or “Is the system, initially prepared in state $|\psi\rangle$, detected in the specific state $|\phi\rangle$?”

In the mathematical logic of quantum theory (the Birkhoff–von Neumann lattice of observables), an elementary proposition corresponds to an orthogonal projection operator onto the one-dimensional ray spanned by the target normalized state $|\phi\rangle$:

$$\hat{P}_\phi \equiv |\phi\rangle \langle \phi|. \quad (1.2.32)$$

The projection operator $\hat{P}_\phi$ is self-adjoint ($\hat{P}_\phi^\dagger = \hat{P}_\phi$) and idempotent ($\hat{P}_\phi^2 = \hat{P}_\phi$). Its spectrum consists of exactly two eigenvalues:

- $\lambda = 1$, representing the outcome “**Yes**” (the particle is detected in state $|\phi\rangle$).
- $\lambda = 0$, representing the outcome “**No**” (the particle is not in state $|\phi\rangle$).

In classical probability theory, for any event E , let $I_E \in \{0, 1\}$ be its indicator random variable ($I_E = 1$ if E occurs, $I_E = 0$ if E does not occur). The statistical expectation value of an indicator variable is *identically the probability of the event*:

$$\mathbb{E}[I_E] = 1 \cdot P(E) + 0 \cdot P(E^c) = P(E). \quad (1.2.33)$$

Therefore, the transition probability $P(\phi | \psi)$ of observing the system in state $|\phi\rangle$, given that it was prepared in state $|\psi\rangle$, **must equal the quantum expectation value of the proposition projector $\hat{P}_\phi$** :

$$P(\phi | \psi) = \mathbb{E}[\hat{P}_\phi] = \langle \hat{P}_\phi \rangle_\psi. \quad (1.2.34)$$

Applying our proven expectation value formula (1.2.20) to the projector $\hat{P}_\phi$:

$$\begin{aligned} P(\phi | \psi) &= \langle \psi | \hat{P}_\phi | \psi \rangle \\ &= \langle \psi | (|\phi\rangle \langle \phi|) | \psi \rangle \\ &= \langle \psi | \phi \rangle \langle \phi | \psi \rangle. \end{aligned} \quad (1.2.35)$$

Using the conjugate symmetry axiom of the inner product, $\langle \psi | \phi \rangle = \langle \phi | \psi \rangle^*$:

$$\boxed{P(\phi | \psi) = \langle \phi | \psi \rangle^* \langle \phi | \psi \rangle = |\langle \phi | \psi \rangle|^2.} \quad (1.2.36)$$

This derivation reveals the operational origin of the inner product:

The inner product $\langle \phi | \psi \rangle$ is defined in this particular way because its modulus squared is the exact statistical expectation value of the physical projection operator $\hat{P}_\phi = |\phi\rangle \langle \phi|$ testing whether state $|\psi\rangle$ passes a filter tailored to state $|\phi\rangle$.

2. The Inner Product as a Probability Amplitude. Equation (1.2.36) demonstrates that $\langle \phi | \psi \rangle$ is not itself a probability (which must be a real number in $[0, 1]$), but a **probability amplitude** residing in the complex plane $\mathbb{C}$. Why does Nature utilize complex probability amplitudes rather than real probabilities?

Consider a physical transition from state $|\psi\rangle$ to state $|\phi\rangle$ that can take place through multiple mutually exclusive intermediate channels $\{|\chi_k\rangle\}$ (for example, paths through multiple slits in an interferometer).

- **Classical Probability (Markovian Addition):** In classical stochastic physics, the law of total probability states that probabilities add directly:

$$P_{\text{classical}}(\phi | \psi) = \sum_k P(\phi | \chi_k) P(\chi_k | \psi). \quad (1.2.37)$$

Because each term is non-negative, intermediate classical channels can *only reinforce* one another; classical probabilities can never cancel to produce destructive interference.

- **Quantum Probability (Superposition of Amplitudes):** In quantum mechanics, by inserting the complete identity $\hat{I} = \sum_k |\chi_k\rangle \langle \chi_k|$ into the inner product, the *amplitudes* add linearly:

$$\langle \phi | \psi \rangle = \langle \phi | \hat{I} | \psi \rangle = \sum_k \langle \phi | \chi_k \rangle \langle \chi_k | \psi \rangle. \quad (1.2.38)$$

Taking the squared modulus to obtain the physical probability:

$$\begin{aligned} P(\phi | \psi) &= \left| \sum_k \langle \phi | \chi_k \rangle \langle \chi_k | \psi \rangle \right|^2 \\ &= \sum_k |\langle \phi | \chi_k \rangle|^2 |\langle \chi_k | \psi \rangle|^2 + \sum_{j \neq k} \langle \psi | \chi_j \rangle \langle \chi_j | \phi \rangle \langle \phi | \chi_k \rangle \langle \chi_k | \psi \rangle \\ &= \sum_k P(\phi | \chi_k) P(\chi_k | \psi) + 2 \sum_{j < k} \text{Re} [\langle \psi | \chi_j \rangle \langle \chi_j | \phi \rangle \langle \phi | \chi_k \rangle \langle \chi_k | \psi \rangle]. \end{aligned} \quad (1.2.39)$$

The first summation in Eq. (1.2.39) reproduces the classical probability sum. The second summation contains the **quantum interference cross-terms!** These cross-terms can be positive (constructive interference), zero, or negative (destructive interference).

Thus, the complex inner product space is the unique mathematical structure that allows physical state vectors to satisfy the superposition principle (producing wave-like interference) while maintaining a rigorous connection to non-negative statistical probabilities through projection expectation values.

3. Why the Inner Product Must Be Antilinear in the Bra. Why must the inner product satisfy antilinearity in the bra, $\langle c\phi | \psi \rangle = c^* \langle \phi | \psi \rangle$, rather than being bilinear over $\mathbb{C}$? If the inner product were bilinear (linear in both arguments over $\mathbb{C}$), then for any scalar $c \in \mathbb{C}$:

$$\langle c\psi | c\psi \rangle_{\text{bilinear}} = c^2 \langle \psi | \psi \rangle. \quad (1.2.40)$$

Choosing $c = i$ would yield $\langle i\psi | i\psi \rangle = i^2 \langle \psi | \psi \rangle = -\langle \psi | \psi \rangle < 0$. This would immediately destroy the positive-definiteness axiom and produce *negative probabilities!*

By contrast, the antilinear (sesquilinear) structure guarantees that:

$$\langle c\psi | c\psi \rangle = c^* c \langle \psi | \psi \rangle = |c|^2 \|\psi\|^2 \geq 0, \quad (1.2.41)$$

ensuring that probabilities remain strictly real and non-negative under arbitrary phase and scale transformations.

4. Probability Conservation (Unitarity). For any complete orthonormal basis $\{|\phi_k\rangle\}$, the completeness relation $\sum_k |\phi_k\rangle \langle \phi_k| = \hat{I}$ guarantees that the total probability of all mutually exclusive measurement outcomes sums to unity:

$$\sum_k P(\phi_k | \psi) = \sum_k |\langle \phi_k | \psi \rangle|^2 = \sum_k \langle \psi | \phi_k \rangle \langle \phi_k | \psi \rangle = \langle \psi | \left(\sum_k |\phi_k\rangle \langle \phi_k| \right) | \psi \rangle = \langle \psi | \psi \rangle = 1. \quad (1.2.42)$$

Normalization $\langle \psi | \psi \rangle = 1$ is therefore nothing other than the statement that the probability of the universe being in *some* physical state is 100%.

Higher Statistical Moments, Variance, and the Vanishing Uncertainty Theorem

A statistical distribution is rarely characterized by its mean alone; complete description requires knowledge of its higher-order moments and dispersion.

1. Higher Statistical Moments. In classical statistics, the k -th raw moment ($k \in \mathbb{N}$) of a random variable X is defined as $\mathbb{E}[X^k] \equiv \sum_i a_i^k P(a_i)$. In quantum mechanics, applying our identity-resolved evaluation to the operator power $\hat{A}^k$:

$$\hat{A} |a_i\rangle = a_i |a_i\rangle \implies \hat{A}^k |a_i\rangle = a_i^k |a_i\rangle. \quad (1.2.43)$$

Evaluating the classical k -th moment using the Born rule:

$$\begin{aligned} \mathbb{E}[X^k] &= \sum_i a_i^k P(a_i) = \sum_i a_i^k \langle \psi | a_i \rangle \langle a_i | \psi \rangle \\ &= \sum_i \langle \psi | (\hat{A}^k |a_i\rangle) \langle a_i | \psi \rangle \\ &= \langle \psi | \hat{A}^k \left(\sum_i |a_i\rangle \langle a_i| \right) | \psi \rangle \\ &= \langle \psi | \hat{A}^k | \psi \rangle \equiv \langle \hat{A}^k \rangle_\psi. \end{aligned} \quad (1.2.44)$$

More generally, for any function $g(\hat{A})$ defined through the spectral functional calculus:

$$\mathbb{E}[g(X)] = \sum_i g(a_i) P(a_i) = \langle \psi | g(\hat{A}) | \psi \rangle \equiv \langle g(\hat{A}) \rangle_\psi. \quad (1.2.45)$$

2. Variance and Quantum Uncertainty. The **variance** of a measurement distribution measures the expected squared deviation from the statistical mean $\mu = \mathbb{E}[X] = \langle \hat{A} \rangle_\psi$:

$$\text{Var}(A) \equiv \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (1.2.46)$$

To express this in operator form, we define the **fluctuation (deviation) operator**:

$$\Delta \hat{A} \equiv \hat{A} - \langle \hat{A} \rangle \hat{I}. \quad (1.2.47)$$

Because $\hat{A}$ is Hermitian and $\langle \hat{A} \rangle$ is a real number, $\Delta \hat{A}$ is self-adjoint: $(\Delta \hat{A})^\dagger = \Delta \hat{A}$. Evaluating its expectation value using Eq. (1.2.44):

$$\begin{aligned} \text{Var}(A) &= \langle (\Delta \hat{A})^2 \rangle = \langle \psi | (\hat{A} - \langle \hat{A} \rangle \hat{I})^2 | \psi \rangle \\ &= \langle \psi | (\hat{A}^2 - 2\langle \hat{A} \rangle \hat{A} + \langle \hat{A} \rangle^2 \hat{I}) | \psi \rangle \\ &= \langle \psi | \hat{A}^2 | \psi \rangle - 2\langle \hat{A} \rangle \langle \psi | \hat{A} | \psi \rangle + \langle \hat{A} \rangle^2 \langle \psi | \psi \rangle \\ &= \langle \hat{A}^2 \rangle - 2\langle \hat{A} \rangle^2 + \langle \hat{A} \rangle^2 \\ &= \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \equiv (\Delta A)^2. \end{aligned} \quad (1.2.48)$$

The **standard deviation** (commonly termed the *quantum uncertainty*) is the square root of the variance:

$$\Delta A \equiv \sigma_A = \sqrt{\text{Var}(A)} = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}. \quad (1.2.49)$$

3. The Vanishing Uncertainty Theorem (Eigenstate Determinism). When does a quantum measurement yield a completely deterministic result with zero statistical spread? Because the deviation operator $\Delta\hat{A}$ is self-adjoint, the variance can be written as the squared Hilbert space norm of the vector $(\Delta\hat{A})|\psi\rangle$:

$$(\Delta A)^2 = \langle\psi|(\Delta\hat{A})^\dagger(\Delta\hat{A})|\psi\rangle = \|(\Delta\hat{A})|\psi\rangle\|^2 \geq 0. \quad (1.2.50)$$

By the positive-definiteness axiom of the Hilbert space norm ($\|\chi\| = 0 \iff |\chi\rangle = 0$), the variance vanishes if and only if:

$$(\Delta A)^2 = 0 \iff \|(\Delta\hat{A})|\psi\rangle\| = 0 \iff (\Delta\hat{A})|\psi\rangle = 0 \iff \hat{A}|\psi\rangle = \langle\hat{A}\rangle|\psi\rangle. \quad (1.2.51)$$

This establishes a rigorous theorem:

The quantum uncertainty vanishes ($\Delta A = 0$) if and only if the state $|\psi\rangle$ is an exact eigenstate of $\hat{A}$. In that case, the expectation value $\langle\hat{A}\rangle$ is identically the eigenvalue a_n , and repeated measurements on the ensemble yield a_n with 100% certainty ($P = 1$).

Statistical Foundation: Why the Quantum Expectation Value and Inner Product Are Defined This Way

1. The Classical Statistics $\longleftrightarrow$ Quantum Operator Mechanics Dictionary:

- **Random Variable $X \longleftrightarrow$ Self-Adjoint Operator $\hat{A}$:** Physical observables are self-adjoint operators whose eigenvalues $\{a_i\}$ constitute the spectrum of possible experimental outcomes.
- **Event / Proposition $E \longleftrightarrow$ Projection Operator $\hat{P} = \hat{P}^\dagger = \hat{P}^2$:** Testing whether a system in state $|\psi\rangle$ possesses property $|\phi\rangle$ corresponds to measuring $\hat{P}_\phi = |\phi\rangle\langle\phi|$.
- **Probability $P(E) = \mathbb{E}[I_E] \longleftrightarrow$ Projector Expectation Value $\langle\hat{P}_\phi\rangle_\psi = |\langle\phi|\psi\rangle|^2$:** The transition probability is the expectation value of the proposition projector:

$$P(\phi | \psi) = \langle\psi|\hat{P}_\phi|\psi\rangle = \langle\psi|(|\phi\rangle\langle\phi|)|\psi\rangle = |\langle\phi|\psi\rangle|^2. \quad (1)$$

This proves that the **inner product $\langle\phi|\psi\rangle$ is the probability amplitude**, whose modulus squared produces the physical transition probability.

- **Statistical Mean $\mathbb{E}[X] = \sum_i a_i P(a_i) \longleftrightarrow$ Quantum Expectation Value $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$:** The operator sandwich is the exact coordinate-free resolution of the classical mean:

$$\sum_i a_i P(a_i) = \sum_i a_i \langle\psi|a_i\rangle\langle a_i|\psi\rangle = \langle\psi|\hat{A}\left(\sum_i |a_i\rangle\langle a_i|\right)|\psi\rangle = \langle\psi|\hat{A}\hat{I}|\psi\rangle = \langle\psi|\hat{A}|\psi\rangle. \quad (2)$$

- **Continuous Mean $\mathbb{E}[X] = \int x \rho(x) dx \longleftrightarrow \langle\hat{x}\rangle = \langle\psi|\hat{x}|\psi\rangle$:** For continuous observables:

$$\int_{-\infty}^{\infty} x |\psi(x)|^2 dx = \int_{-\infty}^{\infty} \langle\psi|(\hat{x}|x\rangle)\langle x|\psi\rangle dx = \langle\psi|\hat{x}\left(\int_{-\infty}^{\infty} |x\rangle\langle x| dx\right)|\psi\rangle = \langle\psi|\hat{x}|\psi\rangle. \quad (3)$$

- **Higher Moments $\mathbb{E}[X^k] \longleftrightarrow \langle\hat{A}^k\rangle = \langle\psi|\hat{A}^k|\psi\rangle$:** The k -th statistical moment of measurement outcomes is the expectation value of the operator power $\hat{A}^k$.

- **Variance** $\text{Var}(X) \longleftrightarrow (\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$: The statistical dispersion equals the squared norm of the fluctuation vector:

$$(\Delta A)^2 = \left\| (\hat{A} - \langle \hat{A} \rangle \hat{I}) |\psi\rangle \right\|^2 \geq 0. \quad (4)$$

2. Foundational Conclusions:

1. **No Arbitrary Conventions:** The expectation value $\langle \psi | \hat{A} | \psi \rangle$ is not an arbitrary postulate; it is the unique coordinate-free mathematical realization of the classical statistical mean under the Born rule.
2. **Origin of the Complex Inner Product:** The bracket $\langle \phi | \psi \rangle$ is linear in $|\psi\rangle$ and antilinear in $\langle \phi |$ to permit quantum interference across intermediate channels ($\langle \phi | \psi \rangle = \sum_k \langle \phi | \chi_k \rangle \langle \chi_k | \psi \rangle$) while strictly guaranteeing positive-definiteness ($\langle \psi | \psi \rangle \geq 0$) and probability conservation ($\sum_k |\langle \phi_k | \psi \rangle|^2 = 1$).
3. **Zero-Variance Criterion:** The measurement variance vanishes ($\Delta A = 0$) if and only if $|\psi\rangle$ is an eigenstate of $\hat{A}$, in which case the expectation value matches the eigenvalue deterministically.

1.3 Continuous Spectra and the Rigged Hilbert Space

When dealing with bound systems, such as a particle in a box or bound states of the hydrogen atom, observables possess discrete eigenvalues labeled by integers $n \in \mathbb{Z}$. However, for unbound systems and fundamental spatial degrees of freedom—most notably the position operator $\hat{x}$ and the linear momentum operator $\hat{p}$ —the spectrum of eigenvalues forms a continuous continuum over the real line $\mathbb{R}$.

1.3.1 Continuous Bases and Dirac Delta Normalization

For continuous observables, the discrete Kronecker delta δ_{mn} is replaced by the Dirac delta function $\delta(x - x')$, and sums over discrete basis states are replaced by integrals:

$$\hat{x} |x'\rangle = x' |x'\rangle, \quad x' \in (-\infty, \infty), \quad (1.3.1)$$

$$\langle x | x' \rangle = \delta(x - x'). \quad (1.3.2)$$

The completeness relation for the continuous position basis becomes:

$$\int_{-\infty}^{\infty} dx |x\rangle \langle x| = \hat{I}. \quad (1.3.3)$$

Expanding an arbitrary physical state $|\psi\rangle$ in the continuous position basis yields:

$$|\psi\rangle = \hat{I} |\psi\rangle = \int_{-\infty}^{\infty} dx |x\rangle \langle x | \psi \rangle \equiv \int_{-\infty}^{\infty} dx \psi(x) |x\rangle, \quad (1.3.4)$$

where $\psi(x) \equiv \langle x | \psi \rangle$ is the familiar **Schrödinger position-space wavefunction**. The inner product of two physical states is expressed as an integral over the spatial domain:

$$\langle \phi | \psi \rangle = \int_{-\infty}^{\infty} dx \langle \phi | x \rangle \langle x | \psi \rangle = \int_{-\infty}^{\infty} dx \phi^*(x) \psi(x). \quad (1.3.5)$$

The normalization condition $\langle \psi | \psi \rangle = 1$ translates into square-integrability:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1. \quad (1.3.6)$$

1.3.2 Why Continuous Eigenstates $|x\rangle$ and $|p\rangle$ Are Not in Hilbert Space

A foundational mathematical subtlety often glossed over in introductory textbooks is that:

The continuous eigenstates $|x\rangle$ and $|p\rangle$ do not belong to the physical Hilbert space $\mathcal{H} = L^2(\mathbb{R})$!

Recall that by definition, the Hilbert space $L^2(\mathbb{R})$ is the space of square-integrable functions with finite norm:

$$\mathcal{H} = L^2(\mathbb{R}) = \left\{ \psi(x) \mid \int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty \right\}. \quad (1.3.7)$$

Let us evaluate the norm of a position eigenket $|x_0\rangle$:

$$\langle x_0 | x_0 \rangle = \delta(x_0 - x_0) = \delta(0) = \infty. \quad (1.3.8)$$

Similarly, consider a momentum eigenstate $|p\rangle$, whose position-space wavefunction is a plane wave $\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$. Evaluating its norm:

$$\langle p | p \rangle = \int_{-\infty}^{\infty} |\langle x | p \rangle|^2 dx = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} 1 dx = \infty. \quad (1.3.9)$$

Neither $|x\rangle$ nor $|p\rangle$ has a finite norm! Since every element of a Hilbert space must have a strictly finite norm, neither $|x\rangle$ nor $|p\rangle$ can be normalized to unity, and neither represents a physically realizable isolated quantum state. A true physical particle is always represented by a normalizable **wave packet**:

$$|\psi\rangle = \int_{-\infty}^{\infty} dp \tilde{\psi}(p) |p\rangle \in L^2(\mathbb{R}), \quad \text{with} \quad \int_{-\infty}^{\infty} |\tilde{\psi}(p)|^2 dp = 1. \quad (1.3.10)$$

1.3.3 The Rigged Hilbert Space (Gelfand Triplet) and Continuous Spectra

The realization that continuous eigenstates have infinite norm led to a historical rift in the foundations of quantum mechanics:

- **The von Neumann Approach (Spectral Measures):** In his 1932 treatise, John von Neumann formulated quantum mechanics strictly within Hilbert space $\mathcal{H} = L^2(\mathbb{R})$. Because continuous eigenstates do not belong to $\mathcal{H}$, von Neumann banished eigenvectors of continuous observables entirely! Instead of $\hat{x}|x\rangle = x|x\rangle$, he formulated the spectral theorem via projection-valued Stieltjes integrals:

$$\hat{x} = \int_{-\infty}^{\infty} \lambda dE(\lambda), \quad E(\lambda)\psi(x) = \theta(\lambda - x)\psi(x). \quad (1.3.11)$$

While mathematically airtight, this framework discarded Dirac's intuitive bra-ket notation, replacing straightforward algebraic manipulations with cumbersome projection-valued measures.

- **The Rigged Hilbert Space Approach (Gelfand Triplet):** In the 1960s, mathematicians I. M. Gelfand, N. Ya. Vilenkin, and K. Maurin proved that Dirac's formalism can be rendered fully rigorous without sacrificing bra-ket calculus by extending the Hilbert space into a **Rigged Hilbert Space** [1], also termed a **Gelfand Triplet**:

$$\Phi \subset \mathcal{H} \subset \Phi^\times. \quad (1.3.12)$$

(As noted by Ballentine [1], the nautical term “rigged” signifies “equipped and ready for action”, in analogy with outfitting a sailing ship with masts, sails, and ropes.)

Ballentine's Algebraic Construction of the Triplet

To grasp the mathematical necessity of the Gelfand triplet, let us examine the transparent algebraic construction presented by Ballentine [1]:

1. **The Universal Space Ξ :** Consider an infinite orthonormal basis $\{|\phi_n\rangle : n = 1, 2, \dots\}$. Let Ξ be the linear space consisting of all formal linear combinations:

$$|\xi\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle, \quad (1.3.13)$$

where *no restrictions whatsoever* are imposed on the complex coefficients c_n . For general vectors in Ξ , the norm and inner product are completely undefined.

2. **The Hilbert Space $\mathcal{H}$:** The Hilbert space is the restricted subspace of Ξ constrained by the condition that the norm is finite:

$$\mathcal{H} = \left\{ |h\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle \left| \langle h|h\rangle = \sum_{n=1}^{\infty} |c_n|^2 < \infty \right. \right\}. \quad (1.3.14)$$

By the Riesz representation theorem, the dual space of continuous linear functionals $\mathcal{H}^*$ is strictly isomorphic to $\mathcal{H}$ itself. Thus, Hilbert space is uniquely self-dual: $\mathcal{H} \cong \mathcal{H}^\times$.

3. **The Nuclear Space Φ (Test Vectors):** Next, we construct a much smaller, highly restricted subspace $\Phi \subset \mathcal{H}$ consisting of vectors $|\omega\rangle = \sum_{n=1}^{\infty} u_n |\phi_n\rangle$ whose expansion coefficients decrease faster than any inverse power of n :

$$\sum_{n=1}^{\infty} |u_n|^2 n^m < \infty \quad \text{for all } m = 0, 1, 2, \dots \quad (1.3.15)$$

The space Φ is an example of a **nuclear space**; its elements represent exceptionally well-behaved physical wave packets that reside well within the domain of all polynomial operator observables.

4. **The Extended Conjugate Space $\Phi^\times$ (Generalized Vectors):** The conjugate space $\Phi^\times$ consists of all formal vectors $|\sigma\rangle = \sum_{n=1}^{\infty} v_n |\phi_n\rangle$ for which the pairing:

$$\langle \sigma | \omega \rangle = \sum_{n=1}^{\infty} v_n^* u_n \quad (1.3.16)$$

is absolutely convergent for *every* test vector $|\omega\rangle \in \Phi$.

Here we discover a fundamental duality theorem of functional analysis:

The smaller and more restricted the test space Φ , the larger and more accommodating its conjugate space $\Phi^\times$!

Because the test coefficients u_n vanish faster than any power of n as $n \rightarrow \infty$, the pairing $\sum_n v_n^* u_n$ converges even if the components v_n diverge polynomially ($v_n \sim n^k$). Consequently, $\Phi^\times$ is vast: it accommodates vectors whose norms are completely infinite!

The three spaces satisfy the nested inclusion hierarchy:

$$\boxed{\Phi \subset \mathcal{H} = \mathcal{H}^\times \subset \Phi^\times.} \quad (1.3.17)$$

Function Space Realization: Schwartz Space and Tempered Distributions

For continuous spatial degrees of freedom, the abstract algebraic triplet $\Phi \subset \mathcal{H} \subset \Phi^\times$ is realized concretely on the real line $\mathbb{R}$:

1. **The Nuclear Test Space $\Phi = \mathcal{S}(\mathbb{R})$ (Schwartz Space):** The space of infinitely differentiable functions $\phi(x) \in C^\infty(\mathbb{R})$ whose values and derivatives fall off faster than any power of $|x|$ at infinity:

$$\int_{-\infty}^{\infty} |\phi(x)|^2 (1 + |x|)^m dx < \infty \quad \text{for all } m = 0, 1, 2, \dots \quad (1.3.18)$$

Elements of $\mathcal{S}(\mathbb{R})$ have the critical property that they remain within $\mathcal{S}(\mathbb{R})$ under the action of any polynomial in $\hat{x}$ and $\hat{p}$:

$$\hat{x}^k \hat{p}^l |\phi\rangle \in \mathcal{S}(\mathbb{R}) \quad \forall k, l \geq 0. \quad (1.3.19)$$

2. **The Hilbert Space $\mathcal{H} = L^2(\mathbb{R})$:** The standard space of square-integrable wavefunctions $\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty$. Physical states are normalizable wave packets residing in $\mathcal{H}$.
3. **The Extended Space $\Phi^\times = \mathcal{S}'(\mathbb{R})$ (Tempered Distributions):** The continuous linear functionals acting on the test space $\mathcal{S}(\mathbb{R})$. Crucially, $\mathcal{S}'(\mathbb{R})$ contains functions that do not vanish at infinity, functions that grow polynomially, and singular distribution kernels!

- The continuous **momentum eigenstates** $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$ are plane waves. While $e^{ipx/\hbar} \notin L^2(\mathbb{R})$, it is bounded ($|e^{ipx/\hbar}| = 1$) and defines a continuous linear functional on any Schwartz test function:

$$\langle p|\phi\rangle = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \phi(x) dx \equiv \tilde{\phi}(p) < \infty. \quad (1.3.20)$$

Thus, **the momentum eigenstate $|p\rangle$ is a legitimate generalized vector in $\Phi^\times$!**

- The continuous **position eigenstates** $\langle x|x_0\rangle = \delta(x - x_0)$ are Dirac delta distributions. While $\delta(x - x_0) \notin L^2(\mathbb{R})$, its action on any test function:

$$\langle x_0|\phi\rangle = \int_{-\infty}^{\infty} \delta(x - x_0) \phi(x) dx = \phi(x_0) < \infty, \quad (1.3.21)$$

is a well-defined continuous linear functional. Thus, **the position eigenstate $|x_0\rangle$ is a legitimate generalized vector in $\Phi^\times$!**

The Gelfand–Maurin Nuclear Spectral Theorem: Justifying Dirac Calculus

The crowning mathematical justification of Dirac's notation is provided by the **Gelfand–Maurin Spectral Theorem**:

Let $\hat{A}$ be a self-adjoint operator on a Hilbert space $\mathcal{H}$ that leaves the nuclear test space Φ invariant ($\hat{A}\Phi \subseteq \Phi$). Then there exists a family of generalized eigenvectors $\{|\lambda\rangle : \lambda \in \sigma(\hat{A})\}$ residing in the extended space $\Phi^\times$ such that:

$$\hat{A}|\lambda\rangle = \lambda|\lambda\rangle \quad \text{in the sense of functionals,} \quad (1)$$

and for any physical test vector $|\phi\rangle \in \Phi$, the spectral expansion and resolution of the identity hold:

$$|\phi\rangle = \int_{\sigma(\hat{A})} d\mu(\lambda) |\lambda\rangle \langle \lambda|\phi\rangle, \quad \hat{A}|\phi\rangle = \int_{\sigma(\hat{A})} d\mu(\lambda) \lambda |\lambda\rangle \langle \lambda|\phi\rangle. \quad (2)$$

This theorem guarantees that Dirac's continuous resolutions of the identity:

$$\int_{-\infty}^{\infty} dx |x\rangle \langle x| = \hat{I}, \quad \int_{-\infty}^{\infty} dp |p\rangle \langle p| = \hat{I}, \quad (3)$$

and the continuous eigenvalue equations $\hat{x}|x\rangle = x|x\rangle$ and $\hat{p}|p\rangle = p|p\rangle$ are **not formal fictions**; they are exact, mathematically rigorous operations valid for all physical states in Φ .

As Ballentine [1] cogently concludes, the Rigged Hilbert Space is the truly natural mathematical habitat for quantum theory: it honors von Neumann's topological rigor while fully preserving the unmatched algebraic power, elegance, and utility of Dirac's bra-ket notation.

1.4 Space Translation Operator and First-Principles Derivation of

$$[x, p] = i\hbar$$

We now turn to the physical origin of the momentum operator $\hat{p}$ and the canonical commutation relation $[x, p] = i\hbar$. Rather than asserting $[x, p] = i\hbar$ as an ad hoc assumption, we derive it from first principles using spatial translation symmetry.

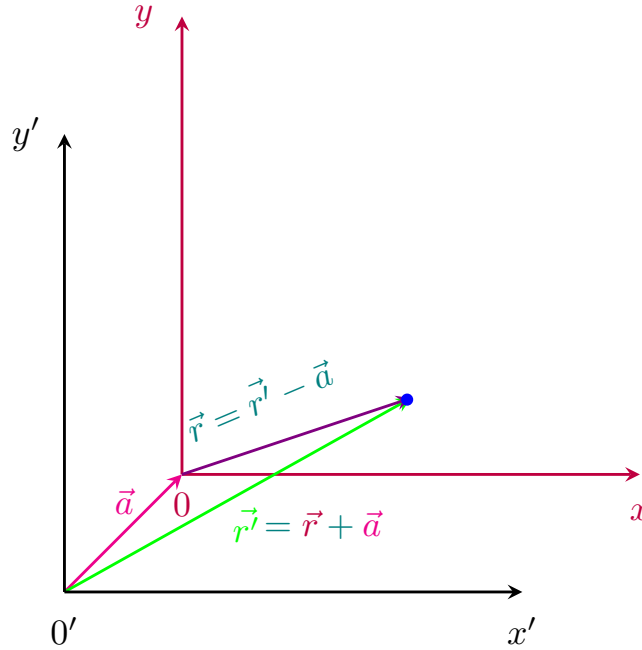


Figure 1.1: Active translation of a physical state by displacement Δx . Translating the state vector forward corresponds to shifting the coordinate evaluation backward: $\langle x | \mathcal{T}(\Delta x) | \psi \rangle = \psi(x - \Delta x)$.

1.4.1 Physical Postulates for the Translation Operator

Consider an operation that translates a localized quantum state in space by an infinitesimal displacement dx . We define the **infinitesimal translation operator** $\mathcal{T}(dx)$ by its action on position eigenkets:

$$\mathcal{T}(dx) |x'\rangle = |x' + dx\rangle. \quad (1.4.1)$$

Physical consistency requires that $\mathcal{T}(dx)$ satisfy four fundamental physical postulates:

1. **Probability Conservation (Unitarity):** Translating a state in space must not change the total probability of finding the particle somewhere in the universe. For any two states $|\alpha\rangle$ and $|\beta\rangle$, the inner product must be invariant:

$$\langle\alpha|\beta\rangle = \langle\mathcal{T}(dx)\alpha|\mathcal{T}(dx)\beta\rangle = \langle\alpha|\mathcal{T}^\dagger(dx)\mathcal{T}(dx)|\beta\rangle. \quad (1.4.2)$$

Since this must hold for all kets, $\mathcal{T}(dx)$ must be a **unitary operator**:

$$\mathcal{T}^\dagger(dx)\mathcal{T}(dx) = \hat{I} \implies \mathcal{T}^{-1}(dx) = \mathcal{T}^\dagger(dx). \quad (1.4.3)$$

2. **Composition Law:** Successive translations by dx_1 followed by dx_2 must equal a single translation by $dx_1 + dx_2$:

$$\mathcal{T}(dx_2)\mathcal{T}(dx_1) = \mathcal{T}(dx_1 + dx_2). \quad (1.4.4)$$

3. **Reversibility / Inverse:** A translation by $-dx$ must undo a translation by $+dx$:

$$\mathcal{T}(-dx)\mathcal{T}(dx) = \mathcal{T}(0) = \hat{I} \implies \mathcal{T}(-dx) = \mathcal{T}^{-1}(dx) = \mathcal{T}^\dagger(dx). \quad (1.4.5)$$

4. **Continuity (Identity Limit):** In the limit of vanishing displacement, the operator must reduce smoothly to the identity operator:

$$\lim_{dx \rightarrow 0} \mathcal{T}(dx) = \hat{I}. \quad (1.4.6)$$

1.4.2 The Generator of Translations

Any linear operator satisfying $\mathcal{T}(0) = \hat{I}$ can be expanded in a Taylor series for infinitesimal dx :

$$\mathcal{T}(dx) = \hat{I} - \frac{i}{\hbar} \hat{p} dx + \mathcal{O}(dx^2), \quad (1.4.7)$$

where the factor of $-i/\hbar$ is chosen by convention so that the generator $\hat{p}$ has the physical dimensions of linear momentum ([action]/[length]), and $\hbar$ is Planck's reduced constant.

Now, we impose the unitarity condition $\mathcal{T}^\dagger(dx)\mathcal{T}(dx) = \hat{I}$ to order dx :

$$\begin{aligned} \hat{I} &= \left(\hat{I} + \frac{i}{\hbar} \hat{p}^\dagger dx \right) \left(\hat{I} - \frac{i}{\hbar} \hat{p} dx \right) \\ &= \hat{I} + \frac{i}{\hbar} (\hat{p}^\dagger - \hat{p}) dx + \mathcal{O}(dx^2). \end{aligned} \quad (1.4.8)$$

For this equation to hold identically, the first-order term in dx must vanish:

$$\hat{p}^\dagger - \hat{p} = 0 \implies \hat{p}^\dagger = \hat{p}. \quad (1.4.9)$$

The generator $\hat{p}$ is necessarily a Hermitian operator! We identify $\hat{p}$ as the physical momentum operator.

A finite spatial translation by displacement Δx can be compounded as an infinite succession of N infinitesimal steps of length $\Delta x/N$:

$$\mathcal{T}(\Delta x) = \lim_{N \rightarrow \infty} \left[\mathcal{T} \left(\frac{\Delta x}{N} \right) \right]^N = \lim_{N \rightarrow \infty} \left(\hat{I} - \frac{i}{\hbar} \hat{p} \frac{\Delta x}{N} \right)^N = \exp \left(-\frac{i}{\hbar} \hat{p} \Delta x \right). \quad (1.4.10)$$

1.4.3 First-Principles Derivation of $[x, p] = i\hbar$

We now derive the fundamental commutator $[x, p] = i\hbar$ purely from the translation property of position kets. Consider the commutator of the position operator $\hat{x}$ with the infinitesimal translation operator $\mathcal{T}(dx)$, acting on an arbitrary position eigenstate $|x'\rangle$:

$$\hat{x}\mathcal{T}(dx)|x'\rangle = \hat{x}|x' + dx\rangle = (x' + dx)|x' + dx\rangle, \quad (1.4.11)$$

$$\mathcal{T}(dx)\hat{x}|x'\rangle = \mathcal{T}(dx)(x'|x'\rangle) = x'\mathcal{T}(dx)|x'\rangle = x'|x' + dx\rangle. \quad (1.4.12)$$

Subtracting Eq. (1.4.12) from Eq. (1.4.11):

$$[\hat{x}, \mathcal{T}(dx)]|x'\rangle = (x' + dx - x')|x' + dx\rangle = dx|x' + dx\rangle. \quad (1.4.13)$$

Expanding $|x' + dx\rangle$ to lowest order in dx :

$$dx|x' + dx\rangle = dx(|x'\rangle + \mathcal{O}(dx)) = dx|x'\rangle + \mathcal{O}(dx^2). \quad (1.4.14)$$

On the other hand, substituting the generator expansion $\mathcal{T}(dx) = \hat{I} - \frac{i}{\hbar}\hat{p}dx$ into the left-hand side commutator gives:

$$[\hat{x}, \mathcal{T}(dx)] = \left[\hat{x}, \hat{I} - \frac{i}{\hbar}\hat{p}dx\right] = -\frac{i}{\hbar}[\hat{x}, \hat{p}]dx. \quad (1.4.15)$$

Equating Eq. (1.4.14) and Eq. (1.4.15) acting on $|x'\rangle$:

$$-\frac{i}{\hbar}[\hat{x}, \hat{p}]dx|x'\rangle = dx|x'\rangle. \quad (1.4.16)$$

Dividing both sides by the infinitesimal scalar dx and multiplying by $i\hbar$:

$$[\hat{x}, \hat{p}]|x'\rangle = i\hbar|x'\rangle. \quad (1.4.17)$$

Because the position eigenkets $\{|x'\rangle\}$ form a complete basis spanning the entire state space, any state $|\psi\rangle = \int dx'\psi(x')|x'\rangle$ satisfies:

$$[\hat{x}, \hat{p}]|\psi\rangle = \int dx'\psi(x')[\hat{x}, \hat{p}]|x'\rangle = i\hbar \int dx'\psi(x')|x'\rangle = i\hbar|\psi\rangle. \quad (1.4.18)$$

Since this holds for any state $|\psi\rangle$, we have derived the operator identity:

$$\boxed{[\hat{x}, \hat{p}] = i\hbar\hat{I}} \quad (1.4.19)$$

This is the canonical commutation relation of quantum mechanics, derived strictly from the geometry of space and the physical requirements of translation invariance!

In three dimensions, with translation vector $\mathbf{a} = (a_1, a_2, a_3)$ and momentum vector $\hat{\mathbf{p}} = (\hat{p}_1, \hat{p}_2, \hat{p}_3)$:

$$\mathcal{T}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\hat{\mathbf{p}} \cdot \mathbf{a}\right), \quad [\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}\hat{I}, \quad [\hat{x}_j, \hat{x}_k] = 0, \quad [\hat{p}_j, \hat{p}_k] = 0. \quad (1.4.20)$$

1.5 Action of Momentum in Position Basis and Fourier Transforms

1.5.1 Differential Representation of Momentum

Having established the operator identity $[\hat{x}, \hat{p}] = i\hbar$, we can now derive the action of the momentum operator on a wavefunction in the position representation. Consider the matrix element $\langle x|\mathcal{T}(dx)|\psi\rangle$.

First, let the translation operator act to the left on the bra. Using $\mathcal{T}^\dagger(dx) = \mathcal{T}(-dx)$:

$$\langle x | \mathcal{T}(dx) = [\mathcal{T}^\dagger(dx) | x \rangle]^\dagger = [\mathcal{T}(-dx) | x \rangle]^\dagger = [|x - dx \rangle]^\dagger = \langle x - dx |. \quad (1.5.1)$$

Therefore:

$$\langle x | \mathcal{T}(dx) | \psi \rangle = \langle x - dx | \psi \rangle = \psi(x - dx). \quad (1.5.2)$$

Expanding $\psi(x - dx)$ in a Taylor series about x :

$$\langle x | \mathcal{T}(dx) | \psi \rangle = \psi(x) - dx \frac{\partial \psi(x)}{\partial x} + \mathcal{O}(dx^2). \quad (1.5.3)$$

Second, evaluate the same matrix element using the generator expansion $\mathcal{T}(dx) = \hat{I} - \frac{i}{\hbar} \hat{p} dx$:

$$\langle x | \mathcal{T}(dx) | \psi \rangle = \langle x | \left(\hat{I} - \frac{i}{\hbar} \hat{p} dx \right) | \psi \rangle = \langle x | \psi \rangle - \frac{i}{\hbar} dx \langle x | \hat{p} | \psi \rangle = \psi(x) - \frac{i}{\hbar} dx \langle x | \hat{p} | \psi \rangle. \quad (1.5.4)$$

Equating the first-order terms in dx from Eq. (1.5.3) and Eq. (1.5.4):

$$-\frac{i}{\hbar} \langle x | \hat{p} | \psi \rangle = -\frac{\partial \psi(x)}{\partial x}. \quad (1.5.5)$$

Multiplying by $i\hbar$:

$$\boxed{\langle x | \hat{p} | \psi \rangle = -i\hbar \frac{\partial}{\partial x} \psi(x)} \quad (1.5.6)$$

Thus, the position-space differential operator representation of the momentum operator is:

$$\hat{p} \longleftrightarrow -i\hbar \frac{\partial}{\partial x}, \quad \text{and in 3D} \quad \hat{\mathbf{p}} \longleftrightarrow -i\hbar \nabla. \quad (1.5.7)$$

1.5.2 Momentum Eigenstates in Position Space

Let $|p'\rangle$ denote an eigenstate of the momentum operator with eigenvalue p' :

$$\hat{p} |p'\rangle = p' |p'\rangle. \quad (1.5.8)$$

Projecting this eigenvalue equation onto the position bra $\langle x|$:

$$\langle x | \hat{p} | p' \rangle = p' \langle x | p' \rangle. \quad (1.5.9)$$

Using the differential representation $\langle x | \hat{p} | p' \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | p' \rangle$:

$$-i\hbar \frac{d}{dx} \langle x | p' \rangle = p' \langle x | p' \rangle \implies \frac{d}{dx} \langle x | p' \rangle = \frac{ip'}{\hbar} \langle x | p' \rangle. \quad (1.5.10)$$

This is a first-order linear ordinary differential equation whose solution is:

$$\langle x | p' \rangle = C e^{ip'x/\hbar}, \quad (1.5.11)$$

where C is a normalization constant. To determine C , we use the Dirac delta normalization condition:

$$\langle p | p' \rangle = \delta(p - p'). \quad (1.5.12)$$

Inserting the completeness relation of the position basis $\int dx |x\rangle \langle x| = \hat{I}$:

$$\langle p | p' \rangle = \int_{-\infty}^{\infty} dx \langle p | x \rangle \langle x | p' \rangle = \int_{-\infty}^{\infty} dx \langle x | p \rangle^* \langle x | p' \rangle$$

$$= |C|^2 \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} e^{ip'x/\hbar} = |C|^2 \int_{-\infty}^{\infty} dx e^{i(p'-p)x/\hbar}. \quad (1.5.13)$$

Recall the standard Fourier integral representation of the Dirac delta function:

$$\int_{-\infty}^{\infty} dx e^{ikx} = 2\pi\delta(k) \implies \int_{-\infty}^{\infty} dx e^{i(p'-p)x/\hbar} = 2\pi\hbar\delta(p-p'). \quad (1.5.14)$$

Substituting this gives:

$$\langle p|p' \rangle = |C|^2 2\pi\hbar\delta(p-p') = \delta(p-p') \implies |C| = \frac{1}{\sqrt{2\pi\hbar}}. \quad (1.5.15)$$

Choosing the standard zero-phase convention (C real and positive), we find:

$$\boxed{\langle x|p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}, \quad \langle p|x \rangle = \langle x|p \rangle^* = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}} \quad (1.5.16)$$

These transformation brackets are the matrix elements connecting the position and momentum representations.

1.5.3 The Fourier Transform Connection

Using these transformation brackets, we can transition seamlessly between position space and momentum space:

$$\psi(x) \equiv \langle x|\psi \rangle = \int_{-\infty}^{\infty} dp \langle x|p \rangle \langle p|\psi \rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} \psi(p), \quad (1.5.17)$$

$$\psi(p) \equiv \langle p|\psi \rangle = \int_{-\infty}^{\infty} dx \langle p|x \rangle \langle x|\psi \rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} \psi(x). \quad (1.5.18)$$

The position-space wavefunction $\psi(x)$ and the momentum-space wavefunction $\psi(p)$ are related by a **Fourier transform**! The symmetric factor of $1/\sqrt{2\pi\hbar}$ ensures that the transformation is unitary (preserving probability norm):

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} |\psi(p)|^2 dp = 1. \quad (1.5.19)$$

1.5.4 Representation of Momentum via the Spectral Decomposition Theorem

In Section 1.1.4, we formulated the spectral decomposition theorem for discrete observables:

$$\hat{A} = \sum_n a_n |a_n\rangle \langle a_n|. \quad (1.5.20)$$

For a self-adjoint operator $\hat{p}$ with continuous spectrum spanning the real line $p \in (-\infty, \infty)$, the spectral theorem (rigorously established in the rigged Hilbert space via Maurin's Nuclear Spectral Theorem) asserts that $\hat{p}$ can be decomposed into an integral over its projection spectral family $d\hat{E}(p) = |p\rangle \langle p| dp$:

$$\boxed{\hat{p} = \int_{-\infty}^{\infty} dp p |p\rangle \langle p|} \quad (1.5.21)$$

where $|p\rangle$ are generalized momentum eigenstates satisfying the orthonormality and completeness relations:

$$\hat{p}|p\rangle = p|p\rangle, \quad \langle p|p'\rangle = \delta(p-p'), \quad \int_{-\infty}^{\infty} dp |p\rangle \langle p| = \hat{I}. \quad (1.5.22)$$

We now use this spectral decomposition to derive both the momentum-space representation and the position-space differential representation of $\hat{p}$.

1. Representation in Momentum Space

In its own eigenbasis, the matrix elements of $\hat{p}$ follow immediately from Eq. (1.5.21):

$$\begin{aligned}\langle p' | \hat{p} | p \rangle &= \langle p' | \left(\int_{-\infty}^{\infty} dp'' p'' | p'' \rangle \langle p'' | \right) | p \rangle \\ &= \int_{-\infty}^{\infty} dp'' p'' \langle p' | p'' \rangle \langle p'' | p \rangle \\ &= \int_{-\infty}^{\infty} dp'' p'' \delta(p' - p'') \delta(p'' - p) = p \delta(p' - p).\end{aligned}\quad (1.5.23)$$

Acting on an arbitrary state $|\psi\rangle$ with momentum wavefunction $\phi(p) \equiv \langle p | \psi \rangle$:

$$\begin{aligned}\langle p | \hat{p} | \psi \rangle &= \int_{-\infty}^{\infty} dp' \langle p | \hat{p} | p' \rangle \langle p' | \psi \rangle \\ &= \int_{-\infty}^{\infty} dp' p \delta(p - p') \phi(p') = p \phi(p).\end{aligned}\quad (1.5.24)$$

Hence, in momentum space, $\hat{p}$ acts simply as a **multiplicative operator**:

$$\hat{p} \phi(p) = p \phi(p). \quad (1.5.25)$$

2. Representation in Position Space (The Integral Kernel)

To determine the action of the momentum operator in the position representation directly from the spectral decomposition, consider the matrix element of $\hat{p}$ between two position eigenstates $\langle x |$ and $| x' \rangle$:

$$\begin{aligned}\langle x | \hat{p} | x' \rangle &= \langle x | \left(\int_{-\infty}^{\infty} dp p | p \rangle \langle p | \right) | x' \rangle \\ &= \int_{-\infty}^{\infty} dp p \langle x | p \rangle \langle p | x' \rangle.\end{aligned}\quad (1.5.26)$$

Inserting the transformation brackets $\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$ and $\langle p | x' \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx'/\hbar}$ obtained in Eq. (1.5.16):

$$\langle x | \hat{p} | x' \rangle = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp p e^{ip(x-x')/\hbar}. \quad (1.5.27)$$

Observe that the factor of p under the integral can be generated by differentiating the plane wave exponential with respect to x :

$$-i\hbar \frac{\partial}{\partial x} e^{ip(x-x')/\hbar} = -i\hbar \left(\frac{ip}{\hbar} \right) e^{ip(x-x')/\hbar} = p e^{ip(x-x')/\hbar}. \quad (1.5.28)$$

Substituting this identity and pulling the spatial derivative outside the momentum integral:

$$\langle x | \hat{p} | x' \rangle = -i\hbar \frac{\partial}{\partial x} \left(\frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp e^{ip(x-x')/\hbar} \right). \quad (1.5.29)$$

Recognizing the integral inside the parentheses as the Fourier representation of the Dirac delta function $\delta(x - x')$:

$$\frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp e^{ip(x-x')/\hbar} = \delta(x - x'), \quad (1.5.30)$$

we arrive at the fundamental distributional kernel for the momentum operator in position space:

$$\boxed{\langle x | \hat{p} | x' \rangle = -i\hbar \frac{\partial}{\partial x} \delta(x - x') = -i\hbar \delta'(x - x')} \quad (1.5.31)$$

where $\delta'(x - x') \equiv \partial/\partial x \delta(x - x')$.

3. Action on an Arbitrary Wavefunction

Now, let $\hat{p}$ act on an arbitrary physical state ket $|\psi\rangle$. In position space, using the completeness of the position basis $\int dx' |x'\rangle \langle x'| = \hat{I}$:

$$\begin{aligned} \langle x|\hat{p}|\psi\rangle &= \langle x|\hat{p}\hat{I}|\psi\rangle = \int_{-\infty}^{\infty} dx' \langle x|\hat{p}|x'\rangle \langle x'|\psi\rangle \\ &= \int_{-\infty}^{\infty} dx' \left[-i\hbar \frac{\partial}{\partial x} \delta(x-x') \right] \psi(x'). \end{aligned} \quad (1.5.32)$$

Because the derivative $\partial/\partial x$ acts on the external observation coordinate x rather than the integration variable x' , we can factor it completely out of the integral:

$$\begin{aligned} \langle x|\hat{p}|\psi\rangle &= -i\hbar \frac{\partial}{\partial x} \int_{-\infty}^{\infty} dx' \delta(x-x') \psi(x') \\ &= \boxed{-i\hbar \frac{\partial \psi(x)}{\partial x}}. \end{aligned} \quad (1.5.33)$$

Alternatively, we can express the action by noticing that $\partial/\partial x \delta(x-x') = -\partial/\partial x' \delta(x-x')$ and integrating by parts with respect to x' :

$$\begin{aligned} \langle x|\hat{p}|\psi\rangle &= \int_{-\infty}^{\infty} dx' \left[i\hbar \frac{\partial}{\partial x'} \delta(x-x') \right] \psi(x') \\ &= \underbrace{[i\hbar \delta(x-x') \psi(x')]_{-\infty}^{\infty}}_{=0} - i\hbar \int_{-\infty}^{\infty} dx' \delta(x-x') \frac{\partial \psi(x')}{\partial x'} \\ &= -i\hbar \frac{\partial \psi(x)}{\partial x}. \end{aligned} \quad (1.5.34)$$

This provides an independent and mathematically transparent derivation of the differential momentum operator $\hat{p} \rightarrow -i\hbar \partial/\partial x$, establishing that the differential operator is nothing other than the position-space manifestation of the continuous spectral projector $\int dp p |p\rangle \langle p|$.

4. Functional Calculus and Functions of Momentum

An immediate and profound consequence of the spectral theorem is the **Borel functional calculus**: for any well-behaved function $f(p)$, the corresponding operator $f(\hat{p})$ is defined spectrally as:

$$f(\hat{p}) = \int_{-\infty}^{\infty} dp f(p) |p\rangle \langle p|. \quad (1.5.35)$$

In position space, the matrix element is:

$$\begin{aligned} \langle x|f(\hat{p})|x'\rangle &= \int_{-\infty}^{\infty} dp f(p) \langle x|p\rangle \langle p|x'\rangle = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp f(p) e^{ip(x-x')/\hbar} \\ &= f\left(-i\hbar \frac{\partial}{\partial x}\right) \delta(x-x'). \end{aligned} \quad (1.5.36)$$

Two vital physical examples follow directly:

1. **Kinetic Energy Operator:** Choosing $f(p) = \frac{p^2}{2m}$:

$$\begin{aligned} \langle x|\frac{\hat{p}^2}{2m}|\psi\rangle &= \int_{-\infty}^{\infty} dp \frac{p^2}{2m} \langle x|p\rangle \langle p|\psi\rangle \\ &= \frac{1}{2m} \left(-i\hbar \frac{\partial}{\partial x}\right)^2 \int_{-\infty}^{\infty} dp \langle x|p\rangle \langle p|\psi\rangle = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2}. \end{aligned} \quad (1.5.37)$$

2. **Free-Particle Propagator:** Choosing the time-evolution function $f(p) = e^{-i\frac{p^2}{2m\hbar}t}$, the position-space matrix element (the Feynman propagator for a free particle) evaluates to an exact Gaussian Fresnel integral:

$$\begin{aligned} K(x, t; x', 0) &\equiv \langle x | e^{-i\hat{p}^2 t / (2m\hbar)} | x' \rangle = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp e^{-i\frac{p^2 t}{2m\hbar}} e^{i\frac{p(x-x')}{\hbar}} \\ &= \sqrt{\frac{m}{2\pi i\hbar t}} \exp\left(\frac{im(x-x')^2}{2\hbar t}\right). \end{aligned} \quad (1.5.38)$$

5. Three-Dimensional Generalization

In three dimensions, the spectral decomposition theorem applies to each Cartesian component $\hat{\mathbf{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$:

$$\hat{\mathbf{p}} = \int_{\mathbb{R}^3} d^3p \mathbf{p} |\mathbf{p}\rangle \langle \mathbf{p}|, \quad \langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p} \cdot \mathbf{x} / \hbar}. \quad (1.5.39)$$

The integral kernel in coordinate space is:

$$\begin{aligned} \langle \mathbf{x} | \hat{\mathbf{p}} | \mathbf{x}' \rangle &= \int d^3p \mathbf{p} \langle \mathbf{x} | \mathbf{p} \rangle \langle \mathbf{p} | \mathbf{x}' \rangle = \frac{1}{(2\pi\hbar)^3} \int d^3p \mathbf{p} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}') / \hbar} \\ &= -i\hbar \nabla_{\mathbf{x}} \left[\frac{1}{(2\pi\hbar)^3} \int d^3p e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}') / \hbar} \right] = -i\hbar \nabla_{\mathbf{x}} \delta^3(\mathbf{x} - \mathbf{x}'). \end{aligned} \quad (1.5.40)$$

Acting on a 3D state $|\psi\rangle$:

$$\langle \mathbf{x} | \hat{\mathbf{p}} | \psi \rangle = \int d^3x' \langle \mathbf{x} | \hat{\mathbf{p}} | \mathbf{x}' \rangle \psi(\mathbf{x}') = -i\hbar \nabla \psi(\mathbf{x}), \quad (1.5.41)$$

confirming $\hat{\mathbf{p}} \longleftrightarrow -i\hbar \nabla$.

1.5.5 Position Operator in Momentum Space

Just as $\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$ in position space, we can determine the action of the position operator $\hat{x}$ in momentum space. Consider the matrix element $\langle p | [\hat{x}, \hat{p}] | \psi \rangle$:

$$\langle p | \hat{x}\hat{p} - \hat{p}\hat{x} | \psi \rangle = \langle p | \hat{x}\hat{p} | \psi \rangle - p \langle p | \hat{x} | \psi \rangle. \quad (1.5.42)$$

Using the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$:

$$i\hbar \langle p | \psi \rangle = \langle p | \hat{x}\hat{p} | \psi \rangle - p \langle p | \hat{x} | \psi \rangle \implies p \langle p | \hat{x} | \psi \rangle = \langle p | \hat{x}\hat{p} | \psi \rangle - i\hbar \psi(p). \quad (1.5.43)$$

Now evaluate $\langle p | \hat{x}\hat{p} | \psi \rangle$ by inserting the resolution of identity in position space:

$$\begin{aligned} \langle p | \hat{x}\hat{p} | \psi \rangle &= \int_{-\infty}^{\infty} dx \langle p | \hat{x} | x \rangle \langle x | \hat{p} | \psi \rangle = \int_{-\infty}^{\infty} dx x \langle p | x \rangle \left(-i\hbar \frac{d\psi(x)}{dx} \right) \\ &= \frac{-i\hbar}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx x e^{-ipx/\hbar} \frac{d\psi(x)}{dx}. \end{aligned} \quad (1.5.44)$$

Integrating by parts:

$$\begin{aligned} \langle p | \hat{x}\hat{p} | \psi \rangle &= \frac{i\hbar}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx \frac{d}{dx} \left(x e^{-ipx/\hbar} \right) \psi(x) \\ &= \frac{i\hbar}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} \psi(x) + \frac{i\hbar}{\sqrt{2\pi\hbar}} \left(-\frac{ip}{\hbar} \right) \int_{-\infty}^{\infty} dx x e^{-ipx/\hbar} \psi(x) \\ &= i\hbar \psi(p) + p \int_{-\infty}^{\infty} dx \left(i\hbar \frac{d}{dp} e^{-ipx/\hbar} \right) \frac{\psi(x)}{\sqrt{2\pi\hbar}} \end{aligned}$$

$$= i\hbar\psi(p) + i\hbar p \frac{d}{dp}\psi(p). \quad (1.5.45)$$

Substituting Eq. (1.5.45) into Eq. (1.5.43):

$$p \langle p|\hat{x}|\psi\rangle = \left(i\hbar\psi(p) + i\hbar p \frac{d}{dp}\psi(p) \right) - i\hbar\psi(p) = i\hbar p \frac{d}{dp}\psi(p). \quad (1.5.46)$$

Dividing both sides by p :

$$\boxed{\langle p|\hat{x}|\psi\rangle = i\hbar \frac{d}{dp}\psi(p)} \quad (1.5.47)$$

Hence, in momentum space, the position operator acts as the differential operator $\hat{x} \longleftrightarrow i\hbar \frac{d}{dp}$. Notice the sign difference: $\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$, whereas $\hat{x} \rightarrow +i\hbar \frac{\partial}{\partial p}$, ensuring $[i\hbar \frac{\partial}{\partial p}, p]\psi(p) = i\hbar\psi(p)$, which preserves $[\hat{x}, \hat{p}] = i\hbar$.

1.6 Coordinate Systems, Curvilinear Coordinates, and Hermiticity

In classical physics, coordinate transformations are routine changes of variables. In quantum mechanics, however, because coordinates become operators and differential operators must satisfy Hermiticity with respect to curvilinear volume measures, switching coordinates requires great care.

1.6.1 Coordinate Surfaces and Curvilinear Geometry

A coordinate system is defined by a set of coordinate surfaces. Specialized curvilinear systems such as elliptic coordinates, bipolar coordinates, or parabolic coordinates are chosen to exploit the symmetries of particular physical problems.

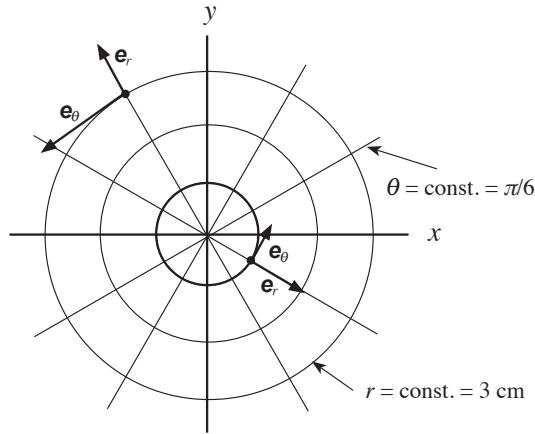


Figure 1.2: In curvilinear coordinate systems, the local unit basis vectors depend explicitly on the spatial coordinates: $\frac{\partial \hat{r}}{\partial \theta} \neq 0$ and $\frac{\partial \hat{\theta}}{\partial r} \neq 0$.

In curvilinear systems, the basis vectors are not spatially uniform (Figure 1.2). For instance, in two-dimensional polar coordinates (r, θ) :

$$\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}, \quad (1.6.1)$$

$$\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}. \quad (1.6.2)$$

Differentiating the basis vectors with respect to the coordinates yields:

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta} \neq 0, \quad \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r} \neq 0. \quad (1.6.3)$$

This coordinate dependence complicates the construction of conjugate momentum operators.

1.6.2 Hermiticity of the Radial Momentum Operator in Spherical Polar Coordinates

In spherical polar coordinates (r, θ, ϕ) , the volume element is:

$$dV = r^2 \sin \theta \, dr d\theta d\phi. \quad (1.6.4)$$

The inner product of two wavefunctions is given by:

$$\langle \psi | \phi \rangle = \int_0^\infty dr \int_0^\pi d\theta \int_0^{2\pi} d\phi \, r^2 \sin \theta \, \psi^*(r, \theta, \phi) \phi(r, \theta, \phi). \quad (1.6.5)$$

We wish to identify the canonical momentum operator P_r conjugate to the radial coordinate r , satisfying $[r, P_r] = i\hbar$. A naive guess from Cartesian intuition would be $P_r \stackrel{?}{=} -i\hbar \frac{\partial}{\partial r}$. Let us test whether this operator is Hermitian under the spherical volume measure:

$$\begin{aligned} \langle \psi | P_r | \phi \rangle &= \int d\Omega \int_0^\infty dr \, r^2 \psi^*(r) \left(-i\hbar \frac{\partial \phi(r)}{\partial r} \right) \\ &= -i\hbar \int d\Omega \left[r^2 \psi^*(r) \phi(r) \Big|_0^\infty - \int_0^\infty dr \, \frac{\partial}{\partial r} (r^2 \psi^*(r) \phi(r)) \right]. \end{aligned} \quad (1.6.6)$$

Assuming wavefunctions vanish sufficiently fast at $r \rightarrow \infty$ and $r \rightarrow 0$, the boundary term vanishes. Applying the product rule to the derivative:

$$\begin{aligned} \langle \psi | P_r | \phi \rangle &= i\hbar \int d\Omega \int_0^\infty dr \left(r^2 \frac{\partial \psi^*(r)}{\partial r} + 2r \psi^*(r) \right) \phi(r) \\ &= \int d\Omega \int_0^\infty dr \, r^2 \left[\left(i\hbar \frac{\partial}{\partial r} + \frac{2i\hbar}{r} \right) \psi(r) \right]^* \phi(r). \end{aligned} \quad (1.6.7)$$

Therefore, the adjoint of $-i\hbar \frac{\partial}{\partial r}$ is:

$$\left(-i\hbar \frac{\partial}{\partial r} \right)^\dagger = -i\hbar \left(\frac{\partial}{\partial r} + \frac{2}{r} \right) \neq -i\hbar \frac{\partial}{\partial r}. \quad (1.6.8)$$

The naive radial derivative operator is **not Hermitian!** To construct a truly self-adjoint radial momentum operator, we symmetrize via the arithmetic mean $\frac{1}{2}(P_r + P_r^\dagger)$:

$$P_r = \frac{1}{2} \left[\frac{\hbar}{i} \frac{\partial}{\partial r} + \frac{\hbar}{i} \left(\frac{\partial}{\partial r} + \frac{2}{r} \right) \right] = \frac{\hbar}{i} \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) = -i\hbar \frac{1}{r} \frac{\partial}{\partial r} r \quad (1.6.9)$$

This operator is strictly Hermitian, and it satisfies the canonical commutation relation:

$$[r, P_r] \psi(r) = r \left[\frac{\hbar}{i} \left(\frac{\partial \psi}{\partial r} + \frac{\psi}{r} \right) \right] - \frac{\hbar}{i} \left[\frac{\partial}{\partial r} + \frac{r\psi}{r} \right] = i\hbar \psi(r). \quad (1.6.10)$$

Moreover, squaring P_r gives the radial kinetic energy:

$$P_r^2 = -\hbar^2 \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) = -\hbar^2 \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) = -\hbar^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right), \quad (1.6.11)$$

which is precisely the radial part of the Laplacian ∇^2 !

1.6.3 Cylindrical Coordinates

In cylindrical polar coordinates (r, θ, z) , the volume element is $dV = r dr d\theta dz$. Repeating the Hermiticity analysis:

$$\begin{aligned}\langle \psi | P_r | \phi \rangle &= \int dz d\theta \int_0^\infty dr r \psi^*(r) \left(-i\hbar \frac{\partial \phi}{\partial r} \right) \\ &= i\hbar \int dz d\theta \int_0^\infty dr \left(r \frac{\partial \psi^*}{\partial r} + \psi^* \right) \phi \\ &= \int dz d\theta \int_0^\infty dr r \left[\frac{\hbar}{i} \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \psi \right]^* \phi.\end{aligned}\quad (1.6.12)$$

Hence, the adjoint is $\left(-i\hbar \frac{\partial}{\partial r} \right)^\dagger = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right)$. Symmetrizing yields the Hermitian radial momentum in cylindrical coordinates:

$$P_r = \frac{\hbar}{i} \left(\frac{\partial}{\partial r} + \frac{1}{2r} \right) = -i\hbar \frac{1}{\sqrt{r}} \frac{\partial}{\partial r} \sqrt{r} \quad (1.6.13)$$

1.6.4 The Phase Operator Problem

Can we define an angular position operator $\hat{\theta}$ conjugate to the angular momentum operator $\hat{L}_z = -i\hbar \frac{\partial}{\partial \theta}$ such that $[\hat{\theta}, \hat{L}_z] = i\hbar$?

In classical mechanics, the angle θ and L_z are canonical conjugates. In quantum mechanics, however, θ is a periodic variable ($\theta \sim \theta + 2\pi$). Wavefunctions must satisfy the periodicity condition $\psi(\theta + 2\pi) = \psi(\theta)$. If a multiplicative operator $\hat{\theta}\psi(\theta) = \theta\psi(\theta)$ were defined, it would map periodic functions into non-periodic ones: $\theta\psi(\theta)$ is not 2π -periodic because $(2\pi)\psi(2\pi) \neq 0 \cdot \psi(0)$.

If one attempts to restrict θ to $[0, 2\pi)$ by defining $\hat{\theta}\psi(\theta) = (\theta - 2\pi \lfloor \frac{\theta}{2\pi} \rfloor)\psi(\theta)$, the operator develops jump discontinuities at the boundary. The commutator $[\hat{\theta}, \hat{L}_z]$ then picks up Dirac delta boundary contributions:

$$[\hat{\theta}, \hat{L}_z] = i\hbar (1 - 2\pi\delta(\theta - 2\pi)) \neq i\hbar. \quad (1.6.14)$$

Furthermore, if $[\hat{\theta}, \hat{L}_z] = i\hbar$ held rigorously, $\hat{\theta}$ would act as a generator of continuous shifts in L_z : $e^{i\alpha\hat{\theta}/\hbar} \hat{L}_z e^{-i\alpha\hat{\theta}/\hbar} = \hat{L}_z - \alpha$, implying that the spectrum of $\hat{L}_z$ must be continuous over all real numbers. But the spectrum of $\hat{L}_z$ is strictly discrete: $m\hbar$ with $m \in \mathbb{Z}$! Therefore, **no well-behaved, self-adjoint phase operator $\hat{\theta}$ conjugate to $\hat{L}_z$ exists**. Instead, one works with periodic trigonometric operators $\sin \hat{\theta}$ and $\cos \hat{\theta}$ or ladder operators $e^{\pm i\hat{\theta}}$.

1.6.5 Vector and Tensor Operators

A vector operator $\hat{\mathbf{V}} = (\hat{V}_1, \hat{V}_2, \hat{V}_3)$ is a set of three operators that transform under spatial rotations like a classical vector. Under an infinitesimal rotation by angle $d\theta$, generated by the total angular momentum operator $\hat{\mathbf{J}}$, a vector operator satisfies the Lie algebraic commutation relations:

$$[\hat{V}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{V}_k. \quad (1.6.15)$$

Some vector operators, such as position $\hat{\mathbf{r}}$ and momentum $\hat{\mathbf{p}}$, have mutually commuting spatial components: $[\hat{x}_i, \hat{x}_j] = 0$ and $[\hat{p}_i, \hat{p}_j] = 0$. Others, such as the orbital angular momentum vector $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ or spin $\hat{\mathbf{S}}$, have non-commuting components:

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k. \quad (1.6.16)$$

1.7 Canonical Quantization Hierarchy

The transition from classical mechanics to quantum mechanics follows a systematic hierarchical procedure known as **Canonical Quantization**, originally formulated by Dirac.

1.7.1 Classical Phase Space and Poisson Brackets

In classical Hamiltonian mechanics, a system with f degrees of freedom is described by generalized coordinates $q = (q_1, \dots, q_f)$ and conjugate generalized momenta $p = (p_1, \dots, p_f)$. The dynamics are governed by the Hamiltonian function $H(q, p, t)$. For any two dynamical functions $A(q, p, t)$ and $B(q, p, t)$ on phase space, the **Poisson bracket** is defined as:

$$\{A, B\}_{\text{PB}} \equiv \sum_{i=1}^f \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right). \quad (1.7.1)$$

The fundamental Poisson brackets among coordinates and conjugate momenta are:

$$\{q_j, p_k\}_{\text{PB}} = \delta_{jk}, \quad \{q_j, q_k\}_{\text{PB}} = 0, \quad \{p_j, p_k\}_{\text{PB}} = 0. \quad (1.7.2)$$

Hamilton's classical equations of motion can be written compactly as:

$$\dot{q}_i = \{q_i, H\}_{\text{PB}}, \quad \dot{p}_i = \{p_i, H\}_{\text{PB}}, \quad (1.7.3)$$

and the total time derivative of any phase-space function $A(q, p, t)$ is:

$$\frac{dA}{dt} = \{A, H\}_{\text{PB}} + \frac{\partial A}{\partial t}. \quad (1.7.4)$$

1.7.2 The Dirac Quantization Rule

Dirac's canonical quantization establishes a correspondence between classical Poisson brackets and quantum commutators:

Promote classical phase-space observables to linear Hermitian operators, replacing Poisson brackets by operator commutators divided by $i\hbar$:

$$\boxed{\{A, B\}_{\text{PB}} \longrightarrow \frac{1}{i\hbar} [\hat{A}, \hat{B}] = -\frac{i}{\hbar} [\hat{A}, \hat{B}]} \quad (1.7.5)$$

Applying this canonical quantization rule to the fundamental phase-space coordinates yields the canonical commutation relations:

$$[\hat{q}_j, \hat{p}_k] = i\hbar \delta_{jk} \hat{I}, \quad [\hat{q}_j, \hat{q}_k] = 0, \quad [\hat{p}_j, \hat{p}_k] = 0. \quad (1.7.6)$$

1.7.3 The Hierarchy of Quantum Mechanics

The canonical quantization program establishes a logical hierarchy:

1. **Classical Hamiltonian Formulation:** Formulate the system in terms of canonical coordinates q_i and conjugate momenta $p_i = \frac{\partial L}{\partial \dot{q}_i}$, with classical Hamiltonian $H(q, p)$.
2. **Promotion to Operators:** Promote coordinates and momenta to Hermitian operators $\hat{q}_i, \hat{p}_i$ acting on Hilbert space, satisfying $[\hat{q}_j, \hat{p}_k] = i\hbar \delta_{jk} \hat{I}$. Replace the classical Hamiltonian $H(q, p)$ by the quantum Hamiltonian operator $\hat{H} = H(\hat{q}, \hat{p})$ (resolving any operator ordering ambiguities by symmetric ordering).
3. **Time Evolution and Dynamics:** The quantum Hamiltonian operator $\hat{H}$ acts as the generator of time translation, governing the dynamic evolution of states and observables.

1.7.4 Semiclassical Physics

In many physical scenarios, certain degrees of freedom can be treated quantum mechanically while others remain classical. In **semiclassical physics**, we promote only selected parameters to quantum operators. For example, in semiclassical radiation theory, an atom is treated with quantized electron operators while the electromagnetic field is treated as a classical wave. Similarly, in semiclassical gravity, matter fields are quantized on a classical curved spacetime background described by the metric tensor $g_{\mu\nu}$:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \langle \psi | \hat{T}_{\mu\nu} | \psi \rangle. \quad (1.7.7)$$

1.8 Coordinate Transformations and Group Symmetries

1.8.1 Jacobians and Integration Measures

When performing coordinate transformations in multi-dimensional integrals, volume elements transform via the Jacobian determinant:

$$dx dy = |J| du dv, \quad \text{where} \quad J = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}. \quad (1.8.1)$$

In the language of differential forms, the volume measure is an exterior wedge product $dx \wedge dy$. Under a transformation to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$dx = \cos \theta dr - r \sin \theta d\theta, \quad (1.8.2)$$

$$dy = \sin \theta dr + r \cos \theta d\theta. \quad (1.8.3)$$

Taking the wedge product:

$$dx \wedge dy = (\cos \theta dr - r \sin \theta d\theta) \wedge (\sin \theta dr + r \cos \theta d\theta) = r(\cos^2 \theta + \sin^2 \theta) dr \wedge d\theta = r dr \wedge d\theta. \quad (1.8.4)$$

Transformations preserving the canonical commutation relations $[\hat{q}_i, \hat{p}_j] = i\hbar \delta_{ij}$ are the quantum analogues of classical **canonical transformations**.

1.8.2 Symmetry Generators and Lie Algebras

In quantum mechanics, continuous symmetries are represented by unitary operators generated by Hermitian operators:

- **Spatial Translations:** Generated by the momentum operator: $\mathcal{T}(\mathbf{a}) = \exp(-i\mathbf{a} \cdot \hat{\mathbf{p}}/\hbar)$.
- **Spatial Rotations:** Generated by the angular momentum operator: $\mathcal{R}(\hat{\mathbf{n}}, \theta) = \exp(-i\theta \hat{\mathbf{n}} \cdot \hat{\mathbf{J}}/\hbar)$.
- **Time Translations:** Generated by the Hamiltonian: $\hat{U}(t) = \exp(-i\hat{H}t/\hbar)$.

According to Noether's theorem, whenever the Hamiltonian commutes with a symmetry generator $\hat{G}$ ($[\hat{H}, \hat{G}] = 0$), the observable corresponding to $\hat{G}$ is a conserved quantity.

The generators T_a of a Lie group satisfy a closed Lie algebra:

$$[T_a, T_b] = if_{abc} T_c, \quad (1.8.5)$$

where f_{abc} are the structure constants of the Lie group. In representation theory:

- The **Fundamental Representation** is the lowest-dimensional non-trivial faithful representation (e.g., 2×2 Pauli matrices for $SU(2)$).

- The **Adjoint Representation** has dimension equal to the number of group generators, with matrix elements given directly by the structure constants: $(T_a)_{bc} = -if_{abc}$.

Having established the mathematical foundations of states, operators, continuous spectra, translations, and canonical quantization, we are now fully prepared to investigate the dynamical flow of time and the three pictures of quantum mechanics in [Chapter 2](#).

Chapter 2

Pictures of Quantum Mechanics & Time Evolution

In classical mechanics, the dynamical state of a system is given by the time-dependent phase-space trajectory $(q(t), p(t))$, while physical observables are functions $f(q(t), p(t))$ evaluated along this path. In quantum mechanics, time plays a profoundly distinct role from spatial coordinates. Most crucially, as we shall prove from first principles, **time is not an operator** in standard quantum mechanics, nor does an eigenket $|t\rangle$ exist in the Hilbert space.

Instead, time evolution is mediated by a family of unitary operators parameterized by an external, classical time parameter t . Because all physical predictions correspond to scalar products and expectation values, there is a fundamental unitary ambiguity in how the dynamical time dependence is allocated between state vectors and physical observables. This freedom leads to the three overarching formulations of quantum dynamics: the **Schrödinger picture**, the **Heisenberg picture**, and the **Interaction (Dirac) picture**.

2.1 Time Evolution from First Principles

2.1.1 Why Time is Not an Operator: The Absence of $|t\rangle$

In Chapter 1, we derived the momentum operator $\hat{p}$ and the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ by constructing the spatial translation operator $\mathcal{T}(dx)$ acting on the complete basis of position eigenkets $\{|x\rangle\}$:

$$\hat{x}|x'\rangle = x'|x'\rangle, \quad \mathcal{T}(dx)|x'\rangle = |x' + dx\rangle. \quad (2.1.1)$$

It is tempting to wonder whether time can be treated on an identical footing: Can we promote time t to a self-adjoint operator $\hat{T}$ conjugate to the Hamiltonian $\hat{H}$, such that $[\hat{T}, \hat{H}] = i\hbar$, in analogy with $[\hat{x}, \hat{p}] = i\hbar$?

The answer is an emphatic **no**. In non-relativistic quantum mechanics, as well as in standard relativistic quantum field theory:

1. **No time operator exists:** Time t is an external, universal evolution parameter (a c -number), not a quantum dynamical observable.
2. **No state $|t\rangle$ exists:** There is no Hilbert space of “time states” $\{|t\rangle\}$ spanned by time eigenvalues. Wavefunctions are functions of coordinates at a given clock time: $\psi(x, t) = \langle x|\psi(t)\rangle$.
3. **No commutation relation $[t, H] = i\hbar$:** Because t is not an operator, we **cannot** define the Hamiltonian operator $\hat{H}$ via a commutation relation $[t, H] = i\hbar$.

2.1.2 Pauli's Theorem: The Impossibility of a Self-Adjoint Time Operator

The physical reason why time cannot be a self-adjoint operator was first pointed out by Wolfgang Pauli in 1933 and is known as **Pauli's Theorem** (or Pauli's Objection).

Theorem 2.1 (Pauli's Theorem). In quantum mechanics, there cannot exist a self-adjoint time operator $\hat{T} = \hat{T}^\dagger$ satisfying the canonical commutation relation with the Hamiltonian:

$$[\hat{T}, \hat{H}] = i\hbar\hat{I}, \quad (2.1.2)$$

if the spectrum of $\hat{H}$ is bounded from below.

Proof. Suppose, for the sake of contradiction, that such a self-adjoint operator $\hat{T}$ exists. For any real parameter $\varepsilon \in \mathbb{R}$, construct the unitary one-parameter family of operators:

$$\hat{V}(\varepsilon) \equiv \exp\left(\frac{i}{\hbar}\varepsilon\hat{T}\right). \quad (2.1.3)$$

Because $\hat{T}$ is self-adjoint, $\hat{V}(\varepsilon)$ is strictly unitary: $\hat{V}^\dagger(\varepsilon)\hat{V}(\varepsilon) = \hat{I}$.

We now examine how $\hat{V}(\varepsilon)$ transforms the Hamiltonian $\hat{H}$ under a similarity transformation. Using the Baker–Campbell–Hausdorff operator identity:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \dots, \quad (2.1.4)$$

setting $\hat{A} = \frac{i\varepsilon}{\hbar}\hat{T}$ and $\hat{B} = \hat{H}$, and evaluating the commutator using Eq. (2.1.2):

$$[\hat{A}, \hat{B}] = \left[\frac{i\varepsilon}{\hbar}\hat{T}, \hat{H}\right] = \frac{i\varepsilon}{\hbar}[\hat{T}, \hat{H}] = \frac{i\varepsilon}{\hbar}(i\hbar\hat{I}) = -\varepsilon\hat{I}. \quad (2.1.5)$$

Because this commutator is a multiple of the identity, all higher nested commutators vanish identically ($[\hat{A}, [\hat{A}, \hat{B}]] = 0$). Therefore, the BCH series terminates after the second term:

$$\hat{V}(\varepsilon)\hat{H}\hat{V}^\dagger(\varepsilon) = \hat{H} - \varepsilon\hat{I}. \quad (2.1.6)$$

Eq. (2.1.6) shows that $\hat{V}(\varepsilon)$ acts as an energy shift operator.

Now, let $|E\rangle$ be any eigenstate of $\hat{H}$ belonging to energy eigenvalue E , such that $\hat{H}|E\rangle = E|E\rangle$. Multiply Eq. (2.1.6) on the right by $\hat{V}(\varepsilon)|E\rangle$:

$$\begin{aligned} \hat{H}(\hat{V}(\varepsilon)|E\rangle) &= (\hat{V}(\varepsilon)\hat{H}\hat{V}^\dagger(\varepsilon) + \varepsilon\hat{I})\hat{V}(\varepsilon)|E\rangle \\ &= \hat{V}(\varepsilon)\hat{H}|E\rangle \\ &= E(\hat{V}(\varepsilon)|E\rangle). \end{aligned} \quad (2.1.7)$$

Acting with $\hat{V}^\dagger(\varepsilon)$ to the left:

$$\hat{H}(\hat{V}^\dagger(\varepsilon)|E\rangle) = (E - \varepsilon)(\hat{V}^\dagger(\varepsilon)|E\rangle). \quad (2.1.8)$$

This reveals that $|E - \varepsilon\rangle \equiv \hat{V}^\dagger(\varepsilon)|E\rangle$ is an exact eigenstate of $\hat{H}$ with eigenvalue $E - \varepsilon$.

Since ε can be *any* arbitrary real number in the continuum $(-\infty, \infty)$, this would imply that if $\hat{H}$ has even a single eigenvalue E , its energy spectrum must be continuous and completely unbounded from below:

$$\text{Spectrum}(\hat{H}) = (-\infty, +\infty). \quad (2.1.9)$$

However, every physical system in nature must possess a stable ground state with finite minimum energy $E_0 > -\infty$ (i.e., the Hamiltonian spectrum must be bounded from below). If the energy spectrum were unbounded from below, particles would emit infinite amounts of radiation by continually cascading into infinitely negative energy states, destroying the stability of matter.

Therefore, a self-adjoint time operator $\hat{T}$ conjugate to $\hat{H}$ **cannot exist** in any physically meaningful quantum theory. $\square$

Remark on Time-Energy Uncertainty: What, then, is the meaning of the famous time-energy uncertainty relation $\Delta E \Delta t \geq \frac{\hbar}{2}$? It is *not* an operator-eigenvalue uncertainty of the Robertson type ($\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$) because there is no operator $\hat{T}$. Rather, it is a dynamical timescale relation (formulated rigorously by Mandelstam and Tamm): Δt represents the characteristic time interval required for the expectation value of an observable $\hat{A}$ to change by one standard deviation:

$$\Delta t \equiv \frac{\Delta A}{\left| \frac{d\langle \hat{A} \rangle}{dt} \right|}, \quad \text{whence} \quad \Delta E \Delta t \geq \frac{\hbar}{2}. \quad (2.1.10)$$

2.1.3 The Unitary Time Translation Operator $\hat{U}(t, t_0)$

Since time is a parameter rather than an operator, time evolution must be defined through a one-parameter mapping between state kets at different times. We define the **time-evolution operator** $\hat{U}(t, t_0)$ as the linear operator that advances any state ket from an initial time t_0 to a later time t :

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle. \quad (2.1.11)$$

Physical consistency requires that $\hat{U}(t, t_0)$ satisfy four fundamental physical postulates:

1. **Probability Conservation (Unitarity):** The total probability of finding the system in some state must remain strictly equal to 1 at all times. For any two evolving states $|\alpha(t)\rangle = \hat{U}(t, t_0) |\alpha(t_0)\rangle$ and $|\beta(t)\rangle = \hat{U}(t, t_0) |\beta(t_0)\rangle$:

$$\langle \alpha(t) | \beta(t) \rangle = \langle \alpha(t_0) | \hat{U}^\dagger(t, t_0) \hat{U}(t, t_0) | \beta(t_0) \rangle = \langle \alpha(t_0) | \beta(t_0) \rangle. \quad (2.1.12)$$

Since this must hold for any arbitrary pair of states, $\hat{U}(t, t_0)$ must be a **unitary operator**:

$$\boxed{\hat{U}^\dagger(t, t_0) \hat{U}(t, t_0) = \hat{I} \iff \hat{U}^{-1}(t, t_0) = \hat{U}^\dagger(t, t_0)} \quad (2.1.13)$$

2. **Composition Law:** Evolving from t_0 to t_1 , and then from t_1 to t_2 , must be equivalent to evolving directly from t_0 to t_2 :

$$\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1) \hat{U}(t_1, t_0), \quad \text{for any } t_0 \leq t_1 \leq t_2. \quad (2.1.14)$$

3. **Reversibility / Inverse:** Evolving backward in time must invert forward evolution:

$$\hat{U}(t_0, t) \hat{U}(t, t_0) = \hat{U}(t_0, t_0) = \hat{I} \implies \hat{U}(t_0, t) = \hat{U}^{-1}(t, t_0) = \hat{U}^\dagger(t, t_0). \quad (2.1.15)$$

4. **Continuity (Identity Limit):** Over an infinitesimal duration, the system state cannot jump discontinuously:

$$\lim_{t \rightarrow t_0} \hat{U}(t, t_0) = \hat{I}. \quad (2.1.16)$$

2.1.4 Derivation of the Schrödinger Equation for State Kets

Consider an infinitesimal time step from t to $t + dt$. Because $\hat{U}(t, t) = \hat{I}$, we expand $\hat{U}(t + dt, t)$ to first order in dt :

$$\hat{U}(t + dt, t) = \hat{I} - \frac{i}{\hbar} \hat{H}(t) dt + \mathcal{O}(dt^2), \quad (2.1.17)$$

where $\hat{H}(t)$ is the generator of infinitesimal time translations. The constant $-i/\hbar$ ensures that $\hat{H}(t)$ has the physical dimensions of energy ([action]/[time]).

Unitarity requires:

$$\hat{I} = \hat{U}^\dagger(t+dt, t) \hat{U}(t+dt, t) = \left(\hat{I} + \frac{i}{\hbar} \hat{H}^\dagger(t) dt \right) \left(\hat{I} - \frac{i}{\hbar} \hat{H}(t) dt \right) = \hat{I} + \frac{i}{\hbar} (\hat{H}^\dagger(t) - \hat{H}(t)) dt + \mathcal{O}(dt^2). \quad (2.1.18)$$

Vanishing of the first-order term mandates:

$$\hat{H}^\dagger(t) = \hat{H}(t). \quad (2.1.19)$$

The generator of time translations is necessarily a Hermitian operator! We identify $\hat{H}$ as the quantum **Hamiltonian operator** of the system.

Now, consider the evolution of the state ket:

$$|\psi(t+dt)\rangle = \hat{U}(t+dt, t) |\psi(t)\rangle = \left(\hat{I} - \frac{i}{\hbar} \hat{H}(t) dt \right) |\psi(t)\rangle. \quad (2.1.20)$$

Subtracting $|\psi(t)\rangle$ and dividing by dt :

$$\frac{|\psi(t+dt)\rangle - |\psi(t)\rangle}{dt} = -\frac{i}{\hbar} \hat{H}(t) |\psi(t)\rangle. \quad (2.1.21)$$

Taking the limit $dt \rightarrow 0$ gives the fundamental equation of motion for state vectors:

$$\boxed{i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle} \quad (2.1.22)$$

This is the abstract, basis-independent **Schrödinger equation**. Similarly, for the time evolution operator itself:

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H}(t) \hat{U}(t, t_0), \quad \text{with initial condition} \quad \hat{U}(t_0, t_0) = \hat{I}. \quad (2.1.23)$$

2.1.5 The Dual Identity of the Hamiltonian and Emergence of the Wave Equation

Right at this juncture, where $\hat{H}$ is introduced as the generator of infinitesimal time translations:

$$\hat{U}(dt) = \hat{I} - \frac{i}{\hbar} \hat{H} dt, \quad (2.1.24)$$

it is of profound conceptual importance to recognize that **the Hamiltonian possesses another, complementary physical form**.

1. **The Generator Form (Dynamical Symmetry):** Abstractly, $\hat{H}$ is the Hermitian generator of temporal translations in Hilbert space. Just as the momentum operator $\hat{\mathbf{p}}$ was derived in Section 1.4 as the generator of spatial translations:

$$\mathcal{T}(d\mathbf{x}) = \hat{I} - \frac{i}{\hbar} \hat{\mathbf{p}} \cdot d\mathbf{x}, \quad (2.1.25)$$

the Hamiltonian $\hat{H}$ is the generator of temporal translations $\hat{U}(dt) = \hat{I} - \frac{i}{\hbar} \hat{H} dt$, establishing the fundamental law of motion:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (2.1.26)$$

2. **The Kinematic Operator Form (Canonical Observables):** Crucially, $\hat{H}$ is not an abstract, inaccessible black box. For any physical system possessing a classical analogue,

the Hamiltonian can be written explicitly in terms of the fundamental kinematic operators—the position operator $\hat{\mathbf{x}}$ and the momentum operator $\hat{\mathbf{p}}$:

$$\hat{H} = \hat{H}(\hat{\mathbf{x}}, \hat{\mathbf{p}}) = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \quad (2.1.27)$$

(or $\hat{H} = \frac{1}{2m}[\hat{\mathbf{p}} - q\mathbf{A}(\hat{\mathbf{x}})]^2 + q\Phi(\hat{\mathbf{x}})$ in an electromagnetic gauge field). Here, the momentum operator $\hat{\mathbf{p}}$ appearing in $\hat{H}$ is none other than the spatial translation generator derived from first principles in Chapter 1!

Fundamental Synthesis: What the Equation Looks Like Projected in the Position Basis

What is the direct consequence of expressing the time translation generator $\hat{H}$ in terms of these kinematic operators?

If we take the abstract dynamical equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad (1)$$

substitute the operator form $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$, and **project the entire equation down onto the continuous position eigenbasis** by taking the inner product with the coordinate bra $\langle \mathbf{x}|$:

$$\langle \mathbf{x}| i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \langle \mathbf{x}| \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) |\psi(t)\rangle. \quad (2)$$

We evaluate each side using the first-principles operator properties established in Chapter 1:

- On the left-hand side, because the static position bra $\langle \mathbf{x}|$ has no explicit time dependence ($\frac{\partial}{\partial t} \langle \mathbf{x}| = 0$), the time derivative acts directly on the position wave function $\psi(\mathbf{x}, t) \equiv \langle \mathbf{x}|\psi(t)\rangle$:

$$\langle \mathbf{x}| i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} \langle \mathbf{x}|\psi(t)\rangle = i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t}. \quad (3)$$

- On the right-hand side, since $\hat{\mathbf{p}}$ is the spatial translation generator, its matrix element in the position representation is the gradient operator:

$$\langle \mathbf{x}| \hat{\mathbf{p}} = -i\hbar \nabla \langle \mathbf{x}| \implies \langle \mathbf{x}| \frac{\hat{\mathbf{p}}^2}{2m} |\psi(t)\rangle = -\frac{\hbar^2}{2m} \nabla^2 \langle \mathbf{x}|\psi(t)\rangle = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t). \quad (4)$$

- Because $|\mathbf{x}\rangle$ is an eigenstate of position ($\hat{\mathbf{x}}|\mathbf{x}\rangle = \mathbf{x}|\mathbf{x}\rangle$), the potential operator acts purely diagonally:

$$\langle \mathbf{x}| V(\hat{\mathbf{x}}) |\psi(t)\rangle = V(\mathbf{x}) \langle \mathbf{x}|\psi(t)\rangle = V(\mathbf{x}) \psi(\mathbf{x}, t). \quad (5)$$

Equating the two sides yields:

$$i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t) \quad (6)$$

This is precisely what the celebrated Schrödinger wave equation is!

The differential Schrödinger equation is **not** an independent postulate guessed from de Broglie waves; it is **precisely the abstract time translation equation of motion projected down into the position eigenbasis, with the Hamiltonian written in terms of the fundamental kinematic operators $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$!**

Fundamental Warning: $\hat{H} = \hat{T} + \hat{V}$ Does NOT Imply $E = T + V$ (The Commutator $[\hat{T}, \hat{V}] \neq 0$ and the “Classically Forbidden” Fallacy)

Writing the Hamiltonian operator as the sum of kinetic and potential parts:

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}), \quad (1)$$

is an exact operator identity on Hilbert space. However, one must strictly avoid the pervasive classical temptation to assume that this translates into a scalar equation between sharp numbers:

$$E \stackrel{?}{=} T + V. \quad (2)$$

In quantum mechanics, **unless** $[\hat{T}, \hat{V}] = 0$, **we can NEVER talk about** $E = T + V$!

Why $[\hat{T}, \hat{V}] \neq 0$ Forbids Simultaneous Energy Partitioning. In quantum mechanics, a physical state possesses a definite, sharp numerical value for an observable if and only if it is an eigenstate of that observable’s operator. For a state $|\psi\rangle$ to have simultaneously sharp total energy E , sharp kinetic energy T , and sharp potential energy V , it must be a simultaneous eigenstate of all three operators:

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad \hat{T}|\psi\rangle = T|\psi\rangle, \quad \hat{V}|\psi\rangle = V|\psi\rangle. \quad (3)$$

By the fundamental spectral theorem of linear operators, a set of Hermitian operators can be simultaneously diagonalized (share a common, complete eigenbasis) **if and only if they mutually commute**.

Evaluating the commutator between kinetic and potential energy operators using $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$:

$$\begin{aligned} [\hat{T}, \hat{V}] &= \frac{1}{2m} [\hat{\mathbf{p}}^2, V(\hat{\mathbf{x}})] = \frac{1}{2m} \sum_j (\hat{p}_j [\hat{p}_j, V(\hat{\mathbf{x}})] + [\hat{p}_j, V(\hat{\mathbf{x}})] \hat{p}_j) \\ &= -\frac{i\hbar}{2m} (\hat{\mathbf{p}} \cdot \nabla V(\hat{\mathbf{x}}) + \nabla V(\hat{\mathbf{x}}) \cdot \hat{\mathbf{p}}) \\ &= -\frac{\hbar^2}{2m} \nabla^2 V(\hat{\mathbf{x}}) - \frac{i\hbar}{m} \nabla V(\hat{\mathbf{x}}) \cdot \hat{\mathbf{p}} \neq 0. \end{aligned} \quad (4)$$

For *any* inhomogeneous potential ($V(\mathbf{x}) \neq \text{const}$), $[\hat{T}, \hat{V}] \neq 0$! Consequently:

$$[\hat{H}, \hat{T}] = [\hat{T} + \hat{V}, \hat{T}] = [\hat{V}, \hat{T}] \neq 0, \quad [\hat{H}, \hat{V}] = [\hat{T} + \hat{V}, \hat{V}] = [\hat{T}, \hat{V}] \neq 0. \quad (5)$$

Because the Hamiltonian does not commute with either operator, **an energy eigenstate $|E\rangle$ is NEVER an eigenstate of $\hat{T}$, nor an eigenstate of $\hat{V}$!**

In an energy stationary state $\hat{H}|E\rangle = E|E\rangle$:

- Total energy is completely sharp: $\Delta H = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = 0$.
- Kinetic energy is quantum mechanically dispersed over a broad continuum of momentum states: $\Delta T = \sqrt{\langle \hat{T}^2 \rangle - \langle \hat{T} \rangle^2} > 0$.
- Potential energy is quantum mechanically dispersed over position space: $\Delta V = \sqrt{\langle \hat{V}^2 \rangle - \langle \hat{V} \rangle^2} > 0$.

While linearity ensures that expectation values sum to the eigenvalue, $\langle \hat{H} \rangle = \langle \hat{T} \rangle + \langle \hat{V} \rangle \implies E = \langle \hat{T} \rangle + \langle \hat{V} \rangle$, this is merely a statistical average over superpositions. There exist no individual scalar numbers T and V such that $E = T + V$.

Why Saying $T > V$ or $T < V$ Is Ill-Defined. Because neither T nor V has a definite numerical value, declaring that “kinetic energy exceeds potential energy” ($T > V$), or “potential energy exceeds kinetic energy” ($T < V$), or that “kinetic energy is negative” ($T < 0$) is **strictly ill-defined in quantum mechanics**:

1. The kinetic energy operator $\hat{T} = \frac{\hat{\mathbf{p}}^2}{2m}$ is a positive semi-definite operator on physical Hilbert space. Its spectrum is strictly non-negative: $\sigma(\hat{T}) = [0, \infty)$, and for any state $|\psi\rangle$, $\langle \hat{T} \rangle = \frac{\hbar^2}{2m} \int |\nabla\psi|^2 d^3x \geq 0$. No measurement of kinetic energy can ever yield a negative outcome!
2. There is no such physical observable as “kinetic energy at position $\mathbf{x}$.” The classical expression $T(\mathbf{x}) \equiv E - V(\mathbf{x})$ assumes that a particle has a simultaneous sharp position $\mathbf{x}$ and a sharp momentum $p(\mathbf{x}) = \sqrt{2m(E - V(\mathbf{x}))}$, which violates the canonical commutation relation $[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}$.

The Demystification of “Classically Forbidden” Regions. Even though such classical scalar analysis ($T = E - V(x) < 0$) is universally used to define “**classically forbidden regions**” in barrier penetration and tunneling problems, this is purely a classical heuristic imported into wave mechanics:

- In the Schrödinger coordinate representation:

$$\frac{\nabla^2\psi(\mathbf{x})}{\psi(\mathbf{x})} = \frac{2m}{\hbar^2}(V(\mathbf{x}) - E), \quad (6)$$

the factor $V(\mathbf{x}) - E$ governs **the spatial curvature of the wavefunction amplitude**, not particle kinetic energy.

- Where $E > V(\mathbf{x})$, the curvature is negative ($\nabla^2\psi/\psi < 0$), forcing the wavefunction to curve toward the axis and produce **oscillatory (propagating)** solutions.
- Where $E < V(\mathbf{x})$, the curvature is positive ($\nabla^2\psi/\psi > 0$), forcing the wavefunction to curve away from the axis and produce **evanescent (exponentially decaying)** solutions.

The particle never enters a region with “negative kinetic energy” ($T < 0$) nor does it “borrow energy from the uncertainty principle.” It simply exhibits an evanescent spatial tail, precisely as electromagnetic waves do in total internal reflection in classical optics! (See Chapter 4, Section 4.3.3 and Section 4.3.4 for the complete proof via off-diagonal matrix elements.)

2.1.6 Integration for Time-Independent Hamiltonians

When the Hamiltonian does not depend explicitly on time ($\frac{\partial \hat{H}}{\partial t} = 0$), the formal solution to the operator differential equation is the operator exponential:

$$\hat{U}(t, t_0) = \exp\left(-\frac{i}{\hbar}\hat{H}(t - t_0)\right) \quad (2.1.28)$$

To evaluate the action of $\hat{U}(t, t_0)$ on a physical state, we insert the completeness relation of the energy eigenbasis $\{|E_n\rangle\}$ satisfying $\hat{H}|E_n\rangle = E_n|E_n\rangle$:

$$\hat{U}(t, t_0) = \hat{U}(t, t_0) \sum_n |E_n\rangle \langle E_n|$$

$$\begin{aligned}
&= \sum_n e^{-i\hat{H}(t-t_0)/\hbar} |E_n\rangle \langle E_n| \\
&= \sum_n e^{-iE_n(t-t_0)/\hbar} |E_n\rangle \langle E_n|.
\end{aligned} \tag{2.1.29}$$

For an arbitrary initial state $|\psi(t_0)\rangle = \sum_n c_n(t_0) |E_n\rangle$, where $c_n(t_0) = \langle E_n|\psi(t_0)\rangle$, the time-evolved state is:

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle = \sum_n c_n(t_0) e^{-iE_n(t-t_0)/\hbar} |E_n\rangle. \tag{2.1.30}$$

Each energy component simply acquires an individual dynamical phase factor $e^{-iE_n(t-t_0)/\hbar}$. If the initial state is an energy eigenstate $|E_k\rangle$, the state remains stationary, picking up only an overall, unobservable global phase: $|\psi(t)\rangle = e^{-iE_k(t-t_0)/\hbar} |E_k\rangle$.

2.2 The Schrödinger Wave Equation as the Position-Basis Projection

A frequent pedagogical pitfall is to present the differential Schrödinger wave equation as an ad hoc axiom guessed by de Broglie wave analogies. We now demonstrate that:

The Schrödinger wave equation is nothing other than the abstract state-vector evolution equation $i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$ projected onto the continuous position basis $\langle \mathbf{x}|$!

2.2.1 Rigorous Bra-Ket Derivation

Let us start from the abstract equation of motion:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \tag{2.2.1}$$

Project both sides directly onto the static coordinate bra $\langle \mathbf{x}|$:

$$\langle \mathbf{x}| i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \langle \mathbf{x}| \hat{H} |\psi(t)\rangle. \tag{2.2.2}$$

Let us analyze each side rigorously in terms of bras and kets:

Left-Hand Side: Because the coordinate bra $\langle \mathbf{x}|$ is an eigenbra of the static position operator, it has no explicit time dependence ($\frac{\partial}{\partial t} \langle \mathbf{x}| = 0$). Hence, the partial time derivative commutes with $\langle \mathbf{x}|$:

$$\langle \mathbf{x}| i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} \langle \mathbf{x}|\psi(t)\rangle = i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t}, \tag{2.2.3}$$

where $\psi(\mathbf{x}, t) \equiv \langle \mathbf{x}|\psi(t)\rangle$ is the position-space wavefunction.

Right-Hand Side: For a non-relativistic particle of mass m moving in an external scalar potential $V(\mathbf{x})$, the quantum Hamiltonian operator is:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}). \tag{2.2.4}$$

Substituting this into the right-hand side of Eq. (2.2.2):

$$\langle \mathbf{x}| \hat{H} |\psi(t)\rangle = \langle \mathbf{x}| \frac{\hat{\mathbf{p}}^2}{2m} |\psi(t)\rangle + \langle \mathbf{x}| V(\hat{\mathbf{x}}) |\psi(t)\rangle. \tag{2.2.5}$$

Now evaluate each term:

1. **Kinetic Energy Term:** In Section 1.5, we proved from translation invariance that $\langle \mathbf{x} | \hat{\mathbf{p}} = -i\hbar \nabla \langle \mathbf{x} |$. Applying this operator twice:

$$\begin{aligned} \langle \mathbf{x} | \hat{\mathbf{p}}^2 | \psi(t) \rangle &= \langle \mathbf{x} | (\hat{\mathbf{p}} \cdot \hat{\mathbf{p}}) | \psi(t) \rangle \\ &= (-i\hbar \nabla) \cdot \langle \mathbf{x} | \hat{\mathbf{p}} | \psi(t) \rangle \\ &= (-i\hbar \nabla) \cdot (-i\hbar \nabla) \langle \mathbf{x} | \psi(t) \rangle \\ &= -\hbar^2 \nabla^2 \psi(\mathbf{x}, t). \end{aligned} \quad (2.2.6)$$

Dividing by $2m$:

$$\langle \mathbf{x} | \frac{\hat{\mathbf{p}}^2}{2m} | \psi(t) \rangle = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t). \quad (2.2.7)$$

2. **Potential Energy Term:** Because $\langle \mathbf{x} |$ is an eigenbra of the position operator $\hat{\mathbf{x}}$, any algebraic function $V(\hat{\mathbf{x}})$ acts on $\langle \mathbf{x} |$ by simple scalar multiplication:

$$\langle \mathbf{x} | V(\hat{\mathbf{x}}) = V(\mathbf{x}) \langle \mathbf{x} |. \quad (2.2.8)$$

Therefore:

$$\langle \mathbf{x} | V(\hat{\mathbf{x}}) | \psi(t) \rangle = V(\mathbf{x}) \langle \mathbf{x} | \psi(t) \rangle = V(\mathbf{x}) \psi(\mathbf{x}, t). \quad (2.2.9)$$

Combining the left-hand and right-hand sides yields:

$$\boxed{i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t)} \quad (2.2.10)$$

The celebrated Schrödinger wave equation is revealed to be nothing more and nothing less than the position-space matrix element of the abstract time evolution law!

2.2.2 Probability Density and the Continuity Equation

From the position wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$, the probability density of finding the particle at position $\mathbf{x}$ at time t is:

$$\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2 = \psi^*(\mathbf{x}, t) \psi(\mathbf{x}, t). \quad (2.2.11)$$

Differentiating ρ with respect to t :

$$\frac{\partial \rho}{\partial t} = \left(\frac{\partial \psi^*}{\partial t} \right) \psi + \psi^* \left(\frac{\partial \psi}{\partial t} \right). \quad (2.2.12)$$

Using the Schrödinger equation $i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$ and its complex conjugate $-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi^* + V\psi^*$:

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \frac{1}{-i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi^* + V\psi^* \right) \psi + \frac{1}{i\hbar} \psi^* \left(-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right) \\ &= \frac{i\hbar}{2m} \left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right) \\ &= -\nabla \cdot \left[\frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right]. \end{aligned} \quad (2.2.13)$$

Defining the **probability current density**:

$$\mathbf{J}(\mathbf{x}, t) \equiv \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) = \frac{1}{m} \text{Re} \left(\psi^* \frac{\hbar}{i} \nabla \psi \right), \quad (2.2.14)$$

we obtain the local conservation of probability:

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0} \quad (2.2.15)$$

Integrating over all space and applying Gauss's divergence theorem, the surface integral vanishes for normalizable wavefunctions, confirming global probability conservation: $\frac{d}{dt} \int \rho d^3x = 0$.

2.3 The Three Pictures of Quantum Dynamics

All physical predictions in quantum mechanics consist of expectation values, transition amplitudes, and matrix elements. Consider the expectation value of an observable $\hat{A}$ at time t :

$$\langle \hat{A} \rangle(t) = \langle \psi(t) | \hat{A} | \psi(t) \rangle = \langle \psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \psi(0) \rangle. \quad (2.3.1)$$

Because matrix and operator multiplication is associative:

$$\langle \hat{A} \rangle(t) = \left(\langle \psi(0) | \hat{U}^\dagger(t) \right) \hat{A} \left(\hat{U}(t) | \psi(0) \rangle \right) = \langle \psi(0) | \left(\hat{U}^\dagger(t) \hat{A} \hat{U}(t) \right) | \psi(0) \rangle. \quad (2.3.2)$$

We have the complete mathematical freedom to attach the time evolution operator $\hat{U}(t)$ to the states, to the operators, or to distribute it between them! This gives rise to three distinct pictures.

2.3.1 The Schrödinger Picture

In the **Schrödinger picture** (denoted by subscript S):

- **States carry all time evolution:**

$$|\psi(t)\rangle_S = \hat{U}(t, 0) |\psi(0)\rangle_S, \quad i\hbar \frac{d}{dt} |\psi(t)\rangle_S = \hat{H}_S |\psi(t)\rangle_S. \quad (2.3.3)$$

- **Operators are stationary:** Physical observables $\hat{A}_S$ do not evolve with time unless they carry explicit parametric time dependence (such as an externally applied, time-dependent electric field):

$$\frac{d\hat{A}_S}{dt} = \frac{\partial \hat{A}_S}{\partial t}. \quad (2.3.4)$$

If the observable is not driven by an explicit external time-dependent parameter, $\frac{d\hat{A}_S}{dt} = 0$.

- **Basis states are fixed:** The eigenkets $|a_n\rangle_S$ of a stationary observable remain frozen in time:

$$\frac{d}{dt} |a_n\rangle_S = 0. \quad (2.3.5)$$

2.3.2 The Heisenberg Picture

In the **Heisenberg picture** (denoted by subscript H):

- **States are frozen in time:** The state vector remains permanently fixed at its initial configuration at $t = 0$:

$$|\psi\rangle_H \equiv |\psi(0)\rangle_S = \hat{U}^\dagger(t) |\psi(t)\rangle_S \implies \frac{d}{dt} |\psi\rangle_H = 0. \quad (2.3.6)$$

- **Operators absorb all dynamical evolution:** Observables evolve via a continuous similarity transformation generated by $\hat{U}(t)$:

$$\hat{A}_H(t) \equiv \hat{U}^\dagger(t) \hat{A}_S(t) \hat{U}(t). \quad (2.3.7)$$

Derivation of the Heisenberg Equation of Motion

Differentiating $\hat{A}_H(t)$ with respect to time using the product rule:

$$\frac{d\hat{A}_H(t)}{dt} = \left(\frac{\partial \hat{U}^\dagger(t)}{\partial t} \right) \hat{A}_S(t) \hat{U}(t) + \hat{U}^\dagger(t) \hat{A}_S(t) \left(\frac{\partial \hat{U}(t)}{\partial t} \right) + \hat{U}^\dagger(t) \left(\frac{\partial \hat{A}_S}{\partial t} \right) \hat{U}(t). \quad (2.3.8)$$

Recall that $\frac{\partial \hat{U}}{\partial t} = -\frac{i}{\hbar} \hat{H}_S \hat{U}$, which upon taking the Hermitian conjugate gives $\frac{\partial \hat{U}^\dagger}{\partial t} = \frac{i}{\hbar} \hat{U}^\dagger \hat{H}_S$. Substituting these:

$$\begin{aligned} \frac{d\hat{A}_H(t)}{dt} &= \left(\frac{i}{\hbar} \hat{U}^\dagger(t) \hat{H}_S \right) \hat{A}_S(t) \hat{U}(t) + \hat{U}^\dagger(t) \hat{A}_S(t) \left(-\frac{i}{\hbar} \hat{H}_S \hat{U}(t) \right) + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H \\ &= \frac{i}{\hbar} \hat{U}^\dagger(t) \hat{H}_S \left(\hat{U}(t) \hat{U}^\dagger(t) \right) \hat{A}_S(t) \hat{U}(t) - \frac{i}{\hbar} \hat{U}^\dagger(t) \hat{A}_S(t) \left(\hat{U}(t) \hat{U}^\dagger(t) \right) \hat{H}_S \hat{U}(t) + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H \\ &= \frac{i}{\hbar} \left(\hat{U}^\dagger \hat{H}_S \hat{U} \right) \left(\hat{U}^\dagger \hat{A}_S \hat{U} \right) - \frac{i}{\hbar} \left(\hat{U}^\dagger \hat{A}_S \hat{U} \right) \left(\hat{U}^\dagger \hat{H}_S \hat{U} \right) + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H. \end{aligned} \quad (2.3.9)$$

Notice that the Heisenberg Hamiltonian is $\hat{H}_H(t) \equiv \hat{U}^\dagger(t) \hat{H}_S \hat{U}(t)$. For a conservative system, $\hat{H}_S$ commutes with $\hat{U}(t) = \exp(-i\hat{H}_S t/\hbar)$, so:

$$\hat{H}_H = \hat{H}_S \equiv \hat{H}. \quad (2.3.10)$$

Therefore:

$$\boxed{\frac{d\hat{A}_H(t)}{dt} = \frac{1}{i\hbar} [\hat{A}_H(t), \hat{H}_H] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H = \frac{i}{\hbar} [\hat{H}_H, \hat{A}_H(t)] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H} \quad (2.3.11)$$

This is the fundamental **Heisenberg Equation of Motion**. Compare this with classical Hamiltonian mechanics:

$$\frac{dA}{dt} = \{A, H\}_{\text{PB}} + \frac{\partial A}{\partial t}. \quad (2.3.12)$$

Under the canonical quantization replacement $\{A, B\}_{\text{PB}} \rightarrow \frac{1}{i\hbar} [\hat{A}, \hat{B}]$, Hamilton's classical equation of motion maps **identically** onto the Heisenberg equation of motion!

Eigenbasis in the Heisenberg Picture

If $\hat{A}_S |a_n\rangle_S = a_n |a_n\rangle_S$ is the eigenvalue equation in the Schrödinger picture, let us examine the eigenbasis of $\hat{A}_H(t)$:

$$\hat{A}_H(t) \left(\hat{U}^\dagger(t) |a_n\rangle_S \right) = \left(\hat{U}^\dagger(t) \hat{A}_S \hat{U}(t) \right) \left(\hat{U}^\dagger(t) |a_n\rangle_S \right) = \hat{U}^\dagger(t) \hat{A}_S |a_n\rangle_S = a_n \left(\hat{U}^\dagger(t) |a_n\rangle_S \right). \quad (2.3.13)$$

Thus, the eigenkets of the Heisenberg operator $\hat{A}_H(t)$ are time-dependent:

$$|a_n, t\rangle_H \equiv \hat{U}^\dagger(t) |a_n\rangle_S. \quad (2.3.14)$$

Notice that the base kets rotate *backwards* in time:

$$i\hbar \frac{d}{dt} |a_n, t\rangle_H = -\hat{H} |a_n, t\rangle_H. \quad (2.3.15)$$

In the Schrödinger picture, the coordinate system (basis) is fixed while the physical state vector moves. In the Heisenberg picture, the physical state vector is fixed while the coordinate axes (eigenbases) rotate in the opposite direction!

2.3.3 The Interaction (Dirac) Picture

The **Interaction picture** (denoted by subscript I) is an intermediate representation indispensable for perturbation theory and quantum field theory. It is applied when the Hamiltonian splits into an unperturbed, exactly solvable part $\hat{H}_0$ and an interaction or perturbation $\hat{V}_S(t)$:

$$\hat{H} = \hat{H}_0 + \hat{V}_S(t). \quad (2.3.16)$$

We define the unitary transformation associated solely with the unperturbed Hamiltonian $\hat{H}_0$:

$$\hat{U}_0(t) \equiv \exp\left(-\frac{i}{\hbar} \hat{H}_0 t\right), \quad i\hbar \frac{d\hat{U}_0}{dt} = \hat{H}_0 \hat{U}_0(t). \quad (2.3.17)$$

States and Operators in the Interaction Picture

- **Interaction-Picture State Vector:** We rotate the Schrödinger state by $\hat{U}_0^\dagger(t)$:

$$|\psi(t)\rangle_I \equiv \hat{U}_0^\dagger(t) |\psi(t)\rangle_S = e^{i\hat{H}_0 t/\hbar} |\psi(t)\rangle_S. \quad (2.3.18)$$

Differentiating with respect to time:

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi(t)\rangle_I &= i\hbar \left(\frac{d\hat{U}_0^\dagger}{dt} \right) |\psi(t)\rangle_S + \hat{U}_0^\dagger(t) \left(i\hbar \frac{d}{dt} |\psi(t)\rangle_S \right) \\ &= i\hbar \left(\frac{i}{\hbar} \hat{U}_0^\dagger \hat{H}_0 \right) |\psi(t)\rangle_S + \hat{U}_0^\dagger(t) (\hat{H}_0 + \hat{V}_S(t)) |\psi(t)\rangle_S \\ &= -\hat{U}_0^\dagger \hat{H}_0 |\psi(t)\rangle_S + \hat{U}_0^\dagger \hat{H}_0 |\psi(t)\rangle_S + \hat{U}_0^\dagger \hat{V}_S(t) |\psi(t)\rangle_S \\ &= (\hat{U}_0^\dagger(t) \hat{V}_S(t) \hat{U}_0(t)) (\hat{U}_0^\dagger(t) |\psi(t)\rangle_S). \end{aligned} \quad (2.3.19)$$

Defining the **interaction potential in the interaction picture**:

$$\hat{V}_I(t) \equiv \hat{U}_0^\dagger(t) \hat{V}_S(t) \hat{U}_0(t) = e^{i\hat{H}_0 t/\hbar} \hat{V}_S(t) e^{-i\hat{H}_0 t/\hbar}, \quad (2.3.20)$$

the state equation of motion becomes:

$$\boxed{i\hbar \frac{d}{dt} |\psi(t)\rangle_I = \hat{V}_I(t) |\psi(t)\rangle_I} \quad (2.3.21)$$

The state vector evolves purely under the interaction $\hat{V}_I(t)$! If the interaction is switched off ($\hat{V} = 0$), the interaction-picture state vector is completely stationary ($\frac{d}{dt} |\psi\rangle_I = 0$).

- **Interaction-Picture Operators:** Observables evolve via the unperturbed Hamiltonian $\hat{H}_0$:

$$\hat{A}_I(t) \equiv \hat{U}_0^\dagger(t) \hat{A}_S(t) \hat{U}_0(t) = e^{i\hat{H}_0 t/\hbar} \hat{A}_S(t) e^{-i\hat{H}_0 t/\hbar}. \quad (2.3.22)$$

Differentiating with respect to time:

$$\boxed{\frac{d\hat{A}_I(t)}{dt} = \frac{1}{i\hbar} [\hat{A}_I(t), \hat{H}_0] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_I} \quad (2.3.23)$$

The Dyson Series

The time-evolution operator in the interaction picture, defined by $|\psi(t)\rangle_I = \hat{U}_I(t, t_0) |\psi(t_0)\rangle_I$, satisfies:

$$i\hbar \frac{\partial \hat{U}_I(t, t_0)}{\partial t} = \hat{V}_I(t) \hat{U}_I(t, t_0), \quad \hat{U}_I(t_0, t_0) = \hat{I}. \quad (2.3.24)$$

Converting this differential equation into an integral equation:

$$\hat{U}_I(t, t_0) = \hat{I} - \frac{i}{\hbar} \int_{t_0}^t dt' \hat{V}_I(t') \hat{U}_I(t', t_0). \quad (2.3.25)$$

Iterating this equation recursively yields the celebrated **Dyson Series**:

$$\begin{aligned} \hat{U}_I(t, t_0) &= \hat{I} - \frac{i}{\hbar} \int_{t_0}^t dt_1 \hat{V}_I(t_1) + \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \hat{V}_I(t_1) \hat{V}_I(t_2) + \dots \\ &= \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_{t_0}^t dt' \hat{V}_I(t') \right), \end{aligned} \quad (2.3.26)$$

where $\mathcal{T}$ is the time-ordering operator. This series provides the formal machinery for time-dependent perturbation theory, Fermi's Golden Rule, and the S -matrix in particle physics.

2.3.4 Comparison of the Three Pictures

Table 2.1 synthesizes the mathematical relationships governing the three dynamical pictures.

Feature	Schrödinger Picture	Heisenberg Picture	Interaction Picture
State Vector	$ \psi(t)\rangle_S = \hat{U} \psi(0)\rangle$	$ \psi\rangle_H = \psi(0)\rangle_S$	$ \psi(t)\rangle_I = \hat{U}_0^\dagger \psi(t)\rangle_S$
Operator	$\hat{A}_S$ (time-independent)	$\hat{A}_H(t) = \hat{U}^\dagger \hat{A}_S \hat{U}$	$\hat{A}_I(t) = \hat{U}_0^\dagger \hat{A}_S \hat{U}_0$
State Evolution	$i\hbar \frac{d}{dt} \psi\rangle_S = \hat{H} \psi\rangle_S$	$\frac{d}{dt} \psi\rangle_H = 0$	$i\hbar \frac{d}{dt} \psi\rangle_I = \hat{V}_I(t) \psi\rangle_I$
Operator Evolution	$\frac{d\hat{A}_S}{dt} = \frac{\partial \hat{A}_S}{\partial t}$	$\frac{d\hat{A}_H}{dt} = \frac{1}{i\hbar} [\hat{A}_H, \hat{H}] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H$	$\frac{d\hat{A}_I}{dt} = \frac{1}{i\hbar} [\hat{A}_I, \hat{H}_0] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_I$
Primary Utility	Intuitive wave mechanics	Classical correspondence	Perturbation theory / QFT

Table 2.1: Systematic comparison of the three fundamental pictures of quantum dynamics.

2.4 Equivalence of Physical Predictions Across Pictures

Because physical predictions are determined by transition amplitudes and expectation values, they are strictly invariant under any unitary transformation connecting the pictures.

2.4.1 Matrix Elements and Expectation Values

Consider the transition amplitude between two physical states $|\phi\rangle$ and $|\psi\rangle$ mediated by an observable $\hat{A}$ at time t . In the Schrödinger picture:

$$\mathcal{M}_S = {}_S \langle \phi(t) | \hat{A}_S | \psi(t) \rangle_S. \quad (2.4.1)$$

Substituting $|\psi(t)\rangle_S = \hat{U}(t) |\psi(0)\rangle$ and $\langle \phi(t)|_S = \langle \phi(0)| \hat{U}^\dagger(t)$:

$$\mathcal{M}_S = \langle \phi(0) | \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t) | \psi(0) \rangle$$

$$= {}_H \langle \phi | \hat{A}_H(t) | \psi \rangle_H \equiv \mathcal{M}_H. \quad (2.4.2)$$

Similarly, in the interaction picture, with $|\psi(t)\rangle_I = \hat{U}_0^\dagger(t) |\psi(t)\rangle_S$ and $\hat{A}_I(t) = \hat{U}_0^\dagger(t) \hat{A}_S \hat{U}_0(t)$:

$$\begin{aligned} \mathcal{M}_I &= {}_I \langle \phi(t) | \hat{A}_I(t) | \psi(t) \rangle_I \\ &= \left({}_S \langle \phi(t) | \hat{U}_0(t) \right) \left(\hat{U}_0^\dagger(t) \hat{A}_S \hat{U}_0(t) \right) \left(\hat{U}_0^\dagger(t) | \psi(t) \rangle_S \right) \\ &= {}_S \langle \phi(t) | \hat{A}_S | \psi(t) \rangle_S \equiv \mathcal{M}_S. \end{aligned} \quad (2.4.3)$$

Thus, for any observable $\hat{A}$ and states $|\phi\rangle, |\psi\rangle$:

$$\boxed{{}_S \langle \phi(t) | \hat{A}_S | \psi(t) \rangle_S = {}_H \langle \phi | \hat{A}_H(t) | \psi \rangle_H = {}_I \langle \phi(t) | \hat{A}_I(t) | \psi(t) \rangle_I} \quad (2.4.4)$$

Consequently, the expectation value is completely independent of the choice of picture:

$$\langle \hat{A} \rangle(t) = {}_S \langle \psi(t) | \hat{A}_S | \psi(t) \rangle_S = {}_H \langle \psi | \hat{A}_H(t) | \psi \rangle_H = {}_I \langle \psi(t) | \hat{A}_I(t) | \psi(t) \rangle_I. \quad (2.4.5)$$

The choice of picture is merely a choice of coordinate frame in Hilbert space. No experiment can distinguish whether states or operators carry the time evolution.

2.5 Ehrenfest's Theorem and the Classical Limit

Ehrenfest's theorem establishes the formal correspondence between quantum mechanics and classical Newtonian mechanics by showing that the quantum expectation values of position and momentum satisfy equations of motion analogous to Newton's second law.

2.5.1 Derivation via the Heisenberg Equation of Motion

Ehrenfest's theorem follows immediately and most transparently from the Heisenberg picture. For any physical observable $\hat{A}_H(t)$ without explicit time dependence ($\frac{\partial \hat{A}_S}{\partial t} = 0$), the time derivative of its expectation value with respect to the time-independent Heisenberg state $|\psi\rangle_H$ is:

$$\frac{d}{dt} \langle \hat{A} \rangle = \frac{d}{dt} {}_H \langle \psi | \hat{A}_H(t) | \psi \rangle_H = {}_H \langle \psi | \frac{d\hat{A}_H(t)}{dt} | \psi \rangle_H. \quad (2.5.1)$$

Substituting the Heisenberg equation of motion $\frac{d\hat{A}_H}{dt} = \frac{1}{i\hbar} [\hat{A}_H, \hat{H}]$:

$$\boxed{\frac{d}{dt} \langle \hat{A} \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle} \quad (2.5.2)$$

2.5.2 Application to a Particle in a Potential

Consider a particle of mass m moving in three dimensions under a scalar potential $V(\hat{\mathbf{x}})$, described by the standard Hamiltonian:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) = \sum_{j=1}^3 \frac{\hat{p}_j^2}{2m} + V(\hat{x}_1, \hat{x}_2, \hat{x}_3). \quad (2.5.3)$$

Position Expectation Value

Let us evaluate the commutator $[\hat{x}_k, \hat{H}]$:

$$[\hat{x}_k, \hat{H}] = \left[\hat{x}_k, \sum_j \frac{\hat{p}_j^2}{2m} + V(\hat{\mathbf{x}}) \right] = \sum_j \frac{1}{2m} [\hat{x}_k, \hat{p}_j^2] + [\hat{x}_k, V(\hat{\mathbf{x}})]. \quad (2.5.4)$$

Since coordinates commute among themselves, $[\hat{x}_k, V(\hat{\mathbf{x}})] = 0$. Using the identity $[A, BC] = [A, B]C + B[A, C]$ and the canonical commutator $[\hat{x}_k, \hat{p}_j] = i\hbar\delta_{kj}$:

$$[\hat{x}_k, \hat{p}_j^2] = [\hat{x}_k, \hat{p}_j]\hat{p}_j + \hat{p}_j[\hat{x}_k, \hat{p}_j] = i\hbar\delta_{kj}\hat{p}_j + \hat{p}_j(i\hbar\delta_{kj}) = 2i\hbar\delta_{kj}\hat{p}_j. \quad (2.5.5)$$

Summing over j :

$$[\hat{x}_k, \hat{H}] = \frac{1}{2m} (2i\hbar\hat{p}_k) = \frac{i\hbar}{m} \hat{p}_k. \quad (2.5.6)$$

Taking the expectation value yields:

$$\boxed{\frac{d\langle\hat{\mathbf{x}}\rangle}{dt} = \frac{\langle\hat{\mathbf{p}}\rangle}{m}} \quad (2.5.7)$$

This is the quantum counterpart of the classical kinematic relation $\frac{d\mathbf{x}}{dt} = \frac{\mathbf{p}}{m}$.

Momentum Expectation Value

Next, evaluate the commutator $[\hat{p}_k, \hat{H}]$:

$$[\hat{p}_k, \hat{H}] = \left[\hat{p}_k, \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right] = \left[\hat{p}_k, \frac{\hat{\mathbf{p}}^2}{2m} \right] + [\hat{p}_k, V(\hat{\mathbf{x}})]. \quad (2.5.8)$$

Because momentum components commute with each other, $[\hat{p}_k, \hat{\mathbf{p}}^2] = 0$. For the potential term, we use the commutator of momentum with an arbitrary function of position:

$$[\hat{p}_k, V(\hat{\mathbf{x}})] = -i\hbar \frac{\partial V(\hat{\mathbf{x}})}{\partial x_k}. \quad (2.5.9)$$

Therefore:

$$\frac{d\langle\hat{p}_k\rangle}{dt} = \frac{1}{i\hbar} \langle[\hat{p}_k, \hat{H}]\rangle = \frac{1}{i\hbar} \left\langle -i\hbar \frac{\partial V(\hat{\mathbf{x}})}{\partial x_k} \right\rangle = - \left\langle \frac{\partial V(\hat{\mathbf{x}})}{\partial x_k} \right\rangle. \quad (2.5.10)$$

In vector notation:

$$\boxed{\frac{d\langle\hat{\mathbf{p}}\rangle}{dt} = -\langle\nabla V(\hat{\mathbf{x}})\rangle} \quad (2.5.11)$$

Differentiating $\langle\hat{\mathbf{x}}\rangle$ a second time with respect to t gives Ehrenfest's second-order equation:

$$\boxed{m \frac{d^2\langle\hat{\mathbf{x}}\rangle}{dt^2} = -\langle\nabla V(\hat{\mathbf{x}})\rangle} \quad (2.5.12)$$

2.5.3 The Quantum-Classical Discrepancy

It is vital to recognize that Ehrenfest's equation is **not** identical to Newton's second law:

$$m \frac{d^2\mathbf{x}_{cl}}{dt^2} = -\nabla V(\mathbf{x}_{cl}). \quad (2.5.13)$$

In Newton's law, the force is evaluated at the particle's position: $\mathbf{F} = -\nabla V(\mathbf{x}_{cl})$. In Ehrenfest's equation, the force is the expectation value of the gradient operator averaged over the entire spatial extent of the wave packet: $\langle\mathbf{F}\rangle = -\langle\nabla V(\hat{\mathbf{x}})\rangle$. In general:

$$\langle\nabla V(\hat{\mathbf{x}})\rangle \neq \nabla V(\langle\hat{\mathbf{x}}\rangle). \quad (2.5.14)$$

To see this quantitatively, Taylor expand $\nabla V(\hat{\mathbf{x}})$ about the mean position $\langle \hat{\mathbf{x}} \rangle$:

$$\begin{aligned} \nabla V(\hat{\mathbf{x}}) &= \nabla V(\langle \hat{\mathbf{x}} \rangle) + \sum_j \left. \frac{\partial \nabla V}{\partial x_j} \right|_{\langle \hat{\mathbf{x}} \rangle} (\hat{x}_j - \langle \hat{x}_j \rangle) \\ &\quad + \frac{1}{2} \sum_{j,k} \left. \frac{\partial^2 \nabla V}{\partial x_j \partial x_k} \right|_{\langle \hat{\mathbf{x}} \rangle} (\hat{x}_j - \langle \hat{x}_j \rangle)(\hat{x}_k - \langle \hat{x}_k \rangle) + \mathcal{O}((\Delta x)^3). \end{aligned} \quad (2.5.15)$$

Taking the expectation value of this expansion:

$$\langle \nabla V(\hat{\mathbf{x}}) \rangle = \nabla V(\langle \hat{\mathbf{x}} \rangle) + \frac{1}{2} \sum_{j,k} \left. \frac{\partial^2 \nabla V}{\partial x_j \partial x_k} \right|_{\langle \hat{\mathbf{x}} \rangle} \text{Cov}(\hat{x}_j, \hat{x}_k) + \dots \quad (2.5.16)$$

where $\text{Cov}(\hat{x}_j, \hat{x}_k) \equiv \langle (\hat{x}_j - \langle \hat{x}_j \rangle)(\hat{x}_k - \langle \hat{x}_k \rangle) \rangle$ is the spatial covariance matrix (with diagonal elements $(\Delta x_j)^2$).

When does quantum mechanics reduce to classical mechanics?

1. **Macroscopic Wave Packets ($\Delta x \rightarrow 0$):** When the spatial width of the wavepacket Δx is negligibly small compared to the characteristic spatial length scale L over which the potential varies:

$$\Delta x \ll L \sim \left| \frac{\nabla V}{\nabla^2 V} \right|. \quad (2.5.17)$$

In this regime, the second-order correction is negligible, and the center of the wave packet follows the classical trajectory.

2. **Harmonic and Linear Potentials (Exact Classicality):** If the potential is at most quadratic:

$$V(\mathbf{x}) = V_0 + \mathbf{a} \cdot \mathbf{x} + \frac{1}{2} \mathbf{x}^T \mathbf{K} \mathbf{x}, \quad (2.5.18)$$

(which encompasses the free particle, uniform gravitational field, and simple harmonic oscillator), all third and higher derivatives of V vanish identically:

$$\frac{\partial^2 \nabla V}{\partial x_j \partial x_k} = \frac{\partial^2 V}{\partial x_j \partial x_k} x_l = 0. \quad (2.5.19)$$

In this remarkable situation, the correction term vanishes **identically**:

$$\boxed{\langle \nabla V(\hat{\mathbf{x}}) \rangle = \nabla V(\langle \hat{\mathbf{x}} \rangle)} \quad (2.5.20)$$

For harmonic systems, the center of mass of the quantum wavepacket moves **exactly** along the classical trajectory for **any** wavefunction, regardless of how broad or delocalized the quantum state is!

2.5.4 Ehrenfest's Theorem for Orbital Angular Momentum and Torque

The time evolution of the expectation value of angular momentum follows directly from the Heisenberg equation of motion. In the position representation, the orbital angular momentum operator is $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$, or in component notation $\hat{L}_i = \epsilon_{ijk} \hat{r}_j \hat{p}_k$.

For a single particle governed by the Hamiltonian $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{r}})$, the time derivative of $\langle \hat{\mathbf{L}} \rangle$ is:

$$\frac{d\langle \hat{L}_i \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{L}_i, \hat{H}] \rangle = \frac{1}{i\hbar} \left\langle \left[\hat{L}_i, \frac{\hat{\mathbf{p}}^2}{2m} \right] \right\rangle + \frac{1}{i\hbar} \langle [\hat{L}_i, V(\hat{\mathbf{r}})] \rangle. \quad (2.5.21)$$

First, consider the kinetic energy term. Because $\hat{\mathbf{p}}^2 = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2$ is a rotationally invariant scalar operator:

$$[\hat{L}_i, \hat{p}_j] = i\hbar \epsilon_{ijk} \hat{p}_k \implies [\hat{L}_i, \hat{\mathbf{p}}^2] = \sum_j \left(\hat{p}_j [\hat{L}_i, \hat{p}_j] + [\hat{L}_i, \hat{p}_j] \hat{p}_j \right) = i\hbar \sum_{j,k} \epsilon_{ijk} (\hat{p}_j \hat{p}_k + \hat{p}_k \hat{p}_j) = 0, \quad (2.5.22)$$

owing to the contraction of the antisymmetric Levi-Civita symbol ϵ_{ijk} with the symmetric tensor $\hat{p}_j \hat{p}_k$. Thus, kinetic energy generates no torque on the quantum particle.

Second, evaluate the commutator with the potential energy $V(\hat{\mathbf{r}})$:

$$\begin{aligned} [\hat{L}_i, V(\hat{\mathbf{r}})] &= \epsilon_{ijk} [\hat{r}_j \hat{p}_k, V(\hat{\mathbf{r}})] = \epsilon_{ijk} \hat{r}_j [\hat{p}_k, V(\hat{\mathbf{r}})] \\ &= \epsilon_{ijk} \hat{r}_j \left(-i\hbar \frac{\partial V}{\partial r_k} \right) = -i\hbar \epsilon_{ijk} \hat{r}_j \frac{\partial V}{\partial r_k}. \end{aligned} \quad (2.5.23)$$

Recognizing $-\frac{\partial V}{\partial r_k} = \hat{F}_k$ as the force operator, we find:

$$\frac{1}{i\hbar} [\hat{L}_i, V(\hat{\mathbf{r}})] = \epsilon_{ijk} \hat{r}_j \left(-\frac{\partial V}{\partial r_k} \right) = (\hat{\mathbf{r}} \times \hat{\mathbf{F}})_i = \hat{\tau}_i, \quad (2.5.24)$$

where $\hat{\boldsymbol{\tau}} \equiv \hat{\mathbf{r}} \times \hat{\mathbf{F}} = -\hat{\mathbf{r}} \times \nabla V(\hat{\mathbf{r}})$ is the quantum **torque operator**.

Therefore, the expectation value of orbital angular momentum obeys the exact quantum analog of the classical torque theorem:

$$\boxed{\frac{d\langle \hat{\mathbf{L}} \rangle}{dt} = \langle \hat{\boldsymbol{\tau}} \rangle = -\langle \hat{\mathbf{r}} \times \nabla V(\hat{\mathbf{r}}) \rangle} \quad (2.5.25)$$

Rotational Invariance and Central Forces. For any central potential $V(\mathbf{r}) = V(r)$, the gradient is purely radial: $\nabla V(r) = V'(r) \frac{\mathbf{r}}{r}$. Consequently:

$$\mathbf{r} \times \nabla V(r) = V'(r) \frac{\mathbf{r} \times \mathbf{r}}{r} = 0. \quad (2.5.26)$$

Hence, for any spherically symmetric potential:

$$[\hat{\mathbf{L}}, \hat{H}] = 0 \implies \frac{d\langle \hat{\mathbf{L}} \rangle}{dt} = 0. \quad (2.5.27)$$

Orbital angular momentum is a conserved constant of motion for any central potential, matching classical Keplerian mechanics.

2.6 Solved Example: Spin Precession in a Magnetic Field (Heisenberg Picture)

To appreciate the mathematical power and elegance of the Heisenberg picture, we solve the dynamics of a spin-1/2 particle placed in a uniform magnetic field.

2.6.1 Physical Setup

Consider an electron (or any spin-1/2 particle) with intrinsic spin angular momentum $\hat{\mathbf{S}} = (\hat{S}_x, \hat{S}_y, \hat{S}_z)$ and magnetic dipole moment:

$$\hat{\boldsymbol{\mu}} = \gamma \hat{\mathbf{S}} = g \frac{q}{2m} \hat{\mathbf{S}}, \quad (2.6.1)$$

where γ is the gyromagnetic ratio ($g \approx 2$ for an electron). The particle is placed in a static, uniform magnetic field aligned along the z -axis:

$$\mathbf{B} = B_0 \hat{\mathbf{z}}. \quad (2.6.2)$$

The interaction Hamiltonian is:

$$\hat{H} = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = -\gamma B_0 \hat{S}_z \equiv \omega_0 \hat{S}_z, \quad (2.6.3)$$

where $\omega_0 \equiv -\gamma B_0$ is the classical **Larmor precession frequency**.

2.6.2 Heisenberg Equations of Motion

In the Heisenberg picture, the spin operators $\hat{S}_{i,H}(t)$ satisfy the equation of motion:

$$\frac{d\hat{S}_{i,H}(t)}{dt} = \frac{1}{i\hbar} [\hat{S}_{i,H}(t), \hat{H}] = \frac{\omega_0}{i\hbar} [\hat{S}_{i,H}(t), \hat{S}_{z,H}]. \quad (2.6.4)$$

Because the unitary transformation preserves all algebraic commutation relations at equal times:

$$[\hat{S}_{j,H}(t), \hat{S}_{k,H}(t)] = \hat{U}^\dagger(t) [\hat{S}_j, \hat{S}_k] \hat{U}(t) = i\hbar \epsilon_{jkl} \hat{U}^\dagger(t) \hat{S}_l \hat{U}(t) = i\hbar \epsilon_{jkl} \hat{S}_{l,H}(t). \quad (2.6.5)$$

Evaluating the commutators for each Cartesian component:

$$\frac{d\hat{S}_{x,H}}{dt} = \frac{\omega_0}{i\hbar} [\hat{S}_{x,H}, \hat{S}_{z,H}] = \frac{\omega_0}{i\hbar} (-i\hbar \hat{S}_{y,H}) = -\omega_0 \hat{S}_{y,H}(t), \quad (2.6.6)$$

$$\frac{d\hat{S}_{y,H}}{dt} = \frac{\omega_0}{i\hbar} [\hat{S}_{y,H}, \hat{S}_{z,H}] = \frac{\omega_0}{i\hbar} (i\hbar \hat{S}_{x,H}) = \omega_0 \hat{S}_{x,H}(t), \quad (2.6.7)$$

$$\frac{d\hat{S}_{z,H}}{dt} = \frac{\omega_0}{i\hbar} [\hat{S}_{z,H}, \hat{S}_{z,H}] = 0. \quad (2.6.8)$$

From Eq. (2.6.8), the z -component is a constant of motion:

$$\hat{S}_{z,H}(t) = \hat{S}_z(0). \quad (2.6.9)$$

2.6.3 Exact Operator Solution

To decouple the transverse components, we define the spin ladder operators in the Heisenberg picture:

$$\hat{S}_{\pm,H}(t) \equiv \hat{S}_{x,H}(t) \pm i\hat{S}_{y,H}(t). \quad (2.6.10)$$

Differentiating $\hat{S}_{+,H}(t)$ with respect to time:

$$\frac{d\hat{S}_{+,H}}{dt} = \frac{d\hat{S}_{x,H}}{dt} + i\frac{d\hat{S}_{y,H}}{dt} = -\omega_0 \hat{S}_{y,H} + i\omega_0 \hat{S}_{x,H} = i\omega_0 (\hat{S}_{x,H} + i\hat{S}_{y,H}) = i\omega_0 \hat{S}_{+,H}(t). \quad (2.6.11)$$

This is a simple linear first-order differential equation whose solution is:

$$\hat{S}_{+,H}(t) = \hat{S}_{+,H}(0) e^{i\omega_0 t}. \quad (2.6.12)$$

Similarly, for $\hat{S}_{-,H}(t) = (\hat{S}_{+,H}(t))^\dagger$:

$$\hat{S}_{-,H}(t) = \hat{S}_{-,H}(0) e^{-i\omega_0 t}. \quad (2.6.13)$$

Reconstructing the Cartesian components from $\hat{S}_{x,H} = \frac{1}{2}(\hat{S}_{+,H} + \hat{S}_{-,H})$ and $\hat{S}_{y,H} = \frac{1}{2i}(\hat{S}_{+,H} - \hat{S}_{-,H})$:

$$\hat{S}_{x,H}(t) = \frac{1}{2} \left[(\hat{S}_x(0) + i\hat{S}_y(0)) e^{i\omega_0 t} + (\hat{S}_x(0) - i\hat{S}_y(0)) e^{-i\omega_0 t} \right]$$

$$\begin{aligned}
&= \hat{S}_x(0) \left(\frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2} \right) - \hat{S}_y(0) \left(\frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i} \right) \\
&= \hat{S}_x(0) \cos(\omega_0 t) - \hat{S}_y(0) \sin(\omega_0 t).
\end{aligned} \tag{2.6.14}$$

Similarly, for $\hat{S}_{y,H}(t)$:

$$\begin{aligned}
\hat{S}_{y,H}(t) &= \frac{1}{2i} \left[\left(\hat{S}_x(0) + i\hat{S}_y(0) \right) e^{i\omega_0 t} - \left(\hat{S}_x(0) - i\hat{S}_y(0) \right) e^{-i\omega_0 t} \right] \\
&= \hat{S}_x(0) \sin(\omega_0 t) + \hat{S}_y(0) \cos(\omega_0 t).
\end{aligned} \tag{2.6.15}$$

In matrix-vector form:

$$\begin{pmatrix} \hat{S}_{x,H}(t) \\ \hat{S}_{y,H}(t) \\ \hat{S}_{z,H}(t) \end{pmatrix} = \begin{pmatrix} \cos(\omega_0 t) & -\sin(\omega_0 t) & 0 \\ \sin(\omega_0 t) & \cos(\omega_0 t) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{S}_x(0) \\ \hat{S}_y(0) \\ \hat{S}_z(0) \end{pmatrix} \tag{2.6.16}$$

The spin operators rotate in the transverse plane at the exact Larmor frequency ω_0 !

2.6.4 Physical Expectation Values

Now suppose that at $t = 0$, the particle's spin is prepared in the eigenstate of $\hat{S}_x$ with eigenvalue $+\hbar/2$:

$$|\psi(0)\rangle = |+\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|+\rangle_z + |-\rangle_z). \tag{2.6.17}$$

The initial expectation values at $t = 0$ are:

$$\langle \hat{S}_x(0) \rangle = \frac{\hbar}{2}, \quad \langle \hat{S}_y(0) \rangle = 0, \quad \langle \hat{S}_z(0) \rangle = 0. \tag{2.6.18}$$

Evaluating the expectation values at any later time t :

$$\langle \hat{S}_x \rangle(t) = \langle \hat{S}_x(0) \rangle \cos(\omega_0 t) - \langle \hat{S}_y(0) \rangle \sin(\omega_0 t) = \frac{\hbar}{2} \cos(\omega_0 t), \tag{2.6.19}$$

$$\langle \hat{S}_y \rangle(t) = \langle \hat{S}_x(0) \rangle \sin(\omega_0 t) + \langle \hat{S}_y(0) \rangle \cos(\omega_0 t) = \frac{\hbar}{2} \sin(\omega_0 t), \tag{2.6.20}$$

$$\langle \hat{S}_z \rangle(t) = \langle \hat{S}_z(0) \rangle = 0. \tag{2.6.21}$$

The expectation value of the spin vector precesses counterclockwise in the xy -plane on the Bloch sphere at the Larmor frequency ω_0 :

$$\langle \hat{\mathbf{S}} \rangle(t) = \frac{\hbar}{2} [\cos(\omega_0 t) \hat{\mathbf{x}} + \sin(\omega_0 t) \hat{\mathbf{y}}]. \tag{2.6.22}$$

This matches the classical precession of a magnetic dipole governed by the torque equation:

$$\frac{d\mathbf{M}}{dt} = \boldsymbol{\mu} \times \mathbf{B}. \tag{2.6.23}$$

2.6.5 Schrödinger Picture Equivalence Check

To confirm the equivalence of pictures, let us solve the identical problem in the Schrödinger picture. The state evolves as:

$$|\psi(t)\rangle_S = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle = e^{-i\omega_0 \hat{S}_z t/\hbar} \frac{1}{\sqrt{2}} (|+\rangle_z + |-\rangle_z)$$

$$= \frac{1}{\sqrt{2}} \left(e^{-i\omega_0 t/2} |+\rangle + e^{i\omega_0 t/2} |-\rangle \right). \quad (2.6.24)$$

Evaluating the expectation value of the Schrödinger operator $\hat{S}_x$:

$$\begin{aligned} {}_S \langle \psi(t) | \hat{S}_x | \psi(t) \rangle_S &= \frac{1}{2} \begin{pmatrix} e^{i\omega_0 t/2} & e^{-i\omega_0 t/2} \end{pmatrix} \left(\frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \begin{pmatrix} e^{-i\omega_0 t/2} \\ e^{i\omega_0 t/2} \end{pmatrix} \\ &= \frac{\hbar}{4} \begin{pmatrix} e^{i\omega_0 t/2} & e^{-i\omega_0 t/2} \end{pmatrix} \begin{pmatrix} e^{i\omega_0 t/2} \\ e^{-i\omega_0 t/2} \end{pmatrix} \\ &= \frac{\hbar}{4} (e^{i\omega_0 t} + e^{-i\omega_0 t}) = \frac{\hbar}{2} \cos(\omega_0 t). \end{aligned} \quad (2.6.25)$$

The Schrödinger picture calculation yields precisely the same physical outcome, confirming the complete mathematical and physical equivalence of the pictures.

2.7 Superselection Rules and Incoherent Subspaces

The standard superposition postulate asserts that if $|\phi\rangle$ and $|\psi\rangle$ are physical state vectors in a Hilbert space $\mathcal{H}$, then any arbitrary linear combination:

$$|\Psi\rangle = c_1 |\phi\rangle + c_2 |\psi\rangle, \quad |c_1|^2 + |c_2|^2 = 1, \quad (2.7.1)$$

represents an equally valid, physically realizable pure state.

However, as Wick, Wightman, and Wigner established in 1952, the unrestricted superposition principle fails in physical reality due to the existence of **superselection rules**.

2.7.1 Definition of a Superselection Sector

A superselection rule operates whenever the full Hilbert space $\mathcal{H}$ decomposes into an orthogonal direct sum of disjoint subspaces, termed **superselection sectors**:

$$\mathcal{H} = \bigoplus_k \mathcal{H}_k = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_3 \oplus \dots \quad (2.7.2)$$

such that for **every physically realizable observable operator** $\hat{A}$, all transition matrix elements between distinct sectors vanish identically:

$$\boxed{\langle \phi | \hat{A} | \psi \rangle = 0 \quad \text{for all } |\phi\rangle \in \mathcal{H}_j, |\psi\rangle \in \mathcal{H}_k \quad (j \neq k)} \quad (2.7.3)$$

In other words, every physical observable $\hat{A}$ is block-diagonal with respect to the sector decomposition:

$$\hat{A} = \begin{pmatrix} \hat{A}_1 & 0 & 0 \\ 0 & \hat{A}_2 & 0 \\ 0 & 0 & \hat{A}_3 \end{pmatrix}. \quad (2.7.4)$$

2.7.2 Impossibility of Measuring Relative Phase

Suppose an experimenter attempts to prepare a coherent superposition of two states drawn from different superselection sectors:

$$|\Psi\rangle = c_1 |\phi\rangle + c_2 |\psi\rangle, \quad |\phi\rangle \in \mathcal{H}_1, |\psi\rangle \in \mathcal{H}_2. \quad (2.7.5)$$

Let us evaluate the expectation value of an arbitrary physical observable $\hat{A}$ in this state:

$$\langle \hat{A} \rangle_\Psi = \langle \Psi | \hat{A} | \Psi \rangle$$

$$\begin{aligned}
&= |c_1|^2 \langle \phi | \hat{A} | \phi \rangle + |c_2|^2 \langle \psi | \hat{A} | \psi \rangle + c_1^* c_2 \langle \phi | \hat{A} | \psi \rangle + c_2^* c_1 \langle \psi | \hat{A} | \phi \rangle \\
&= |c_1|^2 \langle \phi | \hat{A} | \phi \rangle + |c_2|^2 \langle \psi | \hat{A} | \psi \rangle + 2\text{Re} \left(c_1^* c_2 \langle \phi | \hat{A} | \psi \rangle \right).
\end{aligned} \tag{2.7.6}$$

By Eq. (2.7.3), the off-diagonal matrix element vanishes: $\langle \phi | \hat{A} | \psi \rangle \equiv 0$. Therefore, the interference cross-term is strictly zero:

$$\langle \hat{A} \rangle_{\Psi} = |c_1|^2 \langle \phi | \hat{A} | \phi \rangle + |c_2|^2 \langle \psi | \hat{A} | \psi \rangle. \tag{2.7.7}$$

Notice the profound physical consequence:

- Writing $c_1 = |c_1|$ and $c_2 = |c_2|e^{i\theta}$, the expectation value $\langle \hat{A} \rangle$ is **completely independent of the relative phase θ** .
- Because no physical measurement can ever detect the relative phase θ , the pure state $|\Psi\rangle$ is physically indistinguishable from the incoherent statistical mixture described by the density operator:

$$\hat{\rho} = |c_1|^2 |\phi\rangle \langle \phi| + |c_2|^2 |\psi\rangle \langle \psi|. \tag{2.7.8}$$

A coherent superposition between states belonging to different superselection sectors cannot be created, nor does it possess physical reality. The Hilbert space is partitioned into coherent islands within which superpositions can be formed, but across which only statistical mixtures are defined.

2.7.3 Physical Examples of Superselection Rules

1. **Univalence (Spin Superselection):** Under a complete spatial rotation by 2π , a state of total angular momentum j transforms as:

$$\hat{R}(2\pi) |\psi_j\rangle = e^{-i2\pi j} |\psi_j\rangle = (-1)^{2j} |\psi_j\rangle. \tag{2.7.9}$$

For integer spin (bosons, $2j \in \text{even}$), the state is invariant (+1). For half-integer spin (fermions, $2j \in \text{odd}$), the state changes sign (-1). Because a 2π rotation is a physical symmetry that leaves all observables invariant ($[\hat{A}, \hat{R}(2\pi)] = 0$), matrix elements between integer and half-integer states must vanish:

$$\langle \text{boson} | \hat{A} | \text{fermion} \rangle = 0. \tag{2.7.10}$$

One cannot form a coherent superposition of a boson and a fermion!

2. **Electric Charge Superselection:** Because the electromagnetic gauge group $U(1)$ is exact, the total charge operator $\hat{Q}$ commutes with all physical observables: $[\hat{A}, \hat{Q}] = 0$. Consequently, states with different net electrical charge belong to distinct superselection sectors:

$$\langle q_1 | \hat{A} | q_2 \rangle = 0 \quad (q_1 \neq q_2). \tag{2.7.11}$$

A state such as $\frac{1}{\sqrt{2}}(|\text{proton}\rangle + |\text{neutron}\rangle)$ carries an unmeasurable relative phase and cannot exist as a coherent pure state.

3. **Bargmann's Mass Superselection in Galilean Relativity:** In non-relativistic Galilean quantum mechanics, Galilean boosts do not form an ordinary representation of the Galilei group, but a projective ray representation with a central extension proportional to the particle's rest mass M . Bargmann's theorem proves that superposing states of differing total mass produces phase factors that violate Galilean invariance, forbidding coherent superpositions of different masses in non-relativistic QM.

2.8 From Classical Lagrangian to Quantum Hamiltonian: The Path Integral Connection

Throughout this chapter, quantum dynamics has been formulated canonically via the Hamiltonian operator $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$ generating unitary time translations $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$.

However, in 1948, Richard Feynman revealed that quantum time evolution can be equivalently formulated directly in terms of the classical **Lagrangian**:

$$L(x, \dot{x}) = T - V = \frac{1}{2}m\dot{x}^2 - V(x), \quad (2.8.1)$$

illuminating the deep origin of canonical quantization.

2.8.1 Discretization of the Quantum Propagator

The fundamental transition amplitude for a particle to travel from (x_i, t_i) to (x_f, t_f) is the matrix element of the time evolution operator:

$$K(x_f, t_f; x_i, t_i) = \langle x_f | e^{-i\hat{H}(t_f - t_i)/\hbar} | x_i \rangle. \quad (2.8.2)$$

Divide the total time interval $T = t_f - t_i$ into N equal infinitesimal slices of duration $\epsilon \equiv T/N$, with $t_j = t_i + j\epsilon$. Using the completeness relation $\int dx_j |x_j\rangle \langle x_j| = \hat{I}$ at each intermediate time step:

$$K(x_f, t_f; x_i, t_i) = \int dx_1 \dots dx_{N-1} \prod_{j=0}^{N-1} \langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle, \quad (2.8.3)$$

where $x_0 \equiv x_i$ and $x_N \equiv x_f$.

2.8.2 Short-Time Matrix Element and Emergence of the Lagrangian

For an infinitesimal slice $\epsilon \ll 1$, we split the kinetic and potential operators using the Trotter product formula:

$$e^{-i\epsilon \left(\frac{\hat{p}^2}{2m} + V(\hat{x}) \right) / \hbar} = e^{-i\epsilon \frac{\hat{p}^2}{2m} / \hbar} e^{-i\epsilon V(\hat{x}) / \hbar} + \mathcal{O}(\epsilon^2). \quad (2.8.4)$$

Inserting a complete set of momentum eigenstates $\int dp |p\rangle \langle p| = \hat{I}$:

$$\begin{aligned} \langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle &\approx \int_{-\infty}^{\infty} dp \langle x_{j+1} | p \rangle \langle p | e^{-i\epsilon \frac{\hat{p}^2}{2m} / \hbar} e^{-i\epsilon V(\hat{x}) / \hbar} | x_j \rangle \\ &= \int_{-\infty}^{\infty} \frac{dp}{2\pi\hbar} e^{ip(x_{j+1} - x_j)/\hbar} \exp\left(-i\frac{\epsilon p^2}{2m\hbar}\right) \exp\left(-i\frac{\epsilon V(x_j)}{\hbar}\right). \end{aligned} \quad (2.8.5)$$

The momentum integral is an exact Gaussian integral in p :

$$\int_{-\infty}^{\infty} \frac{dp}{2\pi\hbar} \exp\left[-i\frac{\epsilon}{2m\hbar}p^2 + i\frac{(x_{j+1} - x_j)}{\hbar}p\right] = \sqrt{\frac{m}{2\pi i\hbar\epsilon}} \exp\left[\frac{i\epsilon m}{2\hbar} \left(\frac{x_{j+1} - x_j}{\epsilon}\right)^2\right]. \quad (2.8.6)$$

Combining the kinetic and potential exponents:

$$\langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle = \sqrt{\frac{m}{2\pi i\hbar\epsilon}} \exp\left\{ \frac{i}{\hbar} \epsilon \left[\frac{1}{2}m \left(\frac{x_{j+1} - x_j}{\epsilon}\right)^2 - V(x_j) \right] \right\}. \quad (2.8.7)$$

In the continuum limit $\epsilon \rightarrow 0$, the ratio $\frac{x_{j+1} - x_j}{\epsilon} \rightarrow \dot{x}$ is the instantaneous velocity! The term in brackets becomes:

$$\frac{1}{2}m\dot{x}^2 - V(x) = T - V = L(x, \dot{x}), \quad (2.8.8)$$

which is precisely the **classical Lagrangian**!

2.8.3 The Feynman Path Integral

Summing the exponent across all N time slices:

$$\sum_{j=0}^{N-1} \epsilon \left[\frac{1}{2} m \left(\frac{x_{j+1} - x_j}{\epsilon} \right)^2 - V(x_j) \right] \xrightarrow{\epsilon \rightarrow 0} \int_{t_i}^{t_f} L(x, \dot{x}) dt = S[x(t)], \quad (2.8.9)$$

where $S[x(t)]$ is the classical action along the path $x(t)$. Defining the functional integration measure $\mathcal{D}[x(t)] \equiv \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \prod_{j=1}^{N-1} dx_j$, we arrive at Feynman's celebrated path integral:

$$K(x_f, t_f; x_i, t_i) = \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}[x(t)] \exp \left(\frac{i}{\hbar} S[x(t)] \right) \quad (2.8.10)$$

Physical Duality: This derivation illuminates the foundational bridge between the two representations of mechanics:

- **Canonical Quantum Mechanics:** Operates with non-commuting operators $\hat{x}, \hat{p}$ and the **Hamiltonian** $H = T + V$, which generates unitary time translations.
- **Path Integral Quantum Mechanics:** Sums over all possible continuous classical trajectories $x(t)$ weighted by the quantum phase $e^{iS/\hbar}$, governed by the **Lagrangian** $L = T - V$.

In the classical limit $\hbar \rightarrow 0$, the rapid phase oscillations cancel all paths destructively, leaving only the stationary path where $\delta S = 0$ —recovering Hamilton's Principle of Least Action and classical Newtonian trajectories!

With the dynamical principles of quantum evolution firmly established across all pictures and formalisms, we are prepared to explore their direct applications: the harmonic oscillator in Chapter 3, one-dimensional systems in Chapter 4, angular momentum in Chapter 5, and central fields in Chapter 6.

Chapter 3

Harmonic Oscillator

3.1 The Physical Hamiltonian and Potential

The simple harmonic oscillator occupies a central place in theoretical physics. Its ubiquity is neither an accident nor merely a pedagogical convenience: any physical system in the vicinity of a stable equilibrium point behaves, to lowest non-trivial order, as a harmonic oscillator.

Consider an arbitrary, smooth one-dimensional potential $V(x)$ with a local minimum at $x = x_0$. Expanding $V(x)$ in a Taylor series about x_0 , we obtain

$$V(x) = V(x_0) + \left. \frac{dV}{dx} \right|_{x_0} (x - x_0) + \frac{1}{2} \left. \frac{d^2V}{dx^2} \right|_{x_0} (x - x_0)^2 + \mathcal{O}\left((x - x_0)^3\right). \quad (3.1.1)$$

By definition of the equilibrium point, the net force vanishes: $V'(x_0) = 0$. Furthermore, stability requires that the second derivative be strictly positive: $V''(x_0) = k > 0$. Shifting the origin of our coordinate system such that $x_0 = 0$ and choosing the zero of energy such that $V(x_0) = 0$, the leading-order approximation for small displacements $|x| \ll 1$ is the quadratic potential:

$$V(x) \approx \frac{1}{2} k x^2 = \frac{1}{2} m \omega^2 x^2, \quad (3.1.2)$$

where m is the mass of the particle and $\omega \equiv \sqrt{k/m}$ is the classical angular frequency of oscillation.

In quantum mechanics, dynamic observables are elevated to linear operators acting on a Hilbert space $\mathcal{H}$. The quantum Hamiltonian operator $\hat{H}$ is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2, \quad (3.1.3)$$

where the position operator $\hat{x}$ and the momentum operator $\hat{p}$ satisfy the fundamental canonical commutation relation:

$$[\hat{x}, \hat{p}] = i\hbar \hat{I}. \quad (3.1.4)$$

Traditionally, one solves the time-independent Schrödinger equation $\hat{H}\psi(x) = E\psi(x)$ by transforming it into a second-order differential equation for the Hermite polynomials. In this chapter, we adopt a far more powerful and elegant approach: we solve the problem using the **Lie algebra** generated by the dynamical operators. This algebraic method not only yields the full energy spectrum and matrix elements with minimal analytical effort, but it also forms the conceptual blueprint for second quantization, quantum field theory, and quantum optics.

3.2 Lie Algebra and Ladder Operators

3.2.1 Dimensionless Operators and the Heisenberg-Weyl Algebra

To reveal the underlying algebraic symmetry of the Hamiltonian, it is convenient to introduce the characteristic scales of length and momentum:

$$x_0 \equiv \sqrt{\frac{\hbar}{m\omega}}, \quad p_0 \equiv \sqrt{m\hbar\omega} = \frac{\hbar}{x_0}. \quad (3.2.1)$$

The quantity x_0 represents the spatial spread of the quantum ground state, while p_0 represents its momentum spread. In terms of these scales, the Hamiltonian takes the symmetric, dimensionless form

$$\frac{\hat{H}}{\hbar\omega} = \frac{1}{2} \left[\left(\frac{\hat{x}}{x_0} \right)^2 + \left(\frac{\hat{p}}{p_0} \right)^2 \right]. \quad (3.2.2)$$

We now define the dimensionless *annihilation operator* a (also called the lowering operator) and its Hermitian adjoint, the *creation operator* $a^\dagger$ (also called the raising operator):

$$a \equiv \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} = \frac{1}{\sqrt{2}} \left(\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right), \quad (3.2.3)$$

$$a^\dagger \equiv \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} = \frac{1}{\sqrt{2}} \left(\frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right). \quad (3.2.4)$$

Notice that neither a nor $a^\dagger$ is Hermitian: $a \neq a^\dagger$. They are mutually adjoint operators: $(a)^\dagger = a^\dagger$.

We now derive their commutation relation directly from the canonical commutator $[\hat{x}, \hat{p}] = i\hbar$:

$$\begin{aligned} [a, a^\dagger] &= \left[\sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}, \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} \right] \\ &= \frac{m\omega}{2\hbar} [\hat{x}, \hat{x}] - \frac{i}{2\hbar} [\hat{x}, \hat{p}] + \frac{i}{2\hbar} [\hat{p}, \hat{x}] + \frac{1}{2m\hbar\omega} [\hat{p}, \hat{p}]. \end{aligned} \quad (3.2.5)$$

Since any operator commutes with itself ($[\hat{x}, \hat{x}] = [\hat{p}, \hat{p}] = 0$) and $[\hat{p}, \hat{x}] = -[\hat{x}, \hat{p}] = -i\hbar$, this simplifies to:

$$[a, a^\dagger] = -\frac{i}{2\hbar}(i\hbar) + \frac{i}{2\hbar}(-i\hbar) = \frac{1}{2} + \frac{1}{2} = 1. \quad (3.2.6)$$

Similarly, it is trivial to verify that

$$[a, a] = 0, \quad [a^\dagger, a^\dagger] = 0. \quad (3.2.7)$$

The set of operators $\{a, a^\dagger, \hat{I}\}$ closed under the Lie bracket $[\cdot, \cdot]$ forms the three-dimensional nilpotent Lie algebra known as the **Heisenberg-Weyl algebra** $\mathfrak{h}_3$.

3.2.2 Rewriting the Hamiltonian

Next, let us evaluate the operator product $a^\dagger a$:

$$\begin{aligned} a^\dagger a &= \left(\sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} \right) \left(\sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} \right) \\ &= \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{1}{2m\hbar\omega} \hat{p}^2 + \frac{i}{2\hbar} (\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega} \left(\frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 \right) + \frac{i}{2\hbar} [\hat{x}, \hat{p}] \\ &= \frac{\hat{H}}{\hbar\omega} + \frac{i}{2\hbar}(i\hbar) = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}. \end{aligned} \quad (3.2.8)$$

Multiplying through by $\hbar\omega$, we express the Hamiltonian directly in terms of ladder operators:

$$\hat{H} = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right) = \hbar\omega \left(\hat{N} + \frac{1}{2} \right), \quad (3.2.9)$$

where we have introduced the Hermitian **number operator**

$$\hat{N} \equiv a^\dagger a, \quad \hat{N}^\dagger = (a^\dagger a)^\dagger = a^\dagger (a^\dagger)^\dagger = a^\dagger a = \hat{N}. \quad (3.2.10)$$

Alternatively, reversing the order of the product, let us evaluate $aa^\dagger$:

$$\begin{aligned} aa^\dagger &= \left(\sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} \right) \left(\sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} \right) \\ &= \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{1}{2m\hbar\omega} \hat{p}^2 - \frac{i}{2\hbar} (\hat{x}\hat{p} - \hat{p}\hat{x}) \\ &= \frac{\hat{H}}{\hbar\omega} - \frac{i}{2\hbar} (i\hbar) = \frac{\hat{H}}{\hbar\omega} + \frac{1}{2}. \end{aligned} \quad (3.2.11)$$

Equivalently, using the fundamental commutator $[a, a^\dagger] = 1$:

$$aa^\dagger = a^\dagger a + [a, a^\dagger] = \hat{N} + 1. \quad (3.2.12)$$

Multiplying by $\hbar\omega$, the Hamiltonian can equally be expressed in the alternative reversed-order form:

$$\hat{H} = \hbar\omega \left(aa^\dagger - \frac{1}{2} \right) \equiv \hbar\omega \left(\hat{N}' - \frac{1}{2} \right), \quad (3.2.13)$$

where we define the alternative operator product:

$$\hat{N}' \equiv aa^\dagger = \hat{N} + 1. \quad (3.2.14)$$

Notice the striking sign difference in the constant term: when expressed in terms of $a^\dagger a$, the constant is $+\frac{1}{2}\hbar\omega$, whereas in terms of $aa^\dagger$, it is $-\frac{1}{2}\hbar\omega$!

Averaging (symmetrizing) both expressions yields the **Weyl-ordered form**:

$$\hat{H} = \frac{1}{2} \hbar\omega (a^\dagger a + aa^\dagger). \quad (3.2.15)$$

In this symmetric form, the explicit constant scalar cancels out entirely. The non-zero zero-point energy $\frac{1}{2}\hbar\omega$ is a direct quantum manifestation of the non-commutativity of $\hat{x}$ and $\hat{p}$, encoded into the commutator $[a, a^\dagger] = 1$. In Section 3.3.4, we will explore the profound duality between the $a^\dagger a$ and $aa^\dagger$ formulations and prove why both yield the exact same physical energy spectrum.

3.2.3 Fundamental Commutation Relations with the Number Operator

The algebraic action of a and $a^\dagger$ on the eigenstates of $\hat{H}$ is governed by their commutators with $\hat{N}$. Using the derivation identity $[AB, C] = A[B, C] + [A, C]B$, we calculate:

$$[\hat{N}, a] = [a^\dagger a, a] = a^\dagger [a, a] + [a^\dagger, a]a = 0 + (-1)a = -a, \quad (3.2.16)$$

$$[\hat{N}, a^\dagger] = [a^\dagger a, a^\dagger] = a^\dagger [a, a^\dagger] + [a^\dagger, a^\dagger]a = a^\dagger(1) + 0 = a^\dagger. \quad (3.2.17)$$

Since $\hat{H} = \hbar\omega(\hat{N} + 1/2)$, the commutators of a and $a^\dagger$ with the Hamiltonian are:

$$[\hat{H}, a] = \hbar\omega[\hat{N}, a] = -\hbar\omega a, \quad (3.2.18)$$

$$[\hat{H}, a^\dagger] = \hbar\omega[\hat{N}, a^\dagger] = +\hbar\omega a^\dagger. \quad (3.2.19)$$

These relations demonstrate that a and $a^\dagger$ act as energy shift operators.

3.3 Eigenvalue Spectrum and the Fock Basis

3.3.1 The Action of the Ladder Operators

Let $|n\rangle$ denote a normalized eigenstate of the number operator $\hat{N}$ with eigenvalue n :

$$\hat{N}|n\rangle = n|n\rangle, \quad \langle n|n\rangle = 1. \quad (3.3.1)$$

From Eq. (3.2.9), $|n\rangle$ is simultaneously an energy eigenstate with eigenvalue E_n :

$$\hat{H}|n\rangle = \hbar\omega \left(\hat{N} + \frac{1}{2} \right) |n\rangle = \hbar\omega \left(n + \frac{1}{2} \right) |n\rangle \equiv E_n |n\rangle. \quad (3.3.2)$$

Now, let us examine the action of a and $a^\dagger$ on $|n\rangle$. Using the commutation relations (3.2.16) and (3.2.17):

$$\hat{N}(a|n\rangle) = (\hat{N}a)|n\rangle = (a\hat{N} + [\hat{N}, a])|n\rangle = (an - a)|n\rangle = (n-1)(a|n\rangle), \quad (3.3.3)$$

$$\hat{N}(a^\dagger|n\rangle) = (\hat{N}a^\dagger)|n\rangle = (a^\dagger\hat{N} + [\hat{N}, a^\dagger])|n\rangle = (a^\dagger n + a^\dagger)|n\rangle = (n+1)(a^\dagger|n\rangle). \quad (3.3.4)$$

This remarkable result proves that:

- If $|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue n , then $a|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $(n-1)$, and its energy is reduced by $\hbar\omega$:

$$\hat{H}(a|n\rangle) = (E_n - \hbar\omega)(a|n\rangle). \quad (3.3.5)$$

Hence, a is the **annihilation** or **lowering** operator.

- Similarly, $a^\dagger|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $(n+1)$, and its energy is increased by $\hbar\omega$:

$$\hat{H}(a^\dagger|n\rangle) = (E_n + \hbar\omega)(a^\dagger|n\rangle). \quad (3.3.6)$$

Hence, $a^\dagger$ is the **creation** or **raising** operator.

3.3.2 Why n Must Be a Non-Negative Integer

Why does this infinite ladder of states not extend downward into negative infinity? The answer lies in the fundamental postulate of quantum mechanics that the norm of any state vector in Hilbert space must be strictly non-negative:

$$\langle\psi|\psi\rangle \geq 0, \quad \text{with } \langle\psi|\psi\rangle = 0 \iff |\psi\rangle = 0. \quad (3.3.7)$$

Consider the state $|\psi\rangle = a|n\rangle$. Its squared norm is

$$\langle\psi|\psi\rangle = \langle n|a^\dagger a|n\rangle = \langle n|\hat{N}|n\rangle = n\langle n|n\rangle = n. \quad (3.3.8)$$

Since $\langle\psi|\psi\rangle \geq 0$, we conclude that

$$n \geq 0. \quad (3.3.9)$$

Every eigenvalue of the number operator must be greater than or equal to zero!

Now suppose that n were not an integer. We could repeatedly apply the lowering operator a to generate a sequence of states with eigenvalues $n-1, n-2, n-3, \dots$. After $k > n$ applications, we would obtain an eigenstate with a negative eigenvalue $n-k < 0$. The squared norm of this resulting state would be

$$\|a|n-k\rangle\|^2 = (n-k) < 0, \quad (3.3.10)$$

which is a fatal mathematical contradiction in a Hilbert space!

The only possible way this contradiction is avoided is if the descending sequence terminates. Termination requires that there must exist a ground state $|0\rangle$ such that applying the lowering operator yields the null vector:

$$a|0\rangle = 0. \quad (3.3.11)$$

Applying the number operator to this ground state gives

$$\hat{N}|0\rangle = a^\dagger(a|0\rangle) = a^\dagger(0) = 0 = 0|0\rangle. \quad (3.3.12)$$

Therefore, the minimum eigenvalue of $\hat{N}$ is identically zero: $n_0 = 0$.

Ascending from the ground state by repeated action of the creation operator $a^\dagger$, the eigenvalues of $\hat{N}$ must increment by integers:

$$n \in \{0, 1, 2, 3, \dots\} = \mathbb{N}_0. \quad (3.3.13)$$

Consequently, the energy spectrum of the quantum harmonic oscillator is strictly discrete and evenly spaced:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (3.3.14)$$

Even in the lowest possible energy state ($n = 0$), the system retains a non-zero **zero-point energy**:

$$E_0 = \frac{1}{2}\hbar\omega. \quad (3.3.15)$$

This zero-point energy is an inevitable consequence of the Heisenberg uncertainty principle: if the particle were completely at rest at the bottom of the well ($x = 0, p = 0$), both Δx and Δp would be zero, violating $[\hat{x}, \hat{p}] = i\hbar$.

3.3.3 State Normalization and Matrix Elements

We now determine the normalization coefficients when ladder operators act on the Fock states $|n\rangle$.

Let $a|n\rangle = c_n|n-1\rangle$. The squared norm is

$$|c_n|^2 \langle n-1|n-1\rangle = \langle n|a^\dagger a|n\rangle = n \implies |c_n|^2 = n. \quad (3.3.16)$$

Adopting the standard convention that the phase of c_n is chosen real and positive, we have:

$$a|n\rangle = \sqrt{n}|n-1\rangle. \quad (3.3.17)$$

Similarly, let $a^\dagger|n\rangle = d_n|n+1\rangle$. The squared norm is

$$|d_n|^2 \langle n+1|n+1\rangle = \langle n|aa^\dagger|n\rangle = \langle n|(\hat{N} + 1)|n\rangle = n + 1 \implies |d_n|^2 = n + 1. \quad (3.3.18)$$

Choosing the phase to be real and positive:

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (3.3.19)$$

By successive application of $a^\dagger$ starting from the ground state $|0\rangle$, any normalized Fock state $|n\rangle$ can be constructed explicitly as:

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle. \quad (3.3.20)$$

3.3.4 Operator Ordering Duality: Spectral Equivalence of $a^\dagger a$ and $aa^\dagger$

In standard textbook expositions, the quantum harmonic oscillator is almost exclusively solved using the standard number operator $\hat{N} = a^\dagger a$, leading to eigenvalues $n \in \{0, 1, 2, \dots\}$ and energy levels $E_n = \hbar\omega(n + 1/2)$. However, as established in Eq. (3.2.13), the Hamiltonian can equally well be formulated in terms of the reversed operator product:

$$\hat{H} = \hbar\omega \left(aa^\dagger - \frac{1}{2} \right). \quad (3.3.21)$$

This prompts a natural and fundamental question: *What happens if an investigator chooses from the outset to develop the theory using $\hat{N}' \equiv aa^\dagger$ instead of $\hat{N} \equiv a^\dagger a$? Does the reversed ordering change the physical spectrum, and how does the apparent discrepancy in the Hamiltonian's constant term ($+\frac{1}{2}\hbar\omega \longleftrightarrow -\frac{1}{2}\hbar\omega$) resolve itself?*

Let us solve the eigenvalue problem for $\hat{N}' = aa^\dagger$ from first principles.

Action of the Ladder Operators on $aa^\dagger$ Eigenstates

Let $|n'\rangle$ denote a normalized eigenstate of the operator $\hat{N}' = aa^\dagger$ with eigenvalue n' :

$$aa^\dagger |n'\rangle = n' |n'\rangle, \quad \langle n'|n'\rangle = 1. \quad (3.3.22)$$

To determine how the ladder operators act on these states, let us first act on both sides of this eigenvalue equation with the annihilation operator a :

$$a(aa^\dagger |n'\rangle) = n'a |n'\rangle. \quad (3.3.23)$$

Using the fundamental commutator $[a, a^\dagger] = aa^\dagger - a^\dagger a = 1$, we replace the operator product $aa^\dagger$ on the left side with $1 + a^\dagger a$:

$$a(1 + a^\dagger a) |n'\rangle = n'a |n'\rangle. \quad (3.3.24)$$

Distributing a across the sum:

$$a |n'\rangle + a(a^\dagger a) |n'\rangle = n'a |n'\rangle. \quad (3.3.25)$$

Associating the product as $a(a^\dagger a) = (aa^\dagger)a$, we find:

$$a |n'\rangle + (aa^\dagger)(a |n'\rangle) = n'a |n'\rangle. \quad (3.3.26)$$

Subtracting $a |n'\rangle$ from both sides gives the eigenvalue equation:

$$(aa^\dagger)(a |n'\rangle) = (n' - 1)(a |n'\rangle). \quad (3.3.27)$$

Equation (3.3.27) proves that $a |n'\rangle$ is an eigenstate of $aa^\dagger$ with eigenvalue $(n' - 1)$. Thus, a lowers the eigenvalue of $aa^\dagger$ by exactly one unit!

Next, let us act on the eigenvalue equation with the creation operator $a^\dagger$:

$$a^\dagger(aa^\dagger |n'\rangle) = n'a^\dagger |n'\rangle. \quad (3.3.28)$$

Using $a^\dagger a = aa^\dagger - 1$, we rewrite the left side:

$$(a^\dagger a)a^\dagger |n'\rangle = (aa^\dagger - 1)a^\dagger |n'\rangle = n'a^\dagger |n'\rangle. \quad (3.3.29)$$

Expanding and rearranging terms:

$$(aa^\dagger)(a^\dagger |n'\rangle) - a^\dagger |n'\rangle = n'a^\dagger |n'\rangle \implies (aa^\dagger)(a^\dagger |n'\rangle) = (n' + 1)(a^\dagger |n'\rangle). \quad (3.3.30)$$

Equation (3.3.30) proves that $a^\dagger |n'\rangle$ is an eigenstate of $aa^\dagger$ with eigenvalue $(n' + 1)$. Thus, $a^\dagger$ raises the eigenvalue of $aa^\dagger$ by exactly one unit!

Positivity of Norms and the Lower Bound $n' \geq 1$

Why can the ladder of eigenvalues $\{n', n' - 1, n' - 2, \dots\}$ not descend infinitely into the negative domain? As always in quantum mechanics, the answer is dictated by the positive definiteness of the Hilbert space norm:

$$\langle \psi | \psi \rangle \geq 0, \quad \text{with } \langle \psi | \psi \rangle = 0 \iff |\psi\rangle = 0. \quad (3.3.31)$$

Consider the state vector $|\phi\rangle = a |n'\rangle$. Its squared norm must be strictly non-negative:

$$\|a |n'\rangle\|^2 = \langle \phi | \phi \rangle = \langle n' | a^\dagger a |n'\rangle \geq 0. \quad (3.3.32)$$

Now, substitute the commutator identity $a^\dagger a = aa^\dagger - 1$:

$$\langle n' | (aa^\dagger - 1) |n'\rangle \geq 0. \quad (3.3.33)$$

Since $|n'\rangle$ is an eigenstate of $aa^\dagger$ with eigenvalue n' :

$$(n' - 1) \langle n' | n' \rangle \geq 0. \quad (3.3.34)$$

Because the state $|n'\rangle$ is normalized ($\langle n' | n' \rangle = 1 > 0$), we arrive at the pivotal lower bound:

$$n' - 1 \geq 0 \implies n' \geq 1. \quad (3.3.35)$$

The eigenvalues of the operator $aa^\dagger$ are strictly bounded below by 1! Notice the crucial difference:

- For $\hat{N} = a^\dagger a$, the norm condition $\langle n | a^\dagger a |n\rangle = n \geq 0$ establishes that $n \geq 0$.
- For $\hat{N}' = aa^\dagger$, the norm condition $\langle n' | a^\dagger a |n'\rangle = n' - 1 \geq 0$ establishes that $n' \geq 1$.

The operator $aa^\dagger$ is **strictly positive-definite**; its eigenvalues cannot even be zero!

Ground State Termination and the Spectrum

To prevent repeated applications of the lowering operator a from violating the lower bound $n' \geq 1$, the descending ladder must terminate at a ground state $|n'_0\rangle$ such that lowering yields the null vector:

$$a |n'_0\rangle = 0. \quad (3.3.36)$$

Applying the operator $aa^\dagger$ to this ground state and using $aa^\dagger = 1 + a^\dagger a$:

$$(aa^\dagger) |n'_0\rangle = (1 + a^\dagger a) |n'_0\rangle = 1 |n'_0\rangle + a^\dagger (a |n'_0\rangle) = 1 |n'_0\rangle + a^\dagger (0) = 1 |n'_0\rangle. \quad (3.3.37)$$

Therefore, the ground state eigenvalue of $aa^\dagger$ is:

$$n'_0 = 1. \quad (3.3.38)$$

Ascending from this ground state by repeated action of the raising operator $a^\dagger$, the eigenvalues of $aa^\dagger$ increment by integers:

$$n' \in \{1, 2, 3, 4, \dots\} = \mathbb{N}^+. \quad (3.3.39)$$

Normalization Coefficients

Let us compute the normalization coefficients in this representation. Let $a |n'\rangle = c_{n'} |n' - 1\rangle$. Then:

$$|c_{n'}|^2 = \langle n' | a^\dagger a |n'\rangle = \langle n' | (|a\rangle a^\dagger - 1) n' = (n' - 1) \implies a |n'\rangle = \sqrt{n' - 1} |n' - 1\rangle. \quad (3.3.40)$$

Notice that for the ground state ($n' = 1$), this gives $a |1\rangle = \sqrt{1 - 1} |0\rangle = 0$, terminating the sequence automatically and consistently. Similarly, let $a^\dagger |n'\rangle = d_{n'} |n' + 1\rangle$. Then:

$$|d_{n'}|^2 = \langle n' | aa^\dagger |n'\rangle = n' \implies a^\dagger |n'\rangle = \sqrt{n'} |n' + 1\rangle. \quad (3.3.41)$$

The Physical Energy Spectrum and Exact Cancellation

We now calculate the energy eigenvalues using the Hamiltonian expressed in terms of $aa^\dagger$:

$$\hat{H} |n'\rangle = \hbar\omega \left(aa^\dagger - \frac{1}{2} \right) |n'\rangle = \hbar\omega \left(n' - \frac{1}{2} \right) |n'\rangle \equiv E_{n'} |n'\rangle. \quad (3.3.42)$$

Let us evaluate the energy of the lowest states:

- For the ground state ($n' = 1$):

$$E_1 = \hbar\omega \left(1 - \frac{1}{2} \right) = \frac{1}{2} \hbar\omega. \quad (3.3.43)$$

- For the first excited state ($n' = 2$):

$$E_2 = \hbar\omega \left(2 - \frac{1}{2} \right) = \frac{3}{2} \hbar\omega. \quad (3.3.44)$$

- For the second excited state ($n' = 3$):

$$E_3 = \hbar\omega \left(3 - \frac{1}{2} \right) = \frac{5}{2} \hbar\omega. \quad (3.3.45)$$

In general, since the spectrum of $aa^\dagger$ is $n' = n + 1$ with $n \in \{0, 1, 2, \dots\}$, we substitute $n' = n + 1$:

$$E_{n'} = \hbar\omega \left((n + 1) - \frac{1}{2} \right) = \hbar\omega \left(n + \frac{1}{2} \right) = E_n! \quad (3.3.46)$$

The physical energy spectrum is **identically the same!**

Operator Ordering Duality: Why $a^\dagger a$ and $aa^\dagger$ Yield Identical Physics

Why do both operator orderings produce the exact same physical energy levels despite looking different in intermediate algebra? The explanation lies in a beautiful and exact algebraic cancellation between two compensating effects:

Property	Standard Ordering ($a^\dagger a$)	Reversed Ordering ($aa^\dagger$)
Operator Definition	$\hat{N} \equiv a^\dagger a$	$\hat{N}' \equiv aa^\dagger = \hat{N} + 1$
Hamiltonian Form	$\hat{H} = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right)$	$\hat{H} = \hbar\omega \left(aa^\dagger - \frac{1}{2} \right)$
Constant Shift in $\hat{H}$	$+\frac{1}{2}\hbar\omega$	$-\frac{1}{2}\hbar\omega \quad (\Delta C = -\hbar\omega)$
Lower Bound Norm	$\langle n a^\dagger a n \rangle = n \geq 0$	$\langle n' a^\dagger a n' \rangle = n' - 1 \geq 0 \implies n' \geq 1$
Operator Spectrum	$n \in \{0, 1, 2, 3, \dots\}$	$n' \in \{1, 2, 3, 4, \dots\} \quad (\Delta\lambda = +1)$
Ground State Eigenvalue	$n_0 = 0$	$n'_0 = 1$
Lowering Operator Action	$a n \rangle = \sqrt{n} n - 1 \rangle$	$a n' \rangle = \sqrt{n' - 1} n' - 1 \rangle$
Raising Operator Action	$a^\dagger n \rangle = \sqrt{n + 1} n + 1 \rangle$	$a^\dagger n' \rangle = \sqrt{n'} n' + 1 \rangle$
Ground State Condition	$a 0 \rangle = 0 \implies \hat{N} 0 \rangle = 0$	$a 1 \rangle = 0 \implies aa^\dagger 1 \rangle = 1 1 \rangle$
Physical Energy	$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$	$E_{n'} = \hbar\omega \left(n' - \frac{1}{2} \right) = \hbar\omega \left(n + \frac{1}{2} \right)$

The Mechanism of Spectral Invariance:

1. **Hamiltonian Constant Shift:** Moving from $a^\dagger a$ to $aa^\dagger$ introduces the commutator $[a, a^\dagger] = 1$, which shifts the constant term in the Hamiltonian downward by exactly

$\hbar\omega$:

$$\Delta C_H = \left(-\frac{1}{2}\hbar\omega\right) - \left(+\frac{1}{2}\hbar\omega\right) = -\hbar\omega. \quad (1)$$

2. **Spectrum Upward Shift:** Simultaneously, because $aa^\dagger = a^\dagger a + 1$, the eigenvalue spectrum of the operator itself is shifted upward by exactly +1:

$$\Delta\lambda = n' - n = +1 \implies \Delta(\hbar\omega\lambda) = +\hbar\omega. \quad (2)$$

3. **Exact Cancellation:** The upward shift in the operator spectrum ($+\hbar\omega$) and the downward shift in the Hamiltonian constant ($-\hbar\omega$) cancel each other out identically:

$$\Delta E = \Delta(\hbar\omega\lambda) + \Delta C_H = +\hbar\omega - \hbar\omega = 0. \quad (3)$$

4. **State Identification:** The physical states are identical; state $|n' = 1\rangle$ in the $aa^\dagger$ language is the exact same physical vacuum state as $|n = 0\rangle$ in the $a^\dagger a$ language:

$$|n'\rangle \equiv |n' - 1\rangle \equiv |n\rangle. \quad (4)$$

5. **Symmetric Weyl Formulation:** The Weyl-ordered Hamiltonian Eq. (3.2.15) takes the average of both representations:

$$\hat{H} = \frac{1}{2}\hbar\omega(a^\dagger a + aa^\dagger) = \frac{1}{2}\hbar\omega(\hat{N} + \hat{N}'). \quad (5)$$

Its eigenvalue on the n -th physical state is:

$$E_n = \frac{1}{2}\hbar\omega(n + n') = \frac{1}{2}\hbar\omega(n + n + 1) = \hbar\omega\left(n + \frac{1}{2}\right). \quad (6)$$

In this symmetric picture, the zero-point energy $\frac{1}{2}\hbar\omega$ emerges naturally as the average of the ground state eigenvalues of the two dual orderings: $\frac{1}{2}(0 + 1)\hbar\omega = \frac{1}{2}\hbar\omega$.

3.4 Position and Momentum Representations

3.4.1 Expressing $\hat{x}$ and $\hat{p}$ in the Fock Basis

By adding and subtracting the definitions (3.2.3) and (3.2.4), we express the physical position and momentum operators in terms of ladder operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger) = \frac{x_0}{\sqrt{2}}(a + a^\dagger), \quad (3.4.1)$$

$$\hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}}(a - a^\dagger) = \frac{p_0}{i\sqrt{2}}(a - a^\dagger). \quad (3.4.2)$$

Using Eqs. (3.3.17) and (3.3.19), we can compute the matrix elements of $\hat{x}$ and $\hat{p}$ between arbitrary Fock states $|n\rangle$ and $|m\rangle$:

$$\langle m|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n}\delta_{m,n-1} + \sqrt{n+1}\delta_{m,n+1}\right), \quad (3.4.3)$$

$$\langle m|\hat{p}|n\rangle = -i\sqrt{\frac{m\hbar\omega}{2}}\left(\sqrt{n}\delta_{m,n-1} - \sqrt{n+1}\delta_{m,n+1}\right). \quad (3.4.4)$$

Notice immediately that the diagonal matrix elements vanish identically for all n :

$$\langle \hat{x} \rangle_n = \langle n | \hat{x} | n \rangle = 0, \quad \langle \hat{p} \rangle_n = \langle n | \hat{p} | n \rangle = 0. \quad (3.4.5)$$

In every stationary energy eigenstate, the average position is at the origin, and the average momentum is zero.

3.4.2 Variances, the Virial Theorem, and Uncertainty

To find the uncertainties Δx and Δp , we compute the expectation values of $\hat{x}^2$ and $\hat{p}^2$:

$$\begin{aligned} \hat{x}^2 &= \frac{\hbar}{2m\omega} (a + a^\dagger)^2 = \frac{\hbar}{2m\omega} (a^2 + a^{\dagger 2} + aa^\dagger + a^\dagger a) \\ &= \frac{\hbar}{2m\omega} (a^2 + a^{\dagger 2} + 2\hat{N} + 1). \end{aligned} \quad (3.4.6)$$

Since $\langle n | a^2 | n \rangle = \langle n | a^{\dagger 2} | n \rangle = 0$, we find:

$$\langle \hat{x}^2 \rangle_n = \langle n | \hat{x}^2 | n \rangle = \frac{\hbar}{2m\omega} (2n + 1) = \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right). \quad (3.4.7)$$

Similarly, for $\hat{p}^2$:

$$\begin{aligned} \hat{p}^2 &= -\frac{m\hbar\omega}{2} (a - a^\dagger)^2 = -\frac{m\hbar\omega}{2} (a^2 + a^{\dagger 2} - aa^\dagger - a^\dagger a) \\ &= -\frac{m\hbar\omega}{2} (a^2 + a^{\dagger 2} - 2\hat{N} - 1), \end{aligned} \quad (3.4.8)$$

which yields

$$\langle \hat{p}^2 \rangle_n = \langle n | \hat{p}^2 | n \rangle = \frac{m\hbar\omega}{2} (2n + 1) = m\hbar\omega \left(n + \frac{1}{2} \right). \quad (3.4.9)$$

Let us now evaluate the expectation values of the kinetic and potential energies:

$$\langle \hat{T} \rangle_n = \frac{\langle \hat{p}^2 \rangle_n}{2m} = \frac{1}{2} \hbar\omega \left(n + \frac{1}{2} \right) = \frac{1}{2} E_n, \quad (3.4.10)$$

$$\langle \hat{V} \rangle_n = \frac{1}{2} m\omega^2 \langle \hat{x}^2 \rangle_n = \frac{1}{2} \hbar\omega \left(n + \frac{1}{2} \right) = \frac{1}{2} E_n. \quad (3.4.11)$$

Thus, $\langle \hat{T} \rangle_n = \langle \hat{V} \rangle_n = \frac{1}{2} E_n$, in exact agreement with the quantum **Virial Theorem** for a harmonic potential ($2\langle \hat{T} \rangle = \langle x \frac{dV}{dx} \rangle = 2\langle \hat{V} \rangle$).

The position and momentum uncertainties in state $|n\rangle$ are:

$$\Delta x = \sqrt{\langle \hat{x}^2 \rangle_n - \langle \hat{x} \rangle_n^2} = \sqrt{\frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)}, \quad \Delta p = \sqrt{\langle \hat{p}^2 \rangle_n - \langle \hat{p} \rangle_n^2} = \sqrt{m\hbar\omega \left(n + \frac{1}{2} \right)}. \quad (3.4.12)$$

Their product is:

$$\Delta x \Delta p = \hbar \left(n + \frac{1}{2} \right) \geq \frac{\hbar}{2}. \quad (3.4.13)$$

The Heisenberg uncertainty relation $\Delta x \Delta p \geq \hbar/2$ is satisfied for all states, and the lower bound is saturated if and only if $n = 0$:

$$\Delta x \Delta p = \frac{\hbar}{2} \quad (\text{for } n = 0). \quad (3.4.14)$$

The harmonic oscillator ground state is a **minimum-uncertainty wave packet**!

3.4.3 Ground State and Spatial Wavefunctions

The position-space wavefunctions $\psi_n(x) \equiv \langle x|n\rangle$ can be derived purely from the algebraic condition $a|0\rangle = 0$. Projecting Eq. (3.3.11) onto the continuous coordinate basis $\langle x|$:

$$\langle x|a|0\rangle = 0 \implies \langle x|\left(\sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2m\hbar\omega}}\hat{p}\right)|0\rangle = 0. \quad (3.4.15)$$

Using the coordinate representation $\langle x|\hat{x}|\psi\rangle = x\psi(x)$ and $\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{d\psi}{dx}$:

$$\sqrt{\frac{m\omega}{2\hbar}}x\psi_0(x) + \frac{i}{\sqrt{2m\hbar\omega}}\left(-i\hbar\frac{d\psi_0}{dx}\right) = 0. \quad (3.4.16)$$

Multiplying through by $\sqrt{\frac{2m\omega}{\hbar}}$, we obtain a first-order linear differential equation:

$$\frac{d\psi_0}{dx} + \frac{m\omega}{\hbar}x\psi_0(x) = 0. \quad (3.4.17)$$

Separating variables and integrating directly:

$$\frac{d\psi_0}{\psi_0} = -\frac{m\omega}{\hbar}x dx \implies \ln \psi_0(x) = -\frac{m\omega}{2\hbar}x^2 + C, \quad (3.4.18)$$

which gives the Gaussian wavefunction:

$$\psi_0(x) = C_0 \exp\left(-\frac{m\omega}{2\hbar}x^2\right). \quad (3.4.19)$$

Normalizing the ground state via $\int_{-\infty}^{\infty} |\psi_0(x)|^2 dx = 1$:

$$|C_0|^2 \int_{-\infty}^{\infty} e^{-\frac{m\omega}{\hbar}x^2} dx = |C_0|^2 \sqrt{\frac{\pi\hbar}{m\omega}} = 1 \implies C_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}. \quad (3.4.20)$$

Thus, the ground state wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right). \quad (3.4.21)$$

Higher excited states are obtained by repeatedly applying the differential representation of $a^\dagger$:

$$\psi_n(x) = \langle x|n\rangle = \frac{1}{\sqrt{n!}} \langle x|(a^\dagger)^n|0\rangle = \frac{1}{\sqrt{n!}} \left[\frac{1}{\sqrt{2}}\left(\frac{x}{x_0} - x_0\frac{d}{dx}\right)\right]^n \psi_0(x). \quad (3.4.22)$$

Introducing the dimensionless spatial coordinate $\xi \equiv x/x_0 = \sqrt{\frac{m\omega}{\hbar}}x$, this yields the standard Hermite polynomials $H_n(\xi)$:

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) \exp\left(-\frac{m\omega}{2\hbar}x^2\right). \quad (3.4.23)$$

Direct Connection to the Schrödinger Differential Equation

Alternatively, we can verify the ground state directly from the time-independent Schrödinger equation $\hat{H}|0\rangle = E_0|0\rangle$. Projecting into the coordinate representation:

$$\langle x|\hat{H}|0\rangle = \langle x|\left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2\right)|0\rangle = E_0 \langle x|0\rangle. \quad (3.4.24)$$

Using $\langle x|\hat{p}^2|\psi\rangle = -\hbar^2 \frac{d^2\psi}{dx^2}$ and $E_0 = \frac{1}{2}\hbar\omega$, this yields the second-order stationary Schrödinger equation:

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2\right) \psi_0(x) = \frac{1}{2}\hbar\omega \psi_0(x). \quad (3.4.25)$$

Differentiating the Gaussian solution $\psi_0(x) = C_0 \exp(-\frac{m\omega}{2\hbar}x^2)$:

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0(x), \quad (3.4.26)$$

$$\frac{d^2\psi_0}{dx^2} = -\frac{m\omega}{\hbar}\psi_0(x) - \frac{m\omega}{\hbar}x\left(-\frac{m\omega}{\hbar}x\psi_0(x)\right) = \left[-\frac{m\omega}{\hbar} + \left(\frac{m\omega}{\hbar}\right)^2 x^2\right] \psi_0(x). \quad (3.4.27)$$

Substituting this back into the kinetic energy term:

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2\psi_0}{dx^2} &= -\frac{\hbar^2}{2m} \left[-\frac{m\omega}{\hbar} + \left(\frac{m\omega}{\hbar}\right)^2 x^2\right] \psi_0(x) \\ &= \frac{1}{2}\hbar\omega \psi_0(x) - \frac{1}{2}m\omega^2 x^2 \psi_0(x). \end{aligned} \quad (3.4.28)$$

Thus:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_0}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi_0(x) = \frac{1}{2}\hbar\omega \psi_0(x), \quad (3.4.29)$$

in exact and complete agreement with Eq. (3.4.25).

Asymptotic Decay and the Classically Forbidden Region

Equation (3.4.25) provides deep insight into the spatial structure of the harmonic oscillator wavefunctions:

- **Asymptotic Behavior as $x \rightarrow \pm\infty$:** For large $|x|$, the quadratic potential dominates over the total energy:

$$V(x) - E_n = \frac{1}{2}m\omega^2 x^2 - E_n \xrightarrow{|x| \rightarrow \infty} +\infty. \quad (3.4.30)$$

In this asymptotic limit, the stationary Schrödinger equation simplifies to:

$$\frac{d^2\psi}{dx^2} \approx \frac{m^2\omega^2}{\hbar^2} x^2 \psi(x) \implies \psi(x) \sim \exp\left(\pm \frac{m\omega}{2\hbar} x^2\right). \quad (3.4.31)$$

The exponentially growing branch $\exp(+m\omega x^2/2\hbar)$ must be discarded because its norm diverges violently: $\int |\psi|^2 dx \rightarrow \infty$. Therefore, **the wavefunction must decay to zero faster than the quadratic potential grows**:

$$\psi_n(x) \sim \mathcal{P}_n(x) \exp\left(-\frac{m\omega}{2\hbar} x^2\right) \xrightarrow{|x| \rightarrow \infty} 0. \quad (3.4.32)$$

- **Tunneling into the Classically Inaccessible Region:** In classical mechanics, a particle with total energy E_n is strictly bounded by the turning points $x_{\text{turn}} = \pm\sqrt{2E_n/m\omega^2}$, beyond which the kinetic energy would be negative ($T = E_n - V(x) < 0$). In quantum mechanics, however, the rapid Gaussian decay does *not* terminate at the classical turning points! The probability of finding the particle in the classically inaccessible region is strictly positive:

$$P_{\text{forbidden}} = \int_{|x| > x_{\text{turn}}} |\psi_n(x)|^2 dx > 0. \quad (3.4.33)$$

For the ground state ($E_0 = \frac{1}{2}\hbar\omega$), the turning points are $x_{\text{turn}} = \pm x_0 = \pm\sqrt{\frac{\hbar}{m\omega}}$, and the probability of finding the particle beyond the turning points is:

$$P(|x| > x_0) = 2\sqrt{\frac{1}{\pi}} \int_1^\infty e^{-\xi^2} d\xi = \text{erfc}(1) \approx 0.1573 \quad (15.73\%). \quad (3.4.34)$$

There is a substantial $\approx 15.7\%$ probability of finding the quantum particle inside the classically forbidden barrier! As discussed in Chapter 2 (regarding the non-commutativity of $\hat{T}$ and $\hat{V}$), this is not a violation of energy conservation; it simply reflects the wave curvature $\frac{\psi''(x)}{\psi(x)} > 0$ demanded by the differential equation when $V(x) > E_n$.

3.5 What's Oscillating?

3.5.1 The Paradox of Stationary States

The system under study is called the *harmonic oscillator*. In classical physics, an oscillator is defined by periodic spatial motion: a particle released from displacement X_0 oscillates back and forth across the origin according to $x(t) = X_0 \cos(\omega t)$, with momentum $p(t) = -m\omega X_0 \sin(\omega t)$.

Yet, when we examine any energy eigenstate $|n\rangle$ of the quantum harmonic oscillator, we encounter a profound paradox: **absolutely nothing oscillates in space!** Under time evolution, the state $|n(t)\rangle$ evolves as

$$|n(t)\rangle = e^{-i\hat{H}t/\hbar} |n\rangle = e^{-iE_n t/\hbar} |n\rangle. \quad (3.5.1)$$

Let us calculate the expectation value of position as a function of time:

$$\langle \hat{x}(t) \rangle_n = \langle n(t) | \hat{x} | n(t) \rangle = e^{iE_n t/\hbar} \langle n | \hat{x} | n \rangle e^{-iE_n t/\hbar} = \langle n | \hat{x} | n \rangle = 0 \quad \text{for all } t! \quad (3.5.2)$$

Similarly, the momentum expectation value is permanently zero:

$$\langle \hat{p}(t) \rangle_n = \langle n | \hat{p} | n \rangle = 0 \quad \text{for all } t! \quad (3.5.3)$$

Even more striking is the spatial probability density:

$$P_n(x, t) = |\psi_n(x, t)|^2 = |\psi_n(x) e^{-iE_n t/\hbar}|^2 = |\psi_n(x)|^2. \quad (3.5.4)$$

The probability density $P_n(x, t)$ is strictly static. The quantum cloud does not move, breathe, or oscillate. So what is oscillating in the quantum harmonic oscillator?

The answer is that in a single stationary state $|n\rangle$, only the unobservable *quantum phase* oscillates:

$$e^{-i\omega_n t} = \exp \left[-i \left(n + \frac{1}{2} \right) \omega t \right]. \quad (3.5.5)$$

Because quantum probabilities depend only on the absolute square of the wavefunction, a global phase factor cancels out completely. In a pure energy eigenstate, the system does not exhibit any classical macroscopic oscillation whatsoever.

3.5.2 Superposition and Physical Spatial Oscillation

To recover physical oscillation in space, we must create a **superposition of different energy eigenstates**. Observable dynamics in quantum mechanics is always driven by interference between distinct energy levels!

Consider the simplest non-trivial superposition: an equal mixture of the ground state $|0\rangle$ and the first excited state $|1\rangle$:

$$|\Psi(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \quad (3.5.6)$$

According to the time-dependent Schrödinger equation, this state evolves as

$$\begin{aligned}
 |\Psi(t)\rangle &= \frac{1}{\sqrt{2}} \left(e^{-iE_0t/\hbar} |0\rangle + e^{-iE_1t/\hbar} |1\rangle \right) \\
 &= \frac{1}{\sqrt{2}} \left(e^{-i\omega t/2} |0\rangle + e^{-3i\omega t/2} |1\rangle \right) \\
 &= \frac{e^{-i\omega t/2}}{\sqrt{2}} \left(|0\rangle + e^{-i\omega t} |1\rangle \right).
 \end{aligned} \tag{3.5.7}$$

Notice what has happened: the relative phase between $|0\rangle$ and $|1\rangle$ precesses in time with angular frequency

$$\Delta\omega = \frac{E_1 - E_0}{\hbar} = \frac{\frac{3}{2}\hbar\omega - \frac{1}{2}\hbar\omega}{\hbar} = \omega. \tag{3.5.8}$$

This is the Bohr frequency condition. Let us now compute the expectation value of position in this superposition:

$$\begin{aligned}
 \langle \hat{x}(t) \rangle &= \langle \Psi(t) | \hat{x} | \Psi(t) \rangle \\
 &= \frac{1}{2} \left(\langle 0 | + e^{i\omega t} \langle 1 | \right) \hat{x} \left(|0\rangle + e^{-i\omega t} |1\rangle \right) \\
 &= \frac{1}{2} \left[\langle 0 | \hat{x} | 0 \rangle + \langle 1 | \hat{x} | 1 \rangle + e^{-i\omega t} \langle 0 | \hat{x} | 1 \rangle + e^{i\omega t} \langle 1 | \hat{x} | 0 \rangle \right].
 \end{aligned} \tag{3.5.9}$$

Recalling Eq. (3.4.3), the diagonal elements vanish, while the off-diagonal elements are:

$$\langle 0 | \hat{x} | 1 \rangle = \langle 1 | \hat{x} | 0 \rangle = \sqrt{\frac{\hbar}{2m\omega}} = \frac{x_0}{\sqrt{2}}. \tag{3.5.10}$$

Substituting these into our expectation value yields:

$$\langle \hat{x}(t) \rangle = \frac{1}{2} \left(e^{-i\omega t} + e^{i\omega t} \right) \sqrt{\frac{\hbar}{2m\omega}} = \sqrt{\frac{\hbar}{2m\omega}} \cos(\omega t). \tag{3.5.11}$$

The center of mass of the wavefunction now physically oscillates back and forth at the classical frequency ω !

Similarly, computing the momentum expectation value using Eq. (3.4.4):

$$\begin{aligned}
 \langle \hat{p}(t) \rangle &= \frac{1}{2} \left[e^{-i\omega t} \langle 0 | \hat{p} | 1 \rangle + e^{i\omega t} \langle 1 | \hat{p} | 0 \rangle \right] \\
 &= \frac{1}{2} \left[e^{-i\omega t} \left(-i\sqrt{\frac{m\hbar\omega}{2}} \right) + e^{i\omega t} \left(i\sqrt{\frac{m\hbar\omega}{2}} \right) \right] \\
 &= -\sqrt{\frac{m\hbar\omega}{2}} \sin(\omega t) = m \frac{d\langle \hat{x} \rangle}{dt}.
 \end{aligned} \tag{3.5.12}$$

Ehrenfest's theorem is verified: the expectation values obey Newton's second law!

To see the physical motion directly in space, let us write out the spatial probability density $P(x, t) = |\langle x | \Psi(t) \rangle|^2$:

$$\begin{aligned}
 P(x, t) &= \left| \frac{1}{\sqrt{2}} \psi_0(x) + \frac{1}{\sqrt{2}} e^{-i\omega t} \psi_1(x) \right|^2 \\
 &= \frac{1}{2} \psi_0^2(x) + \frac{1}{2} \psi_1^2(x) + \psi_0(x) \psi_1(x) \cos(\omega t).
 \end{aligned} \tag{3.5.13}$$

Recall that $\psi_0(x)$ is a symmetric (even) function, while $\psi_1(x) \propto x\psi_0(x)$ is an antisymmetric (odd) function. Therefore, the cross-term $\psi_0(x)\psi_1(x)$ is an odd function of x :

- At $t = 0$: $\cos(\omega t) = 1$. The cross-term adds constructively for $x > 0$ and destructively for $x < 0$. The probability density is shifted to the right.
- At $t = \frac{\pi}{2\omega}$: $\cos(\omega t) = 0$. The cross-term vanishes. The probability density is symmetric.
- At $t = \frac{\pi}{\omega}$: $\cos(\omega t) = -1$. The cross-term adds constructively for $x < 0$ and destructively for $x > 0$. The probability density is shifted to the left.

The quantum probability fluid physically sloshes back and forth across the origin! Thus, **what is oscillating is the quantum interference cross-term between different energy eigenstates.**

3.5.3 Heisenberg Picture Perspective

This conclusion is illuminated even further in the Heisenberg picture, where state vectors are stationary and operators carry the time dependence. The Heisenberg equations of motion for $\hat{x}_H(t)$ and $\hat{p}_H(t)$ are:

$$\frac{d\hat{x}_H}{dt} = \frac{1}{i\hbar}[\hat{x}_H, \hat{H}] = \frac{\hat{p}_H}{m}, \quad (3.5.14)$$

$$\frac{d\hat{p}_H}{dt} = \frac{1}{i\hbar}[\hat{p}_H, \hat{H}] = -m\omega^2 \hat{x}_H. \quad (3.5.15)$$

Differentiating once more yields the uncoupled second-order operator equations:

$$\frac{d^2\hat{x}_H}{dt^2} + \omega^2 \hat{x}_H = 0, \quad \frac{d^2\hat{p}_H}{dt^2} + \omega^2 \hat{p}_H = 0. \quad (3.5.16)$$

The exact solutions are operator harmonic oscillations identical in form to classical mechanics:

$$\hat{x}_H(t) = \hat{x}(0) \cos(\omega t) + \frac{\hat{p}(0)}{m\omega} \sin(\omega t), \quad (3.5.17)$$

$$\hat{p}_H(t) = \hat{p}(0) \cos(\omega t) - m\omega \hat{x}(0) \sin(\omega t). \quad (3.5.18)$$

Equivalently, for the ladder operator $a_H(t) = e^{i\hat{H}t/\hbar} a e^{-i\hat{H}t/\hbar}$:

$$\frac{da_H}{dt} = \frac{1}{i\hbar}[a_H, \hat{H}] = -i\omega a_H \implies a_H(t) = a(0)e^{-i\omega t}. \quad (3.5.19)$$

Taking the expectation value in any state $|\Psi\rangle$:

$$\langle \hat{x}(t) \rangle = \langle \hat{x}(0) \rangle \cos(\omega t) + \frac{\langle \hat{p}(0) \rangle}{m\omega} \sin(\omega t). \quad (3.5.20)$$

This makes the situation completely clear: the operator itself always oscillates. However, the expectation value $\langle \hat{x}(t) \rangle$ oscillates if and only if either $\langle \hat{x}(0) \rangle \neq 0$ or $\langle \hat{p}(0) \rangle \neq 0$. For any energy eigenstate $|n\rangle$, both initial expectation values vanish, so the oscillation amplitude is zero. Only states with non-vanishing off-diagonal coherence in the energy basis display non-trivial classical motion.

3.6 Coherent States (Glauber States)

3.6.1 Motivation and Definition

While the simple superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ demonstrates oscillation, its probability distribution changes shape as it sloshes, and its oscillation amplitude is locked to the microscopic quantum

scale $x_0/\sqrt{2}$. How can quantum mechanics describe a macroscopic pendulum oscillating with arbitrary amplitude X_0 without deforming or dispersing?

The answer was discovered by Erwin Schrödinger in 1926 and developed into modern quantum optics by Roy Glauber in the 1960s: the **coherent states** (or **Glauber states**) $|\alpha\rangle$. Coherent states are defined as the eigenstates of the non-Hermitian annihilation operator a :

$$a|\alpha\rangle = \alpha|\alpha\rangle, \quad \alpha \in \mathbb{C}. \quad (3.6.1)$$

Because a is not Hermitian, its eigenvalue α is an arbitrary complex number. (Note that the creation operator $a^\dagger$ has no normalizable eigenstates, since $a^\dagger|\psi\rangle = \lambda|\psi\rangle$ leads to divergent coefficients).

3.6.2 Expansion in the Fock Basis and Poisson Statistics

To find the explicit form of $|\alpha\rangle$ in the Fock basis, we expand:

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle. \quad (3.6.2)$$

Applying the annihilation operator to both sides using Eq. (3.3.17):

$$a|\alpha\rangle = \sum_{n=1}^{\infty} c_n \sqrt{n} |n-1\rangle = \sum_{m=0}^{\infty} c_{m+1} \sqrt{m+1} |m\rangle. \quad (3.6.3)$$

Comparing this to $\alpha|\alpha\rangle = \alpha \sum_{m=0}^{\infty} c_m |m\rangle$, we match coefficients:

$$c_{m+1} \sqrt{m+1} = \alpha c_m \implies c_n = \frac{\alpha}{\sqrt{n}} c_{n-1} = \frac{\alpha^2}{\sqrt{n(n-1)}} c_{n-2} = \cdots = \frac{\alpha^n}{\sqrt{n!}} c_0. \quad (3.6.4)$$

The state vector is therefore

$$|\alpha\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (3.6.5)$$

To determine c_0 , we enforce the normalization condition $\langle\alpha|\alpha\rangle = 1$:

$$\langle\alpha|\alpha\rangle = |c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} = |c_0|^2 e^{|\alpha|^2} = 1 \implies c_0 = e^{-|\alpha|^2/2}, \quad (3.6.6)$$

where we have chosen the global phase of c_0 to be real and positive.

Thus, the expansion of a coherent state in the Fock basis is:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (3.6.7)$$

The probability $P(n)$ of measuring n energy quanta (photons/phonons) in a coherent state is

$$P(n) = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} = \frac{\bar{n}^n e^{-\bar{n}}}{n!}, \quad (3.6.8)$$

where $\bar{n} \equiv \langle\hat{N}\rangle = |\alpha|^2$ is the mean number of quanta. This is precisely a **Poisson distribution**! The variance in the number of quanta is

$$(\Delta n)^2 = \langle\hat{N}^2\rangle - \langle\hat{N}\rangle^2 = \bar{n} = |\alpha|^2. \quad (3.6.9)$$

The relative uncertainty in energy decreases inversely with the amplitude:

$$\frac{\Delta E}{\langle E\rangle} \approx \frac{\Delta n}{\bar{n}} = \frac{1}{\sqrt{\bar{n}}} = \frac{1}{|\alpha|} \xrightarrow{|\alpha| \rightarrow \infty} 0. \quad (3.6.10)$$

In the macroscopic limit ($|\alpha|^2 \gg 1$), energy fluctuations become negligible, recovering classical mechanics!

3.6.3 The Displacement Operator

Coherent states can be generated directly from the ground state $|0\rangle$ by the action of the unitary **displacement operator** $D(\alpha)$:

$$D(\alpha) \equiv \exp(\alpha a^\dagger - \alpha^* a). \quad (3.6.11)$$

Notice that the exponent is anti-Hermitian: $(\alpha a^\dagger - \alpha^* a)^\dagger = -(\alpha a^\dagger - \alpha^* a)$, which guarantees that $D(\alpha)$ is unitary:

$$D^\dagger(\alpha) = D(-\alpha) = D(\alpha)^{-1}. \quad (3.6.12)$$

To evaluate the action of $D(\alpha)$, we use the Baker-Campbell-Hausdorff (BCH) formula. For any two operators A and B whose commutator $[A, B]$ commutes with both A and B ($[A, [A, B]] = [B, [A, B]] = 0$):

$$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A, B]}. \quad (3.6.13)$$

Setting $A = \alpha a^\dagger$ and $B = -\alpha^* a$, we compute their commutator:

$$[A, B] = [\alpha a^\dagger, -\alpha^* a] = -|\alpha|^2 [a^\dagger, a] = |\alpha|^2 [a, a^\dagger] = |\alpha|^2. \quad (3.6.14)$$

Since $|\alpha|^2$ is a c-number, Eq. (3.6.13) applies exactly:

$$D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a) = e^{-|\alpha|^2/2} \exp(\alpha a^\dagger) \exp(-\alpha^* a). \quad (3.6.15)$$

Now let this factored operator act on the vacuum state $|0\rangle$. Expanding the exponential $\exp(-\alpha^* a)$:

$$\exp(-\alpha^* a) |0\rangle = \left(1 - \alpha^* a + \frac{(\alpha^*)^2}{2!} a^2 - \dots \right) |0\rangle = |0\rangle, \quad (3.6.16)$$

since $a|0\rangle = 0$. Therefore:

$$\begin{aligned} D(\alpha) |0\rangle &= e^{-|\alpha|^2/2} \exp(\alpha a^\dagger) |0\rangle \\ &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n (a^\dagger)^n}{n!} |0\rangle \\ &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \end{aligned} \quad (3.6.17)$$

Comparing with Eq. (3.6.7), we obtain the fundamental identity:

$$|\alpha\rangle = D(\alpha) |0\rangle. \quad (3.6.18)$$

Furthermore, the displacement operator translates the annihilation and creation operators in phase space:

$$D^\dagger(\alpha) a D(\alpha) = a + \alpha, \quad (3.6.19)$$

$$D^\dagger(\alpha) a^\dagger D(\alpha) = a^\dagger + \alpha^*. \quad (3.6.20)$$

To connect this to spatial and momentum translations, let us write the complex parameter α in terms of classical phase space coordinates (X, P) :

$$\alpha = \frac{1}{\sqrt{2}} \left(\frac{X}{x_0} + i \frac{P}{p_0} \right) = \sqrt{\frac{m\omega}{2\hbar}} X + \frac{i}{\sqrt{2m\hbar\omega}} P. \quad (3.6.21)$$

Substituting this and Eqs. (3.2.3)–(3.2.4) into $D(\alpha)$:

$$\alpha a^\dagger - \alpha^* a = \frac{i}{\hbar} (P\hat{x} - X\hat{p}). \quad (3.6.22)$$

Hence, the displacement operator is nothing other than the unitary phase-space translation operator:

$$D(\alpha) = \exp \left[\frac{i}{\hbar} (P\hat{x} - X\hat{p}) \right]. \quad (3.6.23)$$

A coherent state $|\alpha\rangle$ is physically constructed by translating the ground state wave packet in phase space by displacement X and momentum kick P !

3.6.4 Time Evolution and Classical Trajectories

Let an initial coherent state $|\alpha(0)\rangle = |\alpha_0\rangle$ evolve under the harmonic oscillator Hamiltonian $\hat{H}$. Using the Fock expansion (3.6.7):

$$\begin{aligned} |\alpha(t)\rangle &= e^{-i\hat{H}t/\hbar} |\alpha_0\rangle \\ &= e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{-i\hat{H}t/\hbar} |n\rangle \\ &= e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{-i\omega(n+1/2)t} |n\rangle \\ &= e^{-i\omega t/2} e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha_0 e^{-i\omega t})^n}{\sqrt{n!}} |n\rangle. \end{aligned} \quad (3.6.24)$$

Since $|\alpha_0 e^{-i\omega t}|^2 = |\alpha_0|^2$, this is, up to an overall phase $e^{-i\omega t/2}$, precisely a coherent state with a time-dependent parameter:

$$|\alpha(t)\rangle = e^{-i\omega t/2} |\alpha(t)\rangle_{\text{coh}}, \quad \alpha(t) = \alpha_0 e^{-i\omega t}. \quad (3.6.25)$$

A coherent state remains an exact coherent state under time evolution for all time! In complex phase space, the eigenvalue $\alpha(t)$ simply traces out a circle of radius $|\alpha_0|$ clockwise with constant angular velocity ω :

$$\dot{\alpha}(t) = -i\omega\alpha(t). \quad (3.6.26)$$

Let us write $\alpha_0 = |\alpha_0|e^{i\phi_0}$, so that $\alpha(t) = |\alpha_0|e^{-i(\omega t - \phi_0)}$. Using Eq. (3.4.1) and $a|\alpha\rangle = \alpha|\alpha\rangle$:

$$\begin{aligned} \langle \hat{x}(t) \rangle_{\alpha} &= \langle \alpha(t) | \hat{x} | \alpha(t) \rangle = \sqrt{\frac{\hbar}{2m\omega}} (\alpha(t) + \alpha^*(t)) \\ &= \sqrt{\frac{2\hbar}{m\omega}} |\alpha_0| \cos(\omega t - \phi_0) \equiv X_0 \cos(\omega t - \phi_0). \end{aligned} \quad (3.6.27)$$

Similarly, from Eq. (3.4.2):

$$\begin{aligned} \langle \hat{p}(t) \rangle_{\alpha} &= \langle \alpha(t) | \hat{p} | \alpha(t) \rangle = -i\sqrt{\frac{m\hbar\omega}{2}} (\alpha(t) - \alpha^*(t)) \\ &= -\sqrt{2m\hbar\omega} |\alpha_0| \sin(\omega t - \phi_0) \equiv -m\omega X_0 \sin(\omega t - \phi_0). \end{aligned} \quad (3.6.28)$$

The expectation values $\langle \hat{x}(t) \rangle$ and $\langle \hat{p}(t) \rangle$ follow the exact classical equations of motion with classical oscillation amplitude $X_0 = \sqrt{\frac{2\hbar}{m\omega}} |\alpha_0|$.

3.6.5 Minimum Uncertainty Without Spreading

Finally, let us compute the quantum uncertainties of position and momentum in a coherent state. Using the displacement shift relations (3.6.19) and (3.6.20):

$$\langle \hat{x}^2 \rangle_{\alpha} = \frac{\hbar}{2m\omega} \langle \alpha | (a + a^\dagger)^2 | \alpha \rangle$$

$$\begin{aligned}
&= \frac{\hbar}{2m\omega} \langle 0 | D^\dagger(\alpha) (a + a^\dagger)^2 D(\alpha) | 0 \rangle \\
&= \frac{\hbar}{2m\omega} \langle 0 | (a + \alpha + a^\dagger + \alpha^*)^2 | 0 \rangle \\
&= \frac{\hbar}{2m\omega} \left[\langle 0 | (a + a^\dagger)^2 | 0 \rangle + (\alpha + \alpha^*)^2 \right] \\
&= \frac{\hbar}{2m\omega} + \langle \hat{x} \rangle_\alpha^2.
\end{aligned} \tag{3.6.29}$$

Therefore, the position variance is completely independent of α and constant in time:

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle_\alpha - \langle \hat{x} \rangle_\alpha^2 = \frac{\hbar}{2m\omega} \implies \Delta x = \sqrt{\frac{\hbar}{2m\omega}} = \frac{x_0}{\sqrt{2}}. \tag{3.6.30}$$

By an identical calculation for $\hat{p}$:

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle_\alpha - \langle \hat{p} \rangle_\alpha^2 = \frac{m\hbar\omega}{2} \implies \Delta p = \sqrt{\frac{m\hbar\omega}{2}} = \frac{p_0}{\sqrt{2}}. \tag{3.6.31}$$

Taking their product:

$$\Delta x \Delta p = \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}. \tag{3.6.32}$$

The coherent state maintains the absolute minimum uncertainty permitted by quantum mechanics for all time without spreading!

In coordinate space, the wavefunction of the coherent state is an un-deformed Gaussian packet:

$$\psi_\alpha(x, t) = \langle x | \alpha(t) \rangle = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \exp \left[-\frac{m\omega}{2\hbar} (x - \langle \hat{x}(t) \rangle)^2 + \frac{i}{\hbar} \langle \hat{p}(t) \rangle x + i\Phi(t) \right], \tag{3.6.33}$$

and its spatial probability density is:

$$|\psi_\alpha(x, t)|^2 = \sqrt{\frac{m\omega}{\pi\hbar}} \exp \left[-\frac{m\omega}{\hbar} (x - X_0 \cos(\omega t - \phi_0))^2 \right]. \tag{3.6.34}$$

Unlike a free particle wave packet, which inevitably disperses and spreads in space over time (as we will explore in the next chapter), the coherent state of the harmonic oscillator glides back and forth forever as a rigid, non-dispersive wave packet. In the coherent state, the quantum world and classical world meet in perfect harmony.

Chapter 4

Particle in a Box and One-Dimensional Systems

One-dimensional piecewise-constant potentials are among the most instructive models in quantum mechanics. Despite their elementary geometry, they reveal the core physical and mathematical properties that distinguish wave mechanics from Newtonian dynamics:

- **Spatial Confinement and Discrete Bound States:** Imposing Dirichlet boundary conditions on a finite spatial domain quantizes the permissible energy levels.
- **Continuous Momentum in Bounded Domains:** In hard-walled boxes, the momentum operator is symmetric but fails to be self-adjoint, giving rise to continuous momentum probability distributions with diffraction lobes rather than discrete eigenvalues.
- **Demystification of the “Classically Forbidden” Region:** Evanescent wave penetration is shown to be a property of wave curvature, resolving the paradox of negative kinetic energy by proving that energy eigenstates are not eigenstates of the kinetic energy operator.
- **Quantum Reflection and Barrier Tunneling:** Particles undergo partial reflection even when $E > V_0$ due to de Broglie wave-impedance mismatch, and traverse classically impenetrable barriers ($E < V_0$) with exponentially attenuated transmission.
- **Wave Packet Dispersion:** In free space, matter waves inevitably spread over time due to the non-linear quadratic dispersion relation $\omega(k) = \hbar k^2 / (2m)$.
- **Multi-Path Interference and Quantum Decoherence:** Quantum transition amplitudes add coherently, yielding interference patterns that disappear when path information is acquired through entanglement with a detector.

4.1 Spatial Confinement: The Infinite Square Well

4.1.1 The Dirichlet Box and Boundary Conditions

Consider a quantum particle of mass m confined to a one-dimensional interval of length L bounded by infinitely high potential walls:

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & x \leq 0 \text{ or } x \geq L. \end{cases} \quad (4.1.1)$$

Because the potential energy is infinite outside the interval $[0, L]$, the probability of finding the particle in the exterior region is strictly zero:

$$\psi(x) = 0 \quad \text{for } x \leq 0 \text{ and } x \geq L. \quad (4.1.2)$$

By continuity of the physical wavefunction, $\psi(x)$ must vanish at the boundaries, establishing **Dirichlet boundary conditions**:

$$\psi(0) = 0, \quad \psi(L) = 0. \quad (4.1.3)$$

Within the interior domain $0 < x < L$, $V(x) = 0$, and the time-independent Schrödinger equation reduces to the Helmholtz equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi(x) \implies \frac{d^2\psi}{dx^2} + k^2\psi(x) = 0, \quad k \equiv \frac{\sqrt{2mE}}{\hbar}. \quad (4.1.4)$$

The general solution is a linear combination of oscillatory functions:

$$\psi(x) = A \sin(kx) + B \cos(kx). \quad (4.1.5)$$

Enforcing the boundary conditions (4.1.3):

$$\psi(0) = 0 \implies B = 0, \quad (4.1.6)$$

$$\psi(L) = 0 \implies A \sin(kL) = 0. \quad (4.1.7)$$

To avoid the trivial, unnormalizable solution $\psi(x) \equiv 0$, the amplitude A must be non-zero, requiring the argument of the sine to be an integer multiple of π :

$$k_n L = n\pi \implies k_n = \frac{n\pi}{L}, \quad n \in \{1, 2, 3, \dots\}. \quad (4.1.8)$$

Here $n = 0$ is excluded because it yields $\psi(x) = 0$ everywhere, and negative integers $n < 0$ yield wavefunctions differing only by an unobservable global minus sign ($\sin(-x) = -\sin(x)$), representing the identical quantum state.

4.1.2 Energy Eigenvalues, Normalization, and Orthogonality

The quantization of the wavenumber k_n directly quantizes the allowable energy levels:

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots \quad (4.1.9)$$

Energy quantization in quantum mechanics does not emerge from an ad-hoc postulate; it is the direct mathematical consequence of spatial confinement and boundary matching on the differential operator.

Normalizing the eigenfunctions on the interval $[0, L]$:

$$\langle \psi_n | \psi_n \rangle = \int_0^L |\psi_n(x)|^2 dx = |A|^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = |A|^2 \frac{L}{2} = 1 \implies A = \sqrt{\frac{2}{L}}. \quad (4.1.10)$$

Hence, the orthonormal stationary eigenfunctions are:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad n \in \mathbb{N}^+. \quad (4.1.11)$$

These states form a complete orthonormal basis for the Hilbert space $L^2([0, L])$:

$$\langle \psi_m | \psi_n \rangle = \int_0^L \psi_m^*(x) \psi_n(x) dx = \delta_{mn} \quad (\text{Orthonormality}), \quad (4.1.12)$$

$$\sum_{n=1}^{\infty} \psi_n(x) \psi_n(x') = \frac{2}{L} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi x'}{L}\right) = \delta(x - x') \quad (\text{Completeness}). \quad (4.1.13)$$

4.1.3 Time Evolution and Quantum Revivals

Any arbitrary initial wavepacket $\Psi(x, 0) \in L^2([0, L])$ satisfying the Dirichlet boundary conditions can be uniquely expanded in the energy eigenbasis:

$$\Psi(x, 0) = \sum_{n=1}^{\infty} c_n \psi_n(x), \quad c_n = \langle \psi_n | \Psi(0) \rangle = \int_0^L \psi_n(x) \Psi(x, 0) dx. \quad (4.1.14)$$

According to the time-evolution postulate developed in Chapter 2, each eigenstate evolves with its characteristic phase factor $e^{-iE_n t/\hbar}$:

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) \exp\left(-i \frac{E_n t}{\hbar}\right) = \sum_{n=1}^{\infty} c_n \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \exp\left(-i \frac{n^2 \pi^2 \hbar}{2mL^2} t\right). \quad (4.1.15)$$

Because the energies scale quadratically with n ($E_n = n^2 E_1$), all phases become simultaneous integer multiples of 2π at the **revival time**:

$$T_{\text{rev}} \equiv \frac{2\pi\hbar}{E_1} = \frac{4mL^2}{\pi\hbar}. \quad (4.1.16)$$

At $t = T_{\text{rev}}$, the wavefunction reconstructs its exact initial profile: $\Psi(x, T_{\text{rev}}) = \Psi(x, 0)$. Furthermore, at fractional revival times such as $t = T_{\text{rev}}/2$ or $t = T_{\text{rev}}/4$, the wavepacket splits into discrete, spatially separated quantum mini-packets (Schrödinger cat states).

4.2 The Momentum Distribution in Confinement: Why Momentum is Not Quantized

4.2.1 The Fallacy of Quantized Momentum

A common misconception in introductory quantum physics asserts:

“Because $k_n = n\pi/L$, the particle in the state ψ_n has quantized momentum $p_n = \pm\hbar k_n = \pm \frac{n\pi\hbar}{L}$.”

This statement is completely false!

To verify this, we apply the momentum operator $\hat{p} = -i\hbar \frac{d}{dx}$ directly to the energy eigenstate $\psi_n(x)$:

$$\hat{p}\psi_n(x) = -i\hbar \sqrt{\frac{2}{L}} \frac{d}{dx} \sin\left(\frac{n\pi x}{L}\right) = -i\hbar \sqrt{\frac{2}{L}} \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi x}{L}\right) \neq \lambda \psi_n(x). \quad (4.2.1)$$

The derivative of a sine is a cosine, which is linearly independent of the sine. Therefore, $\psi_n(x)$ is **not** an eigenstate of momentum. A true momentum eigenstate on the real line is a complex exponential $e^{ipx/\hbar}$, which cannot satisfy Dirichlet boundary conditions $\psi(0) = \psi(L) = 0$ without vanishing identically.

If a measurement of momentum is performed on a particle in state $\psi_n(x)$, what outcomes can occur? As Richard Robinett demonstrated [3, 2], momentum measurement outcomes form a **continuous probability distribution** across the entire real line $\mathbb{R}$!

4.2.2 Derivation of the Momentum-Space Wavefunction

The momentum-space wavefunction $\phi_n(p) \equiv \langle p | \psi_n \rangle$ is the projection of the state $|\psi_n\rangle$ onto the momentum basis $\langle p|$:

$$\phi_n(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi_n(x) e^{-ipx/\hbar} dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2}{L}} \int_0^L \sin\left(\frac{n\pi x}{L}\right) e^{-ipx/\hbar} dx. \quad (4.2.2)$$

Using Euler's identity $\sin(k_n x) = \frac{e^{ik_n x} - e^{-ik_n x}}{2i}$ with $k_n = \frac{n\pi}{L}$:

$$\phi_n(p) = \frac{1}{\sqrt{\pi\hbar L}} \frac{1}{2i} \left[\int_0^L e^{i(k_n - p/\hbar)x} dx - \int_0^L e^{-i(k_n + p/\hbar)x} dx \right]. \quad (4.2.3)$$

Evaluating each elementary definite integral:

$$\int_0^L e^{i(k_n - p/\hbar)x} dx = \left[\frac{e^{i(k_n - p/\hbar)x}}{i(k_n - p/\hbar)} \right]_0^L = \frac{e^{ik_n L} e^{-ipL/\hbar} - 1}{i(k_n - p/\hbar)} = \frac{(-1)^n e^{-ipL/\hbar} - 1}{i(k_n - p/\hbar)}, \quad (4.2.4)$$

$$\int_0^L e^{-i(k_n + p/\hbar)x} dx = \left[\frac{e^{-i(k_n + p/\hbar)x}}{-i(k_n + p/\hbar)} \right]_0^L = \frac{(-1)^n e^{-ipL/\hbar} - 1}{-i(k_n + p/\hbar)}, \quad (4.2.5)$$

where we have used $e^{ik_n L} = e^{in\pi} = (-1)^n$ and $e^{-ik_n L} = e^{-in\pi} = (-1)^n$. Factoring out the common numerator:

$$\begin{aligned} \phi_n(p) &= \frac{1}{\sqrt{\pi\hbar L}} \frac{(-1)^n e^{-ipL/\hbar} - 1}{2i} \left[\frac{1}{i(k_n - p/\hbar)} - \frac{1}{-i(k_n + p/\hbar)} \right] \\ &= -\frac{1}{2\sqrt{\pi\hbar L}} \left[(-1)^n e^{-ipL/\hbar} - 1 \right] \left[\frac{1}{k_n - p/\hbar} + \frac{1}{k_n + p/\hbar} \right] \\ &= -\frac{1}{2\sqrt{\pi\hbar L}} \left[(-1)^n e^{-ipL/\hbar} - 1 \right] \frac{2k_n}{k_n^2 - (p/\hbar)^2} \\ &= \frac{1}{\sqrt{\pi\hbar L}} \left[1 - (-1)^n e^{-ipL/\hbar} \right] \frac{k_n}{k_n^2 - (p/\hbar)^2}. \end{aligned} \quad (4.2.6)$$

Multiplying numerator and denominator by $\hbar^2$ and defining $p_n \equiv \hbar k_n = \frac{n\pi\hbar}{L}$:

$$\phi_n(p) = \sqrt{\frac{\hbar}{\pi L}} \frac{p_n}{p_n^2 - p^2} \left[1 - (-1)^n e^{-ipL/\hbar} \right]. \quad (4.2.7)$$

4.2.3 Momentum Probability Density

To compute the momentum probability density $P_n(p) = |\phi_n(p)|^2$, we factor the global phase $e^{-ipL/(2\hbar)}$ from the bracketed factor:

$$1 - (-1)^n e^{-ipL/\hbar} = e^{-ipL/(2\hbar)} \left[e^{ipL/(2\hbar)} - (-1)^n e^{-ipL/(2\hbar)} \right]. \quad (4.2.8)$$

This evaluates differently depending on the parity of n :

- **Odd n** ($(-1)^n = -1$):

$$e^{ipL/(2\hbar)} + e^{-ipL/(2\hbar)} = 2 \cos\left(\frac{pL}{2\hbar}\right). \quad (4.2.9)$$

Therefore:

$$P_n(p) = |\phi_n(p)|^2 = \frac{4\hbar}{\pi L} \frac{p_n^2}{(p^2 - p_n^2)^2} \cos^2\left(\frac{pL}{2\hbar}\right). \quad (4.2.10)$$

- **Even n** ($(-1)^n = +1$):

$$e^{ipL/(2\hbar)} - e^{-ipL/(2\hbar)} = 2i \sin\left(\frac{pL}{2\hbar}\right). \quad (4.2.11)$$

Therefore:

$$P_n(p) = |\phi_n(p)|^2 = \frac{4\hbar}{\pi L} \frac{p_n^2}{(p^2 - p_n^2)^2} \sin^2\left(\frac{pL}{2\hbar}\right). \quad (4.2.12)$$

4.2.4 Physical Anatomy of the Momentum Distribution

Let us analyze the key physical features of $P_n(p)$:

1. **Continuous Spectrum:** $P_n(p)$ is non-zero almost everywhere on $\mathbb{R}$. A momentum measurement on a particle in a box can yield any real value!
2. **Removable Singularity at $p = \pm p_n$:** The apparent divergence at $p = \pm p_n$ is removable. Taking the limit $p \rightarrow p_n$ by setting $p = p_n + \epsilon$: For n odd:

$$\cos\left(\frac{pL}{2\hbar}\right) = \cos\left(\frac{n\pi}{2} + \frac{\epsilon L}{2\hbar}\right) = -(-1)^{(n-1)/2} \sin\left(\frac{\epsilon L}{2\hbar}\right) \approx -(-1)^{(n-1)/2} \frac{\epsilon L}{2\hbar}, \quad (4.2.13)$$

while $p^2 - p_n^2 \approx 2p_n\epsilon$. Substituting these into Eq. (4.2.10):

$$\lim_{p \rightarrow \pm p_n} P_n(p) = \frac{4\hbar}{\pi L} \frac{p_n^2}{(2p_n\epsilon)^2} \left(\frac{\epsilon L}{2\hbar}\right)^2 = \frac{L}{2\pi\hbar}. \quad (4.2.14)$$

An identical limit holds for even n :

$$P_n(\pm p_n) = \frac{L}{2\pi\hbar}. \quad (4.2.15)$$

3. **Diffraction Lobes:** The distribution exhibits two primary peaks centered near $p \approx \pm p_n$ and infinite decaying side-lobes that fall off asymptotically as $1/p^2$ as $|p| \rightarrow \infty$. This is the quantum momentum analog of optical single-slit Fraunhofer diffraction!
4. **Statistical Moments:** Because $P_n(p) = P_n(-p)$ is strictly symmetric:

$$\langle \hat{p} \rangle_n = \int_{-\infty}^{\infty} p P_n(p) dp = 0. \quad (4.2.16)$$

Computing the second moment:

$$\langle \hat{p}^2 \rangle_n = \int_{-\infty}^{\infty} p^2 P_n(p) dp = p_n^2 = \frac{n^2 \pi^2 \hbar^2}{L^2}, \quad (4.2.17)$$

which ensures that the average kinetic energy matches the eigenvalue: $\langle \hat{T} \rangle = \frac{\langle \hat{p}^2 \rangle}{2m} = \frac{p_n^2}{2m} = E_n$.

4.2.5 Robinett's Theorem: Symmetry vs. Self-Adjointness

Why is momentum not quantized into discrete eigenvalues for a particle in a box? The answer is rooted in the mathematical distinction between a **symmetric operator** and a **self-adjoint operator** [3, 2].

In quantum mechanics, physical observables must be represented by self-adjoint operators, because only self-adjoint operators possess real eigenvalues and a complete orthonormal eigenbasis (the Spectral Decomposition Theorem).

Consider the Hilbert space $\mathcal{H} = L^2([0, L])$ and the differential operator $\hat{p} = -i\hbar \frac{d}{dx}$. For any two smooth functions ϕ, ψ , integration by parts yields:

$$\begin{aligned} \langle \phi | \hat{p} | \psi \rangle - \langle \hat{p} \phi | \psi \rangle &= \int_0^L \phi^*(x) \left(-i\hbar \frac{d\psi}{dx} \right) dx - \int_0^L \left(-i\hbar \frac{d\phi}{dx} \right)^* \psi(x) dx \\ &= -i\hbar \int_0^L [\phi^*(x) \psi'(x) + \phi'^*(x) \psi(x)] dx \\ &= -i\hbar [\phi^*(L) \psi(L) - \phi^*(0) \psi(0)]. \end{aligned} \quad (4.2.18)$$

For the particle in a box, the physical domain of wavefunctions is constrained by Dirichlet boundary conditions:

$$\mathcal{D}(\hat{p}) = \left\{ \psi \in H^1([0, L]) : \psi(0) = \psi(L) = 0 \right\}. \quad (4.2.19)$$

Under these boundary conditions, the boundary term in Eq. (4.2.18) identically vanishes for **any** test function $\phi \in H^1([0, L])$, regardless of whether ϕ vanishes at the boundaries! Consequently, the domain of the adjoint operator $\hat{p}^\dagger$ is:

$$\mathcal{D}(\hat{p}^\dagger) = H^1([0, L]) \supsetneq \mathcal{D}(\hat{p}). \quad (4.2.20)$$

Because the domain of $\hat{p}$ is strictly smaller than the domain of its adjoint, **$\hat{p}$ is symmetric, but NOT self-adjoint on the Dirichlet interval!**

According to the von Neumann theory of deficiency indices, $\hat{p}$ on $[0, L]$ has deficiency indices $(1, 1)$. To extend $\hat{p}$ to a self-adjoint operator, one must impose boundary conditions that force the boundary term in Eq. (4.2.18) to vanish for both ψ and ϕ :

$$\phi^*(L)\psi(L) - \phi^*(0)\psi(0) = 0 \implies \psi(L) = e^{i\theta}\psi(0), \quad \theta \in [0, 2\pi). \quad (4.2.21)$$

These are **periodic or twisted boundary conditions**. They describe a particle living on a topological circle S^1 threaded by an Aharonov-Bohm magnetic flux, **not** a particle confined between impenetrable hard walls!

Physical Conclusion: In an infinite square well, the hard walls at $x = 0$ and $x = L$ violently break continuous spatial translation symmetry. Because $\hat{p}$ is the generator of translations, momentum cannot be a conserved quantity (Noether's theorem). During wall collisions, the walls impart impulsive momentum kicks of arbitrary magnitude. Because $\hat{p}$ is not self-adjoint on the Dirichlet domain, it does not commute with the Hamiltonian, nor does it share an eigenbasis with $\hat{H}$.

4.3 The Finite Square Well and the Fallacy of “Negative Kinetic Energy”

4.3.1 Piecewise Constant Potentials and Bound States

We now examine the finite rectangular well of depth $V_0 > 0$ and width $2a$:

$$V(x) = \begin{cases} 0, & |x| > a \quad (\text{Exterior regions}), \\ -V_0, & |x| \leq a \quad (\text{Interior well}). \end{cases} \quad (4.3.1)$$

Equivalently, shifting the energy zero so that the potential is zero inside and V_0 outside:

$$V(x) = \begin{cases} V_0, & |x| > a, \\ 0, & |x| \leq a. \end{cases} \quad (4.3.2)$$

For bound states, the total energy satisfies $0 < E < V_0$ (or $-V_0 < E < 0$ in the negative-depth convention). In the interior piecewise-constant region ($|x| \leq a$), the time-independent Schrödinger equation is:

$$\phi'' + k^2\phi = 0, \quad k \equiv \frac{\sqrt{2mE}}{\hbar}. \quad (4.3.3)$$

In the exterior classically forbidden region ($|x| > a$), the equation is:

$$\phi'' - \kappa^2\phi = 0, \quad \kappa \equiv \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \quad (4.3.4)$$

Crucially, for *any* bound state in this potential, the wavenumbers satisfy:

$$k^2 + \kappa^2 = \frac{2mE}{\hbar^2} + \frac{2m(V_0 - E)}{\hbar^2} = \frac{2mV_0}{\hbar^2} \equiv \lambda \quad (\text{a constant independent of energy level}). \quad (4.3.5)$$

4.3.2 Parity Decomposition and Boundary Conditions

Because the potential $V(x) = V(-x)$ is symmetric under spatial inversion, the parity operator $\hat{\Pi}$ commutes with the Hamiltonian: $[\hat{H}, \hat{\Pi}] = 0$. Consequently, non-degenerate bound states must possess definite parity:

$$\psi(-x) = \pm\psi(x). \quad (4.3.6)$$

The continuous, normalizable solutions across the three regions are:

$$\psi_{\text{even}}(x) = \begin{cases} A \cos(ka) e^{\kappa(x+a)}, & x < -a, \\ A \cos(kx), & -a \leq x \leq a, \\ A \cos(ka) e^{-\kappa(x-a)}, & x > a, \end{cases} \quad (4.3.7)$$

$$\psi_{\text{odd}}(x) = \begin{cases} -B \sin(ka) e^{\kappa(x+a)}, & x < -a, \\ B \sin(kx), & -a \leq x \leq a, \\ B \sin(ka) e^{-\kappa(x-a)}, & x > a. \end{cases} \quad (4.3.8)$$

Matching the logarithmic derivative $\psi'(x)/\psi(x)$ at $x = a$ yields the transcendental quantization conditions:

$$\text{Even states: } \kappa = k \tan(ka), \quad (4.3.9)$$

$$\text{Odd states: } \kappa = -k \cot(ka). \quad (4.3.10)$$

4.3.3 Demystifying the “Classically Forbidden” Region

In classical mechanics, the total energy of a particle is $E = T(x) + V(x) = \frac{p(x)^2}{2m} + V(x)$. In any region where $V(x) > E$, the classical kinetic energy would be:

$$T(x) = E - V(x) < 0 \implies p(x) = \sqrt{2m(E - V(x))} \in i\mathbb{R}. \quad (4.3.11)$$

An imaginary momentum cannot describe classical motion; hence, such regions are termed “classically forbidden.”

The Flawed Uncertainty Principle Heuristic. A common attempt to rationalize quantum penetration into this region appeals to Heisenberg’s uncertainty principle: it is claimed that the particle “borrows” energy $\Delta E \sim \hbar/\Delta t$ from the vacuum, or that localizing the particle within a barrier of width a imparts a momentum uncertainty $\Delta p \sim \hbar/a$, producing an energy disturbance:

$$\Delta E \sim \frac{(\Delta p)^2}{2m} \sim \frac{\hbar^2}{2ma^2} \sim V_0 - E, \quad (4.3.12)$$

which supposedly lifts the particle above the barrier during measurement, “preventing one from ever catching it with negative kinetic energy.”

Why This Heuristic Is Conceptually Wrong: This popular explanation is misleading and physically ungrounded for three fundamental reasons:

1. **$\hat{T}$ is a positive semi-definite operator:** In quantum mechanics, observables are operators, not classical variables. The kinetic energy operator $\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ has a purely non-negative spectrum: $\sigma(\hat{T}) = [0, \infty)$. For *every* physical state $|\psi\rangle \in \mathcal{H}$:

$$\langle \hat{T} \rangle = \langle \psi | \hat{T} | \psi \rangle = \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \left| \frac{d\psi}{dx} \right|^2 dx \geq 0. \quad (4.3.13)$$

No state in Hilbert space has negative kinetic energy, and no measurement of $\hat{T}$ can ever yield a negative outcome. There is no negative kinetic energy to begin with, so no measurement disturbance is needed to “rescue” the particle!

2. **Kinetic energy is a global, non-local operator:** There is no such observable as “kinetic energy at position x .” The classical relation $T(x) = E - V(x)$ assumes that position and momentum are simultaneous numbers, violating $[\hat{x}, \hat{p}] = i\hbar$. Kinetic energy is a differential operator acting across the entire wavefunction.

3. **Wave Curvature vs. Local Energy:** The Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x) \iff \frac{\psi''(x)}{\psi(x)} = \frac{2m}{\hbar^2} (V(x) - E), \quad (4.3.14)$$

is a second-order linear differential equation governing the spatial curvature of the wavefunction. The factor $V(x) - E$ merely sets the sign of this curvature:

- Where $E > V(x)$, $\psi''/\psi < 0$: $\psi(x)$ curves toward the axis, producing oscillatory solutions.
- Where $E < V(x)$, $\psi''/\psi > 0$: $\psi(x)$ curves away from the axis, producing evanescent (exponentially decaying) solutions.

This is mathematically identical to total internal reflection in classical wave optics, where an evanescent electromagnetic wave penetrates into a lower-index dielectric without requiring “negative photon energy.”

4.3.4 The Non-Eigenstate Resolution

The fundamental mathematical truth resolving the paradox is that **an energy eigenstate $|\psi_n\rangle$ is NOT an eigenstate of the kinetic energy operator $\hat{T}$.**

Because the commutator of kinetic and potential energy:

$$[\hat{T}, \hat{V}] = -\frac{\hbar^2}{2m} \left[\frac{d^2}{dx^2}, V(x) \right] = -\frac{\hbar^2}{2m} \left(V''(x) + 2V'(x) \frac{d}{dx} \right) \neq 0 \quad (4.3.15)$$

does not vanish for any inhomogeneous potential, the total Hamiltonian $\hat{H} = \hat{T} + \hat{V}$ and the kinetic energy $\hat{T}$ cannot be simultaneously diagonalized.

In an energy eigenstate $|\psi_n\rangle$:

- The total energy E_n is a sharp eigenvalue of $\hat{H}$: $\Delta H = 0$.
- The kinetic energy $\hat{T}$ is in a quantum dispersion across an entire continuum of momentum states: $\Delta T = \sqrt{\langle \hat{T}^2 \rangle - \langle \hat{T} \rangle^2} > 0$.

Kinetic energy does not have a single numerical value in a bound state.

4.3.5 Rigorous Proof via Off-Diagonal Matrix Elements

We now provide an exact, first-principles proof of this non-eigenstate property by computing the matrix elements of $\hat{H}$ and $\hat{T}$ between different bound states $|\psi_m\rangle$ and $|\psi_n\rangle$ ($m \neq n$) of the finite square well.

Because the system is parity-invariant, matrix elements between opposite-parity states vanish automatically. We therefore consider two distinct odd-parity bound states $\psi_m(x)$ and $\psi_n(x)$ with energies $E_m \neq E_n$, corresponding to distinct parameters (k_m, κ_m) and (k_n, κ_n) . Let $s_i \equiv \sin(k_i a)$ and $c_i \equiv \cos(k_i a)$. From the odd boundary condition (4.3.10):

$$\kappa_i = -k_i \cot(k_i a) \implies c_i = -\frac{\kappa_i}{k_i} s_i. \quad (4.3.16)$$

1. Vanishing of Hamiltonian Off-Diagonal Elements: $\langle \psi_m | \hat{H} | \psi_n \rangle = 0$

Since $\hat{H} | \psi_n \rangle = E_n | \psi_n \rangle$, we have:

$$\langle \psi_m | \hat{H} | \psi_n \rangle = E_n \langle \psi_m | \psi_n \rangle. \quad (4.3.17)$$

The overlap integral decomposes into interior ($|x| < a$) and exterior ($|x| > a$) contributions:

$$\langle \psi_m | \psi_n \rangle = S_{\text{in}} + S_{\text{out}}. \quad (4.3.18)$$

For the interior region:

$$\begin{aligned} S_{\text{in}} &= A_m A_n \int_{-a}^a \sin(k_m x) \sin(k_n x) dx \\ &= A_m A_n \left[\frac{\sin((k_m - k_n)a)}{k_m - k_n} - \frac{\sin((k_m + k_n)a)}{k_m + k_n} \right]. \end{aligned} \quad (4.3.19)$$

Expanding using angle addition and substituting Eq. (4.3.16):

$$\begin{aligned} \sin((k_m \pm k_n)a) &= \sin(k_m a) \cos(k_n a) \pm \cos(k_m a) \sin(k_n a) \\ &= s_m \left(-\frac{\kappa_n}{k_n} s_n \right) \pm \left(-\frac{\kappa_m}{k_m} s_m \right) s_n \\ &= -s_m s_n \left[\frac{\kappa_n k_m \pm \kappa_m k_n}{k_m k_n} \right]. \end{aligned} \quad (4.3.20)$$

Substituting into S_{in} :

$$\begin{aligned} S_{\text{in}} &= -A_m A_n s_m s_n \left[\frac{\kappa_n k_m - \kappa_m k_n}{k_m k_n (k_m - k_n)} - \frac{\kappa_n k_m + \kappa_m k_n}{k_m k_n (k_m + k_n)} \right] \\ &= 2A_m A_n s_m s_n \left[\frac{\kappa_m - \kappa_n}{k_m^2 - k_n^2} \right]. \end{aligned} \quad (4.3.21)$$

For the exterior tails, both tails ($x < -a$ and $x > a$) yield identical positive contributions:

$$\begin{aligned} S_{\text{out}} &= A_m A_n s_m s_n \left[\int_{-\infty}^{-a} e^{(\kappa_m + \kappa_n)(x+a)} dx + \int_a^{\infty} e^{-(\kappa_m + \kappa_n)(x-a)} dx \right] \\ &= A_m A_n s_m s_n \left[\frac{1}{\kappa_m + \kappa_n} + \frac{1}{\kappa_m + \kappa_n} \right] = \frac{2A_m A_n s_m s_n}{\kappa_m + \kappa_n}. \end{aligned} \quad (4.3.22)$$

Combining the interior and exterior contributions:

$$\begin{aligned} \langle \psi_m | \psi_n \rangle &= 2A_m A_n s_m s_n \left[\frac{\kappa_m - \kappa_n}{k_m^2 - k_n^2} + \frac{1}{\kappa_m + \kappa_n} \right] \\ &= 2A_m A_n s_m s_n \left[\frac{(\kappa_m^2 + k_m^2) - (\kappa_n^2 + k_n^2)}{(k_m^2 - k_n^2)(\kappa_m + \kappa_n)} \right]. \end{aligned} \quad (4.3.23)$$

Recalling Eq. (4.3.5), $k_i^2 + \kappa_i^2 = \lambda$ for all bound states in the well. Therefore, the numerator vanishes identically:

$$(\kappa_m^2 + k_m^2) - (\kappa_n^2 + k_n^2) = \lambda - \lambda = 0. \quad (4.3.24)$$

Thus, the bound states are strictly orthogonal:

$$\langle \psi_m | \hat{H} | \psi_n \rangle = E_n \langle \psi_m | \psi_n \rangle = 0 \quad (m \neq n). \quad (4.3.25)$$

2. Non-Vanishing of Kinetic Off-Diagonal Elements: $\langle \psi_m | \hat{T} | \psi_n \rangle \neq 0$

Now evaluate the off-diagonal matrix elements of the kinetic energy operator $\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$. Using integration by parts:

$$T_{mn} \equiv \langle \psi_m | \hat{T} | \psi_n \rangle = \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \psi'_m(x) \psi'_n(x) dx. \quad (4.3.26)$$

In the interior region ($|x| < a$):

$$\psi'_i(x) = A_i k_i \cos(k_i x). \quad (4.3.27)$$

$$\begin{aligned} T_{\text{in}} &= \frac{\hbar^2}{2m} A_m A_n k_m k_n \int_{-a}^a \cos(k_m x) \cos(k_n x) dx \\ &= \frac{\hbar^2}{2m} A_m A_n k_m k_n \left[\frac{\sin((k_m - k_n)a)}{k_m - k_n} + \frac{\sin((k_m + k_n)a)}{k_m + k_n} \right] \\ &= \frac{\hbar^2}{2m} 2A_m A_n s_m s_n \left[\frac{\kappa_m k_m^2 - \kappa_n k_m^2}{k_m^2 - k_n^2} \right]. \end{aligned} \quad (4.3.28)$$

In the exterior region ($|x| > a$): For $x > a$, $\psi'_i(x) = -\kappa_i A_i s_i e^{-\kappa_i(x-a)}$. For $x < -a$, $\psi'_i(x) = -\kappa_i A_i s_i e^{\kappa_i(x+a)}$. Integrating over both symmetric tails:

$$T_{\text{out}} = \frac{\hbar^2}{2m} 2A_m A_n s_m s_n \left[\frac{\kappa_m \kappa_n}{\kappa_m + \kappa_n} \right]. \quad (4.3.29)$$

Combining T_{in} and T_{out} :

$$\begin{aligned} T_{mn} &= \frac{\hbar^2}{2m} 2A_m A_n s_m s_n \left[\frac{\kappa_m k_m^2 - \kappa_n k_m^2}{k_m^2 - k_n^2} + \frac{\kappa_m \kappa_n}{\kappa_m + \kappa_n} \right] \\ &= \frac{\hbar^2}{2m} 2A_m A_n s_m s_n \left[\frac{(\kappa_m k_m^2 - \kappa_n k_m^2)(\kappa_m + \kappa_n) + \kappa_m \kappa_n (k_m^2 - k_n^2)}{(k_m^2 - k_n^2)(\kappa_m + \kappa_n)} \right] \\ &= \frac{\hbar^2}{2m} 2A_m A_n s_m s_n \left[\frac{\kappa_m^2 k_n^2 - \kappa_n^2 k_m^2}{(k_m^2 - k_n^2)(\kappa_m + \kappa_n)} \right]. \end{aligned} \quad (4.3.30)$$

Now, substitute $\kappa_i^2 = \lambda - k_i^2 = \frac{2mV_0}{\hbar^2} - k_i^2$:

$$\begin{aligned} \kappa_m^2 k_n^2 - \kappa_n^2 k_m^2 &= \left(\frac{2mV_0}{\hbar^2} - k_m^2 \right) k_n^2 - \left(\frac{2mV_0}{\hbar^2} - k_n^2 \right) k_m^2 \\ &= \frac{2mV_0}{\hbar^2} (k_n^2 - k_m^2) = -\frac{2mV_0}{\hbar^2} (k_m^2 - k_n^2). \end{aligned} \quad (4.3.31)$$

The factor $(k_m^2 - k_n^2)$ cancels identically with the denominator! Substituting this back into T_{mn} :

$$T_{mn} = \frac{\hbar^2}{2m} 2A_m A_n s_m s_n \left[\frac{-\frac{2mV_0}{\hbar^2} (k_m^2 - k_n^2)}{(k_m^2 - k_n^2)(\kappa_m + \kappa_n)} \right]. \quad (4.3.32)$$

The factors of $\frac{\hbar^2}{2m}$ and $\frac{2m}{\hbar^2}$ cancel, yielding the exact result:

$$\boxed{T_{mn} = \langle \psi_m | \hat{T} | \psi_n \rangle = -\frac{2A_m A_n s_m s_n V_0}{\kappa_m + \kappa_n} \neq 0} \quad (4.3.33)$$

Definitive Physical Conclusion: Eqs. (4.3.25) and (4.3.33) deliver an incontrovertible mathematical theorem:

- In the energy basis, the Hamiltonian is strictly diagonal: $\langle \psi_m | \hat{H} | \psi_n \rangle = E_n \delta_{mn}$.
- In that same basis, the kinetic energy operator is **strictly non-diagonal**: $\langle \psi_m | \hat{T} | \psi_n \rangle \neq 0$ for $m \neq n$.

If the stationary states $|\psi_n\rangle$ were eigenstates of kinetic energy, T_{mn} would vanish for all $m \neq n$. The non-vanishing of T_{mn} proves that $|\psi_n\rangle$ is not an eigenstate of $\hat{T}$.

Therefore, the naive classical comparison $E \geq T$ as point-by-point numerical scalars is a category error and fundamentally ill-defined. Total energy is a sharp eigenvalue of $\hat{H}$, whereas kinetic energy is an operator with wide quantum variance in that very state.

4.4 Scattering States, Potential Steps, and Quantum Tunneling

4.4.1 Boundary Conditions across Potential Discontinuities

When solving the Schrödinger equation across finite potential discontinuities:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x), \quad (4.4.1)$$

integrating across an infinitesimal interval $[x_0 - \epsilon, x_0 + \epsilon]$ centered on the discontinuity yields:

$$-\frac{\hbar^2}{2m} \left[\frac{d\psi}{dx} \Big|_{x_0+\epsilon} - \frac{d\psi}{dx} \Big|_{x_0-\epsilon} \right] + \int_{x_0-\epsilon}^{x_0+\epsilon} [V(x) - E]\psi(x)dx = 0. \quad (4.4.2)$$

Because $V(x)$ is piecewise finite, the integral vanishes as $\epsilon \rightarrow 0$. Therefore:

1. The wavefunction is continuous everywhere: $\psi(x_0^+) = \psi(x_0^-)$.
2. The spatial derivative $\frac{d\psi}{dx}$ is continuous across any finite potential step: $\frac{d\psi}{dx} \Big|_{x_0^+} = \frac{d\psi}{dx} \Big|_{x_0^-}$.

4.4.2 The Step Potential

Consider a potential step of height V_0 :

$$V(x) = \begin{cases} 0, & x < 0 \quad (\text{Region I}), \\ V_0, & x \geq 0 \quad (\text{Region II}). \end{cases} \quad (4.4.3)$$

Let a beam of particles of mass m and energy E be incident from the left ($x < 0$).

Case 1: $E > V_0$ (Quantum Reflection)

In Region I ($x < 0$), the solution consists of incident and reflected plane waves:

$$\psi_I(x) = e^{ik_1x} + Re^{-ik_1x}, \quad k_1 \equiv \frac{\sqrt{2mE}}{\hbar}. \quad (4.4.4)$$

In Region II ($x > 0$), only a transmitted right-moving wave exists:

$$\psi_{II}(x) = Te^{ik_2x}, \quad k_2 \equiv \frac{\sqrt{2m(E - V_0)}}{\hbar}. \quad (4.4.5)$$

Matching boundary conditions at $x = 0$:

$$\psi_I(0) = \psi_{II}(0) \implies 1 + R = T, \quad (4.4.6)$$

$$\psi'_I(0) = \psi'_{II}(0) \implies ik_1(1 - R) = ik_2T \implies 1 - R = \frac{k_2}{k_1}T. \quad (4.4.7)$$

Solving this linear system:

$$T = \frac{2k_1}{k_1 + k_2}, \quad R = \frac{k_1 - k_2}{k_1 + k_2}. \quad (4.4.8)$$

The probability current density $j(x) = \frac{\hbar}{m} \text{Im} \left(\psi^* \frac{d\psi}{dx} \right)$ gives the incident, reflected, and transmitted fluxes:

$$j_{\text{inc}} = \frac{\hbar k_1}{m}, \quad j_{\text{refl}} = -\frac{\hbar k_1}{m}|R|^2, \quad j_{\text{trans}} = \frac{\hbar k_2}{m}|T|^2. \quad (4.4.9)$$

The reflection coefficient $\mathcal{R}$ and transmission coefficient $\mathcal{T}$ are:

$$\mathcal{R} \equiv \frac{|j_{\text{refl}}|}{j_{\text{inc}}} = |R|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2, \quad (4.4.10)$$

$$\mathcal{T} \equiv \frac{j_{\text{trans}}}{j_{\text{inc}}} = \frac{k_2}{k_1}|T|^2 = \frac{4k_1 k_2}{(k_1 + k_2)^2}. \quad (4.4.11)$$

Notice that probability current is strictly conserved:

$$\mathcal{R} + \mathcal{T} = \frac{(k_1 - k_2)^2 + 4k_1 k_2}{(k_1 + k_2)^2} = 1. \quad (4.4.12)$$

Classically, because $E > V_0$, every particle would cross into Region II ($\mathcal{R}_{\text{class}} = 0$). In quantum mechanics, the abrupt change in de Broglie wavelength from $\lambda_1 = 2\pi/k_1$ to $\lambda_2 = 2\pi/k_2$ causes wave-impedance mismatch, producing non-zero reflection $\mathcal{R} > 0$, exactly like light traversing an interface between differing refractive indices.

Case 2: $E < V_0$ (Evanescent Penetration)

When $E < V_0$, k_2 becomes purely imaginary: $k_2 = i\kappa$, where $\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$. The normalizable transmitted solution in Region II must decay as $x \rightarrow +\infty$:

$$\psi_{II}(x) = T e^{-\kappa x}. \quad (4.4.13)$$

Matching boundary conditions at $x = 0$:

$$1 + R = T, \quad ik_1(1 - R) = -\kappa T \implies R = \frac{k_1 - i\kappa}{k_1 + i\kappa}. \quad (4.4.14)$$

The reflection coefficient is:

$$\mathcal{R} = |R|^2 = \frac{k_1^2 + \kappa^2}{k_1^2 + \kappa^2} = 1. \quad (4.4.15)$$

Every particle is completely reflected ($\mathcal{T} = 0$). However, the probability density inside the barrier does not abruptly vanish:

$$|\psi_{II}(x)|^2 = |T|^2 e^{-2\kappa x} = \frac{4k_1^2}{k_1^2 + \kappa^2} e^{-2\kappa x}. \quad (4.4.16)$$

The wave penetrates into the barrier with a characteristic skin depth:

$$\delta = \frac{1}{2\kappa} = \frac{\hbar}{2\sqrt{2m(V_0 - E)}}. \quad (4.4.17)$$

4.4.3 The Rectangular Barrier and Exact Tunneling Transmission

Now consider a finite barrier of height V_0 and width a :

$$V(x) = \begin{cases} 0, & x < 0 \quad (\text{Region I}), \\ V_0, & 0 \leq x \leq a \quad (\text{Region II}), \\ 0, & x > a \quad (\text{Region III}). \end{cases} \quad (4.4.18)$$

For $E < V_0$, the stationary wavefunctions in the three regions are:

$$\psi_I(x) = e^{ikx} + Re^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}, \quad (4.4.19)$$

$$\psi_{II}(x) = Ae^{\kappa x} + Be^{-\kappa x}, \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}, \quad (4.4.20)$$

$$\psi_{III}(x) = Ce^{ikx}. \quad (4.4.21)$$

Matching ψ and ψ' at $x = 0$:

$$1 + R = A + B, \quad ik(1 - R) = \kappa(A - B). \quad (4.4.22)$$

Matching ψ and ψ' at $x = a$:

$$Ae^{\kappa a} + Be^{-\kappa a} = Ce^{ika}, \quad \kappa(Ae^{\kappa a} - Be^{-\kappa a}) = ikCe^{ika}. \quad (4.4.23)$$

Solving for A and B in terms of C :

$$A = \frac{1}{2} \left(1 + \frac{ik}{\kappa} \right) Ce^{(ik-\kappa)a}, \quad (4.4.24)$$

$$B = \frac{1}{2} \left(1 - \frac{ik}{\kappa} \right) Ce^{(ik+\kappa)a}. \quad (4.4.25)$$

Substituting into the $x = 0$ matching equations and eliminating R :

$$2 = Ce^{ika} \left[\cosh(\kappa a) - i \frac{k^2 - \kappa^2}{2k\kappa} \sinh(\kappa a) \right]. \quad (4.4.26)$$

Computing the transmission coefficient $\mathcal{T} = |C|^2$:

$$\begin{aligned} \frac{4}{\mathcal{T}} &= 4 \cosh^2(\kappa a) + \frac{(k^2 - \kappa^2)^2}{k^2 \kappa^2} \sinh^2(\kappa a) \\ &= 4 + \frac{(k^2 + \kappa^2)^2}{k^2 \kappa^2} \sinh^2(\kappa a). \end{aligned} \quad (4.4.27)$$

Using $k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}$, the factor simplifies to $\frac{(k^2 + \kappa^2)^2}{4k^2 \kappa^2} = \frac{V_0^2}{4E(V_0 - E)}$. Hence, the exact transmission coefficient is:

$$\mathcal{T} = \left[1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(\kappa a) \right]^{-1}. \quad (4.4.28)$$

The High/Wide Barrier Limit. For wide or high barriers ($\kappa a \gg 1$), $\sinh(\kappa a) \approx \frac{1}{2}e^{\kappa a} \gg 1$. Eq. (4.4.28) asymptotically reduces to:

$$\mathcal{T} \approx \frac{16E(V_0 - E)}{V_0^2} \exp(-2\kappa a) = \frac{16E(V_0 - E)}{V_0^2} \exp\left(-\frac{2a}{\hbar} \sqrt{2m(V_0 - E)}\right). \quad (4.4.29)$$

4.4.4 Physical Manifestations of Quantum Tunneling

The exponential sensitivity to barrier width a and particle mass m underlies fundamental physical phenomena:

- **Scanning Tunneling Microscopy (STM):** Electrons tunnel through a vacuum gap between an atomically sharp metallic tip and a conducting substrate. Because $\mathcal{T} \propto e^{-2\kappa a}$, changing the gap distance a by 0.1 nm alters the tunneling current by a full order of magnitude, enabling sub-angstrom topographic imaging of individual surface atoms.
- **Gamow Theory of Alpha Decay:** In radioactive nuclei such as ^{238}U , an α -particle is confined by a strong nuclear potential well behind a Coulomb barrier of height ~ 30 MeV. Because the nuclear energy is only ~ 4 – 9 MeV, classical emission is impossible. Quantum tunneling through the Coulomb barrier explains the Geiger-Nuttall law, where a minor factor of 2 change in α energy spans 20 orders of magnitude in nuclear half-life.
- **Stellar Nucleosynthesis:** In the core of the Sun ($T \sim 1.5 \times 10^7$ K), protons possess average kinetic energies of ~ 1 keV, whereas their mutual Coulomb barrier is ~ 1 MeV. Classical protons could never overcome this electrostatic repulsion. Proton-proton fusion occurs exclusively via quantum tunneling through the Coulomb barrier, powering stellar life and the synthesis of the chemical elements.

4.5 The Quantum Bouncer: A Particle in a Linear Gravitational Field

A fascinating application of one-dimensional quantum mechanics is the motion of a quantum particle in a uniform gravitational field bouncing off an impenetrable floor—a system known as the **Quantum Bouncer**.

4.5.1 The Gravitational Potential and Airy Equation

Let a particle of mass m move vertically under uniform gravitational acceleration g along the z -axis, constrained by an impenetrable horizontal mirror at $z = 0$:

$$V(z) = \begin{cases} mgz, & z > 0, \\ \infty, & z \leq 0. \end{cases} \quad (4.5.1)$$

For $z > 0$, the time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dz^2} + mgz\psi(z) = E\psi(z) \implies \frac{d^2\psi}{dz^2} - \frac{2m^2g}{\hbar^2} \left(z - \frac{E}{mg} \right) \psi(z) = 0. \quad (4.5.2)$$

To make this differential equation dimensionless, we introduce the natural quantum gravitational length scale l_g and energy scale $\mathcal{E}_g$:

$$l_g \equiv \left(\frac{\hbar^2}{2m^2g} \right)^{1/3}, \quad \mathcal{E}_g \equiv mgl_g = \left(\frac{mg^2\hbar^2}{2} \right)^{1/3}. \quad (4.5.3)$$

Defining the dimensionless coordinate:

$$\zeta \equiv \frac{z - E/(mg)}{l_g} = \frac{z}{l_g} - \frac{E}{\mathcal{E}_g}, \quad (4.5.4)$$

the Schrödinger equation simplifies to the canonical **Airy differential equation**:

$$\frac{d^2\psi}{d\zeta^2} - \zeta\psi(\zeta) = 0. \quad (4.5.5)$$

4.5.2 Exact Solutions via Airy Functions

The general mathematical solution of Eq. (4.5.5) is a linear combination of the two linearly independent Airy functions:

$$\psi(\zeta) = A \text{Ai}(\zeta) + B \text{Bi}(\zeta). \quad (4.5.6)$$

As $\zeta \rightarrow +\infty$ (corresponding to $z \rightarrow \infty$ far above the floor):

$$\text{Ai}(\zeta) \sim \frac{1}{2\sqrt{\pi}\zeta^{1/4}} \exp\left(-\frac{2}{3}\zeta^{3/2}\right) \rightarrow 0, \quad (4.5.7)$$

$$\text{Bi}(\zeta) \sim \frac{1}{\sqrt{\pi}\zeta^{1/4}} \exp\left(+\frac{2}{3}\zeta^{3/2}\right) \rightarrow \infty. \quad (4.5.8)$$

Because $\text{Bi}(\zeta)$ diverges exponentially as $z \rightarrow \infty$, physical square-integrability demands $B = 0$. Therefore, the normalizable wavefunction is proportional to $\text{Ai}(\zeta)$:

$$\psi(z) = A \text{Ai}\left(\frac{z}{l_g} - \frac{E}{\mathcal{E}_g}\right). \quad (4.5.9)$$

4.5.3 Energy Quantization from the Dirichlet Floor

At the impenetrable floor $z = 0$, the wavefunction must satisfy the Dirichlet boundary condition:

$$\psi(0) = 0 \implies \text{Ai}\left(-\frac{E}{\mathcal{E}_g}\right) = 0. \quad (4.5.10)$$

The Airy function $\text{Ai}(\xi)$ has an infinite sequence of discrete zeros exclusively along the negative real axis:

$$\text{Ai}(-\alpha_n) = 0, \quad n = 1, 2, 3, \dots \quad (4.5.11)$$

where the first four numerical zeros are:

$$\alpha_1 \approx 2.3381, \quad \alpha_2 \approx 4.0879, \quad \alpha_3 \approx 5.5206, \quad \alpha_4 \approx 6.7867. \quad (4.5.12)$$

Equating $-\frac{E}{\mathcal{E}_g} = -\alpha_n$, we obtain the exact discrete energy spectrum of the quantum bouncer:

$$E_n = \mathcal{E}_g \alpha_n = \left(\frac{mg^2\hbar^2}{2}\right)^{1/3} \alpha_n, \quad n = 1, 2, 3, \dots \quad (4.5.13)$$

Using the asymptotic expansion for the zeros of the Airy function:

$$\alpha_n \approx \left[\frac{3\pi}{8}(4n-1)\right]^{2/3}, \quad (4.5.14)$$

we observe that the energy levels scale with the quantum number as $E_n \propto n^{2/3}$. This provides a rich pedagogical comparison across standard one-dimensional confining potentials:

- **Infinite Square Well** ($V \sim \infty$ at walls): $E_n \propto n^2$.
- **Harmonic Oscillator** ($V \sim x^2$): $E_n \propto n^1$.
- **Linear Gravitational Well** ($V \sim z^1$): $E_n \propto n^{2/3}$.
- **Coulomb Potential** ($V \sim -1/r$): $E_n \propto -n^{-2}$.

4.5.4 Experimental Realization: Ultracold Neutrons

In 2002, Valery Nesvizhevsky and collaborators directly verified this gravitational quantization using ultracold neutrons bouncing above a polished horizontal mirror in Earth's gravitational field ($g = 9.81 \text{ m/s}^2$). For a neutron of mass $m_n \approx 1.675 \times 10^{-27} \text{ kg}$, the characteristic quantum gravitational scales are:

$$l_g = \left(\frac{\hbar^2}{2m_n^2 g} \right)^{1/3} \approx 5.87 \mu\text{m}, \quad (4.5.15)$$

$$\mathcal{E}_g = m_n g l_g \approx 0.602 \text{ peV} = 9.64 \times 10^{-32} \text{ J}. \quad (4.5.16)$$

The ground state energy is $E_1 = 2.3381 \mathcal{E}_g \approx 1.41 \text{ peV}$, with a classical turning height of $h_1 = E_1/(m_n g) = \alpha_1 l_g \approx 13.7 \mu\text{m}$. By placing an absorbing roof at height h and measuring neutron transmission, the experimenters observed discrete transmission steps corresponding precisely to the heights of the Airy quantum eigenstates, confirming that gravity quantizes bound states just like electrostatic or nuclear forces!

4.6 The Free Particle and Wave Packet Dispersion

4.6.1 Plane Waves and the Velocity Paradox

For a free particle ($V(x) = 0$), the time-dependent Schrödinger equation is:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}. \quad (4.6.1)$$

Stationary plane-wave solutions with definite momentum $p = \hbar k$ are:

$$\psi_k(x, t) = e^{i(kx - \omega(k)t)}, \quad \omega(k) \equiv \frac{\hbar k^2}{2m} = \frac{E}{\hbar}. \quad (4.6.2)$$

Plane waves suffer from two foundational physical limitations:

1. **Non-Normalizability:** They extend over all space with infinite L^2 norm: $\int_{-\infty}^{\infty} |\psi_k|^2 dx = \infty$. They belong to the distribution space of a rigged Hilbert space, not to $\mathcal{H} = L^2(\mathbb{R})$.
2. **Phase Velocity Mismatch:** The phase velocity of the wave is:

$$v_p \equiv \frac{\omega(k)}{k} = \frac{\hbar k}{2m} = \frac{p}{2m} = \frac{v_{\text{class}}}{2}. \quad (4.6.3)$$

The individual wave crests propagate at only half the classical velocity of the particle!

4.6.2 Wave Packets and Group Velocity

To construct a physically localized particle, we form a continuous linear superposition of plane waves:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi(p) \exp \left[\frac{i}{\hbar} \left(px - \frac{p^2}{2m} t \right) \right] dp, \quad (4.6.4)$$

where $\phi(p) = \langle p | \psi(0) \rangle$ is the momentum-space wavefunction. The envelope of this packet travels at the **group velocity**:

$$v_g \equiv \left. \frac{d\omega}{dk} \right|_{k_0} = \left. \frac{dE}{dp} \right|_{p_0} = \frac{p_0}{m} = v_{\text{class}}. \quad (4.6.5)$$

The group velocity of the quantum wavepacket precisely reproduces the classical particle speed, resolving the phase velocity paradox.

4.6.3 Exact Derivation of Gaussian Wave Packet Dispersion

Let us prepare an initial minimum-uncertainty Gaussian wavepacket at $t = 0$ centered at $x = 0$, with mean momentum $p_0 = \hbar k_0$ and initial spatial width σ_0 :

$$\psi(x, 0) = \frac{1}{(2\pi\sigma_0^2)^{1/4}} \exp\left(-\frac{x^2}{4\sigma_0^2} + \frac{ip_0x}{\hbar}\right). \quad (4.6.6)$$

At $t = 0$, $\Delta x(0) = \sigma_0$ and $\Delta p = \frac{\hbar}{2\sigma_0}$, saturating the Heisenberg bound $\Delta x(0)\Delta p = \frac{\hbar}{2}$. The momentum representation $\phi(p) = \langle p | \psi(0) \rangle$ is obtained via the Fourier transform:

$$\begin{aligned} \phi(p) &= \frac{1}{\sqrt{2\pi\hbar}(2\pi\sigma_0^2)^{1/4}} \int_{-\infty}^{\infty} \exp\left[-\frac{x^2}{4\sigma_0^2} - \frac{i(p-p_0)x}{\hbar}\right] dx \\ &= \left(\frac{2\sigma_0^2}{\pi\hbar^2}\right)^{1/4} \exp\left[-\frac{\sigma_0^2(p-p_0)^2}{\hbar^2}\right]. \end{aligned} \quad (4.6.7)$$

To obtain the wavefunction at any subsequent time t , substitute $\phi(p)$ into Eq. (4.6.4):

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \left(\frac{2\sigma_0^2}{\pi\hbar^2}\right)^{1/4} \int_{-\infty}^{\infty} \exp\left[-\frac{\sigma_0^2(p-p_0)^2}{\hbar^2} + \frac{i}{\hbar}\left(px - \frac{p^2}{2m}t\right)\right] dp. \quad (4.6.8)$$

Completing the square in the exponent and evaluating the Gaussian integral yields:

$$\psi(x, t) = \frac{1}{(2\pi\sigma_0^2)^{1/4}} \frac{1}{\sqrt{1 + i\frac{\hbar t}{2m\sigma_0^2}}} \exp\left[-\frac{(x - \frac{p_0}{m}t)^2}{4\sigma_0^2\left(1 + i\frac{\hbar t}{2m\sigma_0^2}\right)} + \frac{i}{\hbar}\left(p_0x - \frac{p_0^2}{2m}t\right)\right]. \quad (4.6.9)$$

Taking the absolute square $|\psi(x, t)|^2$ gives the spatial probability density:

$$P(x, t) = |\psi(x, t)|^2 = \frac{1}{\sqrt{2\pi}\sigma(t)} \exp\left[-\frac{(x - v_0t)^2}{2\sigma(t)^2}\right], \quad (4.6.10)$$

where $v_0 \equiv p_0/m$, and the wavepacket width $\sigma(t)$ expands according to:

$$\sigma(t) = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2}\right)^2} = \sigma_0 \sqrt{1 + \frac{t^2}{\tau^2}}, \quad \tau \equiv \frac{2m\sigma_0^2}{\hbar}. \quad (4.6.11)$$

The characteristic dispersion timescale is τ .

- For small times $t \ll \tau$: $\sigma(t) \approx \sigma_0$; the packet travels rigidly.
- For large times $t \gg \tau$: $\sigma(t) \approx \frac{\hbar t}{2m\sigma_0} = \frac{\Delta p}{m}t$; the packet spreads linearly at a rate dictated by its initial momentum uncertainty.

The product of uncertainties at time t is:

$$\Delta x(t)\Delta p = \sigma(t)\frac{\hbar}{2\sigma_0} = \frac{\hbar}{2} \sqrt{1 + \frac{t^2}{\tau^2}} > \frac{\hbar}{2} \quad (t > 0). \quad (4.6.12)$$

The packet immediately loses its minimum-uncertainty property as it disperses!

Physical Reason for Dispersion: Dispersion is caused by the strictly non-linear quadratic dispersion relation $\omega(k) = \frac{\hbar k^2}{2m}$. Because group velocity depends on wavenumber ($v_g(k) = \hbar k/m$), the high-momentum Fourier components travel faster than the low-momentum components, stretching the packet apart. This contrasts sharply with the harmonic oscillator coherent states analyzed in Chapter 3: in the oscillator, the parabolic potential $V(x) = \frac{1}{2}m\omega^2x^2$ provides an exact restoring force whose equispaced spectrum prevents dispersion entirely, enabling coherent states to glide indefinitely without spreading.

4.7 Spatial Coherence and Young's Double-Slit Experiment

4.7.1 Two Slits as Coupled Spatial Apertures

Young's double-slit experiment illustrates the fundamental role of quantum probability amplitudes. Let a monochromatic source emit particles of mass m with well-defined momentum $p = \hbar k$ and de Broglie wavelength $\lambda = \frac{h}{p} = \frac{2\pi}{k}$. The particles encounter an opaque barrier containing two narrow slits separated by distance d :

$$\text{Slit 1 at } y_1 = +\frac{d}{2}, \quad \text{Slit 2 at } y_2 = -\frac{d}{2}. \quad (4.7.1)$$

A detection screen is placed at a longitudinal distance $D \gg d$ downstream.

4.7.2 Quantum Transition Amplitudes and Propagators

According to the superposition principle, if an event can occur along multiple indistinguishable pathways, the total transition amplitude $\Psi(y)$ to detect the particle at position y on the screen is the **coherent sum of amplitudes for each path**:

$$\Psi(y) = \psi_1(y) + \psi_2(y), \quad (4.7.2)$$

where $\psi_1(y)$ and $\psi_2(y)$ represent the amplitudes for traversing Slit 1 and Slit 2. In the far-field approximation ($D \gg d, y$):

$$\psi_1(y) = \frac{A_1}{\sqrt{r_1}} e^{ikr_1}, \quad \psi_2(y) = \frac{A_2}{\sqrt{r_2}} e^{ikr_2}, \quad (4.7.3)$$

where r_1 and r_2 are the geometric distances from each slit to point y :

$$r_1 \approx D + \frac{(y - d/2)^2}{2D}, \quad r_2 \approx D + \frac{(y + d/2)^2}{2D}. \quad (4.7.4)$$

The optical path difference is:

$$\Delta r \equiv r_2 - r_1 = \frac{(y + d/2)^2 - (y - d/2)^2}{2D} = \frac{yd}{D}. \quad (4.7.5)$$

The corresponding quantum phase difference between the two amplitudes is:

$$\Delta\phi = k\Delta r = \frac{p}{\hbar} \frac{yd}{D} = \frac{2\pi}{\lambda} \frac{yd}{D}. \quad (4.7.6)$$

4.7.3 Probability and the Interference Cross-Term

The probability of detecting the particle at position y is given by the Born rule:

$$\begin{aligned} P(y) &= |\Psi(y)|^2 = |\psi_1(y) + \psi_2(y)|^2 \\ &= |\psi_1(y)|^2 + |\psi_2(y)|^2 + \psi_1^*(y)\psi_2(y) + \psi_2^*(y)\psi_1(y) \\ &= |\psi_1(y)|^2 + |\psi_2(y)|^2 + 2\text{Re}(\psi_1^*(y)\psi_2(y)). \end{aligned} \quad (4.7.7)$$

Setting $P_1(y) = |\psi_1(y)|^2$ and $P_2(y) = |\psi_2(y)|^2$, and assuming symmetric illumination ($P_1 \approx P_2 \equiv P_0$):

$$P(y) = 2P_0(1 + \cos \Delta\phi) = 4P_0 \cos^2 \left(\frac{\Delta\phi}{2} \right) = 4P_0 \cos^2 \left(\frac{\pi yd}{\lambda D} \right). \quad (4.7.8)$$

The fringe positions are:

- **Bright Fringes (Constructive):** $y_m = m \frac{\lambda D}{d}, \quad m \in \mathbb{Z}.$
- **Dark Fringes (Destructive):** $y_m = \left(m + \frac{1}{2}\right) \frac{\lambda D}{d}.$

The fringe spacing is $\Delta y = \frac{\lambda D}{d} = \frac{hD}{pd}.$

Quantum Interference vs. Classical Corpuscles. For classical particles, each particle definitively passes through one slit or the other; probabilities add directly without interference:

$$P_{\text{classical}}(y) = P_1(y) + P_2(y) = 2P_0. \quad (4.7.9)$$

The hallmark of quantum mechanics is the coherent cross-term $2\text{Re}(\psi_1^*\psi_2) = 2P_0 \cos\left(\frac{2\pi yd}{\lambda D}\right)$. In regions of destructive interference, $P(y) = 0$: opening a second slit reduces the arrival probability at certain points from P_0 to zero!

4.7.4 Which-Way Detection and Quantum Decoherence

What occurs if a detector is installed at the slits to determine which slit the particle traverses?

Let the detector begin in an initial state $|d_0\rangle$. When the particle traverses Slit 1, the detector transitions to state $|d_1\rangle$; when it traverses Slit 2, the detector transitions to state $|d_2\rangle$. By linearity, the combined entangled state of particle plus detector is:

$$|\Psi\rangle = \psi_1(y)|d_1\rangle + \psi_2(y)|d_2\rangle. \quad (4.7.10)$$

The spatial probability density of finding the particle at position y , regardless of the final state of the detector, is obtained by tracing over the detector's Hilbert space:

$$\begin{aligned} P(y) &= \langle\Psi|\Psi\rangle_{\text{detector}} \\ &= |\psi_1(y)|^2 \langle d_1|d_1\rangle + |\psi_2(y)|^2 \langle d_2|d_2\rangle + 2\text{Re}(\psi_1^*(y)\psi_2(y) \langle d_1|d_2\rangle). \end{aligned} \quad (4.7.11)$$

Since the detector states are normalized ($\langle d_1|d_1\rangle = \langle d_2|d_2\rangle = 1$):

$$P(y) = P_1(y) + P_2(y) + 2\text{Re}(\psi_1^*(y)\psi_2(y) \langle d_1|d_2\rangle). \quad (4.7.12)$$

If the detector can distinguish the two paths with complete certainty, the detector states are strictly orthogonal:

$$\langle d_1|d_2\rangle = 0. \quad (4.7.13)$$

Consequently, the interference cross-term **vanishes completely**:

$$P(y) = P_1(y) + P_2(y). \quad (4.7.14)$$

The interference fringes disappear, and the classical sum of probabilities is recovered!

This embodies **Bohr's Principle of Complementarity**: wave-like interference and particle-like path information are mutually exclusive. The loss of interference is not caused by mechanical recoil or momentum kicks, but by the fundamental loss of quantum phase coherence through entanglement with the environment—a universal process known as **decoherence**.

Chapter 5

Angular Momentum, Rotations, and Tensor Operators

In classical physics, angular momentum was defined as the dynamical quantity conserved under rotational symmetry:

$$\vec{L} = \vec{r} \times \vec{p}. \quad (5.0.1)$$

Within the Hamiltonian formulation of classical mechanics, Noether's theorem reveals that the conserved quantities are precisely the generators of canonical transformations. Through the Poisson bracket, the infinitesimal change in any dynamical variable A generated by angular momentum L is given by

$$\delta A = \{A, \vec{n} \cdot \vec{L}\}_{\text{PB}} \delta\theta, \quad (5.0.2)$$

which, when exponentiated for a finite rotation, yields

$$A(\theta) = \exp\left(\theta\{\vec{n} \cdot \vec{L}, \cdot\}_{\text{PB}}\right) A(0). \quad (5.0.3)$$

It is in this precise Lie-algebraic sense that angular momentum is defined in quantum mechanics: angular momentum is the fundamental generator of spatial rotations.

In classical mechanics, a physical object rotating about a point possesses well-defined components (L_x, L_y, L_z) at every instant of time. However, as we explore below, this classical picture breaks down completely in quantum mechanics: the components of angular momentum do not commute with one another. Consequently, they cannot be simultaneously measured or assigned sharp values. Any attempt to interpret quantum-mechanical angular momentum—whether orbital or intrinsic spin—purely through the lens of classical trajectories invariably leads to severe misunderstandings. We therefore dispense with classical analogies and ground our formulation in the immutable bedrock of symmetry principles, Lie group theory, and operator algebra.

5.1 The Rotation Operator from First Principles

5.1.1 Active vs. Passive Transformations

When analyzing spatial transformations, one must distinguish between:

1. **Passive transformations (alias):** The physical system remains untouched, but the observer's coordinate frame is rotated.
2. **Active transformations (alibi):** The coordinate axes remain fixed, but the physical state vector $|\alpha\rangle$ is rotated in Hilbert space into a new state:

$$|\alpha\rangle_R = \hat{R}(\vec{\theta}) |\alpha\rangle. \quad (5.1.1)$$

Because physical probabilities and scalar products must be preserved under any spatial rotation, Wigner's theorem dictates that the rotation operator $\hat{R}(\vec{\theta})$ must be unitary:

$${}_R\langle\alpha|\beta\rangle_R = \langle\alpha|\hat{R}^\dagger\hat{R}|\beta\rangle = \langle\alpha|\beta\rangle \implies \hat{R}^\dagger\hat{R} = \hat{I}, \quad \hat{R}^\dagger = \hat{R}^{-1}. \quad (5.1.2)$$

Furthermore, the composition of rotations requires $\hat{R}(R_1)\hat{R}(R_2) = \hat{R}(R_1R_2)$, and the identity rotation corresponds to $\hat{R}(I) = \hat{I}$.

Consider the transformation of a spatial wavefunction $\psi(\vec{r}) = \langle\vec{r}|\alpha\rangle$. An active rotation rotates the physical state so that the amplitude that was formerly observed at $\vec{r}$ is now observed at the rotated position $\vec{r}' = R\vec{r}$. Therefore, the rotated state $|\alpha\rangle_R$ evaluated at $R\vec{r}$ must equal the original state evaluated at $\vec{r}$:

$$\psi_R(R\vec{r}) = \psi(\vec{r}) \iff \psi_R(\vec{r}) = \psi(R^{-1}\vec{r}). \quad (5.1.3)$$

Equivalently, projecting onto coordinate eigenstates, since the bra transforms as $\langle\vec{r}|\hat{R} = \langle R^{-1}\vec{r}|$, we obtain

$$\langle\vec{r}|\alpha\rangle_R = \langle\vec{r}|\hat{R}|\alpha\rangle = \langle R^{-1}\vec{r}|\alpha\rangle = \psi(R^{-1}\vec{r}). \quad (5.1.4)$$

5.1.2 Infinitesimal Rotations and the Generator

Let us consider an infinitesimal active rotation by an angle $\delta\theta$ about the z -axis, $\vec{n} = \hat{e}_z$. In Cartesian coordinates, the rotation matrix $R_z(\delta\theta)$ is

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \delta\theta & -\sin \delta\theta & 0 \\ \sin \delta\theta & \cos \delta\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \approx \begin{pmatrix} 1 & -\delta\theta & 0 \\ \delta\theta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y\delta\theta \\ y + x\delta\theta \\ z \end{pmatrix}. \quad (5.1.5)$$

The inverse transformation $R_z^{-1}(\delta\theta) = R_z(-\delta\theta)$ acts on coordinates as:

$$x \mapsto x + y\delta\theta, \quad y \mapsto y - x\delta\theta, \quad z \mapsto z. \quad (5.1.6)$$

Applying this to the wavefunction $\psi(\vec{r})$, a first-order Taylor expansion yields

$$\begin{aligned} \psi_R(x, y, z) &= \psi(x + y\delta\theta, y - x\delta\theta, z) \\ &= \psi(x, y, z) + \delta\theta \left(y \frac{\partial\psi}{\partial x} - x \frac{\partial\psi}{\partial y} \right) + \mathcal{O}(\delta\theta^2) \\ &= \left[1 - \frac{i\delta\theta}{\hbar} (x\hat{p}_y - y\hat{p}_x) \right] \psi(x, y, z), \end{aligned} \quad (5.1.7)$$

where we have recognized the canonical momentum operators $\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$ and $\hat{p}_y = -i\hbar \frac{\partial}{\partial y}$. The differential operator appearing in the bracket is precisely the z -component of orbital angular momentum:

$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right). \quad (5.1.8)$$

Thus, for an infinitesimal rotation around an arbitrary unit axis $\hat{n}$ by an angle $\delta\theta$, the unitary rotation operator is

$$\hat{R}(\delta\theta \hat{n}) = \hat{I} - \frac{i}{\hbar} \delta\theta (\hat{n} \cdot \vec{J}), \quad (5.1.9)$$

where $\vec{J}$ is the total angular momentum operator (which encompasses both orbital angular momentum $\vec{L}$ and any intrinsic spin $\vec{S}$).

5.1.3 Finite Rotations: Compounding Infinitesimal Steps

A finite rotation by angle θ about axis $\hat{n}$ can be constructed by subdividing the angle into N equal increments $\delta\theta = \theta/N$ and taking the continuum limit $N \rightarrow \infty$:

$$\hat{R}(\theta \hat{n}) = \lim_{N \rightarrow \infty} \left[\hat{I} - \frac{i}{\hbar} \frac{\theta}{N} (\hat{n} \cdot \vec{J}) \right]^N = \exp \left(-\frac{i}{\hbar} \theta \hat{n} \cdot \vec{J} \right). \quad (5.1.10)$$

Because $\vec{J}$ is a Hermitian observable ($\vec{J}^\dagger = \vec{J}$), Stone's theorem guarantees that $\hat{R}(\theta \hat{n})$ is a strongly continuous one-parameter unitary group.

5.1.4 Non-Abelian Nature and Commutation Relations from Geometry

Unlike spatial translations, 3D rotations do not commute. Consider rotating an ordinary book or a rigid coordinate frame:

- **Operation 1:** Rotate by 90° about the x -axis, then by 90° about the y -axis.
- **Operation 2:** Rotate by 90° about the y -axis, then by 90° about the x -axis.

The final orientations are completely different! Let us analyze this non-commutativity analytically for infinitesimal angles $\delta\theta$.

In $\mathbb{R}^3$, the classical rotation matrices for rotations around the x and y axes to second order in $\delta\theta$ are:

$$R_x(\delta\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 - \frac{1}{2}\delta\theta^2 & -\delta\theta \\ 0 & \delta\theta & 1 - \frac{1}{2}\delta\theta^2 \end{pmatrix}, \quad R_y(\delta\theta) = \begin{pmatrix} 1 - \frac{1}{2}\delta\theta^2 & 0 & \delta\theta \\ 0 & 1 & 0 \\ -\delta\theta & 0 & 1 - \frac{1}{2}\delta\theta^2 \end{pmatrix}. \quad (5.1.11)$$

Multiplying these matrices in both orders and evaluating their group commutator:

$$R_x(\delta\theta)R_y(\delta\theta) = \begin{pmatrix} 1 - \frac{1}{2}\delta\theta^2 & 0 & \delta\theta \\ \delta\theta^2 & 1 - \frac{1}{2}\delta\theta^2 & -\delta\theta \\ -\delta\theta & \delta\theta & 1 - \delta\theta^2 \end{pmatrix}, \quad (5.1.12)$$

$$R_y(\delta\theta)R_x(\delta\theta) = \begin{pmatrix} 1 - \frac{1}{2}\delta\theta^2 & \delta\theta^2 & \delta\theta \\ 0 & 1 - \frac{1}{2}\delta\theta^2 & -\delta\theta \\ -\delta\theta & \delta\theta & 1 - \delta\theta^2 \end{pmatrix}. \quad (5.1.13)$$

Subtracting the two expressions, all zeroth- and first-order terms cancel identically, leaving

$$R_x(\delta\theta)R_y(\delta\theta) - R_y(\delta\theta)R_x(\delta\theta) = \begin{pmatrix} 0 & -\delta\theta^2 & 0 \\ \delta\theta^2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R_z(\delta\theta^2) - I + \mathcal{O}(\delta\theta^3). \quad (5.1.14)$$

This remarkable geometric identity demonstrates that a closed commutator loop around the x and y axes produces an infinitesimal rotation around the z -axis of angle $\delta\theta^2$!

Now, let us evaluate the same commutator using the unitary operators acting on the quantum Hilbert space:

$$\hat{R}_x(\delta\theta) = \hat{I} - \frac{i}{\hbar} \delta\theta J_x - \frac{\delta\theta^2}{2\hbar^2} J_x^2 + \dots, \quad (5.1.15)$$

$$\hat{R}_y(\delta\theta) = \hat{I} - \frac{i}{\hbar} \delta\theta J_y - \frac{\delta\theta^2}{2\hbar^2} J_y^2 + \dots. \quad (5.1.16)$$

The product $\hat{R}_x(\delta\theta)\hat{R}_y(\delta\theta)$ is

$$\hat{R}_x(\delta\theta)\hat{R}_y(\delta\theta) = \hat{I} - \frac{i}{\hbar} \delta\theta (J_x + J_y) - \frac{\delta\theta^2}{\hbar^2} J_x J_y - \frac{\delta\theta^2}{2\hbar^2} (J_x^2 + J_y^2) + \mathcal{O}(\delta\theta^3). \quad (5.1.17)$$

Reversing the order and taking the difference:

$$\hat{R}_x(\delta\theta)\hat{R}_y(\delta\theta) - \hat{R}_y(\delta\theta)\hat{R}_x(\delta\theta) = -\frac{\delta\theta^2}{\hbar^2}[J_x, J_y] + \mathcal{O}(\delta\theta^3). \quad (5.1.18)$$

Matching this quantum operator expression to the geometric result $\hat{R}_z(\delta\theta^2) - \hat{I} = -\frac{i}{\hbar}\delta\theta^2 J_z$:

$$-\frac{\delta\theta^2}{\hbar^2}[J_x, J_y] = -\frac{i}{\hbar}\delta\theta^2 J_z \quad \implies \quad [J_x, J_y] = i\hbar J_z. \quad (5.1.19)$$

Permuting indices cyclically yields the foundational Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$:

$$[J_i, J_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} J_k. \quad (5.1.20)$$

This commutation relation is universal: it is obeyed by orbital angular momentum $\vec{L}$, spin angular momentum $\vec{S}$, and total angular momentum $\vec{J} = \vec{L} + \vec{S}$.

5.2 Vector and Tensor Operator Transformations from First Principles

5.2.1 Transformation of Matrix Elements and Expectation Values

What does it mean for an observable $\vec{V} = (V_x, V_y, V_z)$ to be a **vector operator**?

In quantum mechanics, all physical predictions reside in matrix elements and expectation values. If the physical state of a system is rotated via $|\alpha\rangle \rightarrow |\alpha\rangle_R = \hat{R}|\alpha\rangle$ and $|\beta\rangle \rightarrow |\beta\rangle_R = \hat{R}|\beta\rangle$, the expectation values and transition matrix elements of the components of a physical vector (such as position $\vec{r}$, linear momentum $\vec{p}$, or magnetic moment $\vec{\mu}$) must rotate precisely like the components of a classical 3-vector under the rotation matrix R_{ij} :

$${}_R\langle\alpha| V_i |\beta\rangle_R = \sum_{j=1}^3 R_{ij} \langle\alpha| V_j |\beta\rangle. \quad (5.2.1)$$

Since ${}_R\langle\alpha| = \langle\alpha|\hat{R}^\dagger$, the left-hand side can be re-expressed as

$${}_R\langle\alpha| V_i |\beta\rangle_R = \langle\alpha|\hat{R}^\dagger V_i \hat{R} |\beta\rangle. \quad (5.2.2)$$

Because equation (5.2.1) must hold for *arbitrary* state vectors $|\alpha\rangle$ and $|\beta\rangle$ throughout Hilbert space, the operators themselves must satisfy the operator equality:

$$\hat{R}^\dagger V_i \hat{R} = \sum_{j=1}^3 R_{ij} V_j. \quad (5.2.3)$$

This is the fundamental definition of a vector operator.

5.2.2 Adjoint Action and the Hadamard Expansion

Let us now specialize to a finite rotation by an angle θ around a unit axis $\hat{n}$. The unitary rotation operator is

$$\hat{R}(\theta\hat{n}) = e^{-i\theta\hat{n}\cdot\vec{J}/\hbar}, \quad \hat{R}^\dagger(\theta\hat{n}) = e^{i\theta\hat{n}\cdot\vec{J}/\hbar}. \quad (5.2.4)$$

(In units where $\hbar = 1$, or absorbing $\hbar$ into the generator, we write $\hat{R} = e^{-i\theta\hat{n}\cdot\vec{J}}$.) The transformation law (5.2.3) becomes

$$e^{i\theta\hat{n}\cdot\vec{J}} V_i e^{-i\theta\hat{n}\cdot\vec{J}} = \sum_j (e^{-i\theta\hat{n}\cdot\vec{J}})_{ij} V_j, \quad (5.2.5)$$

where $\mathbf{J}$ on the right-hand side denotes the 3×3 generator matrix in the vector representation of $SO(3)$.

To evaluate the left-hand side, we invoke the **Hadamard Lemma** (also known as the Baker–Campbell–Hausdorff adjoint formula). For any two operators $\hat{A}$ and $\hat{B}$, the adjoint action $\text{ad}_{\hat{A}}(\hat{B}) \equiv [\hat{A}, \hat{B}]$ satisfies

$$\begin{aligned} e^{\hat{A}} \hat{B} e^{-\hat{A}} &= \exp(\text{ad}_{\hat{A}}) \hat{B} \\ &= \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \cdots \end{aligned} \quad (5.2.6)$$

Setting $\hat{A} = i\theta \hat{n} \cdot \vec{J}$, the finite transformation is expressed compactly through the adjoint action:

$$e^{i\theta \hat{n} \cdot [\vec{J}, \cdot]} V_i = \sum_j (e^{-i\theta \hat{n} \cdot \mathbf{J}})_{ij} V_j. \quad (5.2.7)$$

5.2.3 Derivation via the $SO(3)$ Adjoint Representation

Now let us consider an infinitesimal rotation, expanding both sides to first order in the angle θ :

$$\text{Left-hand side: } V_i + i\theta \hat{n} \cdot [\vec{J}, V_i] + \mathcal{O}(\theta^2), \quad (5.2.8)$$

$$\text{Right-hand side: } \sum_j \left(\delta_{ij} - i\theta \hat{n} \cdot \mathbf{J}_{ij} + \mathcal{O}(\theta^2) \right) V_j = V_i - i\theta \sum_j (\hat{n} \cdot \mathbf{J})_{ij} V_j + \mathcal{O}(\theta^2). \quad (5.2.9)$$

Equating the first-order terms in θ :

$$i\theta \sum_a n_a [J_a, V_i] = -i\theta \sum_a n_a \sum_j (\mathbf{J}_a)_{ij} V_j. \quad (5.2.10)$$

Since this must hold for any arbitrary unit axis $\hat{n}$ (i.e. for every component n_a independently):

$$[J_a, V_i] = - \sum_j (\mathbf{J}_a)_{ij} V_j. \quad (5.2.11)$$

In the 3-dimensional vector (adjoint) representation of the Lie algebra $\mathfrak{so}(3)$, the generator of rotation about axis a has matrix elements

$$(\mathbf{J}_a)_{ij} = -i\epsilon_{aij}. \quad (5.2.12)$$

To verify this, note that for rotation about the z -axis ($a = 3$):

$$\mathbf{J}_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (5.2.13)$$

so that $(e^{-i\theta \mathbf{J}_3}) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = R_z(\theta)$. Substituting $(\mathbf{J}_a)_{ij} = -i\epsilon_{aij}$ into the commutator formula gives:

$$[J_a, V_i] = -(-i\epsilon_{aij}) V_j = i\epsilon_{aij} V_j. \quad (5.2.14)$$

Restoring Planck's constant $\hbar$, we arrive at the celebrated commutation relation:

$$\boxed{[J_a, V_i] = i\hbar \sum_j \epsilon_{aij} V_j.} \quad (5.2.15)$$

5.2.4 Alternative Geometric Derivation of the Vector Commutator

We can also derive this fundamental commutator directly from the geometric action of infinitesimal rotations on classical vectors.

Under an active rotation by an infinitesimal angle θ around unit axis $\hat{n}$, any vector $\vec{V}$ transforms as:

$$\vec{V} \longrightarrow R_{\hat{n}} \vec{V} = \vec{V} + \theta(\hat{n} \times \vec{V}) + \mathcal{O}(\theta^2). \quad (5.2.16)$$

In coordinate components:

$$(R_{\hat{n}} \vec{V})_i = V_i + \theta \sum_{j,k} \epsilon_{ijk} n_j V_k = V_i - \theta \sum_{j,k} \epsilon_{jik} n_j V_k. \quad (5.2.17)$$

Now, let $U(R) = e^{-i\theta \hat{n} \cdot \vec{J}/\hbar}$ be the unitary operator that transforms quantum observables:

$$U(R)^\dagger V_i U(R) = (R_{\hat{n}} \vec{V})_i. \quad (5.2.18)$$

Using $U(R) = \hat{I} - \frac{i\theta}{\hbar} \hat{n} \cdot \vec{J} + \dots$ and $U(R)^\dagger = \hat{I} + \frac{i\theta}{\hbar} \hat{n} \cdot \vec{J} + \dots$, the left-hand side expands as

$$\begin{aligned} U(R)^\dagger V_i U(R) &= \left(\hat{I} + \frac{i\theta}{\hbar} \hat{n} \cdot \vec{J} \right) V_i \left(\hat{I} - \frac{i\theta}{\hbar} \hat{n} \cdot \vec{J} \right) \\ &= V_i + \frac{i\theta}{\hbar} [\hat{n} \cdot \vec{J}, V_i] + \mathcal{O}(\theta^2). \end{aligned} \quad (5.2.19)$$

Equating this to the geometric result $(R_{\hat{n}} \vec{V})_i = V_i + \theta \sum_{j,k} \epsilon_{ijk} n_j V_k$:

$$\frac{i\theta}{\hbar} \sum_j n_j [J_j, V_i] = \theta \sum_{j,k} n_j \epsilon_{ijk} V_k. \quad (5.2.20)$$

Multiplying through by $-i\hbar/\theta$ and equating coefficients of each independent unit component n_j :

$$[J_j, V_i] = -i\hbar \sum_k \epsilon_{ijk} V_k = i\hbar \sum_k \epsilon_{jik} V_k. \quad (5.2.21)$$

Relabeling indices $j \rightarrow a$ and $k \rightarrow j$, we recover identically:

$$[J_a, V_i] = i\hbar \sum_j \epsilon_{aij} V_j. \quad (5.2.22)$$

Writing this out component-by-component clarifies the explicit physical mixing:

$$[J_x, V_x] = 0, \quad [J_x, V_y] = i\hbar V_z, \quad [J_x, V_z] = -i\hbar V_y, \quad (5.2.23)$$

$$[J_y, V_x] = -i\hbar V_z, \quad [J_y, V_y] = 0, \quad [J_y, V_z] = i\hbar V_x, \quad (5.2.24)$$

$$[J_z, V_x] = i\hbar V_y, \quad [J_z, V_y] = -i\hbar V_x, \quad [J_z, V_z] = 0. \quad (5.2.25)$$

Notice that when $\vec{V} = \vec{J}$, equation (5.2.15) reproduces the fundamental angular momentum algebra $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$. Thus, the angular momentum vector $\vec{J}$ is itself a vector operator! Other paramount vector operators include position $\vec{r}$, linear momentum $\vec{p}$, and the Laplace–Runge–Lenz vector $\vec{A}$.

5.2.5 Scalar Operators vs. Vector Operators

By definition, a **scalar operator** $\hat{S}$ is completely invariant under spatial rotations:

$$\hat{R}^\dagger \hat{S} \hat{R} = \hat{S} \quad \implies \quad [\hat{S}, \hat{R}] = 0. \quad (5.2.26)$$

Expanding for infinitesimal rotations immediately requires that a scalar operator commutes with every component of angular momentum:

$$\boxed{[\hat{S}, J_i] = 0 \quad (\forall i = x, y, z).} \quad (5.2.27)$$

Examples of scalar operators include:

- The total angular momentum squared: $\vec{J}^2 = J_x^2 + J_y^2 + J_z^2$,
- The dot product of two vector operators: $\vec{V} \cdot \vec{W} = V_x W_x + V_y W_y + V_z W_z$ (such as $\vec{r} \cdot \vec{p}$ or $\vec{L} \cdot \vec{S}$),
- Any central potential Hamiltonian: $\hat{H} = \frac{\hat{p}^2}{2m} + V(r)$.

5.2.6 Cartesian Tensors and Their Reducibility

A classical Cartesian tensor of rank 2 is a quantity T_{ij} with 9 components that transforms under rotation as

$$T'_{ij} = \sum_{k,l} R_{ik} R_{jl} T_{kl}. \quad (5.2.28)$$

The simplest example is the *dyadic* formed from two vectors $\vec{A}$ and $\vec{B}$:

$$T_{ij} = A_i B_j. \quad (5.2.29)$$

Although T_{ij} has 9 components, it forms a **reducible representation** of the rotation group $SO(3)$. That is, the 9-dimensional space can be decomposed into smaller invariant subspaces that do not mix with one another under rotations:

$$T_{ij} = T_{ij}^{(0)} + T_{ij}^{(1)} + T_{ij}^{(2)}, \quad (5.2.30)$$

where:

1. **Rank-0 (Scalar, 1 component):** The trace of the tensor, invariant under rotations:

$$T_{ij}^{(0)} = \frac{1}{3}(\vec{A} \cdot \vec{B}) \delta_{ij} = \frac{1}{3} \left(\sum_k A_k B_k \right) \delta_{ij}. \quad (5.2.31)$$

This corresponds to angular momentum $j = 0$ ($2j + 1 = 1$ state).

2. **Rank-1 (Antisymmetric tensor / Pseudovector, 3 components):**

$$T_{ij}^{(1)} = \frac{1}{2}(A_i B_j - A_j B_i) = \frac{1}{2} \sum_k \epsilon_{ijk} (\vec{A} \times \vec{B})_k. \quad (5.2.32)$$

This corresponds to angular momentum $j = 1$ ($2j + 1 = 3$ states).

3. **Rank-2 (Symmetric traceless tensor, 5 components):**

$$T_{ij}^{(2)} = \frac{1}{2}(A_i B_j + A_j B_i) - \frac{1}{3}(\vec{A} \cdot \vec{B}) \delta_{ij}. \quad (5.2.33)$$

Since a general symmetric 3×3 matrix has 6 independent elements, subtracting the trace leaves $6 - 1 = 5$ independent components. This corresponds precisely to angular momentum $j = 2$ ($2j + 1 = 5$ states).

Summing the multiplicities:

$$3 \times 3 = 1 \oplus 3 \oplus 5 \quad \Longleftrightarrow \quad \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}. \quad (5.2.34)$$

This decomposition demonstrates why Cartesian tensors are cumbersome in quantum mechanics: rotations mix the Cartesian indices in a complicated way. To exploit rotational symmetry cleanly, we must transition to the **spherical tensor** basis, where operators transform as irreducible representations of $SO(3)$, as developed in Section 5.5.

5.3 Angular Momentum Algebra and Matrix Representations

5.3.1 Casimir Operator and Common Eigenbasis

From the fundamental commutation relations $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$, no two distinct components of angular momentum commute:

$$[J_x, J_y] = i\hbar J_z, \quad [J_y, J_z] = i\hbar J_x, \quad [J_z, J_x] = i\hbar J_y. \quad (5.3.1)$$

By the generalized Heisenberg uncertainty principle, the uncertainties satisfy

$$\Delta J_x \Delta J_y \geq \frac{\hbar}{2} |\langle J_z \rangle|. \quad (5.3.2)$$

Thus, a quantum state cannot have well-defined eigenvalues for more than one Cartesian component simultaneously.

However, consider the quadratic Casimir operator:

$$J^2 \equiv \vec{J} \cdot \vec{J} = J_x^2 + J_y^2 + J_z^2. \quad (5.3.3)$$

Evaluating its commutator with J_z :

$$\begin{aligned} [J^2, J_z] &= [J_x^2 + J_y^2 + J_z^2, J_z] = [J_x^2, J_z] + [J_y^2, J_z] \\ &= J_x[J_x, J_z] + [J_x, J_z]J_x + J_y[J_y, J_z] + [J_y, J_z]J_y \\ &= J_x(-i\hbar J_y) + (-i\hbar J_y)J_x + J_y(i\hbar J_x) + (i\hbar J_x)J_y \\ &= -i\hbar(J_x J_y + J_y J_x) + i\hbar(J_y J_x + J_x J_y) = 0. \end{aligned} \quad (5.3.4)$$

By rotational isotropy, J^2 commutes with all three components:

$$\boxed{[J^2, J_i] = 0 \quad (\forall i = x, y, z).} \quad (5.3.5)$$

Therefore, we can choose a Complete Set of Commuting Observables (CSCO) formed by J^2 and any one component, conventionally chosen to be J_z . They share a simultaneous orthonormal eigenbasis, which we label as $|j, m\rangle$:

$$J^2|j, m\rangle = \lambda\hbar^2|j, m\rangle, \quad (5.3.6)$$

$$J_z|j, m\rangle = m\hbar|j, m\rangle. \quad (5.3.7)$$

5.3.2 Ladder Operators and the Spectrum

To determine the spectrum of allowed eigenvalues λ and m , we define the non-Hermitian ladder operators:

$$J_+ \equiv J_x + iJ_y, \quad J_- \equiv J_x - iJ_y = (J_+)^\dagger. \quad (5.3.8)$$

Their commutation relations with J_z and J^2 are:

$$\begin{aligned} [J_z, J_\pm] &= [J_z, J_x \pm iJ_y] = [J_z, J_x] \pm i[J_z, J_y] = i\hbar J_y \pm i(-i\hbar J_x) \\ &= \pm\hbar(J_x \pm iJ_y) = \pm\hbar J_\pm, \end{aligned} \quad (5.3.9)$$

$$[J^2, J_\pm] = 0. \quad (5.3.10)$$

The products of the ladder operators can be expressed in terms of J^2 and J_z :

$$J_- J_+ = (J_x - iJ_y)(J_x + iJ_y) = J_x^2 + J_y^2 + i[J_x, J_y] = J^2 - J_z^2 - \hbar J_z, \quad (5.3.11)$$

$$J_+ J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 - i[J_x, J_y] = J^2 - J_z^2 + \hbar J_z. \quad (5.3.12)$$

Adding both equations yields:

$$J^2 = \frac{1}{2}(J_+J_- + J_-J_+) + J_z^2. \quad (5.3.13)$$

Now consider the action of $J_\pm$ on an eigenstate $|j, m\rangle$:

$$J_z(J_\pm|j, m\rangle) = (J_\pm J_z + [J_z, J_\pm])|j, m\rangle = (m\hbar \pm \hbar)(J_\pm|j, m\rangle) = (m \pm 1)\hbar(J_\pm|j, m\rangle). \quad (5.3.14)$$

Thus, J_+ raises the magnetic quantum number m by 1, and J_- lowers m by 1:

$$J_\pm|j, m\rangle = c_\pm(j, m)|j, m \pm 1\rangle. \quad (5.3.15)$$

5.3.3 Quantization of Angular Momentum

Because J_x and J_y are Hermitian observables, $J_x^2 + J_y^2$ is a positive semi-definite operator. Therefore:

$$\langle j, m|(J^2 - J_z^2)|j, m\rangle = \langle j, m|(J_x^2 + J_y^2)|j, m\rangle \geq 0 \implies \lambda \geq m^2. \quad (5.3.16)$$

Hence, for a given eigenvalue λ , the spectrum of m must be bounded from above by some maximum value j , and bounded from below by some minimum value $j_{\min}$.

For the top state $|j, j\rangle$, raising further must annihilate the state:

$$J_+|j, j\rangle = 0. \quad (5.3.17)$$

Applying J_- from the left:

$$J_-J_+|j, j\rangle = (J^2 - J_z^2 - \hbar J_z)|j, j\rangle = (\lambda - j^2 - j)\hbar^2|j, j\rangle = 0. \quad (5.3.18)$$

Since $|j, j\rangle \neq 0$, we establish that

$$\lambda = j(j+1). \quad (5.3.19)$$

Similarly, for the bottom state $|j, j_{\min}\rangle$, lowering must annihilate the state:

$$J_-|j, j_{\min}\rangle = 0 \implies J_+J_-|j, j_{\min}\rangle = (\lambda - j_{\min}^2 + j_{\min})\hbar^2|j, j_{\min}\rangle = 0. \quad (5.3.20)$$

Thus:

$$j(j+1) = j_{\min}(j_{\min} - 1) \implies (j + j_{\min})(j - j_{\min} + 1) = 0. \quad (5.3.21)$$

Since $j_{\min} \leq j$, the only physically admissible root is

$$j_{\min} = -j. \quad (5.3.22)$$

Starting from $|j, -j\rangle$ and repeatedly applying the raising operator J_+ , we must reach $|j, j\rangle$ in an integer number of steps $n \in \{0, 1, 2, \dots\}$:

$$j - (-j) = 2j = n \implies j = \frac{n}{2}. \quad (5.3.23)$$

Therefore, j can take **integer or half-integer values**:

$$j \in \left\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots\right\}, \quad m \in \{-j, -j+1, \dots, j-1, j\}. \quad (5.3.24)$$

For a given j , the subspace is $(2j+1)$ -dimensional.

To find the matrix elements of the ladder operators, we compute the norm:

$$\begin{aligned} |c_\pm(j, m)|^2 &= \langle j, m|J_\mp J_\pm|j, m\rangle = \langle j, m|(J^2 - J_z^2 \mp \hbar J_z)|j, m\rangle \\ &= \hbar^2[j(j+1) - m(m \pm 1)]. \end{aligned} \quad (5.3.25)$$

Choosing the standard Condon–Shortley phase convention where $c_\pm(j, m)$ is real and positive:

$$J_\pm|j, m\rangle = \hbar\sqrt{j(j+1) - m(m \pm 1)}|j, m \pm 1\rangle. \quad (5.3.26)$$

5.3.4 Matrix Representations: Spin-1/2 and the Derivation of the Pauli Matrices

For the fundamental spinorial representation $j = 1/2$, the Hilbert space $\mathcal{H}_{1/2}$ is two-dimensional ($2j + 1 = 2$). The total angular momentum Casimir operator has the scalar value:

$$J^2|j, m\rangle = \hbar^2 j(j+1)|j, m\rangle = \hbar^2 \left(\frac{1}{2}\right) \left(\frac{1}{2} + 1\right) |j, m\rangle = \frac{3}{4}\hbar^2 |j, m\rangle. \quad (5.3.27)$$

Because J^2 is proportional to the identity operator on this irreducible subspace, its matrix representation in any orthonormal basis is strictly diagonal:

$$J^2 \doteq \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (5.3.28)$$

However, obtaining the explicit matrix representations of the individual Cartesian components J_x, J_y, J_z requires solving a deeper problem. How do the abstract commutation relations $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ translate into concrete 2×2 matrices?

Step 1: Choice of Quantization Axis (Diagonalizing J_z)

The abstract Lie algebra $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ is completely isotropic in three-dimensional space: no spatial direction is algebraically privileged. For any unit vector $\hat{n} \in \mathbb{R}^3$, the projection $\hat{n} \cdot \vec{J}$ possesses the identical eigenvalue spectrum $\{+\hbar/2, -\hbar/2\}$.

To construct a concrete matrix representation, we must make our **first convention**: we *choose* a specific spatial direction—by convention, the z -axis—and declare its corresponding generator J_z to be diagonal. The two orthonormal basis vectors are the eigenstates of J_z :

$$J_z \left| \frac{1}{2}, \frac{1}{2} \right\rangle = +\frac{\hbar}{2} \left| \frac{1}{2}, \frac{1}{2} \right\rangle \equiv +\frac{\hbar}{2} |\uparrow\rangle, \quad (5.3.29)$$

$$J_z \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{\hbar}{2} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \equiv -\frac{\hbar}{2} |\downarrow\rangle. \quad (5.3.30)$$

Adopting the standard descending ordered basis $\{|\uparrow\rangle, |\downarrow\rangle\}$, the basis vectors are represented by the standard orthonormal column vectors:

$$|\uparrow\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (5.3.31)$$

In this basis, the matrix elements of J_z are:

$$\begin{aligned} \langle \uparrow | J_z | \uparrow \rangle &= +\frac{\hbar}{2}, & \langle \uparrow | J_z | \downarrow \rangle &= 0, \\ \langle \downarrow | J_z | \uparrow \rangle &= 0, & \langle \downarrow | J_z | \downarrow \rangle &= -\frac{\hbar}{2}. \end{aligned} \quad (5.3.32)$$

Thus, J_z takes the diagonal form:

$$J_z \doteq \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \equiv \frac{\hbar}{2} \sigma_z. \quad (5.3.33)$$

Step 2: Action of Ladder Operators and the Norm Constraint

Next, consider the ladder operators $J_{\pm} = J_x \pm iJ_y$. From the general commutation relations $[J_z, J_{\pm}] = \pm\hbar J_{\pm}$, applying J_+ raises the magnetic quantum number m by 1, while J_- lowers m by 1:

$$J_+ |\uparrow\rangle = 0 \quad (\text{since } m = +1/2 \text{ is the maximum weight}), \quad (5.3.34)$$

$$J_-|\downarrow\rangle = 0 \quad (\text{since } m = -1/2 \text{ is the minimum weight}). \quad (5.3.35)$$

When J_+ acts on the lower state $|\downarrow\rangle$, it must produce a state proportional to the upper state $|\uparrow\rangle$:

$$J_+|\downarrow\rangle = c_+|\uparrow\rangle, \quad (5.3.36)$$

where $c_+ \in \mathbb{C}$ is a complex transition coefficient. To evaluate c_+ , we take the squared norm of both sides:

$$|c_+|^2 \langle \uparrow | \uparrow \rangle = \langle \downarrow | J_+^\dagger J_+ | \downarrow \rangle = \langle \downarrow | J_- J_+ | \downarrow \rangle. \quad (5.3.37)$$

Recalling the operator identity $J_- J_+ = J^2 - J_z^2 - \hbar J_z$, we evaluate the expectation value:

$$\begin{aligned} J_- J_+ |\downarrow\rangle &= \left[J^2 - J_z^2 - \hbar J_z \right] \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \\ &= \left[\frac{3}{4} \hbar^2 - \left(-\frac{\hbar}{2} \right)^2 - \hbar \left(-\frac{\hbar}{2} \right) \right] \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \\ &= \left[\frac{3}{4} - \frac{1}{4} + \frac{1}{2} \right] \hbar^2 \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \hbar^2 |\downarrow\rangle. \end{aligned} \quad (5.3.38)$$

Therefore, taking the inner product with $\langle \downarrow |$:

$$|c_+|^2 = \hbar^2 \implies |c_+| = \hbar. \quad (5.3.39)$$

Step 3: Why the Lie Algebra Does Not Fix Everything (The Phase Choice)

Here we arrive at a crucial conceptual insight: **the Lie algebra does not uniquely fix the matrix representation.** The algebraic identity $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$ fixes the squared modulus $|c_+|^2 = \hbar^2$, but it leaves the complex phase of c_+ completely arbitrary:

$$c_+ = \hbar e^{i\delta}, \quad \delta \in [0, 2\pi). \quad (5.3.40)$$

Why does the Lie algebra fail to determine δ ? In quantum mechanics, physical states are rays in Hilbert space. If we perform an arbitrary $U(1)$ phase redefinition of the basis vectors:

$$|\uparrow\rangle \rightarrow e^{i\theta_1} |\uparrow\rangle, \quad |\downarrow\rangle \rightarrow e^{i\theta_2} |\downarrow\rangle, \quad (5.3.41)$$

the diagonal matrix elements of J_z are completely invariant:

$$\langle \uparrow | J_z | \uparrow \rangle \rightarrow e^{-i\theta_1} e^{i\theta_1} \langle \uparrow | J_z | \uparrow \rangle = \frac{\hbar}{2}, \quad \langle \downarrow | J_z | \downarrow \rangle \rightarrow e^{-i\theta_2} e^{i\theta_2} \langle \downarrow | J_z | \downarrow \rangle = -\frac{\hbar}{2}. \quad (5.3.42)$$

However, the off-diagonal matrix elements of the ladder operators transform by the relative phase difference:

$$\langle \uparrow | J_+ | \downarrow \rangle \rightarrow e^{-i\theta_1} e^{i\theta_2} \langle \uparrow | J_+ | \downarrow \rangle = e^{i(\theta_2 - \theta_1)} c_+. \quad (5.3.43)$$

Crucially, for *any* chosen relative phase $\delta \equiv \theta_2 - \theta_1$, the resulting operators $J_x = \frac{1}{2}(J_+ + J_-)$ and $J_y = \frac{1}{2i}(J_+ - J_-)$ still satisfy the commutation relations:

$$[J_x, J_y] = i\hbar J_z, \quad [J_y, J_z] = i\hbar J_x, \quad [J_z, J_x] = i\hbar J_y. \quad (5.3.44)$$

The Lie algebra alone is completely satisfied by an infinite family of unitarily equivalent matrix representations!

Step 4: Fixing the Representation via the Condon–Shortley Phase Convention

To obtain concrete, standardized numerical matrices, we must make our **second convention**: we eliminate the relative phase ambiguity by choosing the **Condon–Shortley phase convention**. Under this convention, the matrix elements of the ladder operators between adjacent m -states are chosen to be **strictly real and positive**:

$$e^{i\delta} = +1 \implies c_+ = +\hbar. \quad (5.3.45)$$

With this choice, the action of the raising operator is:

$$J_+|\downarrow\rangle = \hbar|\uparrow\rangle, \quad J_+|\uparrow\rangle = 0. \quad (5.3.46)$$

Evaluating matrix elements $\langle m'|J_+|m\rangle$:

$$J_+ \doteq \begin{pmatrix} \langle\uparrow|J_+|\uparrow\rangle & \langle\uparrow|J_+|\downarrow\rangle \\ \langle\downarrow|J_+|\uparrow\rangle & \langle\downarrow|J_+|\downarrow\rangle \end{pmatrix} = \begin{pmatrix} 0 & \hbar \\ 0 & 0 \end{pmatrix} = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (5.3.47)$$

Since $J_- = J_+^\dagger$, its matrix representation is the conjugate transpose:

$$J_- \doteq \begin{pmatrix} 0 & 0 \\ \hbar & 0 \end{pmatrix} = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (5.3.48)$$

Correspondingly:

$$J_-|\uparrow\rangle = \hbar|\downarrow\rangle, \quad J_-|\downarrow\rangle = 0. \quad (5.3.49)$$

Step 5: Reconstructing J_x and J_y (The Pauli Matrices)

From the definitions $J_\pm = J_x \pm iJ_y$, we invert to recover the Hermitian transverse components:

$$J_x = \frac{J_+ + J_-}{2}, \quad J_y = \frac{J_+ - J_-}{2i}. \quad (5.3.50)$$

Substituting the explicit matrix representations:

$$J_x \doteq \frac{\hbar}{2} \left[\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right] = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \equiv \frac{\hbar}{2} \sigma_x, \quad (5.3.51)$$

$$J_y \doteq \frac{\hbar}{2i} \left[\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right] = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \equiv \frac{\hbar}{2} \sigma_y. \quad (5.3.52)$$

Together with $J_z = \frac{\hbar}{2}\sigma_z$, we have derived the three famous **Pauli spin matrices**:

$$\boxed{\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.} \quad (5.3.53)$$

We can verify their explicit action on the column vectors directly:

$$\sigma_x \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \sigma_x \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad (5.3.54)$$

$$\sigma_y \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \sigma_y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -i \\ 0 \end{pmatrix} = -i \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (5.3.55)$$

Similarly, the raising combination $\sigma_+ = \sigma_x + i\sigma_y = 2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ annihilates the spin-up state and flips spin-down to spin-up:

$$(\sigma_x + i\sigma_y) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (\sigma_x + i\sigma_y) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} -i \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (5.3.56)$$

What If We Chose a Different Convention?

To appreciate why the algebra does not fix everything, suppose an experimenter chose an alternative phase convention, setting $\delta = \pi/2$ so that $c_+ = i\hbar$. Then:

$$J'_+ = \hbar \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix}, \quad J'_- = \hbar \begin{pmatrix} 0 & 0 \\ -i & 0 \end{pmatrix}. \quad (5.3.57)$$

Reconstructing the Cartesian operators yields:

$$J'_x = \frac{J'_+ + J'_-}{2} = \frac{\hbar}{2} \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} = -\frac{\hbar}{2}\sigma_y, \quad J'_y = \frac{J'_+ - J'_-}{2i} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2}\sigma_x. \quad (5.3.58)$$

Checking the commutation relation between these alternative matrices:

$$[J'_x, J'_y] = \left[-\frac{\hbar}{2}\sigma_y, \frac{\hbar}{2}\sigma_x \right] = \frac{\hbar^2}{4}[\sigma_x, \sigma_y] = \frac{\hbar^2}{4}(2i\sigma_z) = i\hbar J'_z. \quad (5.3.59)$$

The Lie algebra is satisfied identically! The alternative representation (J'_x, J'_y, J'_z) is connected to the standard representation (J_x, J_y, J_z) by a unitary similarity transformation:

$$U = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad J'_k = U^\dagger J_k U. \quad (5.3.60)$$

More generally, any phase choice δ corresponds to rotating the Cartesian transverse axes by angle δ around the z -axis:

$$J_x(\delta) = J_x \cos \delta - J_y \sin \delta, \quad J_y(\delta) = J_x \sin \delta + J_y \cos \delta. \quad (5.3.61)$$

Foundational Insight: Lie Algebra vs. Concrete Matrix Representation

A common misconception among students of quantum mechanics is that the commutation relations $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ alone uniquely force the numerical matrices $\sigma_x, \sigma_y, \sigma_z$. As demonstrated above, this is false:

1. What the Lie algebra fixes:

- The dimensionality of irreducible representations: $d = 2j + 1$.
- The spectrum of the Casimir operator: $J^2 = j(j+1)\hbar^2 I$.
- The discrete ladder spacing of eigenvalues: $\Delta m = 1$ with $m \in \{-j, -j+1, \dots, j\}$.
- The *magnitudes* of all matrix elements: $|c_\pm(j, m)| = \hbar\sqrt{j(j+1) - m(m \pm 1)}$.

2. What the Lie algebra leaves free (conventions we must choose):

- **Choice 1 (Quantization axis):** The algebra is spherically symmetric. Diagonalizing J_z is an arbitrary choice of coordinate orientation.
- **Choice 2 (Basis ordering):** Ordering basis states from highest weight $m = +1/2$ to lowest weight $m = -1/2$ is an arbitrary choice of row/column indices.
- **Choice 3 (Phase convention):** The relative phase between basis vectors $|\uparrow\rangle$ and $|\downarrow\rangle$ is an unobservable gauge freedom. Choosing c_+ to be strictly real and positive ($e^{i\delta} = +1$) is the **Condon–Shortley phase convention**.

Conclusion: The Lie algebra determines the invariant geometry and spectrum of the physical observable; human convention fixes the coordinate basis and phase gauge in which concrete matrices are computed.

Algebraic Properties and Finite Rotations of Spin-1/2

The derived Pauli matrices satisfy the fundamental product identity:

$$\sigma_i \sigma_j = \delta_{ij} I + i \sum_k \epsilon_{ijk} \sigma_k. \quad (5.3.62)$$

Taking the symmetric and antisymmetric combinations under index interchange yields the anticommutation (Clifford algebra) and commutation (Lie algebra) relations:

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij} I, \quad [\sigma_i, \sigma_j] = 2i \sum_k \epsilon_{ijk} \sigma_k. \quad (5.3.63)$$

Furthermore, for any two vectors $\vec{A}$ and $\vec{B}$ that commute with the Pauli matrices:

$$(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = (\vec{A} \cdot \vec{B})I + i\vec{\sigma} \cdot (\vec{A} \times \vec{B}). \quad (5.3.64)$$

The algebraic traces and determinants are invariant under unitary changes of basis:

$$\text{Tr}(\sigma_i) = 0, \quad \text{Tr}(\sigma_i \sigma_j) = 2\delta_{ij}, \quad \det(\sigma_i) = -1. \quad (5.3.65)$$

The finite rotation operator for spin-1/2 around axis $\hat{n}$ by angle θ is generated by $\vec{S} = \frac{\hbar}{2}\vec{\sigma}$:

$$\hat{R}(\theta \hat{n}) = \exp\left(-i\frac{\theta}{\hbar} \hat{n} \cdot \vec{S}\right) = \exp\left(-i\frac{\theta}{2} \hat{n} \cdot \vec{\sigma}\right). \quad (5.3.66)$$

Using the property $(\hat{n} \cdot \vec{\sigma})^2 = (\hat{n} \cdot \hat{n})I = I$, expanding the exponential in Taylor series separates into even and odd powers:

$$\begin{aligned} \exp\left(-i\frac{\theta}{2} \hat{n} \cdot \vec{\sigma}\right) &= \sum_{k=0}^{\infty} \frac{(-i\theta/2)^{2k}}{(2k)!} I + \sum_{k=0}^{\infty} \frac{(-i\theta/2)^{2k+1}}{(2k+1)!} (\hat{n} \cdot \vec{\sigma}) \\ &= \cos\left(\frac{\theta}{2}\right) I - i(\hat{n} \cdot \vec{\sigma}) \sin\left(\frac{\theta}{2}\right). \end{aligned} \quad (5.3.67)$$

Notice that for a complete 2π rotation ($\theta = 2\pi$):

$$\hat{R}(2\pi \hat{n}) = \cos(\pi)I = -I. \quad (5.3.68)$$

The state vector changes sign! To return the spinor to its initial state, one must rotate by 4π . This 4π periodicity is the topological hallmark of the double covering $SU(2) \rightarrow SO(3)$.

5.3.5 Spin along an Arbitrary Direction and the Rotating Stern–Gerlach Experiment

A fundamental conceptual question posed in quantum mechanics asks: *If the spin operators $\vec{S}$ act on the two-dimensional complex Hilbert space $\mathbb{C}^2$, and a unit vector $\hat{n}$ belongs to three-dimensional Euclidean space $\mathbb{R}^3$, how is the projection $\vec{S} \cdot \hat{n}$ defined, and how does a Stern–Gerlach apparatus measure spin along a continuously rotating magnetic axis?*

Let $\hat{n} \in \mathbb{R}^3$ denote an arbitrary direction characterized by polar angle θ and azimuthal angle ϕ :

$$\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (5.3.69)$$

The spin operator along direction $\hat{n}$ is the linear combination of Cartesian spin operators weighted by the spatial components of $\hat{n}$:

$$\begin{aligned} \hat{S}_{\hat{n}} &\equiv \vec{S} \cdot \hat{n} = S_x \sin \theta \cos \phi + S_y \sin \theta \sin \phi + S_z \cos \theta \\ &= \frac{\hbar}{2}(\vec{\sigma} \cdot \hat{n}). \end{aligned} \quad (5.3.70)$$

In matrix representation:

$$\begin{aligned}\vec{\sigma} \cdot \hat{n} &= \sin \theta \cos \phi \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin \theta \sin \phi \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \cos \theta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & \sin \theta (\cos \phi - i \sin \phi) \\ \sin \theta (\cos \phi + i \sin \phi) & -\cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}. \quad (5.3.71)\end{aligned}$$

Using the identity $(\vec{\sigma} \cdot \hat{n})^2 = (\hat{n} \cdot \hat{n})I = I$, the eigenvalues of $\vec{\sigma} \cdot \hat{n}$ are strictly ± 1 . Thus, the eigenvalues of $\hat{S}_{\hat{n}}$ are always quantized as:

$$s_{\hat{n}} = \pm \frac{\hbar}{2}, \quad (5.3.72)$$

regardless of which spatial direction $\hat{n}$ is selected!

Solving the eigenvalue equations $(\vec{S} \cdot \hat{n}) |\pm \hat{n}\rangle = \pm \frac{\hbar}{2} |\pm \hat{n}\rangle$ yields the normalized eigenspinors:

$$|+\hat{n}\rangle \equiv |\uparrow_{\hat{n}}\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\phi} \end{pmatrix} = \cos\left(\frac{\theta}{2}\right) |\uparrow\rangle + \sin\left(\frac{\theta}{2}\right) e^{i\phi} |\downarrow\rangle, \quad (5.3.73)$$

$$|-\hat{n}\rangle \equiv |\downarrow_{\hat{n}}\rangle = \begin{pmatrix} -\sin(\theta/2)e^{-i\phi} \\ \cos(\theta/2) \end{pmatrix} = -\sin\left(\frac{\theta}{2}\right) e^{-i\phi} |\uparrow\rangle + \cos\left(\frac{\theta}{2}\right) |\downarrow\rangle. \quad (5.3.74)$$

The Rotating Stern–Gerlach Thought Experiment. Now consider a Stern–Gerlach apparatus whose inhomogeneous magnetic field gradient $\vec{B}(\mathbf{r}, t)$ rotates continuously in space so that its axis of orientation $\hat{n}(t)$ changes over time with angular velocity $\omega_{\text{rot}} = \dot{\theta}$. How will a spin prepared in state $|\uparrow_z\rangle$ behave as the magnet rotates?

1. **Adiabatic Regime** ($\omega_{\text{rot}} \ll \omega_{\text{Larmor}}$): If the rotation rate of the magnetic field is much slower than the electron's Larmor precession frequency ($\omega_{\text{Larmor}} = \frac{eB_0}{m}$), the **Quantum Adiabatic Theorem** applies. The state vector continuously tracks the instantaneous eigenstate $|\uparrow_{\hat{n}(t)}\rangle$ without undergoing transitions to $|\downarrow_{\hat{n}(t)}\rangle$. When $\hat{n}(t)$ traverses a closed loop $\mathcal{C}$ on the unit sphere, returning to its initial orientation after period T , the state acquires a dynamical phase plus a geometric **Berry phase**:

$$|\psi(T)\rangle = e^{i\gamma_{\text{Berry}}} e^{-iE_+T/\hbar} |\psi(0)\rangle, \quad \gamma_{\text{Berry}} = -\frac{1}{2}\Omega(\mathcal{C}) = -\pi(1 - \cos \theta), \quad (5.3.75)$$

where $\Omega(\mathcal{C})$ is the solid angle enclosed by the trajectory of $\hat{n}(t)$ on the Bloch sphere!

2. **Sudden Regime** ($\omega_{\text{rot}} \gg \omega_{\text{Larmor}}$): If the magnet is rotated abruptly compared to the Larmor period, the spin state cannot react in time (the sudden approximation). It remains in its initial state $|\uparrow_z\rangle$. When the particle subsequently passes through the rotated detector oriented along $\hat{n}$, it collapses into $|\uparrow_{\hat{n}}\rangle$ or $|\downarrow_{\hat{n}}\rangle$ with transition probabilities dictated by the Born rule:

$$P(\uparrow_{\hat{n}}) = |\langle \uparrow_{\hat{n}} | \uparrow_z \rangle|^2 = \cos^2\left(\frac{\theta}{2}\right), \quad (5.3.76)$$

$$P(\downarrow_{\hat{n}}) = |\langle \downarrow_{\hat{n}} | \uparrow_z \rangle|^2 = \sin^2\left(\frac{\theta}{2}\right). \quad (5.3.77)$$

5.3.6 Sequential Stern–Gerlach Experiments, Wavefunction Collapse, and the Quantum Markovian Process

The Stern–Gerlach experiment (first performed by Otto Stern and Walther Gerlach in 1922 with neutral silver atoms) provides the definitive empirical laboratory for understanding the physical reality of **wavefunction collapse**, the active nature of quantum measurement, and the deep connection between non-commuting observables and stochastic **Markov processes**.

The Classical Fallacy vs. The Quantum Reality

In classical mechanics, measurement is conceived as an intrinsically passive operation: an apparatus merely reveals a pre-existing physical property (such as the orientation of a magnetic dipole moment $\vec{\mu}$) without altering the state of the particle. Under this classical intuition, if a silver atom is determined to have spin component $S_z = +\hbar/2$, it is assumed to retain $S_z = +\hbar/2$ permanently unless an external mechanical torque deflects it.

Now consider the classic three-stage sequential filter setup shown in Figure 5.1:

$$\text{Oven} \longrightarrow \mathbf{SG}_z \xrightarrow{\text{select } z+} \mathbf{SG}_x \xrightarrow{\text{select } x+} \mathbf{SG}_z \longrightarrow \text{Detectors.} \quad (5.3.78)$$

- In Stage 1 ($\mathbf{SG}_z$), the beam is split vertically along $\hat{z}$. The spin-down beam ($z-$) is physically intercepted and blocked by an absorber. Only the spin-up beam ($z+$) is allowed to propagate forward.
- In Stage 2 ($\mathbf{SG}_x$), this pure $z+$ beam is transmitted through an apparatus oriented horizontally along $\hat{x}$. The beam splits into $x+$ and $x-$. The $x-$ beam is blocked, and only the $x+$ beam is transmitted.
- In Stage 3 ($\mathbf{SG}_z$), the beam is transmitted through an apparatus oriented along $\hat{z}$ for a second time.

What does classical physics predict for Stage 3? Classical physics reasons: “The $z-$ particles were already completely removed in Stage 1. Stage 2 merely filtered for $x+$. Therefore, Stage 3 must yield 100% $z+$ and 0% $z-$.”

Yet experimental reality delivers a profound shock to classical physics: **Stage 3 yields 50% $z+$ and 50% $z-$!** The $z-$ component, which had been definitively eliminated, has **spontaneously reappeared!** How can a particle that was certified to have $S_z = +\hbar/2$ suddenly yield $S_z = -\hbar/2$? The answer lies in the fact that quantum measurement is **active and state-altering**: the measurement of the non-commuting observable $\hat{S}_x$ fundamentally scrambled the quantum state, resetting the physical wavefunction and acting as a memoryless **Markovian transition**.

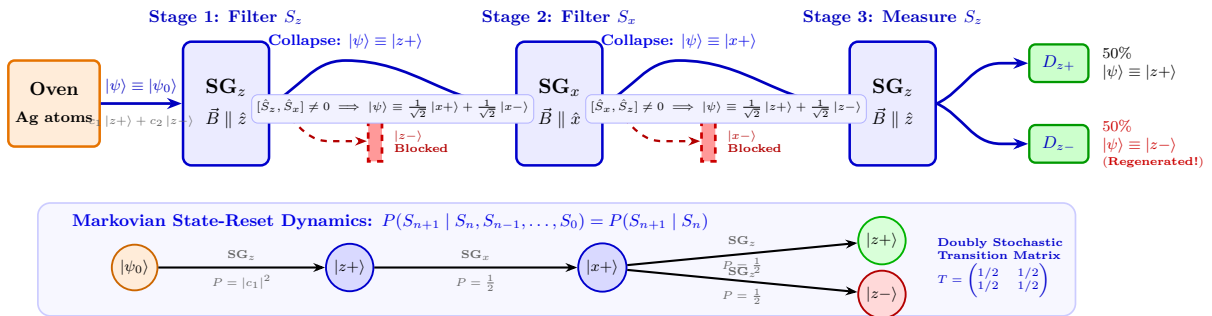


Figure 5.1: Sequential Stern–Gerlach experiment ($\mathbf{SG}_z \rightarrow \mathbf{SG}_x \rightarrow \mathbf{SG}_z$) illustrating wavefunction collapse, mandatory basis rotation between non-commuting observables, and the memoryless Markovian state-reset dynamics of quantum measurement.

Step-by-Step State Redefinition: Tracing $|\psi\rangle \equiv \dots$

Let us trace the physical state vector $|\psi\rangle$ step-by-step through the apparatus, writing $|\psi\rangle \equiv \dots$ at each stage to demonstrate how the wavefunction is dynamically and irreversibly reset.

Step 0: Initial State Entering Detector 1. An unpolarized or partially polarized ensemble of silver atoms emerges from the oven. For an individual atom entering the first apparatus, its spin state can be expanded in the computational z -basis:

$$|\psi\rangle \equiv |\psi_0\rangle = c_1 |z+\rangle + c_2 |z-\rangle, \quad |c_1|^2 + |c_2|^2 = 1. \quad (5.3.79)$$

Step 1: Measurement at Apparatus 1 and First Wavefunction Collapse. The beam enters $\mathbf{SG}_z$, where the magnetic field gradient $\frac{\partial B_z}{\partial z} \hat{z}$ exerts a spin-dependent force $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B}) \approx \mu_z \frac{\partial B_z}{\partial z} \hat{z}$. This interaction entangles the spin degree of freedom with the spatial wavepacket trajectory:

$$|\Psi(t)\rangle = c_1 \phi_{\text{up}}(\mathbf{r}) |z+\rangle + c_2 \phi_{\text{down}}(\mathbf{r}) |z-\rangle. \quad (5.3.80)$$

The beam separates into two macroscopically distinct spatial paths. We place an absorber plate blocking the $z-$ beam and allowing only the $z+$ beam to pass. By the Lüders/von Neumann projection postulate, the physical state vector collapses:

$$|\psi\rangle \equiv |z+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (5.3.81)$$

The state function has been completely redefined! Notice the crucial physics: *the physical system is now described by a brand-new state function* $|\psi\rangle \equiv |z+\rangle$. All historical memory of the initial preparation coefficients c_1, c_2 and their relative phase $\arg(c_1^* c_2)$ has been permanently obliterated.

Step 2: Non-Commuting Observables and Mandatory Basis Re-Expansion. Now, this pure beam enters apparatus 2 ($\mathbf{SG}_x$), whose magnetic field gradient is oriented along the $\hat{x}$ axis. The observable about to be measured is $\hat{S}_x = \frac{\hbar}{2} \sigma_x$. Do the observables commute?

$$[\hat{S}_z, \hat{S}_x] = i\hbar \hat{S}_y \neq 0! \quad (5.3.82)$$

Because $\hat{S}_z$ and $\hat{S}_x$ do not commute, **they cannot be simultaneously diagonalized and do not share a common eigenbasis!** The state $|\psi\rangle \equiv |z+\rangle$ is *not* an eigenstate of the observable $\hat{S}_x$ that apparatus 2 measures! Therefore, to compute what apparatus 2 will observe, we **must rewrite the current wavefunction in the eigenbasis of the observable about to be measured**, namely the $\hat{S}_x$ basis $\{|x+\rangle, |x-\rangle\}$:

$$|x+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|z+\rangle + |z-\rangle), \quad |x-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|z+\rangle - |z-\rangle). \quad (5.3.83)$$

Inverting these expressions to represent the current state $|\psi\rangle \equiv |z+\rangle$ in the $\hat{S}_x$ basis:

$$|\psi\rangle \equiv |z+\rangle = \frac{1}{\sqrt{2}} |x+\rangle + \frac{1}{\sqrt{2}} |x-\rangle. \quad (5.3.84)$$

Equation (5.3.84) reveals the physical situation: the state entering $\mathbf{SG}_x$ is an equal, coherent linear superposition of $x+$ and $x-$.

Step 3: Measurement at Apparatus 2 and Second Wavefunction Collapse. Inside $\mathbf{SG}_x$, the field gradient $\frac{\partial B_x}{\partial x} \hat{x}$ splits the beam horizontally into two distinct components:

- The $x+$ beam ($S_x = +\hbar/2$) with probability $P(x+ | z+) = |\langle x+ | z+ \rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$.
- The $x-$ beam ($S_x = -\hbar/2$) with probability $P(x- | z+) = |\langle x- | z+ \rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$.

We place an absorber plate blocking the $x-$ beam, selecting only the $x+$ beam. The act of measurement/selection projects the state vector onto $|x+\rangle$:

$$|\psi\rangle \equiv |x+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (5.3.85)$$

The state function is once again redefined! The physical system is now described by the new state $|\psi\rangle \equiv |x+\rangle$. What happened to the particle's prior identity of being $S_z = +\hbar/2$? **It has been completely erased!** The state $|\psi\rangle \equiv |x+\rangle$ retains zero memory of the fact that it was previously filtered to be $z+$.

Step 4: Apparatus 3 and the Spontaneous Regeneration of $z-$. Now we send this beam into the third apparatus, oriented along $\hat{z}$ again: **SG_z**. The observable to be measured is $\hat{S}_z$. Once again, $[\hat{S}_x, \hat{S}_z] = -i\hbar\hat{S}_y \neq 0$. The state $|\psi\rangle \equiv |x+\rangle$ is not an eigenstate of $\hat{S}_z$. To predict the outcomes of apparatus 3, we must rewrite $|\psi\rangle$ in the eigenbasis of the observable about to be measured ($\hat{S}_z$):

$$|\psi\rangle \equiv |x+\rangle = \frac{1}{\sqrt{2}} |z+\rangle + \frac{1}{\sqrt{2}} |z-\rangle. \quad (5.3.86)$$

The beam enters **SG_z** and splits vertically into two beams! Evaluating the transition probabilities registered at the final detectors:

$$P(z+ | x+) = |\langle z+ | x+ \rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}, \quad (5.3.87)$$

$$P(z- | x+) = |\langle z- | x+ \rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}. \quad (5.3.88)$$

Upon detection:

$$|\psi\rangle \equiv \begin{cases} |z+\rangle & \text{if detector } D_{z+} \text{ clicks (with probability } 1/2), \\ |z-\rangle & \text{if detector } D_{z-} \text{ clicks (with probability } 1/2). \end{cases} \quad (5.3.89)$$

The Fundamental Lesson: The $z-$ beam, which had been 100% eliminated in Stage 1, has **spontaneously reappeared with 50% probability!** Why? Because measurement in quantum mechanics is not a passive reading of pre-existing attributes. The act of measuring the non-commuting observable $\hat{S}_x$ fundamentally **modified the physical state itself**, actively transforming the state vector from $|\psi\rangle \equiv |z+\rangle$ into $|\psi\rangle \equiv |x+\rangle$, thereby creating a renewed non-zero projection onto $|z-\rangle$.

The Markovian Perspective: Quantum Measurement as a Memoryless Stochastic Process

The sequential Stern–Gerlach experiment provides an exact physical realization of a **Markovian stochastic process** in physics.

1. The Markov Property. In probability theory, a discrete-time stochastic process $\{S_0, S_1, S_2, \dots, S_n, \dots\}$ taking values in a state space is defined to possess the **Markov property** (memorylessness) if the conditional probability distribution of the future state S_{n+1} depends *only* upon the present state S_n , and is completely independent of the past historical trajectory $\{S_0, S_1, \dots, S_{n-1}\}$:

$$P(S_{n+1} = s_{n+1} | S_n = s_n, S_{n-1} = s_{n-1}, \dots, S_0 = s_0) = P(S_{n+1} = s_{n+1} | S_n = s_n). \quad (5.3.90)$$

2. Why Quantum Collapse is Strictly Markovian. According to the von Neumann–Lüders projection postulate, whenever an observable $\hat{A}_n$ is measured at step n and outcome s_n is registered, the state vector collapses to the corresponding eigenstate:

$$|\psi_n\rangle = \frac{\hat{P}_{s_n} |\psi_{n-1}\rangle}{\|\hat{P}_{s_n} |\psi_{n-1}\rangle\|} = |s_n\rangle. \quad (5.3.91)$$

The post-measurement state vector $|\psi_n\rangle \equiv |s_n\rangle$ provides a complete specification of the physical state of the quantum system at step n . When the next measurement of observable $\hat{A}_{n+1}$ is performed at step $n+1$, the transition probability to outcome s_{n+1} is given strictly by the Born rule applied to the current state:

$$P(s_{n+1} | s_n, s_{n-1}, \dots, s_0) = |\langle s_{n+1} | \psi_n \rangle|^2 = |\langle s_{n+1} | s_n \rangle|^2 = P(s_{n+1} | s_n). \quad (5.3.92)$$

The past history $\{s_0, s_1, \dots, s_{n-1}\}$ has **zero predictive power** once the present state $|\psi_n\rangle \equiv |s_n\rangle$ is established! The physical collapse of the wavefunction erases all historical memory of how the system arrived at state $|s_n\rangle$. This is the exact realization of a discrete-time Markov chain.

3. The Unistochastic Transition Matrix. The transition probabilities between successive Stern–Gerlach filters form a transition matrix T :

$$T_{ji} = P(s_{n+1}^{(j)} | s_n^{(i)}) = |\langle s_{n+1}^{(j)} | s_n^{(i)} \rangle|^2. \quad (5.3.93)$$

For transitions between the $\hat{z}$ -basis $\{|z+\rangle, |z-\rangle\}$ and the $\hat{x}$ -basis $|x+\rangle, |x-\rangle$:

$$T_{z \rightarrow x} = \begin{pmatrix} P(x+ | z+) & P(x+ | z-) \\ P(x- | z+) & P(x- | z-) \end{pmatrix} = \begin{pmatrix} |\langle x+ | z+ \rangle|^2 & |\langle x+ | z- \rangle|^2 \\ |\langle x- | z+ \rangle|^2 & |\langle x- | z- \rangle|^2 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}. \quad (5.3.94)$$

Notice that the transition matrix T is **doubly stochastic**:

$$\sum_j T_{ji} = 1 \quad (\text{columns sum to } 1), \quad \sum_i T_{ji} = 1 \quad (\text{rows sum to } 1). \quad (5.3.95)$$

Because each entry $T_{ji} = |U_{ji}|^2$ is the absolute square of a unitary basis transformation matrix U , T is specifically a **unistochastic matrix**.

4. Measurement Collapse vs. Coherent Recombination (The Origin of the Markov Property). To appreciate why wavefunction collapse generates this Markovian behavior, consider what happens if we *do not* perform a measurement at Stage 2! Suppose that instead of blocking the $x-$ beam, we allow both $x+$ and $x-$ beams to travel through Stage 2, and then use an inverted magnetic field to recombine the two beams coherently without detecting which path the particle traversed (a Stern–Gerlach interferometer). What is the state exiting the recombining apparatus? Because no measurement occurred, no wavefunction collapse took place! The quantum probability amplitudes interfere:

$$\langle z+ | \psi_{\text{recombined}} \rangle = \langle z+ | x+ \rangle \langle x+ | z+ \rangle + \langle z+ | x- \rangle \langle x- | z+ \rangle = \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{2} + \frac{1}{2} = 1, \quad (5.3.96)$$

$$\langle z- | \psi_{\text{recombined}} \rangle = \langle z- | x+ \rangle \langle x+ | z+ \rangle + \langle z- | x- \rangle \langle x- | z+ \rangle = \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) + \left(-\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{2} - \frac{1}{2} = 0. \quad (5.3.97)$$

When the beams are recombined coherently without measurement:

$$P(z+) = |1|^2 = 100\%, \quad P(z-) = |0|^2 = 0\%. \quad (5.3.98)$$

The $z-$ beam does *not* reappear! The system remembers its original $z+$ preparation perfectly through quantum phase coherence!

By contrast, when we placed the absorber in Stage 2 to detect/select $x+$, we performed an irreversible measurement:

$$\begin{aligned} P(z+ \rightarrow x+ \rightarrow z-) &= P(z- | x+)P(x+ | z+) \\ &= |\langle z- | x+ \rangle|^2 |\langle x+ | z+ \rangle|^2 = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{1}{4} = 25\%. \end{aligned} \quad (5.3.99)$$

This comparison proves the central theorem of quantum measurement:

Wavefunction collapse destroys quantum phase coherence between alternative branches, converting unitary interference into a memoryless, classical Markov process!

Quantum Measurement as a Markovian State-Reset Process

1. Active Modification of Reality: Measurement in quantum mechanics is not a passive observation of pre-existing attributes. Measuring an observable $\hat{A}$ physically alters the state vector, collapsing it into an eigenstate of $\hat{A}$.

2. Incompatible Observables & Mandatory Basis Rotation: When a system prepared in an eigenstate of $\hat{A}$ is presented to an apparatus measuring an incompatible observable $\hat{B}$ ($[\hat{A}, \hat{B}] \neq 0$), the state must be expanded in the eigenbasis of the observable about to be measured:

$$|\psi\rangle \equiv |a_i\rangle = \sum_j \langle b_j | a_i \rangle |b_j\rangle. \quad (1)$$

3. Exact State-Reset Mechanism ($|\psi\rangle \equiv \dots$): Upon measurement yielding outcome b_j , the physical description is updated:

$$|\psi\rangle \equiv |b_j\rangle. \quad (2)$$

The wavefunction is completely redefined, erasing all memory of the prior state $|a_i\rangle$.

4. The Markov Property of Quantum State Collapse: Because the collapsed state $|\psi_n\rangle \equiv |s_n\rangle$ completely determines all future probabilities via the Born rule:

$$P(s_{n+1} | s_n, s_{n-1}, \dots, s_0) = |\langle s_{n+1} | s_n \rangle|^2 = P(s_{n+1} | s_n), \quad (3)$$

successive quantum measurements constitute an exact **discrete-time Markov chain** with doubly stochastic transition matrix $T_{ji} = |\langle s_{n+1}^{(j)} | s_n^{(i)} \rangle|^2$.

5. Coherence vs. Markovian Transitions: Without measurement, quantum evolution is unitary and reversible; amplitudes add, and interference preserves historical phase memory. Measurement destroys interference cross-terms, replacing unitary superposition with a classical, memoryless stochastic transition.

5.3.7 Two-Level System Perturbation and Avoided Level Crossing

A ubiquitous paradigm in spin physics and quantum optics is the perturbed two-level system. Let an unperturbed Hamiltonian $\hat{H}_0$ possess two orthonormal eigenstates:

$$\hat{H}_0 |\psi_1\rangle = E_1 |\psi_1\rangle, \quad \hat{H}_0 |\psi_2\rangle = E_2 |\psi_2\rangle, \quad \langle \psi_1 | \psi_2 \rangle = 0. \quad (5.3.100)$$

We apply a Hermitian perturbation $\hat{W} = \hat{W}^\dagger$, whose matrix elements in the unperturbed basis are:

$$W_{11} = \langle \psi_1 | \hat{W} | \psi_1 \rangle, \quad W_{22} = \langle \psi_2 | \hat{W} | \psi_2 \rangle, \quad W_{12} = \langle \psi_1 | \hat{W} | \psi_2 \rangle = W_{21}^*. \quad (5.3.101)$$

The total Hamiltonian is represented by the 2×2 Hermitian matrix:

$$\hat{H} = \hat{H}_0 + \hat{W} \doteq \begin{pmatrix} E_1 + W_{11} & W_{12} \\ W_{12}^* & E_2 + W_{22} \end{pmatrix} \equiv \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{pmatrix}. \quad (5.3.102)$$

The secular determinant $\det(\hat{H} - E\hat{I}) = (H_{11} - E)(H_{22} - E) - |H_{12}|^2 = 0$ yields the exact perturbed energy eigenvalues:

$$E_{\pm} = \frac{H_{11} + H_{22}}{2} \pm \sqrt{\left(\frac{H_{11} - H_{22}}{2}\right)^2 + |H_{12}|^2} \quad (5.3.103)$$

Avoided Level Crossing (Level Repulsion): Consider tuning a physical parameter (such as an external electric or magnetic field) so that the diagonal energies cross ($H_{11} = H_{22}$). At the unperturbed crossing point, Eq. (5.3.103) reveals:

$$E_{\pm} = H_{11} \pm |H_{12}| \implies \Delta E \equiv E_+ - E_- = 2|H_{12}| > 0. \quad (5.3.104)$$

Whenever the off-diagonal coupling $H_{12} = \langle \psi_1 | \hat{W} | \psi_2 \rangle$ is non-zero, the two energy levels **refuse to cross!** Instead, they repel each other with an energy gap of $2|H_{12}|$.

Time-Dependent Transitions and Rabi Oscillations: Because the unperturbed states $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal ($\langle \psi_1 | \psi_2 \rangle = 0$), this does *not* imply that their matrix elements under $\hat{W}$ vanish ($\langle \psi_1 | \hat{W} | \psi_2 \rangle \neq 0$). If a system is initialized in state $|\psi_1\rangle$ at $t = 0$, the off-diagonal coupling drives coherent population transfer between the two levels with transition probability:

$$P_{1 \rightarrow 2}(t) = \frac{4|H_{12}|^2}{(H_{11} - H_{22})^2 + 4|H_{12}|^2} \sin^2\left(\frac{\Omega_{\text{Rabi}} t}{2}\right), \quad (5.3.105)$$

where $\Omega_{\text{Rabi}} \equiv \frac{1}{\hbar} \sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}$ is the generalized **Rabi oscillation frequency**.

5.3.8 Matrix Representations: Spin-1

For $j = 1$, the dimension is $2(1) + 1 = 3$, spanned by $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$. Using the ladder formulas $J_{\pm}|1, m\rangle = \hbar\sqrt{2 - m(m \pm 1)}|1, m \pm 1\rangle$:

$$J_+|1, -1\rangle = \sqrt{2}\hbar|1, 0\rangle, \quad J_+|1, 0\rangle = \sqrt{2}\hbar|1, 1\rangle, \quad J_+|1, 1\rangle = 0, \quad (5.3.106)$$

$$J_-|1, 1\rangle = \sqrt{2}\hbar|1, 0\rangle, \quad J_-|1, 0\rangle = \sqrt{2}\hbar|1, -1\rangle, \quad J_-|1, -1\rangle = 0. \quad (5.3.107)$$

In the basis ordered as $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$:

$$J_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad J_+ = \hbar \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}, \quad J_- = \hbar \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}. \quad (5.3.108)$$

Since $J_x = \frac{1}{2}(J_+ + J_-)$ and $J_y = \frac{1}{2i}(J_+ - J_-)$:

$$J_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}. \quad (5.3.109)$$

One readily confirms that $J^2 = J_x^2 + J_y^2 + J_z^2 = 2\hbar^2 I_{3 \times 3}$.

5.3.9 Differential Operator Representation in Spherical Coordinates

For orbital angular momentum $\vec{L} = \vec{r} \times \vec{p}$, we transform from Cartesian coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ) :

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta. \quad (5.3.110)$$

Using the chain rule, the differential operators take the standard form:

$$L_z = -i\hbar \frac{\partial}{\partial \phi}, \quad (5.3.111)$$

$$L_{\pm} = \hbar e^{\pm i\phi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right), \quad (5.3.112)$$

$$L_x = \frac{1}{2}(L_+ + L_-) = i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right), \quad (5.3.113)$$

$$L_y = \frac{1}{2i}(L_+ - L_-) = -i\hbar \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right). \quad (5.3.114)$$

The Casimir operator $L^2 = L_+ L_- + L_z^2 - \hbar L_z$ becomes

$$L^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (5.3.115)$$

The simultaneous eigenfunctions of L^2 and L_z are the **spherical harmonics** $Y_l^m(\theta, \phi)$:

$$L^2 Y_l^m(\theta, \phi) = \hbar^2 l(l+1) Y_l^m(\theta, \phi), \quad L_z Y_l^m(\theta, \phi) = \hbar m Y_l^m(\theta, \phi). \quad (5.3.116)$$

Because spatial rotations by 2π in ϕ return to the exact same physical coordinate point in $\mathbb{R}^3$, the single-valuedness requirement $\psi(\phi + 2\pi) = \psi(\phi)$ forces the orbital quantum number l to be strictly an **integer**:

$$l \in \{0, 1, 2, 3, \dots\}, \quad m \in \{-l, -l+1, \dots, l\}. \quad (5.3.117)$$

Half-integer angular momenta cannot arise as single-valued spatial wavefunctions in $\mathbb{R}^3$; they represent intrinsic quantum degrees of freedom (spin).

5.4 Addition of Angular Momentum and Clebsch–Gordan Coefficients

5.4.1 Composite Systems and Basis Choices

Consider a quantum system composed of two subsystems (e.g. the spin and orbital degrees of freedom of a single particle, or the spins of two distinct electrons). Let $\vec{J}_1$ act on Hilbert space $\mathcal{H}_1$ (dimension $2j_1 + 1$) and $\vec{J}_2$ act on $\mathcal{H}_2$ (dimension $2j_2 + 1$). The total Hilbert space is the tensor product:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2, \quad \dim(\mathcal{H}) = (2j_1 + 1)(2j_2 + 1). \quad (5.4.1)$$

The total angular momentum operator is

$$\vec{J} = \vec{J}_1 \otimes \hat{I}_2 + \hat{I}_1 \otimes \vec{J}_2 \equiv \vec{J}_1 + \vec{J}_2. \quad (5.4.2)$$

Because $\vec{J}_1$ and $\vec{J}_2$ act on distinct spaces, they commute: $[J_{1i}, J_{2j}] = 0$. Consequently, the total angular momentum satisfies the standard Lie algebra:

$$[J_i, J_j] = [J_{1i} + J_{2i}, J_{1j} + J_{2j}] = [J_{1i}, J_{1j}] + [J_{2i}, J_{2j}] = i\hbar \epsilon_{ijk} (J_{1k} + J_{2k}) = i\hbar \epsilon_{ijk} J_k. \quad (5.4.3)$$

There are two natural orthonormal bases for describing $\mathcal{H}$:

1. **Uncoupled Basis:** Eigenstates of $\{J_1^2, J_{1z}, J_2^2, J_{2z}\}$:

$$|j_1, m_1; j_2, m_2\rangle \equiv |j_1, m_1\rangle \otimes |j_2, m_2\rangle. \quad (5.4.4)$$

2. **Coupled Basis:** Eigenstates of $\{J_1^2, J_2^2, J^2, J_z\}$, denoted $|j_1, j_2; J, M\rangle$ (or simply $|J, M\rangle$).

Notice why we cannot diagonalize $\{J^2, J_{1z}\}$ simultaneously. Evaluating their commutator:

$$\begin{aligned} [J^2, J_{1z}] &= [(\vec{J}_1 + \vec{J}_2)^2, J_{1z}] = [J_1^2 + J_2^2 + 2\vec{J}_1 \cdot \vec{J}_2, J_{1z}] \\ &= 2[J_{1x}J_{2x} + J_{1y}J_{2y} + J_{1z}J_{2z}, J_{1z}] = 2[J_{1x}, J_{1z}]J_{2x} + 2[J_{1y}, J_{1z}]J_{2y} \\ &= -2i\hbar J_{1y}J_{2x} + 2i\hbar J_{1x}J_{2y} = 2i\hbar(J_{1x}J_{2y} - J_{1y}J_{2x}) \neq 0. \end{aligned} \quad (5.4.5)$$

Therefore, J^2 does not commute with J_{1z} (or J_{2z}). An interaction that preserves total angular momentum $\vec{J}^2$ mixes the individual projections m_1 and m_2 !

5.4.2 Allowed Values of Total Angular Momentum: The Triangle Rule

Since $J_z = J_{1z} + J_{2z}$, the total magnetic quantum number must be additive:

$$M = m_1 + m_2. \quad (5.4.6)$$

The maximum value of M is $M_{\max} = j_1 + j_2$, obtained uniquely from $m_1 = j_1, m_2 = j_2$. This state must belong to a multiplet with total angular momentum $J = j_1 + j_2$. By applying the total lowering operator $J_- = J_{1-} + J_{2-}$, we generate all $2(j_1 + j_2) + 1$ states of this multiplet down to $M = -(j_1 + j_2)$.

Next, for $M = j_1 + j_2 - 1$, there are two linearly independent uncoupled states:

$$|j_1, j_1 - 1; j_2, j_2\rangle \quad \text{and} \quad |j_1, j_1; j_2, j_2 - 1\rangle. \quad (5.4.7)$$

One linear combination is the $M = j_1 + j_2 - 1$ member of the $J = j_1 + j_2$ multiplet. The orthogonal linear combination is annihilated by J_+ and therefore serves as the highest-weight state of a new multiplet with $J = j_1 + j_2 - 1$.

Continuing this decomposition recursively, the total angular momentum J takes every value in integer steps between $|j_1 - j_2|$ and $j_1 + j_2$:

$$\boxed{|j_1 - j_2| \leq J \leq j_1 + j_2, \quad \Delta J = 1.} \quad (5.4.8)$$

We verify dimension conservation by summing the dimensions of all irreducible multiplets (assuming without loss of generality $j_1 \geq j_2$):

$$\begin{aligned} \sum_{J=|j_1-j_2|}^{j_1+j_2} (2J+1) &= \sum_{k=0}^{2j_2} [2(j_1 + j_2 - k) + 1] \\ &= (2j_2 + 1)(2j_1 + 2j_2 + 1) - 2 \frac{2j_2(2j_2 + 1)}{2} \\ &= (2j_2 + 1)(2j_1 + 1). \end{aligned} \quad (5.4.9)$$

The dimension of the Hilbert space is conserved exactly.

5.4.3 Addition of Two Spin-1/2 Particles: Singlet and Triplet

Let us couple two spin-1/2 particles: $j_1 = 1/2, j_2 = 1/2$. The allowed values of total angular momentum are:

$$J = \frac{1}{2} + \frac{1}{2} = 1 \quad (\text{Triplet}), \quad J = \frac{1}{2} - \frac{1}{2} = 0 \quad (\text{Singlet}). \quad (5.4.10)$$

The total Hilbert space dimension is $2 \times 2 = 4 = 3 \oplus 1$.

The maximum projection state is $|J = 1, M = 1\rangle$:

$$|1, 1\rangle = |\uparrow\uparrow\rangle. \quad (5.4.11)$$

Applying the total lowering operator $J_- = S_{1-} + S_{2-}$:

$$J_-|1, 1\rangle = \hbar\sqrt{1(2) - 1(0)}|1, 0\rangle = \sqrt{2}\hbar|1, 0\rangle. \quad (5.4.12)$$

Meanwhile, acting on the product state:

$$(S_{1-} + S_{2-})|\uparrow\uparrow\rangle = \hbar|\downarrow\uparrow\rangle + \hbar|\uparrow\downarrow\rangle. \quad (5.4.13)$$

Equating both sides yields the normalized $M = 0$ triplet state:

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle). \quad (5.4.14)$$

Lowering once more gives

$$|1, -1\rangle = |\downarrow\downarrow\rangle. \quad (5.4.15)$$

The singlet state $|0, 0\rangle$ must have $M = 0$ and be orthogonal to $|1, 0\rangle$:

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (5.4.16)$$

Notice the profound physical symmetry:

- The **triplet states** ($J = 1$) are **symmetric** under particle exchange $1 \leftrightarrow 2$.
- The **singlet state** ($J = 0$) is **antisymmetric** under particle exchange $1 \leftrightarrow 2$.

By the Pauli exclusion principle, the total wavefunction of two identical fermions must be overall antisymmetric. If the spatial wavefunction is symmetric (e.g. the ground state of helium, or electrons in a covalent bond), the spin state *must* be the antisymmetric singlet $J = 0$. Conversely, if the spatial wavefunction is antisymmetric, the spin state is the symmetric triplet $J = 1$.

5.4.4 Clebsch–Gordan Recursion Relations

The unitary transformation connecting the uncoupled and coupled bases is given by the **Clebsch–Gordan (CG) coefficients**:

$$|j_1, j_2; J, M\rangle = \sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | J, M \rangle |j_1, m_1; j_2, m_2\rangle. \quad (5.4.17)$$

By the Condon–Shortley convention, all CG coefficients are chosen to be strictly real:

$$\langle j_1, m_1; j_2, m_2 | J, M \rangle = \langle J, M | j_1, m_1; j_2, m_2 \rangle \in \mathbb{R}. \quad (5.4.18)$$

They satisfy the orthogonality relations:

$$\sum_{m_1, m_2} \langle J, M | j_1, m_1; j_2, m_2 \rangle \langle j_1, m_1; j_2, m_2 | J', M' \rangle = \delta_{JJ'} \delta_{MM'}, \quad (5.4.19)$$

$$\sum_{J, M} \langle j_1, m_1; j_2, m_2 | J, M \rangle \langle J, M | j_1, m'_1; j_2, m'_2 \rangle = \delta_{m_1 m'_1} \delta_{m_2 m'_2}. \quad (5.4.20)$$

To derive the master recursion relations, we act with the total ladder operator $J_{\pm} = J_{1\pm} + J_{2\pm}$ on the coupled state:

$$J_{\pm}|J, M\rangle = \hbar\sqrt{J(J+1) - M(M \pm 1)}|J, M \pm 1\rangle. \quad (5.4.21)$$

Taking the inner product with an arbitrary uncoupled bra $\langle j_1, m_1; j_2, m_2 |$:

$$\langle j_1, m_1; j_2, m_2 | J_{\pm} | J, M \rangle = \hbar \sqrt{J(J+1) - M(M \pm 1)} \langle j_1, m_1; j_2, m_2 | J, M \pm 1 \rangle. \quad (5.4.22)$$

On the other hand, letting $J_{\pm} = J_{1\pm} + J_{2\pm}$ act to the left onto the bra using $(J_{i\pm})^\dagger = J_{i\mp}$:

$$\begin{aligned} \langle j_1, m_1; j_2, m_2 | (J_{1\pm} + J_{2\pm}) = & \hbar \sqrt{j_1(j_1+1) - m_1(m_1 \mp 1)} \langle j_1, m_1 \mp 1; j_2, m_2 | \\ & + \hbar \sqrt{j_2(j_2+1) - m_2(m_2 \mp 1)} \langle j_1, m_1; j_2, m_2 \mp 1 |. \end{aligned} \quad (5.4.23)$$

Equating the two expressions gives the **Clebsch–Gordan recursion relations**:

$$\begin{aligned} & \sqrt{J(J+1) - M(M \pm 1)} \langle j_1, m_1; j_2, m_2 | J, M \pm 1 \rangle \\ &= \sqrt{j_1(j_1+1) - m_1(m_1 \mp 1)} \langle j_1, m_1 \mp 1; j_2, m_2 | J, M \rangle \\ &+ \sqrt{j_2(j_2+1) - m_2(m_2 \mp 1)} \langle j_1, m_1; j_2, m_2 \mp 1 | J, M \rangle. \end{aligned} \quad (5.4.24)$$

5.4.5 Explicit Construction: Coupling $j_1 = 1$ and $j_2 = 1/2$

Let us apply this machinery to compute the Clebsch–Gordan coefficients for coupling an orbital angular momentum $j_1 = 1$ to a spin $j_2 = 1/2$. The allowed total angular momentum values are

$$J = 1 + \frac{1}{2} = \frac{3}{2}, \quad J = 1 - \frac{1}{2} = \frac{1}{2}. \quad (5.4.25)$$

The total dimension is $(2 \times 1 + 1)(2 \times \frac{1}{2} + 1) = 3 \times 2 = 6 = 4 \oplus 2$.

1. The $J = 3/2$ multiplet: The highest weight state is

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle = |1, 1; \frac{1}{2}, \frac{1}{2}\rangle = |1, 1\rangle |\uparrow\rangle. \quad (5.4.26)$$

Applying the lowering operator $J_- = J_{1-} + J_{2-}$:

$$J_- \left| \frac{3}{2}, \frac{3}{2} \right\rangle = \hbar \sqrt{\frac{3}{2}(\frac{5}{2}) - \frac{3}{2}(\frac{1}{2})} \left| \frac{3}{2}, \frac{1}{2} \right\rangle = \sqrt{3}\hbar \left| \frac{3}{2}, \frac{1}{2} \right\rangle, \quad (5.4.27)$$

$$(J_{1-} + J_{2-})|1, 1\rangle |\uparrow\rangle = \sqrt{2}\hbar |1, 0\rangle |\uparrow\rangle + \hbar |1, 1\rangle |\downarrow\rangle. \quad (5.4.28)$$

Dividing by $\sqrt{3}\hbar$:

$$\left| \frac{3}{2}, \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} |1, 0\rangle |\uparrow\rangle + \frac{1}{\sqrt{3}} |1, 1\rangle |\downarrow\rangle. \quad (5.4.29)$$

Lowering again:

$$J_- \left| \frac{3}{2}, \frac{1}{2} \right\rangle = \hbar \sqrt{\frac{3}{2}(\frac{5}{2}) - \frac{1}{2}(-\frac{1}{2})} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle = 2\hbar \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \quad (5.4.30)$$

$$\begin{aligned} (J_{1-} + J_{2-}) \left| \frac{3}{2}, \frac{1}{2} \right\rangle &= \sqrt{\frac{2}{3}} (\sqrt{2}\hbar |1, -1\rangle |\uparrow\rangle + \hbar |1, 0\rangle |\downarrow\rangle) \\ &+ \frac{1}{\sqrt{3}} (\sqrt{2}\hbar |1, 0\rangle |\downarrow\rangle) \\ &= \frac{2}{\sqrt{3}} \hbar |1, -1\rangle |\uparrow\rangle + \frac{2\sqrt{2}}{\sqrt{3}} \hbar |1, 0\rangle |\downarrow\rangle. \end{aligned} \quad (5.4.31)$$

Dividing by $2\hbar$:

$$\left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} |1, -1\rangle |\uparrow\rangle + \sqrt{\frac{2}{3}} |1, 0\rangle |\downarrow\rangle. \quad (5.4.32)$$

Finally, lowering once more gives:

$$\left| \frac{3}{2}, -\frac{3}{2} \right\rangle = |1, -1\rangle |\downarrow\rangle. \quad (5.4.33)$$

2. The $J = 1/2$ multiplet: The state $|1/2, 1/2\rangle$ must have $M = 1/2$ and be orthonormal to $|3/2, 1/2\rangle$. Setting $|1/2, 1/2\rangle = \alpha|1, 0\rangle |\uparrow\rangle + \beta|1, 1\rangle |\downarrow\rangle$ and using the Condon–Shortley convention (which fixes the coefficient of the state with highest m_1 to be positive, so $\beta > 0$):

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle = -\frac{1}{\sqrt{3}}|1, 0\rangle |\uparrow\rangle + \sqrt{\frac{2}{3}}|1, 1\rangle |\downarrow\rangle. \quad (5.4.34)$$

Lowering with J_- yields:

$$\left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\sqrt{\frac{2}{3}}|1, -1\rangle |\uparrow\rangle + \frac{1}{\sqrt{3}}|1, 0\rangle |\downarrow\rangle. \quad (5.4.35)$$

This completes the exact expansion for all 6 states in the coupled basis.

5.5 Irreducible Spherical Tensors and the Wigner–Eckart Theorem

5.5.1 Spherical Basis and Spherical Tensor Operators

In Section 5.2, we demonstrated that Cartesian tensors are reducible under rotations. To harness the full power of rotational symmetry, we construct operators that transform under rotations according to the *irreducible* representations of $SO(3)$.

Recall that under a spatial rotation, the spherical harmonics $Y_k^q(\theta, \phi)$ transform among themselves according to the Wigner $\mathcal{D}$ -matrices:

$$\hat{R} Y_k^q(\theta, \phi) = \sum_{q'=-k}^k \mathcal{D}_{q'q}^{(k)}(R) Y_k^{q'}(\theta, \phi). \quad (5.5.1)$$

An **irreducible spherical tensor operator of rank k** is defined as a set of $2k + 1$ operators $T_q^{(k)}$ ($q = -k, -k + 1, \dots, k$) that transform under rotations exactly like the spherical harmonics:

$$\hat{R}^\dagger T_q^{(k)} \hat{R} = \sum_{q'=-k}^k \mathcal{D}_{q'q}^{(k)}(R) T_{q'}^{(k)}. \quad (5.5.2)$$

Considering an infinitesimal rotation generated by $\vec{J}$, the finite transformation law (5.5.2) translates into the fundamental commutation relations:

$$[J_z, T_q^{(k)}] = \hbar q T_q^{(k)}, \quad (5.5.3)$$

$$[J_\pm, T_q^{(k)}] = \hbar \sqrt{k(k+1) - q(q \pm 1)} T_{q \pm 1}^{(k)}. \quad (5.5.4)$$

Any set of $2k + 1$ operators satisfying these commutation relations constitutes an irreducible spherical tensor of rank k .

Spherical components of a vector operator ($k = 1$): For any physical vector operator $\vec{V} = (V_x, V_y, V_z)$, the spherical components are defined in direct analogy with the dipole spherical harmonics Y_1^q :

$$T_{+1}^{(1)} \equiv V_{+1} = -\frac{1}{\sqrt{2}}(V_x + iV_y), \quad (5.5.5)$$

$$T_0^{(1)} \equiv V_0 = V_z, \quad (5.5.6)$$

$$T_{-1}^{(1)} \equiv V_{-1} = \frac{1}{\sqrt{2}}(V_x - iV_y). \quad (5.5.7)$$

Let us explicitly verify the commutation relations using $[J_i, V_j] = i\hbar\epsilon_{ijk}V_k$:

$$\begin{aligned} [J_z, V_{+1}] &= -\frac{1}{\sqrt{2}}([J_z, V_x] + i[J_z, V_y]) = -\frac{1}{\sqrt{2}}(i\hbar V_y + i(-i\hbar V_x)) \\ &= -\frac{\hbar}{\sqrt{2}}(V_x + iV_y) = \hbar(+1)V_{+1}, \end{aligned} \quad (5.5.8)$$

$$[J_z, V_0] = [J_z, V_z] = 0 = \hbar(0)V_0, \quad (5.5.9)$$

$$\begin{aligned} [J_z, V_{-1}] &= \frac{1}{\sqrt{2}}([J_z, V_x] - i[J_z, V_y]) = \frac{1}{\sqrt{2}}(i\hbar V_y - i(-i\hbar V_x)) \\ &= -\frac{\hbar}{\sqrt{2}}(V_x - iV_y) = \hbar(-1)V_{-1}. \end{aligned} \quad (5.5.10)$$

Furthermore, checking J_+ on V_0 :

$$\begin{aligned} [J_+, V_0] &= [J_x + iJ_y, V_z] = [J_x, V_z] + i[J_y, V_z] = -i\hbar V_y + i(i\hbar V_x) \\ &= -\hbar(V_x + iV_y) = \sqrt{2}\hbar V_{+1} = \hbar\sqrt{1(2) - 0(1)}V_{+1}. \end{aligned} \quad (5.5.11)$$

The commutation relations match equations (5.5.3)–(5.5.4) with mathematical precision.

5.5.2 Tensor Products of Spherical Tensors

Given two irreducible spherical tensor operators $A^{(k_1)}$ and $B^{(k_2)}$, their direct product can be decomposed into irreducible spherical tensors $T_Q^{(K)}$ of rank K using Clebsch–Gordan coefficients:

$$T_Q^{(K)} = \sum_{q_1, q_2} \langle k_1, q_1; k_2, q_2 | K, Q \rangle A_{q_1}^{(k_1)} B_{q_2}^{(k_2)}. \quad (5.5.12)$$

For example, coupling two vector operators $A^{(1)}$ and $B^{(1)}$ gives ranks $K = 0, 1, 2$:

- **Rank 0 (Scalar):**

$$T_0^{(0)} = \sum_q \langle 1, q; 1, -q | 0, 0 \rangle A_q^{(1)} B_{-q}^{(1)} = -\frac{1}{\sqrt{3}} \vec{A} \cdot \vec{B}. \quad (5.5.13)$$

- **Rank 1 (Cross product):**

$$T_Q^{(1)} = \sum_{q_1, q_2} \langle 1, q_1; 1, q_2 | 1, Q \rangle A_{q_1}^{(1)} B_{q_2}^{(1)} = -\frac{i}{\sqrt{2}} (\vec{A} \times \vec{B})_Q. \quad (5.5.14)$$

- **Rank 2 (Quadrupole tensor):**

$$T_{\pm 2}^{(2)} = A_{\pm 1}^{(1)} B_{\pm 1}^{(1)} = \frac{1}{2} (A_x \pm iA_y)(B_x \pm iB_y), \quad (5.5.15)$$

$$\begin{aligned} T_{\pm 1}^{(2)} &= \frac{1}{\sqrt{2}} (A_{\pm 1}^{(1)} B_0^{(1)} + A_0^{(1)} B_{\pm 1}^{(1)}) \\ &= \mp \frac{1}{2} [(A_x \pm iA_y)B_z + A_z(B_x \pm iB_y)], \end{aligned} \quad (5.5.16)$$

$$\begin{aligned} T_0^{(2)} &= \frac{1}{\sqrt{6}} (A_{+1}^{(1)} B_{-1}^{(1)} + 2A_0^{(1)} B_0^{(1)} + A_{-1}^{(1)} B_{+1}^{(1)}) \\ &= \frac{1}{\sqrt{6}} (2A_z B_z - A_x B_x - A_y B_y). \end{aligned} \quad (5.5.17)$$

5.5.3 The Wigner–Eckart Theorem: Formulation and Proof

We now arrive at one of the crowning achievements of quantum theory: the **Wigner–Eckart Theorem**.

Theorem (Wigner–Eckart Theorem). *The matrix elements of an irreducible spherical tensor operator $T_q^{(k)}$ between angular momentum eigenstates factorize into the product of a Clebsch–Gordan coefficient (which encodes all geometric and rotational symmetry) and a **reduced matrix element** (which depends solely on the physical dynamics and is completely independent of the magnetic quantum numbers m, m', q):*

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = \langle j, m; k, q | j', m' \rangle \frac{\langle \alpha', j' || T^{(k)} || \alpha, j \rangle}{\sqrt{2j' + 1}}, \quad (5.5.18)$$

where α, α' denote any additional quantum numbers (such as principal quantum number n or radial excitation).

Proof. Let us evaluate the matrix element of the commutator $[J_z, T_q^{(k)}] = \hbar q T_q^{(k)}$ between states $\langle \alpha', j', m' |$ and $|\alpha, j, m\rangle$:

$$\langle \alpha', j', m' | (J_z T_q^{(k)} - T_q^{(k)} J_z) | \alpha, j, m \rangle = \hbar q \langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle. \quad (5.5.19)$$

Allowing J_z to act on the bra to the left and ket to the right:

$$(m' - m) \hbar \langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = \hbar q \langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle, \quad (5.5.20)$$

which rearranges to:

$$(m' - m - q) \langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = 0. \quad (5.5.21)$$

Therefore, the matrix element vanishes identically unless:

$$m' = m + q \quad \Longleftrightarrow \quad \Delta m = q. \quad (5.5.22)$$

This is precisely the same selection rule obeyed by the Clebsch–Gordan coefficient $\langle j, m; k, q | j', m' \rangle$.

Next, evaluate the matrix element of the ladder commutator:

$$\langle \alpha', j', m' | [J_{\pm}, T_q^{(k)}] | \alpha, j, m \rangle = \hbar \sqrt{k(k+1) - q(q \pm 1)} \langle \alpha', j', m' | T_{q \pm 1}^{(k)} | \alpha, j, m \rangle. \quad (5.5.23)$$

Expanding $[J_{\pm}, T_q^{(k)}] = J_{\pm} T_q^{(k)} - T_q^{(k)} J_{\pm}$ and letting $J_{\pm}$ act on the outer states:

$$\begin{aligned} \langle \alpha', j', m' | J_{\pm} T_q^{(k)} | \alpha, j, m \rangle &= \hbar \sqrt{j'(j' + 1) - m'(m' \mp 1)} \\ &\quad \times \langle \alpha', j', m' \mp 1 | T_q^{(k)} | \alpha, j, m \rangle, \end{aligned} \quad (5.5.24)$$

$$\begin{aligned} \langle \alpha', j', m' | T_q^{(k)} J_{\pm} | \alpha, j, m \rangle &= \hbar \sqrt{j(j+1) - m(m \pm 1)} \\ &\quad \times \langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \pm 1 \rangle. \end{aligned} \quad (5.5.25)$$

Subtracting these two terms:

$$\begin{aligned} &\sqrt{j'(j' + 1) - m'(m' \mp 1)} \langle \alpha', j', m' \mp 1 | T_q^{(k)} | \alpha, j, m \rangle \\ &\quad - \sqrt{j(j+1) - m(m \pm 1)} \langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \pm 1 \rangle \\ &= \sqrt{k(k+1) - q(q \pm 1)} \langle \alpha', j', m' | T_{q \pm 1}^{(k)} | \alpha, j, m \rangle. \end{aligned} \quad (5.5.26)$$

Now, compare equation (5.5.26) with the Clebsch–Gordan recursion relation (5.4.24). Setting $J = j'$, $M = m'$, $j_1 = j$, $m_1 = m$, $j_2 = k$, $m_2 = q$, equation (5.4.24) reads:

$$\begin{aligned} & \sqrt{j'(j'+1) - m'(m' \mp 1)} \langle j, m; k, q | j', m' \mp 1 \rangle \\ & - \sqrt{j(j+1) - m(m \pm 1)} \langle j, m \pm 1; k, q | j', m' \rangle \\ & = \sqrt{k(k+1) - q(q \pm 1)} \langle j, m; k, q \pm 1 | j', m' \rangle. \end{aligned} \quad (5.5.27)$$

Both sets of equations represent an identical system of coupled linear homogeneous equations for the m, m', q dependence! A fundamental theorem of linear algebra guarantees that any two non-trivial solutions to the same system of linear homogeneous equations must be proportional:

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = C(\alpha', j'; \alpha, j) \langle j, m; k, q | j', m' \rangle. \quad (5.5.28)$$

The constant of proportionality $C(\alpha', j'; \alpha, j)$ depends only on the dynamical labels (α', j', α, j) and the operator $T^{(k)}$, but is *completely independent* of m, m' , and q . Defining this constant as

$$C(\alpha', j'; \alpha, j) \equiv \frac{\langle \alpha', j' | T^{(k)} | \alpha, j \rangle}{\sqrt{2j'+1}}, \quad (5.5.29)$$

we complete the proof of the Wigner–Eckart theorem. ■

5.5.4 Selection Rules

The Wigner–Eckart theorem immediately yields rigorous **selection rules** for all quantum matrix elements:

1. **m -selection rule:** The matrix element vanishes unless

$$m' = m + q \quad \Longleftrightarrow \quad \Delta m = q. \quad (5.5.30)$$

2. **j -selection rule (Triangle inequality):** The matrix element vanishes unless (j, k, j') form the sides of a closed triangle:

$$|j - k| \leq j' \leq j + k. \quad (5.5.31)$$

For a scalar operator ($k = 0$): $j' = j$ and $m' = m$. For a vector operator ($k = 1$): $\Delta j \in \{-1, 0, +1\}$ (with $j = 0 \rightarrow j' = 0$ strictly forbidden).

3. **Parity selection rule:** If the operator has definite parity under spatial inversion $\hat{P}^\dagger T_q^{(k)} \hat{P} = \pi_T T_q^{(k)}$ (where $\pi_T = \pm 1$), the matrix element vanishes unless

$$\pi_{\alpha'} \pi_T \pi_\alpha = +1. \quad (5.5.32)$$

For electric dipole transitions (where the position operator $\vec{r}$ has odd parity $\pi = -1$), transitions between states of the same parity (e.g. $s \rightarrow s$ or $p \rightarrow p$) are strictly forbidden (Laporte's rule).

5.5.5 The Projection Theorem for Vector Operators

An indispensable corollary of the Wigner–Eckart theorem is the **Projection Theorem** for vector operators ($k = 1$) evaluated within a subspace of constant total angular momentum j ($j' = j$).

Theorem (The Projection Theorem for Vector Operators). *For any vector operator $\vec{V}$, its matrix elements between states in the same angular momentum multiplet $|j, m\rangle$ are directly proportional to the matrix elements of $\vec{J}$:*

$$\langle \alpha, j, m' | \vec{V} | \alpha, j, m \rangle = \frac{\langle \alpha, j, m | \vec{J} \cdot \vec{V} | \alpha, j, m \rangle}{\hbar^2 j(j+1)} \langle \alpha, j, m' | \vec{J} | \alpha, j, m \rangle. \quad (5.5.33)$$

Proof. Applying the Wigner–Eckart theorem to the spherical components of $\vec{V}$ and $\vec{J}$ within the manifold of fixed j :

$$\langle \alpha, j, m' | V_q | \alpha, j, m \rangle = \langle j, m; 1, q | j, m' \rangle \frac{\langle \alpha, j | \vec{V} | \alpha, j \rangle}{\sqrt{2j+1}}, \quad (5.5.34)$$

$$\langle \alpha, j, m' | J_q | \alpha, j, m \rangle = \langle j, m; 1, q | j, m' \rangle \frac{\langle \alpha, j | \vec{J} | \alpha, j \rangle}{\sqrt{2j+1}}. \quad (5.5.35)$$

Taking the ratio of these two equations, the Clebsch–Gordan coefficient cancels completely:

$$\frac{\langle \alpha, j, m' | V_q | \alpha, j, m \rangle}{\langle \alpha, j, m' | J_q | \alpha, j, m \rangle} = \frac{\langle \alpha, j | \vec{V} | \alpha, j \rangle}{\langle \alpha, j | \vec{J} | \alpha, j \rangle} \equiv c_{\alpha, j}. \quad (5.5.36)$$

Thus, $\vec{V} = c_{\alpha, j} \vec{J}$ as an effective operator within the subspace.

To determine the scalar proportionality constant $c_{\alpha, j}$, we take the dot product with $\vec{J}$:

$$\vec{J} \cdot \vec{V} = c_{\alpha, j} \vec{J} \cdot \vec{J} = c_{\alpha, j} \vec{J}^2. \quad (5.5.37)$$

Taking the expectation value in any state $|j, m\rangle$:

$$\langle \alpha, j, m | \vec{J} \cdot \vec{V} | \alpha, j, m \rangle = c_{\alpha, j} \hbar^2 j(j+1). \quad (5.5.38)$$

Solving for $c_{\alpha, j}$:

$$c_{\alpha, j} = \frac{\langle \alpha, j, m | \vec{J} \cdot \vec{V} | \alpha, j, m \rangle}{\hbar^2 j(j+1)}. \quad (5.5.39)$$

Substituting $c_{\alpha, j}$ back into the matrix element yields equation (5.5.33). ■

5.5.6 Physical Applications: The Landé g -Factor

A celebrated application of the Projection Theorem is the calculation of the **Landé g -factor** in the anomalous Zeeman effect.

Consider an atom in an external magnetic field $\vec{B} = B\hat{e}_z$. The magnetic interaction Hamiltonian is

$$\hat{H}_Z = -\vec{\mu} \cdot \vec{B} = \frac{\mu_B}{\hbar} (\vec{L} + g_s \vec{S}) \cdot \vec{B} = \frac{\mu_B}{\hbar} (L_z + 2S_z)B, \quad (5.5.40)$$

where $\mu_B = \frac{e\hbar}{2m_e}$ is the Bohr magneton, and $g_s \approx 2$ is the electron spin g -factor. In the coupled basis $|L, S; J, M_J\rangle$ where spin-orbit coupling $\vec{L} \cdot \vec{S}$ is diagonal, neither L_z nor S_z is diagonal on its own.

However, the operator $\vec{V} \equiv \vec{L} + 2\vec{S} = \vec{J} + \vec{S}$ is a vector operator! By the Projection Theorem, its matrix elements within the multiplet of fixed J are:

$$\langle J, M_J | (L_z + 2S_z) | J, M_J \rangle = g_J \langle J, M_J | J_z | J, M_J \rangle = g_J \hbar M_J, \quad (5.5.41)$$

where the Landé g -factor is

$$g_J = \frac{\langle \vec{J} \cdot (\vec{J} + \vec{S}) \rangle}{\hbar^2 J(J+1)} = 1 + \frac{\langle \vec{J} \cdot \vec{S} \rangle}{\hbar^2 J(J+1)}. \quad (5.5.42)$$

Using the identity $\vec{L}^2 = (\vec{J} - \vec{S})^2 = \vec{J}^2 + \vec{S}^2 - 2\vec{J} \cdot \vec{S}$:

$$\vec{J} \cdot \vec{S} = \frac{1}{2}(\vec{J}^2 + \vec{S}^2 - \vec{L}^2). \quad (5.5.43)$$

Evaluating the expectation value in the state $|L, S; J, M_J\rangle$:

$$\boxed{g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}}. \quad (5.5.44)$$

The Zeeman energy shift of each magnetic sub-level is therefore simply

$$\Delta E = g_J \mu_B B M_J. \quad (5.5.45)$$

Without computing a single wave-function integral, the Wigner–Eckart theorem and the Projection Theorem have completely solved the energy splitting of the anomalous Zeeman effect!

Chapter 6

Central Potentials and Hidden Symmetries

6.1 The General Central Potential in Spherical Coordinates

In classical mechanics, whenever a particle moves under a central force $\vec{F}(\vec{r}) = -\vec{\nabla}V(r) = f(r)\hat{r}$, the torque $\vec{\tau} = \vec{r} \times \vec{F} = 0$ vanishes identically. As an immediate consequence, the orbital angular momentum $\vec{L} = \vec{r} \times \vec{p}$ is a strictly conserved constant of motion:

$$\frac{d\vec{L}}{dt} = \vec{v} \times \vec{p} + \vec{r} \times \vec{F} = 0. \quad (6.1.1)$$

Conservation of $\vec{L}$ confines the classical motion to a two-dimensional plane perpendicular to the fixed angular momentum vector.

In quantum mechanics, the Hamiltonian of a particle of reduced mass μ moving in a spherically symmetric potential $V(r)$ is given by

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + V(\hat{r}). \quad (6.1.2)$$

Because the potential depends exclusively on the radial coordinate $r = |\vec{r}|$, it is invariant under arbitrary spatial rotations generated by the angular momentum operator $\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$. Recalling the fundamental commutator $[r^2, \hat{L}_i] = 0$ and $[\hat{p}^2, \hat{L}_i] = 0$, we find that the Hamiltonian commutes with all three components of angular momentum:

$$[\hat{H}, \hat{L}_i] = 0 \quad (i = x, y, z). \quad (6.1.3)$$

Consequently, $\hat{H}$ also commutes with the total angular momentum squared $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$:

$$[\hat{H}, \hat{L}^2] = 0, \quad [\hat{H}, \hat{L}_z] = 0. \quad (6.1.4)$$

Because $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$ form a Complete Set of Commuting Observables (CSCO) for the spatial degrees of freedom, there exists a simultaneous eigenbasis $|n, l, m\rangle$ satisfying:

$$\hat{H} |n, l, m\rangle = E_{nl} |n, l, m\rangle, \quad (6.1.5)$$

$$\hat{L}^2 |n, l, m\rangle = \hbar^2 l(l+1) |n, l, m\rangle, \quad (6.1.6)$$

$$\hat{L}_z |n, l, m\rangle = \hbar m |n, l, m\rangle, \quad (6.1.7)$$

where $l \in \{0, 1, 2, \dots\}$ and $m \in \{-l, -l+1, \dots, +l\}$. Notice that the energy eigenvalues E_{nl} cannot depend on the magnetic quantum number m ; because $[\hat{H}, \hat{L}_{\pm}] = 0$, the ladder operators $\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y$ map eigenstates of different m into one another without shifting the energy. Thus, every energy level with angular momentum l possesses an unavoidable, rotational degeneracy of at least $(2l+1)$.

6.1.1 Decomposition of Kinetic Energy and the Radial Momentum

To express the Hamiltonian in spherical polar coordinates (r, θ, ϕ) , we examine the position basis representation. In 3D coordinate space, the position eigenbasis is the tensor product

$$|\vec{r}\rangle = |r\rangle \otimes |\theta\rangle \otimes |\phi\rangle, \quad (6.1.8)$$

with the volume element $d^3r = r^2 \sin \theta dr d\theta d\phi$. In vector calculus, the Laplacian ∇^2 decomposes into radial and angular parts:

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (6.1.9)$$

Recognizing the angular differential operator as $-\hat{L}^2/\hbar^2$:

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right], \quad (6.1.10)$$

the Laplacian can be written operatorially as

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2}. \quad (6.1.11)$$

A natural question arises: what is the quantum operator conjugate to the radial coordinate r ? A naive definition $\hat{p}_r = -i\hbar \partial/\partial r$ is *not* Hermitian on the Hilbert space $L^2(\mathbb{R}^3, r^2 dr d\Omega)$ because of the r^2 weighting factor in the integration measure. The correct Hermitian radial momentum operator is constructed via symmetric ordering:

$$\hat{p}_r \equiv \frac{1}{2} \left[\hat{\vec{p}} \cdot \left(\frac{\vec{r}}{r} \right) + \left(\frac{\vec{r}}{r} \right) \cdot \hat{\vec{p}} \right] = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right). \quad (6.1.12)$$

One can readily verify that $\hat{p}_r = \hat{p}_r^\dagger$. Computing its square yields:

$$\begin{aligned} \hat{p}_r^2 &= -\hbar^2 \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \\ &= -\hbar^2 \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) = -\hbar^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right). \end{aligned} \quad (6.1.13)$$

Therefore, the quantum kinetic energy operator decouples neatly into purely radial and rotational components, matching the classical expression:

$$\hat{H} = \frac{\hat{p}_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} + V(r). \quad (6.1.14)$$

6.1.2 The Radial Schrödinger Equation and the Effective Potential

Projecting the state vector into coordinate space, the common eigenfunctions factorize into a radial function and the spherical harmonics:

$$\psi_{nlm}(r, \theta, \phi) = \langle r, \theta, \phi | n, l, m \rangle = R_{nl}(r) Y_{lm}(\theta, \phi). \quad (6.1.15)$$

Because $\hat{L}^2 Y_{lm}(\theta, \phi) = \hbar^2 l(l+1) Y_{lm}(\theta, \phi)$, acting with $\hat{H}$ on ψ_{nlm} removes all angular dependence and produces an ordinary differential equation for $R_{nl}(r)$:

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r) \right] R_{nl}(r) = E_{nl} R_{nl}(r). \quad (6.1.16)$$

To eliminate the first-derivative term and bring the radial equation into the form of a one-dimensional Schrödinger equation, we define the reduced radial wavefunction:

$$u_{nl}(r) \equiv rR_{nl}(r). \quad (6.1.17)$$

Differentiating $u_{nl}(r)$ with respect to r :

$$\frac{du}{dr} = R + r \frac{dR}{dr}, \quad \frac{d^2u}{dr^2} = 2 \frac{dR}{dr} + r \frac{d^2R}{dr^2} = r \left(\frac{d^2R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right). \quad (6.1.18)$$

Substituting this back into the radial equation gives the canonical 1D radial Schrödinger equation:

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + V_{\text{eff}}(r) \right] u_{nl}(r) = E_{nl} u_{nl}(r), \quad (6.1.19)$$

where the **effective potential** is defined by

$$V_{\text{eff}}(r) \equiv V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2}. \quad (6.1.20)$$

The additional term $\frac{\hbar^2 l(l+1)}{2\mu r^2}$ represents the quantum *centrifugal barrier*. For any state with non-zero orbital angular momentum ($l > 0$), this repulsive barrier diverges as $r \rightarrow 0$, pushing the particle away from the origin.

6.1.3 Boundary Conditions and Asymptotics at the Origin

The normalization condition for the three-dimensional wavefunction requires

$$\int_{\mathbb{R}^3} |\psi_{nlm}(\vec{r})|^2 d^3r = \int_0^\infty r^2 |R_{nl}(r)|^2 dr \int |Y_{lm}|^2 d\Omega = \int_0^\infty |u_{nl}(r)|^2 dr = 1. \quad (6.1.21)$$

Thus $u_{nl}(r)$ must be square-integrable on the half-line $r \in [0, \infty)$, demanding $\lim_{r \rightarrow \infty} u_{nl}(r) = 0$ for bound states.

Near the origin ($r \rightarrow 0$), assuming the physical potential is less singular than $1/r^2$ (that is, $\lim_{r \rightarrow 0} r^2 V(r) = 0$), the centrifugal term dominates:

$$\frac{d^2u}{dr^2} \approx \frac{l(l+1)}{r^2} u. \quad (6.1.22)$$

Postulating an ansatz $u(r) \sim r^s$, we find $s(s-1) = l(l+1)$, yielding two independent roots:

$$s = l+1 \quad \text{or} \quad s = -l. \quad (6.1.23)$$

The irregular solution $u(r) \sim r^{-l}$ leads to $R(r) \sim r^{-(l+1)}$, which for $l \geq 1$ violates square integrability $\int r^2 |R|^2 dr < \infty$ near $r = 0$. Even for $l = 0$, where $R(r) \sim 1/r$, computing $\nabla^2(1/r) = -4\pi\delta^3(\vec{r})$ would inject an unphysical delta-function singularity into the Schrödinger equation unless canceled by a singular potential source. Thus, the physical boundary condition is strictly:

$$u_{nl}(r) \sim r^{l+1} \implies R_{nl}(r) \sim r^l \quad \text{as } r \rightarrow 0. \quad (6.1.24)$$

In particular, $u_{nl}(0) = 0$ for all l , and the spatial wavefunction $\psi(\vec{r})$ vanishes at the origin for all states except *s*-waves ($l = 0$).

6.2 The Spherical Cavity: Free Particle in an Infinite Spherical Well

A fundamental 3D model that provides the prototype for nuclear shell models and quantum dots is the **spherical cavity** (or 3D spherical box), consisting of a particle of mass μ confined within an impenetrable spherical boundary of radius a :

$$V(r) = \begin{cases} 0, & r < a, \\ \infty, & r \geq a. \end{cases} \quad (6.2.1)$$

6.2.1 Radial Equation and Spherical Bessel Functions

Inside the cavity ($r < a$), the potential vanishes ($V(r) = 0$), and the radial Schrödinger equation (4.1.4) becomes:

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \left[k^2 - \frac{l(l+1)}{r^2} \right] R(r) = 0, \quad k \equiv \frac{\sqrt{2\mu E}}{\hbar}. \quad (6.2.2)$$

Introducing the dimensionless radial variable $\rho \equiv kr$, Eq. (6.2.2) takes the canonical form of the **Spherical Bessel Differential Equation**:

$$\frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} + \left[1 - \frac{l(l+1)}{\rho^2} \right] R = 0. \quad (6.2.3)$$

The two linearly independent solutions are the **spherical Bessel functions** $j_l(\rho)$ and the **spherical Neumann functions** $n_l(\rho)$. As detailed in Siegfried Flügge's *Practical Quantum Mechanics* (Problem 19), they are generated by **Rayleigh's formulas**:

$$j_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{\sin \rho}{\rho} \right), \quad (6.2.4)$$

$$n_l(\rho) = -(-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho} \right)^l \left(\frac{\cos \rho}{\rho} \right). \quad (6.2.5)$$

The lowest-order explicit functions are:

$$j_0(\rho) = \frac{\sin \rho}{\rho}, \quad n_0(\rho) = -\frac{\cos \rho}{\rho}, \quad (6.2.6)$$

$$j_1(\rho) = \frac{\sin \rho}{\rho^2} - \frac{\cos \rho}{\rho}, \quad n_1(\rho) = -\frac{\cos \rho}{\rho^2} - \frac{\sin \rho}{\rho}, \quad (6.2.7)$$

$$j_2(\rho) = \left(\frac{3}{\rho^3} - \frac{1}{\rho} \right) \sin \rho - \frac{3}{\rho^2} \cos \rho, \quad n_2(\rho) = -\left(\frac{3}{\rho^3} - \frac{1}{\rho} \right) \cos \rho - \frac{3}{\rho^2} \sin \rho. \quad (6.2.8)$$

6.2.2 Regularity at the Origin and Boundary Matching

The general radial solution is a linear superposition:

$$R_{nl}(r) = A j_l(kr) + B n_l(kr). \quad (6.2.9)$$

Examining the asymptotic behavior near the origin ($\rho \rightarrow 0$):

$$j_l(\rho) \xrightarrow{\rho \rightarrow 0} \frac{\rho^l}{(2l+1)!!} \quad (\text{finite and regular}), \quad (6.2.10)$$

$$n_l(\rho) \xrightarrow{\rho \rightarrow 0} -(2l-1)!! \rho^{-(l+1)} \rightarrow -\infty \quad (\text{singular}). \quad (6.2.11)$$

Because $n_l(\rho)$ diverges severely at $r = 0$, physical square-integrability demands $B = 0$. Hence, the acceptable radial solutions are strictly:

$$R_{nl}(r) = A_{nl}j_l(k_{nl}r). \quad (6.2.12)$$

At the hard spherical wall $r = a$, the wavefunction must vanish continuously:

$$R_{nl}(a) = 0 \implies j_l(k_{nl}a) = 0. \quad (6.2.13)$$

Let ξ_{ln} denote the n -th positive zero of the spherical Bessel function $j_l(\xi)$:

$$j_l(\xi_{ln}) = 0, \quad n = 1, 2, 3, \dots \quad (6.2.14)$$

The quantized wavenumbers are $k_{ln} = \frac{\xi_{ln}}{a}$, yielding the discrete energy eigenvalues:

$$E_{ln} = \frac{\hbar^2 \xi_{ln}^2}{2\mu a^2}, \quad l \in \{0, 1, 2, \dots\}, \quad n \in \{1, 2, 3, \dots\} \quad (6.2.15)$$

6.2.3 Energy Levels and Zeros of Spherical Bessel Functions

Let us examine the energy spectrum for the lowest angular momenta:

1. **s-states** ($l = 0$):

$$j_0(\xi) = \frac{\sin \xi}{\xi} = 0 \implies \xi_{0n} = n\pi \implies E_{0n} = \frac{n^2 \pi^2 \hbar^2}{2\mu a^2}. \quad (6.2.16)$$

Notice that the reduced radial function $u_{0n}(r) = rR_{0n}(r) \propto \sin(n\pi r/a)$ is identical to the odd-parity bound states of a one-dimensional box of length $2a$!

2. **p-states** ($l = 1$):

$$j_1(\xi) = 0 \iff \tan \xi = \xi. \quad (6.2.17)$$

The transcendental equation $\tan \xi = \xi$ yields the numerical zeros:

$$\xi_{11} \approx 4.4934, \quad \xi_{12} \approx 7.7253, \quad \xi_{13} \approx 10.9041. \quad (6.2.18)$$

The lowest p -state energy is $E_{11} = \frac{(4.4934)^2 \hbar^2}{2\mu a^2} \approx 20.19 \frac{\hbar^2}{2\mu a^2}$, which lies substantially above the ground state $E_{01} = \pi^2 \frac{\hbar^2}{2\mu a^2} \approx 9.87 \frac{\hbar^2}{2\mu a^2}$ due to the repulsive centrifugal energy $\frac{\hbar^2(1)(2)}{2\mu r^2}$.

3. **d-states** ($l = 2$): The zeros of $j_2(\xi) = 0$ occur at:

$$\xi_{21} \approx 5.7635, \quad \xi_{22} \approx 9.0950, \quad \xi_{23} \approx 12.3229. \quad (6.2.19)$$

The complete energy ordering of the lowest bound states is:

$$E_{01} < E_{11} < E_{21} < E_{02} < E_{31} < E_{12} < E_{41} < E_{22} \dots \quad (6.2.20)$$

Each energy level E_{ln} is $(2l + 1)$ -fold degenerate with respect to the magnetic quantum number $m \in \{-l, \dots, +l\}$. Using the Lommel integral $\int_0^1 x^2 [j_l(\xi_{ln}x)]^2 dx = \frac{1}{2} [j_{l+1}(\xi_{ln})]^2$, the normalized complete eigenfunctions are:

$$\psi_{nlm}(r, \theta, \phi) = \frac{\sqrt{2}}{a^{3/2} |j_{l+1}(\xi_{ln})|} j_l\left(\frac{\xi_{ln}r}{a}\right) Y_l^m(\theta, \phi). \quad (6.2.21)$$

6.3 The 3D Isotropic Harmonic Oscillator

The three-dimensional isotropic harmonic oscillator is governed by the Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\mu\omega^2 r^2 = \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\mu\omega^2 (x^2 + y^2 + z^2). \quad (6.3.1)$$

This system serves as a prime archetype of a superintegrable system possessing a dynamical symmetry larger than its manifest geometric rotation group $SO(3)$.

6.3.1 Separation in Cartesian Coordinates

In Cartesian coordinates, the Hamiltonian decomposes into three uncoupled, independent one-dimensional harmonic oscillators:

$$\hat{H} = \hat{H}_x + \hat{H}_y + \hat{H}_z = \hbar\omega \left(a_x^\dagger a_x + a_y^\dagger a_y + a_z^\dagger a_z + \frac{3}{2} \right), \quad (6.3.2)$$

where the ladder operators are defined standardly for each axis $k \in \{x, y, z\}$:

$$a_k = \sqrt{\frac{\mu\omega}{2\hbar}} \left(\hat{x}_k + \frac{i}{\mu\omega} \hat{p}_k \right), \quad [a_j, a_k^\dagger] = \delta_{jk}. \quad (6.3.3)$$

The energy eigenstates are product states $|n_x, n_y, n_z\rangle = |n_x\rangle \otimes |n_y\rangle \otimes |n_z\rangle$ with energy eigenvalues

$$E_N = \hbar\omega \left(n_x + n_y + n_z + \frac{3}{2} \right) = \hbar\omega \left(N + \frac{3}{2} \right), \quad (6.3.4)$$

where $N = n_x + n_y + n_z \in \{0, 1, 2, \dots\}$ is the total principal quantum number.

The degeneracy of the energy level E_N is the number of distinct partitions of the integer N into three non-negative integers (n_x, n_y, n_z) . By elementary combinatorics (stars and bars):

$$g_N = \binom{N+3-1}{3-1} = \binom{N+2}{2} = \frac{(N+1)(N+2)}{2}. \quad (6.3.5)$$

For $N = 0$, $g_0 = 1$; for $N = 1$, $g_1 = 3$; for $N = 2$, $g_2 = 6$; and for $N = 3$, $g_3 = 10$.

6.3.2 Separation in Spherical Coordinates and Angular Momentum Multiplets

Because the potential $V(r) = \frac{1}{2}\mu\omega^2 r^2$ is spherically symmetric, the system can equally be solved in spherical coordinates. The radial Schrödinger equation for $u(r) = rR(r)$ is

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{1}{2}\mu\omega^2 r^2 + \frac{\hbar^2 l(l+1)}{2\mu r^2} \right] u(r) = E u(r). \quad (6.3.6)$$

Introducing the natural length scale $x_0 = \sqrt{\hbar/(\mu\omega)}$ and the dimensionless variable $\xi = r/x_0$, the asymptotic behaviors are:

- As $r \rightarrow 0$, $u \sim \xi^{l+1}$.
- As $r \rightarrow \infty$, $u \sim e^{-\xi^2/2}$.

Factoring out the asymptotics, we write $u(\xi) = \xi^{l+1} e^{-\xi^2/2} v(\xi)$. Transforming to the variable $z = \xi^2 = \mu\omega r^2/\hbar$, the differential equation for $v(z)$ becomes the confluent hypergeometric (Kummer's) equation:

$$z \frac{d^2 v}{dz^2} + \left(l + \frac{3}{2} - z \right) \frac{dv}{dz} - \left(\frac{2l+3-2E/(\hbar\omega)}{4} \right) v = 0. \quad (6.3.7)$$

To prevent $v(z)$ from growing as e^z at infinity (which would destroy square integrability), the power series must terminate into a polynomial of degree $n_r \in \{0, 1, 2, \dots\}$, where n_r is the radial quantum number:

$$\frac{2l + 3 - 2E/(\hbar\omega)}{4} = -n_r \implies E = \hbar\omega \left(2n_r + l + \frac{3}{2} \right). \quad (6.3.8)$$

Matching this with the Cartesian energy eigenvalue $E_N = \hbar\omega(N + 3/2)$, we identify:

$$N = 2n_r + l \implies l = N - 2n_r. \quad (6.3.9)$$

Because $n_r \geq 0$ is an integer, for a given energy level N , the allowed values of the orbital angular momentum l run over:

$$l = N, N - 2, N - 4, \dots, 1 \text{ or } 0. \quad (6.3.10)$$

Crucially, all angular momentum states belonging to a given shell N share the *exact same parity* $\Pi = (-1)^l = (-1)^N$. For example:

- $N = 0 \implies l = 0$ (s -state, 1 state).
- $N = 1 \implies l = 1$ (p -state, $2(1) + 1 = 3$ states).
- $N = 2 \implies l = 2, 0$ (d and s states, $5 + 1 = 6$ states).
- $N = 3 \implies l = 3, 1$ (f and p states, $7 + 3 = 10$ states).

Summing the $(2l + 1)$ degeneracies over all allowed l :

$$\sum_{n_r=0}^{\lfloor N/2 \rfloor} [2(N - 2n_r) + 1] = \frac{(N + 1)(N + 2)}{2} = g_N. \quad (6.3.11)$$

The Cartesian and spherical bases yield precisely the same spectrum and degeneracies.

6.3.3 Dynamical $SU(3)$ Symmetry

Why should states with completely different angular momentum values (such as $l = 2$ and $l = 0$ for $N = 2$) have identical energy? This “accidental” degeneracy is explained by a dynamical symmetry group larger than $SO(3)$.

Consider the 9 bilinear operators formed from the Cartesian creation and annihilation operators:

$$A_{ij} \equiv a_i^\dagger a_j \quad (i, j \in \{x, y, z\}). \quad (6.3.12)$$

Using the commutator identity $[AB, C] = A[B, C] + [A, C]B$, their commutation relations are

$$[A_{ij}, A_{kl}] = [a_i^\dagger a_j, a_k^\dagger a_l] = \delta_{jk} a_i^\dagger a_l - \delta_{il} a_k^\dagger a_j = \delta_{jk} A_{il} - \delta_{il} A_{kj}. \quad (6.3.13)$$

These are precisely the commutation relations of the Lie algebra $\mathfrak{u}(3)$. Because the Hamiltonian is proportional to the trace $\hat{N} = \sum_k A_{kk}$:

$$\hat{H} = \hbar\omega \left(\sum_{k=1}^3 A_{kk} + \frac{3}{2} \right), \quad (6.3.14)$$

and $[A_{kk}, A_{ij}] = A_{ik}\delta_{kj} - A_{kj}\delta_{ik}$, summing over k yields:

$$[\hat{H}, A_{ij}] = 0 \quad \text{for all } i, j \in \{x, y, z\}. \quad (6.3.15)$$

The total number operator $\hat{N}$ commutes with all generators and generates a $U(1)$ phase symmetry. The remaining $9 - 1 = 8$ traceless, Hermitian generators:

$$T_a \equiv \frac{1}{2} \sum_{i,j} a_i^\dagger (\lambda_a)_{ij} a_j \quad (a = 1, \dots, 8), \quad (6.3.16)$$

where λ_a are the Gell-Mann matrices, generate the special unitary Lie algebra $\mathfrak{su}(3)$.

The angular momentum operators $\hat{\vec{L}}$ are embedded within this algebra as the antisymmetric combinations:

$$\hat{L}_k = -i\hbar \epsilon_{ijk} a_i^\dagger a_j. \quad (6.3.17)$$

The remaining symmetric generators of $SU(3)$ are quadrupole-like operators that change the angular momentum l by ± 2 while conserving the total quanta N . Each energy eigenspace corresponding to quantum number N constitutes a completely symmetric, irreducible representation $(N, 0)$ of $SU(3)$, whose dimension is rigorously $\frac{1}{2}(N+1)(N+2)$. Thus, what seemed an “accidental” degeneracy is a direct manifestation of the dynamical $SU(3)$ symmetry of the isotropic harmonic oscillator.

6.4 The Hydrogen Atom: Analytic Solution

The non-relativistic hydrogen-like atom consists of an electron of charge $-e$ interacting via the Coulomb potential with a nucleus of charge $+Ze$:

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}. \quad (6.4.1)$$

In the center-of-mass frame, the relative motion is governed by the reduced mass $\mu = \frac{m_e m_N}{m_e + m_N} \approx m_e$. For bound states ($E < 0$), we introduce the inverse length scale

$$\kappa \equiv \sqrt{\frac{-2\mu E}{\hbar^2}}, \quad (6.4.2)$$

and the dimensionless radial coordinate

$$\rho \equiv 2\kappa r. \quad (6.4.3)$$

Substituting ρ into the radial Schrödinger equation for $u(\rho) = rR(r)$ yields:

$$\frac{d^2 u}{d\rho^2} + \left[-\frac{1}{4} + \frac{\lambda}{\rho} - \frac{l(l+1)}{\rho^2} \right] u = 0, \quad (6.4.4)$$

where the dimensionless parameter λ is defined by

$$\lambda \equiv \frac{Ze^2}{4\pi\epsilon_0 \hbar} \sqrt{\frac{\mu}{-2E}} = \frac{Ze^2 \mu}{4\pi\epsilon_0 \hbar^2 \kappa}. \quad (6.4.5)$$

6.4.1 Asymptotic Behavior and Associated Laguerre Polynomials

Examining the asymptotic limits:

- As $\rho \rightarrow \infty$: $d^2 u / d\rho^2 \approx \frac{1}{4}u \implies u(\rho) \sim e^{-\rho/2}$ (rejecting $e^{+\rho/2}$ which diverges).
- As $\rho \rightarrow 0$: $d^2 u / d\rho^2 \approx \frac{l(l+1)}{\rho^2}u \implies u(\rho) \sim \rho^{l+1}$.

We therefore factor out these asymptotic behaviors:

$$u(\rho) = \rho^{l+1} e^{-\rho/2} w(\rho) \implies R(\rho) = \rho^l e^{-\rho/2} w(\rho). \quad (6.4.6)$$

Substituting this ansatz into the differential equation yields the standard Kummer equation for $w(\rho)$:

$$\rho \frac{d^2 w}{d\rho^2} + (2l + 2 - \rho) \frac{dw}{d\rho} - (l + 1 - \lambda) w = 0. \quad (6.4.7)$$

Expanding $w(\rho) = \sum_{k=0}^{\infty} c_k \rho^k$ in a Frobenius series gives the recurrence relation:

$$c_{k+1} = \frac{k + l + 1 - \lambda}{(k + 1)(k + 2l + 2)} c_k. \quad (6.4.8)$$

For large k , $c_{k+1}/c_k \approx 1/k$, which matches the series expansion of e^ρ . If the series did not terminate, $w(\rho)$ would behave as e^ρ as $\rho \rightarrow \infty$, rendering $u(\rho) \sim \rho^{l+1} e^{+\rho/2}$, which is not normalizable. Therefore, the numerator must vanish for some non-negative integer $k = n_r \in \{0, 1, 2, \dots\}$:

$$n_r + l + 1 - \lambda = 0 \implies \lambda = n_r + l + 1 \equiv n, \quad (6.4.9)$$

where $n \in \{1, 2, 3, \dots\}$ is the **principal quantum number**, and $n_r = n - l - 1 \geq 0$ is the radial quantum number (number of radial nodes).

6.4.2 Energy Eigenvalues and Degeneracy

Equating $\lambda = n$ with its physical definition:

$$n = \frac{Ze^2}{4\pi\epsilon_0\hbar} \sqrt{\frac{\mu}{-2E}} \implies E_n = -\frac{\mu(Ze^2)^2}{2(4\pi\epsilon_0)^2\hbar^2 n^2} = -\frac{Z^2 \text{Ry}}{n^2}, \quad (6.4.10)$$

where the Rydberg energy unit is

$$\text{Ry} \equiv \frac{\mu e^4}{2(4\pi\epsilon_0)^2\hbar^2} = \frac{1}{2} \mu c^2 \alpha^2 \approx 13.606 \text{ eV}, \quad (6.4.11)$$

and $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036}$ is the fine-structure constant. The characteristic length scale is the Bohr radius:

$$a_0 \equiv \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} \approx 0.529 \text{ \AA}, \quad \kappa = \frac{Z}{na_0}, \quad \rho = \frac{2Zr}{na_0}. \quad (6.4.12)$$

The terminating polynomial $w(\rho)$ is proportional to the associated Laguerre polynomial $L_{n-l-1}^{2l+1}(\rho)$. The normalized radial wavefunctions are given explicitly by:

$$R_{nl}(r) = -\sqrt{\left(\frac{2Z}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}} e^{-Zr/(na_0)} \left(\frac{2Zr}{na_0}\right)^l L_{n+l}^{2l+1}\left(\frac{2Zr}{na_0}\right). \quad (6.4.13)$$

6.4.3 Spatial and Spin Degeneracy

For a fixed principal quantum number n , the orbital angular momentum is bounded by $n_r = n - l - 1 \geq 0$:

$$l \in \{0, 1, 2, \dots, n-1\}. \quad (6.4.14)$$

For each l , there are $2l + 1$ magnetic sublevels $m \in \{-l, \dots, +l\}$. The total spatial degeneracy of the n -th energy level is:

$$g_n = \sum_{l=0}^{n-1} (2l + 1) = 2 \sum_{l=0}^{n-1} l + \sum_{l=0}^{n-1} 1 = 2 \frac{(n-1)n}{2} + n = n^2. \quad (6.4.15)$$

Accounting for the two intrinsic spin states of the electron ($s = 1/2$, $m_s = \pm 1/2$), the total degeneracy of level n is

$$g_n^{(\text{total})} = 2n^2. \quad (6.4.16)$$

Just as in the harmonic oscillator, the fact that states with different values of l share the same energy E_n points to a remarkable hidden symmetry.

6.5 Radial Ladder Operators and the Factorization Method

Before turning to the global Lie algebra of the Runge-Lenz vector, we present the powerful *Infeld-Hull factorization method* (the foundation of Supersymmetric Quantum Mechanics, or SUSY QM) applied directly to the radial Schrödinger equation. This method constructs ladder operators that step between angular momentum channels $l \rightarrow l + 1$.

For a fixed l , the radial Hamiltonian operating on $u(r) = rR(r)$ is

$$\hat{H}_l = -\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{Ze^2}{4\pi\epsilon_0 r}. \quad (6.5.1)$$

We seek first-order differential operators A_l and $A_l^\dagger$ of the form:

$$A_l \equiv \frac{\hbar}{\sqrt{2\mu}} \frac{d}{dr} - \frac{\hbar(l+1)}{\sqrt{2\mu}r} + \frac{\mu Ze^2}{4\pi\epsilon_0(l+1)\hbar\sqrt{2\mu}}, \quad (6.5.2)$$

$$A_l^\dagger \equiv -\frac{\hbar}{\sqrt{2\mu}} \frac{d}{dr} - \frac{\hbar(l+1)}{\sqrt{2\mu}r} + \frac{\mu Ze^2}{4\pi\epsilon_0(l+1)\hbar\sqrt{2\mu}}. \quad (6.5.3)$$

Let us define the convenient constants $\alpha \equiv \frac{\hbar}{\sqrt{2\mu}}$, $\beta_l \equiv \frac{\hbar(l+1)}{\sqrt{2\mu}r}$, and $\gamma_l \equiv \frac{\mu Ze^2}{4\pi\epsilon_0(l+1)\hbar\sqrt{2\mu}}$. Then

$$A_l = \alpha \frac{d}{dr} - \frac{\beta_l}{r} + \gamma_l, \quad A_l^\dagger = -\alpha \frac{d}{dr} - \frac{\beta_l}{r} + \gamma_l. \quad (6.5.4)$$

We now directly compute the product $A_l^\dagger A_l$:

$$\begin{aligned} A_l^\dagger A_l &= \left(-\alpha \frac{d}{dr} - \frac{\beta_l}{r} + \gamma_l \right) \left(\alpha \frac{d}{dr} - \frac{\beta_l}{r} + \gamma_l \right) \\ &= -\alpha^2 \frac{d^2}{dr^2} + \alpha \left[\frac{d}{dr}, \frac{\beta_l}{r} - \gamma_l \right] + \left(\frac{\beta_l}{r} - \gamma_l \right)^2 \\ &= -\alpha^2 \frac{d^2}{dr^2} - \frac{\alpha\beta_l}{r^2} + \frac{\beta_l^2}{r^2} - \frac{2\beta_l\gamma_l}{r} + \gamma_l^2 \\ &= -\alpha^2 \frac{d^2}{dr^2} + \frac{\beta_l(\beta_l - \alpha)}{r^2} - \frac{2\beta_l\gamma_l}{r} + \gamma_l^2. \end{aligned} \quad (6.5.5)$$

Evaluating each coefficient in this expression:

$$\alpha^2 = \frac{\hbar^2}{2\mu}, \quad (6.5.6)$$

$$\beta_l(\beta_l - \alpha) = \frac{\hbar^2(l+1)l}{2\mu} = \frac{\hbar^2 l(l+1)}{2\mu}, \quad (6.5.7)$$

$$2\beta_l\gamma_l = 2\left(\frac{\hbar(l+1)}{\sqrt{2\mu}}\right)\left(\frac{\mu Ze^2}{4\pi\epsilon_0(l+1)\hbar\sqrt{2\mu}}\right) = \frac{Ze^2}{4\pi\epsilon_0}, \quad (6.5.8)$$

$$\gamma_l^2 = \frac{\mu(Ze^2)^2}{2(4\pi\epsilon_0)^2(l+1)^2\hbar^2} = -E_{l+1}^{(0)}. \quad (6.5.9)$$

Notice that these terms match $\hat{H}_l$ with flawless precision! We have established:

$$A_l^\dagger A_l = \hat{H}_l - E_{l+1}^{(0)}, \quad \text{where } E_{l+1}^{(0)} \equiv -\frac{\mu(Ze^2)^2}{2(4\pi\epsilon_0)^2\hbar^2(l+1)^2}. \quad (6.5.10)$$

Now, let us compute the reversed product $A_l A_l^\dagger$:

$$\begin{aligned} A_l A_l^\dagger &= \left(\alpha \frac{d}{dr} - \frac{\beta_l}{r} + \gamma_l\right) \left(-\alpha \frac{d}{dr} - \frac{\beta_l}{r} + \gamma_l\right) \\ &= -\alpha^2 \frac{d^2}{dr^2} - \alpha \left[\frac{d}{dr}, \frac{\beta_l}{r} - \gamma_l\right] + \left(\frac{\beta_l}{r} - \gamma_l\right)^2 \\ &= -\alpha^2 \frac{d^2}{dr^2} + \frac{\beta_l(\beta_l + \alpha)}{r^2} - \frac{2\beta_l\gamma_l}{r} + \gamma_l^2. \end{aligned} \quad (6.5.11)$$

Remarkably:

$$\beta_l(\beta_l + \alpha) = \frac{\hbar^2(l+1)(l+2)}{2\mu}. \quad (6.5.12)$$

This is precisely the centrifugal barrier for angular momentum $l+1$! Therefore,

$$A_l A_l^\dagger = \hat{H}_{l+1} - E_{l+1}^{(0)}. \quad (6.5.13)$$

6.5.1 Energy Hierarchy from Intertwining Relations

The relations $A_l^\dagger A_l = \hat{H}_l - E_{l+1}^{(0)}$ and $A_l A_l^\dagger = \hat{H}_{l+1} - E_{l+1}^{(0)}$ yield the fundamental **intertwining relation**:

$$\hat{H}_{l+1} A_l = (A_l A_l^\dagger + E_{l+1}^{(0)}) A_l = A_l (A_l^\dagger A_l + E_{l+1}^{(0)}) = A_l \hat{H}_l. \quad (6.5.14)$$

Let $u_{E,l}$ be an eigenstate of $\hat{H}_l$ with energy eigenvalue E :

$$\hat{H}_l u_{E,l} = E u_{E,l}. \quad (6.5.15)$$

Applying A_l and invoking the intertwining relation:

$$\hat{H}_{l+1}(A_l u_{E,l}) = A_l \hat{H}_l u_{E,l} = E(A_l u_{E,l}). \quad (6.5.16)$$

Thus, A_l maps an eigenstate of angular momentum l to an eigenstate of angular momentum $l+1$ with the *identical energy* E !

Because $A_l^\dagger A_l$ is a positive semi-definite operator:

$$\langle u_{E,l} | A_l^\dagger A_l | u_{E,l} \rangle = \|A_l u_{E,l}\|^2 = E - E_{l+1}^{(0)} \geq 0 \implies E \geq E_{l+1}^{(0)}. \quad (6.5.17)$$

The lowest possible energy state in the l -channel saturates this bound:

$$A_l u_{l,\text{ground}}(r) = 0 \implies \hat{H}_l u_{l,\text{ground}} = E_{l+1}^{(0)} u_{l,\text{ground}}. \quad (6.5.18)$$

The state annihilated by A_l cannot be stepped up to $l+1$ because $A_l u = 0$; it terminates the tower at the maximum possible l for that energy. Defining $n \equiv l+1$:

$$E_n = -\frac{\mu(Ze^2)^2}{2(4\pi\epsilon_0)^2\hbar^2 n^2} = -\frac{Z^2 R_y}{n^2}. \quad (6.5.19)$$

For any such energy E_n , applying the lowering operators $A_{l-1}^\dagger, A_{l-2}^\dagger, \dots, A_0^\dagger$ generates corresponding energy eigenstates with angular momenta $l = n-2, n-3, \dots, 0$. Thus, the entire spectrum and the degeneracy over $l \in \{0, \dots, n-1\}$ are derived completely algebraically from first-order radial differential operators.

6.6 Hidden $SO(4)$ Symmetry and the Quantum Runge-Lenz Vector

We now present Wolfgang Pauli's 1926 algebraic derivation of the hydrogen spectrum, which predated Schrödinger's wave equation solution and revealed the hidden $\mathfrak{so}(4)$ Lie algebra of the Coulomb potential.

6.6.1 Classical Laplace-Runge-Lenz Vector

In celestial mechanics, the Kepler problem possesses an additional conserved vector pointing from the focus to the perihelion, known as the Laplace-Runge-Lenz (LRL) vector:

$$\vec{A}_{\text{cl}} = \frac{1}{\mu}(\vec{p} \times \vec{L}) - \kappa \frac{\vec{r}}{r}, \quad \left(\kappa \equiv \frac{Ze^2}{4\pi\epsilon_0} \right). \quad (6.6.1)$$

Its conservation $d\vec{A}_{\text{cl}}/dt = 0$ prevents the perihelion from precessing, ensuring that classical Keplerian orbits remain closed ellipses.

6.6.2 Quantum Symmetrization and Hermiticity

To promote $\vec{A}_{\text{cl}}$ to a quantum observable, we encounter an operator ordering ambiguity. Classically, $\vec{p} \times \vec{L} = -\vec{L} \times \vec{p}$. In quantum mechanics, however:

$$(\vec{p} \times \vec{L})^\dagger = -\vec{L}^\dagger \times \vec{p}^\dagger = -\vec{L} \times \vec{p} \neq \vec{p} \times \vec{L}. \quad (6.6.2)$$

To determine the difference, let us evaluate the i -th component:

$$\begin{aligned} (\vec{p} \times \vec{L} + \vec{L} \times \vec{p})_i &= \epsilon_{ijk}(p_j L_k + L_j p_k) = \epsilon_{ijk}[p_j, L_k] \\ &= \epsilon_{ijk}(i\hbar\epsilon_{jkm}p_m) = i\hbar(\epsilon_{kij}\epsilon_{kmj})p_m \\ &= i\hbar(\delta_{im}\delta_{jj} - \delta_{ij}\delta_{jm})p_m = i\hbar(3p_i - p_i) = 2i\hbar p_i. \end{aligned} \quad (6.6.3)$$

Therefore:

$$\vec{p} \times \vec{L} + \vec{L} \times \vec{p} = 2i\hbar\vec{p} \implies -\vec{L} \times \vec{p} = \vec{p} \times \vec{L} - 2i\hbar\vec{p}. \quad (6.6.4)$$

We define the Hermitian quantum Runge-Lenz operator by Weyl symmetrization:

$$\begin{aligned} \hat{A} &\equiv \frac{1}{2\mu} \left(\hat{\vec{p}} \times \hat{\vec{L}} - \hat{\vec{L}} \times \hat{\vec{p}} \right) - \kappa \frac{\hat{\vec{r}}}{r} \\ &= \frac{1}{\mu} \left(\hat{\vec{p}} \times \hat{\vec{L}} - i\hbar\hat{\vec{p}} \right) - \kappa \frac{\hat{\vec{r}}}{r}. \end{aligned} \quad (6.6.5)$$

Taking the Hermitian conjugate:

$$\hat{A}^\dagger = \frac{1}{2\mu} \left(-\hat{\vec{L}} \times \hat{\vec{p}} + \hat{\vec{p}} \times \hat{\vec{L}} \right) - \kappa \frac{\hat{\vec{r}}}{r} = \hat{A}. \quad (6.6.6)$$

The quantum Runge-Lenz vector is manifestly Hermitian.

6.6.3 Conservation of the Runge-Lenz Vector: $[\hat{H}, \hat{A}] = 0$

We now demonstrate that $\hat{A}$ is a constant of motion:

$$[\hat{H}, \hat{A}] = \left[\frac{p^2}{2\mu} - \frac{\kappa}{r}, \hat{A} \right] = 0. \quad (6.6.7)$$

Let us evaluate the commutators of each piece:

1. Kinetic part with $\hat{\vec{p}} \times \hat{\vec{L}}$: Since $[p^2, p_j] = 0$ and $[p^2, L_k] = 0$ (because p^2 is a scalar under rotations), we have:

$$\left[\frac{p^2}{2\mu}, \frac{1}{\mu} (\hat{\vec{p}} \times \hat{\vec{L}} - i\hbar \hat{\vec{p}}) \right] = 0. \quad (6.6.8)$$

The only non-vanishing contribution from the kinetic term comes from $[p^2, \vec{r}/r]$:

$$\left[\frac{p^2}{2\mu}, -\kappa \frac{\vec{r}}{r} \right]_i = -\frac{\kappa}{2\mu} [p_j p_j, \frac{r_i}{r}] = -\frac{\kappa}{2\mu} \left(p_j [p_j, \frac{r_i}{r}] + [p_j, \frac{r_i}{r}] p_j \right). \quad (6.6.9)$$

Using $[p_j, f(\vec{r})] = -i\hbar \frac{\partial f}{\partial r_j}$ and $\frac{\partial}{\partial r_j} \left(\frac{r_i}{r} \right) = \frac{\delta_{ij}}{r} - \frac{r_i r_j}{r^3}$:

$$\left[\frac{p^2}{2\mu}, -\kappa \frac{\vec{r}}{r} \right] = \frac{i\hbar\kappa}{2\mu} \left[\vec{p} \left(\frac{1}{r} \right) + \left(\frac{1}{r} \right) \vec{p} - \frac{\vec{r}(\vec{r} \cdot \vec{p}) + (\vec{p} \cdot \vec{r})\vec{r}}{r^3} \right]. \quad (6.6.10)$$

2. Potential part with $\hat{\vec{p}} \times \hat{\vec{L}}$: Using $[-\frac{\kappa}{r}, \vec{p}] = i\hbar\kappa \frac{\vec{r}}{r^3}$ and $[\frac{1}{r}, \vec{L}] = 0$:

$$\left[-\frac{\kappa}{r}, \frac{1}{\mu} (\hat{\vec{p}} \times \hat{\vec{L}} - i\hbar \hat{\vec{p}}) \right] = -\frac{i\hbar\kappa}{\mu r^3} [\vec{r} \times (\vec{r} \times \vec{p}) - i\hbar \vec{r}]. \quad (6.6.11)$$

Using the vector triple product $\vec{r} \times (\vec{r} \times \vec{p}) = \vec{r}(\vec{r} \cdot \vec{p}) - r^2 \vec{p}$, this potential commutator precisely cancels the kinetic commutator!

Hence, $[\hat{H}, \hat{\vec{A}}] = 0$, confirming that $\hat{\vec{A}}$ is an exact quantum invariant of the Coulomb problem.

6.6.4 The Commutation Relations and the $SO(4)$ Algebra

Because $\hat{\vec{A}}$ is a vector operator under spatial rotations, its commutator with the angular momentum $\hat{\vec{L}}$ is standard:

$$[\hat{L}_i, \hat{A}_j] = i\hbar \epsilon_{ijk} \hat{A}_k. \quad (6.6.12)$$

A straightforward but lengthy operator calculation reveals the commutator between components of $\hat{\vec{A}}$:

$$[\hat{A}_i, \hat{A}_j] = -i\hbar \left(\frac{2\hat{H}}{\mu} \right) \epsilon_{ijk} \hat{L}_k. \quad (6.6.13)$$

Notice that the right-hand side is proportional to the Hamiltonian $\hat{H}$. Furthermore, the scalar product of $\hat{\vec{L}}$ and $\hat{\vec{A}}$ vanishes:

$$\hat{\vec{L}} \cdot \hat{\vec{A}} = \hat{\vec{A}} \cdot \hat{\vec{L}} = 0, \quad (6.6.14)$$

reflecting the classical orthogonality of the orbital plane and the perihelion vector.

Next, computing the magnitude squared $\hat{\vec{A}}^2$:

$$\hat{\vec{A}}^2 = \frac{2\hat{H}}{\mu} \left(\hat{\vec{L}}^2 + \hbar^2 \right) + \kappa^2. \quad (6.6.15)$$

6.6.5 Decoupling into Two Independent $SU(2)$ Algebras

For bound states, the energy is negative ($E < 0$). In an energy subspace where $\hat{H}$ can be replaced by its eigenvalue $E < 0$, the factor $-2E/\mu$ is strictly positive. We define the rescaled Runge-Lenz operator:

$$\hat{\vec{M}} \equiv \sqrt{-\frac{\mu}{2E}} \hat{\vec{A}}. \quad (6.6.16)$$

Then the commutation relations become:

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k, \quad (6.6.17)$$

$$[\hat{L}_i, \hat{M}_j] = i\hbar\epsilon_{ijk}\hat{M}_k, \quad (6.6.18)$$

$$[\hat{M}_i, \hat{M}_j] = i\hbar\epsilon_{ijk}\hat{L}_k. \quad (6.6.19)$$

These six generators ($\hat{\vec{L}}, \hat{\vec{M}}$) form the Lie algebra of the four-dimensional rotation group $\mathfrak{so}(4)$.

To uncouple this algebra into familiar ladder operators, we define two linear combinations:

$$\hat{j}_1 \equiv \frac{1}{2}(\hat{\vec{L}} + \hat{\vec{M}}), \quad \hat{j}_2 \equiv \frac{1}{2}(\hat{\vec{L}} - \hat{\vec{M}}). \quad (6.6.20)$$

Let us check their mutual commutators:

$$\begin{aligned} [j_{1a}, j_{2b}] &= \frac{1}{4}[L_a + M_a, L_b - M_b] \\ &= \frac{1}{4}([L_a, L_b] - [L_a, M_b] + [M_a, L_b] - [M_a, M_b]) \\ &= \frac{i\hbar}{4}\epsilon_{abc}(L_c - M_c - M_c - L_c) = 0! \end{aligned} \quad (6.6.21)$$

And their self-commutators:

$$\begin{aligned} [j_{1a}, j_{1b}] &= \frac{1}{4}([L_a, L_b] + [L_a, M_b] + [M_a, L_b] + [M_a, M_b]) \\ &= \frac{i\hbar}{4}\epsilon_{abc}(L_c + M_c + M_c + L_c) = i\hbar\epsilon_{abc}j_{1c}. \end{aligned} \quad (6.6.22)$$

Similarly, $[j_{2a}, j_{2b}] = i\hbar\epsilon_{abc}j_{2c}$.

The Lie algebra $\mathfrak{so}(4)$ factors into the direct sum of two completely commuting $\mathfrak{su}(2)$ algebras:

$$\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2). \quad (6.6.23)$$

Each $\mathfrak{su}(2)$ factor possesses standard ladder operators:

$$j_{1\pm} = j_{1x} \pm ij_{1y}, \quad j_{2\pm} = j_{2x} \pm ij_{2y}, \quad (6.6.24)$$

which step between the m_{j1} and m_{j2} projections independently.

6.6.6 Algebraic Derivation of the Energy Spectrum

The irreducible representations of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ are labeled by two independent angular momentum quantum numbers (j_1, j_2) , where $j_1, j_2 \in \{0, 1/2, 1, 3/2, \dots\}$.

However, the physical operators $\hat{\vec{L}}$ and $\hat{\vec{M}}$ are constrained by the orthogonality relation $\hat{\vec{L}} \cdot \hat{\vec{M}} = 0$. Evaluating $\hat{j}_1^2 - \hat{j}_2^2$:

$$\hat{j}_1^2 - \hat{j}_2^2 = \frac{1}{4}(\hat{\vec{L}} + \hat{\vec{M}})^2 - \frac{1}{4}(\hat{\vec{L}} - \hat{\vec{M}})^2 = \hat{\vec{L}} \cdot \hat{\vec{M}} = 0. \quad (6.6.25)$$

Applying this to an irreducible representation (j_1, j_2) requires:

$$\hbar^2 j_1(j_1 + 1) - \hbar^2 j_2(j_2 + 1) = 0 \implies j_1 = j_2 \equiv j. \quad (6.6.26)$$

The physical bound states must belong to diagonal representations $(j, j)!$

Now evaluate the sum of the Casimirs:

$$\hat{j}_1^2 + \hat{j}_2^2 = \frac{1}{2} \left(\hat{L}^2 + \hat{M}^2 \right) = 2\hbar^2 j(j + 1). \quad (6.6.27)$$

Substituting the definition of $\hat{M}^2 = -\frac{\mu}{2E} \hat{A}^2$:

$$\hat{M}^2 = -\frac{\mu}{2E} \left[\frac{2E}{\mu} (\hat{L}^2 + \hbar^2) + \kappa^2 \right] = -(\hat{L}^2 + \hbar^2) - \frac{\mu\kappa^2}{2E}. \quad (6.6.28)$$

Adding $\hat{L}^2$:

$$\hat{L}^2 + \hat{M}^2 = -\hbar^2 - \frac{\mu\kappa^2}{2E}. \quad (6.6.29)$$

Equating this to $4\hbar^2 j(j + 1)$:

$$-\hbar^2 - \frac{\mu\kappa^2}{2E} = 4\hbar^2 j(j + 1) \implies -\frac{\mu\kappa^2}{2E} = \hbar^2 (4j^2 + 4j + 1) = \hbar^2 (2j + 1)^2. \quad (6.6.30)$$

Defining the integer

$$n \equiv 2j + 1 \in \{1, 2, 3, \dots\}, \quad (6.6.31)$$

we solve directly for the energy:

$$E_n = -\frac{\mu\kappa^2}{2\hbar^2 n^2} = -\frac{\mu(Ze^2)^2}{2(4\pi\epsilon_0)^2 \hbar^2 n^2}. \quad (6.6.32)$$

This is the celebrated Bohr-Rydberg formula, derived purely from symmetry without ever solving a differential equation!

6.6.7 Representation Degeneracy and Angular Momentum Composition

The dimension of the irreducible representation (j, j) is

$$\dim(j, j) = (2j_1 + 1)(2j_2 + 1) = (2j + 1)^2 = n^2. \quad (6.6.33)$$

Since $\hat{L} = \hat{j}_1 + \hat{j}_2$, standard angular momentum addition dictates that the total orbital angular momentum l runs over:

$$l \in \{|j - j|, |j - j| + 1, \dots, j + j\} = \{0, 1, 2, \dots, 2j = n - 1\}. \quad (6.6.34)$$

Each value of l appears exactly once, and each contains $2l + 1$ magnetic states m . The hidden $SO(4)$ symmetry explains both the n^2 degeneracy and the allowed range of l .

6.7 Parabolic Coordinates and the Stark Effect

The hidden symmetry of the Coulomb potential also manifests as separability in coordinates other than spherical polar coordinates. Specifically, the hydrogen atom separates cleanly in **parabolic coordinates** (ξ, η, ϕ) .

6.7.1 Definition of Parabolic Coordinates

Parabolic coordinates are defined by:

$$\xi = r + z, \quad \eta = r - z, \quad \phi = \arctan\left(\frac{y}{x}\right), \quad (6.7.1)$$

where $\xi, \eta \in [0, \infty)$ and $\phi \in [0, 2\pi)$. Inverting these relations:

$$r = \frac{\xi + \eta}{2}, \quad z = \frac{\xi - \eta}{2}, \quad x = \sqrt{\xi\eta} \cos \phi, \quad y = \sqrt{\xi\eta} \sin \phi. \quad (6.7.2)$$

Surfaces of constant ξ are paraboloids of revolution opening along the $-z$ direction with focus at the origin, while surfaces of constant η are confocal paraboloids opening along $+z$ (see Figure 6.1). The volume element is:

$$d^3r = \frac{1}{4}(\xi + \eta)d\xi d\eta d\phi. \quad (6.7.3)$$

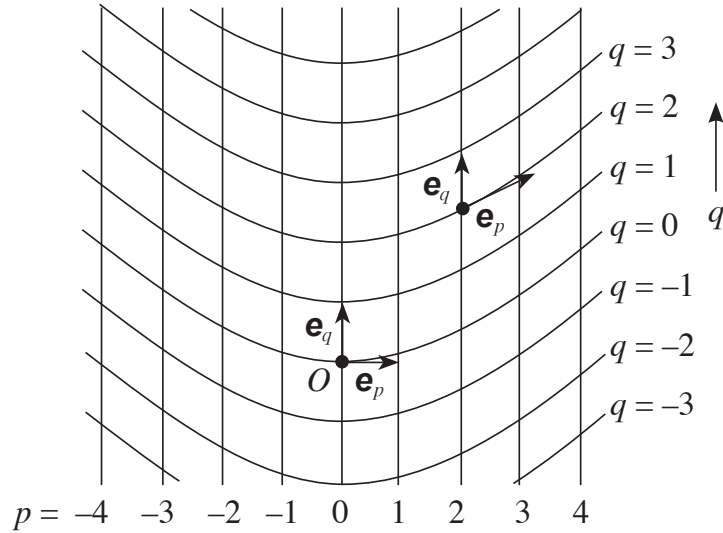


Figure 6.1: Confocal parabolic coordinate system (ξ, η) . Constant ξ curves (blue) and constant η curves (red) form an orthogonal grid of parabolas with a shared focus at the origin.

6.7.2 Separation of Variables in Parabolic Coordinates

The Laplacian in parabolic coordinates is:

$$\nabla^2 = \frac{4}{\xi + \eta} \left[\frac{\partial}{\partial \xi} \left(\xi \frac{\partial}{\partial \xi} \right) + \frac{\partial}{\partial \eta} \left(\eta \frac{\partial}{\partial \eta} \right) \right] + \frac{1}{\xi\eta} \frac{\partial^2}{\partial \phi^2}. \quad (6.7.4)$$

The Schrödinger equation $\left[-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi = E\psi$ becomes:

$$\left\{ \frac{4}{\xi + \eta} \left[\frac{\partial}{\partial \xi} \left(\xi \frac{\partial}{\partial \xi} \right) + \frac{\partial}{\partial \eta} \left(\eta \frac{\partial}{\partial \eta} \right) \right] + \frac{1}{\xi\eta} \frac{\partial^2}{\partial \phi^2} + \frac{2\mu}{\hbar^2} \left[E + \frac{2Ze^2}{4\pi\epsilon_0(\xi + \eta)} \right] \right\} \psi = 0. \quad (6.7.5)$$

We employ the product ansatz:

$$\psi(\xi, \eta, \phi) = f(\xi)g(\eta) \frac{e^{im\phi}}{\sqrt{2\pi}}. \quad (6.7.6)$$

Substituting this ansatz and multiplying by $\frac{\xi+\eta}{4}$, the equation decouples into two identical ODEs linked by separation constants Z_1 and Z_2 :

$$\frac{d}{d\xi} \left(\xi \frac{df}{d\xi} \right) + \left[\frac{\mu E}{2\hbar^2} \xi + \frac{\mu Z_1 e^2}{4\pi\epsilon_0 \hbar^2} - \frac{m^2}{4\xi} \right] f(\xi) = 0, \quad (6.7.7)$$

$$\frac{d}{d\eta} \left(\eta \frac{dg}{d\eta} \right) + \left[\frac{\mu E}{2\hbar^2} \eta + \frac{\mu Z_2 e^2}{4\pi\epsilon_0 \hbar^2} - \frac{m^2}{4\eta} \right] g(\eta) = 0, \quad (6.7.8)$$

with the constraint:

$$Z_1 + Z_2 = Z. \quad (6.7.9)$$

Introducing $\epsilon = \sqrt{-2\mu E/\hbar^2}$ and requiring normalizability at infinity and regularity at the origin leads to polynomial solutions with parabolic quantum numbers $n_1, n_2 \in \{0, 1, 2, \dots\}$:

$$\frac{\mu Z_1 e^2}{4\pi\epsilon_0 \hbar^2 \epsilon} = n_1 + \frac{|m| + 1}{2}, \quad \frac{\mu Z_2 e^2}{4\pi\epsilon_0 \hbar^2 \epsilon} = n_2 + \frac{|m| + 1}{2}. \quad (6.7.10)$$

Summing these two equations:

$$\frac{\mu(Z_1 + Z_2)e^2}{4\pi\epsilon_0 \hbar^2 \epsilon} = \frac{\mu Z e^2}{4\pi\epsilon_0 \hbar^2 \epsilon} = n_1 + n_2 + |m| + 1 \equiv n. \quad (6.7.11)$$

The principal quantum number is $n = n_1 + n_2 + |m| + 1$.

6.7.3 Connection with the Runge-Lenz Component A_z

Why is the Coulomb problem separable in parabolic coordinates? Because the separation of variables in parabolic coordinates diagonalizes the z -component of the Runge-Lenz vector $\hat{A}_z$! In the parabolic basis $|n_1, n_2, m\rangle$, $\hat{H}$, $\hat{L}_z$, and $\hat{A}_z$ are simultaneously diagonal:

$$\hat{H} |n_1, n_2, m\rangle = E_n |n_1, n_2, m\rangle, \quad (6.7.12)$$

$$\hat{L}_z |n_1, n_2, m\rangle = \hbar m |n_1, n_2, m\rangle, \quad (6.7.13)$$

$$\hat{A}_z |n_1, n_2, m\rangle = \frac{Ze^2}{4\pi\epsilon_0 n} (n_1 - n_2) |n_1, n_2, m\rangle. \quad (6.7.14)$$

When an external electric field $\vec{\mathcal{E}} = \mathcal{E}\hat{z}$ is applied along the z -axis (the **Stark effect**), the spherical symmetry is destroyed, but cylindrical symmetry about z is preserved. The perturbation Hamiltonian is:

$$\hat{H}' = e\mathcal{E}z = \frac{1}{2}e\mathcal{E}(\xi - \eta). \quad (6.7.15)$$

Because $\hat{H}'$ depends on $\xi - \eta$, it commutes with $\hat{A}_z$ and is strictly diagonal in the parabolic basis $|n_1, n_2, m\rangle$! Therefore, the first-order Stark energy shift can be evaluated without diagonalizing degenerate perturbation matrices:

$$\Delta E^{(1)} = \langle n_1, n_2, m | e\mathcal{E}z | n_1, n_2, m \rangle = \frac{3}{2}e\mathcal{E}a_0 n (n_1 - n_2). \quad (6.7.16)$$

The linear Stark splitting is directly proportional to $(n_1 - n_2)$, revealing the profound physical utility of parabolic coordinates and the quantum Runge-Lenz vector.

6.8 The Hydrogen Atom in Momentum Space: Fock's Stereographic Projection

In 1935, Vladimir Fock published a groundbreaking paper demonstrating that the “accidental” $SO(4)$ degeneracy of the Hydrogen atom is purely geometric when viewed in **momentum space**.

6.8.1 Fourier Transform of the Coulomb Equation

In coordinate space, the stationary Schrödinger equation for the Hydrogen atom (with $Z = 1$) is:

$$\left(\frac{\hat{p}^2}{2\mu} - E\right)\psi(\vec{r}) = \frac{e^2}{4\pi\epsilon_0 r}\psi(\vec{r}). \quad (6.8.1)$$

The momentum-space wavefunction $\phi(\vec{p}) = \langle \vec{p} | \psi \rangle$ is defined by the three-dimensional Fourier transform:

$$\phi(\vec{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \psi(\vec{r}) e^{-i\vec{p}\cdot\vec{r}/\hbar} d^3r. \quad (6.8.2)$$

Taking the Fourier transform of the Coulomb potential term requires the Fourier transform of $1/r$:

$$\int_{\mathbb{R}^3} \frac{e^{-i\vec{q}\cdot\vec{r}/\hbar}}{r} d^3r = \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^3} \frac{e^{-\epsilon r - i\vec{q}\cdot\vec{r}/\hbar}}{r} d^3r = \frac{4\pi\hbar^2}{q^2}. \quad (6.8.3)$$

By the convolution theorem, the position-space multiplication $\frac{1}{r}\psi(\vec{r})$ transforms into a momentum-space integral convolution:

$$\left(\frac{p^2}{2\mu} - E\right)\phi(\vec{p}) = \frac{e^2}{4\pi\epsilon_0} \frac{1}{2\pi^2\hbar} \int_{\mathbb{R}^3} \frac{\phi(\vec{p}')}{|\vec{p} - \vec{p}'|^2} d^3p'. \quad (6.8.4)$$

For bound states, $E < 0$. We define the characteristic momentum scale:

$$p_0 \equiv \sqrt{-2\mu E} > 0. \quad (6.8.5)$$

Then the kinetic factor becomes $\frac{p^2}{2\mu} - E = \frac{p^2 + p_0^2}{2\mu}$. Dividing by this factor yields the homogeneous Fredholm integral equation:

$$\phi(\vec{p}) = \frac{\mu e^2}{4\pi\epsilon_0 \pi^2 \hbar} \frac{1}{p^2 + p_0^2} \int_{\mathbb{R}^3} \frac{\phi(\vec{p}')}{|\vec{p} - \vec{p}'|^2} d^3p'. \quad (6.8.6)$$

6.8.2 Stereographic Projection onto the 3-Sphere S^3

Fock made the brilliant realization that Eq. (6.8.6) can be mapped onto the surface of a four-dimensional hypersphere S^3 of unit radius embedded in Euclidean four-space $\mathbb{R}^4$:

$$S^3 = \left\{ (u_1, u_2, u_3, u_4) \in \mathbb{R}^4 : u_1^2 + u_2^2 + u_3^2 + u_4^2 = 1 \right\}. \quad (6.8.7)$$

The mapping from 3D momentum space $\vec{p} \in \mathbb{R}^3$ to S^3 is a **stereographic projection**:

$$\vec{u} = \frac{2p_0\vec{p}}{p^2 + p_0^2}, \quad u_4 = \frac{p^2 - p_0^2}{p^2 + p_0^2}. \quad (6.8.8)$$

One immediately verifies that $\vec{u}^2 + u_4^2 = 1$:

$$\sum_{\alpha=1}^4 u_\alpha^2 = \frac{4p_0^2 p^2 + (p^2 - p_0^2)^2}{(p^2 + p_0^2)^2} = \frac{(p^2 + p_0^2)^2}{(p^2 + p_0^2)^2} = 1. \quad (6.8.9)$$

Under this projection, the origin $\vec{p} = 0$ maps to the south pole $\vec{u} = (0, 0, 0, -1)$, while the infinite momentum sphere $|\vec{p}| \rightarrow \infty$ maps to the north pole $(0, 0, 0, +1)$.

The chordal Euclidean distance between two points $\vec{u}$ and $\vec{u}'$ on the hypersphere S^3 satisfies:

$$\sum_{\alpha=1}^4 (u_\alpha - u'_\alpha)^2 = 2(1 - \vec{u} \cdot \vec{u}' - u_4 u'_4) = \frac{4p_0^2 |\vec{p} - \vec{p}'|^2}{(p^2 + p_0^2)(p'^2 + p_0^2)}. \quad (6.8.10)$$

Moreover, the differential surface area element $d\Omega_4$ on S^3 is related to d^3p by:

$$d\Omega_4 = \left(\frac{2p_0}{p^2 + p_0^2} \right)^3 d^3p. \quad (6.8.11)$$

6.8.3 Hyperspherical Harmonics and the $SO(4)$ Spectrum

We introduce a rescaled wavefunction on the sphere S^3 :

$$\Phi(\vec{u}, u_4) \equiv \frac{\pi}{\sqrt{8}} p_0^{-5/2} (p^2 + p_0^2)^2 \phi(\vec{p}). \quad (6.8.12)$$

Substituting this transformation into Eq. (6.8.6), the momentum factors cancel, producing an isotropic integral equation on the 3-sphere:

$$\Phi(u) = \lambda \frac{1}{2\pi^2} \int_{S^3} \frac{\Phi(u')}{|u - u'|^2} d\Omega_4(u'), \quad \lambda \equiv \frac{\mu e^2}{4\pi\epsilon_0 \hbar p_0}. \quad (6.8.13)$$

Equation (6.8.13) is manifestly invariant under the full four-dimensional rotation group $SO(4)$! The integral kernel $1/|u - u'|^2$ is the Green's function for the four-dimensional Laplacian Δ_{S^3} . Its eigenfunctions are the **four-dimensional hyperspherical harmonics** $Y_{nlm}(u)$, with eigenvalues:

$$\lambda = n, \quad n \in \{1, 2, 3, \dots\}. \quad (6.8.14)$$

Equating $\lambda = n$:

$$\frac{\mu e^2}{4\pi\epsilon_0 \hbar p_0} = n \implies p_0 = \frac{\mu e^2}{4\pi\epsilon_0 \hbar n} = \frac{\hbar}{a_0 n}. \quad (6.8.15)$$

Since $E = -\frac{p_0^2}{2\mu}$, we immediately obtain the complete Bohr energy spectrum:

$$E_n = -\frac{p_0^2}{2\mu} = -\frac{\mu e^4}{2(4\pi\epsilon_0)^2 \hbar^2 n^2} = -\frac{\text{Ry}}{n^2} \quad (6.8.16)$$

The n^2 degeneracy of each Hydrogen energy level is nothing other than the dimension of the degree- n irreducible representation of the 4D rotation group $SO(4)$ on S^3 !

6.8.4 Ground-State Momentum Wavefunction

For the ground state ($n = 1, l = 0, m = 0$), the hyperspherical harmonic on S^3 is a constant: $\Phi(u) = \Phi_0 = \frac{1}{\sqrt{2\pi^2}}$. Inverting the transformation:

$$\phi_{100}(\vec{p}) = \sqrt{\frac{8}{\pi}} \frac{p_0^{5/2}}{(p^2 + p_0^2)^2}, \quad p_0 = \frac{\hbar}{a_0}. \quad (6.8.17)$$

The probability density of measuring momentum $\vec{p}$ in the ground state of Hydrogen is:

$$P(\vec{p}) = |\phi_{100}(\vec{p})|^2 = \frac{8}{\pi} \frac{p_0^5}{(p^2 + p_0^2)^4}. \quad (6.8.18)$$

Unlike a Gaussian wavepacket, this distribution decays asymptotically as p^{-8} at large momentum:

$$P(p) \propto \frac{1}{p^8} \quad \text{as } p \rightarrow \infty. \quad (6.8.19)$$

This slow power-law decay is the Fourier signature of the sharp Coulomb cusp in coordinate space ($\psi(r) \sim 1 - r/a_0$ as $r \rightarrow 0$), directly confirming that high-momentum Fourier components originate from the singularity at the atomic nucleus.

Chapter 7

The Dirac Equation, Relativistic Quantum Mechanics, and Identical Particles

7.1 From the Klein-Gordon Equation to the Dirac Equation

Non-relativistic quantum mechanics, as codified in the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right) \psi, \quad (7.1.1)$$

is built upon the classical Galilean dispersion relation $E = \vec{p}^2/(2m)$. Consequently, it treats space and time on fundamentally asymmetric footings: the equation contains a first-order derivative with respect to time t , but second-order derivatives with respect to the spatial coordinates $\vec{r}$. Such an asymmetry is fundamentally incompatible with Einstein's Special Theory of Relativity, in which space and time transform into each other under Lorentz transformations.

7.1.1 Shortcomings of the Klein-Gordon Equation

The most direct attempt to construct a relativistic wave equation begins by promoting the Einstein energy-momentum relation for a free particle of rest mass m :

$$E^2 = \vec{p}^2 c^2 + m^2 c^4, \quad (7.1.2)$$

into differential operators via the standard quantum substitutions $E \rightarrow i\hbar \partial/\partial t$ and $\vec{p} \rightarrow -i\hbar \vec{\nabla}$. This yields the **Klein-Gordon equation**:

$$-\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = \left(-\hbar^2 c^2 \nabla^2 + m^2 c^4 \right) \psi \implies \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2 c^2}{\hbar^2} \right) \psi = 0. \quad (7.1.3)$$

In relativistic 4-vector notation ($x^\mu = (ct, \vec{r})$, $\partial_\mu = (\frac{1}{c} \partial/\partial t, \vec{\nabla})$), this is compactly written as:

$$\left(\partial_\mu \partial^\mu + \frac{m^2 c^2}{\hbar^2} \right) \psi = 0, \quad (\partial_\mu \partial^\mu = \square). \quad (7.1.4)$$

While manifestly Lorentz invariant, the Klein-Gordon equation suffered from what appeared at the time to be fatal defects for a single-particle probability theory:

1. **Second Order in Time:** A second-order differential equation in time requires specifying both $\psi(\vec{r}, 0)$ and its time derivative $\dot{\psi}(\vec{r}, 0)$ as initial conditions. This violates the fundamental postulate of quantum mechanics that the state of a system at $t = 0$ is completely specified by a single state vector $|\psi(0)\rangle$.

2. **Indefinite Probability Density:** By multiplying the equation by ψ^* and subtracting its complex conjugate, one obtains the continuity equation $\partial\rho/\partial t + \vec{\nabla} \cdot \vec{j} = 0$, where

$$\rho = \frac{i\hbar}{2mc^2} \left(\psi^* \frac{\partial\psi}{\partial t} - \psi \frac{\partial\psi^*}{\partial t} \right). \quad (7.1.5)$$

Because ψ and $\partial\psi/\partial t$ can be chosen independently at $t = 0$, $\rho(\vec{r}, t)$ can take arbitrarily negative values! An observable that can become negative cannot represent a physical probability density.

3. **Negative Energy Solutions:** Plane wave solutions $\psi \propto e^{i(\vec{p}\cdot\vec{r}-Et)/\hbar}$ yield two branches of energy:

$$E = \pm \sqrt{\vec{p}^2 c^2 + m^2 c^4}. \quad (7.1.6)$$

Without an exclusion principle (since Klein-Gordon was formulated for spin-0 scalars), electrons would continuously emit radiation and cascade into infinitely negative energy states, destroying the stability of matter.

7.1.2 Dirac's Linearization Ansatz

In 1928, Paul Dirac recognized that to preserve a positive-definite probability density, the wave equation must be *first order* in time:

$$i\hbar \frac{\partial\psi}{\partial t} = \hat{H}_D \psi. \quad (7.1.7)$$

To satisfy the requirements of Special Relativity, space and time must enter symmetrically. Therefore, the Dirac Hamiltonian $\hat{H}_D$ must also be strictly *linear* in the spatial momentum operators $\hat{\vec{p}} = -i\hbar\vec{\nabla}$:

$$\hat{H}_D = c\vec{\alpha} \cdot \hat{\vec{p}} + \beta mc^2 = c(\alpha_1 \hat{p}_x + \alpha_2 \hat{p}_y + \alpha_3 \hat{p}_z) + \beta mc^2, \quad (7.1.8)$$

where the coefficients $\alpha_1, \alpha_2, \alpha_3$, and β cannot be ordinary numbers, but must be linear operators (matrices) acting on the multicomponent state vector ψ .

7.1.3 The Dirac Matrix Algebra

To ensure that a free relativistic particle satisfies the Einstein relation $\hat{H}^2 = \vec{p}^2 c^2 + m^2 c^4$, we square the Dirac Hamiltonian:

$$\begin{aligned} \hat{H}_D^2 &= \left(c \sum_{i=1}^3 \alpha_i p_i + \beta mc^2 \right) \left(c \sum_{j=1}^3 \alpha_j p_j + \beta mc^2 \right) \\ &= c^2 \sum_{i=1}^3 \alpha_i^2 p_i^2 + c^2 \sum_{i<j} (\alpha_i \alpha_j + \alpha_j \alpha_i) p_i p_j + mc^3 \sum_{i=1}^3 (\alpha_i \beta + \beta \alpha_i) p_i + \beta^2 m^2 c^4. \end{aligned} \quad (7.1.9)$$

Comparing this directly with $\hat{H}_D^2 = c^2 \sum_i p_i^2 + m^2 c^4$, we find that the operators must satisfy the **Clifford algebra**:

$$\alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} I, \quad (7.1.10)$$

$$\alpha_i \beta + \beta \alpha_i = 0, \quad (7.1.11)$$

$$\beta^2 = I, \quad (7.1.12)$$

for all $i, j \in \{1, 2, 3\}$. Furthermore, because $\hat{H}_D$ is an observable, it must be Hermitian:

$$\alpha_i^\dagger = \alpha_i, \quad \beta^\dagger = \beta. \quad (7.1.13)$$

These algebraic conditions impose powerful constraints on the nature of the matrices:

1. **Eigenvalues:** Since $\alpha_i^2 = I$ and $\beta^2 = I$, the eigenvalues of α_i and β are exclusively ± 1 .
2. **Tracelessness:** Using the anticommutation relation $\alpha_i\beta = -\beta\alpha_i$ and the cyclic property of the trace:

$$\text{Tr}(\alpha_i) = \text{Tr}(\alpha_i\beta^2) = \text{Tr}(\beta\alpha_i\beta) = -\text{Tr}(\alpha_i\beta^2) = -\text{Tr}(\alpha_i) \implies \text{Tr}(\alpha_i) = 0. \quad (7.1.14)$$

Identical reasoning shows that $\text{Tr}(\beta) = 0$.

3. **Dimension:** Because the eigenvalues are ± 1 and must sum to zero (tracelessness), the dimension N of the matrices must be an *even integer*.
4. **Why $N = 4$:** In $N = 2$ dimensions, the maximum number of mutually anticommuting Hermitian matrices is three (the three Pauli matrices $\sigma_x, \sigma_y, \sigma_z$). However, the Dirac algebra requires *four* mutually anticommuting matrices ($\alpha_1, \alpha_2, \alpha_3, \beta$). Therefore, $N = 2$ is mathematically insufficient, and the minimum possible dimension is $N = 4$!

Thus, relativistic quantum mechanics inevitably demands that the wavefunction of an electron is a four-component column vector, known as a **Dirac 4-spinor**:

$$\psi(\vec{r}, t) = \begin{pmatrix} \psi_1(\vec{r}, t) \\ \psi_2(\vec{r}, t) \\ \psi_3(\vec{r}, t) \\ \psi_4(\vec{r}, t) \end{pmatrix}. \quad (7.1.15)$$

7.1.4 The Dirac-Pauli Representation

The standard 4×4 representation (the Dirac-Pauli representation) is constructed using 2×2 blocks containing the Pauli spin matrices $\vec{\sigma}$ and the 2×2 identity matrix I_2 :

$$\vec{\alpha} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad (7.1.16)$$

where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (7.1.17)$$

7.1.5 Covariant Form and Gamma Matrices

Multiplying the Dirac equation $i\hbar \partial\psi/\partial t = (-i\hbar c\vec{\alpha} \cdot \vec{\nabla} + \beta mc^2)\psi$ from the left by β/c yields:

$$\left(i\hbar\beta \frac{1}{c} \frac{\partial}{\partial t} + i\hbar\beta\vec{\alpha} \cdot \vec{\nabla} - mc \right) \psi = 0. \quad (7.1.18)$$

We define the four **Dirac gamma matrices** γ^μ ($\mu = 0, 1, 2, 3$):

$$\gamma^0 \equiv \beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^k \equiv \beta\alpha_k = \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix}. \quad (7.1.19)$$

Notice that $(\gamma^0)^\dagger = \gamma^0$, whereas $(\gamma^k)^\dagger = (\beta\alpha_k)^\dagger = \alpha_k\beta = -\beta\alpha_k = -\gamma^k$. The gamma matrices satisfy the covariant anticommutation relations:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I_4, \quad g^{\mu\nu} = \text{diag}(+1, -1, -1, -1). \quad (7.1.20)$$

The Dirac equation then assumes its celebrated covariant form:

$$(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0 \iff (i\hbar\not{\partial} - mc)\psi = 0, \quad (7.1.21)$$

where Feynman slash notation denotes $\not{A} \equiv \gamma^\mu A_\mu$.

7.1.6 Continuity Equation and Positive-Definite Probability Density

To derive the conserved current, we start from the Dirac equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -i\hbar c \sum_{k=1}^3 \alpha_k \frac{\partial \psi}{\partial x_k} + \beta mc^2 \psi. \quad (7.1.22)$$

Taking the Hermitian conjugate (and using $\alpha_k^\dagger = \alpha_k, \beta^\dagger = \beta$):

$$-i\hbar \frac{\partial \psi^\dagger}{\partial t} = i\hbar c \sum_{k=1}^3 \frac{\partial \psi^\dagger}{\partial x_k} \alpha_k + \psi^\dagger \beta mc^2. \quad (7.1.23)$$

Multiplying the first equation from the left by $\psi^\dagger$ and the second from the right by ψ , and subtracting:

$$i\hbar \left(\psi^\dagger \frac{\partial \psi}{\partial t} + \frac{\partial \psi^\dagger}{\partial t} \psi \right) = -i\hbar c \sum_{k=1}^3 \left[\psi^\dagger \alpha_k \frac{\partial \psi}{\partial x_k} + \left(\frac{\partial \psi^\dagger}{\partial x_k} \right) \alpha_k \psi \right]. \quad (7.1.24)$$

Dividing by $i\hbar$, we arrive at the exact continuity equation:

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0, \quad (7.1.25)$$

where the probability density ρ is given by

$$\rho(\vec{r}, t) = \psi^\dagger \psi = \sum_{a=1}^4 |\psi_a(\vec{r}, t)|^2 \geq 0, \quad (7.1.26)$$

and the probability current density is

$$\vec{j}(\vec{r}, t) = c\psi^\dagger \vec{\alpha} \psi. \quad (7.1.27)$$

Unlike the Klein-Gordon case, the Dirac probability density $\rho = \psi^\dagger \psi$ is **strictly positive-definite** everywhere and for all time! In covariant form, defining the Dirac adjoint $\bar{\psi} \equiv \psi^\dagger \gamma^0$, the 4-current is $j^\mu = c\bar{\psi} \gamma^\mu \psi$, which obeys the Lorentz-invariant conservation law:

$$\partial_\mu j^\mu = 0. \quad (7.1.28)$$

7.2 Non-Relativistic Limit: Derivation of the Pauli Equation

To understand how the Dirac equation connects with non-relativistic quantum mechanics, we introduce an electromagnetic field $(\Phi, \vec{A})$ via the gauge-invariant **minimal coupling**:

$$i\hbar \frac{\partial}{\partial t} \rightarrow i\hbar \frac{\partial}{\partial t} - q\Phi, \quad \hat{\vec{p}} \rightarrow \hat{\vec{\pi}} = \hat{\vec{p}} - q\vec{A}, \quad (7.2.1)$$

where q is the particle charge (for an electron, $q = -e$). The Dirac equation in an electromagnetic field is:

$$\left(i\hbar \frac{\partial}{\partial t} - q\Phi \right) \psi = \left[c\vec{\alpha} \cdot (\vec{p} - q\vec{A}) + \beta mc^2 \right] \psi. \quad (7.2.2)$$

7.2.1 Large and Small Components

In the Dirac-Pauli representation, we split the 4-spinor ψ into two 2-component spinors ϕ (upper two components) and χ (lower two components). For low energy, the rest mass energy mc^2 dominates the time evolution. We explicitly factor out this rapid phase oscillation:

$$\psi(\vec{r}, t) = e^{-imc^2 t/\hbar} \begin{pmatrix} \phi(\vec{r}, t) \\ \chi(\vec{r}, t) \end{pmatrix}. \quad (7.2.3)$$

Differentiating with respect to time:

$$i\hbar \frac{\partial \psi}{\partial t} = e^{-imc^2 t/\hbar} \left[mc^2 \begin{pmatrix} \phi \\ \chi \end{pmatrix} + i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \phi \\ \chi \end{pmatrix} \right]. \quad (7.2.4)$$

Substituting the explicit block forms of $\vec{\alpha} = \begin{pmatrix} 0 & \vec{\sigma} \\ \vec{\sigma} & 0 \end{pmatrix}$ and $\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$:

$$mc^2 \begin{pmatrix} \phi \\ \chi \end{pmatrix} + i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \phi \\ \chi \end{pmatrix} - q\Phi \begin{pmatrix} \phi \\ \chi \end{pmatrix} = c \begin{pmatrix} 0 & \vec{\sigma} \cdot \vec{\pi} \\ \vec{\sigma} \cdot \vec{\pi} & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \chi \end{pmatrix} + mc^2 \begin{pmatrix} \phi \\ -\chi \end{pmatrix}. \quad (7.2.5)$$

Subtracting $mc^2\phi$ from the upper row and adding $mc^2\chi$ to the lower row yields the exact coupled system:

$$\left(i\hbar \frac{\partial}{\partial t} - q\Phi \right) \phi = c(\vec{\sigma} \cdot \vec{\pi})\chi, \quad (7.2.6)$$

$$\left(i\hbar \frac{\partial}{\partial t} - q\Phi + 2mc^2 \right) \chi = c(\vec{\sigma} \cdot \vec{\pi})\phi. \quad (7.2.7)$$

In the non-relativistic regime, the kinetic and potential energies are negligible compared to the electron rest energy:

$$\left| i\hbar \frac{\partial \chi}{\partial t} \right| \ll 2mc^2 |\chi|, \quad |q\Phi \chi| \ll 2mc^2 |\chi|. \quad (7.2.8)$$

Under this approximation, equation (7.2.7) simplifies dramatically:

$$2mc^2 \chi \approx c(\vec{\sigma} \cdot \vec{\pi})\phi \implies \chi \approx \frac{\vec{\sigma} \cdot \vec{\pi}}{2mc} \phi \sim \mathcal{O}\left(\frac{v}{c}\right) \phi. \quad (7.2.9)$$

Because χ is suppressed relative to ϕ by a factor of $v/c \ll 1$, ϕ is termed the **large component** (representing the electronic degree of freedom) and χ the **small component** (mixing with positronic degrees of freedom).

7.2.2 The Pauli Equation and the Landé g -Factor

Substituting $\chi \approx \frac{\vec{\sigma} \cdot \vec{\pi}}{2mc} \phi$ into equation (7.2.6):

$$i\hbar \frac{\partial \phi}{\partial t} = \left[\frac{(\vec{\sigma} \cdot \vec{\pi})^2}{2m} + q\Phi \right] \phi. \quad (7.2.10)$$

To evaluate $(\vec{\sigma} \cdot \vec{\pi})^2$, we employ the fundamental identity for Pauli matrices:

$$(\vec{\sigma} \cdot \vec{u})(\vec{\sigma} \cdot \vec{v}) = (\vec{u} \cdot \vec{v})I + i\vec{\sigma} \cdot (\vec{u} \times \vec{v}). \quad (7.2.11)$$

Setting $\vec{u} = \vec{v} = \vec{\pi} = \vec{p} - q\vec{A}$:

$$(\vec{\sigma} \cdot \vec{\pi})^2 = \vec{\pi}^2 + i\vec{\sigma} \cdot (\vec{\pi} \times \vec{\pi}). \quad (7.2.12)$$

Because $\vec{p}$ and $\vec{A}$ do not commute, the cross-product $\vec{\pi} \times \vec{\pi}$ does not vanish! Let us evaluate its z -component:

$$\begin{aligned} (\vec{\pi} \times \vec{\pi})_z &= \pi_x \pi_y - \pi_y \pi_x = [p_x - qA_x, p_y - qA_y] \\ &= -q[p_x, A_y] - q[A_x, p_y] = -q(-i\hbar \frac{\partial A_y}{\partial x}) - q(i\hbar \frac{\partial A_x}{\partial y}) \\ &= i\hbar q \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) = i\hbar q (\vec{\nabla} \times \vec{A})_z = i\hbar q B_z. \end{aligned} \quad (7.2.13)$$

Thus, in full generality:

$$\vec{\pi} \times \vec{\pi} = i\hbar q \vec{B}. \quad (7.2.14)$$

Substituting this back into the operator square gives:

$$(\vec{\sigma} \cdot \vec{\pi})^2 = (\vec{p} - q\vec{A})^2 + i\vec{\sigma} \cdot (i\hbar q \vec{B}) = (\vec{p} - q\vec{A})^2 - q\hbar \vec{\sigma} \cdot \vec{B}. \quad (7.2.15)$$

We have arrived directly at the **Pauli equation**:

$$i\hbar \frac{\partial \phi}{\partial t} = \left[\frac{(\vec{p} - q\vec{A})^2}{2m} - \frac{q\hbar}{2m} \vec{\sigma} \cdot \vec{B} + q\Phi \right] \phi. \quad (7.2.16)$$

7.2.3 Natural Emergence of Spin and $g = 2$

The term $-\frac{q\hbar}{2m} \vec{\sigma} \cdot \vec{B}$ describes the Zeeman interaction between the magnetic field $\vec{B}$ and the intrinsic magnetic dipole moment $\vec{\mu}$ of the particle:

$$\hat{H}_{\text{Zeeman}} = -\vec{\mu} \cdot \vec{B}. \quad (7.2.17)$$

Defining the quantum spin operator $\vec{S} \equiv \frac{\hbar}{2} \vec{\sigma}$, the Zeeman Hamiltonian becomes:

$$-\vec{\mu} \cdot \vec{B} = -\frac{q\hbar}{2m} \vec{\sigma} \cdot \vec{B} = -2 \left(\frac{q}{2m} \right) \vec{S} \cdot \vec{B} = -g \left(\frac{q}{2m} \right) \vec{S} \cdot \vec{B}. \quad (7.2.18)$$

The gyromagnetic ratio (Landé g -factor) is revealed to be:

$$g = 2! \quad (7.2.19)$$

For an electron with charge $q = -e$:

$$\vec{\mu}_e = -g \frac{e}{2m} \vec{S} = -2\mu_B \frac{\vec{S}}{\hbar}, \quad \left(\mu_B \equiv \frac{e\hbar}{2m} \right). \quad (7.2.20)$$

In non-relativistic physics, both the existence of spin-1/2 and its anomalous g -factor of 2 had to be bolted on ad-hoc (the Uhlenbeck-Goudsmit hypothesis). Dirac's derivation proved that electron spin and $g = 2$ are not arbitrary dynamical assumptions, but necessary kinematic consequences of combining quantum mechanics with special relativity!

7.3 Fine Structure and the Foldy-Wouthuysen Transformation

When we carry the non-relativistic expansion of the Dirac equation to next order in $(v/c)^2 \sim 1/c^2$, we obtain the relativistic fine-structure corrections to the hydrogen atom.

Systematically, this expansion is achieved through a unitary transformation $\psi \rightarrow e^{iS}\psi$ that block-diagonalizes the Hamiltonian to order $1/c^2$, decoupling the positive- and negative-energy states. This is the celebrated **Foldy-Wouthuysen transformation**. For a particle moving

in a central electrostatic potential $V(r) = q\Phi(r)$, the effective Hamiltonian acting on the large component ϕ is:

$$\hat{H}_{\text{FW}} \approx mc^2 + \frac{\vec{p}^2}{2m} - \frac{\vec{p}^4}{8m^3c^2} + V(r) + \frac{1}{2m^2c^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \cdot \vec{S} + \frac{\hbar^2}{8m^2c^2} \nabla^2 V(r). \quad (7.3.1)$$

Aside from the rest energy mc^2 and the standard Schrödinger Hamiltonian $\frac{p^2}{2m} + V(r)$, three distinct fine-structure terms appear:

7.3.1 1. Relativistic Kinetic Correction

The term

$$\hat{H}'_{\text{kin}} = -\frac{\vec{p}^4}{8m^3c^2}, \quad (7.3.2)$$

originates from the relativistic kinetic energy expansion:

$$T = \sqrt{\vec{p}^2c^2 + m^2c^4} - mc^2 = mc^2 \left(\sqrt{1 + \frac{p^2}{m^2c^2}} - 1 \right) = \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + \mathcal{O}(p^6/c^4). \quad (7.3.3)$$

It represents the first relativistic mass-velocity correction.

7.3.2 2. Spin-Orbit Coupling and Thomas Precession

The term

$$\hat{H}_{\text{SO}} = \frac{1}{2m^2c^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \cdot \vec{S}, \quad (7.3.4)$$

couples the electron's spin $\vec{S}$ to its orbital angular momentum $\vec{L}$.

- A classical electron moving with velocity $\vec{v}$ through an electrostatic field $\vec{E} = -\vec{\nabla}V = -\frac{1}{q} \frac{1}{r} \frac{dV}{dr} \vec{r}$ experiences in its rest frame an effective magnetic field:

$$\vec{B}' \approx -\frac{\vec{v} \times \vec{E}}{c} = \frac{1}{qmc} \frac{1}{r} \frac{dV}{dr} = \frac{1}{qmc} \frac{1}{r} \frac{dV}{dr} \vec{L}. \quad (7.3.5)$$

Interacting with the magnetic moment $\vec{\mu} = 2\frac{q}{2m} \vec{S}$ yields an apparent interaction energy $-\vec{\mu} \cdot \vec{B}' = \frac{1}{m^2c^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \cdot \vec{S}$.

- This is twice as large as the correct quantum result! In 1926, Llewellyn Thomas realized that because the electron's rest frame is continuously accelerating, its coordinate axes rotate via relativistic kinematic precession (Thomas precession) at frequency $\vec{\omega}_T \approx -\frac{1}{2c^2} (\vec{v} \times \vec{a})$, reducing the net interaction by exactly a factor of 1/2.
- The Dirac equation produces this Thomas precession factor of 1/2 **completely automatically**, without requiring any ad-hoc relativistic boost corrections.

7.3.3 3. The Darwin Term

The third correction is the **Darwin term**:

$$\hat{H}_{\text{Darwin}} = \frac{\hbar^2}{8m^2c^2} \nabla^2 V(r). \quad (7.3.6)$$

For the Coulomb potential $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$, using Poisson's equation $\nabla^2(1/r) = -4\pi\delta^3(\vec{r})$:

$$\hat{H}_{\text{Darwin}} = \frac{\hbar^2}{8m^2c^2} \left(\frac{4\pi Ze^2}{4\pi\epsilon_0} \right) \delta^3(\vec{r}) = \frac{\pi Ze^2\hbar^2}{2m^2c^2(4\pi\epsilon_0)} \delta^3(\vec{r}). \quad (7.3.7)$$

Because $\delta^3(\vec{r})$ has support solely at the origin $r = 0$, and spatial wavefunctions with $l > 0$ vanish as $r^l \rightarrow 0$, the Darwin term contributes **only to s-states** ($l = 0$)! It provides an upward shift in energy for s-electrons, explaining why s-states do not experience the spin-orbit splitting yet are shifted relative to other states.

7.4 Zitterbewegung and Relativistic Causality

7.4.1 Zitterbewegung: The Rapid Quantum Trembling

To uncover the physical origin of the Darwin term, we examine the velocity operator in the Heisenberg picture:

$$\hat{v} \equiv \frac{d\vec{x}}{dt} = \frac{i}{\hbar} [\hat{H}_D, \vec{x}] = \frac{i}{\hbar} [c\vec{\alpha} \cdot \vec{p} + \beta mc^2, \vec{x}]. \quad (7.4.1)$$

Using $[p_j, x_k] = -i\hbar\delta_{jk}$, we find:

$$\hat{v} = c\vec{\alpha}. \quad (7.4.2)$$

Because $\alpha_k^2 = I$, the eigenvalues of each velocity component are strictly $\pm c$! An instantaneous measurement of the electron velocity always yields the speed of light.

To understand how this relates to subluminal group velocity, we calculate the acceleration:

$$i\hbar \frac{d\vec{\alpha}}{dt} = [\vec{\alpha}, \hat{H}_D] = \vec{\alpha}\hat{H}_D - \hat{H}_D\vec{\alpha} = 2\vec{\alpha}\hat{H}_D - \{\vec{\alpha}, \hat{H}_D\}. \quad (7.4.3)$$

Using $\{\alpha_k, \hat{H}_D\} = 2cp_k$, this differential equation becomes:

$$i\hbar \frac{d\vec{\alpha}}{dt} = 2\vec{\alpha}\hat{H}_D - 2c\vec{p}. \quad (7.4.4)$$

Since $\hat{H}_D$ and $\vec{p}$ are constants of motion for a free particle, we can integrate this equation:

$$\vec{\alpha}(t) = \frac{c\vec{p}}{\hat{H}_D} + \left(\vec{\alpha}(0) - \frac{c\vec{p}}{\hat{H}_D} \right) e^{-2i\hat{H}_D t/\hbar}. \quad (7.4.5)$$

Integrating once more with respect to time yields the position operator trajectory:

$$\vec{x}(t) = \vec{x}(0) + \frac{c^2\vec{p}}{\hat{H}_D} t + \frac{i\hbar c}{2\hat{H}_D} \left(\vec{\alpha}(0) - \frac{c\vec{p}}{\hat{H}_D} \right) e^{-2i\hat{H}_D t/\hbar}. \quad (7.4.6)$$

The position operator consists of two distinct parts:

- A classical drift term $\vec{x}(0) + \frac{c^2\vec{p}}{E}t$, describing rectilinear motion at the group velocity $\vec{v}_g = c^2\vec{p}/E$.
- An ultrafast oscillatory term, named by Schrödinger as **Zitterbewegung** (trembling motion), with characteristic frequency:

$$\omega_Z = \frac{2mc^2}{\hbar} \approx 1.55 \times 10^{21} \text{ rad/s}, \quad (7.4.7)$$

and spatial amplitude:

$$\delta r \approx \frac{\hbar}{2mc} = \frac{\lambda_c}{2} \approx 1.93 \times 10^{-13} \text{ m}, \quad (7.4.8)$$

where $\lambda_c = \hbar/(mc)$ is the reduced Compton wavelength of the electron.

7.4.2 Physical Demystification of the Darwin Term

Zitterbewegung is a direct consequence of the quantum interference between positive-energy ($E > 0$) and negative-energy ($E < 0$) components in the wavepacket. Because the electron trembles rapidly over a sphere of radius $\sim \lambda_c$, it does not feel the electrostatic potential $V(\vec{r})$ at a single point. Instead, it experiences a spatially averaged potential:

$$\langle V(\vec{r} + \delta\vec{r}) \rangle \approx V(\vec{r}) + \langle \delta\vec{r} \rangle \cdot \vec{\nabla} V(\vec{r}) + \frac{1}{2} \sum_{i,j} \langle \delta r_i \delta r_j \rangle \frac{\partial^2 V}{\partial x_i \partial x_j}. \quad (7.4.9)$$

Assuming isotropic, zero-mean fluctuations ($\langle \delta\vec{r} \rangle = 0$, $\langle \delta r_i \delta r_j \rangle = \frac{1}{3} \langle (\delta r)^2 \rangle \delta_{ij}$):

$$\langle V(\vec{r} + \delta\vec{r}) \rangle \approx V(\vec{r}) + \frac{1}{6} \langle (\delta r)^2 \rangle \nabla^2 V(\vec{r}). \quad (7.4.10)$$

Substituting the Zitterbewegung amplitude $\langle (\delta r)^2 \rangle \sim 3 \left(\frac{\hbar}{2mc} \right)^2$:

$$\Delta V = \frac{1}{6} \cdot 3 \left(\frac{\hbar}{2mc} \right)^2 \nabla^2 V(\vec{r}) = \frac{\hbar^2}{8m^2 c^2} \nabla^2 V(\vec{r})! \quad (7.4.11)$$

The Darwin term is nothing other than the effective electrostatic smearing of the electron due to relativistic Zitterbewegung!

7.4.3 Relativistic Causality: From Single-Particle QM to QFT

In non-relativistic quantum mechanics, the spatial propagator for a free particle is

$$K(\vec{x}, t; \vec{x}_0, 0) = \langle \vec{x} | e^{-i\hat{H}t/\hbar} | \vec{x}_0 \rangle = \left(\frac{m}{2\pi i \hbar t} \right)^{3/2} \exp \left(\frac{im|\vec{x} - \vec{x}_0|^2}{2\hbar t} \right). \quad (7.4.12)$$

For any $t > 0$, no matter how infinitesimal, $K(\vec{x}, t; \vec{x}_0, 0) \neq 0$ for all points $\vec{x}$ across the entire universe, including regions where $|\vec{x} - \vec{x}_0| \gg ct$! Thus, non-relativistic quantum mechanics inherently permits superluminal propagation and violates Einstein causality.

In relativistic quantum mechanics, attempting to strictly localize a particle within a region smaller than its Compton wavelength ($\Delta x < \hbar/mc$) implies a momentum uncertainty $\Delta p > mc$ by Heisenberg's principle. This corresponds to an energy uncertainty:

$$\Delta E \sim c \Delta p > mc^2. \quad (7.4.13)$$

This energy is sufficient to create real particle-antiparticle pairs from the vacuum! Consequently, any attempt to define a single-particle localized position state $|\vec{x}\rangle$ inevitably destabilizes the single-particle Hilbert space into an infinite-dimensional Fock space.

In **Quantum Field Theory (QFT)**, causality is rigorously preserved through the principle of **microcausality**: space-like separated field operators must commute (for bosons) or anticommute (for fermions):

$$[\phi(x), \phi(y)] = 0 \quad \text{or} \quad \{\psi(x), \bar{\psi}(y)\} = 0 \quad \text{for } (x - y)^2 < 0. \quad (7.4.14)$$

The quantum transition amplitude for a particle to propagate from x to y outside the forward light cone is precisely cancelled by the destructive quantum interference of an antiparticle propagating from y to x ! Relativistic causality is preserved in nature exclusively because antimatter exists.

7.4.4 The Non-Relativistic Limit, Galilean Relativity, and the Return of Position Eigenkets

A profound conceptual question arises at this juncture: *Why and how are we justified in treating the Dirac equation as a single-particle wave equation equipped with coordinate operators $\hat{\vec{x}}$, momentum operators $\hat{\vec{p}}$, and position-space wavefunctions $\psi(\vec{x}, t)$?*

In full relativistic Quantum Field Theory (QFT), position is famously **not** a dynamical operator, but merely a continuous classical parameter or spacetime coordinate $x^\mu = (ct, \vec{x})$ that labels the quantum field operators $\hat{\psi}(x)$. Furthermore, as shown above, attempting to define a strictly localized position eigenket $|\vec{x}\rangle$ in a relativistic theory fails because localizing an excitation within a Compton volume ($\Delta x \lesssim \hbar/(mc)$) injects sufficient energy ($\Delta E \gtrsim mc^2$) to create real particle-antiparticle pairs from the vacuum, immediately forcing us out of a single-particle Hilbert space into an infinite-dimensional Fock space.

However, an essential thing to notice is that **in the non-relativistic limit, we deliberately drop the strict causality argument inherent to quantum field theory**. Once the requirement of relativistic microcausality across all space-like intervals is relaxed (or taken in the formal limit $c \rightarrow \infty$):

1. **Decoupling of the Vacuum:** Pair production is completely suppressed because the energy required to produce a particle-antiparticle pair, $2mc^2$, becomes infinitely large compared to the kinetic and potential energies available in the non-relativistic regime ($E - mc^2 \ll mc^2$). The particle number becomes a strictly conserved superselection quantity.
2. **Decoupling from Fock Space:** Consequently, the single-particle sector decouples entirely from the infinite-dimensional many-body Fock space.
3. **Return of Position Eigenkets:** Within this decoupled single-particle Hilbert space, **sharp position eigenkets $|\vec{x}\rangle$ are once again legitimately allowed to exist!**
4. **Restoration of Canonical Operators:** And with the resurrection of position eigenkets $|\vec{x}\rangle$, the position operator $\hat{\vec{x}}$, the conjugate momentum operator $\hat{\vec{p}}$, and their fundamental canonical commutator:

$$[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}\hat{I} \quad (7.4.15)$$

make their triumphant and necessary way back into the theory!

The Galilean Analogy: Emergence of a Global Vector Space. The physical rationale for dropping the QFT causality constraint to restore position eigenkets is directly and beautifully analogous to how a universal vector space emerges in classical mechanics when transitioning from Special Relativity to Galilean Relativity:

- In **Special Relativity**, spacetime is a four-dimensional pseudo-Riemannian manifold (Minkowski space). Because the speed of light c is finite and invariant, each inertial observer possesses their own distinct reference frame with its own local spacetime slicing $(t, \vec{x})$. Transforming between frames via Lorentz boosts mixes space and time, leading to relativistic length contraction and time dilation:

$$\Delta x' = \gamma(\Delta x - v\Delta t), \quad \Delta t' = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right). \quad (7.4.16)$$

Because spatial intervals and directions are observer-dependent, there is no observer-independent, universal three-dimensional vector space: spatial vectors do not transform by simple Euclidean rotations, and the naive Euclidean triangle law of vector addition

fails between reference frames without accounting for Lorentz contractions and Wigner rotations. Each observer essentially has their own spatial vector space, and transforming between them alters the vectors.

- In **Galilean Relativity**, however, we take the formal limit $c \rightarrow \infty$. In this limit, time becomes absolute ($t' = t$), and length contraction and time dilation are completely ignored ($\gamma \rightarrow 1$). By dropping these relativistic spacetime distortions, space completely decouples from time, yielding a single, universal, **global vector space** $\mathbb{R}^3$ common to all inertial observers. Within this global vector space, spatial vectors exist independently of the observer, and the elementary Euclidean triangle law of vector addition:

$$\vec{r}_{\text{total}} = \vec{r}_1 + \vec{r}_2 \quad (7.4.17)$$

holds unconditionally.

In precisely the same manner, by taking the non-relativistic limit ($c \rightarrow \infty$) of relativistic quantum theory, we ignore the local light-cone causality restrictions that govern field operators in QFT. This allows us to construct a single, well-defined, global single-particle Hilbert space where position eigenkets $|\vec{x}\rangle$ form a complete basis:

$$\int d^3x |\vec{x}\rangle \langle \vec{x}| = \hat{I}. \quad (7.4.18)$$

It is this foundational step that enables us to treat the Dirac equation not merely as a classical field equation, but as an effective single-particle quantum Hamiltonian $\hat{H}_D = c\vec{\alpha} \cdot \hat{\vec{p}} + \beta mc^2 + V(\hat{\vec{x}})$, systematically reducing it via the Pauli and Foldy-Wouthuysen approximations to describe non-relativistic electrons with spin and relativistic fine-structure corrections.

7.5 Identical Particles and Exchange Symmetry

7.5.1 Quantum Indistinguishability

In classical mechanics, identical particles (such as identical billiard balls) are always fundamentally distinguishable because we can continuously trace their phase-space trajectories $(\vec{r}_i(t), \vec{p}_i(t))$ without disturbing their motion.

In quantum mechanics, the Heisenberg uncertainty principle prohibits continuous phase-space tracking: any attempt to measure the position of a particle creates an unpredictable momentum disturbance that destroys trajectory continuity. When two identical particles (particles possessing identical intrinsic properties such as mass, electric charge, and spin) interact and separate, it is physically impossible, even in principle, to determine which particle emerged where. Identical particles in quantum mechanics are **fundamentally indistinguishable**.

7.5.2 The Permutation Operator $\hat{P}_{12}$ and the Origin of the $\pm$ Sign

Consider a system of two identical particles (such as two identical atoms in an optical lattice, or two electrons in a multi-atom molecule). Let $|\psi_1, \psi_2\rangle \equiv |\psi_1\rangle \otimes |\psi_2\rangle$ denote the product state where particle 1 is in state $|\psi_1\rangle$ and particle 2 is in state $|\psi_2\rangle$. We define the **permutation (or exchange) operator** $\hat{P}_{12}$ by its action of interchanging the labels and physical coordinates of the two particles:

$$\hat{P}_{12} |\psi_1, \psi_2\rangle = |\psi_2, \psi_1\rangle. \quad (7.5.1)$$

Applying $\hat{P}_{12}$ twice restores the original order of the state vector:

$$\hat{P}_{12}^2 |\psi_1, \psi_2\rangle = \hat{P}_{12} |\psi_2, \psi_1\rangle = |\psi_1, \psi_2\rangle \implies \boxed{\hat{P}_{12}^2 = \hat{I}} \quad (7.5.2)$$

The permutation operator is therefore an **involution**: it is its own inverse ($\hat{P}_{12}^{-1} = \hat{P}_{12}$). Furthermore, because it preserves inner products ($\langle \hat{P}_{12}\Phi | \hat{P}_{12}\Psi \rangle = \langle \Phi | \Psi \rangle$), $\hat{P}_{12}$ is unitary ($\hat{P}_{12}^\dagger \hat{P}_{12} = \hat{I}$), which together with $\hat{P}_{12}^2 = \hat{I}$ proves that $\hat{P}_{12}$ is also **Hermitian**:

$$\hat{P}_{12}^\dagger = \hat{P}_{12}^{-1} = \hat{P}_{12}. \quad (7.5.3)$$

Because $\hat{P}_{12}$ is a Hermitian operator, its eigenvalues must be real.

Exchange Invariance of the Multi-Particle Hamiltonian: Now consider the Hamiltonian of a two-body or multi-atom system:

$$\hat{H}(1, 2) = \frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} + V_{\text{ext}}(\hat{r}_1) + V_{\text{ext}}(\hat{r}_2) + V_{\text{int}}(|\hat{r}_1 - \hat{r}_2|). \quad (7.5.4)$$

Because the two particles possess identical masses, identical charges, and experience identical external potentials, exchanging particle 1 and particle 2 leaves the physical Hamiltonian completely invariant:

$$\hat{P}_{12}\hat{H}(1, 2)\hat{P}_{12}^{-1} = \hat{H}(2, 1) = \hat{H}(1, 2). \quad (7.5.5)$$

Multiplying from the right by $\hat{P}_{12}$, we obtain the fundamental commutation relation:

$$\boxed{[\hat{H}, \hat{P}_{12}] = 0} \quad (7.5.6)$$

More generally, because identical particles are physically indistinguishable, *any* physically observable operator $\hat{O}(1, 2)$ must treat both particles on an equal footing, meaning $[\hat{O}, \hat{P}_{12}] = 0$.

Simultaneous Diagonalization: How the $\pm$ Sign Enters Quantum Mechanics: In linear algebra and quantum mechanics, the fundamental spectral theorem dictates: **two commuting Hermitian operators can be simultaneously diagonalized**. That is, they share a complete, common set of eigenstates.

Because $[\hat{H}, \hat{P}_{12}] = 0$, we are not merely allowed, but physically *demand*ed by the indistinguishability of identical particles, to seek stationary states $|\Psi\rangle$ that are simultaneous eigenstates of both the Hamiltonian and the exchange operator:

$$\hat{H}|\Psi\rangle = E|\Psi\rangle, \quad \hat{P}_{12}|\Psi\rangle = \lambda|\Psi\rangle. \quad (7.5.7)$$

What are the allowed eigenvalues λ ? Applying the operator identity (7.5.2) to the state $|\Psi\rangle$:

$$\hat{P}_{12}^2|\Psi\rangle = \hat{P}_{12}(\lambda|\Psi\rangle) = \lambda^2|\Psi\rangle = \hat{I}|\Psi\rangle = |\Psi\rangle \implies \lambda^2 = 1. \quad (7.5.8)$$

Therefore, the eigenvalues of the permutation operator are mathematically constrained to be strictly:

$$\boxed{\lambda = \pm 1} \quad (7.5.9)$$

This is the exact reason why the $\pm$ sign enters multi-particle quantum mechanics!

A simple unsymmetrized product state $|\psi_a, \psi_b\rangle = |\psi_a\rangle_1 \otimes |\psi_b\rangle_2$ is generally *not* an eigenstate of $\hat{P}_{12}$, because $\hat{P}_{12}|\psi_a, \psi_b\rangle = |\psi_b, \psi_a\rangle \neq \lambda|\psi_a, \psi_b\rangle$ whenever $|\psi_a\rangle \neq |\psi_b\rangle$. However, because $[\hat{H}, \hat{P}_{12}] = 0$, the physical eigenstates of the Hamiltonian must lie in the eigenspaces of $\hat{P}_{12}$. We project any product state into these definite eigenspaces using the **symmetrization** and **antisymmetrization projection operators**:

$$\hat{\Pi}_+ \equiv \frac{1}{2}(\hat{I} + \hat{P}_{12}), \quad \hat{\Pi}_- \equiv \frac{1}{2}(\hat{I} - \hat{P}_{12}). \quad (7.5.10)$$

These projection operators satisfy the complete set of projection properties:

$$\hat{\Pi}_{\pm}^2 = \hat{\Pi}_{\pm}, \quad \hat{\Pi}_{+}\hat{\Pi}_{-} = \hat{\Pi}_{-}\hat{\Pi}_{+} = 0, \quad \hat{\Pi}_{+} + \hat{\Pi}_{-} = \hat{I}. \quad (7.5.11)$$

Applying $\hat{\Pi}_{\pm}$ to our product state generates the physical simultaneous eigenstates corresponding to eigenvalues $\lambda = \pm 1$:

$$|\Psi_{\pm}\rangle \equiv \sqrt{2} \hat{\Pi}_{\pm} |\psi_a, \psi_b\rangle = \frac{1}{\sqrt{2}} (|\psi_a\rangle_1 |\psi_b\rangle_2 \pm |\psi_b\rangle_1 |\psi_a\rangle_2). \quad (7.5.12)$$

In the coordinate representation $\langle \vec{r}_1, \vec{r}_2 | \Psi_{\pm} \rangle$, this yields the familiar symmetric (+) and antisymmetric (−) wavefunctions:

$$\Psi_{\pm}(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}} [\psi_a(\vec{r}_1)\psi_b(\vec{r}_2) \pm \psi_b(\vec{r}_1)\psi_a(\vec{r}_2)]. \quad (7.5.13)$$

Conservation of Permutation Symmetry and Superselection: By the Heisenberg equation of motion, the expectation value of $\hat{P}_{12}$ evolves as:

$$\frac{d\langle \hat{P}_{12} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{P}_{12}] \rangle = 0. \quad (7.5.14)$$

Furthermore, because $[\hat{H}, \hat{P}_{12}] = 0$, the Hamiltonian also commutes with the projectors: $[\hat{H}, \hat{\Pi}_{\pm}] = 0$. Consequently, the Hamiltonian has vanishing matrix elements between the symmetric and antisymmetric subspaces:

$$\langle \Psi_{+} | \hat{H} | \Psi_{-} \rangle = \langle \Psi_{+} | \hat{\Pi}_{+} \hat{H} \hat{\Pi}_{-} | \Psi_{-} \rangle = \langle \Psi_{+} | \hat{H} \hat{\Pi}_{+} \hat{\Pi}_{-} | \Psi_{-} \rangle = 0. \quad (7.5.15)$$

No physical interaction or external field can ever induce a transition between the + and − sectors! The full Hilbert space permanently splits into two uncoupled **superselection sectors**:

$$\mathcal{H} = \mathcal{H}_{+} \oplus \mathcal{H}_{-}. \quad (7.5.16)$$

Conceptual Foundation: Why $[\hat{H}, \hat{P}_{12}] = 0$ Demands Simultaneous Eigenstates and Generates the $\pm$ Sign

Students often ask: “Where does the $\pm$ sign in $\psi_a(\vec{r}_1)\psi_b(\vec{r}_2) \pm \psi_b(\vec{r}_1)\psi_a(\vec{r}_2)$ come from? Why not $e^{i\theta}$, or something else?”

The derivation above reveals the deep mathematical and physical chain of reasoning:

1. **Physical Indistinguishability:** In a multi-particle or multi-atom system, nature does not possess individual particle “tags”. Swapping particle coordinates and momenta leaves the system Hamiltonian identical: $\hat{H}(1, 2) = \hat{H}(2, 1)$, which translates to $[\hat{H}, \hat{P}_{12}] = 0$.
2. **Simultaneous Diagonalization:** Because the Hamiltonian commutes with the exchange operator $\hat{P}_{12}$, the stationary states of the system must be simultaneous eigenstates of both operators: $\hat{H}|\Psi\rangle = E|\Psi\rangle$ and $\hat{P}_{12}|\Psi\rangle = \lambda|\Psi\rangle$.
3. **The Involutory Constraint $\hat{P}_{12}^2 = \hat{I}$:** Exchanging two particles and then immediately exchanging them again brings the system back to its original state. Therefore, $\hat{P}_{12}^2 = \hat{I}$. Applying this to the eigenstate gives $\hat{P}_{12}^2|\Psi\rangle = \lambda^2|\Psi\rangle = |\Psi\rangle$, which enforces:

$$\lambda^2 = 1 \implies \lambda = \pm 1.$$

The $\pm$ sign is neither arbitrary nor a mere convenience; it is the **only algebraic solution** compatible with indistinguishability and two-particle exchange!

4. **Superselection and Stability:** Because $[\hat{O}, \hat{P}_{12}] = 0$ for all physical observables $\hat{O}$, no perturbation can mix $\lambda = +1$ states with $\lambda = -1$ states. If a system is created as a boson ($\lambda = +1$) or a fermion ($\lambda = -1$), it remains in that symmetry sector under all evolution for all time.

7.5.3 The Symmetrization Postulate

Nature restricts the physically realized states of identical particles according to the **Symmetrization Postulate** (proven in relativistic QFT as the **Spin-Statistics Theorem**):

- **Bosons (Integer Spin: $S = 0, 1, 2, \dots$):** The total state vector must be completely *symmetric* under the exchange of any two particles:

$$\hat{P}_{12} |\Psi\rangle = + |\Psi\rangle. \quad (7.5.17)$$

Bosons obey Bose-Einstein statistics. Any number of bosons may occupy the exact same quantum state, leading to phenomena such as Bose-Einstein condensation, lasers, and superfluidity.

- **Fermions (Half-Integer Spin: $S = 1/2, 3/2, \dots$):** The total state vector must be completely *antisymmetric* under the exchange of any two particles:

$$\hat{P}_{12} |\Psi\rangle = - |\Psi\rangle. \quad (7.5.18)$$

Fermions obey Fermi-Dirac statistics. Electrons, protons, neutrons, and quarks are all fermions.

7.5.4 The Pauli Exclusion Principle and Slater Determinants

For two identical fermions in single-particle states $|a\rangle$ and $|b\rangle$, the physical antisymmetrized state is

$$|\Psi_{12}\rangle = \frac{1}{\sqrt{2}} (|a\rangle_1 |b\rangle_2 - |b\rangle_1 |a\rangle_2). \quad (7.5.19)$$

If both fermions attempt to occupy the same quantum state ($|a\rangle = |b\rangle$):

$$|\Psi_{12}\rangle = \frac{1}{\sqrt{2}} (|a\rangle_1 |a\rangle_2 - |a\rangle_1 |a\rangle_2) = 0! \quad (7.5.20)$$

The state vanishes identically. This is the **Pauli Exclusion Principle**: *no two identical fermions can ever occupy the same quantum state.*

For an N -fermion system occupying single-particle orbitals $\phi_1, \phi_2, \dots, \phi_N$, the normalized, totally antisymmetric wavefunction is expressed as the **Slater Determinant**:

$$\Psi(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \phi_1(\vec{x}_1) & \phi_2(\vec{x}_1) & \cdots & \phi_N(\vec{x}_1) \\ \phi_1(\vec{x}_2) & \phi_2(\vec{x}_2) & \cdots & \phi_N(\vec{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\vec{x}_N) & \phi_2(\vec{x}_N) & \cdots & \phi_N(\vec{x}_N) \end{pmatrix}. \quad (7.5.21)$$

Exchanging any two particle coordinates $\vec{x}_i \leftrightarrow \vec{x}_j$ corresponds to swapping two rows of the matrix, which changes the sign of the determinant ($\hat{P}_{ij}\Psi = -\Psi$). If any two single-particle states are identical ($\phi_i = \phi_j$), two columns are identical and the determinant vanishes identically.

7.5.5 Space and Spin Decomposition for Two Electrons

For electrons (spin-1/2), the total state vector decomposes into the tensor product of a spatial wavefunction and a spin state:

$$|\Psi\rangle = |\psi_{\text{space}}\rangle \otimes |\chi_{\text{spin}}\rangle. \quad (7.5.22)$$

The permutation operator acts on both sectors simultaneously:

$$\hat{P}_{12} |\Psi\rangle = (\hat{P}_{12}^{\text{space}} |\psi_{\text{space}}\rangle) \otimes (\hat{P}_{12}^{\text{spin}} |\chi_{\text{spin}}\rangle) = -|\Psi\rangle. \quad (7.5.23)$$

The spin Hilbert space of two spin-1/2 particles decomposes as $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$:

1. **Triplet States** ($S = 1$, $M_S = +1, 0, -1$):

$$|1, 1\rangle = |\uparrow\uparrow\rangle, \quad (7.5.24)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad (7.5.25)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle. \quad (7.5.26)$$

These states are **symmetric** under spin exchange: $\hat{P}_{12}^{\text{spin}} |1, M_S\rangle = +|1, M_S\rangle$. Therefore, to maintain total antisymmetry, the spatial wavefunction must be strictly **antisymmetric**:

$$\psi_A(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}} [\phi_a(\vec{r}_1)\phi_b(\vec{r}_2) - \phi_b(\vec{r}_1)\phi_a(\vec{r}_2)]. \quad (7.5.27)$$

2. **Singlet State** ($S = 0$, $M_S = 0$):

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (7.5.28)$$

This state is **antisymmetric** under spin exchange: $\hat{P}_{12}^{\text{spin}} |0, 0\rangle = -|0, 0\rangle$. Therefore, the spatial wavefunction must be strictly **symmetric**:

$$\psi_S(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}} [\phi_a(\vec{r}_1)\phi_b(\vec{r}_2) + \phi_b(\vec{r}_1)\phi_a(\vec{r}_2)]. \quad (7.5.29)$$

7.5.6 Exchange Interaction and the Origin of Ferromagnetism

Consider the electrostatic Coulomb repulsion between two electrons:

$$\hat{V}_C = \frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|}. \quad (7.5.30)$$

Notice that the electrostatic interaction $\hat{V}_C$ is purely spatial and contains no spin operators whatsoever. However, because the spatial wavefunction symmetry is locked to the spin state by the Symmetrization Postulate, the expectation value of $\hat{V}_C$ depends strongly on whether the electrons form a spin singlet or triplet!

Calculating the expectation value:

$$\langle \psi_{S/A} | \hat{V}_C | \psi_{S/A} \rangle = J \pm K, \quad (7.5.31)$$

where:

- J is the **Direct Coulomb Integral** (classical electrostatic repulsion of charge distributions):

$$J \equiv \iint d^3r_1 d^3r_2 |\phi_a(\vec{r}_1)|^2 \frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|} |\phi_b(\vec{r}_2)|^2 > 0. \quad (7.5.32)$$

- K is the **Exchange Integral** (a purely quantum interference effect due to orbital overlap):

$$K \equiv \iint d^3r_1 d^3r_2 \phi_a^*(\vec{r}_1) \phi_b^*(\vec{r}_2) \frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|} \phi_a(\vec{r}_2) \phi_b(\vec{r}_1) > 0. \quad (7.5.33)$$

The total energies of the singlet and triplet configurations are:

$$E_{\text{singlet}} = E_a + E_b + J + K, \quad (7.5.34)$$

$$E_{\text{triplet}} = E_a + E_b + J - K. \quad (7.5.35)$$

The energy difference between the triplet and singlet states is:

$$\Delta E = E_{\text{triplet}} - E_{\text{singlet}} = -2K < 0! \quad (7.5.36)$$

The triplet state (parallel spins, $S = 1$) has *lower* energy than the singlet state (antiparallel spins, $S = 0$).

The Fermi Hole

Why does the triplet state have lower electrostatic energy? Consider the probability density of finding both electrons at the exact same spatial location $\vec{r}_1 = \vec{r}_2 = \vec{r}$:

$$\psi_A(\vec{r}, \vec{r}) = \frac{1}{\sqrt{2}} [\phi_a(\vec{r}) \phi_b(\vec{r}) - \phi_b(\vec{r}) \phi_a(\vec{r})] = 0! \quad (7.5.37)$$

The antisymmetric spatial wavefunction vanishes identically at $\vec{r}_1 = \vec{r}_2$. Around each electron in a triplet state, there exists a spatial exclusion zone known as the **Fermi hole**, which keeps electrons with parallel spins apart. In contrast, the symmetric spatial wavefunction ψ_S enhances the probability of finding the electrons close together. Because electrons in the triplet state avoid each other spatially, their Coulomb repulsion is substantially smaller, lowering their total energy by $2K$. This is the fundamental physical origin of **Hund's first rule** in atomic physics.

The Heisenberg Exchange Hamiltonian

We can express this exchange energy as an effective operator acting on the spin degrees of freedom. Recall that the total spin operator is $\vec{S} = \vec{S}_1 + \vec{S}_2$:

$$\vec{S}^2 = \vec{S}_1^2 + \vec{S}_2^2 + 2\vec{S}_1 \cdot \vec{S}_2 = \frac{3}{2}\hbar^2 + 2\vec{S}_1 \cdot \vec{S}_2 \implies \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} \left(\vec{S}^2 - \frac{3}{2}\hbar^2 \right). \quad (7.5.38)$$

Evaluating for both states:

$$\text{Triplet } (S = 1): \quad \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} \left(2\hbar^2 - \frac{3}{2}\hbar^2 \right) = +\frac{1}{4}\hbar^2, \quad (7.5.39)$$

$$\text{Singlet } (S = 0): \quad \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2} \left(0 - \frac{3}{2}\hbar^2 \right) = -\frac{3}{4}\hbar^2. \quad (7.5.40)$$

The spin exchange operator can be written identically as:

$$\hat{P}_{12}^{\text{spin}} = \frac{1}{2} \hat{I} + \frac{2}{\hbar^2} \vec{S}_1 \cdot \vec{S}_2. \quad (7.5.41)$$

The effective Hamiltonian in spin space takes the form:

$$\hat{H}_{\text{eff}} = \left(E_a + E_b + J - \frac{1}{2}K \right) \hat{I} - \frac{2K}{\hbar^2} \vec{S}_1 \cdot \vec{S}_2 = \text{const} - 2J_{\text{ex}} \vec{S}_1 \cdot \vec{S}_2, \quad (7.5.42)$$

where $J_{\text{ex}} \equiv K/\hbar^2 > 0$ is the exchange coupling constant.

This is the celebrated **Heisenberg Exchange Hamiltonian**. It reveals one of the most magnificent syntheses in modern physics:

The interaction responsible for the alignment of electron spins in ferromagnets (such as iron) is NOT the magnetic dipole-dipole interaction (which is far too weak, corresponding to Curie temperatures of only ~ 1 K). Rather, it is the enormous electrostatic Coulomb repulsion ($\sim 10^3$ K), coupled to the spin configuration via the Pauli exclusion principle and exchange symmetry!

Chapter 8

Time-Independent Perturbation Theory and Atomic Fine Structure

In realistic quantum systems, exact analytic solutions to the stationary Schrödinger equation $\hat{H}|\psi\rangle = E|\psi\rangle$ are exceptionally rare, essentially restricted to idealized models such as the harmonic oscillator, the particle in an infinite box, and the unperturbed Coulomb potential. For virtually all physical problems of experimental relevance—ranging from relativistic corrections in atoms to external electromagnetic fields and multi-electron interactions—one must resort to systematic approximation techniques.

Among these, **time-independent perturbation theory** (historically formulated by Lord Rayleigh in classical acoustics and extended to quantum mechanics by Erwin Schrödinger) provides an algebraic and operator-based method for obtaining systematically improvable asymptotic expansions of eigenvalues and eigenstates.

8.1 The Perturbative Paradigm and Non-Degenerate Formulations

8.1.1 Motivation and the Tracking Parameter λ

Suppose we have a quantum system whose total Hamiltonian $\hat{H}$ cannot be diagonalized exactly, but can be partitioned into two parts:

$$\hat{H} = \hat{H}_0 + \hat{V}, \quad (8.1.1)$$

where $\hat{H}_0$ is an unperturbed, exactly solvable Hamiltonian whose discrete spectrum and complete orthonormal eigenbasis are known:

$$\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle, \quad \langle m^{(0)} | n^{(0)} \rangle = \delta_{mn}, \quad \sum_n |n^{(0)}\rangle \langle n^{(0)}| = \hat{I}. \quad (8.1.2)$$

The operator $\hat{V}$ (often denoted $\delta\hat{H}$) represents a small perturbation. In general, $\hat{V}$ is not diagonal in the unperturbed eigenbasis $\{|n^{(0)}\rangle\}$; its off-diagonal matrix elements $V_{mn} = \langle m^{(0)} | \hat{V} | n^{(0)} \rangle$ couple different unperturbed states together.

To systematically keep track of the relative magnitude of corrections, we introduce a formal, dimensionless tracking parameter $\lambda \in [0, 1]$ and consider the family of Hamiltonians:

$$\hat{H}(\lambda) = \hat{H}_0 + \lambda\hat{V} = \hat{H}_0 + \lambda\delta\hat{H}. \quad (8.1.3)$$

Setting $\lambda = 0$ smoothly recovers the unperturbed system $\hat{H}_0$, while setting $\lambda = 1$ corresponds to the full physical Hamiltonian $\hat{H}$. We assume that the true energy eigenvalues $E_n(\lambda)$ and eigenstates $|n(\lambda)\rangle$ are analytic functions of λ in a neighborhood of $\lambda = 0$:

$$(\hat{H}_0 + \lambda\hat{V}) |n\rangle = E_n |n\rangle. \quad (8.1.4)$$

8.1.2 Formal Series Expansions

We expand the energy eigenvalue E_n and the state vector $|n\rangle$ as formal power series in powers of λ :

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \lambda^3 E_n^{(3)} + \dots = \sum_{k=0}^{\infty} \lambda^k E_n^{(k)}, \quad (8.1.5)$$

$$|n\rangle = c |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \lambda^3 |n^{(3)}\rangle + \dots = c |n^{(0)}\rangle + \sum_{k=1}^{\infty} \lambda^k |n^{(k)}\rangle. \quad (8.1.6)$$

Here c is a normalization constant that will be determined by the normalization condition.

8.1.3 Normalization Conditions: Intermediate vs. Total Normalization

There are two standard conventions for normalizing the perturbed state $|n\rangle$:

Method 1: Total Normalization ($\langle n|n\rangle = 1$). Evaluating the inner product using the series expansion (8.1.6):

$$\begin{aligned} \langle n|n\rangle &= \left(c^* \langle n^{(0)}| + \lambda \langle n^{(1)}| + \lambda^2 \langle n^{(2)}| + \dots \right) \left(c |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right) \\ &= |c|^2 \langle n^{(0)}|n^{(0)}\rangle + \lambda \left(c^* \langle n^{(0)}|n^{(1)}\rangle + c \langle n^{(1)}|n^{(0)}\rangle \right) \\ &\quad + \lambda^2 \left(c^* \langle n^{(0)}|n^{(2)}\rangle + c \langle n^{(2)}|n^{(0)}\rangle + \langle n^{(1)}|n^{(1)}\rangle \right) + \mathcal{O}(\lambda^3) = 1. \end{aligned} \quad (8.1.7)$$

Since this must hold identically for all values of λ :

$$\mathcal{O}(\lambda^0) : |c|^2 = 1 \implies c = 1 \quad (\text{choosing overall phase real and positive}), \quad (8.1.8)$$

$$\mathcal{O}(\lambda^1) : \langle n^{(0)}|n^{(1)}\rangle + \langle n^{(1)}|n^{(0)}\rangle = 2 \operatorname{Re} \langle n^{(0)}|n^{(1)}\rangle = 0, \quad (8.1.9)$$

$$\mathcal{O}(\lambda^2) : \langle n^{(0)}|n^{(2)}\rangle + \langle n^{(2)}|n^{(0)}\rangle + \langle n^{(1)}|n^{(1)}\rangle = 0 \implies 2 \operatorname{Re} \langle n^{(0)}|n^{(2)}\rangle = -\langle n^{(1)}|n^{(1)}\rangle. \quad (8.1.10)$$

Since the overall phase of $|n\rangle$ can be chosen freely, setting the imaginary part of $\langle n^{(0)}|n^{(1)}\rangle$ to zero enforces:

$$\langle n^{(0)}|n^{(1)}\rangle = 0. \quad (8.1.11)$$

Method 2: Intermediate Normalization ($\langle n^{(0)}|n\rangle = 1$). Because any scalar multiple of an eigenstate is equally an eigenstate, it is mathematically more convenient to impose the **intermediate normalization condition**:

$$\langle n^{(0)}|n\rangle = 1. \quad (8.1.12)$$

Substituting the power series (8.1.6) with $c = 1$ into (8.1.12):

$$\langle n^{(0)}|n^{(0)}\rangle + \lambda \langle n^{(0)}|n^{(1)}\rangle + \lambda^2 \langle n^{(0)}|n^{(2)}\rangle + \dots = 1. \quad (8.1.13)$$

Since $\langle n^{(0)}|n^{(0)}\rangle = 1$, matching coefficients of λ^k for each order $k \geq 1$ yields:

$$\boxed{\langle n^{(0)}|n^{(k)}\rangle = 0, \quad \forall k \geq 1} \quad (8.1.14)$$

Thus, under intermediate normalization, every higher-order correction $|n^{(k)}\rangle$ ($k \geq 1$) lies strictly in the subspace orthogonal to the unperturbed state $|n^{(0)}\rangle$.

8.1.4 Order-by-Order Matching

Substituting the power series (8.1.5) and (8.1.6) into the stationary Schrödinger equation $(\hat{H}_0 + \lambda \hat{V})|n\rangle = E_n|n\rangle$:

$$\begin{aligned} & (\hat{H}_0 + \lambda \hat{V}) \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right) \\ &= \left(E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \right) \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right). \end{aligned} \quad (8.1.15)$$

Expanding both sides and equating coefficients of identical powers of λ :

$$\mathcal{O}(\lambda^0): \quad (\hat{H}_0 - E_n^{(0)}) |n^{(0)}\rangle = 0, \quad (8.1.16)$$

$$\mathcal{O}(\lambda^1): \quad (\hat{H}_0 - E_n^{(0)}) |n^{(1)}\rangle = (E_n^{(1)} - \hat{V}) |n^{(0)}\rangle, \quad (8.1.17)$$

$$\mathcal{O}(\lambda^2): \quad (\hat{H}_0 - E_n^{(0)}) |n^{(2)}\rangle = E_n^{(2)} |n^{(0)}\rangle + E_n^{(1)} |n^{(1)}\rangle - \hat{V} |n^{(1)}\rangle, \quad (8.1.18)$$

$$\mathcal{O}(\lambda^3): \quad (\hat{H}_0 - E_n^{(0)}) |n^{(3)}\rangle = E_n^{(3)} |n^{(0)}\rangle + E_n^{(2)} |n^{(1)}\rangle + E_n^{(1)} |n^{(2)}\rangle - \hat{V} |n^{(2)}\rangle. \quad (8.1.19)$$

8.1.5 First-Order Corrections

To extract the first-order energy shift $E_n^{(1)}$, project equation (8.1.17) onto the unperturbed bra $\langle n^{(0)}|$:

$$\langle n^{(0)} | (\hat{H}_0 - E_n^{(0)}) |n^{(1)}\rangle = \langle n^{(0)} | (E_n^{(1)} - \hat{V}) |n^{(0)}\rangle. \quad (8.1.20)$$

Because $\hat{H}_0$ is Hermitian and $\langle n^{(0)} | \hat{H}_0 = E_n^{(0)} \langle n^{(0)} |$, the left-hand side vanishes identically:

$$\langle n^{(0)} | \hat{H}_0 |n^{(1)}\rangle - E_n^{(0)} \langle n^{(0)} |n^{(1)}\rangle = E_n^{(0)} \langle n^{(0)} |n^{(1)}\rangle - E_n^{(0)} \langle n^{(0)} |n^{(1)}\rangle = 0. \quad (8.1.21)$$

The right-hand side becomes:

$$E_n^{(1)} \langle n^{(0)} |n^{(0)}\rangle - \langle n^{(0)} | \hat{V} |n^{(0)}\rangle = 0 \implies \boxed{E_n^{(1)} = \langle n^{(0)} | \hat{V} |n^{(0)}\rangle = V_{nn}} \quad (8.1.22)$$

The first-order energy correction is simply the expectation value of the perturbing potential in the unperturbed state.

To determine the first-order state correction $|n^{(1)}\rangle$, we expand $|n^{(1)}\rangle$ in the complete orthonormal eigenbasis of $\hat{H}_0$:

$$|n^{(1)}\rangle = \sum_k c_k^{(1)} |k^{(0)}\rangle, \quad c_k^{(1)} = \langle k^{(0)} |n^{(1)}\rangle. \quad (8.1.23)$$

From intermediate normalization (8.1.14), $c_n^{(1)} = \langle n^{(0)} |n^{(1)}\rangle = 0$. Projecting (8.1.17) onto an orthogonal bra $\langle k^{(0)} |$ ($k \neq n$):

$$\langle k^{(0)} | (\hat{H}_0 - E_n^{(0)}) |n^{(1)}\rangle = E_n^{(1)} \langle k^{(0)} |n^{(0)}\rangle - \langle k^{(0)} | \hat{V} |n^{(0)}\rangle. \quad (8.1.24)$$

Using $\langle k^{(0)} | \hat{H}_0 = E_k^{(0)} \langle k^{(0)} |$ and $\langle k^{(0)} |n^{(0)}\rangle = 0$:

$$(E_k^{(0)} - E_n^{(0)}) c_k^{(1)} = -V_{kn} \implies c_k^{(1)} = \frac{\langle k^{(0)} | \hat{V} |n^{(0)}\rangle}{E_n^{(0)} - E_k^{(0)}}. \quad (8.1.25)$$

Thus, the first-order state vector is:

$$\boxed{|n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k^{(0)} | \hat{V} |n^{(0)}\rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle} \quad (8.1.26)$$

8.1.6 Second-Order Corrections and Level Repulsion

Projecting the second-order equation (8.1.18) onto $\langle n^{(0)} |$:

$$0 = E_n^{(2)} \langle n^{(0)} | n^{(0)} \rangle + E_n^{(1)} \langle n^{(0)} | n^{(1)} \rangle - \langle n^{(0)} | \hat{V} | n^{(1)} \rangle. \quad (8.1.27)$$

Since $\langle n^{(0)} | n^{(1)} \rangle = 0$:

$$E_n^{(2)} = \langle n^{(0)} | \hat{V} | n^{(1)} \rangle. \quad (8.1.28)$$

Inserting the expansion (8.1.26) for $|n^{(1)}\rangle$:

$$E_n^{(2)} = \sum_{k \neq n} \frac{\langle n^{(0)} | \hat{V} | k^{(0)} \rangle \langle k^{(0)} | \hat{V} | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} = \sum_{k \neq n} \frac{|\langle k^{(0)} | \hat{V} | n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}} \quad (8.1.29)$$

Theorem 8.1 (Universal Ground-State Depression). For the ground state ($n = 0$), $E_0^{(0)}$ is the absolute minimum eigenvalue of $\hat{H}_0$. Consequently, $E_0^{(0)} - E_k^{(0)} < 0$ for all $k \neq 0$. It follows that:

$$E_0^{(2)} = \sum_{k \neq 0} \frac{|\langle k^{(0)} | \hat{V} | 0^{(0)} \rangle|^2}{E_0^{(0)} - E_k^{(0)}} \leq 0. \quad (8.1.30)$$

Every non-vanishing static perturbation shifts the ground-state energy downward at second order. This is the fundamental quantum principle of **level repulsion**: interaction between states pushes lower-energy states downward and higher-energy states upward.

8.2 Analytical Verification on Exactly Solvable Models

8.2.1 The Two-Level Matrix Model: Exact vs. Perturbative Expansion

To rigorously test the validity and convergence of perturbation theory, consider an exactly solvable two-state system with unperturbed Hamiltonian $\hat{H}_0$ and perturbation $\lambda \delta \hat{H}$:

$$\hat{H}_0 = \begin{pmatrix} E_1^{(0)} & 0 \\ 0 & E_2^{(0)} \end{pmatrix}, \quad \lambda \delta \hat{H} = \begin{pmatrix} 0 & \lambda V \\ \lambda V & 0 \end{pmatrix}, \quad \hat{H}(\lambda) = \begin{pmatrix} E_1^{(0)} & \lambda V \\ \lambda V & E_2^{(0)} \end{pmatrix}, \quad (8.2.1)$$

where $E_1^{(0)} \neq E_2^{(0)}$ and V is real.

Exact Diagonalization

The exact eigenvalues are the roots of the characteristic equation:

$$\det(\hat{H} - E\hat{I}) = (E_1^{(0)} - E)(E_2^{(0)} - E) - \lambda^2 V^2 = 0. \quad (8.2.2)$$

Expanding the quadratic:

$$E^2 - (E_1^{(0)} + E_2^{(0)})E + E_1^{(0)}E_2^{(0)} - \lambda^2 V^2 = 0. \quad (8.2.3)$$

Solving via the quadratic formula:

$$E_{\pm} = \frac{E_1^{(0)} + E_2^{(0)}}{2} \pm \frac{1}{2} \sqrt{(E_1^{(0)} - E_2^{(0)})^2 + 4\lambda^2 V^2}. \quad (8.2.4)$$

Factoring out $|E_1^{(0)} - E_2^{(0)}|$ from the square root (assuming without loss of generality $E_1^{(0)} > E_2^{(0)}$):

$$E_{\pm} = \frac{E_1^{(0)} + E_2^{(0)}}{2} \pm \frac{E_1^{(0)} - E_2^{(0)}}{2} \sqrt{1 + \frac{4\lambda^2 V^2}{(E_1^{(0)} - E_2^{(0)})^2}}. \quad (8.2.5)$$

Now, performing a Taylor (binomial) expansion of $\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots$ for $x = \frac{4\lambda^2 V^2}{(E_1^{(0)} - E_2^{(0)})^2} \ll 1$:

$$\sqrt{1 + \frac{4\lambda^2 V^2}{(E_1^{(0)} - E_2^{(0)})^2}} = 1 + \frac{2\lambda^2 V^2}{(E_1^{(0)} - E_2^{(0)})^2} - \frac{2\lambda^4 V^4}{(E_1^{(0)} - E_2^{(0)})^4} + \dots \quad (8.2.6)$$

Substituting this back into E_+ :

$$\begin{aligned} E_+ &= \frac{E_1^{(0)} + E_2^{(0)}}{2} + \frac{E_1^{(0)} - E_2^{(0)}}{2} \left[1 + \frac{2\lambda^2 V^2}{(E_1^{(0)} - E_2^{(0)})^2} - \dots \right] \\ &= E_1^{(0)} + \frac{\lambda^2 V^2}{E_1^{(0)} - E_2^{(0)}} - \frac{\lambda^4 V^4}{(E_1^{(0)} - E_2^{(0)})^3} + \dots \end{aligned} \quad (8.2.7)$$

Similarly for the lower state E_- :

$$E_- = E_2^{(0)} - \frac{\lambda^2 V^2}{E_1^{(0)} - E_2^{(0)}} + \frac{\lambda^4 V^4}{(E_1^{(0)} - E_2^{(0)})^3} + \dots \quad (8.2.8)$$

Perturbation Theory Comparison

Let us apply Rayleigh–Schrödinger perturbation theory directly to state $|1^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$E_1^{(1)} = \langle 1^{(0)} | \delta \hat{H} | 1^{(0)} \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & V \\ V & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0, \quad (8.2.9)$$

$$E_1^{(2)} = \frac{|\langle 2^{(0)} | \delta \hat{H} | 1^{(0)} \rangle|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{V^2}{E_1^{(0)} - E_2^{(0)}}. \quad (8.2.10)$$

The perturbation result $E_1 = E_1^{(0)} + \lambda^2 \frac{V^2}{E_1^{(0)} - E_2^{(0)}}$ matches the exact binomial expansion (8.2.7) **term-by-term!**

8.2.2 Harmonic Oscillator with Perturbed Spring Constant

Consider a 1D harmonic oscillator whose spring constant $k = m\omega^2$ is modified by an amount Δk :

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} (k + \Delta k) \hat{x}^2 = \hat{H}_0 + \frac{1}{2} \Delta k \hat{x}^2. \quad (8.2.11)$$

Exact Solution

The exact Hamiltonian is simply an oscillator with new frequency $\omega' = \sqrt{\frac{k+\Delta k}{m}} = \omega\sqrt{1 + \frac{\Delta k}{k}}$. Expanding ω' in powers of $\Delta k/k$:

$$\omega' = \omega \left(1 + \frac{\Delta k}{2k} - \frac{(\Delta k)^2}{8k^2} + \frac{(\Delta k)^3}{16k^3} - \dots \right). \quad (8.2.12)$$

The exact energy levels are:

$$E_n^{\text{exact}} = \left(n + \frac{1}{2} \right) \hbar \omega' = \left(n + \frac{1}{2} \right) \hbar \omega \left[1 + \frac{\Delta k}{2k} - \frac{(\Delta k)^2}{8k^2} + \dots \right]. \quad (8.2.13)$$

Perturbation Theory via Ladder Operators

We treat $\delta\hat{H} = \frac{1}{2}\Delta k \hat{x}^2$ as a perturbation. Expressing $\hat{x}$ in terms of creation and annihilation operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \implies \hat{x}^2 = \frac{\hbar}{2m\omega} (\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) = \frac{\hbar}{2m\omega} (\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{N} + 1). \quad (8.2.14)$$

1. First-Order Energy Shift:

$$E_n^{(1)} = \langle n | \delta\hat{H} | n \rangle = \frac{1}{2}\Delta k \frac{\hbar}{2m\omega} \langle n | (2\hat{N} + 1) | n \rangle = \frac{\Delta k \hbar}{4m\omega} (2n + 1) = \frac{\Delta k}{2k} \left(n + \frac{1}{2} \right) \hbar \omega. \quad (8.2.15)$$

This matches the linear term in (8.2.13) identically!

2. Second-Order Energy Shift: The non-zero off-diagonal matrix elements connect only to $n \pm 2$:

$$\langle n+2 | \hat{x}^2 | n \rangle = \frac{\hbar}{2m\omega} \sqrt{(n+1)(n+2)}, \quad (8.2.16)$$

$$\langle n-2 | \hat{x}^2 | n \rangle = \frac{\hbar}{2m\omega} \sqrt{n(n-1)}. \quad (8.2.17)$$

Substituting into the second-order formula (8.1.29):

$$\begin{aligned} E_n^{(2)} &= \frac{|\langle n+2 | \delta\hat{H} | n \rangle|^2}{E_n^{(0)} - E_{n+2}^{(0)}} + \frac{|\langle n-2 | \delta\hat{H} | n \rangle|^2}{E_n^{(0)} - E_{n-2}^{(0)}} \\ &= \left(\frac{\Delta k}{2} \right)^2 \left(\frac{\hbar}{2m\omega} \right)^2 \left[\frac{(n+1)(n+2)}{-2\hbar\omega} + \frac{n(n-1)}{+2\hbar\omega} \right] \\ &= \frac{(\Delta k)^2 \hbar}{16m^2\omega^3} [-n^2 - 3n - 2 + n^2 - n] = -\frac{(\Delta k)^2 \hbar}{16m^2\omega^3} (4n + 2) \\ &= -\frac{(\Delta k)^2}{8k^2} \left(n + \frac{1}{2} \right) \hbar \omega. \end{aligned} \quad (8.2.18)$$

This matches the quadratic term in (8.2.13) with absolute precision!

8.2.3 Harmonic Oscillator Driven by a Uniform Force

Consider an oscillator perturbed by a constant external force F :

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - F\hat{x}. \quad (8.2.19)$$

Exact Solution by Completing the Square

Completing the square in coordinate space:

$$\frac{1}{2}m\omega^2 x^2 - Fx = \frac{1}{2}m\omega^2 \left(x - \frac{F}{m\omega^2}\right)^2 - \frac{F^2}{2m\omega^2}. \quad (8.2.20)$$

Defining the shifted coordinate $x' = x - x_0$, where $x_0 = \frac{F}{m\omega^2}$:

$$E_n^{\text{exact}} = \left(n + \frac{1}{2}\right) \hbar\omega - \frac{F^2}{2m\omega^2}. \quad (8.2.21)$$

Perturbation Theory and Wavefunction Displacement

Treating $\hat{V} = -F\hat{x}$ as a perturbation:

$$E_n^{(1)} = -F \langle n | \hat{x} | n \rangle = 0, \quad (8.2.22)$$

$$E_n^{(2)} = F^2 \left[\frac{|\langle n+1 | \hat{x} | n \rangle|^2}{-\hbar\omega} + \frac{|\langle n-1 | \hat{x} | n \rangle|^2}{+\hbar\omega} \right] = F^2 \left(\frac{\hbar}{2m\omega} \right) \left[-\frac{n+1}{\hbar\omega} + \frac{n}{\hbar\omega} \right] = -\frac{F^2}{2m\omega^2}. \quad (8.2.23)$$

The second-order perturbation reproduces the exact energy shift; all higher orders $E_n^{(k \geq 3)}$ vanish identically.

Now examine the **state correction** to first order:

$$|n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k | -F\hat{x} | n \rangle}{E_n^{(0)} - E_k^{(0)}} |k\rangle = \frac{F}{\hbar\omega} \sqrt{\frac{\hbar}{2m\omega}} \left[\sqrt{n+1} |n+1\rangle - \sqrt{n} |n-1\rangle \right]. \quad (8.2.24)$$

Notice that the momentum operator is $\hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger)$. Acting on $|n\rangle$:

$$\hat{p} |n\rangle = -i\sqrt{\frac{m\hbar\omega}{2}} \left(\sqrt{n} |n-1\rangle - \sqrt{n+1} |n+1\rangle \right) = i\sqrt{\frac{m\hbar\omega}{2}} \left(\sqrt{n+1} |n+1\rangle - \sqrt{n} |n-1\rangle \right). \quad (8.2.25)$$

Therefore:

$$|n^{(1)}\rangle = \frac{F}{\hbar\omega} \sqrt{\frac{\hbar}{2m\omega}} \left(\frac{\hat{p}}{i\sqrt{m\hbar\omega/2}} \right) |n\rangle = -\frac{iF}{\hbar m\omega^2} \hat{p} |n\rangle = -\frac{ix_0}{\hbar} \hat{p} |n\rangle. \quad (8.2.26)$$

In the position representation, $\hat{p} = -i\hbar \frac{d}{dx}$, so:

$$\psi_n^{(1)}(x) = -x_0 \frac{d\psi_n^{(0)}(x)}{dx}. \quad (8.2.27)$$

The total first-order state is:

$$\psi_n(x) \approx \psi_n^{(0)}(x) - x_0 \frac{d\psi_n^{(0)}(x)}{dx} \approx \psi_n^{(0)}(x - x_0). \quad (8.2.28)$$

First-order perturbation theory precisely reconstructs the leading Taylor series expansion of the physically displaced wavefunction $\psi_n(x - x_0)$!

8.3 The Quadratic Stark Effect and the Unsöld Bound

Consider the ground state $|100\rangle$ of a hydrogenic atom in an external uniform electric field $\vec{\mathcal{E}} = \mathcal{E}\hat{z}$:

$$\hat{V} = e\mathcal{E}\hat{z} = e\mathcal{E}r \cos \theta. \quad (8.3.1)$$

8.3.1 Parity Proof for Vanishing First-Order Shift

Under a spatial parity inversion $\hat{\Pi}$ ($\vec{r} \rightarrow -\vec{r}$):

$$r \rightarrow r, \quad \theta \rightarrow \pi - \theta, \quad \phi \rightarrow \phi + \pi. \quad (8.3.2)$$

The spherical harmonics transform as:

$$Y_{lm}(\pi - \theta, \phi + \pi) = (-1)^l Y_{lm}(\theta, \phi). \quad (8.3.3)$$

For the ground state ($l = 0$), $\psi_{100}(-\vec{r}) = +\psi_{100}(\vec{r})$ is strictly even. However, the perturbation is odd:

$$z = r \cos \theta \implies \hat{\Pi}^\dagger z \hat{\Pi} = -z. \quad (8.3.4)$$

Evaluating the first-order energy shift:

$$E_1^{(1)} = e\mathcal{E} \langle 100 | z | 100 \rangle = e\mathcal{E} \int d^3r |\psi_{100}(r)|^2 z = 0, \quad (8.3.5)$$

because the integrand is the product of an even function and an odd function over symmetric integration bounds.

8.3.2 Second-Order Energy Shift and the Unsöld Approximation

The second-order shift is:

$$E_1^{(2)} = e^2 \mathcal{E}^2 \sum_{n>1, l, m} \frac{|\langle nlm | z | 100 \rangle|^2}{E_1^{(0)} - E_n^{(0)}}. \quad (8.3.6)$$

From angular momentum selection rules, $\langle nlm | z | 100 \rangle$ is non-zero only for $l = 1$ and $m = 0$ (p_z orbitals). The nearest accessible state is $n = 2$ ($2p$ level $|210\rangle$), where $E_1^{(0)} - E_2^{(0)} = -13.6(1 - 1/4) \text{ eV} = -\frac{3}{4} \text{ Ry}$.

For all higher excited states $n \geq 3$, the energy denominators $|E_1^{(0)} - E_n^{(0)}|$ are strictly larger than $|E_1^{(0)} - E_2^{(0)}|$. Following **Albrecht Unsöld** (1927), we replace the energy denominators with the minimum energy gap $\Delta E = E_1^{(0)} - E_2^{(0)} = -\frac{3}{8} \frac{e^2}{4\pi\epsilon_0 a_0}$:

$$|E_1^{(2)}| < \frac{e^2 \mathcal{E}^2}{|E_1^{(0)} - E_2^{(0)}|} \sum_{k \neq 1} |\langle k | z | 1 \rangle|^2. \quad (8.3.7)$$

Using completeness $\sum_k |k\rangle \langle k| = \hat{I}$ and noting that $\langle 1 | z | 1 \rangle = 0$:

$$\sum_{k \neq 1} |\langle k | z | 1 \rangle|^2 = \sum_k \langle 1 | z | k \rangle \langle k | z | 1 \rangle - |\langle 1 | z | 1 \rangle|^2 = \langle 1 | z^2 | 1 \rangle = \langle z^2 \rangle_{1s}. \quad (8.3.8)$$

Because the $1s$ orbital is spherically symmetric:

$$\langle x^2 \rangle = \langle y^2 \rangle = \langle z^2 \rangle = \frac{1}{3} \langle r^2 \rangle. \quad (8.3.9)$$

For the hydrogenic ground state, $\langle r^2 \rangle_{1s} = 3a_0^2$, which immediately implies:

$$\langle z^2 \rangle_{1s} = a_0^2. \quad (8.3.10)$$

Substituting this expectation value into the Unsöld bound:

$$E_1^{(2)} \approx -\frac{e^2 \mathcal{E}^2 a_0^2}{\frac{3}{8} \frac{e^2}{4\pi\epsilon_0 a_0}} = -\frac{8}{3} (4\pi\epsilon_0) a_0^3 \mathcal{E}^2 \approx -2.67 (4\pi\epsilon_0) a_0^3 \mathcal{E}^2. \quad (8.3.11)$$

Comparing this simple algebraic estimate with the exact result obtained by Dalgarno and Lewis:

$$E_1^{(2)\text{exact}} = -\frac{9}{4} (4\pi\epsilon_0) a_0^3 \mathcal{E}^2 = -2.25 (4\pi\epsilon_0) a_0^3 \mathcal{E}^2. \quad (8.3.12)$$

The Unsöld closure approximation captures the atomic polarizability $\alpha_{\text{pol}} = \frac{9}{2} a_0^3$ to within 18% without computing a single continuum integral!

8.4 Degenerate Perturbation Theory via Projection Operators

8.4.1 Projection Operators and the Feshbach Formalism

Let the unperturbed Hamiltonian $\hat{H}_0$ possess an N -fold degenerate eigenvalue $E^{(0)}$ spanning the eigenspace $\mathcal{D}$. We introduce the orthogonal projection operators:

$$\hat{P}_0 \equiv \sum_{i \in \mathcal{D}} |n_i^{(0)}\rangle \langle n_i^{(0)}|, \quad \hat{Q}_0 \equiv \hat{I} - \hat{P}_0 = \sum_{m \notin \mathcal{D}} |m^{(0)}\rangle \langle m^{(0)}|, \quad (8.4.1)$$

which satisfy:

$$\hat{P}_0^2 = \hat{P}_0, \quad \hat{Q}_0^2 = \hat{Q}_0, \quad \hat{P}_0 \hat{Q}_0 = \hat{Q}_0 \hat{P}_0 = 0, \quad [\hat{P}_0, \hat{H}_0] = [\hat{Q}_0, \hat{H}_0] = 0. \quad (8.4.2)$$

The full eigenvalue equation is:

$$(E_n - \hat{H}_0 - \lambda \hat{V}) |n\rangle = 0. \quad (8.4.3)$$

Inserting $\hat{I} = \hat{P}_0 + \hat{Q}_0$ and projecting from the left by $\hat{P}_0$ and $\hat{Q}_0$ decomposes the Hilbert space into two coupled equations:

$$\hat{P}_0 (E_n - \hat{H}_0 - \lambda \hat{V}) \hat{P}_0 |n\rangle + \hat{P}_0 (E_n - \hat{H}_0 - \lambda \hat{V}) \hat{Q}_0 |n\rangle = 0, \quad (8.4.4)$$

$$\hat{Q}_0 (E_n - \hat{H}_0 - \lambda \hat{V}) \hat{P}_0 |n\rangle + \hat{Q}_0 (E_n - \hat{H}_0 - \lambda \hat{V}) \hat{Q}_0 |n\rangle = 0. \quad (8.4.5)$$

Since $\hat{P}_0 \hat{H}_0 \hat{Q}_0 = 0$ and $\hat{P}_0 \hat{H}_0 \hat{P}_0 = E^{(0)} \hat{P}_0$:

$$(E_n - E^{(0)} - \lambda \hat{P}_0 \hat{V} \hat{P}_0) \hat{P}_0 |n\rangle - \lambda \hat{P}_0 \hat{V} \hat{Q}_0 |n\rangle = 0, \quad (8.4.6)$$

$$-\lambda \hat{Q}_0 \hat{V} \hat{P}_0 |n\rangle + (E_n - \hat{H}_0 - \lambda \hat{Q}_0 \hat{V} \hat{Q}_0) \hat{Q}_0 |n\rangle = 0. \quad (8.4.7)$$

8.4.2 Order-by-Order Matching in Subspaces

Expanding $E_n = E^{(0)} + \lambda E^{(1)} + \lambda^2 E^{(2)} + \dots$ and $|n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots$:

1. Order λ^0 in Q -space:

$$(E^{(0)} - \hat{H}_0) \hat{Q}_0 |n^{(0)}\rangle = 0 \implies \hat{Q}_0 |n^{(0)}\rangle = 0. \quad (8.4.8)$$

The zeroth-order state lies strictly in the degenerate subspace: $|n^{(0)}\rangle \in \mathcal{D}$.

2. Order λ^1 in P -space:

$$(E^{(1)} - \hat{P}_0 \hat{V} \hat{P}_0) |n^{(0)}\rangle = 0 \implies \boxed{\hat{P}_0 \hat{V} \hat{P}_0 |n^{(0)}\rangle = E^{(1)} |n^{(0)}\rangle} \quad (8.4.9)$$

This is the operator statement of the **secular equation!** The first-order energy shifts $E^{(1)}$ are the eigenvalues of the perturbation operator restricted to the degenerate subspace, and the corresponding eigenvectors are the “good” zeroth-order states.

3. Order λ^1 in Q -space:

$$(E^{(0)} - \hat{H}_0) \hat{Q}_0 |n^{(1)}\rangle = \hat{Q}_0 \hat{V} \hat{P}_0 |n^{(0)}\rangle \implies \hat{Q}_0 |n^{(1)}\rangle = \sum_{m \notin \mathcal{D}} \frac{\langle m^{(0)} | \hat{V} | n^{(0)} \rangle}{E^{(0)} - E_m^{(0)}} |m^{(0)}\rangle. \quad (8.4.10)$$

4. Order λ^2 in P -space (Second-Order Energy in Degenerate Theory):

$$E^{(2)} |n^{(0)}\rangle = \hat{P}_0 \hat{V} \hat{Q}_0 |n^{(1)}\rangle \implies \boxed{E_{n_i}^{(2)} = \sum_{m \notin \mathcal{D}} \frac{|\langle m^{(0)} | \hat{V} | n_i^{(0)} \rangle|^2}{E^{(0)} - E_m^{(0)}}} \quad (8.4.11)$$

8.4.3 Application: 2D Isotropic Harmonic Oscillator with Coupling

Consider a two-dimensional harmonic oscillator perturbed by cross-coupling:

$$\hat{H} = \frac{\hat{p}_x^2 + \hat{p}_y^2}{2m} + \frac{1}{2}m\omega^2 (\hat{x}^2 + \hat{y}^2) + \lambda\hat{x}\hat{y}. \quad (8.4.12)$$

The unperturbed eigenvalues are $E_{n_x, n_y}^{(0)} = (n_x + n_y + 1)\hbar\omega$. The first excited level ($E = 2\hbar\omega$) is twofold degenerate, spanned by $|1, 0\rangle$ and $|0, 1\rangle$.

Degenerate Perturbation Matrix

We compute the matrix elements of $\hat{V} = \lambda\hat{x}\hat{y}$ in the basis $\{|1, 0\rangle, |0, 1\rangle\}$:

$$V_{11} = \lambda \langle 1, 0 | \hat{x}\hat{y} | 1, 0 \rangle = \lambda \langle 1 | \hat{x} | 1 \rangle \langle 0 | \hat{y} | 0 \rangle = 0, \quad (8.4.13)$$

$$V_{22} = \lambda \langle 0, 1 | \hat{x}\hat{y} | 0, 1 \rangle = \lambda \langle 0 | \hat{x} | 0 \rangle \langle 1 | \hat{y} | 1 \rangle = 0, \quad (8.4.14)$$

$$V_{12} = \lambda \langle 1, 0 | \hat{x}\hat{y} | 0, 1 \rangle = \lambda \langle 1 | \hat{x} | 0 \rangle \langle 0 | \hat{y} | 1 \rangle = \lambda \left(\sqrt{\frac{\hbar}{2m\omega}} \right) \left(\sqrt{\frac{\hbar}{2m\omega}} \right) = \frac{\lambda\hbar}{2m\omega}. \quad (8.4.15)$$

The secular determinant is:

$$\det \begin{pmatrix} -E^{(1)} & \frac{\lambda\hbar}{2m\omega} \\ \frac{\lambda\hbar}{2m\omega} & -E^{(1)} \end{pmatrix} = 0 \implies (E^{(1)})^2 - \left(\frac{\lambda\hbar}{2m\omega} \right)^2 = 0. \quad (8.4.16)$$

The first-order energy corrections are:

$$E_{\pm}^{(1)} = \pm \frac{\lambda\hbar}{2m\omega}. \quad (8.4.17)$$

The corresponding good zeroth-order states are the symmetric and antisymmetric superpositions:

$$|\psi_{\pm}^{(0)}\rangle = \frac{1}{\sqrt{2}} (|1, 0\rangle \pm |0, 1\rangle). \quad (8.4.18)$$

Exact Comparison via Normal Modes

We can solve this Hamiltonian exactly by rotating coordinates by 45° :

$$x' = \frac{x+y}{\sqrt{2}}, \quad y' = \frac{x-y}{\sqrt{2}} \implies x = \frac{x'+y'}{\sqrt{2}}, \quad y = \frac{x'-y'}{\sqrt{2}}. \quad (8.4.19)$$

The potential transforms into decoupled uncoupled oscillators:

$$\begin{aligned} V(x', y') &= \frac{1}{2}m\omega^2(x'^2 + y'^2) + \frac{\lambda}{2}(x'^2 - y'^2) \\ &= \frac{1}{2} \left(m\omega^2 + \lambda \right) x'^2 + \frac{1}{2} \left(m\omega^2 - \lambda \right) y'^2 \equiv \frac{1}{2}m\omega_+^2 x'^2 + \frac{1}{2}m\omega_-^2 y'^2. \end{aligned} \quad (8.4.20)$$

The normal mode frequencies are:

$$\omega_{\pm} = \sqrt{\omega^2 \pm \frac{\lambda}{m}} = \omega \sqrt{1 \pm \frac{\lambda}{m\omega^2}} \approx \omega \pm \frac{\lambda}{2m\omega}. \quad (8.4.21)$$

The exact excited-state energies are:

$$E_+^{\text{exact}} = \left(1 + \frac{1}{2} \right) \hbar\omega_+ + \frac{1}{2} \hbar\omega_- \approx 2\hbar\omega + \frac{\lambda\hbar}{2m\omega}, \quad (8.4.22)$$

$$E_-^{\text{exact}} = \frac{1}{2} \hbar\omega_+ + \left(1 + \frac{1}{2} \right) \hbar\omega_- \approx 2\hbar\omega - \frac{\lambda\hbar}{2m\omega}. \quad (8.4.23)$$

Degenerate perturbation theory reproduces the exact normal-mode splitting!

8.5 Fine Structure of the Hydrogen Atom

The unperturbed Bohr-Schrödinger Hamiltonian of hydrogen-like ions (nuclear charge Z):

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r}, \quad (8.5.1)$$

exhibits an energy spectrum depending strictly on the principal quantum number n :

$$E_n^{(0)} = -\frac{Z^2\alpha^2 mc^2}{2n^2} = -\frac{Z^2 \text{Ry}}{n^2}, \quad \alpha \equiv \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137.036}. \quad (8.5.2)$$

Each energy level $E_n^{(0)}$ has an n^2 -fold orbital degeneracy (or $2n^2$ when including the electron spin degree of freedom). High-resolution spectroscopy reveals that these degenerate spectral lines are split into fine multiplets of relative order $\alpha^2 \sim 10^{-4}$.

This **fine structure** arises from three distinct relativistic corrections derived from the non-relativistic expansion of the Dirac Hamiltonian:

$$\hat{H}_{\text{FS}} = \hat{H}_{\text{rel}} + \hat{H}_{\text{SO}} + \hat{H}_{\text{Darwin}}. \quad (8.5.3)$$

8.5.1 Relativistic Kinetic Energy Correction

The relativistic energy-momentum relation is $E = \sqrt{\vec{p}^2 c^2 + m^2 c^4}$. Expanding the kinetic energy $T = E - mc^2$ in powers of $p/(mc)$:

$$\begin{aligned} T &= mc^2 \sqrt{1 + \frac{p^2}{m^2 c^2}} - mc^2 = mc^2 \left[1 + \frac{1}{2} \frac{p^2}{m^2 c^2} - \frac{1}{8} \frac{p^4}{m^4 c^4} + \dots \right] - mc^2 \\ &= \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \dots \end{aligned} \quad (8.5.4)$$

The leading relativistic correction to the Hamiltonian is therefore:

$$\hat{H}_{\text{rel}} = -\frac{\hat{p}^4}{8m^3 c^2}. \quad (8.5.5)$$

Since $\hat{p}^4$ is spherically symmetric, it commutes with $\hat{L}^2$ and $\hat{L}_z$, so the standard basis $|nlm\rangle$ remains diagonal. We compute the first-order energy shift:

$$E_{\text{rel}}^{(1)} = -\frac{1}{8m^3 c^2} \langle nlm | \hat{p}^4 | nlm \rangle = -\frac{1}{8m^3 c^2} \langle nlm | \hat{p}^2 \hat{p}^2 | nlm \rangle. \quad (8.5.6)$$

To avoid calculating four spatial derivatives, we express $\hat{p}^2$ in terms of the unperturbed Hamiltonian:

$$\frac{\hat{p}^2}{2m} = \hat{H}_0 - V(r) \implies \hat{p}^2 = 2m \left(\hat{H}_0 + \frac{Ze^2}{4\pi\epsilon_0 r} \right). \quad (8.5.7)$$

Because $\hat{H}_0$ is Hermitian and acts on its eigenstate $|nlm\rangle$ as $E_n^{(0)}$:

$$\begin{aligned} \langle nlm | \hat{p}^4 | nlm \rangle &= 4m^2 \langle nlm | \left(E_n^{(0)} + \frac{Ze^2}{4\pi\epsilon_0 r} \right)^2 | nlm \rangle \\ &= 4m^2 \left[(E_n^{(0)})^2 + 2E_n^{(0)} \left(\frac{Ze^2}{4\pi\epsilon_0} \right) \left\langle \frac{1}{r} \right\rangle + \left(\frac{Ze^2}{4\pi\epsilon_0} \right)^2 \left\langle \frac{1}{r^2} \right\rangle \right]. \end{aligned} \quad (8.5.8)$$

From standard Hydrogen radial matrix integrals (Kramers' relations):

$$\left\langle \frac{1}{r} \right\rangle = \frac{Z}{a_0 n^2}, \quad \left\langle \frac{1}{r^2} \right\rangle = \frac{Z^2}{a_0^2 n^3 (l + 1/2)}, \quad (8.5.9)$$

where $a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}$ is the Bohr radius. Noting that $E_n^{(0)} = -\frac{Ze^2}{8\pi\epsilon_0 a_0 n^2} \implies \frac{Ze^2}{4\pi\epsilon_0} = -2a_0 n^2 E_n^{(0)}$:

$$\langle nlm | \hat{p}^4 | nlm \rangle = 4m^2 (E_n^{(0)})^2 \left[1 - 4 + \frac{4n}{l + 1/2} \right] = 4m^2 (E_n^{(0)})^2 \left[\frac{4n}{l + 1/2} - 3 \right]. \quad (8.5.10)$$

Substituting back into $E_{\text{rel}}^{(1)}$:

$$E_{\text{rel}}^{(1)} = -\frac{(E_n^{(0)})^2}{2mc^2} \left[\frac{4n}{l + 1/2} - 3 \right]. \quad (8.5.11)$$

This correction is strictly negative and lifts the degeneracy with respect to the orbital angular momentum l .

8.5.2 Spin-Orbit Coupling

To understand the physical origin of spin-orbit coupling, it is crucial to analyze the motion from the **rest frame of the electron**:

1. **Laboratory Frame (Nuclear Rest Frame):** The positively charged nucleus of charge $+Ze$ is stationary at the origin, producing a purely electrostatic, spherically symmetric Coulomb field:

$$\vec{E} = -\frac{1}{e} \frac{dV}{dr} \vec{r} = \frac{Ze}{4\pi\epsilon_0 r^3} \vec{r}, \quad \vec{B} = 0. \quad (8.5.12)$$

The electron orbits the nucleus with velocity $\vec{v} = \vec{p}/m$.

2. **Electron's Rest Frame:** In the instantaneous rest frame of the orbiting electron, the situation is reversed: the electron is at rest, while the positively charged nucleus is circling around it with velocity $-\vec{v}$! This orbiting positive charge constitutes an electric current loop, producing an intense **internal magnetic field** $\vec{B}_{\text{int}}$ at the location of the electron.

Applying the Lorentz transformation of electromagnetic fields for $v \ll c$:

$$\vec{B}_{\text{int}} \approx -\frac{1}{c^2} \vec{v} \times \vec{E} = \frac{1}{mec^2} \frac{1}{r} \frac{dV}{dr} = \frac{1}{mec^2} \frac{1}{r} \frac{dV}{dr} \vec{L} = \frac{Ze}{4\pi\epsilon_0 mc^2 r^3} \vec{L}. \quad (8.5.13)$$

This internal magnetic field is enormous: for a hydrogen electron in a $2p$ orbital ($n = 2, l = 1$), evaluating $\langle 1/r^3 \rangle$ gives:

$$\langle B_{\text{int}} \rangle \approx \frac{\hbar}{e} \frac{e^2}{4\pi\epsilon_0 mc^2} \left\langle \frac{1}{r^3} \right\rangle \sim 0.4 \text{ to } 1.6 \text{ Tesla!} \quad (8.5.14)$$

In heavier ions with nuclear charge Z , because $\langle 1/r^3 \rangle \propto Z^3$, this internal field scales as $\langle B_{\text{int}} \rangle \propto Z^4$, reaching hundreds to thousands of Tesla!

3. **Dipole Interaction and Thomas Precession:** The electron possesses an intrinsic magnetic dipole moment due to its spin:

$$\vec{\mu}_s = -g_s \frac{e}{2m} \vec{S} \approx -\frac{e}{m} \vec{S} \quad (\text{since } g_s \approx 2). \quad (8.5.15)$$

The magnetic potential energy of this dipole in the internal magnetic field of the orbiting nucleus is:

$$U_{\text{mag}} = -\vec{\mu}_s \cdot \vec{B}_{\text{int}} = \frac{e}{m} \vec{S} \cdot \left(\frac{1}{mec^2} \frac{1}{r} \frac{dV}{dr} \vec{L} \right) = \frac{1}{m^2 c^2} \frac{1}{r} \frac{dV}{dr} (\vec{L} \cdot \vec{S}). \quad (8.5.16)$$

However, because the electron accelerates continuously in its curved orbit, its rest frame is non-inertial and undergoes relativistic kinematic precession (**Thomas precession**) at frequency $\vec{\omega}_T \approx -\frac{1}{2c^2}(\vec{v} \times \vec{a})$. Kinematics in an accelerated reference frame reduces the effective magnetic interaction by exactly a factor of 1/2:

$$\hat{H}_{\text{SO}} = \frac{1}{2m^2 c^2} \frac{1}{r} \frac{dV}{dr} \hat{\vec{L}} \cdot \hat{\vec{S}} = \frac{Ze^2}{8\pi\epsilon_0 m^2 c^2 r^3} \hat{\vec{L}} \cdot \hat{\vec{S}}. \quad (8.5.17)$$

Eigenbasis and Angular Coupling

The operator $\hat{\vec{L}} \cdot \hat{\vec{S}}$ does not commute with $\hat{L}_z$ or $\hat{S}_z$. Therefore, the uncoupled basis $|n, l, m_l, s, m_s\rangle$ does not diagonalize $\hat{H}_{\text{SO}}$. However, defining the total angular momentum:

$$\hat{\vec{J}} = \hat{\vec{L}} + \hat{\vec{S}} \implies \hat{J}^2 = \hat{L}^2 + \hat{S}^2 + 2\hat{\vec{L}} \cdot \hat{\vec{S}}, \quad (8.5.18)$$

we express the scalar product as:

$$\hat{\vec{L}} \cdot \hat{\vec{S}} = \frac{1}{2} (\hat{J}^2 - \hat{L}^2 - \hat{S}^2). \quad (8.5.19)$$

The coupled basis states $|n, j, m_j, l, s\rangle$ (where $s = 1/2$ and $j = l \pm 1/2$) are simultaneous eigenstates of $\hat{J}^2$, $\hat{J}_z$, $\hat{L}^2$, and $\hat{S}^2$. In this basis, $\hat{\vec{L}} \cdot \hat{\vec{S}}$ is strictly diagonal:

$$\langle j, m_j, l, s | \hat{\vec{L}} \cdot \hat{\vec{S}} | j, m_j, l, s \rangle = \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)] = \frac{\hbar^2}{2} \left[j(j+1) - l(l+1) - \frac{3}{4} \right]. \quad (8.5.20)$$

Explicitly:

$$\langle \hat{\vec{L}} \cdot \hat{\vec{S}} \rangle = \begin{cases} \frac{\hbar^2}{2} l, & \text{for } j = l + 1/2, \\ -\frac{\hbar^2}{2} (l+1), & \text{for } j = l - 1/2. \end{cases} \quad (8.5.21)$$

Radial Expectation Value and Divergence at $l = 0$

For $l > 0$, the expectation value of $1/r^3$ in Hydrogen is:

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{Z^3}{a_0^3 n^3 l(l+1/2)(l+1)}. \quad (8.5.22)$$

Thus, the first-order spin-orbit energy shift for $l > 0$ is:

$$\begin{aligned} E_{\text{SO}}^{(1)} &= \frac{Ze^2 \hbar^2}{16\pi\epsilon_0 m^2 c^2} \frac{Z^3}{a_0^3 n^3 l(l+1/2)(l+1)} \left[j(j+1) - l(l+1) - \frac{3}{4} \right] \\ &= \frac{(E_n^{(0)})^2 n}{mc^2} \frac{[j(j+1) - l(l+1) - 3/4]}{l(l+1/2)(l+1)}. \end{aligned} \quad (8.5.23)$$

Notice that for s -states ($l = 0$), $\langle 1/r^3 \rangle$ diverges, yet $\vec{L} = 0$. A rigorous treatment requires the Darwin term.

8.5.3 The Darwin Term

As derived in Chapter 7 from the Foldy-Wouthuysen transformation of the Dirac equation, a relativistic electron undergoes rapid oscillatory motion (**Zitterbewegung**) over a Compton wavelength scale $\delta r \sim \frac{\hbar}{mc}$. Consequently, the point electron does not feel the bare potential $V(\vec{r})$, but a spatial average over its fluctuating position:

$$\langle V(\vec{r} + \delta\vec{r}) \rangle \approx V(\vec{r}) + \langle \delta\vec{r} \rangle \cdot \vec{\nabla} V(\vec{r}) + \frac{1}{2} \sum_{i,j} \langle \delta r_i \delta r_j \rangle \partial_i \partial_j V(\vec{r}). \quad (8.5.24)$$

For an isotropic fluctuation, $\langle \delta\vec{r} \rangle = 0$ and $\langle \delta r_i \delta r_j \rangle = \frac{1}{3} \langle (\delta r)^2 \rangle \delta_{ij} \sim \frac{\hbar^2}{m^2 c^2} \delta_{ij}$. This gives the effective Hamiltonian shift known as the **Darwin term**:

$$\hat{H}_{\text{Darwin}} = \frac{\pi \hbar^2}{2m^2 c^2} \left(\frac{Ze^2}{4\pi\epsilon_0} \right) \delta^3(\vec{r}). \quad (8.5.25)$$

Because of the Dirac delta function $\delta^3(\vec{r})$, this operator acts exclusively on states with non-vanishing probability density at the origin $\vec{r} = 0$. Only s -wave wavefunctions ($l = 0$) are non-zero at $r = 0$:

$$|\psi_{n00}(0)|^2 = \frac{Z^3}{\pi a_0^3 n^3}. \quad (8.5.26)$$

For all states with $l > 0$, $|\psi_{nlm}(0)|^2 = 0$, so $E_{\text{Darwin}}^{(1)} = 0$. For $l = 0$:

$$E_{\text{Darwin}}^{(1)} = \frac{\pi \hbar^2}{2m^2 c^2} \left(\frac{Ze^2}{4\pi\epsilon_0} \right) \frac{Z^3}{\pi a_0^3 n^3} = \frac{(E_n^{(0)})^2}{2mc^2} (2n). \quad (8.5.27)$$

8.5.4 Grand Synthesis: The Universal Fine Structure Formula

We now assemble the total fine structure energy shift $E_{\text{FS}}^{(1)} = E_{\text{rel}}^{(1)} + E_{\text{SO}}^{(1)} + E_{\text{Darwin}}^{(1)}$:

1. **Case 1:** $l = 0$ ($j = 1/2$): Here $E_{\text{SO}}^{(1)} = 0$, while $E_{\text{rel}}^{(1)} = -\frac{(E_n^{(0)})^2}{2mc^2} (8n - 3)$ and $E_{\text{Darwin}}^{(1)} = \frac{(E_n^{(0)})^2}{2mc^2} (2n)$. Summing:

$$E_{\text{FS}}^{(1)} = \frac{(E_n^{(0)})^2}{2mc^2} [3 - 8n + 2n] = \frac{(E_n^{(0)})^2}{2mc^2} \left(3 - \frac{4n}{1/2 + 1/2} \right). \quad (8.5.28)$$

2. **Case 2:** $l > 0$ ($j = l + 1/2$): Here $j(j+1) - l(l+1) - 3/4 = l$. Substituting into (8.5.23):

$$E_{\text{SO}}^{(1)} = \frac{(E_n^{(0)})^2}{mc^2} \frac{n}{(l + 1/2)(l + 1)}. \quad (8.5.29)$$

Adding $E_{\text{rel}}^{(1)}$:

$$\begin{aligned} E_{\text{FS}}^{(1)} &= \frac{(E_n^{(0)})^2}{2mc^2} \left[3 - \frac{4n}{l + 1/2} + \frac{2n}{(l + 1/2)(l + 1)} \right] \\ &= \frac{(E_n^{(0)})^2}{2mc^2} \left[3 - \frac{4n(l + 1) - 2n}{(l + 1/2)(l + 1)} \right] = \frac{(E_n^{(0)})^2}{2mc^2} \left[3 - \frac{4n(l + 1/2)}{(l + 1/2)(l + 1)} \right] \\ &= \frac{(E_n^{(0)})^2}{2mc^2} \left[3 - \frac{4n}{j + 1/2} \right]. \end{aligned} \quad (8.5.30)$$

3. **Case 3:** $l > 0$ ($j = l - 1/2$): An identical algebraic cancellation yields precisely $\frac{(E_n^{(0)})^2}{2mc^2} \left(3 - \frac{4n}{j+1/2}\right)$.

All three cases unify into a single, breathtakingly elegant formula:

$$E_{n,j} = E_n^{(0)} \left[1 + \frac{(Z\alpha)^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right) \right] \quad (8.5.31)$$

- The dependence on the orbital quantum number l has completely vanished! States with the same principal quantum number n and the same total angular momentum j remain strictly degenerate.
- In particular, the $2s_{1/2}$ ($n = 2, l = 0, j = 1/2$) and $2p_{1/2}$ ($n = 2, l = 1, j = 1/2$) levels are exactly degenerate in Dirac theory. The minute splitting of 1057.8 MHz observed between them is the **Lamb shift**, explained solely by Quantum Electrodynamics (vacuum polarization and self-energy).

8.6 The Zeeman Effect

8.6.1 Minimal Coupling and Vector Potential for Uniform Field

When an atom is placed in an external uniform magnetic field $\vec{B} = B\hat{z}$, we describe the field via the symmetric vector potential:

$$\vec{A} = -\frac{1}{2} (\vec{r} \times \vec{B}) = \frac{1}{2} (-By\hat{x} + Bx\hat{y}). \quad (8.6.1)$$

Checking the Coulomb gauge:

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{2} \left(\frac{\partial(-By)}{\partial x} + \frac{\partial(Bx)}{\partial y} \right) = 0. \quad (8.6.2)$$

The kinetic Hamiltonian for an electron ($q = -e$) is:

$$\hat{H}_{\text{kin}} = \frac{(\hat{\vec{p}} + e\vec{A})^2}{2m} = \frac{\hat{\vec{p}}^2}{2m} + \frac{e}{2m} (\hat{\vec{p}} \cdot \vec{A} + \vec{A} \cdot \hat{\vec{p}}) + \frac{e^2}{2m} \vec{A}^2. \quad (8.6.3)$$

Because $\vec{\nabla} \cdot \vec{A} = 0$, $[\hat{\vec{p}}, \vec{A}] = -i\hbar \vec{\nabla} \cdot \vec{A} = 0$, so $\hat{\vec{p}} \cdot \vec{A} = \vec{A} \cdot \hat{\vec{p}}$. Evaluating the linear paramagnetic term:

$$\vec{A} \cdot \hat{\vec{p}} = \frac{1}{2} (\vec{B} \times \vec{r}) \cdot \hat{\vec{p}} = \frac{1}{2} \vec{B} \cdot (\vec{r} \times \hat{\vec{p}}) = \frac{1}{2} \vec{B} \cdot \hat{\vec{L}}. \quad (8.6.4)$$

Thus, the linear term is:

$$\frac{e}{m} \vec{A} \cdot \hat{\vec{p}} = \frac{e}{2m} \vec{B} \cdot \hat{\vec{L}} = -\hat{\mu}_L \cdot \vec{B}, \quad \hat{\mu}_L = -\frac{e}{2m} \hat{\vec{L}} = -\frac{\mu_B}{\hbar} \hat{\vec{L}}, \quad (8.6.5)$$

where $\mu_B \equiv \frac{e\hbar}{2m}$ is the Bohr magneton.

The quadratic term is the **diamagnetic Hamiltonian**:

$$\hat{H}_{\text{dia}} = \frac{e^2}{2m} \vec{A}^2 = \frac{e^2 B^2}{8m} (x^2 + y^2). \quad (8.6.6)$$

The ratio of diamagnetic to paramagnetic interaction is:

$$\frac{\langle \hat{H}_{\text{dia}} \rangle}{\langle \hat{H}_{\text{para}} \rangle} \sim \frac{\frac{e^2 B^2 a_0^2}{8m}}{\mu_B B} = \frac{e B a_0^2}{4\hbar} \approx 10^{-6} B \text{ [Tesla]}. \quad (8.6.7)$$

For laboratory magnetic fields $B \ll 10^5$ T, the diamagnetic term is completely negligible.

Adding the interaction of the electron's intrinsic spin magnetic dipole moment $\hat{\mu}_S = -g_s \frac{\mu_B}{\hbar} \hat{S} \approx -\frac{2\mu_B}{\hbar} \hat{S}$, the total linear **Zeeman Hamiltonian** is:

$$\hat{H}_Z = -(\hat{\mu}_L + \hat{\mu}_S) \cdot \vec{B} = \frac{\mu_B B}{\hbar} (\hat{L}_z + 2\hat{S}_z). \quad (8.6.8)$$

8.6.2 The Master Hierarchy of Atomic Interactions: When Each Effect Dominates

To navigate atomic spectra with precision, an investigator must master the relative energy scales and understand precisely **when each physical effect begins to dominate**.

Interaction	Physical Origin	Energy Scale (ΔE)	B_{eff}	Dominance Criterion & Good Basis
Gross Structure	Central Coulomb attraction	$\sim \frac{1}{2}mc^2(Z\alpha)^2 \sim 10$ eV	—	Dominates for all bound states. Basis: $ n, l, m_l, s, m_s\rangle$.
Fine Structure (<i>LS</i> Coupling)	Internal $\vec{B}_{\text{int}}$ from orbiting nucleus + relativistic mass + Darwin	$\sim \alpha^2 E_n^{(0)} \sim 10^{-4}$ eV	1 to 10 T	Dominates over external fields when $B_{\text{ext}} \ll B_{\text{int}}$. Coupled basis: $ j, m_j, l, s\rangle$.
Weak-Field (Anomalous) Zeeman	External field couples to $\vec{L} + 2\vec{S}$	$\mu_B B_{\text{ext}} \ll \Delta E_{\text{FS}}$	$B_{\text{ext}} \ll 1$ T	$B_{\text{ext}} \ll B_{\text{int}}$. $\vec{L} \cdot \vec{S}$ intact; $\vec{J}$ conserved. Shift: $g_j \mu_B B_{\text{ext}} m_j$.
Intermediate Zeeman	$\hat{H}_{\text{FS}} \sim \hat{H}_Z$ (competing torques)	$\mu_B B_{\text{ext}} \sim \Delta E_{\text{FS}}$	1–10 T	Neither $\vec{J}$ nor (L_z, S_z) conserved. Exact secular diagonalization (Breit–Rabi).
Strong-Field (Paschen–Back)	External torque overpowers internal $\vec{B}_{\text{int}}$	$\mu_B B_{\text{ext}} \gg \Delta E_{\text{FS}}$	$B_{\text{ext}} \gg 10$ T	$B_{\text{ext}} \gg B_{\text{int}}$. $\vec{L}$ and $\vec{S}$ uncoupled. Basis: $ l, m_l, s, m_s\rangle$.
Hyperfine Structure	Nuclear magnetic dipole $\vec{\mu}_I$ couples to electron spin	$\sim \frac{m_e}{M_p} \alpha^2 E_n^{(0)} \sim 10^{-6}$ eV	0.05–0.1 T	Isolated atom ($B_{\text{ext}} \rightarrow 0$). Total angular momentum $\vec{F} = \vec{J} + \vec{I}$.
Back–Goudsmit Effect	External field uncouples nuclear spin $\vec{I}$ from $\vec{J}$	$\mu_B B_{\text{ext}} \gg \Delta E_{\text{HFS}}$	$B_{\text{ext}} \gtrsim 0.1$ T	Nuclear Paschen–Back effect. $\vec{I}$ and $\vec{J}$ precess independently.

Conceptual Clarification: Disentangling *LS* Coupling from the Anomalous Zeeman Effect

A pervasive point of confusion among physics students arises from the apparent overlap between *LS* coupling and the Anomalous Zeeman effect: both involve the electron's spin $\vec{S}$, both invoke magnetic dipole interactions, and both employ the Landé *g*-factor and coupled angular momentum states $|j, m_j\rangle$. Why are they distinct?

- ***LS* Coupling is an INTERNAL Relativistic Effect ($B_{\text{ext}} = 0$):** Spin-orbit coupling exists in an isolated atom in complete vacuum without any external magnetic field! In the electron's rest frame, the positively charged nucleus is circling around it, generating an **internal magnetic field** $\vec{B}_{\text{int}} \sim 1$ to 10 Tesla. The coupling $-\vec{\mu}_s \cdot \vec{B}_{\text{int}} \propto \vec{L} \cdot \vec{S}$ locks the orbital and spin angular momenta together into $\vec{J} = \vec{L} + \vec{S}$, splitting the gross energy levels into the fine-structure multiplets

$$(^2S_{1/2}, ^2P_{1/2}, ^2P_{3/2}).$$

- **The Anomalous Zeeman Effect is the Response to a WEAK EXTERNAL Field ($B_{\text{ext}} \ll B_{\text{int}}$):** When we turn on a laboratory magnetic field $\vec{B}_{\text{ext}}$ that is much weaker than the internal field ($B_{\text{ext}} \ll 1 \text{ T}$), the external torque is insufficient to break the internal $\vec{L} \cdot \vec{S}$ bond. The vectors $\vec{L}$ and $\vec{S}$ continue to precess rapidly about $\vec{J}$ at frequency $\omega_{LS} \sim \Delta E_{\text{FS}}/\hbar \sim 10^{11} \text{ rad/s}$. The weak external field merely exerts a slow torque on the time-averaged total magnetic moment, causing $\vec{J}$ to slowly precess about $\vec{B}_{\text{ext}}$ at Larmor frequency $\omega_L \sim \mu_B B_{\text{ext}}/\hbar \ll \omega_{LS}$.

Because the electron's gyromagnetic ratio is $g_s \approx 2$ while the orbital ratio is $g_l = 1$, the total magnetic moment $\vec{\mu} = -\frac{\mu_B}{\hbar}(\vec{L} + 2\vec{S})$ is **not parallel** to total angular momentum $\vec{J} = \vec{L} + \vec{S}$! Only its projection along $\vec{J}$ survives the rapid time-average, yielding the Landé g -factor:

$$\langle \vec{\mu} \rangle = -g_j \frac{\mu_B}{\hbar} \vec{J}. \quad (1)$$

This is called “anomalous” purely for historical reasons: before the discovery of electron spin, classical electrodynamics predicted that every atom should exhibit a simple triplet splitting (the normal Zeeman effect with $g = 1$). The multi-line splitting patterns caused by $g_j \neq 1$ were deemed “anomalous.”

- **The Paschen–Back Effect is the STRONG EXTERNAL Field Limit ($B_{\text{ext}} \gg B_{\text{int}}$):** When $B_{\text{ext}} \gg 10 \text{ T}$, the external magnetic torque overpowers the internal field $\vec{B}_{\text{int}}$, ripping $\vec{L}$ and $\vec{S}$ apart. They decouple and precess independently around $\vec{B}_{\text{ext}}$, restoring the simple equidistant pattern.

8.6.3 Weak-Field Limit: The Anomalous Zeeman Effect

In weak magnetic fields ($B \ll 1 \text{ T}$), fine structure dominates: $\hat{H}_Z \ll \hat{H}_{\text{FS}}$. The coupled states $|n, j, m_j, l, s\rangle$ remain the appropriate basis.

Using the Projection Theorem for the vector operator $\vec{L} + 2\vec{S} = \vec{J} + \vec{S}$:

$$\langle j, m_j, l, s | (\hat{L}_z + 2\hat{S}_z) | j, m_j, l, s \rangle = \frac{\langle (\hat{\vec{J}} + \hat{\vec{S}}) \cdot \hat{\vec{J}} \rangle}{j(j+1)\hbar^2} m_j \hbar. \quad (8.6.9)$$

To evaluate the scalar product:

$$(\vec{J} + \vec{S}) \cdot \vec{J} = J^2 + \vec{S} \cdot \vec{J}. \quad (8.6.10)$$

Since $\vec{L} = \vec{J} - \vec{S} \implies L^2 = J^2 + S^2 - 2\vec{S} \cdot \vec{J}$, we have:

$$\vec{S} \cdot \vec{J} = \frac{1}{2} (J^2 + S^2 - L^2). \quad (8.6.11)$$

Therefore:

$$(\vec{J} + \vec{S}) \cdot \vec{J} = J^2 + \frac{1}{2} (J^2 + S^2 - L^2) = \frac{3}{2} J^2 + \frac{1}{2} S^2 - \frac{1}{2} L^2. \quad (8.6.12)$$

Taking the expectation value:

$$\langle (\vec{J} + \vec{S}) \cdot \vec{J} \rangle = \frac{\hbar^2}{2} [3j(j+1) + s(s+1) - l(l+1)]. \quad (8.6.13)$$

Dividing by $j(j+1)\hbar^2$:

$$g_j \equiv \frac{\langle (\vec{J} + \vec{S}) \cdot \vec{J} \rangle}{j(j+1)\hbar^2} = 1 + \frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)}. \quad (8.6.14)$$

For a single electron with $s = 1/2$, $s(s+1) = 3/4$, giving the celebrated **Landé g -factor**:

$$g_j = 1 + \frac{j(j+1) - l(l+1) + 3/4}{2j(j+1)} \quad (8.6.15)$$

The weak-field Zeeman energy shift is:

$$E_Z^{(1)} = g_j \mu_B B m_j \quad (8.6.16)$$

Let us calculate the Landé g -factors explicitly for the Hydrogen multiplets:

- $1s_{1/2}$ ($l = 0, j = 1/2$):

$$g_{1/2} = 1 + \frac{3/4 - 0 + 3/4}{2(3/4)} = 1 + 1 = 2 \implies \Delta E = \pm \mu_B B. \quad (8.6.17)$$

- $2s_{1/2}$ ($l = 0, j = 1/2$):

$$g_{1/2} = 2 \implies \Delta E = \pm \mu_B B. \quad (8.6.18)$$

- $2p_{1/2}$ ($l = 1, j = 1/2$):

$$g_{1/2} = 1 + \frac{3/4 - 2 + 3/4}{2(3/4)} = 1 + \frac{-1/2}{3/2} = 1 - \frac{1}{3} = \frac{2}{3} \implies \Delta E = \pm \frac{1}{3} \mu_B B. \quad (8.6.19)$$

- $2p_{3/2}$ ($l = 1, j = 3/2$):

$$g_{3/2} = 1 + \frac{15/4 - 2 + 3/4}{2(15/4)} = 1 + \frac{10/4}{30/4} = 1 + \frac{1}{3} = \frac{4}{3}. \quad (8.6.20)$$

The four Zeeman sublevels are:

$$m_j = \pm \frac{3}{2} : \quad \Delta E = \pm \left(\frac{4}{3}\right) \left(\frac{3}{2}\right) \mu_B B = \pm 2 \mu_B B, \quad (8.6.21)$$

$$m_j = \pm \frac{1}{2} : \quad \Delta E = \pm \left(\frac{4}{3}\right) \left(\frac{1}{2}\right) \mu_B B = \pm \frac{2}{3} \mu_B B. \quad (8.6.22)$$

8.6.4 Strong-Field Limit: The Paschen–Back Effect

When the external magnetic field is so strong ($B \gg 1$ T) that $\hat{H}_Z \gg \hat{H}_{\text{FS}}$, the external magnetic torque uncouples the orbital angular momentum $\vec{L}$ and electron spin $\vec{S}$. They precess independently about the external field axis $\hat{z}$.

In this regime, the uncoupled basis $|n, l, m_l, s, m_s\rangle$ is the appropriate zeroth-order basis. The Zeeman Hamiltonian is already completely diagonal:

$$E_Z = \mu_B B (m_l + 2m_s). \quad (8.6.23)$$

The spin-orbit interaction $\hat{H}_{\text{SO}} = A \vec{L} \cdot \vec{S}$ is now treated as a small perturbation on top of the dominant Zeeman splitting:

$$\langle \hat{H}_{\text{SO}} \rangle = A \langle l, m_l, s, m_s | (L_x S_x + L_y S_y + L_z S_z) | l, m_l, s, m_s \rangle = A \hbar^2 m_l m_s, \quad (8.6.24)$$

since the off-diagonal terms $L_{\pm}S_{\mp}$ have vanishing diagonal matrix elements. The total energy in the Paschen–Back regime is:

$$E = E_n^{(0)} + \mu_B B (m_l + 2m_s) + A\hbar^2 m_l m_s. \quad (8.6.25)$$

Let us evaluate the primary Zeeman splittings for the lowest levels:

1. **1s state** ($n = 1, l = 0, m_l = 0, m_s = \pm 1/2$):

$$E_Z = 2\mu_B B m_s = \pm \mu_B B. \quad (8.6.26)$$

The single level splits into a doublet separated by $2\mu_B B$.

2. **2s state** ($n = 2, l = 0, m_l = 0, m_s = \pm 1/2$):

$$E_Z = \pm \mu_B B. \quad (8.6.27)$$

3. **2p state** ($n = 2, l = 1, m_l \in \{+1, 0, -1\}, m_s = \pm 1/2$): There are $2 \times 3 = 6$ uncoupled states. Evaluating $m_l + 2m_s$:

$$|m_l = +1, m_s = +1/2\rangle : E_Z = (+1 + 1)\mu_B B = +2\mu_B B, \quad (8.6.28)$$

$$|m_l = +1, m_s = -1/2\rangle : E_Z = (+1 - 1)\mu_B B = 0, \quad (8.6.29)$$

$$|m_l = 0, m_s = +1/2\rangle : E_Z = (0 + 1)\mu_B B = +\mu_B B, \quad (8.6.30)$$

$$|m_l = 0, m_s = -1/2\rangle : E_Z = (0 - 1)\mu_B B = -\mu_B B, \quad (8.6.31)$$

$$|m_l = -1, m_s = +1/2\rangle : E_Z = (-1 + 1)\mu_B B = 0, \quad (8.6.32)$$

$$|m_l = -1, m_s = -1/2\rangle : E_Z = (-1 - 1)\mu_B B = -2\mu_B B. \quad (8.6.33)$$

The 6 degenerate states split into **5 equidistant levels** at energies $\{-2, -1, 0, +1, +2\}\mu_B B$, with the central level at $E = 0$ being doubly degenerate.

8.6.5 Hyperfine Structure: Nuclear Spin Coupling and the Back–Goudsmit Effect

Beyond the fine structure, high-precision atomic spectroscopy reveals an even finer splitting: the **hyperfine structure**.

Physical Origin: The Nuclear Magnetic Dipole

The atomic nucleus is not an inert point charge; it possesses an intrinsic nuclear spin $\vec{I}$ and an associated nuclear magnetic dipole moment:

$$\hat{\vec{\mu}}_I = g_I \frac{e}{2M_p} \hat{\vec{I}} = g_I \frac{\mu_N}{\hbar} \hat{\vec{I}}, \quad (8.6.34)$$

where M_p is the proton mass, g_I is the nuclear g -factor ($g_p \approx 5.586$ for a proton), and $\mu_N \equiv \frac{e\hbar}{2M_p}$ is the **nuclear magneton**. Because the proton is 1836 times heavier than the electron:

$$\mu_N = \frac{m_e}{M_p} \mu_B \approx \frac{1}{1836} \mu_B \approx 5.051 \times 10^{-27} \text{ J/T}. \quad (8.6.35)$$

Consequently, nuclear magnetic interactions are suppressed by a factor of $m_e/M_p \sim 10^{-3}$ relative to electron fine structure:

$$\Delta E_{\text{HFS}} \sim \frac{m_e}{M_p} \Delta E_{\text{FS}} \sim 10^{-6} \text{ eV}. \quad (8.6.36)$$

The Hyperfine Hamiltonian and the Fermi Contact Interaction

The electron moves in the magnetic dipole field generated by the nucleus:

$$\vec{B}_{\text{nuc}}(\vec{r}) = \frac{\mu_0}{4\pi} \left[\frac{3(\vec{\mu}_I \cdot \hat{r})\hat{r} - \vec{\mu}_I}{r^3} + \frac{8\pi}{3} \vec{\mu}_I \delta^3(\vec{r}) \right]. \quad (8.6.37)$$

The interaction energy with the electron magnetic moment $\vec{\mu}_e = -g_s \frac{\mu_B}{\hbar} \vec{S}$ consists of two distinct terms:

$$\hat{H}_{\text{HFS}} = -\vec{\mu}_e \cdot \vec{B}_{\text{nuc}} = \hat{H}_{\text{dipole}} + \hat{H}_{\text{contact}}. \quad (8.6.38)$$

1. **Dipole–Dipole Term** ($r > 0$): For states with orbital angular momentum $l > 0$, the dipolar field averages over spherical angles, coupling $\vec{L}$, $\vec{S}$, and $\vec{I}$.
2. **Fermi Contact Term** ($\vec{r} = 0$): For s -states ($l = 0$), the spherical dipole term averages identically to zero, and the interaction is dominated entirely by the singular delta-function term:

$$\hat{H}_{\text{contact}} = \frac{2}{3} \mu_0 g_s \mu_B g_I \mu_N \frac{\hat{S} \cdot \hat{I}}{\hbar^2} \delta^3(\vec{r}). \quad (8.6.39)$$

The 21 cm Hydrogen Line

For the ground state of Hydrogen ($1s_{1/2}$, $n = 1, l = 0, j = 1/2, s = 1/2, I = 1/2$), the spatial density at the nucleus is $|\psi_{100}(0)|^2 = \frac{1}{\pi a_0^3}$. Taking the spatial expectation value yields the effective spin-spin coupling:

$$\hat{H}_{\text{HFS}} = A_{\text{HFS}} \hat{S} \cdot \hat{I}, \quad A_{\text{HFS}} = \frac{2}{3} \mu_0 g_s \mu_B g_I \mu_N \frac{1}{\pi a_0^3 \hbar^2} = \frac{4}{3} g_p \frac{m_e}{M_p} \frac{\alpha^4 m c^2}{\hbar^2}. \quad (8.6.40)$$

We couple the electron spin $\vec{S}$ and nuclear spin $\vec{I}$ into the **total atomic angular momentum**:

$$\hat{F} \equiv \hat{J} + \hat{I} = \hat{S} + \hat{I} \implies \hat{S} \cdot \hat{I} = \frac{1}{2} \left(\hat{F}^2 - \hat{S}^2 - \hat{I}^2 \right). \quad (8.6.41)$$

Since $s = 1/2$ and $I = 1/2$, the allowed values of total angular momentum are:

$$F = 1 \quad (\text{triplet state, } 2F + 1 = 3), \quad F = 0 \quad (\text{singlet state, } 2F + 1 = 1). \quad (8.6.42)$$

Evaluating the eigenvalue of $\hat{S} \cdot \hat{I}$:

$$\left\langle \hat{S} \cdot \hat{I} \right\rangle = \frac{\hbar^2}{2} \left[F(F+1) - \frac{3}{4} - \frac{3}{4} \right] = \begin{cases} +\frac{1}{4} \hbar^2, & F = 1 \text{ (parallel spins),} \\ -\frac{3}{4} \hbar^2, & F = 0 \text{ (antiparallel spins).} \end{cases} \quad (8.6.43)$$

The hyperfine energy splitting between the triplet and singlet ground states is:

$$\Delta E_{\text{HFS}} = E(F=1) - E(F=0) = A_{\text{HFS}} \hbar^2 = \frac{4}{3} g_p \frac{m_e}{M_p} \alpha^4 m c^2 \approx 5.874 \times 10^{-6} \text{ eV}. \quad (8.6.44)$$

The transition frequency is:

$$\nu = \frac{\Delta E_{\text{HFS}}}{h} \approx 1420.40575 \text{ MHz} \implies \lambda = \frac{c}{\nu} \approx 21.106 \text{ cm}. \quad (8.6.45)$$

This is the universally famous **21 cm hydrogen line**, the premier observational probe of neutral hydrogen gas in radio astronomy and cosmology!

Hyperfine Zeeman Effect and the Back–Goudsmit Effect

Because the hyperfine splitting corresponds to an effective internal nuclear magnetic field:

$$B_{\text{nuc}} \sim \frac{\Delta E_{\text{HFS}}}{\mu_B} \approx \frac{5.87 \times 10^{-6} \text{ eV}}{5.79 \times 10^{-5} \text{ eV/T}} \approx 0.1 \text{ Tesla} \quad (1000 \text{ Gauss}), \quad (8.6.46)$$

an external laboratory magnetic field begins to perturb hyperfine structure at extremely modest field strengths:

- **Weak Field** ($B_{\text{ext}} \ll 0.1 \text{ T}$): Hyperfine coupling dominates. Total angular momentum $\vec{F} = \vec{J} + \vec{I}$ remains conserved. The $F = 1$ triplet splits into 3 equidistant sublevels ($m_F \in \{+1, 0, -1\}$) with effective g -factor g_F :

$$\Delta E_Z = g_F \mu_B B_{\text{ext}} m_F. \quad (8.6.47)$$

- **Strong Field** ($B_{\text{ext}} \gg 0.1 \text{ T}$, **The Back–Goudsmit Effect**): The external magnetic field completely uncouples $\vec{I}$ and $\vec{J}$. The electron angular momentum $\vec{J}$ and nuclear spin $\vec{I}$ precess independently about $\vec{B}_{\text{ext}}$. The good quantum numbers become (m_J, m_I) , and the energy levels split as:

$$E = g_J \mu_B B_{\text{ext}} m_J - g_I \mu_N B_{\text{ext}} m_I + A_{\text{HFS}} m_J m_I. \quad (8.6.48)$$

8.7 The Linear Stark Effect in the $n = 2$ Subspace

At the first excited level $n = 2$, the unperturbed Bohr state is 4-fold degenerate (disregarding spin):

$$|2s\rangle \equiv |200\rangle, \quad |2p_0\rangle \equiv |210\rangle, \quad |2p_1\rangle \equiv |211\rangle, \quad |2p_{-1}\rangle \equiv |21, -1\rangle. \quad (8.7.1)$$

When an external electric field $\vec{\mathcal{E}} = \mathcal{E} \hat{z}$ is applied, the perturbation is $\hat{V} = e\mathcal{E}z$.

8.7.1 Evaluation of Matrix Elements

Because z commutes with $\hat{L}_z$, the selection rule $\Delta m = 0$ is strict: matrix elements between different m states vanish identically. Furthermore, by parity, diagonal elements vanish: $\langle 2s|z|2s\rangle = \langle 2p_m|z|2p_m\rangle = 0$. The only non-vanishing matrix element connects $|200\rangle$ and $|210\rangle$:

$$\langle 200|z|210\rangle = \int_0^\infty dr r^3 R_{20}(r) R_{21}(r) \int_0^\pi \sin \theta d\theta \cos \theta \int_0^{2\pi} d\phi Y_{00}^* Y_{10}. \quad (8.7.2)$$

Using the radial wavefunctions:

$$R_{20}(r) = \frac{1}{\sqrt{2}a_0^{3/2}} \left(1 - \frac{r}{2a_0}\right) e^{-r/(2a_0)}, \quad R_{21}(r) = \frac{1}{2\sqrt{6}a_0^{3/2}} \frac{r}{a_0} e^{-r/(2a_0)}, \quad (8.7.3)$$

the angular integral evaluates to $1/\sqrt{3}$, and the radial integral yields:

$$\int_0^\infty dr r^3 R_{20}(r) R_{21}(r) = -3\sqrt{3} a_0. \quad (8.7.4)$$

Therefore:

$$\langle 200|z|210\rangle = \frac{1}{\sqrt{3}} (-3\sqrt{3} a_0) = -3a_0. \quad (8.7.5)$$

8.7.2 Secular Determinant and Energy Splittings

The 4×4 perturbation matrix in the basis $\{|200\rangle, |210\rangle, |211\rangle, |21, -1\rangle\}$ is block-diagonal:

$$\mathbf{V} = \begin{pmatrix} 0 & -3e\mathcal{E}a_0 & 0 & 0 \\ -3e\mathcal{E}a_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (8.7.6)$$

The states $|211\rangle$ and $|21, -1\rangle$ are already uncoupled and have zero first-order shift ($E^{(1)} = 0$). For the 2×2 block:

$$\det \begin{pmatrix} -E^{(1)} & -3e\mathcal{E}a_0 \\ -3e\mathcal{E}a_0 & -E^{(1)} \end{pmatrix} = 0 \implies (E^{(1)})^2 - 9e^2\mathcal{E}^2a_0^2 = 0. \quad (8.7.7)$$

The first-order energy shifts are:

$$\boxed{E^{(1)} = \pm 3e\mathcal{E}a_0, \quad E^{(1)} = 0 \quad (\text{doubly degenerate})} \quad (8.7.8)$$

The corresponding good zeroth-order eigenstates are the sp -hybrid orbitals:

$$|\psi_{\pm}^{(0)}\rangle = \frac{1}{\sqrt{2}} (|200\rangle \mp |210\rangle). \quad (8.7.9)$$

The electron cloud is polarized along the z -axis, acquiring a permanent electric dipole moment $\vec{d} = \mp 3ea_0\hat{z}$ whose interaction energy $-\vec{d} \cdot \vec{\mathcal{E}}$ yields the **linear Stark effect**.

Chapter 9

The Variational Principle and the Semi-classical WKB Approximation

When a quantum system is subjected to strong interactions, perturbation theory fails because there is no small dimensionless parameter $\lambda \ll 1$ in which to expand. In such strongly coupled regimes, non-perturbative approximation methods are essential.

This chapter presents two foundational non-perturbative analytical techniques:

1. **The Variational Method (Rayleigh–Ritz):** An energy-minimization technique based on functional calculus that provides rigorous upper bounds on ground- and excited-state energies.
2. **The Semiclassical WKB Approximation (Wentzel–Kramers–Brillouin):** An asymptotic expansion in powers of Planck’s constant $\hbar$ that connects quantum mechanics to classical phase-space trajectories, providing accurate quantization rules and tunneling transmission probabilities for arbitrary potentials.

9.1 The Rayleigh–Ritz Variational Principle

9.1.1 Fundamental Theorem and Ground-State Upper Bound

Let $\hat{H}$ be a time-independent Hamiltonian that is bounded from below, possessing a discrete spectrum with unknown true ground-state energy E_0 and corresponding normalized eigenstate $|\psi_0\rangle$:

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle, \quad E_0 \leq E_1 \leq E_2 \leq \dots \quad (9.1.1)$$

Let $|\psi\rangle$ be an arbitrary normalizable trial state in the domain of $\hat{H}$. We define the **Rayleigh quotient functional**:

$$E[\psi] \equiv \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}. \quad (9.1.2)$$

Theorem 9.1 (Rayleigh–Ritz Variational Principle). For any trial state vector $|\psi\rangle$, the Rayleigh functional is an absolute upper bound to the true ground-state energy:

$$E[\psi] \geq E_0, \quad (9.1.3)$$

with equality holding if and only if $|\psi\rangle$ is an exact ground-state eigenstate of $\hat{H}$.

Proof. Since the eigenstates $\{|\psi_n\rangle\}$ form a complete orthonormal basis of the Hilbert space ($\sum_n |\psi_n\rangle \langle\psi_n| = \hat{I}$ and $\langle\psi_m|\psi_n\rangle = \delta_{mn}$), we expand the trial state $|\psi\rangle$ as:

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle, \quad c_n = \langle\psi_n|\psi\rangle. \quad (9.1.4)$$

The inner product and Hamiltonian expectation value become:

$$\langle\psi|\psi\rangle = \sum_n |c_n|^2, \quad (9.1.5)$$

$$\langle\psi|\hat{H}|\psi\rangle = \sum_n |c_n|^2 E_n. \quad (9.1.6)$$

Subtracting $E_0 \langle\psi|\psi\rangle$ from both sides of the expectation value:

$$\langle\psi|\hat{H}|\psi\rangle - E_0 \langle\psi|\psi\rangle = \sum_n |c_n|^2 (E_n - E_0). \quad (9.1.7)$$

Since $E_n \geq E_0$ for every n , every term in the summation is strictly non-negative: $|c_n|^2 (E_n - E_0) \geq 0$. Therefore:

$$\langle\psi|\hat{H}|\psi\rangle - E_0 \langle\psi|\psi\rangle \geq 0 \implies \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} \geq E_0. \quad (9.1.8)$$

Equality requires $\sum_n |c_n|^2 (E_n - E_0) = 0$. For any non-degenerate state with $E_n > E_0$ ($n \geq 1$), this forces $c_n = 0$ for all $n \geq 1$, which implies $|\psi\rangle = c_0 |\psi_0\rangle$. $\square$

9.1.2 Quadratic Error Suppression

A remarkable mathematical feature of the variational method is that a trial wavefunction that is accurate only to first order $\mathcal{O}(\epsilon)$ produces an energy expectation value that is accurate to **second order** $\mathcal{O}(\epsilon^2)$.

To prove this, let the trial state $|\psi\rangle$ differ from the true normalized ground state $|\psi_0\rangle$ by a small orthogonal deviation $\epsilon |\psi_\perp\rangle$ ($\langle\psi_0|\psi_\perp\rangle = 0$, $\langle\psi_\perp|\psi_\perp\rangle = 1$):

$$|\psi\rangle = \frac{|\psi_0\rangle + \epsilon |\psi_\perp\rangle}{\sqrt{1 + \epsilon^2}}. \quad (9.1.9)$$

Evaluating the expectation value:

$$\langle\psi|\hat{H}|\psi\rangle = \frac{\langle\psi_0|\hat{H}|\psi_0\rangle + \epsilon \langle\psi_0|\hat{H}|\psi_\perp\rangle + \epsilon \langle\psi_\perp|\hat{H}|\psi_0\rangle + \epsilon^2 \langle\psi_\perp|\hat{H}|\psi_\perp\rangle}{1 + \epsilon^2}. \quad (9.1.10)$$

Because $\hat{H}|\psi_0\rangle = E_0 |\psi_0\rangle$ and $\langle\psi_0|\psi_\perp\rangle = 0$:

$$\langle\psi_\perp|\hat{H}|\psi_0\rangle = E_0 \langle\psi_\perp|\psi_0\rangle = 0, \quad \langle\psi_0|\hat{H}|\psi_\perp\rangle = E_0 \langle\psi_0|\psi_\perp\rangle = 0. \quad (9.1.11)$$

The linear terms in ϵ vanish identically! Expanding $(1 + \epsilon^2)^{-1} = 1 - \epsilon^2 + \dots$:

$$\begin{aligned} E[\psi] &= \left[E_0 + \epsilon^2 \langle\psi_\perp|\hat{H}|\psi_\perp\rangle \right] \left[1 - \epsilon^2 + \mathcal{O}(\epsilon^4) \right] \\ &= E_0 + \epsilon^2 \left(\langle\psi_\perp|\hat{H}|\psi_\perp\rangle - E_0 \right) + \mathcal{O}(\epsilon^4). \end{aligned} \quad (9.1.12)$$

Thus:

$$\boxed{E[\psi] - E_0 = \mathcal{O}(\epsilon^2)} \quad (9.1.13)$$

An error of 10% in the trial wavefunction ($\epsilon \sim 0.1$) yields an energy error of only 1% ($\epsilon^2 \sim 0.01$).

9.1.3 Variational Bounds on Excited States

The variational principle can be extended to find upper bounds on the first excited state energy E_1 .

Theorem 9.2 (Excited-State Variational Theorem). If a trial state $|\psi_\perp\rangle$ is strictly orthogonal to the true ground state $|\psi_0\rangle$:

$$\langle\psi_0|\psi_\perp\rangle = 0, \quad (9.1.14)$$

then the Rayleigh quotient provides an upper bound to the first excited state energy:

$$\frac{\langle\psi_\perp|\hat{H}|\psi_\perp\rangle}{\langle\psi_\perp|\psi_\perp\rangle} \geq E_1. \quad (9.1.15)$$

Proof. Expand $|\psi_\perp\rangle = \sum_n c_n |\psi_n\rangle$. The orthogonality condition demands $c_0 = \langle\psi_0|\psi_\perp\rangle = 0$. Thus:

$$\langle\psi_\perp|\hat{H}|\psi_\perp\rangle - E_1 \langle\psi_\perp|\psi_\perp\rangle = \sum_{n=1}^{\infty} |c_n|^2 (E_n - E_1) \geq 0, \quad (9.1.16)$$

since $E_n \geq E_1$ for all $n \geq 1$. $\square$

In practice, if the Hamiltonian commutes with a symmetry operator such as parity $\hat{\Pi}$ ($[\hat{H}, \hat{\Pi}] = 0$), the true ground state is invariably even ($\Pi = +1$). By deliberately choosing an *odd* trial wavefunction ($\psi(-x) = -\psi(x)$), orthogonality $\langle\psi_0|\psi_\perp\rangle = 0$ is guaranteed automatically by symmetry, yielding a rigorous bound on the lowest odd state.

9.2 Applications of the Variational Method

9.2.1 The Quartic Anharmonic Oscillator

Consider the 1D anharmonic oscillator:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 + \lambda\hat{x}^4. \quad (9.2.1)$$

We propose a normalized Gaussian trial wavefunction with variational parameter α :

$$\psi(x; \alpha) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}. \quad (9.2.2)$$

Step-by-Step Expectation Values

1. Kinetic Energy:

$$\langle\hat{T}\rangle = -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} dx \psi^*(x) \frac{d^2\psi}{dx^2} = \frac{\hbar^2\alpha}{4m}. \quad (9.2.3)$$

2. Harmonic Potential Energy: Using the Gaussian integral $\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \frac{\sqrt{\pi}}{2\alpha^{3/2}}$:

$$\left\langle \frac{1}{2}m\omega^2 x^2 \right\rangle = \frac{1}{2}m\omega^2 \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \frac{m\omega^2}{4\alpha}. \quad (9.2.4)$$

3. Quartic Potential Energy: Using $\int_{-\infty}^{\infty} x^4 e^{-\alpha x^2} dx = \frac{3\sqrt{\pi}}{4\alpha^{5/2}}$:

$$\langle\lambda x^4\rangle = \lambda \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{\infty} x^4 e^{-\alpha x^2} dx = \frac{3\lambda}{4\alpha^2}. \quad (9.2.5)$$

Minimization

The total variational energy is:

$$E(\alpha) = \frac{\hbar^2 \alpha}{4m} + \frac{m\omega^2}{4\alpha} + \frac{3\lambda}{4\alpha^2}. \quad (9.2.6)$$

Differentiating with respect to α and setting the derivative to zero:

$$\frac{dE}{d\alpha} = \frac{\hbar^2}{4m} - \frac{m\omega^2}{4\alpha^2} - \frac{3\lambda}{2\alpha^3} = 0. \quad (9.2.7)$$

Multiplying by $4\alpha^3$:

$$\frac{\hbar^2}{m}\alpha^3 - m\omega^2\alpha - 6\lambda = 0. \quad (9.2.8)$$

In the harmonic limit $\lambda \rightarrow 0$, this yields $\alpha = \frac{m\omega}{\hbar}$, reproducing the exact ground-state energy $E = \frac{1}{2}\hbar\omega$. For non-zero $\lambda > 0$, the cubic equation can be solved analytically or numerically, yielding an upper bound valid across all coupling strengths from weak to ultra-strong coupling.

9.2.2 Ground State of the Helium Atom

The Helium atom consists of a nucleus of charge $Z = 2$ and two indistinguishable electrons with positions $\vec{r}_1$ and $\vec{r}_2$. In atomic units ($\hbar = m = e = 4\pi\epsilon_0 = 1$), the Hamiltonian is:

$$\hat{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - \frac{Z}{r_1} - \frac{Z}{r_2} + \frac{1}{|\vec{r}_1 - \vec{r}_2|}. \quad (9.2.9)$$

The electron-electron repulsion term $\hat{V}_{ee} = \frac{1}{|\vec{r}_1 - \vec{r}_2|}$ prevents separation of variables. If we neglect $\hat{V}_{ee}$ completely, each electron sees the bare nuclear charge $Z = 2$, giving an unperturbed ground state energy of:

$$E^{(0)} = -Z^2 - Z^2 = -8 \text{ a.u.} = -108.8 \text{ eV}, \quad (9.2.10)$$

which significantly overestimates the experimentally measured ground state energy $E_{\text{exp}} = -78.98 \text{ eV} = -2.9037 \text{ a.u.}$

First-Order Perturbation Theory Baseline

Treating $\hat{V}_{ee} = \frac{1}{r_{12}}$ as a perturbation to the unshielded state $\psi^{(0)}(\vec{r}_1, \vec{r}_2) = \frac{Z^3}{\pi} e^{-Z(r_1+r_2)}$:

$$E^{(1)} = \left\langle \psi^{(0)} \left| \frac{1}{r_{12}} \right| \psi^{(0)} \right\rangle = \frac{5}{8}Z = \frac{5}{4} \text{ a.u.} = +34.0 \text{ eV}. \quad (9.2.11)$$

The first-order energy estimate is:

$$E = E^{(0)} + E^{(1)} = -8 + 1.25 = -6.75 \text{ a.u.} = -74.8 \text{ eV}. \quad (9.2.12)$$

This is closer, but still exhibits a 5.3% error.

Variational Screening and Effective Nuclear Charge

Physically, each electron clouds the nucleus, partially screening the positive charge seen by the other electron. An ad-hoc model where one electron sees $Z = 2$ and the other sees $Z = 2 - \sigma$ is unphysical because the electrons are indistinguishable fermions and must occupy symmetric spatial orbitals in the spin-singlet ground state.

We therefore introduce a symmetric **effective nuclear charge** $Z_{\text{eff}} = \alpha$ as our variational parameter:

$$\psi(\vec{r}_1, \vec{r}_2; \alpha) = \phi_{1s}(\vec{r}_1)\phi_{1s}(\vec{r}_2) = \frac{\alpha^3}{\pi} e^{-\alpha(r_1+r_2)}. \quad (9.2.13)$$

This wavefunction is normalized: $\langle \psi | \psi \rangle = 1$.

We rewrite the exact Hamiltonian (9.2.9) as:

$$\hat{H} = \left(-\frac{1}{2}\nabla_1^2 - \frac{\alpha}{r_1} \right) + \left(-\frac{1}{2}\nabla_2^2 - \frac{\alpha}{r_2} \right) + (\alpha - Z) \left(\frac{1}{r_1} + \frac{1}{r_2} \right) + \frac{1}{|\vec{r}_1 - \vec{r}_2|}. \quad (9.2.14)$$

Using the virial theorem for a single hydrogenic orbital of charge α , the kinetic energy is $\langle T \rangle = \frac{1}{2}\alpha^2$ and the potential energy is $\langle -\alpha/r \rangle = -\alpha^2$. Furthermore:

$$\left\langle \frac{1}{r_1} \right\rangle = \left\langle \frac{1}{r_2} \right\rangle = \alpha. \quad (9.2.15)$$

The expectation value of the bare single-particle terms is:

$$\langle \hat{T}_1 + \hat{T}_2 + \hat{V}_{\text{nuc}} \rangle = 2 \left(\frac{1}{2}\alpha^2 \right) - 2Z \left\langle \frac{1}{r_1} \right\rangle = \alpha^2 - 2Z\alpha. \quad (9.2.16)$$

The Electron–Electron Repulsion Integral

The expectation value of the mutual Coulomb repulsion is:

$$\left\langle \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right\rangle = \frac{\alpha^6}{\pi^2} \iint d^3r_1 d^3r_2 \frac{e^{-2\alpha(r_1+r_2)}}{|\vec{r}_1 - \vec{r}_2|}. \quad (9.2.17)$$

Using the multipole expansion for the inverse distance:

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \gamma), \quad r_{<} = \min(r_1, r_2), \quad r_{>} = \max(r_1, r_2), \quad (9.2.18)$$

where γ is the angle between $\vec{r}_1$ and $\vec{r}_2$. Because the trial state is spherically symmetric ($l=0$), only the monopole term $l=0$ survives the angular integration:

$$\iint d\Omega_1 d\Omega_2 \frac{1}{|\vec{r}_1 - \vec{r}_2|} = (4\pi)^2 \frac{1}{r_{>}}. \quad (9.2.19)$$

The radial integral splits into two symmetric domains ($r_1 < r_2$ and $r_1 > r_2$):

$$\begin{aligned} \left\langle \frac{1}{|\vec{r}_1 - \vec{r}_2|} \right\rangle &= 16\alpha^6 \cdot 2 \int_0^\infty dr_1 r_1^2 e^{-2\alpha r_1} \int_{r_1}^\infty dr_2 r_2^2 \frac{1}{r_2} e^{-2\alpha r_2} \\ &= 32\alpha^6 \int_0^\infty dr_1 r_1^2 e^{-2\alpha r_1} \left[\frac{2\alpha r_1 + 1}{4\alpha^2} e^{-2\alpha r_1} \right] \\ &= 8\alpha^4 \int_0^\infty dr_1 (2\alpha r_1^3 + r_1^2) e^{-4\alpha r_1} \\ &= 8\alpha^4 \left[2\alpha \frac{3!}{(4\alpha)^4} + \frac{2!}{(4\alpha)^3} \right] = \frac{5}{8}\alpha. \end{aligned} \quad (9.2.20)$$

Optimal Energy and Physical Interpretation

Assembling the full expectation value:

$$E(\alpha) = \alpha^2 - 2Z\alpha + \frac{5}{8}\alpha = \alpha^2 - \left(2Z - \frac{5}{8}\right)\alpha. \quad (9.2.21)$$

We minimize with respect to α :

$$\frac{dE}{d\alpha} = 2\alpha - \left(2Z - \frac{5}{8}\right) = 0 \implies \alpha = Z - \frac{5}{16}. \quad (9.2.22)$$

For Helium ($Z = 2$):

$$\boxed{Z_{\text{eff}} = 2 - \frac{5}{16} = \frac{27}{16} = 1.6875} \quad (9.2.23)$$

Each electron screens the nuclear charge by precisely $\frac{5}{16}$ of a proton charge! Substituting $Z_{\text{eff}} = 27/16$ back into the energy functional:

$$E_{\text{min}} = -\left(Z - \frac{5}{16}\right)^2 = -\left(\frac{27}{16}\right)^2 \text{ a.u.} = -2.8477 \text{ a.u.} = -77.49 \text{ eV}. \quad (9.2.24)$$

Comparing with experiment:

- Unperturbed ($Z = 2$ bare): -108.8 eV (38% error).
- First-order perturbation theory ($Z_{\text{eff}} = 2$ fixed): -74.8 eV (5.3% error).
- Variational screening ($Z_{\text{eff}} = 27/16$): -77.49 eV (**1.9% error**).

9.2.3 Bound States of the Yukawa (Screened Coulomb) Potential

In nuclear physics and plasma physics, interactions are modeled by the Yukawa potential:

$$V(r) = -g^2 \frac{e^{-\alpha r}}{r}, \quad (9.2.25)$$

where α is the inverse screening radius. In the limit $\alpha \rightarrow 0$, $V(r)$ reduces to the attractive Coulomb potential, which always admits bound states. However, for finite screening $\alpha > 0$, the potential becomes shallower and short-ranged; beyond a critical screening α_{crit} , bound states cease to exist.

We choose a normalized exponential trial wavefunction with parameter β :

$$\psi(r; \beta) = \sqrt{\frac{\beta^3}{\pi}} e^{-\beta r}. \quad (9.2.26)$$

The kinetic energy expectation value is:

$$\langle \hat{T} \rangle = \frac{\hbar^2 \beta^2}{2m}. \quad (9.2.27)$$

The potential energy expectation value is:

$$\langle \hat{V} \rangle = -4\pi g^2 \frac{\beta^3}{\pi} \int_0^\infty dr r^2 \frac{e^{-\alpha r}}{r} e^{-2\beta r} = -4g^2 \beta^3 \int_0^\infty dr r e^{-(2\beta+\alpha)r} = -\frac{4g^2 \beta^3}{(2\beta+\alpha)^2}. \quad (9.2.28)$$

The total variational energy is:

$$E(\beta) = \frac{\hbar^2 \beta^2}{2m} - \frac{4g^2 \beta^3}{(2\beta+\alpha)^2}. \quad (9.2.29)$$

For a bound state to exist, $E(\beta)$ must be strictly negative for some $\beta > 0$:

$$\frac{\hbar^2 \beta^2}{2m} < \frac{4g^2 \beta^3}{(2\beta + \alpha)^2} \implies (2\beta + \alpha)^2 < \frac{8mg^2}{\hbar^2} \beta. \quad (9.2.30)$$

Expanding the quadratic:

$$4\beta^2 + \left(4\alpha - \frac{8mg^2}{\hbar^2}\right)\beta + \alpha^2 < 0. \quad (9.2.31)$$

A positive real solution for β exists if and only if the discriminant of this quadratic is positive:

$$\Delta = \left(4\alpha - \frac{8mg^2}{\hbar^2}\right)^2 - 16\alpha^2 = \frac{64m^2g^4}{\hbar^4} \left(1 - \frac{\hbar^2\alpha}{mg^2}\right) > 0. \quad (9.2.32)$$

This gives a rigorous upper bound on the critical screening parameter for bound state formation:

$$\boxed{\alpha \leq \alpha_{\text{crit}} = \frac{mg^2}{\hbar^2}} \quad (9.2.33)$$

If the screening α exceeds $\frac{mg^2}{\hbar^2}$, the screened Coulomb field cannot support a bound state.

9.3 The Semiclassical WKB Approximation

9.3.1 The Semiclassical Expansion in Powers of $\hbar$

Consider a particle of mass m and total energy E moving in a one-dimensional potential $V(x)$. The stationary Schrödinger equation is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \implies \frac{d^2\psi}{dx^2} + \frac{p^2(x)}{\hbar^2}\psi = 0, \quad (9.3.1)$$

where the **classical momentum** $p(x)$ is defined by:

$$p(x) \equiv \sqrt{2m(E - V(x))}. \quad (9.3.2)$$

In the semiclassical limit ($\hbar \rightarrow 0$), the quantum wavefunction must rapidly oscillate like a classical plane wave $e^{ipx/\hbar}$ with position-dependent phase and amplitude. We introduce the exact exponential ansatz:

$$\psi(x) = \exp\left(\frac{i}{\hbar}S(x)\right). \quad (9.3.3)$$

Differentiating (9.3.3):

$$\frac{d\psi}{dx} = \frac{i}{\hbar}S'(x)\psi, \quad (9.3.4)$$

$$\frac{d^2\psi}{dx^2} = \left[\frac{i}{\hbar}S''(x) - \frac{1}{\hbar^2}(S'(x))^2\right]\psi. \quad (9.3.5)$$

Substituting into (9.3.1) and dividing through by $\psi(x) \neq 0$:

$$(S'(x))^2 - i\hbar S''(x) = p^2(x). \quad (9.3.6)$$

Equation (9.3.6) is a non-linear differential equation of Riccati type, completely equivalent to the Schrödinger equation.

We now expand the action function $S(x)$ as a formal asymptotic power series in $\hbar$:

$$S(x) = S_0(x) + \left(\frac{\hbar}{i}\right) S_1(x) + \left(\frac{\hbar}{i}\right)^2 S_2(x) + \dots \quad (9.3.7)$$

Substituting into (9.3.6):

$$\left(S'_0 + \frac{\hbar}{i} S'_1 + \dots\right)^2 + \hbar \left(S''_0 + \frac{\hbar}{i} S''_1 + \dots\right) = p^2(x). \quad (9.3.8)$$

Collecting like powers of $\hbar$:

$$\mathcal{O}(\hbar^0): \quad (S'_0(x))^2 = p^2(x), \quad (9.3.9)$$

$$\mathcal{O}(\hbar^1): \quad 2S'_0(x)S'_1(x) + S''_0(x) = 0. \quad (9.3.10)$$

Solution to Leading and Next-to-Leading Order

From the zeroth-order equation (9.3.9):

$$S'_0(x) = \pm p(x) \implies S_0(x) = \pm \int^x p(x') dx'. \quad (9.3.11)$$

$S_0(x)$ is simply the classical action along the trajectory.

From the first-order equation (9.3.10):

$$S'_1(x) = -\frac{S''_0(x)}{2S'_0(x)} = -\frac{p'(x)}{2p(x)} = -\frac{1}{2} \frac{d}{dx} (\ln p(x)). \quad (9.3.12)$$

Integrating immediately gives:

$$S_1(x) = -\frac{1}{2} \ln p(x) + \text{const} \implies \exp\left(\frac{i}{\hbar} S_1(x)\right) = \frac{1}{\sqrt{p(x)}}. \quad (9.3.13)$$

Combining S_0 and S_1 , the leading-order semiclassical WKB wavefunction is:

$$\psi_{\text{WKB}}(x) \approx \frac{A}{\sqrt{p(x)}} \exp\left(\frac{i}{\hbar} \int^x p(x') dx'\right) + \frac{B}{\sqrt{p(x)}} \exp\left(-\frac{i}{\hbar} \int^x p(x') dx'\right) \quad (9.3.14)$$

Physical Significance of the Pre-Factor

Notice the amplitude factor $1/\sqrt{p(x)}$. The probability density in the classically allowed region is:

$$|\psi_{\text{WKB}}(x)|^2 \propto \frac{1}{p(x)} = \frac{1}{mv(x)}. \quad (9.3.15)$$

The probability of finding the quantum particle in an interval dx is inversely proportional to its classical velocity $v(x)$. The particle spends more time where it moves slowly ($dt = dx/v$), reproducing the exact classical probability distribution!

9.3.2 Validity Criterion and Breakdown at Turning Points

The WKB expansion is an asymptotic expansion whose validity requires the second term in (9.3.6) to be negligible compared to the first:

$$|\hbar S''(x)| \ll |S'(x)|^2 \approx p^2(x). \quad (9.3.16)$$

Using $S'_0(x) = p(x) \implies S''_0(x) = p'(x)$:

$$\hbar|p'(x)| \ll p^2(x) \implies \left| \frac{d}{dx} \left(\frac{\hbar}{p(x)} \right) \right| \ll 1. \quad (9.3.17)$$

Defining the local reduced de Broglie wavelength:

$$\bar{\lambda}(x) \equiv \frac{\hbar}{p(x)} = \frac{\lambda(x)}{2\pi}, \quad (9.3.18)$$

the validity criterion becomes:

$$\left| \frac{d\bar{\lambda}}{dx} \right| \ll 1 \quad (9.3.19)$$

The local de Broglie wavelength must vary slowly over a distance of one wavelength.

At a **classical turning point** x_0 where $E = V(x_0)$, the momentum vanishes: $p(x_0) = 0$. Consequently, the de Broglie wavelength diverges ($\bar{\lambda} \rightarrow \infty$), and the pre-factor $1/\sqrt{p(x)} \rightarrow \infty$. The WKB approximation *breaks down catastrophically* near turning points.

9.3.3 Classically Forbidden Regions

In a classically forbidden region where $E < V(x)$, the momentum is purely imaginary:

$$p(x) = i\kappa(x), \quad \kappa(x) \equiv \sqrt{2m(V(x) - E)}. \quad (9.3.20)$$

Here, the oscillatory exponentials transform into decaying and growing exponentials:

$$\psi_{\text{WKB}}(x) = \frac{C}{\sqrt{\kappa(x)}} \exp\left(-\frac{1}{\hbar} \int_{x_0}^x \kappa(x') dx'\right) + \frac{D}{\sqrt{\kappa(x)}} \exp\left(+\frac{1}{\hbar} \int_{x_0}^x \kappa(x') dx'\right) \quad (9.3.21)$$

9.4 Connection Formulas across Turning Points

Because the WKB solutions diverge at turning points, one cannot directly match the oscillatory solution (9.3.14) to the exponential solution (9.3.21).

To bridge the gap, we construct an exact boundary layer solution by linearizing the potential in the immediate vicinity of the turning point x_0 :

$$V(x) \approx V(x_0) + V'(x_0)(x - x_0) = E + V'(x_0)(x - x_0). \quad (9.4.1)$$

Let us consider a **right-hand turning point** where $V'(x_0) > 0$ (allowed region for $x < x_0$, forbidden region for $x > x_0$).

9.4.1 The Airy Differential Equation

Near x_0 , the Schrödinger equation becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V'(x_0)(x - x_0)\psi = 0 \implies \frac{d^2\psi}{dx^2} - \frac{2mV'(x_0)}{\hbar^2}(x - x_0)\psi = 0. \quad (9.4.2)$$

Introducing the dimensionless coordinate:

$$z \equiv \left(\frac{2mV'(x_0)}{\hbar^2} \right)^{1/3} (x - x_0), \quad (9.4.3)$$

the equation transforms into the canonical **Airy differential equation**:

$$\frac{d^2\psi}{dz^2} - z\psi = 0. \quad (9.4.4)$$

The linearly independent solutions are the Airy functions $\text{Ai}(z)$ and $\text{Bi}(z)$. In the classically forbidden region $x > x_0$ ($z > 0$), physical boundary conditions require $\psi(x) \rightarrow 0$ as $x \rightarrow \infty$. Because $\text{Bi}(z)$ diverges exponentially as $z \rightarrow +\infty$, the physical solution across the turning point must be purely $\text{Ai}(z)$:

$$\psi(z) = C_0 \text{Ai}(z). \quad (9.4.5)$$

9.4.2 Asymptotic Matching and Connection Matrix

The Airy function $\text{Ai}(z)$ possesses well-known asymptotic expansions for large $|z|$:

- **Deep in the forbidden region** ($z \gg 1$, $x > x_0$):

$$\text{Ai}(z) \approx \frac{1}{2\sqrt{\pi}z^{1/4}} \exp\left(-\frac{2}{3}z^{3/2}\right). \quad (9.4.6)$$

Expressing z in terms of $\kappa(x') \approx \sqrt{2mV'(x_0)(x' - x_0)}$:

$$\frac{2}{3}z^{3/2} = \frac{1}{\hbar} \int_{x_0}^x \kappa(x') dx'. \quad (9.4.7)$$

Therefore:

$$\psi_{\text{forbidden}}(x) \propto \frac{1}{2\sqrt{\kappa(x)}} \exp\left(-\frac{1}{\hbar} \int_{x_0}^x \kappa(x') dx'\right). \quad (9.4.8)$$

- **Deep in the allowed region** ($-z \gg 1$, $x < x_0$):

$$\text{Ai}(z) \approx \frac{1}{\sqrt{\pi}(-z)^{1/4}} \cos\left(\frac{2}{3}(-z)^{3/2} - \frac{\pi}{4}\right). \quad (9.4.9)$$

Expressing in terms of $p(x') \approx \sqrt{2mV'(x_0)(x_0 - x')}$:

$$\frac{2}{3}(-z)^{3/2} = \frac{1}{\hbar} \int_x^{x_0} p(x') dx'. \quad (9.4.10)$$

Therefore:

$$\psi_{\text{allowed}}(x) \propto \frac{1}{\sqrt{p(x)}} \cos\left(\frac{1}{\hbar} \int_x^{x_0} p(x') dx' - \frac{\pi}{4}\right). \quad (9.4.11)$$

Comparing (9.4.8) and (9.4.11), we establish the celebrated **Kramers connection formula**:

$$\boxed{\frac{1}{2\sqrt{\kappa(x)}} \exp\left(-\frac{1}{\hbar} \int_{x_0}^x \kappa(x') dx'\right) \longleftrightarrow \frac{1}{\sqrt{p(x)}} \cos\left(\frac{1}{\hbar} \int_x^{x_0} p(x') dx' - \frac{\pi}{4}\right)} \quad (9.4.12)$$

The matching across a turning point introduces a fundamental **Berry-like phase loss of $\pi/4$** upon reflection from a smooth potential barrier.

For a **left-hand turning point** x_0 ($V'(x_0) < 0$, allowed for $x > x_0$):

$$\boxed{\frac{1}{\sqrt{p(x)}} \cos\left(\frac{1}{\hbar} \int_x^{x_0} p(x') dx' - \frac{\pi}{4}\right) \longleftrightarrow \frac{1}{2\sqrt{\kappa(x)}} \exp\left(-\frac{1}{\hbar} \int_x^{x_0} \kappa(x') dx'\right)} \quad (9.4.13)$$

9.5 The Bohr–Sommerfeld Quantization Rule

Consider a bound particle in a potential well with two classical turning points x_1 and x_2 ($x_1 < x_2$), such that $V(x_1) = V(x_2) = E$.

Connecting from the left turning point x_1 into the well using (9.4.13):

$$\psi(x) = \frac{C}{\sqrt{p(x)}} \cos\left(\frac{1}{\hbar} \int_{x_1}^x p(x') dx' - \frac{\pi}{4}\right). \quad (9.5.1)$$

Connecting from the right turning point x_2 into the well using (9.4.12):

$$\psi(x) = \frac{C'}{\sqrt{p(x)}} \cos\left(\frac{1}{\hbar} \int_x^{x_2} p(x') dx' - \frac{\pi}{4}\right). \quad (9.5.2)$$

For these two expressions to describe the identical physical wavefunction throughout the allowed region $x_1 < x < x_2$, their arguments must sum to an integer multiple of π :

$$\left(\frac{1}{\hbar} \int_{x_1}^x p(x') dx' - \frac{\pi}{4}\right) + \left(\frac{1}{\hbar} \int_x^{x_2} p(x') dx' - \frac{\pi}{4}\right) = n\pi, \quad n = 0, 1, 2, \dots \quad (9.5.3)$$

The variable x cancels inside the integrals, yielding:

$$\frac{1}{\hbar} \int_{x_1}^{x_2} p(x') dx' - \frac{\pi}{2} = n\pi \implies \int_{x_1}^{x_2} p(x) dx = \left(n + \frac{1}{2}\right) \pi \hbar. \quad (9.5.4)$$

In terms of the complete closed classical phase-space orbit integral $\oint p dx = 2 \int_{x_1}^{x_2} p dx$:

$$\boxed{\oint p dx = \left(n + \frac{1}{2}\right) 2\pi \hbar = \left(n + \frac{1}{2}\right) h, \quad n = 0, 1, 2, \dots} \quad (9.5.5)$$

Equation (9.5.5) is the modernized **Bohr–Sommerfeld quantization rule**, where the zero-point energy Maslov index $+1/2$ arises rigorously from the two $\pi/4$ phase shifts at the smooth turning points.

9.5.1 Phase Shifts for Different Boundary Walls

- **Two soft turning points:** Each contributes $\pi/4$, giving:

$$\int_{x_1}^{x_2} p dx = \left(n + \frac{1}{2}\right) \pi \hbar. \quad (9.5.6)$$

- **One soft turning point and one rigid vertical wall** (e.g., $V(x) = \infty$ for $x \leq 0$): At a hard wall, $\psi(0) = 0$, requiring a phase shift of $\pi/2$ instead of $\pi/4$:

$$\int_0^{x_2} p dx = \left(n + \frac{3}{4}\right) \pi \hbar, \quad n = 0, 1, 2, \dots \quad (9.5.7)$$

- **Two rigid vertical walls** (infinite square well): Both walls force $\psi = 0$, contributing zero phase shift to the cosine argument:

$$\int_0^L p dx = (n+1)\pi \hbar \implies p = \frac{(n+1)\pi \hbar}{L}, \quad (9.5.8)$$

reproducing the exact infinite square well energy spectrum $E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$.

9.5.2 Exact Harmonic Oscillator Spectrum from WKB

For $V(x) = \frac{1}{2}m\omega^2x^2$, the turning points are $x_{1,2} = \pm\sqrt{\frac{2E}{m\omega^2}}$. The momentum is:

$$p(x) = \sqrt{2m\left(E - \frac{1}{2}m\omega^2x^2\right)} = m\omega\sqrt{x_0^2 - x^2}, \quad x_0 = \sqrt{\frac{2E}{m\omega^2}}. \quad (9.5.9)$$

The action integral is:

$$\int_{-x_0}^{x_0} m\omega\sqrt{x_0^2 - x^2} dx = m\omega\left(\frac{\pi x_0^2}{2}\right) = \frac{\pi E}{\omega}. \quad (9.5.10)$$

Equating to $\left(n + \frac{1}{2}\right)\pi\hbar$:

$$\frac{\pi E}{\omega} = \left(n + \frac{1}{2}\right)\pi\hbar \implies \boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega} \quad (9.5.11)$$

The WKB approximation produces the *exact* energy spectrum of the quantum harmonic oscillator for all $n \geq 0$.

9.6 WKB Barrier Penetration and Quantum Tunneling

Consider an incident particle of energy E tunneling through a smooth, wide potential barrier $V(x) > E$ spanning between turning points x_1 and x_2 .

9.6.1 The Gamow Transmission Formula

Tracing the exponential decay of the WKB wavefunction (9.3.21) across the barrier from x_1 to x_2 :

$$\psi(x_2) \sim \psi(x_1) \exp\left(-\frac{1}{\hbar} \int_{x_1}^{x_2} \kappa(x') dx'\right). \quad (9.6.1)$$

The quantum transmission probability $\mathcal{T}$ is proportional to the transmitted flux ratio $|j_{\text{trans}}|/|j_{\text{inc}}| \approx |\psi(x_2)/\psi(x_1)|^2$:

$$\boxed{\mathcal{T} \approx \exp\left(-\frac{2}{\hbar} \int_{x_1}^{x_2} \sqrt{2m(V(x) - E)} dx\right)} \quad (9.6.2)$$

Equation (9.6.2) is **Gamow's tunneling formula**.

9.6.2 Gamow's Theory of Alpha Decay and the Geiger–Nuttall Law

In 1928, George Gamow applied this formula to solve the puzzle of alpha decay in heavy radioactive nuclei (such as $^{238}\text{U} \rightarrow ^{234}\text{Th} + \alpha$). While the kinetic energy of emitted α -particles varies by only a factor of 2 across isotopes ($4 \text{ MeV} \leq E_\alpha \leq 9 \text{ MeV}$), their half-lives span an astonishing **23 orders of magnitude** (10^{-7} s to 10^{10} years)!

The α -particle ($q_1 = 2e$) is trapped inside the nucleus of radius R ($Z_2 = Z - 2$) by the strong attractive nuclear well. Beyond R , the particle encounters a high repulsive Coulomb barrier:

$$V(r) = \frac{2(Z-2)e^2}{4\pi\epsilon_0 r}. \quad (9.6.3)$$

The outer turning point r_c occurs where $V(r_c) = E_\alpha$:

$$r_c = \frac{2(Z-2)e^2}{4\pi\epsilon_0 E_\alpha}. \quad (9.6.4)$$

The tunneling exponent is:

$$\gamma = \frac{2}{\hbar} \int_R^{r_c} \sqrt{2m \left(\frac{2(Z-2)e^2}{4\pi\epsilon_0 r} - E_\alpha \right)} dr. \quad (9.6.5)$$

Substituting $r = r_c \cos^2 \theta$:

$$\begin{aligned} \gamma &= \frac{2}{\hbar} \sqrt{2mE_\alpha} \int_0^{\arccos \sqrt{R/r_c}} \sqrt{\frac{1}{\cos^2 \theta} - 1} (-2r_c \sin \theta \cos \theta d\theta) \\ &= \frac{4r_c}{\hbar} \sqrt{2mE_\alpha} \int_0^{\arccos \sqrt{R/r_c}} \sin^2 \theta d\theta \\ &= \frac{2r_c}{\hbar} \sqrt{2mE_\alpha} \left[\arccos \sqrt{\frac{R}{r_c}} - \sqrt{\frac{R}{r_c}} \left(1 - \frac{R}{r_c} \right) \right]. \end{aligned} \quad (9.6.6)$$

For heavy nuclei, the barrier is wide ($r_c \gg R$), so $\arccos \sqrt{R/r_c} \approx \frac{\pi}{2} - \sqrt{R/r_c}$:

$$\gamma \approx \frac{2r_c}{\hbar} \sqrt{2mE_\alpha} \left[\frac{\pi}{2} - 2\sqrt{\frac{R}{r_c}} \right] = \frac{2\pi(Z-2)e^2}{4\pi\epsilon_0 \hbar} \sqrt{\frac{2m}{E_\alpha}} - \frac{8}{\hbar} \sqrt{\frac{m(Z-2)e^2 R}{4\pi\epsilon_0}}. \quad (9.6.7)$$

The decay rate $\lambda = 1/\tau = \nu \mathcal{T}$ (where $\nu = v/(2R) \sim 10^{21} \text{ s}^{-1}$ is the collision frequency with the barrier). Taking the natural logarithm:

$$\ln \tau_{1/2} = C_1 + C_2 \frac{Z-2}{\sqrt{E_\alpha}}. \quad (9.6.8)$$

This is the celebrated empirical **Geiger–Nuttall law**. The exponential sensitivity of WKB tunneling to $1/\sqrt{E_\alpha}$ explains why minute variations in α -particle energy produce enormous astronomical variations in nuclear lifetimes.

In George Gamow's original 1928 formulation, this decaying state was described by assigning a **complex energy eigenvalue**:

$$E = E_0 - \frac{i\hbar\lambda}{2} = E_0 - \frac{i\hbar}{2\tau}, \quad (9.6.9)$$

subject to purely outgoing wave (Siegert) boundary conditions $\psi(r) \propto e^{+ikr}$ as $r \rightarrow \infty$. Because probability continuously escapes through the Coulomb barrier into the spatial continuum, the effective Hamiltonian restricted to the nuclear interior is non-Hermitian, and the time evolution of the nuclear state is non-unitary ($|\langle \psi(0) | \psi(t) \rangle|^2 = e^{-\lambda t}$)—a spatial tunneling counterpart to the non-unitary radiative decay analyzed in Chapter 10.

Chapter 10

Time-Dependent Perturbation Theory, Transitions, and Geometric Phases

In physical systems, perturbations are rarely static. Atoms are illuminated by laser pulses, nuclei decay via sudden collisions, and quantum bits are driven by oscillating microwave fields. When the Hamiltonian depends explicitly on time:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t), \quad (10.0.1)$$

energy is no longer conserved, and stationary states do not exist. Instead, the central physical questions concern **transition probabilities** between unperturbed eigenstates, emission and absorption rates, and the phase accumulated along time-dependent quantum trajectories.

10.1 The Interaction Picture and the Dyson Series

10.1.1 Comparing the Pictures of Quantum Dynamics

Recall from Chapter 2 how dynamics can be distributed between states and operators:

1. **Schrödinger Picture:** State vectors carry the entire time evolution:

$$i\hbar \frac{\partial}{\partial t} |\psi_S(t)\rangle = \hat{H}(t) |\psi_S(t)\rangle, \quad \frac{d\hat{A}_S}{dt} = 0. \quad (10.1.1)$$

2. **Heisenberg Picture:** State vectors are frozen at $t = 0$, while operators evolve:

$$|\psi_H\rangle = |\psi_S(0)\rangle, \quad \frac{d\hat{A}_H}{dt} = \frac{i}{\hbar} [\hat{H}_H, \hat{A}_H] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H. \quad (10.1.2)$$

3. **Dirac (Interaction) Picture:** The trivial, fast dynamics generated by $\hat{H}_0$ is shifted into the operators, leaving the state vector to evolve purely under the influence of the interaction $\hat{V}(t)$:

$$|\psi_I(t)\rangle \equiv e^{i\hat{H}_0 t/\hbar} |\psi_S(t)\rangle, \quad \hat{A}_I(t) \equiv e^{i\hat{H}_0 t/\hbar} \hat{A}_S e^{-i\hat{H}_0 t/\hbar}. \quad (10.1.3)$$

10.1.2 Equation of Motion in the Interaction Picture

Differentiating the interaction-picture state vector $|\psi_I(t)\rangle$ with respect to time:

$$i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle = i\hbar \left(\frac{i\hat{H}_0}{\hbar} \right) e^{i\hat{H}_0 t/\hbar} |\psi_S(t)\rangle + e^{i\hat{H}_0 t/\hbar} \left(i\hbar \frac{\partial}{\partial t} |\psi_S(t)\rangle \right)$$

$$\begin{aligned}
&= -e^{i\hat{H}_0 t/\hbar} \hat{H}_0 |\psi_S(t)\rangle + e^{i\hat{H}_0 t/\hbar} (\hat{H}_0 + \hat{V}_S(t)) |\psi_S(t)\rangle \\
&= e^{i\hat{H}_0 t/\hbar} \hat{V}_S(t) |\psi_S(t)\rangle = e^{i\hat{H}_0 t/\hbar} \hat{V}_S(t) e^{-i\hat{H}_0 t/\hbar} |\psi_I(t)\rangle.
\end{aligned} \tag{10.1.4}$$

Defining the interaction-picture perturbation operator:

$$\hat{V}_I(t) \equiv e^{i\hat{H}_0 t/\hbar} \hat{V}_S(t) e^{-i\hat{H}_0 t/\hbar}, \tag{10.1.5}$$

we obtain the fundamental equation of motion:

$$\boxed{i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle} \tag{10.1.6}$$

If $\hat{V} = 0$, $|\psi_I(t)\rangle = |\psi_I(0)\rangle$ is strictly constant.

10.1.3 The Time Evolution Operator and the Dyson Series

We define the unitary time-evolution operator $\hat{U}_I(t, t_0)$ in the interaction picture:

$$|\psi_I(t)\rangle = \hat{U}_I(t, t_0) |\psi_I(t_0)\rangle, \quad \hat{U}_I(t_0, t_0) = \hat{I}. \tag{10.1.7}$$

Substituting into (10.1.6):

$$i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) = \hat{V}_I(t) \hat{U}_I(t, t_0). \tag{10.1.8}$$

Integrating both sides from t_0 to t converts this into a Volterra **integral equation**:

$$\hat{U}_I(t, t_0) = \hat{I} - \frac{i}{\hbar} \int_{t_0}^t dt' \hat{V}_I(t') \hat{U}_I(t', t_0). \tag{10.1.9}$$

Substituting $\hat{U}_I$ back into its own integrand generates the **Dyson series**:

$$\begin{aligned}
\hat{U}_I(t, t_0) &= \hat{I} - \frac{i}{\hbar} \int_{t_0}^t dt_1 \hat{V}_I(t_1) + \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \hat{V}_I(t_1) \hat{V}_I(t_2) + \dots \\
&= \sum_{n=0}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \dots \int_{t_0}^{t_{n-1}} dt_n \hat{V}_I(t_1) \dots \hat{V}_I(t_n).
\end{aligned} \tag{10.1.10}$$

Using the time-ordering operator $\mathcal{T}$:

$$\boxed{\hat{U}_I(t, t_0) = \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(t') dt' \right)} \tag{10.1.11}$$

10.1.4 First-Order Transition Amplitudes

Suppose the system is prepared initially at $t = 0$ in an unperturbed energy eigenstate $|i^{(0)}\rangle$ ($E_i^{(0)}$). The probability amplitude to find the system in state $|f^{(0)}\rangle$ ($E_f^{(0)} \neq E_i^{(0)}$) at time t is:

$$c_f(t) = \langle f^{(0)} | \hat{U}_I(t, 0) | i^{(0)} \rangle. \tag{10.1.12}$$

To first order in the Dyson series:

$$c_f^{(1)}(t) = -\frac{i}{\hbar} \int_0^t dt' \langle f^{(0)} | \hat{V}_I(t') | i^{(0)} \rangle. \tag{10.1.13}$$

Using $\hat{V}_I(t') = e^{i\hat{H}_0 t'/\hbar} \hat{V}_S(t') e^{-i\hat{H}_0 t'/\hbar}$,

$$\langle f^{(0)} | \hat{V}_I(t') | i^{(0)} \rangle = e^{i(E_f^{(0)} - E_i^{(0)})t'/\hbar} \langle f^{(0)} | \hat{V}_S(t') | i^{(0)} \rangle = V_{fi}(t') e^{i\omega_{fi} t'}, \quad (10.1.14)$$

where $\omega_{fi} \equiv \frac{E_f^{(0)} - E_i^{(0)}}{\hbar}$ is the **Bohr transition frequency**.

Thus:

$$c_f^{(1)}(t) = -\frac{i}{\hbar} \int_0^t dt' V_{fi}(t') e^{i\omega_{fi} t'} \quad (10.1.15)$$

The transition amplitude is proportional to the Fourier transform of the matrix element at the transition frequency!

10.2 Two-Level Systems and Exact Rabi Oscillations

When an external oscillating field drives transitions between two levels, first-order perturbation theory breaks down near resonance due to secular terms. Fortunately, a two-level system under a rotating magnetic field admits an **exact non-perturbative solution**.

10.2.1 Hamiltonian in a Rotating Magnetic Field

Consider a spin-1/2 particle with gyromagnetic ratio γ placed in a static magnetic field $\vec{B}_0 = B_0 \hat{z}$ and a perpendicular field $\vec{B}_1(t)$ rotating at angular frequency ω in the xy -plane:

$$\vec{B}(t) = B_1 \cos(\omega t) \hat{x} + B_1 \sin(\omega t) \hat{y} + B_0 \hat{z}. \quad (10.2.1)$$

The Hamiltonian is:

$$\hat{H}(t) = -\hat{\vec{\mu}} \cdot \vec{B}(t) = -\frac{\hbar}{2} [\omega_0 \sigma_z + \omega_1 (\sigma_x \cos \omega t + \sigma_y \sin \omega t)], \quad (10.2.2)$$

where $\omega_0 \equiv \gamma B_0$ is the Larmor frequency and $\omega_1 \equiv \gamma B_1$ is the coupling strength. In matrix form:

$$\hat{H}(t) = -\frac{\hbar}{2} \begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix}. \quad (10.2.3)$$

10.2.2 Transformation to the Rotating Frame

To eliminate the explicit time dependence, we perform a unitary transformation to a reference frame rotating with the magnetic field at frequency ω :

$$|\psi_R(t)\rangle \equiv \hat{U}_R(t) |\psi(t)\rangle, \quad \hat{U}_R(t) = e^{i\omega t \sigma_z/2} = \begin{pmatrix} e^{i\omega t/2} & 0 \\ 0 & e^{-i\omega t/2} \end{pmatrix}. \quad (10.2.4)$$

Differentiating $|\psi_R(t)\rangle$:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\psi_R(t)\rangle &= i\hbar \dot{\hat{U}}_R |\psi(t)\rangle + \hat{U}_R \left(i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle \right) \\ &= \left(-\frac{\hbar\omega}{2} \sigma_z + \hat{U}_R \hat{H}(t) \hat{U}_R^\dagger \right) |\psi_R(t)\rangle \equiv \hat{H}_{\text{eff}} |\psi_R(t)\rangle. \end{aligned} \quad (10.2.5)$$

Evaluating the rotated Hamiltonian:

$$\hat{U}_R \hat{H}(t) \hat{U}_R^\dagger = -\frac{\hbar}{2} \begin{pmatrix} \omega_0 & \omega_1 \\ \omega_1 & -\omega_0 \end{pmatrix}. \quad (10.2.6)$$

Combining terms, the effective Hamiltonian in the rotating frame is **strictly time-independent**:

$$\hat{H}_{\text{eff}} = -\frac{\hbar}{2} \begin{pmatrix} \omega_0 - \omega & \omega_1 \\ \omega_1 & -(\omega_0 - \omega) \end{pmatrix} = -\frac{\hbar}{2} [\Delta \sigma_z + \omega_1 \sigma_x], \quad (10.2.7)$$

where $\Delta \equiv \omega_0 - \omega$ is the **detuning** from resonance.

10.2.3 Exact Transition Probability and Rabi's Formula

Because $\hat{H}_{\text{eff}}$ is time-independent, the time evolution operator in the rotating frame is simply the matrix exponential:

$$\hat{U}_R(t, 0) = \exp\left(-\frac{i}{\hbar} \hat{H}_{\text{eff}} t\right) = \exp\left(\frac{it}{2} [\Delta\sigma_z + \omega_1\sigma_x]\right). \quad (10.2.8)$$

Defining the **generalized Rabi frequency**:

$$\Omega_R \equiv \sqrt{\Delta^2 + \omega_1^2} = \sqrt{(\omega_0 - \omega)^2 + \omega_1^2}, \quad (10.2.9)$$

and using the Pauli matrix identity $e^{i\theta(\hat{n} \cdot \vec{\sigma})} = I \cos \theta + i(\hat{n} \cdot \vec{\sigma}) \sin \theta$:

$$\hat{U}_R(t, 0) = \cos\left(\frac{\Omega_R t}{2}\right) \hat{I} + i \frac{\Delta\sigma_z + \omega_1\sigma_x}{\Omega_R} \sin\left(\frac{\Omega_R t}{2}\right). \quad (10.2.10)$$

If the system begins in the spin-up state $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, the amplitude to find it in the spin-down state $|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ at time t is:

$$\langle \downarrow | \psi_R(t) \rangle = \langle \downarrow | \hat{U}_R(t, 0) | \uparrow \rangle = i \frac{\omega_1}{\Omega_R} \sin\left(\frac{\Omega_R t}{2}\right). \quad (10.2.11)$$

The transition probability is the modulus squared:

$$P_{\uparrow \rightarrow \downarrow}(t) = \frac{\omega_1^2}{(\omega_0 - \omega)^2 + \omega_1^2} \sin^2\left(\frac{\sqrt{(\omega_0 - \omega)^2 + \omega_1^2} t}{2}\right) \quad (10.2.12)$$

Equation (10.2.12) is the exact **Rabi formula**.

- **On Resonance** ($\omega = \omega_0$, $\Delta = 0$): $P_{\uparrow \rightarrow \downarrow}(t) = \sin^2\left(\frac{\omega_1 t}{2}\right)$. At time $t = \pi/\omega_1$ (a “ π -pulse”), the spin completely flips from $|\uparrow\rangle$ to $|\downarrow\rangle$ ($P = 1$). At $t = \pi/(2\omega_1)$ (a “ $\pi/2$ -pulse”), the state becomes an equal coherent superposition $\frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle)$.
- **Off Resonance** ($\Delta \gg \omega_1$): The oscillations become faster ($\Omega_R \approx |\Delta|$), but the maximum transition probability drops sharply as $\frac{\omega_1^2}{\Delta^2} \ll 1$.

10.3 Constant and Harmonic Perturbations: Fermi's Golden Rule

10.3.1 Constant Perturbation Switched on at $t = 0$

Suppose a static potential $\hat{V}$ is abruptly turned on at $t = 0$:

$$\hat{V}(t) = \begin{cases} 0, & t < 0, \\ \hat{V}, & t \geq 0. \end{cases} \quad (10.3.1)$$

The first-order transition amplitude to state $|f\rangle \neq |i\rangle$ is:

$$c_f^{(1)}(t) = -\frac{i}{\hbar} V_{fi} \int_0^t dt' e^{i\omega_{fi}t'} = -\frac{V_{fi}}{\hbar\omega_{fi}} (e^{i\omega_{fi}t} - 1) = -\frac{2iV_{fi}}{\hbar\omega_{fi}} e^{i\omega_{fi}t/2} \sin\left(\frac{\omega_{fi}t}{2}\right). \quad (10.3.2)$$

The transition probability is:

$$P_{i \rightarrow f}(t) = |c_f^{(1)}(t)|^2 = \frac{4|V_{fi}|^2}{\hbar^2} \frac{\sin^2\left(\frac{\omega_{fi}t}{2}\right)}{\omega_{fi}^2}. \quad (10.3.3)$$

- **Short times** ($t \ll 1/\omega_{fi}$): $\sin(\omega_{fi}t/2) \approx \omega_{fi}t/2 \implies P(t) \approx \frac{|V_{fi}|^2}{\hbar^2} t^2$. The probability grows quadratically.
- **Long times** ($t \rightarrow \infty$): The diffraction sinc function narrows into a delta function:

$$\lim_{t \rightarrow \infty} \frac{\sin^2(\omega_{fi}t/2)}{\omega_{fi}^2 t} = \frac{\pi}{2} \delta(\omega_{fi}) = \frac{\pi \hbar}{2} \delta(E_f - E_i). \quad (10.3.4)$$

10.3.2 Harmonic Perturbations

For a monochromatic harmonic potential:

$$\hat{V}(t) = \hat{V} e^{-i\omega t} + \hat{V}^\dagger e^{i\omega t}, \quad (t \geq 0). \quad (10.3.5)$$

The first-order transition amplitude is:

$$c_f^{(1)}(t) = -\frac{1}{\hbar} \left[\langle f | \hat{V} | i \rangle \frac{e^{i(\omega_{fi}-\omega)t} - 1}{\omega_{fi} - \omega} + \langle f | \hat{V}^\dagger | i \rangle \frac{e^{i(\omega_{fi}+\omega)t} - 1}{\omega_{fi} + \omega} \right]. \quad (10.3.6)$$

The two terms correspond to **absorption** ($E_f \approx E_i + \hbar\omega$) and **stimulated emission** ($E_f \approx E_i - \hbar\omega$).

10.3.3 Continuum Transitions and Fermi's Golden Rule

When the final states form a continuum with density of states $\rho(E_f)$, integrating over final energies yields:

$$P_{\text{tot}}(t) = \int dE_f \rho(E_f) P_{i \rightarrow f}(t) \xrightarrow{t \rightarrow \infty} \frac{2\pi}{\hbar} |\langle f | \hat{V} | i \rangle|^2 \rho(E_f = E_i \pm \hbar\omega) t. \quad (10.3.7)$$

The constant **transition rate** $W_{i \rightarrow f} \equiv \frac{dP_{\text{tot}}}{dt}$ is:

$$W_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f | \hat{V} | i \rangle|^2 \rho(E_f) \quad (10.3.8)$$

Equation (10.3.8) is **Fermi's Golden Rule**.

10.3.4 Interaction of Radiation with Matter and E1 Transitions

In the electric dipole approximation ($e^{i\vec{k} \cdot \vec{r}} \approx 1$):

$$\langle f | \hat{V} | i \rangle = \frac{eA_0}{m} \hat{\epsilon} \cdot \langle f | \hat{p} | i \rangle = -i\omega_{fi} A_0 \hat{\epsilon} \cdot \langle f | \hat{d} | i \rangle, \quad (10.3.9)$$

where $\hat{d} = -e\vec{r}$ is the electric dipole operator. The **Einstein A coefficient** for spontaneous emission is:

$$A_{i \rightarrow f} = \frac{4\alpha\omega_{if}^3}{3c^2} |\langle f | \vec{r} | i \rangle|^2 \quad (10.3.10)$$

with the dipole selection rules $\Delta l = \pm 1$ and $\Delta m = 0, \pm 1$.

Electrodynamic Foundation: The Classical Lorentz Oscillator Model vs. Quantum Spontaneous Emission

Why does the spontaneous emission rate in Eq. (10.3.10) scale as ω_{if}^3 and the square of the transition dipole matrix element? The deep physical origin becomes transparent when viewed through classical electrodynamics: **classically, spontaneous emission is simply radiation reaction damping from an oscillating dipole.**

1. The Classical Lorentz Oscillator Model: In H. A. Lorentz's classical model of light-matter interaction, an optical electron of mass m_e and charge $-e$ is bound elastically to a massive nucleus with natural resonance frequency ω_0 . When subjected to an external driving electric field $\vec{E}(t) = \vec{E}_0 e^{-i\omega t}$, Newton's equation of motion reads:

$$m_e (\ddot{\vec{x}} + \gamma \dot{\vec{x}} + \omega_0^2 \vec{x}) = -e\vec{E}(t), \quad (1)$$

where γ is a damping coefficient. In an isolated atom (with no collisions or mechanical friction), what causes damping? **Electromagnetic radiation reaction!** An accelerating charge continuously radiates electromagnetic energy into space. According to the classical **Larmor formula**, the instantaneous radiated power is:

$$P(t) = \frac{e^2 \|\ddot{\vec{x}}\|^2}{6\pi\epsilon_0 c^3}. \quad (2)$$

This continuous loss of kinetic energy generates an effective self-force known as the **Abraham-Lorentz radiation reaction force**:

$$\vec{F}_{\text{rad}} = \frac{e^2}{6\pi\epsilon_0 c^3} \ddot{\vec{x}}. \quad (3)$$

For an electron undergoing quasi-harmonic oscillation at frequency ω_0 , $\vec{x}(t) \approx \vec{x}_0 e^{-i\omega_0 t}$, so $\ddot{\vec{x}} = -\omega_0^2 \vec{x}$. The radiation reaction force thus takes the exact form of a viscous damping force:

$$\vec{F}_{\text{rad}} \approx - \left(\frac{e^2 \omega_0^2}{6\pi\epsilon_0 c^3} \right) \dot{\vec{x}} \equiv -m_e \gamma_{\text{rad}} \dot{\vec{x}}, \quad (4)$$

where the **classical radiative damping rate** is:

$$\gamma_{\text{rad}} = \frac{e^2 \omega_0^2}{6\pi\epsilon_0 m_e c^3}. \quad (5)$$

2. Radiated Power of an Oscillating Dipole: The accelerating electron constitutes a time-varying electric dipole $\vec{p}(t) = -e\vec{x}(t) = \vec{p}_0 \cos(\omega_0 t)$. The cycle-averaged power radiated by an oscillating dipole in free space is:

$$\langle P_{\text{rad}} \rangle = \frac{\omega_0^4 \|\vec{p}_0\|^2}{12\pi\epsilon_0 c^3}. \quad (6)$$

Classically, an excited atom radiates a continuous electromagnetic wave at frequency ω_0 , continuously diminishing its mechanical oscillation energy at rate $dE_{\text{mech}}/dt = -\langle P_{\text{rad}} \rangle = -\gamma_{\text{rad}} E_{\text{mech}}$.

3. Quantum Correspondence and the Einstein A Coefficient: In quantum mechanics, energy cannot be drained continuously; it must be emitted in discrete lumps (photons) of energy $\hbar\omega_{if} = E_i - E_f$. What corresponds to the classical oscillating dipole $\vec{p}_0$?

Consider an atom prepared in a coherent superposition of initial state $|i\rangle$ and final state $|f\rangle$:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{-iE_it/\hbar} |i\rangle + e^{-iE_ft/\hbar} |f\rangle \right). \quad (7)$$

The expectation value of the quantum electric dipole operator $\hat{\vec{d}} = -e\vec{r}$ in this superposition state is:

$$\left\langle \hat{\vec{d}} \right\rangle (t) = \langle \psi(t) | (-e\vec{r}) | \psi(t) \rangle = \text{Re} \left[\langle f | (-e\vec{r}) | i \rangle e^{-i\omega_{if}t} \right] = 2 |\langle f | (-e\vec{r}) | i \rangle| \cos(\omega_{if}t + \phi). \quad (8)$$

Remarkably, the quantum atom in a transition state is *literally an oscillating electric dipole* with amplitude:

$$\|\vec{p}_0\| = 2e |\langle f | \vec{r} | i \rangle|. \quad (9)$$

Substituting this dipole amplitude into the classical Larmor radiated power formula (6) yields:

$$\langle P_{\text{rad}} \rangle = \frac{\omega_{if}^4 (4e^2 |\langle f | \vec{r} | i \rangle|^2)}{12\pi\epsilon_0 c^3} = \frac{e^2 \omega_{if}^4}{3\pi\epsilon_0 c^3} |\langle f | \vec{r} | i \rangle|^2. \quad (10)$$

Because each transition emits exactly one photon of energy $\hbar\omega_{if}$, the quantum transition rate $A_{i \rightarrow f}$ (number of photons radiated per unit time) is simply the classical radiated power divided by the quantum energy of a single photon:

$$A_{i \rightarrow f} \equiv \frac{\langle P_{\text{rad}} \rangle}{\hbar\omega_{if}} = \frac{e^2 \omega_{if}^3}{3\pi\epsilon_0 \hbar c^3} |\langle f | \vec{r} | i \rangle|^2 = \frac{4\alpha \omega_{if}^3}{3c^2} |\langle f | \vec{r} | i \rangle|^2, \quad (11)$$

where we used the definition of the fine-structure constant $\alpha \equiv \frac{e^2}{4\pi\epsilon_0 \hbar c}$.

Conclusion: Quantum spontaneous emission is the quantum-electrodynamic realization of classical Larmor radiation from an oscillating dipole. In absorption, the same Lorentz oscillator produces a Lorentzian resonance cross section $\sigma(\omega)$, whose integrated frequency strength reproduces the quantum **oscillator strength** $f_{if} = \frac{2m_e \omega_{if}}{3\hbar} |\langle f | \vec{r} | i \rangle|^2$ and obeys the fundamental **Thomas–Reiche–Kuhn (TRK) sum rule** $\sum_f f_{if} = 1$.

10.3.5 Decaying States, Complex Energies, and Non-Unitary Time Evolution

Rather than postulating a decaying state phenomenologically, we can **derive it directly from first principles** by writing down the microscopic Hamiltonian of the coupled atom–field system and solving the time-dependent Schrödinger equation (the celebrated **Weisskopf–Wigner theory of spontaneous emission**).

The Coupled Atom–Continuum Hamiltonian

Consider an atom with an excited state $|i\rangle$ of energy E_i and a ground state $|g\rangle$ of energy $E_g = 0$, coupled to a continuous reservoir of environmental electromagnetic modes $\{|k\rangle\}$ with energies

$E_k = \hbar\omega_k$. In the single-excitation subspace of the universe, the state $|k\rangle \equiv |g; 1_k\rangle$ represents the atom in the ground state with one emitted photon in mode k , while $|i\rangle \equiv |e; 0\rangle$ represents the excited atom in the vacuum.

The unperturbed Hamiltonian is:

$$\hat{H}_0 = E_i |i\rangle \langle i| + \sum_k E_k |k\rangle \langle k|. \quad (10.3.11)$$

The interaction Hamiltonian (dipole coupling) mediates transitions between $|i\rangle$ and the continuum modes $\{|k\rangle\}$:

$$\hat{V} = \sum_k (V_k |i\rangle \langle k| + V_k^* |k\rangle \langle i|), \quad V_k \equiv \langle i | \hat{V} | k \rangle. \quad (10.3.12)$$

The total coupled Hamiltonian is:

$$\boxed{\hat{H} = E_i |i\rangle \langle i| + \sum_k E_k |k\rangle \langle k| + \sum_k (V_k |i\rangle \langle k| + V_k^* |k\rangle \langle i|)} \quad (10.3.13)$$

Notice a critical fact: in the full composite Hilbert space $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{atom}} \otimes \mathcal{H}_{\text{field}}$, the total Hamiltonian $\hat{H}$ is strictly Hermitian ($\hat{H}^\dagger = \hat{H}$).

Solving the Coupled Equations of Motion: Derivation of the Decaying State

Let the total state vector in the single-excitation subspace evolve as:

$$|\Psi(t)\rangle = c_i(t) e^{-iE_i t/\hbar} |i\rangle + \sum_k c_k(t) e^{-iE_k t/\hbar} |k\rangle, \quad (10.3.14)$$

subject to the initial condition that the atom is initially in state $|i\rangle$ and the radiation field is in the vacuum state:

$$c_i(0) = 1, \quad c_k(0) = 0 \quad (\forall k). \quad (10.3.15)$$

Substituting $|\Psi(t)\rangle$ into the time-dependent Schrödinger equation $i\hbar \partial |\Psi\rangle / \partial t = \hat{H} |\Psi\rangle$, the unperturbed diagonal energy contributions cancel, leaving:

$$i\hbar \dot{c}_i(t) e^{-iE_i t/\hbar} |i\rangle + i\hbar \sum_k \dot{c}_k(t) e^{-iE_k t/\hbar} |k\rangle = \hat{V} |\Psi(t)\rangle. \quad (10.3.16)$$

Projecting onto $\langle i|$ and $\langle k|$ yields the coupled system of equations of motion:

$$i\hbar \dot{c}_i(t) = \sum_k V_k c_k(t) e^{-i(E_k - E_i)t/\hbar}, \quad (10.3.17)$$

$$i\hbar \dot{c}_k(t) = V_k^* c_i(t) e^{i(E_k - E_i)t/\hbar}. \quad (10.3.18)$$

Integrating Eq. (10.3.18) with the initial condition $c_k(0) = 0$:

$$c_k(t) = -\frac{i}{\hbar} V_k^* \int_0^t dt' c_i(t') e^{i(E_k - E_i)t'/\hbar}. \quad (10.3.19)$$

Substituting this expression for $c_k(t)$ directly back into Eq. (10.3.17) yields an exact integro-differential equation with memory kernel for the excited state amplitude:

$$\dot{c}_i(t) = -\frac{1}{\hbar^2} \sum_k |V_k|^2 \int_0^t dt' c_i(t') e^{-i(E_k - E_i)(t-t')/\hbar}. \quad (10.3.20)$$

The Continuum Limit and Markov Approximation: In the continuum limit (large quantization volume), the discrete modes merge into a continuous density of states $\rho(E)$:

$$\sum_k |V_k|^2 \cdots \longrightarrow \int_0^\infty dE \rho(E) |V(E)|^2 \cdots \quad (10.3.21)$$

Introducing the elapsed time $\tau \equiv t - t'$:

$$\dot{c}_i(t) = -\frac{1}{\hbar^2} \int_0^\infty dE \rho(E) |V(E)|^2 \int_0^t d\tau c_i(t - \tau) e^{-i(E - E_i)\tau/\hbar}. \quad (10.3.22)$$

Because the continuum coupling $\rho(E) |V(E)|^2$ is a smooth, broadband function with bandwidth $\Delta E_{\text{bath}} \gg \hbar\Gamma$, the phase factor $e^{-i(E - E_i)\tau/\hbar}$ dephases over a microscopic bath memory time:

$$\tau_{\text{bath}} \sim \frac{\hbar}{\Delta E_{\text{bath}}} \ll \frac{1}{\Gamma}. \quad (10.3.23)$$

Because the memory time of the bath is negligible compared to the radiative lifetime $\tau = 1/\Gamma$, the atomic amplitude $c_i(t - \tau)$ remains essentially constant over the non-zero range of the kernel. This establishes the **Markov approximation** (or **Weisskopf–Wigner approximation**):

$$c_i(t - \tau) \approx c_i(t). \quad (10.3.24)$$

For times $t \gg \tau_{\text{bath}}$, the upper limit of the τ -integral can be extended to $+\infty$:

$$\int_0^\infty d\tau e^{-i(E - E_i)\tau/\hbar} = \lim_{\epsilon \rightarrow 0^+} \int_0^\infty d\tau e^{-i(E - E_i - i\epsilon)\tau/\hbar} = \lim_{\epsilon \rightarrow 0^+} \frac{-i\hbar}{E - E_i - i\epsilon}. \quad (10.3.25)$$

Applying the **Sokhotski–Plemelj theorem**:

$$\lim_{\epsilon \rightarrow 0^+} \frac{1}{E - E_i - i\epsilon} = \mathcal{P} \left(\frac{1}{E - E_i} \right) + i\pi\delta(E - E_i), \quad (10.3.26)$$

we evaluate the integral:

$$\int_0^\infty d\tau e^{-i(E - E_i)\tau/\hbar} = \pi\hbar\delta(E - E_i) - i\hbar\mathcal{P} \left(\frac{1}{E - E_i} \right), \quad (10.3.27)$$

where $\mathcal{P}$ denotes the Cauchy principal value.

Analytical Solution and Emergence of Complex Energy: Substituting this result into our differential equation for $\dot{c}_i(t)$:

$$\dot{c}_i(t) = -\frac{1}{\hbar^2} c_i(t) \int_0^\infty dE \rho(E) |V(E)|^2 \left[\pi\hbar\delta(E - E_i) - i\hbar\mathcal{P} \left(\frac{1}{E - E_i} \right) \right]. \quad (10.3.28)$$

This simplifies directly into the linear differential equation:

$$\boxed{\frac{dc_i(t)}{dt} = - \left(\frac{\Gamma}{2} + \frac{i}{\hbar} \Delta E_{\text{shift}} \right) c_i(t)} \quad (10.3.29)$$

where:

$$\boxed{\Gamma \equiv \frac{2\pi}{\hbar} \rho(E_i) |V(E_i)|^2} \quad (\text{Fermi's Golden Rule emerges dynamically!}), \quad (10.3.30)$$

and the imaginary pole contribution generates the radiative level shift (**Lamb shift**):

$$\boxed{\Delta E_{\text{shift}} \equiv \mathcal{P} \int_0^\infty dE \frac{\rho(E) |V(E)|^2}{E_i - E}} \quad (10.3.31)$$

Integrating Eq. (10.3.29) with initial condition $c_i(0) = 1$:

$$c_i(t) = \exp \left[-\frac{i}{\hbar} \Delta E_{\text{shift}} t - \frac{\Gamma}{2} t \right] \quad (10.3.32)$$

The state vector in the atomic subspace evolves as:

$$|\psi_{\text{atom}}(t)\rangle = c_i(t) e^{-iE_i t/\hbar} |i\rangle = \exp \left[-\frac{i}{\hbar} \left(E_i + \Delta E_{\text{shift}} - \frac{i\hbar\Gamma}{2} \right) t \right] |i\rangle. \quad (10.3.33)$$

Defining the shifted physical resonance energy $E'_i \equiv E_i + \Delta E_{\text{shift}}$, we have analytically derived the **complex energy**:

$$E_{\text{eff}} = E'_i - \frac{i\hbar\Gamma}{2} = E'_i - \frac{i\hbar}{2\tau} \quad (10.3.34)$$

and the survival probability:

$$P_i(t) = |c_i(t)|^2 = e^{-\Gamma t} = e^{-t/\tau} \quad (10.3.35)$$

The exponential decay and complex energy are not ad-hoc phenomenological postulates; **they are the rigorous mathematical solution of the coupled atom–field Hamiltonian!**

The Non-Hermitian Effective Hamiltonian and Projection onto the Atom

On the atomic subspace spanned by $|i\rangle$, the derived equation of motion (10.3.29) can be written as:

$$i\hbar \frac{\partial |\psi_{\text{atom}}(t)\rangle}{\partial t} = \hat{H}_{\text{eff}} |\psi_{\text{atom}}(t)\rangle, \quad \boxed{\hat{H}_{\text{eff}} = E'_i - \frac{i}{2}\hbar\Gamma} \quad (10.3.36)$$

Taking the Hermitian adjoint:

$$\hat{H}_{\text{eff}}^\dagger = E'_i + \frac{i}{2}\hbar\Gamma \neq \hat{H}_{\text{eff}}. \quad (10.3.37)$$

Because $\Gamma > 0$, **the effective Hamiltonian is explicitly non-Hermitian!**

Why the Hamiltonian is Not Supposed to Be Unitary: Treating Radiation Classically

In standard closed-system quantum mechanics, the time evolution operator $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ generated by a Hermitian Hamiltonian $\hat{H}$ is strictly **unitary**:

$$\hat{U}^\dagger(t)\hat{U}(t) = e^{i\hat{H}^\dagger t/\hbar} e^{-i\hat{H}t/\hbar} = e^{i\hat{H}t/\hbar} e^{-i\hat{H}t/\hbar} = \hat{I}, \quad (10.3.38)$$

which guarantees the strict conservation of total probability:

$$\frac{d}{dt} \langle \psi(t) | \psi(t) \rangle = 0 \implies \langle \psi(t) | \psi(t) \rangle = 1 \quad \forall t. \quad (10.3.39)$$

For our decaying state, however, the effective time evolution operator is:

$$\hat{U}_{\text{eff}}(t) = e^{-i\hat{H}_{\text{eff}}t/\hbar} = e^{-i\hat{H}_0t/\hbar} e^{-\hat{\Gamma}t/2}. \quad (10.3.40)$$

Evaluating the product $\hat{U}_{\text{eff}}^\dagger \hat{U}_{\text{eff}}$:

$$\hat{U}_{\text{eff}}^\dagger(t) \hat{U}_{\text{eff}}(t) = e^{-\hat{\Gamma}t/2} e^{i\hat{H}_0t/\hbar} e^{-i\hat{H}_0t/\hbar} e^{-\hat{\Gamma}t/2} = e^{-\hat{\Gamma}t} \neq \hat{I}. \quad (10.3.41)$$

The time evolution operator is **strictly non-unitary**, and the norm of the state vector decays continuously:

$$\frac{d}{dt} \langle \psi(t) | \psi(t) \rangle = -\langle \psi(t) | \hat{\Gamma} | \psi(t) \rangle < 0. \quad (10.3.42)$$

Conceptual Foundation: Treating Radiation Classically and the Origin of Non-Unitary Evolution

Students often experience cognitive dissonance upon first encountering decaying states: “We were taught that the Hamiltonian is always Hermitian and time evolution is always unitary. How can $\hat{H}_{\text{eff}}$ have an imaginary part, and how can probability disappear?”

The answer lies in recognizing the physical approximation being made: **we are treating radiation classically (or semiclassically).**

1. **Classical Radiation as an Irreversible Sink:** In classical electrodynamics (such as the Lorentz oscillator model with Abraham–Lorentz radiation reaction $\vec{F}_{\text{rad}} = -m_e \gamma_{\text{rad}} \dot{\vec{x}}$), accelerating charges emit electromagnetic waves that carry energy and momentum away to spatial infinity ($r \rightarrow \infty$). That radiated energy escapes and never returns. Classical radiation is an intrinsically dissipative, non-conservative process.
2. **The Reduced Atomic Hilbert Space:** When we describe the atom decaying without explicitly tracking every quantum state of the emitted electromagnetic field, we restrict our Hilbert space solely to the atomic states $\mathcal{H}_{\text{atom}}$. Because energy and probability leak out of the atom into the outgoing classical radiation field, **the Hamiltonian on this reduced space is not supposed to be unitary!** If the effective Hamiltonian were unitary, probability in the atomic subspace would be rigidly conserved ($\langle \psi(t) | \psi(t) \rangle = 1$), which would physically prohibit the excited state from ever decaying.
3. **The Full Quantum (QED) Resolution:** In full Quantum Electrodynamics (QED), the radiation field is quantized into photon Fock states. The total Hilbert space is $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{atom}} \otimes \mathcal{H}_{\text{field}}$, and the total Hamiltonian:

$$\hat{H}_{\text{total}} = \hat{H}_{\text{atom}} + \hat{H}_{\text{field}} + \hat{H}_{\text{int}}$$

is strictly **Hermitian** ($\hat{H}_{\text{total}}^\dagger = \hat{H}_{\text{total}}$), generating strictly **unitary** time evolution. The state vector evolves as a coherent entangled superposition:

$$|\Psi(t)\rangle = c_i(t) |i; 0_{\text{photon}}\rangle + \sum_{\vec{k}, \lambda} c_{f, \vec{k}, \lambda}(t) |f; 1_{\vec{k}, \lambda}\rangle.$$

Total probability is 100% conserved at all times: $|c_i(t)|^2 + \sum_{\vec{k}, \lambda} |c_{f, \vec{k}, \lambda}(t)|^2 = 1$. The non-unitary decaying state $c_i(t) \sim e^{-\Gamma t/2}$ is obtained precisely by projecting $|\Psi(t)\rangle$ onto the no-photon atomic subspace $\mathcal{H}_{\text{atom}}$.

Takeaway: Non-unitary evolution and complex energies are not failures of quantum mechanics; they are the exact, mathematically consistent consequence of projecting out the radiative degrees of freedom and treating radiation classically as an open dissipation channel.

Natural Linewidth and the Breit–Wigner (Lorentzian) Profile

What is the physical spectrum of radiation emitted by a decaying state? Because the amplitude decays exponentially for $t \geq 0$:

$$\psi_i(t) = \Theta(t) e^{-i(E_i - i\hbar\Gamma/2)t/\hbar}, \quad (10.3.43)$$

the state is not monochromatic. Its energy distribution is obtained by taking the Fourier transform into the frequency/energy domain:

$$\begin{aligned}\phi(E) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi_i(t) e^{iEt/\hbar} dt \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int_0^{\infty} \exp \left[\frac{i}{\hbar} \left(E - E_i + \frac{i\hbar\Gamma}{2} \right) t \right] dt \\ &= \frac{1}{\sqrt{2\pi\hbar}} \left[\frac{\exp \left[\frac{i}{\hbar} \left(E - E_i + \frac{i\hbar\Gamma}{2} \right) t \right]}{\frac{i}{\hbar} \left(E - E_i + \frac{i\hbar\Gamma}{2} \right)} \right]_0^{\infty}.\end{aligned}\quad (10.3.44)$$

Because $\Gamma > 0$, the upper limit at $t \rightarrow \infty$ vanishes exponentially due to the damping factor $e^{-\Gamma t/2} \rightarrow 0$. Evaluating at $t = 0$:

$$\phi(E) = \frac{i\sqrt{\hbar}}{\sqrt{2\pi}} \frac{1}{(E - E_i) + \frac{i\hbar\Gamma}{2}}. \quad (10.3.45)$$

The spectral intensity (the probability distribution of finding an emitted photon with energy $E = \hbar\omega$) is given by $I(E) = |\phi(E)|^2$:

$$I(E) = \frac{\hbar}{2\pi} \frac{1}{(E - E_i)^2 + \left(\frac{\hbar\Gamma}{2}\right)^2} \quad (10.3.46)$$

Normalizing so that $\int_{-\infty}^{\infty} \mathcal{P}(E) dE = 1$, we obtain the celebrated **Breit–Wigner (or Lorentzian) distribution**:

$$\mathcal{P}(E) = \frac{1}{\pi} \frac{\frac{\hbar\Gamma}{2}}{(E - E_i)^2 + \left(\frac{\hbar\Gamma}{2}\right)^2}. \quad (10.3.47)$$

In terms of angular frequency $\omega = E/\hbar$ and resonance frequency $\omega_0 = E_i/\hbar$:

$$\mathcal{P}(\omega) = \frac{1}{\pi} \frac{\Gamma/2}{(\omega - \omega_0)^2 + (\Gamma/2)^2}. \quad (10.3.48)$$

This is precisely the **natural lineshape** observed in high-resolution atomic emission spectroscopy!

Full Width at Half Maximum (FWHM) and the Energy–Time Uncertainty Relation:

At the center of the resonance ($E = E_i$), the intensity is maximal: $I(E_i) \propto \frac{1}{(\hbar\Gamma/2)^2}$. The intensity drops to half its peak value ($I(E) = \frac{1}{2}I(E_i)$) when:

$$(E - E_i)^2 = \left(\frac{\hbar\Gamma}{2}\right)^2 \implies E - E_i = \pm \frac{\hbar\Gamma}{2}. \quad (10.3.49)$$

The total **Full Width at Half Maximum (natural linewidth)** is:

$$\Delta E = \hbar\Gamma = \frac{\hbar}{\tau} \iff \Delta\omega = \Gamma = \frac{1}{\tau} \quad (10.3.50)$$

Multiplying the energy width ΔE by the state lifetime $\tau \equiv \Delta t$:

$$\Delta E \cdot \Delta t = \hbar \quad (10.3.51)$$

The Fundamental Mechanism of Line Broadening: Environmental Perturbations and the Uncertainty Principle

Why is an atomic spectral line broadened into a continuous frequency distribution instead of radiating an infinitely sharp Dirac delta spike $\delta(E - E_0)$? The answer reveals one of the most profound physical insights connecting open quantum systems to the uncertainty principle.

1. The Isolated Atom: Exact Energy Eigenstates: Consider an idealized atom completely isolated from the rest of the universe, governed solely by the bare atomic Hamiltonian $\hat{H}_0$. If the atom occupies an unperturbed state $|n\rangle$, it is an **exact eigenstate** of the Hamiltonian:

$$\hat{H}_0 |n\rangle = E_n |n\rangle. \quad (10.3.52)$$

In this isolated state, the quantum variance of energy is strictly zero:

$$\Delta E = \sqrt{\langle n | \hat{H}_0^2 | n \rangle - \langle n | \hat{H}_0 | n \rangle^2} = \sqrt{E_n^2 - E_n^2} = 0. \quad (10.3.53)$$

The state undergoes trivial, purely stationary phase evolution $|n(t)\rangle = e^{-iE_n t/\hbar} |n\rangle$. Because no perturbations exist to cause transitions, the state lives forever: its lifetime is infinite ($\tau \rightarrow \infty$). Any radiative transition between two such strictly isolated stationary states would produce an infinitely sharp, zero-width spectral line:

$$I(E) \propto \delta(E - (E_i - E_f)). \quad (10.3.54)$$

2. The Real World: Environmental Coupling as a Perturbation: In reality, no physical atom can ever be completely isolated from its environment. The atom is unavoidably coupled to environmental degrees of freedom through an interaction Hamiltonian:

$$\hat{H}_{\text{total}} = \hat{H}_0 + \hat{V}_{\text{env}}. \quad (10.3.55)$$

Depending on the physical experimental context, the environmental perturbation $\hat{V}_{\text{env}}$ arises from:

- **Quantum Vacuum Fluctuations (Natural Broadening):** Even at zero temperature in complete darkness, the atom is coupled to the infinite continuum of vacuum electromagnetic field modes $\hat{\vec{E}}_{\text{vac}}$. The dipole interaction $\hat{V}_{\text{env}} = -\hat{\vec{d}} \cdot \hat{\vec{E}}_{\text{vac}}$ induces spontaneous emission.
- **Atomic Collisions (Pressure / Collisional Broadening):** In any gas, surrounding atoms collide with or pass near the radiating atom, creating time-dependent electrostatic perturbations $\hat{V}_{\text{coll}}(t)$ that disrupt the phase of the electron oscillation.
- **Thermal Photons and Stray Fields (Thermal / Stark Broadening):** Ambient blackbody radiation or fluctuating stray electric and magnetic fields dynamically perturb atomic levels.

3. The Core Realization: The State is NO LONGER an Eigenstate of the Hamiltonian: Because the environment introduces the interaction term $\hat{V}_{\text{env}}$, the atomic state $|n\rangle$ is **no longer an eigenstate of the total Hamiltonian**:

$$\boxed{\hat{H}_{\text{total}} |n\rangle = (\hat{H}_0 + \hat{V}_{\text{env}}) |n\rangle \neq E |n\rangle} \quad (10.3.56)$$

Because $\hat{H}_0$ and $\hat{V}_{\text{env}}$ do not commute ($[\hat{H}_0, \hat{V}_{\text{env}}] \neq 0$), the state $|n\rangle$ is *not* a stationary state of the full physical system! When expanded in the true, continuous energy eigenbasis $\{|\Phi(E)\rangle\}$ of the coupled system (atom + environment), $|n\rangle$ is a continuous wavepacket superposing an entire band of true energy eigenstates:

$$|n\rangle = \int dE c(E) |\Phi(E)\rangle. \quad (10.3.57)$$

4. The Direct Consequence of the Uncertainty Principle: Because $|n\rangle$ is no longer an eigenstate of the Hamiltonian, its energy is fundamentally **not sharply defined!** The quantum variance of the total Hamiltonian in the state $|n\rangle$ is strictly non-zero:

$$\Delta E \equiv \sqrt{\langle n|\hat{H}_{\text{total}}^2|n\rangle - \langle n|\hat{H}_{\text{total}}|n\rangle^2} > 0. \quad (10.3.58)$$

How large is this energy spread ΔE ? This is precisely what the **Mandelstam–Tamm energy–time uncertainty principle** dictates! As established in Chapter 2:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}, \quad \text{where} \quad \Delta t \equiv \frac{\Delta A}{|d\langle A\rangle/dt|}. \quad (10.3.59)$$

Choosing the projection operator onto the initial atomic state $\hat{P}_n = |n\rangle\langle n|$ as our dynamical observable, the characteristic time required for the state to undergo significant change or decay into the environment is precisely its physical lifetime:

$$\Delta t \sim \tau = \frac{1}{\Gamma}. \quad (10.3.60)$$

Therefore, the uncertainty principle demands that the state must possess an intrinsic, irreducible energy spread:

$$\Delta E \sim \frac{\hbar}{\Delta t} = \frac{\hbar}{\tau} = \hbar\Gamma \quad (10.3.61)$$

5. Physical Manifestation as Spectral Line Broadening: This intrinsic energy spread ΔE is not an observational flaw or detector imperfection; **it is the exact physical origin of spectral line broadening!** Every distinct broadening mechanism simply represents a different environmental interaction that curtails the coherent lifetime τ of the quantum state:

1. **Natural Linewidth** ($\Delta E_{\text{nat}} = \hbar/\tau_{\text{spont}}$): Caused by coupling to the quantum electromagnetic vacuum. For typical optical dipole transitions, $\tau_{\text{spont}} \sim 10^{-8}$ s, which fixes the fundamental lower bound on linewidth at $\Delta E_{\text{nat}} \sim 10^{-7}$ eV ($\Delta\nu \sim 10\text{--}20$ MHz). No matter how cold or isolated an atom is kept, it can never have a narrower linewidth than this quantum vacuum limit.
2. **Collisional / Pressure Broadening** ($\Delta E_{\text{coll}} = \hbar/\tau_{\text{coll}}$): Collisions with gas particles interrupt the phase coherence of the radiating dipole after an average collision time $\tau_{\text{coll}} \propto 1/P$. At high gas pressures P , $\tau_{\text{coll}} \ll \tau_{\text{spont}}$, dramatically shortening the state's coherent lifetime and proportionally expanding the Lorentzian spectral width: $\Delta E_{\text{coll}} \gg \Delta E_{\text{nat}}$.
3. **Thermal / Environmental Decoherence:** Ambient thermal baths dynamically scramble the quantum phase over a decoherence time τ_{deph} , broadening the transition lineshape into a Lorentzian profile of width $\hbar/\tau_{\text{deph}}$.

Fundamental Synthesis: Environmental Perturbations, Loss of the Eigenstate Property, and the Uncertainty Principle Origin of Line Broadening

The physics of spectral line broadening can be summarized in a unified conceptual sequence:

Isolated Atom: $\hat{H}_0 |n\rangle = E_n |n\rangle \implies \Delta E = 0, \quad \tau \rightarrow \infty \implies$ **Sharp Delta Peak**
 $\delta(E - E_0)$

$\Downarrow$ *Environment introduces perturbation $\hat{V}_{\text{env}}$*

Coupled System: $\hat{H}_{\text{total}} |n\rangle = (\hat{H}_0 + \hat{V}_{\text{env}}) |n\rangle \neq E |n\rangle \implies |n\rangle$ **is NOT an eigenstate**

$\Downarrow$ *State acquires finite lifetime τ before transitioning into environmental continuum*
Uncertainty Principle: $\Delta E \cdot \Delta t \geq \hbar/2 \implies$ **Intrinsic Energy Spread** $\Delta E \sim \hbar/\tau$
 $\Downarrow$
Spectral Line Broadening: Lorentzian profile $\mathcal{P}(E) \propto \frac{\hbar\Gamma}{(E-E_0)^2 + (\hbar\Gamma/2)^2}$ with FWHM
 $\Delta E = \hbar\Gamma = \hbar/\tau$.

Unified Physical Law: *Spectral line broadening is the direct, unavoidable manifestation of the energy–time uncertainty principle: because environmental interactions perturb the atom and prevent it from being an exact eigenstate of the total Hamiltonian, the state possesses an intrinsic energy spread $\Delta E \sim \hbar/\tau$ proportional to its transition rate.*

10.4 The Sudden Approximation

10.4.1 General Theory

Suppose the Hamiltonian undergoes an abrupt change from $\hat{H}_-$ to $\hat{H}_+$ over an infinitesimal time interval $[-\epsilon, +\epsilon]$ ($\epsilon \rightarrow 0$):

$$\hat{H}(t) = \begin{cases} \hat{H}_-, & t < 0, \\ \hat{H}_+, & t > 0. \end{cases} \quad (10.4.1)$$

Integrating the Schrödinger equation across the jump:

$$i\hbar \int_{-\epsilon}^{+\epsilon} \frac{\partial |\psi\rangle}{\partial t} dt = \int_{-\epsilon}^{+\epsilon} \hat{H}(t) |\psi(t)\rangle dt. \quad (10.4.2)$$

Because $\hat{H}(t)$ is finite, the right-hand side vanishes as $\epsilon \rightarrow 0$:

$$i\hbar \left(|\psi(0^+)\rangle - |\psi(0^-)\rangle \right) = 0 \implies \boxed{|\psi(0^+)\rangle = |\psi(0^-)\rangle} \quad (10.4.3)$$

The physical state vector is **strictly continuous** across the sudden discontinuity.

For $t > 0$, the state evolves according to the new eigenstates $\{|\phi_n^{(+)}\rangle\}$ of $\hat{H}_+$:

$$|\psi(t > 0)\rangle = \sum_n c_n e^{-iE_n^{(+)}t/\hbar} |\phi_n^{(+)}\rangle, \quad c_n = \langle \phi_n^{(+)} | \psi(0^-) \rangle. \quad (10.4.4)$$

The transition probability to land in state $|\phi_n^{(+)}\rangle$ is:

$$\boxed{P_{i \rightarrow n} = \left| \langle \phi_n^{(+)} | \psi_i^{(-)} \rangle \right|^2} \quad (10.4.5)$$

10.4.2 Example: Sudden Expansion of an Infinite Square Well

Consider a particle in the ground state of an infinite square well of width L :

$$\psi(x, 0^-) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right), & 0 \leq x \leq L, \\ 0, & \text{otherwise.} \end{cases} \quad (10.4.6)$$

At $t = 0$, the right wall is instantaneously moved from $x = L$ to $x = 2L$. The new eigenstates of the well of width $2L$ are:

$$\phi_n(x) = \sqrt{\frac{2}{2L}} \sin\left(\frac{n\pi x}{2L}\right) = \frac{1}{\sqrt{L}} \sin\left(\frac{n\pi x}{2L}\right), \quad (0 \leq x \leq 2L). \quad (10.4.7)$$

Evaluation of Overlap Amplitudes

The expansion coefficients are:

$$c_n = \int_0^{2L} \phi_n^*(x) \psi(x, 0^-) dx = \frac{\sqrt{2}}{L} \int_0^L \sin\left(\frac{n\pi x}{2L}\right) \sin\left(\frac{\pi x}{L}\right) dx. \quad (10.4.8)$$

Using the trigonometric product identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$:

$$c_n = \frac{1}{\sqrt{2}L} \int_0^L \left[\cos\left(\frac{(n-2)\pi x}{2L}\right) - \cos\left(\frac{(n+2)\pi x}{2L}\right) \right] dx. \quad (10.4.9)$$

1. **For $n = 2$:** The first cosine argument is zero ($\cos 0 = 1$):

$$c_2 = \frac{1}{\sqrt{2}L} \left[L - \frac{2L}{4\pi} \sin\left(\frac{4\pi L}{2L}\right) \right] = \frac{1}{\sqrt{2}}. \quad (10.4.10)$$

The probability of finding the particle in the $n = 2$ state of the expanded well is:

$$P_2 = |c_2|^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} = 50\%. \quad (10.4.11)$$

2. **For $n \neq 2$:**

$$\begin{aligned} c_n &= \frac{1}{\sqrt{2}L} \left[\frac{2L}{(n-2)\pi} \sin\left(\frac{(n-2)\pi}{2}\right) - \frac{2L}{(n+2)\pi} \sin\left(\frac{(n+2)\pi}{2}\right) \right] \\ &= \frac{\sqrt{2}}{\pi} \left[\frac{-\cos(n\pi/2)}{n-2} - \frac{\cos(n\pi/2)}{n+2} \right] = -\frac{2\sqrt{2}}{\pi} \cos\left(\frac{n\pi}{2}\right) \left[\frac{1}{n-2} + \frac{1}{n+2} \right] \\ &= -\frac{4\sqrt{2}}{\pi} \frac{n \cos(n\pi/2)}{n^2 - 4}. \end{aligned} \quad (10.4.12)$$

- For **odd** n : $\cos(n\pi/2) = 0$, so $c_n = 0$ identically! Transitions to odd states are forbidden by the asymmetry of the initial state relative to the new well.
- For **even** $n = 2k$ ($k \neq 1$): $\cos(k\pi) = (-1)^k$:

$$c_{2k} = -\frac{4\sqrt{2}}{\pi} \frac{2k(-1)^k}{4k^2 - 4} = \frac{2\sqrt{2}(-1)^{k+1}}{\pi} \frac{k}{k^2 - 1}. \quad (10.4.13)$$

For example, for $n = 4$ ($k = 2$):

$$c_4 = -\frac{2\sqrt{2}}{\pi} \frac{2}{3} = -\frac{4\sqrt{2}}{3\pi} \approx -0.600 \implies P_4 = \frac{32}{9\pi^2} \approx 36.0\%. \quad (10.4.14)$$

The sum of probabilities satisfies completeness: $P_2 + P_4 + P_6 + \dots = 1$.

10.5 The Adiabatic Approximation and the Adiabatic Theorem

10.5.1 Instantaneous Basis Expansion

In the opposite physical limit, the parameters of the Hamiltonian $\hat{H}(t)$ vary exceedingly slowly with time. At each instant t , we define the **instantaneous eigenstates**:

$$\hat{H}(t) |n(t)\rangle = E_n(t) |n(t)\rangle, \quad \langle m(t) | n(t) \rangle = \delta_{mn}. \quad (10.5.1)$$

We expand the true state vector $|\psi(t)\rangle$ in this instantaneous basis, extracting the dynamical phase factor $\theta_n(t) \equiv \frac{1}{\hbar} \int_0^t E_n(t') dt'$:

$$|\psi(t)\rangle = \sum_n c_n(t) \exp\left(-\frac{i}{\hbar} \int_0^t E_n(t') dt'\right) |n(t)\rangle = \sum_n c_n(t) e^{-i\theta_n(t)} |n(t)\rangle. \quad (10.5.2)$$

Substituting into the Schrödinger equation $i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$:

$$i\hbar \sum_n \left[\dot{c}_n e^{-i\theta_n} |n\rangle - \frac{i}{\hbar} E_n c_n e^{-i\theta_n} |n\rangle + c_n e^{-i\theta_n} |\dot{n}\rangle \right] = \sum_n c_n e^{-i\theta_n} E_n |n\rangle. \quad (10.5.3)$$

The dynamical energy terms cancel:

$$\sum_n \dot{c}_n(t) e^{-i\theta_n(t)} |n(t)\rangle = - \sum_n c_n(t) e^{-i\theta_n(t)} |\dot{n}(t)\rangle. \quad (10.5.4)$$

Projecting onto an instantaneous bra $\langle m(t)|$:

$$\dot{c}_m(t) = - \sum_n c_n(t) e^{i(\theta_m(t) - \theta_n(t))} \langle m(t) | \dot{n}(t) \rangle. \quad (10.5.5)$$

10.5.2 Evaluation of $\langle m | \dot{n} \rangle$ and the Adiabatic Criterion

To compute $\langle m(t) | \dot{n}(t) \rangle$ for $m \neq n$, we differentiate the instantaneous eigenvalue equation $\hat{H}(t) |n(t)\rangle = E_n(t) |n(t)\rangle$ with respect to time:

$$\dot{\hat{H}}(t) |n(t)\rangle + \hat{H}(t) |\dot{n}(t)\rangle = \dot{E}_n(t) |n(t)\rangle + E_n(t) |\dot{n}(t)\rangle. \quad (10.5.6)$$

Projecting from the left with $\langle m(t) |$ ($m \neq n$):

$$\langle m(t) | \dot{\hat{H}}(t) | n(t) \rangle + E_m(t) \langle m(t) | \dot{n}(t) \rangle = 0 + E_n(t) \langle m(t) | \dot{n}(t) \rangle. \quad (10.5.7)$$

Solving for the matrix element:

$$\boxed{\langle m(t) | \dot{n}(t) \rangle = \frac{\langle m(t) | \dot{\hat{H}}(t) | n(t) \rangle}{E_n(t) - E_m(t)}, \quad (m \neq n)} \quad (10.5.8)$$

Substituting into (10.5.5), the coupling between different states $m \neq n$ is negligible provided:

$$\boxed{\frac{|\langle m(t) | \dot{\hat{H}}(t) | n(t) \rangle|}{|E_m(t) - E_n(t)|^2} \ll \frac{1}{\hbar}} \quad (10.5.9)$$

This is the **Adiabatic Condition**. If the Hamiltonian evolves slowly enough that (10.5.9) holds, no transitions between different levels occur.

10.6 Geometric Phases and Berry's Phase

10.6.1 Derivation of the Geometric Phase

When the adiabatic condition is satisfied and the system is initialized in state $|n(0)\rangle$ ($c_n(0) = 1$, $c_{m \neq n} = 0$), equation (10.5.5) decouples completely:

$$\dot{c}_n(t) = -c_n(t) \langle n(t) | \dot{n}(t) \rangle. \quad (10.6.1)$$

Integrating directly:

$$c_n(t) = \exp(i\gamma_n(t)) c_n(0), \quad (10.6.2)$$

where $\gamma_n(t)$ is the **geometric phase**:

$$\gamma_n(t) \equiv i \int_0^t \langle n(t') | \dot{n}(t') \rangle dt'. \quad (10.6.3)$$

Notice that $\langle n | n \rangle = 1 \implies \frac{d}{dt} \langle n | n \rangle = \langle \dot{n} | n \rangle + \langle n | \dot{n} \rangle = 0 \implies 2 \operatorname{Re} \langle n | \dot{n} \rangle = 0$. Thus, $\langle n | \dot{n} \rangle$ is purely imaginary, ensuring that $\gamma_n(t)$ is **strictly real**.

10.6.2 Berry's Phase for a Closed Path

Suppose the Hamiltonian depends on time through a set of parameters $\vec{R}(t) = (R_1, R_2, \dots)$ that trace a closed curve C in parameter space over period T ($\vec{R}(T) = \vec{R}(0)$). Using the chain rule:

$$|\dot{n}(t)\rangle = \dot{\vec{R}} \cdot \vec{\nabla}_{\vec{R}} |n(\vec{R})\rangle. \quad (10.6.4)$$

Equation (10.6.3) transforms into a geometric line integral:

$$\gamma_n(C) = i \oint_C \langle n(\vec{R}) | \vec{\nabla}_{\vec{R}} n(\vec{R}) \rangle \cdot d\vec{R} \quad (10.6.5)$$

Equation (10.6.5) is the celebrated **Berry Phase** (Sir Michael Berry, 1984). It depends purely on the geometry and topology of the circuit C , not on how fast the path was traversed.

10.6.3 Berry Connection and Berry Curvature

We define the **Berry connection** (a gauge vector potential in parameter space):

$$\vec{\mathcal{A}}_n(\vec{R}) \equiv i \langle n(\vec{R}) | \vec{\nabla}_{\vec{R}} n(\vec{R}) \rangle. \quad (10.6.6)$$

Applying Stokes' theorem, the line integral around C equals the flux of the **Berry curvature** $\vec{\Omega}_n$ through any surface S bounded by C :

$$\gamma_n(C) = \iint_S \vec{\Omega}_n(\vec{R}) \cdot d\vec{S}, \quad \vec{\Omega}_n(\vec{R}) \equiv \vec{\nabla}_{\vec{R}} \times \vec{\mathcal{A}}_n(\vec{R}) \quad (10.6.7)$$

Inserting resolution of the identity $\sum_m |m\rangle \langle m| = \hat{I}$ and using $\langle m | \vec{\nabla} n \rangle = \frac{\langle m | \vec{\nabla} \hat{H} | n \rangle}{E_n - E_m}$ from (10.5.8):

$$\vec{\Omega}_n(\vec{R}) = -\operatorname{Im} \sum_{m \neq n} \frac{\langle n(\vec{R}) | \vec{\nabla}_{\vec{R}} \hat{H} | m(\vec{R}) \rangle \times \langle m(\vec{R}) | \vec{\nabla}_{\vec{R}} \hat{H} | n(\vec{R}) \rangle}{(E_m(\vec{R}) - E_n(\vec{R}))^2} \quad (10.6.8)$$

Level degeneracies ($E_m = E_n$) act as **magnetic monopoles** of Berry curvature in parameter space!

10.6.4 Physical Realization: Spin-1/2 in a Conical Magnetic Field

Consider a spin-1/2 particle in a magnetic field $\vec{B}(t)$ of constant magnitude B_0 whose direction slowly traces out a cone of half-angle θ , subtending a solid angle $\Omega_{\text{solid}} = 2\pi(1 - \cos\theta)$.

Evaluating (10.6.8) for $\hat{H} = -\frac{1}{2}\hbar\gamma\vec{\sigma} \cdot \vec{B}$ yields the field of a point magnetic monopole of strength $-1/2$ located at $\vec{B} = 0$:

$$\vec{\Omega}_+(\vec{B}) = -\frac{1}{2} \frac{\vec{B}}{B^3}. \quad (10.6.9)$$

The Berry phase is the total flux through the spherical cap:

$$\boxed{\gamma_+(C) = \iint_S \left(-\frac{1}{2} \frac{\hat{B}}{B^2} \right) \cdot (B^2 d\Omega \hat{B}) = -\frac{1}{2} \Omega_{\text{solid}} = -\pi(1 - \cos\theta)} \quad (10.6.10)$$

The acquired geometric phase is directly proportional to the solid angle subtended by the circuit on the unit sphere.

Chapter 11

Quantum Scattering Theory

Scattering experiments represent the primary tool of experimental physics for probing the structure of matter on microscopic and subatomic scales, from Rutherford's discovery of the atomic nucleus to deep inelastic scattering at the Large Hadron Collider.

In bound-state problems, one determines discrete energy eigenvalues $E_n < 0$. In scattering theory, one considers the continuous spectrum $E > 0$, where an incident wavepacket prepared far in the past collides with a localized target potential $V(\vec{r})$ and disperses into outgoing spherical waves detected far in the future.

11.1 Cross Sections and Asymptotic Boundary Conditions

11.1.1 Definition of Cross Sections

Consider a beam of monoenergetic particles of mass μ , wavevector $\vec{k} = k\hat{z}$, and energy $E = \frac{\hbar^2 k^2}{2\mu}$ incident upon a stationary target potential $V(\vec{r})$ localized near the origin.

The **incident probability current density** is:

$$\vec{j}_{\text{inc}} = \frac{\hbar}{\mu} \text{Im} [\psi_{\text{inc}}^* \vec{\nabla} \psi_{\text{inc}}] = \frac{\hbar \vec{k}}{\mu} = v \hat{z}, \quad (11.1.1)$$

where $v = \hbar k / \mu$ is the incident particle velocity.

Detectors placed far away at polar coordinates (r, θ, ϕ) subtend a small solid angle $d\Omega = \sin \theta d\theta d\phi$. The number of particles scattered into $d\Omega$ per unit time is $dN(\theta, \phi)$. The **differential cross section** $\frac{d\sigma}{d\Omega}$ is defined as the scattered rate per unit solid angle divided by the incident flux:

$$\frac{d\sigma}{d\Omega} \equiv \frac{1}{|\vec{j}_{\text{inc}}|} \frac{dN}{d\Omega} = \frac{r^2 |\vec{j}_{\text{scat}}|}{|\vec{j}_{\text{inc}}|}. \quad (11.1.2)$$

The **total scattering cross section** is the integral over all solid angles:

$$\sigma_{\text{tot}} \equiv \int \frac{d\sigma}{d\Omega} d\Omega = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \frac{d\sigma}{d\Omega}. \quad (11.1.3)$$

11.1.2 The Asymptotic Scattering Wavefunction

Because the potential $V(\vec{r})$ decays asymptotically ($V(r) \rightarrow 0$ faster than $1/r$ as $r \rightarrow \infty$), the stationary Schrödinger equation must satisfy the **Sommerfeld radiation boundary condition**:

$$\psi(\vec{r}) \xrightarrow{r \rightarrow \infty} e^{i\vec{k} \cdot \vec{r}} + f(\theta, \phi) \frac{e^{ikr}}{r}. \quad (11.1.4)$$

The first term is the unperturbed incident plane wave e^{ikz} , while the second term represents an outgoing scattered spherical wave with angular amplitude $f(\theta, \phi)$, known as the **scattering amplitude**.

Evaluating the scattered radial current density for large r :

$$j_{\text{scat},r} = \frac{\hbar}{\mu} \text{Im} \left[\psi_{\text{scat}}^* \frac{\partial \psi_{\text{scat}}}{\partial r} \right] = \frac{\hbar k}{\mu r^2} |f(\theta, \phi)|^2 + \mathcal{O}(r^{-3}). \quad (11.1.5)$$

Substituting into (11.1.2):

$$\boxed{\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2} \quad (11.1.6)$$

The central objective of quantum scattering theory is the systematic calculation of the scattering amplitude $f(\theta, \phi)$.

11.2 The Lippmann–Schwinger Equation and Green’s Functions

11.2.1 Operator Formulation in Hilbert Space

Let the total Hamiltonian be partitioned as $\hat{H} = \hat{H}_0 + \hat{V}$, where $\hat{H}_0 = \frac{\hat{p}^2}{2\mu}$ describes the free particle. The free-particle eigenstates are plane waves $|\phi\rangle$ with continuous eigenvalue E :

$$\hat{H}_0 |\phi\rangle = E |\phi\rangle, \quad \langle \vec{r} | \phi \rangle = \frac{1}{(2\pi)^{3/2}} e^{i\vec{k} \cdot \vec{r}}. \quad (11.2.1)$$

The full scattering state $|\psi\rangle$ satisfies:

$$(\hat{H}_0 + \hat{V}) |\psi\rangle = E |\psi\rangle \implies (E - \hat{H}_0) |\psi\rangle = \hat{V} |\psi\rangle. \quad (11.2.2)$$

Because E lies in the continuous spectrum of $\hat{H}_0$, the operator $(E - \hat{H}_0)$ is singular and cannot be directly inverted. To select outgoing (+) or incoming (−) boundary conditions, we add an infinitesimal imaginary term $\pm i\epsilon$ ($\epsilon \rightarrow 0^+$):

$$\boxed{|\psi^{(\pm)}\rangle = |\phi\rangle + \frac{1}{E - \hat{H}_0 \pm i\epsilon} \hat{V} |\psi^{(\pm)}\rangle} \quad (11.2.3)$$

Equation (11.2.3) is the celebrated **Lippmann–Schwinger equation**. It incorporates both the differential equation and the asymptotic boundary conditions into a single implicit operator equation.

11.2.2 The Free Green’s Function in Coordinate Space

Projecting (11.2.3) onto the position eigenbra $\langle \vec{r} |$:

$$\langle \vec{r} | \psi^{(\pm)} \rangle = \langle \vec{r} | \phi \rangle + \int d^3 r' \langle \vec{r} | \frac{1}{E - \hat{H}_0 \pm i\epsilon} | \vec{r}' \rangle \langle \vec{r}' | \hat{V} | \psi^{(\pm)} \rangle. \quad (11.2.4)$$

Assuming a local potential $\langle \vec{r}' | \hat{V} | \vec{r}'' \rangle = V(\vec{r}') \delta^3(\vec{r}' - \vec{r}'')$, we define the **free Green’s function**:

$$G_{\pm}(\vec{r}, \vec{r}') \equiv \frac{\hbar^2}{2\mu} \langle \vec{r} | \frac{1}{E - \hat{H}_0 \pm i\epsilon} | \vec{r}' \rangle. \quad (11.2.5)$$

In terms of $G_{\pm}$, the Lippmann–Schwinger equation becomes an integral equation:

$$\psi^{(\pm)}(\vec{r}) = \phi(\vec{r}) + \frac{2\mu}{\hbar^2} \int d^3 r' G_{\pm}(\vec{r}, \vec{r}') V(\vec{r}') \psi^{(\pm)}(\vec{r}'). \quad (11.2.6)$$

Acting with $(\nabla^2 + k^2)$ on the Green’s operator, where $k^2 = \frac{2\mu E}{\hbar^2}$, reveals that $G_{\pm}$ is the fundamental solution to the Helmholtz equation:

$$(\nabla^2 + k^2) G_{\pm}(\vec{r}, \vec{r}') = \delta^3(\vec{r} - \vec{r}'). \quad (11.2.7)$$

11.2.3 Evaluation of the Outgoing Green's Function via Residue Calculus

Because free space is translationally invariant, $G_{\pm}(\vec{r}, \vec{r}') = G_{\pm}(\vec{r} - \vec{r}')$. Setting $\vec{R} = \vec{r} - \vec{r}'$, we insert a complete basis of momentum states $|\vec{q}\rangle$:

$$G_{\pm}(\vec{R}) = \frac{\hbar^2}{2\mu} \int d^3q \langle \vec{r} | \frac{1}{E - \frac{\vec{p}^2}{2\mu} \pm i\epsilon} | \vec{q} \rangle \langle \vec{q} | \vec{r}' \rangle = \frac{1}{(2\pi)^3} \int d^3q \frac{e^{i\vec{q} \cdot \vec{R}}}{k^2 - q^2 \pm i\epsilon}. \quad (11.2.8)$$

Integrating over the angular coordinates of $\vec{q}$:

$$\begin{aligned} G_{\pm}(\vec{R}) &= \frac{1}{(2\pi)^3} \int_0^\infty q^2 dq \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{e^{iqR \cos \theta}}{k^2 - q^2 \pm i\epsilon} \\ &= \frac{1}{4\pi^2 i R} \int_0^\infty q dq \frac{e^{iqR} - e^{-iqR}}{k^2 - q^2 \pm i\epsilon} = -\frac{1}{8\pi^2 i R} \int_{-\infty}^\infty dq \frac{q (e^{iqR} - e^{-iqR})}{q^2 - (k \pm i\epsilon')^2}. \end{aligned} \quad (11.2.9)$$

The integrand has simple poles in the complex q -plane at $q = \pm(k \pm i\epsilon')$. For the outgoing solution ($+i\epsilon$), the poles lie at:

$$q = k + i\epsilon', \quad q = -k - i\epsilon'. \quad (11.2.10)$$

Closing the contour in the upper half-plane for e^{iqR} (enclosing $q = k + i\epsilon'$) and in the lower half-plane for e^{-iqR} (enclosing $q = -k - i\epsilon'$), the Cauchy Residue Theorem yields:

$$\boxed{G_+(\vec{r} - \vec{r}') = -\frac{1}{4\pi} \frac{e^{ik|\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|}} \quad (11.2.11)$$

The $+i\epsilon$ prescription rigorously produces the **outgoing spherical wave**.

11.2.4 Far-Field Asymptotics and the Exact Scattering Amplitude

In the radiation zone far from the target ($r \gg r'$):

$$|\vec{r} - \vec{r}'| = \sqrt{r^2 - 2\vec{r} \cdot \vec{r}' + r'^2} \approx r \left(1 - \frac{\hat{r} \cdot \vec{r}'}{r} \right) = r - \hat{r} \cdot \vec{r}'. \quad (11.2.12)$$

Substituting into the exponential and denominator of (11.2.11):

$$\frac{e^{ik|\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|} \xrightarrow{r \rightarrow \infty} \frac{e^{ikr}}{r} e^{-ik\hat{r} \cdot \vec{r}'} = \frac{e^{ikr}}{r} e^{-i\vec{k}' \cdot \vec{r}'}, \quad (11.2.13)$$

where $\vec{k}' \equiv k\hat{r}$ is the **scattered wavevector** pointing toward the detector ($|\vec{k}'| = |\vec{k}| = k$).

Substituting this asymptotic expansion into the coordinate Lippmann–Schwinger equation (11.2.6):

$$\psi^{(+)}(\vec{r}) \xrightarrow{r \rightarrow \infty} e^{i\vec{k} \cdot \vec{r}} - \frac{\mu}{2\pi\hbar^2} \frac{e^{ikr}}{r} \int d^3r' e^{-i\vec{k}' \cdot \vec{r}'} V(\vec{r}') \psi^{(+)}(\vec{r}'). \quad (11.2.14)$$

Comparing directly with the Sommerfeld boundary condition (11.1.4), we extract the **exact scattering amplitude**:

$$\boxed{f(\vec{k}', \vec{k}) = -\frac{\mu}{2\pi\hbar^2} \int d^3r' e^{-i\vec{k}' \cdot \vec{r}'} V(\vec{r}') \psi^{(+)}(\vec{r}') = -\frac{4\pi^2\mu}{\hbar^2} \langle \vec{k}' | \hat{V} | \psi^{(+)} \rangle} \quad (11.2.15)$$

11.3 The Born Approximation

11.3.1 The Born Series

Equation (11.2.15) is not an explicit solution because the unknown exact scattering wavefunction $\psi^{(+)}(\vec{r})$ appears inside the integral. We generate an iterative perturbation expansion by repeatedly substituting (11.2.6) into itself:

$$\psi^{(+)}(\vec{r}) = \phi(\vec{r}) + \int G_+ V \phi + \iint G_+ V G_+ V \phi + \dots \quad (11.3.1)$$

This expansion in powers of the potential $\hat{V}$ is the **Born series**.

11.3.2 The First Born Approximation

To lowest order in the perturbation, we approximate the true scattering state inside the interaction volume by the incident unperturbed plane wave:

$$\psi^{(+)}(\vec{r}) \approx \phi(\vec{r}) = e^{i\vec{k} \cdot \vec{r}}. \quad (11.3.2)$$

Substituting into (11.2.15) yields the **First Born Approximation**:

$$f^{(1)}(\theta, \phi) = -\frac{\mu}{2\pi\hbar^2} \int d^3r' e^{-i(\vec{k}' - \vec{k}) \cdot \vec{r}'} V(\vec{r}') \quad (11.3.3)$$

Defining the **momentum transfer vector**:

$$\vec{q} \equiv \vec{k}' - \vec{k}, \quad (11.3.4)$$

the scattering amplitude is simply proportional to the **3D spatial Fourier transform** of the potential evaluated at $\vec{q}$:

$$f^{(1)}(\vec{q}) = -\frac{\mu}{2\pi\hbar^2} \tilde{V}(\vec{q}). \quad (11.3.5)$$

Since elastic scattering conserves energy ($|\vec{k}'| = |\vec{k}| = k$), the magnitude of the momentum transfer is purely a function of the scattering angle θ :

$$q = |\vec{k}' - \vec{k}| = \sqrt{k^2 + k^2 - 2k^2 \cos \theta} = 2k \sin\left(\frac{\theta}{2}\right). \quad (11.3.6)$$

11.3.3 Spherically Symmetric Potentials

When $V(\vec{r}) = V(r)$ is central, the azimuthal integral is trivial:

$$\begin{aligned} f^{(1)}(\theta) &= -\frac{\mu}{2\pi\hbar^2} \int_0^\infty r'^2 dr' V(r') \int_0^\pi \sin \theta' d\theta' \int_0^{2\pi} d\phi' e^{-iqr' \cos \theta'} \\ &= -\frac{\mu}{\hbar^2} \int_0^\infty r'^2 dr' V(r') \left[\frac{e^{-iqr' \cos \theta'}}{-iqr'} \right]_0^\pi. \end{aligned} \quad (11.3.7)$$

Thus:

$$f^{(1)}(\theta) = -\frac{2\mu}{\hbar^2 q} \int_0^\infty dr r V(r) \sin(qr) \quad (11.3.8)$$

11.3.4 Applications of the Born Approximation

The Yukawa Potential

Consider the Yukawa potential representing screened nuclear interactions:

$$V(r) = V_0 \frac{e^{-\alpha r}}{r}, \quad (\alpha > 0). \quad (11.3.9)$$

Using (11.3.8):

$$f^{(1)}(\theta) = -\frac{2\mu V_0}{\hbar^2 q} \int_0^\infty dr e^{-\alpha r} \sin(qr) = -\frac{2\mu V_0}{\hbar^2 q} \frac{q}{\alpha^2 + q^2} = -\frac{2\mu V_0}{\hbar^2 (\alpha^2 + q^2)}. \quad (11.3.10)$$

Substituting $q = 2k \sin(\theta/2)$:

$$f^{(1)}(\theta) = -\frac{2\mu V_0}{\hbar^2 [4k^2 \sin^2(\theta/2) + \alpha^2]}. \quad (11.3.11)$$

The differential cross section is:

$$\frac{d\sigma}{d\Omega} = \frac{4\mu^2 V_0^2}{\hbar^4 [4k^2 \sin^2(\theta/2) + \alpha^2]^2}. \quad (11.3.12)$$

Rutherford Scattering

The Coulomb potential between two charges $Z_1 e$ and $Z_2 e$ is the zero-screening limit $\alpha \rightarrow 0$ of the Yukawa potential with $V_0 = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0}$. Taking $\alpha \rightarrow 0$:

$$f_{\text{Coulomb}}(\theta) = -\frac{2\mu \left(\frac{Z_1 Z_2 e^2}{4\pi\epsilon_0} \right)}{\hbar^2 [4k^2 \sin^2(\theta/2)]} = -\frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 (4E) \sin^2(\theta/2)}. \quad (11.3.13)$$

The differential cross section is:

$$\boxed{\frac{d\sigma}{d\Omega} = \left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 E} \right)^2 \frac{1}{\sin^4(\theta/2)}} \quad (11.3.14)$$

Remarkably, the first Born quantum approximation reproduces the **exact classical Rutherford formula** without quantum corrections!

Scattering by an Electric Dipole in the Born Approximation

Consider an incident electron (mass μ , charge $-e$, incident along $\vec{k} = k\hat{z}$) scattering elastically off a fixed electric dipole consisting of charge $+e$ located at $+\vec{a}$ and charge $-e$ located at $-\vec{a}$ along the $\hat{x}$ -axis ($\vec{a} = a\hat{x}$).

The net electrostatic potential energy seen by the electron is non-central:

$$V(\vec{r}) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{|\vec{r} - \vec{a}|} + \frac{e^2}{4\pi\epsilon_0} \frac{1}{|\vec{r} + \vec{a}|}. \quad (11.3.15)$$

In the first Born approximation, the scattering amplitude is:

$$f^{(1)}(\vec{k}', \vec{k}) = -\frac{\mu}{2\pi\hbar^2} \int d^3r e^{-i\vec{q}\cdot\vec{r}} V(\vec{r}), \quad (11.3.16)$$

where $\vec{q} = \vec{k}' - \vec{k}$ is the momentum transfer vector.

To compute the 3D spatial Fourier transform of each displaced Coulomb center, we make a change of variables $\vec{r}' = \vec{r} \mp \vec{a} \implies \vec{r} = \vec{r}' \pm \vec{a}$:

$$\int d^3r \frac{e^{-i\vec{q}\cdot\vec{r}}}{|\vec{r} \mp \vec{a}|} = e^{\mp i\vec{q}\cdot\vec{a}} \int d^3r' \frac{e^{-i\vec{q}\cdot\vec{r}'}}{r'}. \quad (11.3.17)$$

Evaluating the standard regularized Coulomb Fourier transform:

$$\int d^3r' \frac{e^{-i\vec{q}\cdot\vec{r}'}}{r'} = \lim_{\mu_0 \rightarrow 0} \int d^3r' \frac{e^{-\mu_0 r' - i\vec{q}\cdot\vec{r}'}}{r'} = \frac{4\pi}{q^2}. \quad (11.3.18)$$

Consequently:

$$\int d^3r e^{-i\vec{q}\cdot\vec{r}} \left[-\frac{1}{|\vec{r} - \vec{a}|} + \frac{1}{|\vec{r} + \vec{a}|} \right] = \frac{4\pi}{q^2} \left(-e^{-i\vec{q}\cdot\vec{a}} + e^{+i\vec{q}\cdot\vec{a}} \right) = \frac{8\pi i}{q^2} \sin(\vec{q} \cdot \vec{a}). \quad (11.3.19)$$

Substituting into the Born amplitude:

$$f^{(1)}(\vec{q}) = -\frac{\mu}{2\pi\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right) \left[\frac{8\pi i}{q^2} \sin(\vec{q} \cdot \vec{a}) \right] = -\frac{i\mu e^2}{\pi\epsilon_0\hbar^2} \frac{\sin(\vec{q} \cdot \vec{a})}{q^2} \quad (11.3.20)$$

Now, evaluate the scalar product $\vec{q} \cdot \vec{a}$. With incident wavevector $\vec{k} = k\hat{z}$ and scattered wavevector directed toward polar angles (θ, ϕ) :

$$\vec{k} = (0, 0, k), \quad \vec{k}' = (k \sin \theta \cos \phi, k \sin \theta \sin \phi, k \cos \theta). \quad (11.3.21)$$

The momentum transfer is:

$$\vec{q} = \vec{k}' - \vec{k} = (k \sin \theta \cos \phi, k \sin \theta \sin \phi, k(\cos \theta - 1)). \quad (11.3.22)$$

Its magnitude is $q^2 = 4k^2 \sin^2(\theta/2)$. For a dipole aligned along $\vec{a} = a\hat{x}$:

$$\vec{q} \cdot \vec{a} = ka \sin \theta \cos \phi. \quad (11.3.23)$$

The resulting differential scattering cross section is:

$$\frac{d\sigma}{d\Omega} = |f^{(1)}(\vec{q})|^2 = \left(\frac{\mu e^2}{\pi\epsilon_0\hbar^2} \right)^2 \frac{\sin^2(ka \sin \theta \cos \phi)}{[4k^2 \sin^2(\theta/2)]^2} = \left(\frac{\mu e^2}{4\pi\epsilon_0\hbar^2 k^2} \right)^2 \frac{\sin^2(ka \sin \theta \cos \phi)}{\sin^4(\theta/2)} \quad (11.3.24)$$

This result illustrates several profound physical features:

1. **Azimuthal Anisotropy:** Because the target lacks spherical symmetry, the cross section depends explicitly on the azimuthal angle ϕ . Scattering is maximal in the plane containing the dipole axis ($\phi = 0, \pi$) and strictly vanishes at right angles to the dipole axis ($\phi = \pi/2, 3\pi/2$), where $\vec{q} \cdot \vec{a} = 0$.
2. **Softening of the Forward Singularity:** In the forward direction $\theta \rightarrow 0$, $ka \sin \theta \cos \phi \approx ka\theta \cos \phi$, while $q^2 \approx k^2\theta^2$. The amplitude scales as:

$$f^{(1)}(\theta \rightarrow 0) \sim \frac{ka\theta \cos \phi}{k^2\theta^2} = \frac{a \cos \phi}{k\theta} \propto \frac{1}{\theta} \implies \frac{d\sigma}{d\Omega} \propto \frac{1}{\theta^2}. \quad (11.3.25)$$

The cross section diverges only as $1/\theta^2$, compared to the much harsher $1/\theta^4$ Rutherford divergence. The physical reason is that the equal and opposite charges of the dipole screen each other at large distances, reducing the far field from $1/r$ (monopole) to $1/r^2$ (dipole).

3. **The Point-Dipole Limit ($ka \ll 1$):** When the de Broglie wavelength is much larger than the dipole separation ($\lambda \gg a$), we can expand $\sin(\vec{q} \cdot \vec{a}) \approx \vec{q} \cdot \vec{a}$. The dipole moment vector is $\vec{p}_{\text{dip}} = 2e\vec{a}$. The scattering amplitude becomes:

$$f^{(1)}(\vec{q}) \approx -\frac{i\mu}{2\pi\epsilon_0\hbar^2} \frac{\vec{q} \cdot \vec{p}_{\text{dip}}}{q^2}, \quad (11.3.26)$$

which is purely directional and independent of the short-distance charge separation.

11.3.5 Validity of the Born Approximation

The Born approximation assumes that the perturbation to the incident wave is small throughout the scattering potential. At the origin $\vec{r} = 0$, evaluating $|\psi^{(+)}(0) - \phi(0)| \ll 1$ leads to two distinct criteria:

- **Low-Energy Limit** ($ka \ll 1$): The potential must be very weak compared to the zero-point kinetic energy:

$$|V_0| \ll \frac{\hbar^2}{\mu a^2}. \quad (11.3.27)$$

- **High-Energy Limit** ($ka \gg 1$): The potential energy must be small compared to the geometric mean of kinetic energy and barrier height:

$$|V_0| \ll \frac{\hbar^2 k}{\mu a} = \frac{\hbar v}{a}. \quad (11.3.28)$$

The Born approximation always improves as the incident beam energy increases ($k \rightarrow \infty$).

11.4 Partial Wave Analysis

At low incident energies, the Born approximation fails whenever the potential is strong enough to support bound states (such as nuclear forces or hard-core potentials). In such cases, one must exploit spherical symmetry by decomposing the wavefunction into angular momentum eigenstates: **Partial Wave Analysis**.

11.4.1 Partial Wave Expansion

Because $[\hat{L}, \hat{H}] = 0$ for a central potential $V(r)$, the total orbital angular momentum l and its projection m are strictly conserved. The scattering amplitude $f(\theta)$ depends only on θ and can be expanded in Legendre polynomials $P_l(\cos \theta)$:

$$f(\theta) = \sum_{l=0}^{\infty} (2l+1) f_l(k) P_l(\cos \theta), \quad (11.4.1)$$

where $f_l(k)$ is the **partial wave scattering amplitude** for the l -th channel.

The incident plane wave is decomposed using Rayleigh's expansion:

$$e^{ikz} = e^{ikr \cos \theta} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos \theta), \quad (11.4.2)$$

where $j_l(kr)$ are spherical Bessel functions. Using the large- r asymptotic form $j_l(kr) \rightarrow \frac{\sin(kr - l\pi/2)}{kr}$:

$$e^{ikz} \xrightarrow{r \rightarrow \infty} \sum_{l=0}^{\infty} (2l+1) i^l \frac{e^{i(kr - l\pi/2)} - e^{-i(kr - l\pi/2)}}{2ikr} P_l(\cos \theta). \quad (11.4.3)$$

Notice that the incident plane wave is a superposition of incoming spherical waves ($\propto e^{-ikr}/r$) and outgoing spherical waves ($\propto e^{ikr}/r$).

11.4.2 Phase Shifts and the Radial Equation

Inside the potential, the radial Schrödinger equation for $u_l(r) = rR_l(r)$ is:

$$\frac{d^2 u_l}{dr^2} + \left[k^2 - \frac{2\mu}{\hbar^2} V(r) - \frac{l(l+1)}{r^2} \right] u_l = 0, \quad u_l(0) = 0. \quad (11.4.4)$$

Outside the range of the potential ($r > a$, where $V(r) = 0$), $u_l(r)$ satisfies the free spherical Bessel equation:

$$R_l(r) = B_l [\cos \delta_l j_l(kr) - \sin \delta_l n_l(kr)], \quad (11.4.5)$$

where $n_l(kr)$ is the spherical Neumann function, and $\delta_l(k)$ is the **phase shift**. For $r \rightarrow \infty$, $n_l(kr) \rightarrow -\frac{\cos(kr - l\pi/2)}{kr}$. Combining terms:

$$R_l(r) \xrightarrow{r \rightarrow \infty} \frac{B_l}{kr} \sin \left(kr - \frac{l\pi}{2} + \delta_l \right). \quad (11.4.6)$$

The potential shifts the phase of the outgoing spherical wave by an angle δ_l :

- If the potential is **attractive** ($V(r) < 0$), the local wavelength is shortened, pulling the wavefunction inward: $\delta_l > 0$.
- If the potential is **repulsive** ($V(r) > 0$), the wavefunction is pushed outward: $\delta_l < 0$.

11.4.3 Determining the Partial Wave Amplitude

Matching the asymptotic full solution $\psi(r, \theta) = \sum_l (2l+1) i^l R_l(r) P_l(\cos \theta)$ with $e^{ikz} + f(\theta) \frac{e^{ikr}}{r}$ shows that the incoming spherical wave must remain unperturbed (flux conservation), which fixes $B_l = e^{i\delta_l}$. The scattered amplitude for channel l is:

$$f_l(k) = \frac{e^{2i\delta_l} - 1}{2ik} = \frac{e^{i\delta_l} \sin \delta_l}{k} = \frac{1}{k (\cot \delta_l - i)} \quad (11.4.7)$$

Defining the **S-matrix element** for elastic scattering:

$$S_l(k) \equiv e^{2i\delta_l}, \quad |S_l(k)| = 1. \quad (11.4.8)$$

Unitarity of the S -matrix ($|S_l| = 1$) reflects probability conservation.

11.4.4 Connecting Phase Shifts to the Potential via Born Approximation

By projecting the Lippmann–Schwinger equation onto angular momentum channels, one derives a direct relation between the phase shift δ_l and the potential $V(r)$ in the weak-scattering limit:

$$\tan \delta_l \approx -\frac{2\mu k}{\hbar^2} \int_0^\infty r^2 V(r) [j_l(kr)]^2 dr \quad (11.4.9)$$

For s -wave scattering ($l = 0$, where $j_0(kr) = \frac{\sin(kr)}{kr}$):

$$\tan \delta_0 \approx -\frac{2\mu}{\hbar^2 k} \int_0^\infty V(r) \sin^2(kr) dr. \quad (11.4.10)$$

Equation (11.4.9) connects the Born approximation directly to partial-wave phase shifts.

11.4.5 Total Cross Section and the Optical Theorem

Substituting (11.4.1) into the total cross section and using the orthogonality of Legendre polynomials $\int_{-1}^1 P_l(x)P_{l'}(x)dx = \frac{2}{2l+1}\delta_{ll'}$:

$$\sigma_{\text{tot}} = 2\pi \int_{-1}^1 |f(\theta)|^2 d(\cos \theta) = 4\pi \sum_{l=0}^{\infty} (2l+1) |f_l(k)|^2. \quad (11.4.11)$$

Using $|f_l(k)|^2 = \frac{\sin^2 \delta_l}{k^2}$:

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l \quad (11.4.12)$$

Now evaluate the forward scattering amplitude $f(\theta = 0)$ where $P_l(1) = 1$:

$$f(0) = \sum_{l=0}^{\infty} (2l+1) f_l(k) = \sum_{l=0}^{\infty} (2l+1) \left[\frac{\sin \delta_l \cos \delta_l}{k} + i \frac{\sin^2 \delta_l}{k} \right]. \quad (11.4.13)$$

Taking the imaginary part:

$$\text{Im}[f(0)] = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l. \quad (11.4.14)$$

Comparing directly with (11.4.12) proves the **Optical Theorem**:

$$\sigma_{\text{tot}} = \frac{4\pi}{k} \text{Im}[f(0)] \quad (11.4.15)$$

Deep Conceptual Insight: Exact Non-Perturbative Nature of the Optical Theorem vs. Perturbative Cross Sections

A frequent and dangerous misconception is to attempt to evaluate the optical theorem using a low-order perturbative scattering amplitude:

$$\sigma_{\text{tot}} \stackrel{?}{=} \frac{4\pi}{k} \text{Im} [f^{(1)}(0)]. \quad (1)$$

This expression is **identically zero!** In the first Born approximation, the forward scattering amplitude for any real, local potential $V^*(\vec{r}) = V(\vec{r})$ is:

$$f^{(1)}(0) = -\frac{\mu}{2\pi\hbar^2} \int d^3r V(\vec{r}) \in \mathbb{R} \implies \text{Im} [f^{(1)}(0)] = 0. \quad (2)$$

Yet the first Born total cross section is strictly non-zero and positive:

$$\sigma^{(1)} = \int |f^{(1)}(\theta, \phi)|^2 d\Omega = \frac{\mu^2}{4\pi^2\hbar^4} \int d\Omega \left| \int d^3r e^{-i\vec{q}\cdot\vec{r}} V(\vec{r}) \right|^2 > 0. \quad (3)$$

Does the optical theorem fail in perturbation theory? **No!** The optical theorem is an **exact, non-perturbative relation**:

$$\sigma_{\text{tot}}^{\text{exact}} = \frac{4\pi}{k} \text{Im} [f^{\text{exact}}(0)] \quad (4)$$

which connects terms of **different orders** in the perturbation expansion.

Order-by-Order Matching from the Lippmann–Schwinger Equation

Expanding the exact scattered state vector $|\psi_k^+\rangle$ in powers of the potential V :

$$|\psi_k^+\rangle = |k\rangle + G_0^+ V |k\rangle + \mathcal{O}(V^2), \quad G_0^+ = \frac{1}{E - H_0 + i\epsilon}. \quad (5)$$

The exact scattering amplitude is:

$$\begin{aligned} f(\vec{k}_f, \vec{k}_i) &= -\frac{4\pi^2 m}{\hbar^2} \langle k_f | V | \psi_{k_i}^+ \rangle \\ &= -\frac{4\pi^2 m}{\hbar^2} \left[\langle k_f | V | k_i \rangle + \langle k_f | V \frac{1}{E - H_0 + i\epsilon} V | k_i \rangle + \mathcal{O}(V^3) \right] \\ &\equiv f^{(1)}(\vec{k}_f, \vec{k}_i) + f^{(2)}(\vec{k}_f, \vec{k}_i) + \mathcal{O}(V^3). \end{aligned} \quad (6)$$

For forward scattering ($\vec{k}_f = \vec{k}_i = \vec{k}$):

$$f(0) = f^{(1)}(0) + f^{(2)}(0) + \mathcal{O}(V^3). \quad (7)$$

Because $f^{(1)}(0) \sim \mathcal{O}(V)$ is strictly real, the imaginary part begins only at **second order**:

$$\text{Im}[f(0)] = \text{Im}[f^{(2)}(0)] + \mathcal{O}(V^3). \quad (8)$$

Meanwhile, the exact total cross section expanded in powers of V is:

$$\sigma_{\text{tot}} = \int |f|^2 d\Omega = \int |f^{(1)} + \mathcal{O}(V^2)|^2 d\Omega = \int |f^{(1)}|^2 d\Omega + \mathcal{O}(V^3) = \sigma^{(1)} + \mathcal{O}(V^3). \quad (9)$$

Notice that $\sigma^{(1)}$ is of order $\mathcal{O}(V^2)$! Therefore, order-by-order consistency of the optical theorem demands:

$$\boxed{\sigma^{(1)} = \frac{4\pi}{k} \text{Im}[f^{(2)}(0)]} \quad (10)$$

The total cross section calculated from the **first-order** amplitude is determined by the imaginary part of the **second-order** forward amplitude!

Explicit Proof via the Sokhotski–Plemelj Formula

To compute $\text{Im}[f^{(2)}(0)]$ explicitly, we use the distribution identity:

$$\frac{1}{x + i\epsilon} = \mathcal{P} \left(\frac{1}{x} \right) - i\pi\delta(x), \quad (11)$$

so the free Green's operator decomposes into:

$$G_0^+ = \frac{1}{E - H_0 + i\epsilon} = \mathcal{P} \left(\frac{1}{E - H_0} \right) - i\pi\delta(E - H_0). \quad (12)$$

The Cauchy principal value operator is Hermitian and contributes solely to the real part of $f^{(2)}(0)$. The imaginary part arises entirely from the Dirac delta function:

$$\text{Im} \left[\langle k | V \frac{1}{E - H_0 + i\epsilon} V | k \rangle \right] = -\pi \langle k | V \delta(E - H_0) V | k \rangle. \quad (13)$$

Inserting a complete set of intermediate free-momentum plane wave states $\int d^3q |q\rangle \langle q| = I$:

$$\begin{aligned} \text{Im} [f^{(2)}(0)] &= -\frac{4\pi^2 m}{\hbar^2} \left(-\pi \int d^3q \langle k|V|q\rangle \langle q|\delta(E - H_0)V|k\rangle \right) \\ &= \frac{4\pi^3 m}{\hbar^2} \int d^3q |\langle k|V|q\rangle|^2 \delta(E_k - E_q). \end{aligned} \quad (14)$$

Using the dispersion relation $E_q = \frac{\hbar^2 q^2}{2m}$ and $E_k = \frac{\hbar^2 k^2}{2m}$:

$$\delta(E_k - E_q) = \delta\left(\frac{\hbar^2}{2m}(k^2 - q^2)\right) = \frac{2m}{\hbar^2} \delta(k^2 - q^2) = \frac{2m}{\hbar^2} \frac{\delta(q - k) + \delta(q + k)}{2k}. \quad (15)$$

Since q and k are positive norms ($q, k > 0$), $\delta(q + k)$ has no support and drops out:

$$\delta(E_k - E_q) = \frac{m}{\hbar^2 k} \delta(q - k). \quad (16)$$

Writing the momentum volume element in spherical coordinates $d^3q = q^2 dq d\Omega$:

$$\begin{aligned} \text{Im} [f^{(2)}(0)] &= \frac{4\pi^3 m}{\hbar^2} \int q^2 dq d\Omega |\langle k|V|q\rangle|^2 \left[\frac{m}{\hbar^2 k} \delta(q - k) \right] \\ &= \frac{4\pi^3 m^2 k}{\hbar^4} \int d\Omega |\langle k|V|q\rangle|_{q=k}^2. \end{aligned} \quad (17)$$

Recalling the definition of the first Born scattering amplitude $f^{(1)}(\theta, \phi) = -\frac{4\pi^2 m}{\hbar^2} \langle k|V|q\rangle$ on the energy shell $q = k$:

$$\left| f^{(1)}(\theta, \phi) \right|^2 = \frac{16\pi^4 m^2}{\hbar^4} |\langle k|V|q\rangle|^2 \implies \frac{4\pi^3 m^2 k}{\hbar^4} |\langle k|V|q\rangle|^2 = \frac{k}{4\pi} \left| f^{(1)}(\theta, \phi) \right|^2. \quad (18)$$

Integrating over the solid angle $d\Omega$:

$$\text{Im} [f^{(2)}(0)] = \frac{k}{4\pi} \int \left| f^{(1)}(\theta, \phi) \right|^2 d\Omega = \frac{k}{4\pi} \sigma^{(1)}. \quad (19)$$

Multiplying by $4\pi/k$:

$$\boxed{\sigma^{(1)} = \frac{4\pi}{k} \text{Im} [f^{(2)}(0)]} \quad (20)$$

This rigorous identity resolves the dilemma completely:

1. The Optical Theorem is an **exact non-perturbative relation** for the **exact total cross section** σ_{tot} .
2. If calculated perturbatively, the imaginary part of the forward amplitude at order $n + 1$ determines the total cross section at order n .
3. At no point does the optical theorem imply that $\text{Im}[f^{(1)}(0)]$ gives the cross section; the first Born forward amplitude is strictly real, and the cross section emerges only when the second-order forward amplitude is evaluated through the Sokhotski–Plemelj singularity.

11.4.6 Explicit Verification for an Exponential Potential

To see this order-by-order mechanism in concrete detail, consider a particle of mass m scattering off an exponential potential:

$$V(r) = g^2 e^{-\mu r}, \quad (\mu > 0). \quad (11.4.16)$$

The first Born scattering amplitude is given by the radial integral (11.3.8):

$$f^{(1)}(q) = -\frac{2m}{\hbar^2 q} \int_0^\infty dr r \left(g^2 e^{-\mu r} \right) \sin(qr). \quad (11.4.17)$$

Using the standard parametric derivative of the Laplace transform:

$$\int_0^\infty dr r e^{-\mu r} \sin(qr) = -\frac{\partial}{\partial \mu} \left[\int_0^\infty dr e^{-\mu r} \sin(qr) \right] = -\frac{\partial}{\partial \mu} \left(\frac{q}{\mu^2 + q^2} \right) = \frac{2\mu q}{(\mu^2 + q^2)^2}. \quad (11.4.18)$$

Substituting this into $f^{(1)}(q)$:

$$f^{(1)}(q) = -\frac{2mg^2}{\hbar^2 q} \left[\frac{2\mu q}{(\mu^2 + q^2)^2} \right] = -\frac{4mg^2\mu}{\hbar^2(\mu^2 + q^2)^2}. \quad (11.4.19)$$

In the forward direction $\theta = 0 \implies q = 0$:

$$f^{(1)}(0) = -\frac{4mg^2}{\hbar^2\mu^3} \in \mathbb{R}. \quad (11.4.20)$$

As guaranteed, $f^{(1)}(0)$ is strictly real, so $\text{Im}[f^{(1)}(0)] = 0$.

Now compute the total cross section at order $\mathcal{O}(g^4)$ by integrating over all scattering angles:

$$\sigma^{(1)} = \int |f^{(1)}(q)|^2 d\Omega = 2\pi \int_0^\pi \left[\frac{4mg^2\mu}{\hbar^2} \right]^2 \frac{\sin \theta d\theta}{[\mu^2 + 4k^2 \sin^2(\theta/2)]^4}. \quad (11.4.21)$$

Using the change of variables $q = 2k \sin(\theta/2) \implies q dq = k^2 \sin \theta d\theta$, with integration range $q \in [0, 2k]$:

$$\begin{aligned} \sigma^{(1)} &= \frac{2\pi}{k^2} \left(\frac{4mg^2\mu}{\hbar^2} \right)^2 \int_0^{2k} \frac{q dq}{(\mu^2 + q^2)^4} \\ &= \frac{32\pi m^2 g^4 \mu^2}{\hbar^4 k^2} \left[-\frac{1}{6(\mu^2 + q^2)^3} \right]_0^{2k} \\ &= \frac{16\pi m^2 g^4 \mu^2}{3\hbar^4 k^2} \left[\frac{1}{\mu^6} - \frac{1}{(\mu^2 + 4k^2)^3} \right] \end{aligned} \quad (11.4.22)$$

Now evaluate the imaginary part of the second-order forward amplitude using the Sokhotski–Plemelj formula:

$$\begin{aligned} \text{Im} [f^{(2)}(0)] &= -\frac{4\pi^2 m}{\hbar^2} \text{Im} \left\langle \vec{k} \left| \hat{V} \frac{1}{E_k - \hat{H}_0 + i\epsilon} \hat{V} \right| \vec{k} \right\rangle \\ &= -\frac{4\pi^2 m}{\hbar^2} \left(-\pi \int d^3 q' \left| \left\langle \vec{k} \left| \hat{V} \right| \vec{q}' \right\rangle \right|^2 \delta(E_k - E_{q'}) \right). \end{aligned} \quad (11.4.23)$$

Substituting the density of states $\delta(E_k - E_{q'}) = \frac{m}{\hbar^2 k} \delta(q' - k)$ and $d^3 q' = q'^2 dq' d\Omega$:

$$\text{Im} [f^{(2)}(0)] = \frac{4\pi^3 m^2 k}{\hbar^4} \int d\Omega \left| \left\langle \vec{k} \left| \hat{V} \right| \vec{q}' \right\rangle \right|_{q'=k}^2 = \frac{k}{4\pi} \int |f^{(1)}(\theta, \phi)|^2 d\Omega = \frac{k}{4\pi} \sigma^{(1)}. \quad (11.4.24)$$

Multiplying by $4\pi/k$:

$$\boxed{\frac{4\pi}{k} \operatorname{Im} [f^{(2)}(0)] = \sigma^{(1)} = \frac{16\pi m^2 g^4 \mu^2}{3\hbar^4 k^2} \left[\frac{1}{\mu^6} - \frac{1}{(\mu^2 + 4k^2)^3} \right]} \quad (11.4.25)$$

This explicit calculation demonstrates with complete algebraic transparency that the optical theorem is satisfied order by order: the cross section from the first Born term is identically equal to the imaginary part of the second Born forward scattering amplitude.

11.4.7 Threshold Behavior and Scattering Length

For low incident energies $k \rightarrow 0$, particles with orbital angular momentum l cannot overcome the centrifugal barrier $\frac{\hbar^2 l(l+1)}{2\mu r^2}$ unless $ka \gtrsim \sqrt{l(l+1)}$. Consequently:

$$\tan \delta_l \propto (ka)^{2l+1} \xrightarrow{k \rightarrow 0} 0. \quad (11.4.26)$$

For $ka \ll 1$, all higher partial waves ($l \geq 1$) are completely frozen out. Scattering is purely ***s*-wave** ($l = 0$), and the cross section is completely isotropic:

$$f(\theta) \approx f_0(k) = \frac{1}{k \cot \delta_0 - ik}. \quad (11.4.27)$$

We define the **scattering length** a_s :

$$a_s \equiv -\lim_{k \rightarrow 0} \frac{\tan \delta_0}{k}. \quad (11.4.28)$$

The low-energy *s*-wave cross section is:

$$\boxed{\sigma_0 = 4\pi a_s^2} \quad (11.4.29)$$

11.4.8 Hard Sphere Scattering

Consider a completely impenetrable sphere of radius a :

$$V(r) = \begin{cases} \infty, & r \leq a, \\ 0, & r > a. \end{cases} \quad (11.4.30)$$

Boundary condition at $r = a$ requires $u_l(a) = 0$:

$$\cos \delta_l j_l(ka) - \sin \delta_l n_l(ka) = 0 \implies \tan \delta_l = \frac{j_l(ka)}{n_l(ka)}. \quad (11.4.31)$$

Low-Energy Limit ($ka \ll 1$): For $l = 0$, $j_0(x) = \frac{\sin x}{x} \approx 1$ and $n_0(x) = -\frac{\cos x}{x} \approx -\frac{1}{x}$:

$$\tan \delta_0 = \frac{j_0(ka)}{n_0(ka)} \approx -ka \implies \delta_0 \approx -ka. \quad (11.4.32)$$

The scattering length is $a_s = a$. The total cross section is:

$$\boxed{\sigma = 4\pi a^2} \quad (11.4.33)$$

This quantum result is **four times the classical geometric cross section** πa^2 ! The factor of 4 arises from wave diffraction bending around the entire sphere.

High-Energy Limit ($ka \gg 1$): Summing over all contributing partial waves up to $l_{\max} \approx ka$:

$$\boxed{\sigma \rightarrow 2\pi a^2} \quad (11.4.34)$$

This is **twice the classical cross section!** One factor of πa^2 comes from classical geometric reflection; the second factor of πa^2 comes from sharp forward shadow diffraction required by the Optical Theorem.

11.4.9 The Attractive Square Well and Resonances

Consider an attractive spherical square well of depth V_0 and radius a :

$$V(r) = \begin{cases} -V_0, & r < a, \\ 0, & r > a. \end{cases} \quad (11.4.35)$$

The radial s -wave equation ($l = 0$) has the solutions:

$$u_0(r) = \begin{cases} A \sin(Kr), & r < a, & K = \sqrt{\frac{2\mu(E+V_0)}{\hbar^2}}, \\ C \sin(kr + \delta_0), & r > a, & k = \sqrt{\frac{2\mu E}{\hbar^2}}. \end{cases} \quad (11.4.36)$$

Matching the logarithmic derivative at $r = a$:

$$\left. \frac{u'_0(r)}{u_0(r)} \right|_{r=a} \implies K \cot(Ka) = k \cot(ka + \delta_0). \quad (11.4.37)$$

Solving for the phase shift δ_0 :

$$\tan(ka + \delta_0) = \frac{k}{K} \tan(Ka) \implies \delta_0 = \arctan\left(\frac{k}{K} \tan(Ka)\right) - ka. \quad (11.4.38)$$

Taking the zero-energy limit $k \rightarrow 0$ (where $K \rightarrow K_0 = \sqrt{2\mu V_0}/\hbar$):

$$\tan \delta_0 \approx \left(\frac{\tan(K_0 a)}{K_0 a} - 1 \right) ka. \quad (11.4.39)$$

The **scattering length** is:

$$\boxed{a_s = a \left(1 - \frac{\tan(K_0 a)}{K_0 a} \right)} \quad (11.4.40)$$

- If $\tan(K_0 a) < K_0 a$, $a_s > 0$ (effective repulsive core).
- When the well depth reaches the threshold to bind a new bound state, $K_0 a = (n+1/2)\pi \implies \tan(K_0 a) \rightarrow \infty$. The scattering length **diverges to infinity** ($a_s \rightarrow \pm\infty$): a **zero-energy resonance!**

11.4.10 The Ramsauer–Townsend Effect

When low-energy electrons (~ 0.7 eV) scatter off heavy noble gas atoms (such as Argon or Krypton), the scattering cross section drops almost to zero: the gas becomes **completely transparent** to electrons!

This is the **Ramsauer–Townsend effect**. From the square well formula, when the energy and depth satisfy $Ka = \pi$:

$$\tan(Ka) = \tan \pi = 0 \implies \delta_0 = -ka \approx 0 \pmod{\pi}. \quad (11.4.41)$$

At this critical energy:

$$\sin^2 \delta_0 = 0 \implies \sigma_0 \approx 0. \quad (11.4.42)$$

Because higher partial waves ($l \geq 1$) are suppressed at low energies, the total cross section virtually vanishes ($\sigma \rightarrow 0$), analogous to optical transmission through an anti-reflective coating.

11.4.11 Resonance Scattering and the Breit–Wigner Formula

When a potential barrier supports a quasi-bound metastable state at energy E_R with lifetime $\tau = \hbar/\Gamma$, an incident particle can temporarily tunnel into the well and form a long-lived resonance.

Across the resonance, the phase shift $\delta_l(E)$ increases rapidly by π over energy window Γ :

$$\delta_l(E) = \delta_{\text{bg}} + \arctan\left(\frac{\Gamma/2}{E_R - E}\right). \quad (11.4.43)$$

Assuming background phase $\delta_{\text{bg}} \approx 0$, the partial cross section is:

$$\sigma_l(E) = \frac{4\pi(2l+1)}{k^2} \sin^2 \delta_l = \frac{4\pi(2l+1)}{k^2} \frac{\Gamma^2/4}{(E - E_R)^2 + \Gamma^2/4}. \quad (11.4.44)$$

This is the celebrated **Breit–Wigner resonance formula**. At peak energy $E = E_R$, the partial cross section reaches the maximum theoretical unitarity limit:

$$\sigma_l(E_R) = \frac{4\pi(2l+1)}{k^2}. \quad (11.4.45)$$

11.4.12 The Repulsive Spherical Potential Barrier

Consider the repulsive counterpart to the square well: a finite spherical barrier of height $V_0 > 0$ and radius a :

$$V(r) = \begin{cases} V_0 > 0, & r \leq a, \\ 0, & r > a. \end{cases} \quad (11.4.46)$$

For incident particles with low energy $E < V_0$, the radial s -wave equation for $\chi_0(r) = rR_0(r)$ is:

$$\frac{d^2\chi_0}{dr^2} - \alpha^2\chi_0(r) = 0 \quad (r \leq a), \quad \alpha \equiv \sqrt{\frac{2\mu(V_0 - E)}{\hbar^2}}. \quad (11.4.47)$$

Imposing the regular boundary condition at the origin $\chi_0(0) = 0$, the internal solution is hyperbolic:

$$\chi_0(r) = A \sinh(\alpha r), \quad (r \leq a). \quad (11.4.48)$$

Outside the barrier ($r > a$, where $V(r) = 0$), the solution is the free spherical wave:

$$\chi_0(r) = C \sin(kr + \delta_0), \quad k \equiv \sqrt{\frac{2\mu E}{\hbar^2}}. \quad (11.4.49)$$

Matching the logarithmic derivative at the boundary $r = a$:

$$\left. \frac{\chi'_0(r)}{\chi_0(r)} \right|_{r=a^-} = \alpha \coth(\alpha a), \quad \left. \frac{\chi'_0(r)}{\chi_0(r)} \right|_{r=a^+} = k \cot(ka + \delta_0). \quad (11.4.50)$$

Equating them gives:

$$k \cot(ka + \delta_0) = \alpha \coth(\alpha a) \implies \tan(ka + \delta_0) = \frac{k}{\alpha} \tanh(\alpha a). \quad (11.4.51)$$

Using the trigonometric angle-subtraction identity for the phase shift:

$$\tan \delta_0 = \tan[(ka + \delta_0) - ka] = \frac{\tan(ka + \delta_0) - \tan(ka)}{1 + \tan(ka + \delta_0) \tan(ka)}. \quad (11.4.52)$$

In the low-energy limit $k \rightarrow 0$ ($ka \ll 1$, $\tan(ka) \approx ka$):

$$\tan \delta_0 \approx \frac{\frac{k}{\alpha} \tanh(\alpha a) - ka}{1 + \mathcal{O}(k^2)} = ka \left[\frac{\tanh(\alpha a)}{\alpha a} - 1 \right]. \quad (11.4.53)$$

The **scattering length** a_s is:

$$a_s \equiv -\lim_{k \rightarrow 0} \frac{\tan \delta_0}{k} = a \left(1 - \frac{\tanh(\alpha a)}{\alpha a} \right) \quad (11.4.54)$$

where $\alpha \xrightarrow{k \rightarrow 0} \sqrt{2\mu V_0}/\hbar$. The low-energy total cross section is:

$$\sigma_0 = 4\pi a_s^2 = 4\pi a^2 \left(1 - \frac{\tanh(\alpha a)}{\alpha a} \right)^2 \quad (11.4.55)$$

Physical Limits:

- **Impenetrable Hard-Core Limit** ($V_0 \rightarrow \infty$): Here $\alpha a \rightarrow \infty$, so $\tanh(\alpha a) \rightarrow 1$. We obtain:

$$a_s \rightarrow a \left(1 - \frac{1}{\alpha a} \right) \xrightarrow{V_0 \rightarrow \infty} a, \quad \sigma_0 \rightarrow 4\pi a^2, \quad (11.4.56)$$

smoothly recovering the hard-sphere scattering length and cross section.

- **Weak Barrier Limit** ($V_0 \rightarrow 0$): Expanding the hyperbolic tangent for $\alpha a \ll 1$:

$$\tanh(\alpha a) \approx \alpha a - \frac{1}{3}(\alpha a)^3 + \dots \implies 1 - \frac{\tanh(\alpha a)}{\alpha a} \approx \frac{1}{3}(\alpha a)^2. \quad (11.4.57)$$

The scattering length reduces to:

$$a_s \approx \frac{1}{3}a(\alpha a)^2 = \frac{2\mu V_0 a^3}{3\hbar^2}, \quad (11.4.58)$$

which is strictly positive (repulsive) and scales linearly with V_0 , in agreement with the first Born approximation.

11.4.13 *s*-Wave Scattering from a Delta-Function Shell and the Continuity Fallacy

Consider scattering of a particle of mass μ by an infinitely thin spherical delta-function shell of radius a and coupling strength α :

$$V(r) = \alpha \delta(r - a), \quad (\alpha \in \mathbb{R}). \quad (11.4.59)$$

Mathematical Warning: The Fallacy of the Taylor Series "Proof" of Wavefunction Continuity

When evaluating boundary conditions across a singular potential such as (11.4.59), a widespread textbook fallacy consists of writing Taylor expansions from the right and left of the singularity:

$$\psi^+(x) = \psi(a) + \frac{\partial \psi}{\partial x} \Big|_{x=a^+} (x - a) + \frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} \Big|_{x=a^+} (x - a)^2 + \dots \quad (1)$$

$$\psi^-(x) = \psi(a) + \left. \frac{\partial \psi}{\partial x} \right|_{x=a^-} (x-a) + \frac{1}{2} \left. \frac{\partial^2 \psi}{\partial x^2} \right|_{x=a^-} (x-a)^2 + \dots \quad (2)$$

and arguing: “Even though the derivatives do not match at $x = a$, all higher-order terms vanish due to the $(x-a)$ factor, so $\psi^+(a) = \psi^-(a) = \psi(a)$, proving continuity.”

Why this argument is completely false (Circular Reasoning):

1. Taylor’s theorem **presupposes** that the function $\psi(x)$ is continuous and differentiable at $x = a$ in the first place! If $\psi(x)$ has an actual step discontinuity at $x = a$, the quantity $\psi(a)$ and its one-sided derivatives at a cannot even be evaluated within a standard Taylor expansion.
2. If this Taylor series argument were mathematically legitimate, one could apply the exact same series expansion to *any* discontinuous function (e.g., the Heaviside step function $\Theta(x-a)$) and falsely “prove” that it is continuous everywhere! Taking the limit $\epsilon \rightarrow 0$ in a Taylor series is illegal when the expansion point is itself a point of discontinuity.

The Rigorous Physical and Mathematical Proof of Continuity: The true proof of continuity comes from **integrating the Schrödinger equation across the singularity**:

$$\lim_{\epsilon \rightarrow 0} \int_{a-\epsilon}^{a+\epsilon} \chi_0''(r) dr = \lim_{\epsilon \rightarrow 0} \int_{a-\epsilon}^{a+\epsilon} \left[\frac{2\mu}{\hbar^2} \alpha \delta(r-a) - k^2 \right] \chi_0(r) dr. \quad (3)$$

Because $\delta(r-a)$ is a Dirac measure, the integral of the delta function produces a finite constant, while the continuous $k^2 \chi_0(r)$ term vanishes in the limit $\epsilon \rightarrow 0$:

$$\chi_0'(a^+) - \chi_0'(a^-) = \frac{2\mu\alpha}{\hbar^2} \chi_0(a). \quad (4)$$

This proves that the first derivative $\chi_0'(r)$ has at most a **finite step jump discontinuity**. Now integrate the derivative once more across $[a-\epsilon, a+\epsilon]$:

$$\chi_0(a+\epsilon) - \chi_0(a-\epsilon) = \int_{a-\epsilon}^{a+\epsilon} \chi_0'(r) dr. \quad (5)$$

Since $\chi_0'(r)$ is piecewise continuous and bounded by a finite maximum $M = \max(|\chi_0'(a^+)|, |\chi_0'(a^-)|)$ on $[a-\epsilon, a+\epsilon]$:

$$|\chi_0(a+\epsilon) - \chi_0(a-\epsilon)| \leq \int_{a-\epsilon}^{a+\epsilon} |\chi_0'(r)| dr \leq M(2\epsilon) \xrightarrow{\epsilon \rightarrow 0} 0. \quad (6)$$

Therefore:

$$\boxed{\lim_{\epsilon \rightarrow 0} \chi_0(a+\epsilon) = \lim_{\epsilon \rightarrow 0} \chi_0(a-\epsilon) = \chi_0(a)} \quad (7)$$

Wavefunction continuity is not an assumption or a consequence of Taylor expansions; it is an unavoidable consequence of the Schrödinger equation itself: a finite derivative jump guarantees that the integral across an infinitesimal interval vanishes identically.

Exact Radial Solution for the Delta Shell

The radial Schrödinger equation for the s -wave channel ($l = 0$) with $\chi_0(r) = rR_0(r)$ is:

$$\frac{d^2\chi_0}{dr^2} + k^2\chi_0(r) = \frac{2\mu\alpha}{\hbar^2}\delta(r-a)\chi_0(r). \quad (11.4.60)$$

- **Interior Region ($r < a$):** The potential vanishes and the physical boundary condition requires $\chi_0(0) = 0$:

$$\chi_0(r) = A \sin(kr), \quad (r < a). \quad (11.4.61)$$

- **Exterior Region ($r > a$):** The general free radial solution is:

$$\chi_0(r) = C \sin(kr + \delta_0), \quad (r > a). \quad (11.4.62)$$

Boundary Matching at the Shell $r = a$

Applying the two rigorous boundary conditions at $r = a$:

1. **Continuity of the Wavefunction:**

$$C \sin(ka + \delta_0) = A \sin(ka). \quad (11.4.63)$$

2. **Derivative Jump Discontinuity:**

$$\begin{aligned} \left. \frac{d\chi_0}{dr} \right|_{r=a^+} - \left. \frac{d\chi_0}{dr} \right|_{r=a^-} &= \frac{2\mu\alpha}{\hbar^2} \chi_0(a) \\ kC \cos(ka + \delta_0) - kA \cos(ka) &= \frac{2\mu\alpha}{\hbar^2} A \sin(ka). \end{aligned} \quad (11.4.64)$$

To eliminate the unknown normalization constants A and C , we divide the derivative jump equation (11.4.64) by the continuity equation (11.4.63):

$$\frac{kC \cos(ka + \delta_0)}{C \sin(ka + \delta_0)} - \frac{kA \cos(ka)}{A \sin(ka)} = \frac{\frac{2\mu\alpha}{\hbar^2} A \sin(ka)}{A \sin(ka)}. \quad (11.4.65)$$

This simplifies to:

$$k \cot(ka + \delta_0) - k \cot(ka) = \frac{2\mu\alpha}{\hbar^2}. \quad (11.4.66)$$

Solving for $\cot(ka + \delta_0)$:

$$\cot(ka + \delta_0) = \cot(ka) + \frac{2\mu\alpha}{\hbar^2 k} = \frac{k \cos(ka) + \frac{2\mu\alpha}{\hbar^2} \sin(ka)}{k \sin(ka)}. \quad (11.4.67)$$

Taking the reciprocal yields the exact expression for $\tan(ka + \delta_0)$:

$$\boxed{\tan(ka + \delta_0) = \frac{k \sin(ka)}{\frac{2\mu\alpha}{\hbar^2} \sin(ka) + k \cos(ka)}} \quad (11.4.68)$$

The exact s -wave phase shift is:

$$\delta_0(k) = \arctan \left[\frac{k \sin(ka)}{\frac{2\mu\alpha}{\hbar^2} \sin(ka) + k \cos(ka)} \right] - ka. \quad (11.4.69)$$

Low-Energy Limit and Scattering Length

In the low-energy limit ($ka \ll 1$), we expand the trigonometric functions: $\sin(ka) \approx ka$ and $\cos(ka) \approx 1$. Equation (11.4.68) becomes:

$$\tan(ka + \delta_0) \approx \frac{k(ka)}{\frac{2\mu\alpha}{\hbar^2}(ka) + k} = \frac{ka}{\frac{2\mu\alpha a}{\hbar^2} + 1}. \quad (11.4.70)$$

We define the dimensionless coupling parameter:

$$\beta \equiv \frac{2\mu\alpha a}{\hbar^2}. \quad (11.4.71)$$

Then:

$$\tan(ka + \delta_0) \approx \frac{ka}{\beta + 1}. \quad (11.4.72)$$

We now use the angle-subtraction formula $\tan \delta_0 = \tan[(ka + \delta_0) - ka]$:

$$\tan \delta_0 = \frac{\tan(ka + \delta_0) - \tan(ka)}{1 + \tan(ka + \delta_0) \tan(ka)} \approx \frac{\frac{ka}{\beta+1} - ka}{1 + \mathcal{O}(k^2 a^2)} = ka \left(\frac{1}{\beta+1} - 1 \right) = -ka \frac{\beta}{\beta+1}. \quad (11.4.73)$$

The **scattering length** a_s is:

$$a_s \equiv -\lim_{k \rightarrow 0} \frac{\tan \delta_0}{k} = a \frac{\beta}{\beta+1} = \frac{a}{1 + \frac{1}{\beta}} \quad (11.4.74)$$

The total low-energy scattering cross section is:

$$\sigma_0 = 4\pi a_s^2 = 4\pi a^2 \left(\frac{\beta}{\beta+1} \right)^2 = 4\pi \frac{\beta^2 a^2}{(\beta+1)^2} \quad (11.4.75)$$

Analysis of Physical Regimes

1. The Hard-Sphere Limit ($\alpha \rightarrow +\infty \implies \beta \rightarrow +\infty$):

$$a_s = \lim_{\beta \rightarrow \infty} \frac{a}{1 + 1/\beta} = a \implies \sigma_0 = 4\pi a^2. \quad (11.4.76)$$

An infinitely strong delta shell becomes an impenetrable spherical wall, identically reproducing the quantum hard-sphere result.

2. The Weak-Potential Limit ($\beta \ll 1$):

$$a_s \approx a\beta = \frac{2\mu\alpha a^2}{\hbar^2} \implies \sigma_0 \approx 4\pi a^2 \beta^2 = \frac{16\pi\mu^2 \alpha^2 a^4}{\hbar^4}. \quad (11.4.77)$$

The cross section scales quadratically with the potential strength α^2 , exactly matching the first Born approximation.

3. Attractive Shell ($\alpha < 0 \implies \beta < 0$) and Zero-Energy Resonance: When the potential is attractive, β is negative. As the attraction increases toward the critical value:

$$\beta = -1 \iff \alpha_c = -\frac{\hbar^2}{2\mu a}, \quad (11.4.78)$$

the denominator $(\beta + 1)$ vanishes, causing the scattering length to **diverge to infinity**:

$$a_s \longrightarrow \pm\infty. \quad (11.4.79)$$

This divergence marks a **zero-energy resonance**: at $\alpha = \alpha_c$, an attractive spherical delta shell develops its very first bound state at exactly $E = 0$!

Chapter 12

Covariant Relativistic Quantum Mechanics and Field Equations

The unification of quantum mechanics with Einstein's Special Relativity represents one of the highest summits of theoretical physics. Yet the earliest attempts to construct a relativistic single-particle wave mechanics encountered immediate, insurmountable contradictions: negative probability densities in the Klein–Gordon equation, negative energy spectra in the Dirac equation, superluminal wave packet spreading, and the Klein paradox at potential barriers.

As established by Heisenberg's uncertainty principle $\Delta x \Delta p \gtrsim \hbar$ and mass-energy equivalence $E = mc^2$, confining a particle of mass m to within its **reduced Compton wavelength**:

$$\Delta x \leq \frac{\hbar}{mc} \implies \Delta p \geq mc \implies \Delta E \sim \Delta p c \geq mc^2, \quad (12.0.1)$$

imparts sufficient quantum momentum fluctuations to create particle-antiparticle pairs out of the vacuum. Consequently, **particle number is not conserved** in relativistic quantum regimes. Any consistent relativistic quantum theory cannot describe an isolated single particle; it must describe an infinite number of degrees of freedom capable of creation and annihilation: a **relativistic quantum field theory**.

This chapter presents the complete, rigorous development of relativistic wave equations for spin-0 (Klein–Gordon) and spin-1/2 (Dirac) systems in manifest four-vector covariance, their exact algebraic structures, the Dirac adjoint and conserved currents, explicit four-spinors, Lorentz covariance in spinor space, bilinear covariants, helicity and chirality, trace technology, the Gordon decomposition, the Klein paradox, and the inevitable transition to second quantization.

12.1 Failure of Single-Particle Relativistic Mechanics

12.1.1 Non-Conservation of Particle Species and Particle Number

In non-relativistic quantum mechanics, the number of particles is strictly conserved. The Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle = \left(\frac{\hat{p}^2}{2m} + V(\hat{x}) \right) |\psi(t)\rangle, \quad (12.1.1)$$

governs a state vector $|\psi(t)\rangle$ residing in a fixed single-particle Hilbert space $\mathcal{H}_1$. The spatial probability density $\rho(\vec{r}, t) = |\psi(\vec{r}, t)|^2$ satisfies the continuity equation:

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0, \quad \vec{J} = \frac{\hbar}{2mi} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*), \quad (12.1.2)$$

ensuring that the total probability is strictly time-independent:

$$\frac{d}{dt} \int_{\mathbb{R}^3} \rho(\vec{r}, t) d^3r = - \oint_{\partial V} \vec{J} \cdot d\vec{S} = 0 \implies \int_{\mathbb{R}^3} |\psi(\vec{r}, t)|^2 d^3r = 1. \quad (12.1.3)$$

For a plane wave state $\psi(\vec{r}, t) = N e^{-i(Et - \vec{p} \cdot \vec{r})/\hbar}$, the probability density is spatially uniform, $\rho = |N|^2$ (number of particles per unit volume), and the current density is simply:

$$\vec{J} = |N|^2 \frac{\vec{p}}{m} = \rho \vec{v}, \quad (12.1.4)$$

representing a uniform flux of non-relativistic particles moving with velocity $\vec{v} = \vec{p}/m$.

However, in relativistic regimes where kinetic energies exceed rest mass energies ($E_{\text{kin}} \geq 2mc^2$), collision processes routinely produce new particles:

$$p + p \longrightarrow p + p + \pi^0 \quad (\text{Threshold kinetic energy: } E_{\text{kin}} \geq 2m_\pi c^2 \approx 280 \text{ MeV}), \quad (12.1.5)$$

$$p + p \longrightarrow p + p + p + \bar{p} \quad (\text{Threshold kinetic energy: } E_{\text{kin}} \geq 2m_p c^2 \approx 1876 \text{ MeV}). \quad (12.1.6)$$

Neither particle number nor particle species is conserved. A single-particle wave function $\psi(\vec{r}, t)$ cannot describe a physical state that evolves into a three-particle or four-particle state.

12.1.2 The Compton Wavelength as the Lower Bound of Single-Particle Localization

To see precisely when single-particle quantum mechanics breaks down, consider attempting to localize a particle of mass m inside a spatial box of linear dimension L :

$$\Delta x \sim L. \quad (12.1.7)$$

By Heisenberg's uncertainty principle, the momentum uncertainty is bounded by:

$$\Delta p \sim \frac{\hbar}{L}. \quad (12.1.8)$$

In the extreme relativistic limit, energy and momentum are related by $\Delta E \sim c\Delta p$. The energy fluctuation associated with confining the particle to size L is therefore:

$$\Delta E \sim \frac{\hbar c}{L}. \quad (12.1.9)$$

If this energy uncertainty exceeds the threshold required to create a particle-antiparticle pair of mass m :

$$\Delta E \geq 2mc^2 \implies \frac{\hbar c}{L} \gtrsim mc^2, \quad (12.1.10)$$

virtual pairs are spontaneously created out of the vacuum. Solving for the critical length scale L :

$$L \lesssim \frac{\hbar}{mc} \equiv \bar{\lambda}_c. \quad (12.1.11)$$

Here $\bar{\lambda}_c = \hbar/(mc)$ is the **reduced Compton wavelength** of the particle ($\approx 3.86 \times 10^{-13}$ m for an electron, and $\approx 0.21 \times 10^{-15}$ m for a proton).

Any measurement attempting to localize a particle to a region smaller than its Compton wavelength imparts sufficient energy to create additional particles. Instead of observing a single localized particle, one observes an indeterminate number of particles. Hence, **the concept of a single-particle position wave function $\psi(\vec{r}, t)$ is physically meaningless below $\bar{\lambda}_c$.**

12.1.3 The Myth of the Dirac "Cure": Why Dirac Theory Still Cannot Describe a Single-Particle Wavefunction

A widespread pedagogical misconception—frequently repeated in introductory quantum texts—proceeds along the following narrative:

“The Klein–Gordon equation failed because it was second order in time, leading to negative probability densities. Dirac brilliantly rescued single-particle relativistic quantum mechanics by constructing a first-order wave equation, which yielded a strictly positive probability density $\rho = \psi^\dagger \psi > 0$, thereby successfully solving the relativistic single-particle problem.”

This narrative is **fundamentally false** on both historical and physical grounds:

1. **The Negative-Energy Continuum and Vacuum Instability:** While Dirac’s first-order Hamiltonian $\hat{H} = c\vec{\alpha} \cdot \vec{p} + \beta mc^2$ yields a positive-definite probability density $\rho = \psi^\dagger \psi \geq 0$, its energy spectrum is **unbounded from below**:

$$E = \pm \sqrt{\vec{p}^2 c^2 + m^2 c^4}. \quad (12.1.12)$$

Half of the available eigenstates possess negative energies $E \leq -mc^2$. In any realistic single-particle theory where an electron couples to the quantized electromagnetic field, an electron in the ground state of a hydrogen atom ($E \approx mc^2 - 13.6$ eV) would emit an infinite cascade of photons and plunge endlessly toward $E \rightarrow -\infty$ on a timescale of $\sim 10^{-10}$ s! A single-particle relativistic system has **no stable ground state**.

2. **The Dirac Sea is an Infinite-Body Theory, Not Single-Particle Mechanics:** To prevent this catastrophic vacuum collapse, Dirac proposed his famous **Hole Theory** (1930): all negative-energy states are completely filled with an infinite, unobservable sea of electrons obeying the Pauli exclusion principle. But filling infinitely many states means that the “vacuum” contains an infinite number of particles! The moment one invokes the Dirac Sea, one has **abandoned the single-particle paradigm entirely** and adopted an infinite-body field theory in disguise.
3. **Complete Failure of the Sea Hypothesis for Bosons:** The Dirac Sea mechanism relies crucially on the Pauli exclusion principle: each negative-energy state can hold at most one electron. However, **bosons (spin-0 pions, spin-1 gauge bosons, Higgs bosons) obey Bose–Einstein statistics and cannot be excluded!** A sea of negative-energy bosons would suffer infinite pileup into negative energies. Yet relativistic spin-0 and spin-1 particles exist abundantly in nature. Therefore, the Dirac Sea cannot be a universal physical resolution.
4. **The Klein Paradox and Spontaneous Pair Production:** When a relativistic electron encounters a sharp electrostatic potential step of height $V_0 > E + mc^2$, the reflection coefficient is strictly greater than unity: $R > 1$. The incident wave does not simply reflect; it knocks electrons and positrons out of the vacuum! Single-particle probability is not conserved because particles are being created and destroyed.
5. **The Localization–Causality Conflict (Hegerfeldt’s Theorem):** If one attempts to define a relativistic single-particle theory by projecting solely onto positive-energy states ($\hat{H} = +\sqrt{\vec{p}^2 c^2 + m^2 c^4}$), the resulting theory violently violates relativistic causality: the position-space propagator $\langle \vec{x}(t) | \vec{x}_0(0) \rangle$ is strictly non-zero outside the forward light cone ($|\vec{x} - \vec{x}_0| > ct$). A positive-energy wavepacket localized within a finite volume instantly spreads over all space!

The Inevitable Transition to Quantum Field Theory

These paradoxes are not mathematical defects of the Klein–Gordon or Dirac equations; they are nature’s way of demonstrating that **there is no such thing as a relativistic single-particle wave mechanics**.

In relativity, spacetime coordinates $x^\mu = (ct, \vec{x})$ are not eigenvalues of quantum position operators; they are **continuous parameters** labeling local operator fields $\hat{\phi}(x)$ and $\hat{\psi}(x)$. A state with a particle localized at position x is defined by acting on the vacuum $|0\rangle$ with the field operator:

$$|x\rangle \equiv \hat{\phi}(x) |0\rangle \quad (\text{or } |x, s\rangle \equiv \hat{\psi}_s^\dagger(x) |0\rangle). \quad (12.1.13)$$

Projecting an arbitrary physical state $|\Psi\rangle$ residing in multi-particle Fock space onto this localized state yields:

$$\langle x|\Psi\rangle = \langle 0|\hat{\phi}(x)|\Psi\rangle. \quad (12.1.14)$$

Acting with the wave operator:

$$\left(\square + \frac{m^2 c^2}{\hbar^2}\right) \langle x|\Psi\rangle = \langle 0|\left(\square + \frac{m^2 c^2}{\hbar^2}\right) \hat{\phi}(x)|\Psi\rangle = 0. \quad (12.1.15)$$

Because this relation must hold for **any** physical state $|\Psi\rangle$, all dynamical spacetime dependence is stripped from the state vectors and transferred to the field operator itself:

$$\boxed{\left(\square + \frac{m^2 c^2}{\hbar^2}\right) \hat{\phi}(x) = 0, \quad (i\hbar\gamma^\mu \partial_\mu - mc) \hat{\psi}(x) = 0} \quad (12.1.16)$$

The Klein–Gordon and Dirac equations are **classical field equations** (analogous to Maxwell’s field equations $\partial_\mu F^{\mu\nu} = j^\nu$), which must be quantized through **Second Quantization**. Only then do the positive- and negative-frequency modes represent particle and antiparticle creation/annihilation operators, ensuring that the physical energy spectrum is strictly positive definite, causality is preserved, and dynamic pair production is naturally described.

12.2 The Klein–Gordon Equation

12.2.1 Covariant Notation and Metric Conventions

Throughout this chapter, we work in Minkowski spacetime with coordinates:

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \vec{r}) = (ct, x, y, z), \quad (12.2.1)$$

using the standard relativistic metric tensor signature $(+, -, -, -)$:

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \text{diag}(+1, -1, -1, -1). \quad (12.2.2)$$

Indices are raised and lowered via the metric: $x_\mu = g_{\mu\nu} x^\nu = (ct, -\vec{r})$. The spacetime partial derivative operators are defined as:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, \vec{\nabla}\right) = (\partial_0, \partial_1, \partial_2, \partial_3), \quad (12.2.3)$$

$$\partial^\mu \equiv g^{\mu\nu} \partial_\nu = \frac{\partial}{\partial x_\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\vec{\nabla} \right) = (\partial^0, -\vec{\nabla}). \quad (12.2.4)$$

The four-momentum vector is:

$$p^\mu = (p^0, \vec{p}) = \left(\frac{E}{c}, \vec{p} \right), \quad p_\mu = g_{\mu\nu} p^\nu = \left(\frac{E}{c}, -\vec{p} \right). \quad (12.2.5)$$

The relativistic invariant scalar product is:

$$p^\mu p_\mu = (p^0)^2 - \vec{p}^2 = \frac{E^2}{c^2} - \vec{p}^2 = m^2 c^2. \quad (12.2.6)$$

The relativistic d'Alembertian wave operator is the Lorentz scalar:

$$\square \equiv \partial_\mu \partial^\mu = g^{\mu\nu} \partial_\mu \partial_\nu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2. \quad (12.2.7)$$

12.2.2 Derivation of the Klein–Gordon Wave Equation

To construct a relativistic wave equation, we promote the four-momentum observables to differential operators acting on a single-particle state vector $|\psi\rangle$:

$$p_\mu \longrightarrow i\hbar \partial_\mu \implies \frac{E}{c} \rightarrow i\hbar \frac{1}{c} \frac{\partial}{\partial t}, \quad \vec{p} \rightarrow -i\hbar \vec{\nabla}. \quad (12.2.8)$$

The classical dispersion relation (12.2.6) is written as:

$$\frac{E^2}{c^2} = \vec{p}^2 + m^2 c^2 \implies \frac{\hat{H}^2}{c^2} |\psi\rangle = (\hat{p}^2 + m^2 c^2) |\psi\rangle. \quad (12.2.9)$$

Projecting onto the coordinate representation $\langle x|\psi\rangle = \psi(\vec{r}, t)$ yields:

$$-\frac{\hbar^2}{c^2} \frac{\partial^2 \psi}{\partial t^2} = (-\hbar^2 \nabla^2 + m^2 c^2) \psi(\vec{r}, t). \quad (12.2.10)$$

Rearranging all terms to one side:

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2 c^2}{\hbar^2} \right) \psi(\vec{r}, t) = 0. \quad (12.2.11)$$

In manifest four-vector covariant notation, this is the **Klein–Gordon equation** (derived independently by Schrödinger, Gordon, Klein, and Fock in 1926):

$$\boxed{\left(\partial_\mu \partial^\mu + \frac{m^2 c^2}{\hbar^2} \right) \psi(x) = 0} \quad (12.2.12)$$

In natural units ($\hbar = c = 1$), this takes the elegant form:

$$(\square + m^2) \psi(x) = 0. \quad (12.2.13)$$

12.2.3 Conserved Relativistic Four-Current Density

To investigate whether $\psi(x)$ can be interpreted as a probability amplitude, we derive its continuity equation from first principles. Consider a complex scalar field $\psi(x)$ satisfying (12.2.12) (setting $\hbar = c = 1$ for intermediate algebra):

$$\frac{\partial^2 \psi}{\partial t^2} - \nabla^2 \psi + m^2 \psi = 0. \quad (12.2.14)$$

Taking the complex conjugate of (12.2.14):

$$\frac{\partial^2 \psi^*}{\partial t^2} - \nabla^2 \psi^* + m^2 \psi^* = 0. \quad (12.2.15)$$

Multiplying (12.2.14) from the left by ψ^* and (12.2.15) from the left by ψ :

$$\psi^* \frac{\partial^2 \psi}{\partial t^2} - \psi^* \nabla^2 \psi + m^2 \psi^* \psi = 0, \quad (12.2.16)$$

$$\psi \frac{\partial^2 \psi^*}{\partial t^2} - \psi \nabla^2 \psi^* + m^2 \psi \psi^* = 0. \quad (12.2.17)$$

Subtracting the second equation from the first cancels the mass term $m^2|\psi|^2$:

$$\left(\psi^* \frac{\partial^2 \psi}{\partial t^2} - \psi \frac{\partial^2 \psi^*}{\partial t^2} \right) - \left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right) = 0. \quad (12.2.18)$$

Using the product rule for derivatives:

$$\frac{\partial}{\partial t} \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right) = \psi^* \frac{\partial^2 \psi}{\partial t^2} + \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial t} - \psi \frac{\partial^2 \psi^*}{\partial t^2} - \frac{\partial \psi}{\partial t} \frac{\partial \psi^*}{\partial t} = \psi^* \frac{\partial^2 \psi}{\partial t^2} - \psi \frac{\partial^2 \psi^*}{\partial t^2}, \quad (12.2.19)$$

and similarly for the spatial Laplacian:

$$\vec{\nabla} \cdot \left(\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^* \right) = \psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*. \quad (12.2.20)$$

Therefore:

$$\frac{\partial}{\partial t} \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right) - \vec{\nabla} \cdot \left(\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^* \right) = 0. \quad (12.2.21)$$

Multiplying by the normalization factor $\frac{i\hbar}{2m}$ (reinserting physical dimensions):

$$\frac{\partial}{\partial t} \underbrace{\left[\frac{i\hbar}{2mc^2} \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right) \right]}_{\rho(x)} + \vec{\nabla} \cdot \underbrace{\left[-\frac{i\hbar}{2m} \left(\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^* \right) \right]}_{\vec{J}(x)} = 0. \quad (12.2.22)$$

In manifest four-vector form, we define the **Klein–Gordon four-current**:

$$j^\mu(x) \equiv \frac{i\hbar}{2m} (\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*), \quad \partial_\mu j^\mu = 0 \quad (12.2.23)$$

where $j^0 = c\rho$ and $\vec{j} = \vec{J}$.

12.2.4 The Negative Probability Density Crisis

Evaluating the probability density $\rho(x)$ for a relativistic free plane wave solution:

$$\psi(x) = N e^{-i(Et - \vec{p} \cdot \vec{r})/\hbar} = N e^{-ip_\mu x^\mu/\hbar}, \quad (12.2.24)$$

we compute the temporal derivatives:

$$\frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar} N e^{-ip \cdot x/\hbar} = -\frac{iE}{\hbar} \psi, \quad (12.2.25)$$

$$\frac{\partial \psi^*}{\partial t} = +\frac{iE}{\hbar} N^* e^{+ip \cdot x/\hbar} = +\frac{iE}{\hbar} \psi^*. \quad (12.2.26)$$

Substituting into (12.2.22):

$$\begin{aligned}\rho(x) &= \frac{i\hbar}{2mc^2} \left[\psi^* \left(-\frac{iE}{\hbar} \psi \right) - \psi \left(\frac{iE}{\hbar} \psi^* \right) \right] \\ &= \frac{i\hbar}{2mc^2} \left(-\frac{2iE}{\hbar} |\psi|^2 \right) = \frac{E}{mc^2} |\psi|^2 = \frac{E}{mc^2} |N|^2.\end{aligned}\quad (12.2.27)$$

Because the Klein–Gordon dispersion relation is quadratic ($E^2 = \vec{p}^2 c^2 + m^2 c^4$), it admits two mathematical branches of solutions:

$$E = \pm \sqrt{\vec{p}^2 c^2 + m^2 c^4}. \quad (12.2.28)$$

- For the **positive-energy branch** ($E > 0$): $\rho = \frac{|E|}{mc^2} |N|^2 > 0$.
- For the **negative-energy branch** ($E < 0$): $\rho = -\frac{|E|}{mc^2} |N|^2 < 0$.

A negative probability density is physically nonsensical in single-particle quantum mechanics. Furthermore, because the Klein–Gordon equation is second order in time, $\psi(\vec{r}, 0)$ and $\partial\psi(\vec{r}, 0)/\partial t$ constitute completely independent Cauchy initial data. One can arbitrarily choose initial configurations such that $\rho(\vec{r}, 0)$ is positive in one region of space and negative in another!

12.2.5 The Pauli–Weisskopf Reinterpretation

In 1934, Wolfgang Pauli and Victor Weisskopf resolved this fundamental dilemma by demonstrating that $\psi(x)$ is not a single-particle Schrödinger probability wave, but a classical field that must be quantized as an operator. Multiplying the four-current by the elementary electric charge e :

$$j_{\text{EM}}^\mu(x) \equiv e j^\mu(x) = \frac{ie\hbar}{2m} (\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*), \quad \partial_\mu j_{\text{EM}}^\mu = 0. \quad (12.2.29)$$

Here, $\rho_{\text{EM}} = j_{\text{EM}}^0/c$ is the **conserved electromagnetic charge density**, not a probability density. A positive value ($\rho_{\text{EM}} > 0$) represents a region dominated by positively charged bosons (such as π^+ mesons), while a negative value ($\rho_{\text{EM}} < 0$) represents a region dominated by negatively charged antiparticles (π^- mesons).

12.2.6 Classical Field Equations and the Emergence of the Yukawa Potential

To understand how the classical Klein–Gordon field mediates physical forces between particles, we couple the real scalar field $\phi(x)$ to an external static source density $\rho(\vec{r})$. Working in natural units ($\hbar = c = 1$) for clarity, consider the Lagrangian density:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \rho(x) \phi(x). \quad (12.2.30)$$

Applying the Euler–Lagrange field equations:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \implies \partial_\mu \partial^\mu \phi + m^2 \phi = \rho(x), \quad (12.2.31)$$

which is the inhomogeneous Klein–Gordon equation:

$$(\square + m^2) \phi(x) = \rho(x). \quad (12.2.32)$$

The Static Field Equation and Green's Function

In the static, time-independent limit where the source is stationary ($\partial_t \phi = 0$ and $\square = -\nabla^2$):

$$(-\nabla^2 + m^2)\phi(\vec{r}) = \rho(\vec{r}) \implies (\nabla^2 - m^2)\phi(\vec{r}) = -\rho(\vec{r}). \quad (12.2.33)$$

Equation (12.2.33) is the **screened Poisson equation**. The static Green's function $G(\vec{r} - \vec{r}')$ represents the scalar field configuration produced by a point source of unit strength located at $\vec{r}'$:

$$(\nabla^2 - m^2)G(\vec{r}) = -\delta^3(\vec{r}). \quad (12.2.34)$$

Solving by 3D Fourier transform:

$$G(\vec{r}) = \int \frac{d^3q}{(2\pi)^3} \frac{e^{i\vec{q}\cdot\vec{r}}}{q^2 + m^2} = \frac{1}{4\pi^2 i r} \int_{-\infty}^{\infty} \frac{q e^{iqr}}{q^2 + m^2} dq. \quad (12.2.35)$$

Closing the contour in the upper half-plane around the simple pole at $q = +im$, the residue theorem yields:

$$\boxed{G(\vec{r}) = \frac{1}{4\pi r} e^{-mr}} \quad (12.2.36)$$

Reinserting physical units with $\mu \equiv mc/\hbar = 1/\bar{\lambda}_c$:

$$G(\vec{r}) = \frac{1}{4\pi r} e^{-\mu r}. \quad (12.2.37)$$

For a point particle of coupling strength g_1 located at the origin with source density $\rho_1(\vec{r}) = g_1 \delta^3(\vec{r})$, the resulting static scalar field is:

$$\phi_1(\vec{r}) = g_1 G(\vec{r}) = \frac{g_1}{4\pi r} e^{-\mu r}. \quad (12.2.38)$$

Emergence of the Two-Body Interaction Potential Energy

Crucially, the solution $\phi_1(\vec{r})$ of the field equation of motion is the **scalar field configuration**, not a potential energy.

The physical potential energy $V(\vec{r})$ emerges only when a second particle with source density $\rho_2(\vec{x}) = g_2 \delta^3(\vec{x} - \vec{r})$ is placed inside this scalar field. The interaction Hamiltonian is:

$$H_{\text{int}} = - \int d^3x \mathcal{L}_{\text{int}} = - \int d^3x \rho_2(\vec{x}) \phi_1(\vec{x}) = -g_2 \phi_1(\vec{r}). \quad (12.2.39)$$

Substituting the field generated by the first particle yields the mutual **interaction potential energy**:

$$\boxed{V(r) = -g_2 \phi_1(r) = -\frac{g_1 g_2}{4\pi} \frac{e^{-\mu r}}{r} = -\frac{g_1 g_2}{4\pi r} e^{-mcr/\hbar}} \quad (12.2.40)$$

This clarifies the conceptual chain:

1. The equation of motion $(\square + m^2)\phi = \rho$ determines the **spacetime scalar field** $\phi(x)$.
2. The static Green's function determines the **field configuration** $\phi_1(\vec{r}) = \frac{g_1}{4\pi r} e^{-\mu r}$ sourced by a charge.
3. The interaction energy between two static sources mediated by this scalar exchange defines the **Yukawa potential energy** $V(r)$.
4. The minus sign in $H_{\text{int}} = -g_2 \phi_1$ demonstrates that the exchange of a spin-0 scalar particle produces a force that is **universally attractive** between like scalar charges ($g_1 g_2 > 0$).

Hideki Yukawa (1935) used this exact mechanism to model the strong nuclear force binding nucleons: the finite range of nuclear forces ($R \sim 1.4$ fm) corresponds to the exchange of a massive scalar boson whose Compton wavelength is $\bar{\lambda}_c = \hbar/(mc) \sim 1.4$ fm, predicting the existence of the pion (π^0) with mass $m \approx 140$ MeV/ c^2 .

12.3 The Dirac Equation from First Principles

12.3.1 Dirac's Linearization Ansatz and Modern Framing

Driven by the desire to eliminate the negative probability densities that plagued the single-particle interpretation of the Klein–Gordon equation, Paul Dirac (1928) searched for a wave equation that was **strictly first order in time**:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi. \quad (12.3.1)$$

To preserve relativistic Lorentz covariance, temporal and spatial coordinates must appear symmetrically on strictly equal footing. Because the temporal derivative appears only to first order, the spatial gradient operators must likewise enter only linearly:

$$\hat{H} = c\vec{\alpha} \cdot \hat{\vec{p}} + \beta mc^2 = c \sum_{i=1}^3 \alpha_i \hat{p}^i + \beta mc^2, \quad (12.3.2)$$

where $\hat{p}^i = -i\hbar \partial / \partial x^i = -i\hbar \partial_i$. Here, $\alpha_1, \alpha_2, \alpha_3$ and β cannot be ordinary commuting c -numbers; they must be linear operators (matrices) acting on a multicomponent spinor ψ .

Conceptual Framing: Field Equation vs. Wavefunction

While Dirac originally conceived $\psi(\vec{r}, t)$ as a single-particle probability wave (analogous to the non-relativistic Schrödinger wavefunction), we must emphasize from the outset that this single-particle interpretation is physically unsustainable:

- The Hamiltonian (12.3.2) inevitably possesses negative-energy eigenvalues $E \leq -mc^2$, destabilizing any single-particle system.
- The Dirac equation is correctly understood as the **classical field equation for relativistic spin-1/2 matter** (the fermionic counterpart of Maxwell's equations for the electromagnetic field).
- The transition to physical predictive power occurs when $\psi(x)$ is elevated to an **operator-valued quantum field** acting on Fock space, where negative frequencies describe the creation and annihilation of antiparticles (positrons) carrying positive physical energy.

We now derive the exact algebraic constraints governing α_i and β .

12.3.2 Clifford Constraints on α_i and β

For a free relativistic particle, any physical state must satisfy the relativistic energy-momentum dispersion relation $\hat{H}^2 = \hat{p}^2 c^2 + m^2 c^4$. Squaring the Hamiltonian (12.3.2):

$$\begin{aligned} \hat{H}^2 &= (c\alpha_i \hat{p}^i + \beta mc^2) (c\alpha_j \hat{p}^j + \beta mc^2) \\ &= c^2 \alpha_i \alpha_j \hat{p}^i \hat{p}^j + mc^3 (\alpha_i \beta + \beta \alpha_i) \hat{p}^i + \beta^2 m^2 c^4. \end{aligned} \quad (12.3.3)$$

Using the symmetry of the momentum product $\hat{p}^i \hat{p}^j = \hat{p}^j \hat{p}^i$, we write $\alpha_i \alpha_j \hat{p}^i \hat{p}^j = \frac{1}{2}(\alpha_i \alpha_j + \alpha_j \alpha_i) \hat{p}^i \hat{p}^j$. Therefore:

$$\hat{H}^2 = \frac{c^2}{2} (\alpha_i \alpha_j + \alpha_j \alpha_i) \hat{p}^i \hat{p}^j + mc^3 (\alpha_i \beta + \beta \alpha_i) \hat{p}^i + \beta^2 m^2 c^4. \quad (12.3.4)$$

Comparing this directly to the required relation:

$$\hat{H}^2 = c^2 \sum_{i=1}^3 (\hat{p}^i)^2 I + m^2 c^4 I = c^2 \delta_{ij} \hat{p}^i \hat{p}^j I + m^2 c^4 I, \quad (12.3.5)$$

we find that the coefficients α_i and β must satisfy the fundamental algebraic conditions:

$$\boxed{\alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} I, \quad \alpha_i \beta + \beta \alpha_i = 0, \quad \beta^2 = I} \quad (12.3.6)$$

Furthermore, to ensure that the Hamiltonian $\hat{H}$ is self-adjoint ($\hat{H}^\dagger = \hat{H}$), the matrices must be Hermitian:

$$\alpha_i^\dagger = \alpha_i, \quad \beta^\dagger = \beta. \quad (12.3.7)$$

12.3.3 Properties of α_i and β Matrices

From the algebraic relations (12.3.6), several crucial properties follow:

1. **Eigenvalues are ± 1 :** Because $\alpha_i^2 = I$ and $\beta^2 = I$, their eigenvalues λ satisfy $\lambda^2 = 1 \implies \lambda = \pm 1$.
2. **Matrices are strictly traceless:** Using $\beta^2 = I \implies \beta = \beta^{-1}$ and the anticommutator $\alpha_i \beta = -\beta \alpha_i$:

$$\text{Tr}(\alpha_i) = \text{Tr}(\alpha_i \beta \beta^{-1}) = \text{Tr}(\beta^{-1} \alpha_i \beta) = \text{Tr}(-\beta^{-1} \beta \alpha_i) = -\text{Tr}(\alpha_i) \implies \text{Tr}(\alpha_i) = 0. \quad (12.3.8)$$

Similarly, using $(\alpha_i)^2 = I$:

$$\text{Tr}(\beta) = \text{Tr}(\beta \alpha_i \alpha_i^{-1}) = \text{Tr}(\alpha_i^{-1} \beta \alpha_i) = -\text{Tr}(\alpha_i^{-1} \alpha_i \beta) = -\text{Tr}(\beta) \implies \text{Tr}(\beta) = 0. \quad (12.3.9)$$

3. **Matrix dimension must be even:** Since the eigenvalues are ± 1 and the sum of eigenvalues (the trace) is zero, there must be an equal number of $+1$ and -1 eigenvalues. Thus, the dimension N must be even: $N = 2, 4, 6, \dots$

12.3.4 Why 2×2 Matrices Fail: The Ashok Das Theorem

Can we satisfy these conditions with 2×2 matrices ($N = 2$)? As proven on page 22 of Ashok Das (*Lectures on Quantum Mechanics*), any arbitrary 2×2 complex matrix M can be uniquely decomposed into the basis of the identity matrix I_2 and the three Pauli matrices $\vec{\sigma}$:

$$M = c_0 I_2 + c_1 \sigma_x + c_2 \sigma_y + c_3 \sigma_z = c_0 I_2 + \vec{c} \cdot \vec{\sigma}. \quad (12.3.10)$$

The Pauli matrices satisfy:

$$\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} I_2. \quad (12.3.11)$$

There are only **three** mutually anticommuting 2×2 matrices ($\sigma_1, \sigma_2, \sigma_3$). If we identify $\alpha_i = \sigma_i$, there is no fourth anticommuting 2×2 matrix available for β . Any fourth matrix M commuting or anticommuting with all three must be proportional to the identity I_2 , which cannot anticommute with σ_i .

Therefore, 2×2 matrices cannot represent the Dirac algebra. The minimal dimension for a relativistic spin-1/2 particle is:

$$N = 4. \quad (12.3.12)$$

12.3.5 Explicit Construction of the 4×4 Dirac–Pauli Representation

We choose a basis where β is diagonal. Since its eigenvalues are $+1, +1, -1, -1$:

$$\beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (12.3.13)$$

We write α_i in general 2×2 block form:

$$\alpha_i = \begin{pmatrix} A_i & B_i \\ B_i^\dagger & C_i \end{pmatrix}, \quad (12.3.14)$$

where Hermiticity requires $A_i^\dagger = A_i$ and $C_i^\dagger = C_i$. Evaluating the anticommutator with β :

$$\begin{aligned} \alpha_i \beta + \beta \alpha_i &= \begin{pmatrix} A_i & B_i \\ B_i^\dagger & C_i \end{pmatrix} \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix} + \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix} \begin{pmatrix} A_i & B_i \\ B_i^\dagger & C_i \end{pmatrix} \\ &= \begin{pmatrix} A_i & -B_i \\ B_i^\dagger & -C_i \end{pmatrix} + \begin{pmatrix} A_i & B_i \\ -B_i^\dagger & -C_i \end{pmatrix} = \begin{pmatrix} 2A_i & 0 \\ 0 & -2C_i \end{pmatrix} = 0. \end{aligned} \quad (12.3.15)$$

Thus:

$$A_i = 0, \quad C_i = 0. \quad (12.3.16)$$

The condition $\alpha_i^2 = I_4$ requires:

$$\begin{pmatrix} 0 & B_i \\ B_i^\dagger & 0 \end{pmatrix} \begin{pmatrix} 0 & B_i \\ B_i^\dagger & 0 \end{pmatrix} = \begin{pmatrix} B_i B_i^\dagger & 0 \\ 0 & B_i^\dagger B_i \end{pmatrix} = \begin{pmatrix} I_2 & 0 \\ 0 & I_2 \end{pmatrix} \implies B_i B_i^\dagger = I_2. \quad (12.3.17)$$

Furthermore, $\alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} I_4$ requires $B_i B_j^\dagger + B_j B_i^\dagger = 2\delta_{ij} I_2$. The canonical standard choice is $B_i = B_i^\dagger = \sigma_i$.

This establishes the standard **Dirac–Pauli representation**:

$$\boxed{\beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \quad (i = 1, 2, 3)} \quad (12.3.18)$$

12.4 The Covariant Dirac Equation and Clifford Algebra

12.4.1 Definition of Gamma Matrices

To cast Dirac's equation into manifestly covariant form, we multiply (12.3.2) from the left by β/c :

$$\frac{\beta}{c} \left(i\hbar \frac{\partial \psi}{\partial t} \right) = \beta \left(\sum_{i=1}^3 \alpha_i \hat{p}^i + \beta mc \right) \psi. \quad (12.4.1)$$

Using $\hat{p}^i = -i\hbar \partial_i = i\hbar \partial^i$ and $\beta/c \partial_t = \beta \partial_0$:

$$i\hbar \beta \partial_0 \psi + i\hbar \beta \alpha_i \partial^i \psi - mc \beta^2 \psi = 0. \quad (12.4.2)$$

Using $\beta^2 = I_4$ and noting that $\partial_i = -\partial^i$, this becomes:

$$(i\hbar \beta \partial_0 + i\hbar \beta \alpha_i \partial_i - mc) \psi = 0. \quad (12.4.3)$$

We define the four **Dirac gamma matrices**:

$$\gamma^0 \equiv \beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^i \equiv \beta \alpha_i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \quad (i = 1, 2, 3) \quad (12.4.4)$$

With this definition, the Dirac equation takes the manifestly covariant form:

$$(i\hbar\gamma^\mu\partial_\mu - mc)\psi(x) = 0 \quad (12.4.5)$$

In Feynman slash notation ($\not{A} \equiv \gamma^\mu A_\mu = \gamma^0 A^0 - \vec{\gamma} \cdot \vec{A}$):

$$(i\hbar\not{\partial} - mc)\psi(x) = 0 \quad \text{or} \quad (\not{p} - mc)\psi = 0. \quad (12.4.6)$$

12.4.2 Direct Derivation of the Clifford Algebra

We now verify the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I_4$ directly from the definitions:

1. Temporal square:

$$(\gamma^0)^2 = \beta^2 = I_4 = g^{00}I_4. \quad (12.4.7)$$

2. Spatial squares:

$$(\gamma^i)^2 = (\beta\alpha_i)(\beta\alpha_i) = \beta(\alpha_i\beta)\alpha_i = \beta(-\beta\alpha_i)\alpha_i = -\beta^2\alpha_i^2 = -(I_4)(I_4) = -I_4 = g^{ii}I_4. \quad (12.4.8)$$

3. Spatial anticommutators ($i \neq j$):

$$\begin{aligned} \gamma^i\gamma^j + \gamma^j\gamma^i &= (\beta\alpha_i)(\beta\alpha_j) + (\beta\alpha_j)(\beta\alpha_i) \\ &= -\beta^2\alpha_i\alpha_j - \beta^2\alpha_j\alpha_i = -(\alpha_i\alpha_j + \alpha_j\alpha_i) = -2\delta_{ij}I_4 = 2g^{ij}I_4. \end{aligned} \quad (12.4.9)$$

4. Mixed temporal-spatial anticommutators:

$$\gamma^0\gamma^i + \gamma^i\gamma^0 = \beta(\beta\alpha_i) + (\beta\alpha_i)\beta = \beta^2\alpha_i + \beta(-\alpha_i\beta) = \alpha_i - \beta^2\alpha_i = \alpha_i - \alpha_i = 0 = 2g^{0i}I_4. \quad (12.4.10)$$

Combining all four relations yields the foundational **Clifford algebra**:

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}I_4 \quad (12.4.11)$$

12.4.3 Hermitian Conjugation of Gamma Matrices

From (12.4.4), we evaluate the Hermitian conjugate of each gamma matrix:

$$(\gamma^0)^\dagger = \beta^\dagger = \beta = \gamma^0, \quad (12.4.12)$$

$$(\gamma^i)^\dagger = (\beta\alpha_i)^\dagger = \alpha_i^\dagger\beta^\dagger = \alpha_i\beta = -\beta\alpha_i = -\gamma^i. \quad (12.4.13)$$

Thus, γ^0 is Hermitian while the spatial matrices $\gamma^1, \gamma^2, \gamma^3$ are anti-Hermitian. Using $\gamma^0\gamma^0 = I_4$ and $\gamma^0\gamma^i = -\gamma^i\gamma^0$, we express this in a single covariant identity:

$$(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0 \quad (12.4.14)$$

12.4.4 The Dirac Adjoint and Strictly Positive Probability Density

Taking the Hermitian conjugate of the covariant Dirac equation (12.4.5):

$$[(i\hbar\gamma^\mu\partial_\mu - mc)\psi]^\dagger = -i\hbar(\partial_\mu\psi^\dagger)(\gamma^\mu)^\dagger - mc\psi^\dagger = 0. \quad (12.4.15)$$

Using $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$:

$$-i\hbar(\partial_\mu\psi^\dagger)\gamma^0\gamma^\mu\gamma^0 - mc\psi^\dagger = 0. \quad (12.4.16)$$

Multiplying from the right by γ^0 and using $(\gamma^0)^2 = I_4$:

$$-i\hbar(\partial_\mu\psi^\dagger)\gamma^0\gamma^\mu - mc\psi^\dagger\gamma^0 = 0. \quad (12.4.17)$$

We define the **Dirac adjoint spinor**:

$$\boxed{\bar{\psi}(x) \equiv \psi^\dagger(x)\gamma^0} \quad (12.4.18)$$

The adjoint Dirac equation then reads:

$$\boxed{i\hbar\partial_\mu\bar{\psi}\gamma^\mu + mc\bar{\psi} = 0 \quad \Longleftrightarrow \quad \bar{\psi}\left(i\hbar\overleftarrow{\partial} + mc\right) = 0} \quad (12.4.19)$$

where $\overleftarrow{\partial}_\mu$ acts on the bra to its left.

To derive the conserved current, we multiply (12.4.5) from the left by $\bar{\psi}$ and (12.4.19) from the right by ψ :

$$i\hbar\bar{\psi}\gamma^\mu(\partial_\mu\psi) - mc\bar{\psi}\psi = 0, \quad (12.4.20)$$

$$i\hbar(\partial_\mu\bar{\psi})\gamma^\mu\psi + mc\bar{\psi}\psi = 0. \quad (12.4.21)$$

Adding these two equations:

$$i\hbar\left[\bar{\psi}\gamma^\mu(\partial_\mu\psi) + (\partial_\mu\bar{\psi})\gamma^\mu\psi\right] = 0 \implies \partial_\mu(\bar{\psi}\gamma^\mu\psi) = 0. \quad (12.4.22)$$

We define the **Dirac four-current**:

$$\boxed{j^\mu \equiv c\bar{\psi}\gamma^\mu\psi, \quad \partial_\mu j^\mu = 0} \quad (12.4.23)$$

The time component $j^0 = c\rho$ defines the spatial probability density:

$$\rho(x) = \frac{j^0}{c} = \bar{\psi}\gamma^0\psi = \psi^\dagger\gamma^0\gamma^0\psi = \psi^\dagger\psi = \sum_{a=1}^4 |\psi_a(x)|^2. \quad (12.4.24)$$

Because $\rho(x)$ is a sum of absolute squares, **the Dirac probability density is strictly non-negative**:

$$\boxed{\rho(\vec{r}, t) \geq 0} \quad (12.4.25)$$

everywhere in space and at all times, overcoming the negative-density catastrophe of the Klein–Gordon equation.

12.5 Rest Frame Solutions and Pauli Spin Algebra

12.5.1 Solutions at Rest ($\vec{p} = 0$)

For a particle at rest ($\vec{p} = 0$), the spatial derivatives vanish ($\vec{\nabla}\psi = 0$), and the Dirac Hamiltonian simplifies to:

$$\hat{H}\psi = \beta mc^2\psi = E\psi. \quad (12.5.1)$$

In the Dirac–Pauli representation:

$$\begin{pmatrix} mc^2 I_2 & 0 \\ 0 & -mc^2 I_2 \end{pmatrix} \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = E \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}. \quad (12.5.2)$$

This matrix yields four orthogonal eigenvectors:

- **Positive energy** ($E = +mc^2$):

$$\omega^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \omega^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}. \quad (12.5.3)$$

These correspond to an electron at rest with spin up ($s_z = +\hbar/2$) and spin down ($s_z = -\hbar/2$).

- **Negative energy** ($E = -mc^2$):

$$\omega^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \omega^{(4)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (12.5.4)$$

These correspond to the negative-energy states that Dirac reinterpreted as holes in the negative-energy sea (positrons).

12.5.2 Explicit Matrix Verification: $(\vec{\sigma} \cdot \vec{p})^2 = \vec{p}^2 I_2$

We verify the Pauli spin identity $(\vec{\sigma} \cdot \vec{p})^2 = \vec{p}^2 I_2$ by direct 2×2 matrix multiplication. Writing:

$$\begin{aligned} \vec{\sigma} \cdot \vec{p} &= \sigma_x p_x + \sigma_y p_y + \sigma_z p_z \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} p_x + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} p_y + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} p_z \\ &= \begin{pmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{pmatrix} \equiv \begin{pmatrix} p_z & p_- \\ p_+ & -p_z \end{pmatrix} \end{aligned} \quad (12.5.5)$$

where $p_{\pm} \equiv p_x \pm ip_y$. Squaring this matrix:

$$\begin{aligned} (\vec{\sigma} \cdot \vec{p})^2 &= \begin{pmatrix} p_z & p_- \\ p_+ & -p_z \end{pmatrix} \begin{pmatrix} p_z & p_- \\ p_+ & -p_z \end{pmatrix} \\ &= \begin{pmatrix} p_z^2 + p_- p_+ & p_z p_- - p_- p_z \\ p_+ p_z - p_z p_+ & p_+ p_- + p_z^2 \end{pmatrix}. \end{aligned} \quad (12.5.6)$$

Evaluating the products:

$$p_- p_+ = (p_x - ip_y)(p_x + ip_y) = p_x^2 + p_y^2, \quad (12.5.7)$$

$$p_+ p_- = (p_x + ip_y)(p_x - ip_y) = p_x^2 + p_y^2. \quad (12.5.8)$$

The off-diagonal elements vanish identically because c -numbers commute ($p_z p_- - p_- p_z = 0$). Thus:

$$(\vec{\sigma} \cdot \vec{p})^2 = \begin{pmatrix} p_z^2 + p_x^2 + p_y^2 & 0 \\ 0 & p_x^2 + p_y^2 + p_z^2 \end{pmatrix} = \begin{pmatrix} \vec{p}^2 & 0 \\ 0 & \vec{p}^2 \end{pmatrix} = \vec{p}^2 I_2. \quad (12.5.9)$$

12.5.3 Matrix Decomposition Exercise

Consider decomposing an arbitrary Hermitian 2×2 matrix into the Pauli basis:

$$M = \begin{pmatrix} z & x - iy \\ x + iy & t \end{pmatrix}. \quad (12.5.10)$$

Expanding in terms of elementary matrix units:

$$M = z \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + t \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (12.5.11)$$

Noting that:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{I_2 + \sigma_z}{2}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I_2 - \sigma_z}{2}, \quad (12.5.12)$$

we obtain:

$$\begin{aligned} M &= z \left(\frac{I_2 + \sigma_z}{2} \right) + t \left(\frac{I_2 - \sigma_z}{2} \right) + x \sigma_x + y \sigma_y \\ &= \left(\frac{z+t}{2} \right) I_2 + x \sigma_x + y \sigma_y + \left(\frac{z-t}{2} \right) \sigma_z. \end{aligned} \quad (12.5.13)$$

12.6 General Plane Wave Solutions and Explicit Spinors

12.6.1 Coupled Block Equations

We seek free plane wave solutions of the covariant Dirac equation $(\not{p} - mc)\psi = 0$:

$$\psi(x) = \omega(\vec{p}) e^{-ip \cdot x / \hbar} = \begin{pmatrix} \omega_A \\ \omega_B \end{pmatrix} e^{-i(Et - \vec{p} \cdot \vec{r}) / \hbar}, \quad (12.6.1)$$

where ω_A and ω_B are two-component spinors. In the Dirac–Pauli representation:

$$\not{p} = \gamma^0 p^0 - \vec{\gamma} \cdot \vec{p} = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix} \frac{E}{c} - \begin{pmatrix} 0 & \vec{\sigma} \cdot \vec{p} \\ -\vec{\sigma} \cdot \vec{p} & 0 \end{pmatrix} = \begin{pmatrix} E/c & -\vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -E/c \end{pmatrix}. \quad (12.6.2)$$

The Dirac equation $(\not{p} - mc)\omega = 0$ becomes:

$$\begin{pmatrix} \frac{E}{c} - mc & -\vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -\frac{E}{c} - mc \end{pmatrix} \begin{pmatrix} \omega_A \\ \omega_B \end{pmatrix} = 0. \quad (12.6.3)$$

Multiplying by c , this yields the coupled linear system:

$$(E - mc^2)\omega_A - c(\vec{\sigma} \cdot \vec{p})\omega_B = 0 \implies \omega_A = \frac{c(\vec{\sigma} \cdot \vec{p})}{E - mc^2}\omega_B, \quad (12.6.4)$$

$$c(\vec{\sigma} \cdot \vec{p})\omega_A - (E + mc^2)\omega_B = 0 \implies \omega_B = \frac{c(\vec{\sigma} \cdot \vec{p})}{E + mc^2}\omega_A. \quad (12.6.5)$$

Substituting (12.6.5) into (12.6.4):

$$\omega_A = \frac{c^2(\vec{\sigma} \cdot \vec{p})^2}{E^2 - m^2c^4}\omega_A = \frac{c^2\vec{p}^2}{E^2 - m^2c^4}\omega_A. \quad (12.6.6)$$

For non-trivial solutions ($\omega_A \neq 0$), the scalar coefficient must equal unity:

$$\frac{c^2\vec{p}^2}{E^2 - m^2c^4} = 1 \implies E^2 = c^2\vec{p}^2 + m^2c^4 \implies E = \pm\sqrt{c^2\vec{p}^2 + m^2c^4} \equiv \pm E_p. \quad (12.6.7)$$

12.6.2 Positive-Energy Spinors $u^{(s)}(\vec{p})$

For $E = +E_p > 0$, the upper component ω_A dominates in the non-relativistic limit ($\omega_B \rightarrow 0$ as $c \rightarrow \infty$). Choosing the standard orthonormal Pauli two-spinors:

$$\chi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (12.6.8)$$

we set $\omega_A = \chi^{(s)}$. The lower component is then determined by (12.6.5):

$$\omega_B = \frac{c(\vec{\sigma} \cdot \vec{p})}{E_p + mc^2}\chi^{(s)}. \quad (12.6.9)$$

Using $\vec{\sigma} \cdot \vec{p} = \begin{pmatrix} p_z & p_- \\ p_+ & -p_z \end{pmatrix}$:

$$(\vec{\sigma} \cdot \vec{p})\chi^{(1)} = \begin{pmatrix} p_z & p_- \\ p_+ & -p_z \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p_z \\ p_+ \end{pmatrix}, \quad (12.6.10)$$

$$(\vec{\sigma} \cdot \vec{p})\chi^{(2)} = \begin{pmatrix} p_z & p_- \\ p_+ & -p_z \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} p_- \\ -p_z \end{pmatrix}. \quad (12.6.11)$$

The two positive-energy four-spinors are therefore:

$$u^{(1)}(\vec{p}) = N \begin{pmatrix} 1 \\ 0 \\ \frac{cp_z}{E_p + mc^2} \\ \frac{cp_+}{E_p + mc^2} \end{pmatrix}, \quad u^{(2)}(\vec{p}) = N \begin{pmatrix} 0 \\ 1 \\ \frac{cp_-}{E_p + mc^2} \\ \frac{-cp_z}{E_p + mc^2} \end{pmatrix} \quad (12.6.12)$$

12.6.3 Negative-Energy Spinors and Physical Antiparticle Spinors $v^{(s)}(\vec{p})$

For $E = -E_p < 0$, the lower component ω_B dominates. Setting $\omega_B = \chi^{(s)}$, equation (12.6.4) gives:

$$\omega_A = \frac{c(\vec{\sigma} \cdot \vec{p})}{-E_p - mc^2}\chi^{(s)} = -\frac{c(\vec{\sigma} \cdot \vec{p})}{E_p + mc^2}\chi^{(s)}. \quad (12.6.13)$$

In quantum field theory, an antiparticle of physical momentum $\vec{p}$ and physical energy $+E_p$ corresponds to a negative-energy plane wave state carrying four-momentum $-p^\mu = (-E_p/c, -\vec{p})$. Replacing $\vec{p} \rightarrow -\vec{p}$ reverses the sign of $\vec{\sigma} \cdot \vec{p}$. Following the standard convention:

$$v^{(s)}(\vec{p}) \equiv \omega^{(s+2)}(-\vec{p}). \quad (12.6.14)$$

Explicitly:

$$v^{(1)}(\vec{p}) = N \begin{pmatrix} \frac{cp_-}{E_p + mc^2} \\ \frac{-cp_z}{E_p + mc^2} \\ 0 \\ 1 \end{pmatrix}, \quad v^{(2)}(\vec{p}) = N \begin{pmatrix} \frac{cp_z}{E_p + mc^2} \\ \frac{cp_+}{E_p + mc^2} \\ 1 \\ 0 \end{pmatrix} \quad (12.6.15)$$

The antiparticle spinors satisfy:

$$(\not{p} + mc)v^{(s)}(\vec{p}) = 0. \quad (12.6.16)$$

12.6.4 Derivation of the Normalization Constant

We determine the normalization constant N from first principles. First, we prove the fundamental relation connecting $u^\dagger u$ and $\bar{u}u$:

$$(\not{p} - mc)u = 0 \implies \bar{u}\gamma^\mu \not{p}u = mc\bar{u}\gamma^\mu u, \quad (12.6.17)$$

$$\bar{u}(\not{p} - mc) = 0 \implies \bar{u}\not{p}\gamma^\mu u = mc\bar{u}\gamma^\mu u. \quad (12.6.18)$$

Adding (12.6.17) and (12.6.18):

$$\bar{u}\{\gamma^\mu, \not{p}\}u = 2mc\bar{u}\gamma^\mu u. \quad (12.6.19)$$

Using $\{\gamma^\mu, \not{p}\} = \{\gamma^\mu, \gamma^\nu p_\nu\} = 2g^{\mu\nu}p_\nu = 2p^\mu$:

$$2p^\mu \bar{u}u = 2mc\bar{u}\gamma^\mu u \implies \bar{u}\gamma^\mu u = \frac{p^\mu}{mc} \bar{u}u. \quad (12.6.20)$$

Writing the left-hand side as $u^\dagger \gamma^0 \gamma^\mu u$:

$$u^\dagger \gamma^0 \gamma^\mu u = \frac{p^\mu}{mc} u^\dagger \gamma^0 u. \quad (12.6.21)$$

For $\mu = 0$, since $(\gamma^0)^2 = I_4$ and $p^0 = E/c$:

$$u^\dagger u = \frac{E}{mc^2} \bar{u}u. \quad (12.6.22)$$

Now we calculate $u^{(1)\dagger}u^{(1)}$ directly from (12.6.12):

$$\begin{aligned} u^{(1)\dagger}(\vec{p})u^{(1)}(\vec{p}) &= |N|^2 \begin{pmatrix} 1 & 0 & \frac{cp_z}{E_p + mc^2} & \frac{cp_+^*}{E_p + mc^2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \frac{cp_z}{E_p + mc^2} \\ \frac{cp_+}{E_p + mc^2} \end{pmatrix} \\ &= |N|^2 \left[1 + \frac{c^2 p_z^2}{(E_p + mc^2)^2} + \frac{c^2 (p_x^2 + p_y^2)}{(E_p + mc^2)^2} \right] \\ &= |N|^2 \left[1 + \frac{c^2 \vec{p}^2}{(E_p + mc^2)^2} \right]. \end{aligned} \quad (12.6.23)$$

Using the dispersion relation $c^2 \vec{p}^2 = E_p^2 - m^2 c^4 = (E_p - mc^2)(E_p + mc^2)$:

$$1 + \frac{c^2 \vec{p}^2}{(E_p + mc^2)^2} = 1 + \frac{E_p - mc^2}{E_p + mc^2} = \frac{(E_p + mc^2) + (E_p - mc^2)}{E_p + mc^2} = \frac{2E_p}{E_p + mc^2}. \quad (12.6.24)$$

Thus:

$$u^\dagger u = |N|^2 \frac{2E_p}{E_p + mc^2}. \quad (12.6.25)$$

Comparing (12.6.25) with (12.6.22):

$$|N|^2 \frac{2E_p}{E_p + mc^2} = \frac{E_p}{mc^2} \bar{u}u \implies |N|^2 = \frac{E_p + mc^2}{2mc^2} \bar{u}u. \quad (12.6.26)$$

- **Non-relativistic normalization** ($\bar{u}_s u_{s'} = \delta_{ss'}$):

$$N = \sqrt{\frac{E_p + mc^2}{2mc^2}}. \quad (12.6.27)$$

- **Covariant relativistic normalization** ($\bar{u}_s u_{s'} = 2mc \delta_{ss'}$):

$$N = \sqrt{\frac{E_p + mc^2}{c}}. \quad (12.6.28)$$

In natural units ($\hbar = c = 1$) with covariant normalization $\bar{u}_s(p) u_{s'}(p) = 2m \delta_{ss'}$:

$$\boxed{N = \sqrt{E_p + m}, \quad u^\dagger(p) u(p) = 2E_p, \quad v^\dagger(p) v(p) = 2E_p} \quad (12.6.29)$$

12.6.5 Orthogonality and Completeness Relations

The covariant orthogonality relations for the four-spinors are:

$$\bar{u}_r(p) u_s(p) = 2m \delta_{rs}, \quad (12.6.30)$$

$$\bar{v}_r(p) v_s(p) = -2m \delta_{rs}, \quad (12.6.31)$$

$$\bar{u}_r(p) v_s(p) = \bar{v}_r(p) u_s(p) = 0. \quad (12.6.32)$$

Summing over the two physical spin states yields the fundamental **completeness relations** (spin sums):

$$\boxed{\sum_{s=1}^2 u_s(p) \bar{u}_s(p) = \not{p} + m, \quad \sum_{s=1}^2 v_s(p) \bar{v}_s(p) = \not{p} - m} \quad (12.6.33)$$

We define the covariant **energy projection operators**:

$$\boxed{\Lambda_+(p) \equiv \frac{\not{p} + m}{2m}, \quad \Lambda_-(p) \equiv \frac{-\not{p} + m}{2m}} \quad (12.6.34)$$

Using $\not{p}\not{p} = p^2 I_4 = m^2 I_4$, we verify their projection properties:

$$\Lambda_+^2 = \frac{(\not{p} + m)^2}{4m^2} = \frac{\not{p}^2 + 2m\not{p} + m^2}{4m^2} = \frac{2m^2 + 2m\not{p}}{4m^2} = \frac{\not{p} + m}{2m} = \Lambda_+, \quad (12.6.35)$$

$$\Lambda_+ \Lambda_- = \frac{(\not{p} + m)(-\not{p} + m)}{4m^2} = \frac{m^2 - \not{p}^2}{4m^2} = \frac{m^2 - m^2}{4m^2} = 0, \quad (12.6.36)$$

$$\Lambda_+ + \Lambda_- = \frac{\not{p} + m}{2m} + \frac{-\not{p} + m}{2m} = \frac{2m}{2m} = I_4. \quad (12.6.37)$$

12.7 Lorentz Covariance of the Dirac Equation

12.7.1 Transformation Laws and Index Structure

Under a proper orthochronous Lorentz transformation:

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}, \quad x^{\nu} = (\Lambda^{-1})^{\nu}_{\mu} x'^{\mu}. \quad (12.7.1)$$

The metric tensor invariant requires:

$$g^{\alpha\beta} = g^{\mu\nu} \Lambda_{\nu}^{\alpha} \Lambda_{\mu}^{\beta}. \quad (12.7.2)$$

Multiplying by the inverse transformation matrix $(\Lambda^{-1})_{\mu}^{\beta}$:

$$g^{\alpha\beta} (\Lambda^{-1})_{\mu}^{\beta} = g^{\mu\nu} \Lambda_{\nu}^{\alpha} \implies (\Lambda^{-1})_{\mu}^{\alpha} = g^{\alpha\beta} g_{\mu\nu} \Lambda_{\beta}^{\nu} \equiv \Lambda_{\mu}^{\alpha}. \quad (12.7.3)$$

The partial derivative transforms as a covariant four-vector:

$$\frac{\partial}{\partial x^{\mu}} = \frac{\partial x'^{\nu}}{\partial x^{\mu}} \frac{\partial}{\partial x'^{\nu}} = \Lambda^{\nu}_{\mu} \frac{\partial}{\partial x'^{\nu}} \implies \partial_{\mu} = \Lambda^{\nu}_{\mu} \partial'_{\nu}, \quad (12.7.4)$$

$$\frac{\partial}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial}{\partial x^{\nu}} = (\Lambda^{-1})^{\nu}_{\mu} \frac{\partial}{\partial x^{\nu}} \implies \partial'_{\mu} = (\Lambda^{-1})^{\nu}_{\mu} \partial_{\nu}. \quad (12.7.5)$$

12.7.2 Spinor Transformation and Intertwining Relation

Under a Lorentz transformation $x \rightarrow x' = \Lambda x$, the Dirac spinor transforms via an invertible 4×4 matrix $S(\Lambda)$:

$$\psi'(x') = S(\Lambda) \psi(x) = S(\Lambda) \psi(\Lambda^{-1} x'). \quad (12.7.6)$$

The unprimed and primed Dirac equations are:

$$(i\gamma^{\mu} \partial_{\mu} - m) \psi(x) = 0, \quad (12.7.7)$$

$$(i\gamma^{\mu} \partial'_{\mu} - m) \psi'(x') = 0. \quad (12.7.8)$$

Expressing (12.7.7) in terms of primed coordinates:

$$(i\gamma^{\mu} \Lambda^{\nu}_{\mu} \partial'_{\nu} - m) S^{-1}(\Lambda) \psi'(x') = 0. \quad (12.7.9)$$

Operating with $S(\Lambda)$ from the left:

$$(iS(\Lambda) \gamma^{\mu} S^{-1}(\Lambda) \Lambda^{\nu}_{\mu} \partial'_{\nu} - m) \psi'(x') = 0. \quad (12.7.10)$$

For this equation to be identical to (12.7.8) for any derivative ∂'_{ν} , the matrix in front of ∂'_{ν} must equal γ^{ν} :

$$S(\Lambda) \gamma^{\mu} S^{-1}(\Lambda) \Lambda^{\nu}_{\mu} = \gamma^{\nu}. \quad (12.7.11)$$

Multiplying by Λ_{ν}^{ρ} and inverting:

$$\boxed{S^{-1}(\Lambda) \gamma^{\mu} S(\Lambda) = \Lambda^{\mu}_{\nu} \gamma^{\nu}} \quad (12.7.12)$$

12.7.3 Generators of the Spinor Representation

For an infinitesimal Lorentz transformation:

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu}, \quad (12.7.13)$$

we expand $S(\Lambda)$ to first order in the infinitesimal parameters $\omega_{\mu\nu}$:

$$S(\Lambda) = I_4 - \frac{i}{4} \sigma_{\alpha\beta} \omega^{\alpha\beta}, \quad S^{-1}(\Lambda) = I_4 + \frac{i}{4} \sigma_{\alpha\beta} \omega^{\alpha\beta}. \quad (12.7.14)$$

Substituting into (12.7.12):

$$\left(I_4 + \frac{i}{4} \sigma_{\alpha\beta} \omega^{\alpha\beta} \right) \gamma^\mu \left(I_4 - \frac{i}{4} \sigma_{\alpha\beta} \omega^{\alpha\beta} \right) = \gamma^\mu + \omega^\mu{}_\nu \gamma^\nu. \quad (12.7.15)$$

Expanding to first order:

$$\gamma^\mu + \frac{i}{4} \omega^{\alpha\beta} [\sigma_{\alpha\beta}, \gamma^\mu] = \gamma^\mu + \omega^\mu{}_\nu \gamma^\nu. \quad (12.7.16)$$

Subtracting γ^μ :

$$\omega^{\alpha\beta} [\sigma_{\alpha\beta}, \gamma^\mu] = -4i \omega^\mu{}_\nu \gamma^\nu = -4i g^{\mu\alpha} \omega_{\alpha\beta} \gamma^\beta. \quad (12.7.17)$$

Using the antisymmetry $\omega_{\alpha\beta} = -\omega_{\beta\alpha} = \frac{1}{2}(\omega_{\alpha\beta} - \omega_{\beta\alpha})$:

$$-4i \omega^\mu{}_\nu \gamma^\nu = -2i \omega^{\alpha\beta} (g_\alpha{}^\mu \gamma_\beta - g_\beta{}^\mu \gamma_\alpha). \quad (12.7.18)$$

Thus, the generators $\sigma_{\alpha\beta}$ must satisfy the commutation relation:

$$\boxed{[\sigma_{\alpha\beta}, \gamma^\mu] = -2i (g_\alpha{}^\mu \gamma_\beta - g_\beta{}^\mu \gamma_\alpha)} \quad (12.7.19)$$

We verify that this relation is satisfied by:

$$\boxed{\sigma_{\alpha\beta} \equiv \frac{i}{2} [\gamma_\alpha, \gamma_\beta]} \quad (12.7.20)$$

Using the commutator identity $[[A, B], C] = A\{B, C\} - \{A, C\}B$:

$$\begin{aligned} \left[\frac{i}{2} [\gamma_\alpha, \gamma_\beta], \gamma^\mu \right] &= \frac{i}{2} (\gamma_\alpha \{\gamma_\beta, \gamma^\mu\} - \{\gamma_\alpha, \gamma^\mu\} \gamma_\beta) \\ &= \frac{i}{2} (\gamma_\alpha (2g_\beta{}^\mu) - (2g_\alpha{}^\mu) \gamma_\beta) = -2i (g_\alpha{}^\mu \gamma_\beta - g_\beta{}^\mu \gamma_\alpha). \end{aligned} \quad (12.7.21)$$

The finite transformation matrix for any finite Lorentz transformation is:

$$\boxed{S(\Lambda) = \exp \left(-\frac{i}{4} \sigma_{\mu\nu} \omega^{\mu\nu} \right)} \quad (12.7.22)$$

12.7.4 Hermitian Adjoint of Spinor Transformations

We determine the transformation law for the Dirac adjoint $\bar{\psi} = \psi^\dagger \gamma^0$. First, evaluate $\gamma^0 \sigma_{\mu\nu}^\dagger \gamma^0$:

$$\begin{aligned} \gamma^0 \sigma_{\mu\nu}^\dagger \gamma^0 &= \gamma^0 \left(\frac{i}{2} [\gamma_\mu, \gamma_\nu] \right)^\dagger \gamma^0 = -\frac{i}{2} \gamma^0 [\gamma_\nu^\dagger, \gamma_\mu^\dagger] \gamma^0 \\ &= -\frac{i}{2} (\gamma^0 \gamma_\nu^\dagger \gamma^0 \gamma^0 \gamma_\mu^\dagger \gamma^0 - \gamma^0 \gamma_\mu^\dagger \gamma^0 \gamma^0 \gamma_\nu^\dagger \gamma^0). \end{aligned} \quad (12.7.23)$$

Using $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$:

$$\gamma^0 \sigma_{\mu\nu}^\dagger \gamma^0 = -\frac{i}{2} [\gamma_\nu, \gamma_\mu] = \frac{i}{2} [\gamma_\mu, \gamma_\nu] = \sigma_{\mu\nu}. \quad (12.7.24)$$

Now evaluate $S^\dagger(\Lambda)$:

$$S^\dagger(\Lambda) = \exp\left(+\frac{i}{4} \sigma_{\mu\nu}^\dagger \omega^{\mu\nu}\right). \quad (12.7.25)$$

Multiplying by γ^0 from both sides:

$$\gamma^0 S^\dagger(\Lambda) \gamma^0 = \exp\left(+\frac{i}{4} \gamma^0 \sigma_{\mu\nu}^\dagger \gamma^0 \omega^{\mu\nu}\right) = \exp\left(+\frac{i}{4} \sigma_{\mu\nu} \omega^{\mu\nu}\right) = S^{-1}(\Lambda). \quad (12.7.26)$$

Multiplying from the right by γ^0 :

$$\boxed{S^\dagger(\Lambda) \gamma^0 = \gamma^0 S^{-1}(\Lambda)} \quad (12.7.27)$$

Therefore, the Dirac adjoint $\bar{\psi}'(x')$ transforms with the inverse matrix:

$$\boxed{\bar{\psi}'(x') = \psi'^\dagger(x') \gamma^0 = \psi^\dagger(x) S^\dagger(\Lambda) \gamma^0 = \psi^\dagger(x) \gamma^0 S^{-1}(\Lambda) = \bar{\psi}(x) S^{-1}(\Lambda)} \quad (12.7.28)$$

12.8 Bilinear Covariants and Chirality

12.8.1 Transformation of the 16 Dirac Bilinears

Using $\psi'(x') = S(\Lambda)\psi(x)$ and $\bar{\psi}'(x') = \bar{\psi}(x)S^{-1}(\Lambda)$, we prove the Lorentz transformation properties of all sixteen independent bilinears:

1. Scalar (1 component):

$$\bar{\psi}'(x')\psi'(x') = \bar{\psi}(x)S^{-1}(\Lambda)S(\Lambda)\psi(x) = \bar{\psi}(x)\psi(x). \quad (12.8.1)$$

It is invariant under all Lorentz transformations.

2. Vector (4 components):

$$\bar{\psi}'(x')\gamma^\mu\psi'(x') = \bar{\psi}(x)S^{-1}(\Lambda)\gamma^\mu S(\Lambda)\psi(x) = \Lambda^\mu_\nu \bar{\psi}(x)\gamma^\nu\psi(x). \quad (12.8.2)$$

It transforms as a true four-vector (conserved probability current).

3. Tensor (6 components):

$$\bar{\psi}'(x')\sigma^{\mu\nu}\psi'(x') = \bar{\psi}(x)S^{-1}(\Lambda)\sigma^{\mu\nu}S(\Lambda)\psi(x) = \Lambda^\mu_\alpha \Lambda^\nu_\beta \bar{\psi}(x)\sigma^{\alpha\beta}\psi(x). \quad (12.8.3)$$

It transforms as an antisymmetric rank-2 Lorentz tensor.

12.8.2 The γ^5 Matrix and Chiral Projections

We define the fifth Dirac matrix γ^5 (or γ_5):

$$\boxed{\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}} \quad (12.8.4)$$

1. Hermiticity:

$$\begin{aligned}
(\gamma^5)^\dagger &= -i(\gamma^3)^\dagger(\gamma^2)^\dagger(\gamma^1)^\dagger(\gamma^0)^\dagger = -i(-\gamma^3)(-\gamma^2)(-\gamma^1)(\gamma^0) \\
&= i\gamma^3\gamma^2\gamma^1\gamma^0 = i(-1)^6\gamma^0\gamma^1\gamma^2\gamma^3 = \gamma^5.
\end{aligned} \tag{12.8.5}$$

2. Involution:

$$(\gamma^5)^2 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} = \begin{pmatrix} I_2 & 0 \\ 0 & I_2 \end{pmatrix} = I_4. \tag{12.8.6}$$

3. Anticommutation with all γ^μ :

$$\{\gamma^\mu, \gamma^5\} = 0 \quad (\mu = 0, 1, 2, 3). \tag{12.8.7}$$

For $\mu = 0$: $\gamma^0\gamma^5 = i\gamma^0\gamma^0\gamma^1\gamma^2\gamma^3 = i\gamma^1\gamma^2\gamma^3$, whereas $\gamma^5\gamma^0 = i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0 = -i\gamma^0\gamma^0\gamma^1\gamma^2\gamma^3 = -i\gamma^1\gamma^2\gamma^3$.

With γ^5 , we complete the list of sixteen bilinear covariants:

$$\text{4. Axial Vector (4 components): } \bar{\psi}\gamma^\mu\gamma^5\psi \rightarrow \det(\Lambda)\Lambda^\mu_\nu\bar{\psi}\gamma^\nu\gamma^5\psi.$$

$$\text{5. Pseudoscalar (1 component): } \bar{\psi}\gamma^5\psi \rightarrow \det(\Lambda)\bar{\psi}\gamma^5\psi.$$

Total components: $1 + 4 + 6 + 4 + 1 = 16$, spanning the space of all 4×4 complex matrices.

We define the **chiral projection operators**:

$$P_R \equiv \frac{I_4 + \gamma^5}{2} = \frac{1}{2} \begin{pmatrix} I_2 & I_2 \\ I_2 & I_2 \end{pmatrix}, \quad P_L \equiv \frac{I_4 - \gamma^5}{2} = \frac{1}{2} \begin{pmatrix} I_2 & -I_2 \\ -I_2 & I_2 \end{pmatrix} \tag{12.8.8}$$

They satisfy $P_R^2 = P_R$, $P_L^2 = P_L$, $P_R P_L = P_L P_R = 0$, and $P_R + P_L = I_4$.

12.8.3 Massless Fermions: Equivalence of Chirality and Helicity

For a massless Dirac particle ($m = 0$), the Dirac equation $E\omega = c(\vec{\alpha} \cdot \vec{p})\omega$ in two-component blocks reads:

$$E \begin{pmatrix} \omega_A \\ \omega_B \end{pmatrix} = c \begin{pmatrix} 0 & \vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & 0 \end{pmatrix} \begin{pmatrix} \omega_A \\ \omega_B \end{pmatrix} \implies \begin{cases} E\omega_A = c(\vec{\sigma} \cdot \vec{p})\omega_B, \\ E\omega_B = c(\vec{\sigma} \cdot \vec{p})\omega_A. \end{cases} \tag{12.8.9}$$

We define the chiral eigenstates:

$$\omega_R \equiv \frac{1}{2}(\omega_A + \omega_B), \quad \omega_L \equiv \frac{1}{2}(\omega_A - \omega_B). \tag{12.8.10}$$

Adding and subtracting the coupled equations:

$$E(\omega_A + \omega_B) = c(\vec{\sigma} \cdot \vec{p})(\omega_A + \omega_B) \implies E\omega_R = c(\vec{\sigma} \cdot \vec{p})\omega_R, \tag{12.8.11}$$

$$E(\omega_A - \omega_B) = -c(\vec{\sigma} \cdot \vec{p})(\omega_A - \omega_B) \implies E\omega_L = -c(\vec{\sigma} \cdot \vec{p})\omega_L. \tag{12.8.12}$$

Dividing by $E = c|\vec{p}|$:

$$\left(\frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|}\right)\omega_R = +1\omega_R, \quad \left(\frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|}\right)\omega_L = -1\omega_L. \tag{12.8.13}$$

The helicity operator is the projection of spin along momentum:

$$\hat{h} \equiv \frac{\vec{\Sigma} \cdot \vec{p}}{|\vec{p}|}, \quad \vec{\Sigma} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}. \tag{12.8.14}$$

Thus, **for massless fermions, chirality is identically equal to helicity**:

- Right-handed chiral state ($P_R\psi$): positive helicity ($h = +1$).
- Left-handed chiral state ($P_L\psi$): negative helicity ($h = -1$).

12.9 Trace Technology and Contraction Identities

12.9.1 Gamma Matrix Contractions

Evaluating contracted products of gamma matrices in four spacetime dimensions ($g^\mu_\mu = 4$):

1. **Single contraction:**

$$\gamma^\mu \gamma_\mu = g_{\mu\nu} \gamma^\mu \gamma^\nu = \frac{1}{2} g_{\mu\nu} \{\gamma^\mu, \gamma^\nu\} = \frac{1}{2} g_{\mu\nu} (2g^{\mu\nu} I_4) = g_{\mu\nu} g^{\mu\nu} I_4 = 4I_4. \quad (12.9.1)$$

2. **Contraction with one intermediate gamma:**

$$\gamma^\mu \gamma^\nu \gamma_\mu = (2g^{\mu\nu} I_4 - \gamma^\nu \gamma^\mu) \gamma_\mu = 2\gamma^\nu - \gamma^\nu (\gamma^\mu \gamma_\mu) = 2\gamma^\nu - 4\gamma^\nu = -2\gamma^\nu. \quad (12.9.2)$$

3. **Contraction with a slashed vector:**

$$\gamma^\mu \not{a} \gamma_\mu = \gamma^\mu (\gamma^\nu a_\nu) \gamma_\mu = (\gamma^\mu \gamma^\nu \gamma_\mu) a_\nu = -2\gamma^\nu a_\nu = -2\not{a}. \quad (12.9.3)$$

4. **Contraction with two intermediate gammas:**

$$\begin{aligned} \gamma^\mu \gamma^\nu \gamma^\rho \gamma_\mu &= (2g^{\mu\nu} I_4 - \gamma^\nu \gamma^\mu) \gamma^\rho \gamma_\mu = 2\gamma^\rho \gamma^\nu - \gamma^\nu (\gamma^\mu \gamma^\rho \gamma_\mu) \\ &= 2\gamma^\rho \gamma^\nu - \gamma^\nu (-2\gamma^\rho) = 2(\gamma^\rho \gamma^\nu + \gamma^\nu \gamma^\rho) = 4g^{\nu\rho} I_4. \end{aligned} \quad (12.9.4)$$

5. **Square of a slashed vector:**

$$\not{a} \not{a} = \gamma^\mu a_\mu \gamma^\nu a_\nu = a_\mu a_\nu \gamma^\mu \gamma^\nu = \frac{1}{2} a_\mu a_\nu \{\gamma^\mu, \gamma^\nu\} = a_\mu a_\nu g^{\mu\nu} I_4 = a^2 I_4. \quad (12.9.5)$$

12.9.2 Trace Theorems

Using the cyclic property of the trace ($\text{Tr}(ABC) = \text{Tr}(BCA)$):

1. $\text{Tr}(I_4) = 4.$

2. **Trace of an odd number of gamma matrices vanishes:** Using $(\gamma^5)^2 = I_4$ and $\{\gamma^\mu, \gamma^5\} = 0$:

$$\begin{aligned} \text{Tr}(\gamma^{\mu_1} \dots \gamma^{\mu_n}) &= \text{Tr}(\gamma^{\mu_1} \dots \gamma^{\mu_n} \gamma^5 \gamma^5) = \text{Tr}(\gamma^5 \gamma^{\mu_1} \dots \gamma^{\mu_n} \gamma^5) \\ &= (-1)^n \text{Tr}(\gamma^{\mu_1} \dots \gamma^{\mu_n} \gamma^5 \gamma^5) = (-1)^n \text{Tr}(\gamma^{\mu_1} \dots \gamma^{\mu_n}). \end{aligned} \quad (12.9.6)$$

If n is odd, $(-1)^n = -1 \implies \text{Tr}(\gamma^{\mu_1} \dots \gamma^{\mu_n}) = 0.$

3. $\text{Tr}(\gamma^\mu) = 0.$

4. $\text{Tr}(\gamma^5) = \text{Tr}(\gamma^0 \gamma^0 \gamma^5) = -\text{Tr}(\gamma^0 \gamma^5 \gamma^0) = -\text{Tr}(\gamma^5 \gamma^0 \gamma^0) = -\text{Tr}(\gamma^5) \implies \text{Tr}(\gamma^5) = 0.$

5. **Trace of two gamma matrices:**

$$\text{Tr}(\gamma^\mu \gamma^\nu) = \frac{1}{2} \text{Tr}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) = \frac{1}{2} \text{Tr}(2g^{\mu\nu} I_4) = g^{\mu\nu} \text{Tr}(I_4) = 4g^{\mu\nu}. \quad (12.9.7)$$

6. **Trace of four gamma matrices:** Using $\gamma^\mu \gamma^\nu = 2g^{\mu\nu} I_4 - \gamma^\nu \gamma^\mu$:

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}). \quad (12.9.8)$$

7. **Traces with γ^5 :**

$$\text{Tr}(\gamma^5 \gamma^\mu \gamma^\nu) = 0, \quad (12.9.9)$$

$$\text{Tr}(\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = -4i\epsilon^{\mu\nu\rho\sigma}. \quad (12.9.10)$$

12.10 The Gordon Decomposition and $g = 2$

Let $u(p)$ and $u(p')$ be two positive-energy solutions satisfying:

$$\not{p}u(p) = mc u(p), \quad \bar{u}(p')\not{p}' = mc \bar{u}(p'). \quad (12.10.1)$$

Using the gamma identity $\gamma^\mu \gamma^\nu = g^{\mu\nu} - i\sigma^{\mu\nu}$, where $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$:

$$\gamma^\mu \not{p} = \gamma^\mu \gamma^\nu p_\nu = p^\mu - i\sigma^{\mu\nu} p_\nu, \quad (12.10.2)$$

$$\not{p}' \gamma^\mu = \gamma^\nu \gamma^\mu p'_\nu = p'^\mu + i\sigma^{\mu\nu} p'_\nu. \quad (12.10.3)$$

Sandwiching between spinors:

$$\bar{u}(p')\gamma^\mu \not{p}u(p) = mc \bar{u}(p')\gamma^\mu u(p) = \bar{u}(p') (p^\mu - i\sigma^{\mu\nu} p_\nu) u(p), \quad (12.10.4)$$

$$\bar{u}(p')\not{p}' \gamma^\mu u(p) = mc \bar{u}(p')\gamma^\mu u(p) = \bar{u}(p') (p'^\mu + i\sigma^{\mu\nu} p'_\nu) u(p). \quad (12.10.5)$$

Adding (12.10.4) and (12.10.5):

$$2mc \bar{u}(p')\gamma^\mu u(p) = \bar{u}(p') [(p + p')^\mu + i\sigma^{\mu\nu}(p' - p)_\nu] u(p). \quad (12.10.6)$$

Dividing by $2mc$:

$$\boxed{\bar{u}(p')\gamma^\mu u(p) = \bar{u}(p') \left[\frac{(p + p')^\mu}{2mc} + \frac{i\sigma^{\mu\nu} q_\nu}{2mc} \right] u(p)} \quad (12.10.7)$$

where $q = p' - p$ is the four-momentum transfer.

- The first term $\frac{(p+p')^\mu}{2mc}$ is the **convection current** of a spinless charged particle (identical to the Klein–Gordon current).
- The second term $\frac{i\sigma^{\mu\nu} q_\nu}{2mc}$ is the **spin-magnetization current**, which couples directly to the electromagnetic field tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

When coupled minimally to an electromagnetic four-potential A_μ , the spin-magnetization interaction energy in the rest frame is:

$$H_{\text{spin}} = -\vec{\mu} \cdot \vec{B} = -g \frac{e}{2m} \vec{S} \cdot \vec{B}. \quad (12.10.8)$$

Comparing with the coefficient in (12.10.7) reveals:

$$\boxed{g = 2} \quad (12.10.9)$$

The Dirac theory predicts the gyromagnetic ratio $g = 2$ purely from relativity and linearity.

12.11 The Klein Paradox and Dirac Hole Theory

12.11.1 Relativistic Potential Step

Consider a relativistic electron of energy $E > mc^2$ incident from the left ($x < 0$) on an electrostatic step potential:

$$V(x) = \begin{cases} 0, & x < 0 \quad (\text{Region I}), \\ V_0, & x > 0 \quad (\text{Region II}). \end{cases} \quad (12.11.1)$$

In Region I ($x < 0$), the free Dirac equation has momentum $p = \sqrt{E^2/c^2 - m^2c^2}$. In Region II ($x > 0$), minimal coupling $p_0 \rightarrow (E - V_0)/c$ modifies the dispersion relation to:

$$(E - V_0)^2 = p'^2 c^2 + m^2 c^4 \implies p' = \pm \frac{1}{c} \sqrt{(E - V_0)^2 - m^2 c^4}. \quad (12.11.2)$$

We examine three energy regimes:

1. **Low Step** ($V_0 < E - mc^2$): p' is real, standard transmission into positive-energy continuum states.
2. **Intermediate Step** ($E - mc^2 < V_0 < E + mc^2$): $(E - V_0)^2 < m^2 c^4$, so $p' = i\kappa$ is imaginary. The transmitted wave decays exponentially ($e^{-\kappa x}$), giving total reflection $R = 1$ (ordinary quantum barrier penetration).
3. **Supercritical Step** ($V_0 > E + mc^2$): Because $V_0 - E > mc^2$, $(E - V_0)^2 > m^2 c^4$, so p' becomes **strictly real**:

$$p' = \pm \frac{1}{c} \sqrt{(V_0 - E)^2 - m^2 c^4}. \quad (12.11.3)$$

The wave inside the barrier does not decay; it **propagates oscillatory**!

12.11.2 Current Matching and $R > 1$

To select the physically correct sign for p' , we examine the group velocity inside Region II:

$$v_g = \frac{\partial E}{\partial p'} = \frac{p' c^2}{E - V_0}. \quad (12.11.4)$$

Because $E - V_0 < -mc^2 < 0$, the denominator is negative. For energy to flow away from the step to $+x$, we must have $v_g > 0$, which forces us to choose:

$$p' < 0. \quad (12.11.5)$$

The phase velocity points toward the left, while the group velocity carries flux toward $+x$.

Matching the four-component wavefunctions at the boundary $x = 0$, the reflection coefficient R (ratio of reflected to incident probability flux) is:

$$R = \frac{|j_{\text{refl}}|}{|j_{\text{inc}}|} = \left(\frac{1 - r}{1 + r} \right)^2, \quad r \equiv \frac{p'}{p} \frac{E + mc^2}{(E - V_0) + mc^2}. \quad (12.11.6)$$

Since $p' < 0$ and $(E - V_0) + mc^2 < 0$, the ratio r is strictly negative:

$$r < 0 \implies \left| \frac{1 - r}{1 + r} \right| > 1 \implies \boxed{R > 1} \quad (12.11.7)$$

More particles are reflected back into Region I than were originally incident! This is **Klein's paradox** (Oskar Klein, 1929).

12.11.3 Resolution via Pair Production and Hole Theory

The apparent paradox reflects the breakdown of the single-particle picture. A potential step exceeding $V_0 > 2mc^2$ over a distance $d \lesssim \bar{\lambda}_c$ creates an electrostatic field exceeding the **Schwinger critical field**:

$$\mathcal{E}_c = \frac{m^2 c^3}{e\hbar} \approx 1.3 \times 10^{18} \text{ V/m}. \quad (12.11.8)$$

This electric field tilts the vacuum energy levels so severely that the negative-energy sea in Region II overlaps with the positive-energy continuum of Region I. Electron-positron pairs are spontaneously created from the vacuum:

- The created electrons are repelled by the positive potential, joining the reflected beam in Region I and making $R > 1$.
- The created positrons (holes carrying charge $+e$) carry positive energy into Region II.

Deep Conceptual Synthesis: Wave Equations as Field Equations, Not Single-Particle Wavefunctions

The overarching arc of relativistic quantum mechanics reveals a profound epistemological truth:

1. **The Breakdown of Fixed-Particle Hilbert Spaces:** In non-relativistic physics, the conservation of mass forbids particle creation. The single-particle Schrödinger wavefunction $\psi(\vec{r}, t)$ describes a solitary particle for all time within a fixed Hilbert space $\mathcal{H}_1$.

In relativity, Einstein's equivalence $E = mc^2$ combined with Heisenberg's principle $\Delta x \Delta p \gtrsim \hbar$ guarantees that whenever a particle is confined to a region smaller than its reduced Compton wavelength $\bar{\lambda}_c = \hbar/(mc)$, the energy uncertainty $\Delta E \geq 2mc^2$ is sufficient to create new particle-antiparticle pairs from the vacuum. **Particle number cannot be conserved in relativistic quantum mechanics.**

2. **Why Dirac's Equation Still Breaks Down as a Wavefunction:** Dirac's brilliant linearization eliminated negative probability densities ($\rho = \psi^\dagger \psi \geq 0$), but could not eliminate negative-energy eigenvalues ($E \leq -mc^2$). If ψ is treated as a single-particle probability wave:

- Matter is unstable against radiative cascade into negative-energy states.
- The “Dirac Sea” resolution requires an infinite number of background particles, thereby abandoning single-particle physics.
- The Sea hypothesis completely fails for bosons (photons, pions, Higgs bosons), which obey Bose–Einstein statistics and cannot be excluded.
- The Klein paradox shows that sharp potential barriers ($V_0 > E + mc^2$) inevitably generate particle pairs ($R > 1$).

3. **The Inevitability of Quantum Field Theory:** In modern quantum field theory, the Klein–Gordon and Dirac equations are **not** Schrödinger equations for single particles; they are **classical relativistic field equations** governing local field amplitudes.

In Quantum Field Theory:

- Spacetime coordinates $x^\mu = (ct, \vec{x})$ are continuous parameters labeling operator fields, not quantum operators.
- States reside in multi-particle **Fock space** $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$.
- The localized state is created from the vacuum: $|x\rangle = \hat{\phi}(x)|0\rangle$ or $|x, s\rangle = \hat{\psi}_s^\dagger(x)|0\rangle$.
- Negative frequencies are reinterpreted as positive-energy antiparticles via creation/annihilation operators.
- Canonical anticommutation guarantees positive-definite energy, vacuum stability, microcausality, and the spin-statistics theorem.

12.12 The Transition to Second Quantization

12.12.1 The Dirac Field Operator

To incorporate pair creation and annihilation naturally, we elevate the Dirac wave function to a **quantum field operator** $\hat{\psi}(x)$ acting on multi-particle Fock space:

$$\hat{\psi}(x) = \sum_{s=1}^2 \int \frac{d^3p}{(2\pi\hbar)^3 \sqrt{2E_p}} \left[\hat{b}_s(\vec{p}) u_s(\vec{p}) e^{-ip \cdot x / \hbar} + \hat{d}_s^\dagger(\vec{p}) v_s(\vec{p}) e^{+ip \cdot x / \hbar} \right] \quad (12.12.1)$$

where:

- $\hat{b}_s(\vec{p})$ annihilates an electron of momentum $\vec{p}$ and spin s .
- $\hat{d}_s^\dagger(\vec{p})$ creates a positron (antiparticle) of momentum $\vec{p}$ and spin s .

12.12.2 Canonical Anticommutation and the Spin-Statistics Theorem

If the field operators are quantized using commutators, the Hamiltonian spectrum is unbounded from below ($\hat{H} \sim \sum E_p (\hat{b}^\dagger \hat{b} - \hat{d} \hat{d}^\dagger)$), allowing infinite cascades of negative-energy emission. To guarantee vacuum stability (positive energy spectrum) and microcausality ($\{\hat{\psi}(x), \hat{\psi}(y)\} = 0$ for spacelike separation $(x - y)^2 < 0$), the creation and annihilation operators must obey **canonical anticommutation relations**:

$$\{\hat{b}_s(\vec{p}), \hat{b}_{s'}^\dagger(\vec{p}')\} = (2\pi\hbar)^3 \delta_{ss'} \delta^3(\vec{p} - \vec{p}'), \quad \{\hat{d}_s(\vec{p}), \hat{d}_{s'}^\dagger(\vec{p}')\} = (2\pi\hbar)^3 \delta_{ss'} \delta^3(\vec{p} - \vec{p}') \quad (12.12.2)$$

with all other anticommutators vanishing.

Applying normal ordering ($: \hat{O} :$), which places annihilation operators to the right and accounts for the minus sign under fermion anticommutation ($: \hat{d} \hat{d}^\dagger := -\hat{d}^\dagger \hat{d}$), the field Hamiltonian is:

$$\hat{H} =: \int d^3r \hat{\psi}^\dagger \left(-i\hbar c \vec{\alpha} \cdot \vec{\nabla} + \beta m c^2 \right) \hat{\psi} : = \sum_{s=1}^2 \int \frac{d^3p}{(2\pi\hbar)^3} E_p \left[\hat{b}_s^\dagger(\vec{p}) \hat{b}_s(\vec{p}) + \hat{d}_s^\dagger(\vec{p}) \hat{d}_s(\vec{p}) \right] \quad (12.12.3)$$

Both electrons ($\hat{b}^\dagger \hat{b}$) and positrons ($\hat{d}^\dagger \hat{d}$) contribute strictly positive energies $E_p > 0$. The Pauli exclusion principle ($\hat{b}_s^\dagger{}^2 = 0$), the stability of the vacuum, the existence of antimatter, and the spin-statistics theorem are thereby fully reconciled.

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